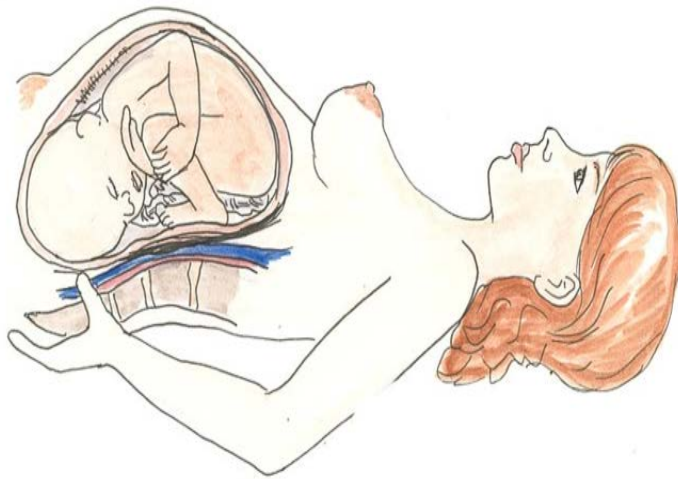


Obstetrics and Gynecology AIIMS 2017

Question 1:

A pregnant lady with PIH was being monitored with bed rest in supine position. Which of the following is a complication of the below representation?



- a) Ascending aorta syndrome
- b) Superior vena caval syndrome
- c) Abdominal aorta syndrome
- d) Supine venacava syndrome

Question 2:

Which of the following is part of the WHO guidelines for the management of second stage of labor?

- a) Do a routine episiotomy
- b) Manual support to perineum to maintain flexion of head
- c) Never give warm compress to the perineum
- d) Perform labor in lithotomy position

Question 3:

Identify the best parameter to assess in case of male infertility

- a) The quantity of sperms
- b) The motility of sperms
- c) The morphology of sperms
- d) The number of sperm

Question 4:

A 26-year-old woman underwent a workup for primary infertility. On hysterosalpingography, there was a bilateral cornual block. What is the next step in management?

- a) Tuboplasty
- b) Hysteroscopy and laparoscopy-guided surgery
- c) Hydrotubation
- d) In-vitro fertilization

Question 5:

A 40-year-old woman comes with a history of postcoital bleeding and a visible mass seen in the cervix per speculum. The next step to be done:

- a) Conization
- b) Cryotherapy
- c) Targeted biopsy
- d) Pap smear

Question 6:

For emergency contraception, a single tab of 0.75 mg of LNG was taken, it should be followed by which of the following?

- a) Single dose of 0.75 after 12 hrs
- b) Two doses 12 hrs apart
- c) Three doses 12 hrs apart
- d) None of the above

Question 7:

A primigravida came to the labor room at 40 weeks + 5 days gestation for induction of labor. On per vaginal examination, the cervix is 1 cm dilated and is 30% effaced. The vertex is at -1 station and the cervix is soft and posterior. What will be the bishop score for this lady?

- a) 0
- b) 3
- c) 5
- d) 8

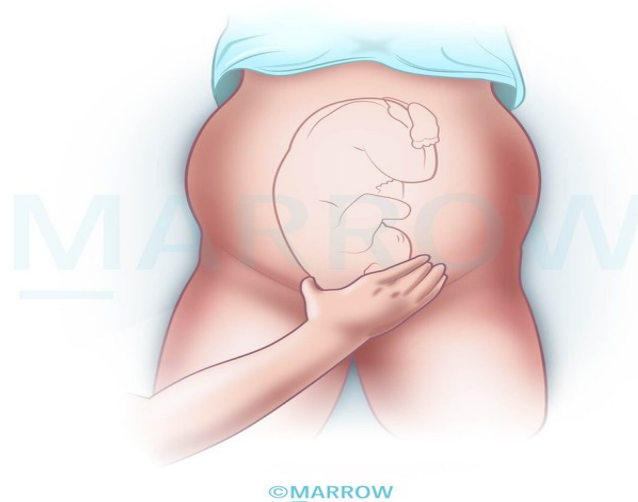
Question 8:

A woman whose obstetric score is G2 P1 L1 had an LMP of Jan 9th 2017. Calculate the EDD according to Naegele's rule.

- a) September 19
- b) October 19
- c) October 16
- d) September 16

Question 9:

Identify the grip in the image given below:



- a) Second Pelvic grip
- b) Abdominal grip
- c) Pawlik grip
- d) Fundal grip

Question 10:

A primigravida came with 6 cm cervical dilatation and a contraction rate of 3/10 minutes. Which stage of labor is she in?

- a) 1st stage
- b) 2nd stage
- c) 3rd stage
- d) 4th stage

Question 11:

Identify the following instrument commonly used in obstetrics.



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- a) Mayo scissors
- b) Curved scissors
- c) Suture remover
- d) Episiotomy scissors

Question 12:

A 53-year-old postmenopausal woman is found to have endometrial hyperplasia with atypia. What is the next step in management?

- a) Progesterone
- b) Hysterectomy
- c) Mirena
- d) Dilatation and curettage

Question 13:

A 16-year-old female patient presents with an ovarian mass. On further evaluation, her serum biomarkers are found to be normal except for LDH, which is found to be elevated. The most likely diagnosis is:

- a) Dysgerminoma
- b) Endodermal sinus tumor
- c) Malignant teratoma
- d) Mucinous cystadenocarcinoma

Question 14:

A pregnant woman has a fetus with Down's syndrome. Triple test done at 15-16 weeks will have _____.

- a) Low MSAFP, high hCG, low UE₃
- b) High MSAFP, low hCG, high UE₃
- c) Low MSAFP, high hCG, high UE₃
- d) High MSAFP, low hCG, low UE₃

Question 15:

In a partogram, where do you start plotting with 1st contractions?

- a) Before the action line
- b) After the action line
- c) Before the alert line

- d) After the alert line

Question 16:

Which of the following drug must always be available for emergency use in labor ward if a patient is on opioid analgesic?

- a) Fentanyl
- b) Naloxone
- c) Morphine
- d) Bupivacaine

Question 17:

A pregnant lady of 49 kg is in her first trimester. What are her additional calorie requirements in this trimester?

- a) 500
- b) 300
- c) Continue same diet
- d) 600

Question 18:

A 18 year old girl with primary amenorrhea, no breast development and Prader 3 genitalia was found to have raised FSH. Her karyotype revealed 45 XO/46 XY and USG revealed Mullerian structures with poorly visualized gonads. What would be the next line of management?

- a) Vaginoplasty
- b) Bilateral gonadectomy
- c) HRT to induce puberty
- d) Clitoroplasty

Question 19:

Fixative for PAP smear:

- a) 10% buffered formalin
- b) 95% ethyl alcohol
- c) Air dried
- d) None of the above

Question 20:

The deepest part of perineal body injury, which causes cystocoele, enterocoele and urethral descent is _____

- a) Pubococcygeous
- b) Bulbospongiosus
- c) Ischiocavernous
- d) Superficial transverse perineal

Question 21:

Which is not seen in pregnancy induced hypertension:

- a) Systemic hypertension within 10 weeks of gestation
- b) Diabetic nephropathy
- c) Diabetic retinopathy
- d) Thrombocytopenia

Question 22:

A couple with 2 children who are not able to conceive for over an year now present to the OPD. The male partner is diagnosed to be having hypogonadotrophic hypogonadism. Which of the following is not true ?

- a) Low LH AND FSH
- b) Low testosterone
- c) Oligospermia
- d) Low prolactin levels

Question 23:

HPV nonavalent vaccine is active against which of the following serotypes?

- a) 6, 11, 16, 18, 20, 23, 35, 42, 52
- b) 6, 11, 16, 19, 23, 33, 45, 58
- c) 6, 11, 16, 18, 32, 34, 45, 52, 58
- d) 6, 11, 16, 18, 31, 33, 45, 52, 58

Question 24:

A 30 year old woman came to OPD with a twin pregnancy at 10 weeks of gestation. She has a 3 year old female child who was born at term and had 2 first trimester abortions previously. Which of the following is the correct representation of her obstetric score/formula. G=gravid, P= para:

- a) G4P2L1A2
- b) G4P1 A2 L1
- c) G5P0 L1 A 2
- d) G5P1 A2 L1

Question 25:

The partogram is plotted in regular intervals after _ cm dilatation:

- a) 5 cm
- b) 8 cm
- c) 4 cm
- d) 6 cm

Question 26:

In fetal alcohol syndrome, which of the following is not seen?

- a) Thinning of corpus callosum
- b) Microcephaly
- c) Holoprosencephaly
- d) Macrocephaly

Question 27:

A 32 year old female nearing her due date has been admitted following regular contractions. The resident doctor did an examination and told the intern that the head is at +1 station. Where is the exact position of head?

- a) At the perineum
- b) Just below ischial spine
- c) Just above ischial spine
- d) High up in the false pelvis

Question 28:

Ureteric injury is most likely associated with which of the following?

- a) Wertheim's hysterectomy
- b) Abdominal hysterectomy
- c) Vaginal hysterectomy
- d) Anterior colporrhaphy

Question 29:

A lady was discharged after a normal vaginal delivery. On third day she came back with fever, tachycardia and seizures. Fundus showed papilledema with no focal deficits. Diagnosis is:

- a) Subarachnoid haemorrhage
- b) Meningitis
- c) Acute migraine
- d) Cortical vein thrombosis

Question 30:

Rh negative mother, ICT negative was given Anti D during 28 week of pregnancy. Which of the following is the ideal one

- a) No need additional dose since she is ICT negative
- b) Give another dose of Anti D 72 hour postpartum irrespective of baby blood group
- c) Give another dose of Anti D 72 hour postpartum depending on the baby blood group

- d) All of the above**

Question 31:

A 25 year old female came with complaints of 4 months amenorrhea. Her FSH and LH were elevated above the normal value, thyroid function tests were normal. Which is the next step in management ?

- a) Estimate serum estradiol values**
- b) Give progesterone and stop after 10 days for withdrawal bleeding**
- c) Check USG for polycystic ovaries**
- d) Do a urine pregnancy test**

Question 32:

A 25-year-old female came with complaints of greenish discharge per vaginum and on examination she had a strawberry cervix. Which of the following is the treatment for her

- a) Doxycycline**
- b) Amoxicillin**
- c) Fluconazole**
- d) Metronidazole**

Question 33:

A 30 year old female came for routine PAP smear testing. The report came as carcinoma in situ. What is the next step?

- a) Conisation**
- b) Hysterectomy**
- c) Colposcopy and biopsy**
- d) HPV DNA testing**

Question 34:

A 62-year-old female with history of ovarian carcinoma in the mother achieved menopause with smooth transition. And she is on hormone replacement therapy. She developed

abdominal pain and on examination the right adnexal mass was palpable, non tender. On USG examination she has a 3 X 3 cm smooth cyst in her ovary. Which is the next step

- a) Reassure and advice regular follow up once in 6 months
- b) Confirm USG findings by ordering CT scan
- c) Do an exploratory laparoscopy to visualise the nature of cyst
- d) Do CA-125 levels, if normal advise regular check up

Question 35:

In endometriotic lesion, histology represents its:

- a) High cholesterol
- b) High estrogen
- c) High levels of prolactin
- d) High progesterone

Answer Key

Question No.	Correct Option
1	d
2	b
3	c
4	b
5	c
6	a
7	c
8	c
9	c
10	a
11	d
12	b
13	a
14	a
15	c

16	b
17	c
18	b
19	b
20	a
21	a
22	d
23	d
24	b
25	c
26	d
27	b
28	a
29	d
30	c
31	a
32	d
33	c
34	d
35	b

Detailed Explanations

Solution to Question 1:

Supine vena cava syndrome is the complication of the given representation. The image shows compression of the inferior vena cava by the gravid uterus which can sometimes lead to supine hypotension.

During late pregnancy, the gravid uterus compresses the great vessels against the hard spine, when the patient is in the supine position. Of the two vessels, the vena cava is compressed, but the aorta, owing to its structure is relatively resistant.

Normally, the vena cava compression results in opening up collateral circulation by means of paravertebral and azygous veins but in some cases, when the collateral circulation fails to open up, the venous return of the heart may be seriously curtailed resulting in hypotension, tachycardia, and syncope and is called supine vena cava syndrome.

The normal blood pressure is quickly restored by turning the patient to the lateral position.

Solution to Question 2:

The practice of guarding the perineum during delivery of the fetal head is part of the WHO management guidelines for the second stage of labor.

The fingers of one hand support the perineum, while the second hand applies pressure to the fetal head to control the speed of crowning. The advantage of this is to prevent perineal tears.

The second stage of labor begins with full dilatation of the cervix and ends with the delivery of the fetus.

The main events include the descent of the fetus (bearing down by the mother, uterine contractions), cardinal movements of labor, and delivery of the fetus.

Management of the second stage of labor:

- The most common position used is the dorsal position with the hips flexed, knees flexed and abduction of the thighs. WHO recommends any position of choice including the upright position.
- WHO recommendations for the prevention of perineal tears include warm compresses to the perineum, perineal massage, guarding the perineum, and modified Ritgen's maneuver.
- Modified Ritgen's maneuver: A sterile towel or pad is held in one hand and placed over the perineum; the other hand applies gentle downward pressure on the occiput to promote flexion and controlled delivery of the head (ensuring the smallest diameter of the head passes through the outlet). Once the occiput is born—with gentle pressure, the head is extended—forehead, nose, and mouth are then delivered.
- WHO doesn't recommend fundal pressure and routine episiotomy.

Other options:

Option A: A liberal or routine episiotomy is discouraged. Episiotomy is indicated in cases of:

- Shoulder dystocia
- Breech delivery
- Fetal macrosomia
- Operative vaginal deliveries
- Persistent occiput posterior positions

Option C: Manual techniques, such as massage or application of warm compresses, are recommended for healthy pregnant women requesting pain relief during labor.

Option D: The position that is most comfortable to the mother i.e. upright/squatting position is encouraged. This position gives:

- less discomfort and difficulty in bearing down
- less labor pain
- less perineal/vaginal trauma and wound infections

Second stage	Nulliparous	Multiparous
Mean/average duration	50 minutes	30 minutes
Maximum duration	2 hours (+1 with epidural)	1 hour (+1 with epidural)

Second stage	Nulliparous	Multiparous
WHO maximum duration of 2nd stage/diagnosis of arrest	3 hours	2 hours

Solution to Question 3:

The morphology of sperms is the best parameter to assess male infertility.

Characterization of normal sperm morphology shows improved correlation with fertilization rates during IVF cycles.

The criteria requires careful analysis of the following:

- The shape and size of the sperm head,
- The relative size of the acrosome in proportion to the head,
- And characteristics of the tail, including length, coiling, or presence of two tails.

Significantly decreased fertilization rates are observed when normal morphology of the sample falls below 4 percent.

Abnormal sperm morphology is termed teratospermia or teratozoospermia. Whereas sperm concentration and progressive motility have value for distinguishing fertile from infertile men, strict sperm morphology (as determined by an individual having extensive training and experience) was the one most discriminating value.

Parameters In Semen Analysis	Reference Limits
Volume	>1.5mL
Count	>15 million/mL
Total motility	>40%
Morphology	>4% normal forms
WBCs	< 1 million/mL
Round cells	<5 million/mL

Solution to Question 4:

In the given clinical scenario, the next step in management will be hysteroscopy and laparoscopy-guided surgery.

Tubal contractions, particularly cornual spasms, can give an incorrect impression of proximal fallopian tube obstruction (a false-positive result) on hysterosalpingography. Laparoscopy with chromotubation is therefore considered to confirm the diagnosis of cornual spasm. Laparoscopy is both diagnostic and therapeutic in this condition.

Hysteroscopy is an operative procedure that provides an endoscopic view of the endometrial cavity and tubal ostia via the cervix, performed in the preovulatory phase. It can be diagnostic or therapeutic. It is the gold standard investigation for visualizing the uterine cavity and tubal ostia.

Contraindications to hysteroscopy are as follows:

- Acute genital tract infection (Eg: genital herpes infection)
- Viable intrauterine pregnancy
- During menstruation, the view is obscured, and the infection rate is high.
- Scarred uterus and enlarged uterus of size ≥ 12 weeks are relative contraindications.
- Cervical stenosis can cause cervical tears and uterine perforation.
- Cardiopulmonary disorder due to increased risk of fluid overload and pulmonary edema.
- Known cases of cervical cancer

Laparoscopy is the gold standard for the evaluation of tubal factors in infertility, commonly done in the proliferative phase. It can be diagnostic or therapeutic.

Benefits of laparoscopic evaluation:

- Tubes: For detection of tubal patency, block (site and side), motility, hydrosalpinx change, peritubal adhesions, fimbrial agglutination
- Ovaries: PCOS changes, endometriosis, PID, adhesions
- Uterus: Anomalies, fibroids
- Peritoneal factors: Adhesions, PID, endometriosis, tuberculosis
- Therapy at the same sitting (as appropriate)

Therapeutic indications of hysteroscopy	Diagnostic indications of hysteroscopy
Uterine septum Asherman syndrome Retained IUCD Polyp ectomy Submucous fibroid Abnormal uterine bleeding Tubal sterilization using sclerosing agents	To study the endocervical mucosal lining Congenital malformations of the uterus Endometrial tuberculosis Misplaced IUCD Endometrial lesions Endometrial polyp Asherman syndrome Cornual tubal blockage

Diagnostic	Therapeutic (Operative)
Age >35 years History of pelvic pain, dyspareunia, pelvic surgery, adhesions Abnormal HSG Failure to conceive after a reasonable period (6 months) with normal HSG Women with comorbid pelvic pathology (PID, endometriosis)	Unexplained infertility Reconstructive tubal surgery Adhesiolysis Fulguration of endometriotic implants Ovarian drilling in PCOS GIFT and ZIFT procedures

Solution to Question 5:

A targeted biopsy is the next line of management in the patient with a visible cervical mass and postcoital bleeding. A pap smear is done when there are symptoms with no identifiable lesion.

Other options:

Option A: Conization is an invasive procedure where the cervix is cut in a cone-shaped manner. It is both diagnostic and therapeutic. It is indicated for diagnosis in women with HSIL or adenocarcinoma in situ and may be considered under the following conditions:

- The limits of the lesion cannot be visualized with colposcopy.
- SCJ (squamocolumnar junction) is not seen in colposcopy.
- Colposcopy is positive for CIN 2/CIN 3.
- Lack of correlation between cytology, biopsy, and colposcopy.
- Suspected micro invasion in cytology, colposcopy, or biopsy.
- Colposcopist is unable to rule out invasive cancer.

Hence it is very clear that conization is not indicated for all symptomatic women and it should be done following either a PAP, colposcopy or a biopsy with an abnormal or inconclusive result.

Option B: Cryotherapy is ideal only for small superficial lesions. It is indicated for biopsy-proven lesions. Cryotherapy is appropriate for CIN 1 which has persisted for more than 24 months and is confined to the ectocervix.

Option D: The false negative rate of pap smear in the presence of invasive cancer is 50% and a negative pap test should never be relied on in a symptomatic patient, especially one with a visible lesion.

CIN: Cervical intraepithelial neoplasia, HSIL: High squamous intraepithelial lesion, LSIL: Low squamous intraepithelial lesion

Solution to Question 6:

In this case, the 2nd dose of LNG 0.75 mg (levonorgestrel) is taken after 12 hours.

Levonorgestrel can be used in emergency contraception without major side effects. One tablet of LNG 0.75 mg should be taken within 72 hours of unprotected intercourse, and another after 12 hours. Alternatively, two tablets can be as a single dose (1.5 mg).

The tablets can be offered up to 120 hours but the efficacy decreases with the longer coital-drug interval.

Advantages of LNG tablets:

- It has no estrogen and its associated side effects.
- It can be offered to hypertensive, cardiac and diabetic women.

- It can be offered to a lactating woman.
- It can be given as late as 120 hours after unprotected intercourse (but efficacy decreases).
- Single-dose therapy is an advantage.

Contraindications to LNG tablets include liver disease, a history of thrombophlebitis and migraine.

Indications for emergency contraception:

- Unprotected intercourse
- Condom rupture
- Missed pill
- Delay in taking progesterone-only pill for ≥ 3 hours
- Rape/Sexual assault

Drugs used in emergency contraception:

- Mifepristone (RU486) is a progesterone antagonist, which blocks the hormonal action on the endometrium, causing the sloughing and shedding of decidua, preventing implantation. A single dose of 25-50 mg is effective in pregnancy in 99.1% of cases. It causes delayed menstruation and expensive than LNG.
- Ulipristal is a synthetic progesterone hormone receptor modulator. It binds to progesterone receptors and prevents/delays ovulation and suppresses endometrium, preventing implantation. A 30 mg tablet should be taken within 5 days. A two per cent pregnancy rate has been reported. Side effects are headaches and mood changes.
- Centchroman: 2 tablets (60 mg) taken twice in 24 hours, within 24 hours of intercourse can prevent implantation in 99% of women.
- Copper T IUCD: inserted within 5 days of unprotected intercourse can prevent implantation of a fertilized ovum. The advantages of copper T IUCD are- it is cheap, has a failure rate of 0.1%, inserted as late as 5 days after unprotected intercourse and remains an ongoing contraceptive method for 3-5 years.
- Yuzpe method: ethinyl estradiol 50 μ g + norgestrel 0.25 mg. It is given as 2 tab stat and 2 after 12 hours.

Solution to Question 7:

The Bishop's score, in this case, is 5.

Bishop's pre-induction Cervical Scoring System.

Total score = 13; Favorable score = 6-13; Unfavorable score = 0-5

In the modified Bishop score (1991) the effacement is replaced by cervical length in cm.

From the data given in the question:

- Cervical dilatation is 1 cm, hence score = 1.

- Effacement is 30%, hence score = 0
- Station of the head is -1, hence score = 2
- Consistency of cervix is soft, hence score = 2
- Position of the cervix is posterior, hence score = 0

Hence, Bishop's score is $1+0+2+2+0 = 5$.

ScoreCervix	0	1	2	3
Dilatation (cm)	Closed	1-2	3-4	>5
Effacement (%)	0-30	40-50	60-70	≥80
Consistency	Firm	Medium	Soft	-
Position	Posterior	Midline	Anterior	-
Head station	-3	-2	-1, 0	+1, +2

Score	0	1	2	3
Cervical length in (cm)	More than 4	2-4	1-2	Less than 1

Solution to Question 8:

As per Naegele's rule, the EDD (expected date of delivery) of this woman is October 16, 2017.

Naegele's rule states that EDD is obtained by adding 7 days and subtracting 3 months from the LMP (the 1st day of the last menstrual period). Another method is adding 9 months and 7 days to LMP. Less than 5% of births follow Naegele's rule.

It is based on a 28-day regular cycle. If the cycle is shorter/longer than 28 days, corrected EDD is calculated as follows:

- For a 35-day cycle, add another 7 ($35-28=7$) days to the calculated EDD to get the corrected EDD.
- For a 25-day cycle, subtract 3 ($28-25=3$) days from the calculated EDD to get the corrected EDD.

Naegele's rule cannot be followed in IVF pregnancies. Considering the human embryo takes 266 days to develop, EDD is calculated in IVF pregnancies by the following method:

- Same day as fertilization (day 0): EDD = date of transfer + 266 days
- 3 days after fertilization (day 3): EDD = date of transfer + 263 days
- 5 days after fertilization (day 5): EDD = date of transfer + 261 days

Solution to Question 9:

Leopold's third maneuver (also called the Pawlik's grip) is performed by grasping with the thumb and fingers, the lower portion of the maternal abdomen just above the symphysis pubis. If the presenting part is not engaged, a movable mass will be felt.

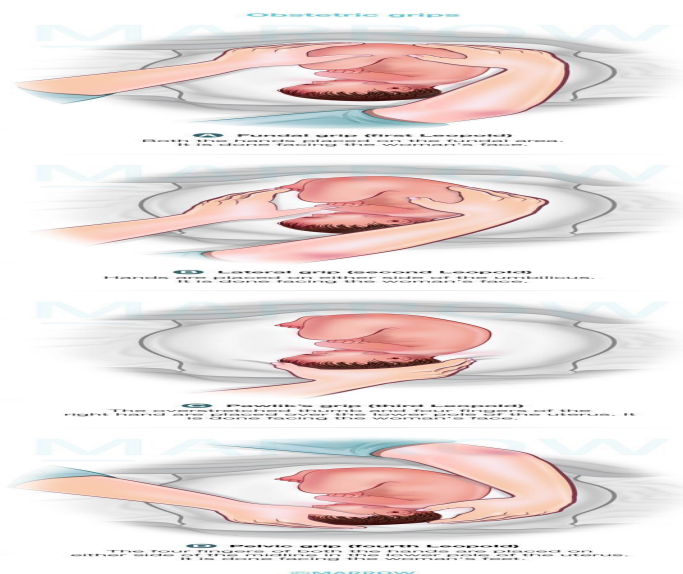
Abdominal palpation- Leopold maneuvers:

In the abdominal examination, the mother is made to lie supine and comfortably positioned with her abdomen bared. Leopold maneuvers can be performed throughout the latter months of pregnancy and during and between the contractions of labor.

Leopold's maneuvers may be difficult if not impossible to perform and interpret if the patient is obese, if the amniotic fluid volume is excessive, or if the placenta is anteriorly implanted.

With experience, it is possible to estimate the size of the fetus.

- 1st Leopold (fundal grip): identifies which fetal pole (cephalic or podalic) occupies the uterine fundus. The fetal head is hard and round and is more mobile and ballotable whereas the breech is felt as a large, soft and irregular mass.
- 2nd Leopold (lateral/umbilical grip): palms are placed on either side of the abdomen and gentle, but deep pressure is exerted. The back is felt as a hard and resistant structure. The fetal limbs are felt as numerous, small, irregular parts. By noting whether the back is directed anteriorly, transversely or posteriorly, the fetal orientation can be determined.
- 3rd Leopold (pawlik's grip): it aids in confirmation of fetal presentation. In transverse lie, pawlik's grip is empty.
- 4th Leopold (pelvic grip): it helps to determine the degree of descent.



Solution to Question 10:

In this case, cervical dilatation of 6 cm and contraction rate of 3/10 minutes is indicative of first stage of labor.

The first stage of labor is characterized by dilatation and effacement of cervix and can be divided into latent phase and active labor.

- The onset of latent labor is the point at which the mother perceives regular contractions.
- Cervical dilatation of ≥ 6 cm in the presence of uterine contractions can be taken to reliably represent the threshold for active labor.

The second stage of labor begins with complete cervical dilatation and ends with fetal delivery.

The third stage of labor is characterized by the delivery of the placenta.

The fourth stage is a stage of observation for at least an hour after the full expulsion of the placenta.

Solution to Question 11:

The instrument given in the picture is episiotomy scissors. The angulation in the scissor is to prevent the extension of pelvic tears into the anal margins and obstetric anal sphincter injuries / complete perineal tears.

Complications of episiotomy:

They can be immediate or remote.

Immediate complications include an extension of the incision into the rectum, vulval hematoma, wound dehiscence and injury to anal sphincter.

Remote complications include dyspareunia, chances of perineal lacerations in subsequent labor and scar endometriosis (rare).

Mayo scissors are long and stout scissors. They can be blunt-tipped/pointed tip/straight or curved. Straight mayo scissors are designed for cutting body tissues near the surface of a wound. They are also used to cut sutures and are also known as suture-cutting scissors.

Curved mayo scissors allow much deeper penetration into the wound and are used to cut thick tissues such as those found in the uterus, muscles, breast, and foot. Mayo scissors used for dissection are placed in tissue with the tips closed. The scissors are then opened so that the tips open and spread out the tissue during the dissection process.

Episiotomy scissor



©Marrow

Normal Mayo scissors



©Marrow

Solution to Question 12:

Among the given options, for post-menopausal women with atypical endometrial hyperplasia, the next step in management would be a hysterectomy because the risk of progression to cancer over time approximates 29 per cent.

Due to a high rate of finding concurrent invasive malignancy coexistent with atypical hyperplasia, the ideal management is to perform a fractional curettage with hysteroscopy before proceeding with a hysterectomy (with removal of both tubes and ovaries).

Endometrial hyperplasia is endometrial thickening with a proliferation of irregularly sized and shaped glands along with an increased gland-to-stroma ratio.

Classification of endometrial hyperplasia:

- Simple hyperplasia is characterised by dilated or cystic glands with round to slightly irregular shapes, an increased glandular-to-stromal ratio without glandular crowding, and no cytologic atypia.
- Complex hyperplasia has architecturally complex (budding and infolding) crowded glands, with less intervening stroma without atypia.
- Atypical hyperplasia refers to cytologic atypia and can be categorised as simple or complex, depending on the corresponding glandular architecture. Criteria for cytologic atypia include
 -
 - large nuclei of variable size and shape with lost polarity
 - increased nuclear-to-cytoplasmic ratios
 - prominent nucleoli
 - irregularly clumped chromatin with parachromatin clearing

Type of hyperplasia	Progression to cancer (%)
Without atypia	
Simple cystic	1
Complex adenomatous	3
With atypia	
Simple cystic	8
Complex adenomatous	29

Solution to Question 13:

The given clinical history of a 16-year-old female presenting with an ovarian mass and raised LDH, points towards the diagnosis of dysgerminoma.

Dysgerminoma corresponds to the seminoma of the testis and accounts for 3-5% of all ovarian tumors. It usually arises in young girls or in children, with an average age of 20 years. It is the most common malignant tumor of the ovary. It is usually unilateral, but they are the only germ cell malignancy with a significant rate of bilateral ovarian involvement (15-20%) and can be found at gonadal or extragonadal sites.

Genetics:

- Express OCT-3, OCT-4 and NANOG
- Activating mutations in the KIT gene in one-third of cases

Pathology:

- Solid neoplasm with areas of softening due to degeneration with fleshy consistency

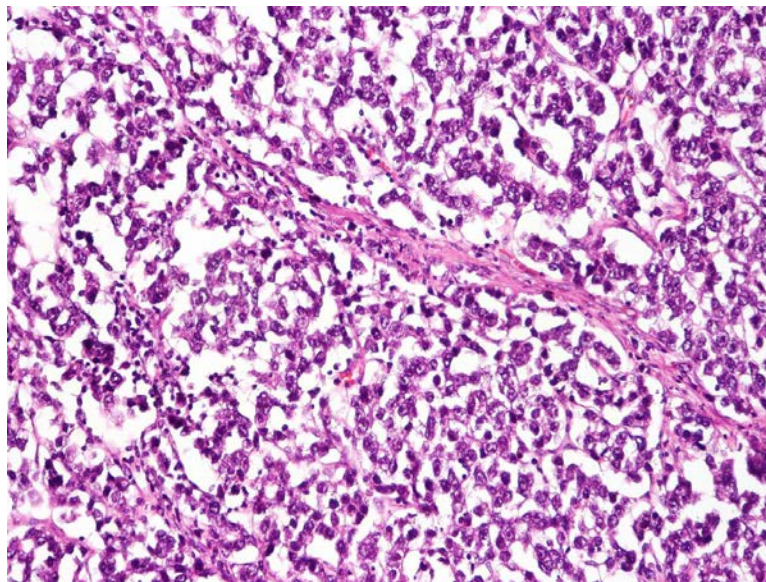
- Mimics the pattern of the primitive gonad, lymphocytic infiltration is seen

It presents with abdominal pain, distension, a palpable mass and ascites. Some tumors occur in patients with gonadal dysgenesis, including pseudohermaphroditism. It secretes placental alkaline phosphatase, lactate dehydrogenase and beta-HCG, but no sex hormones.

Morphology and histology:

- Most dysgerminomas (80% to 90%) are unilateral tumors ranging in size from barely visible nodules to masses that virtually fill the abdomen.
- On cut surfaces, they have a solid yellow-white to gray-pink appearance and are often soft and fleshy.
- Like seminoma, it is composed of large vesicular cells with clear cytoplasm, well-defined cell boundaries, and centrally placed regular nuclei.
- The tumor cells grow in sheets or cords separated by scant fibrous stroma, which is infiltrated by lymphocytes and may contain noncaseating granulomas

The image below shows the histopathology of a dysgerminoma, characterized by uniform cells resembling primordial germ cells separated by fibrous septa with lymphocytes.



Management:

- Though dysgerminomas are radiosensitive, they are contraindicated due to ovarian destruction in young girls. It is employed only for residual and recurrent tumors.
- Treatment includes surgical removal and postoperative chemotherapy.

Chemotherapy comprises of

- Injection bleomycin 15 mg i.v (or) i.m weekly for 12 weeks
- Injection etoposide 100mg/m² 1-5 days every 3 weeks
- Injection cisplatin 20 mg/m² 1-5 days 3 weeks

Solution to Question 14:

The triple test findings in pregnancies with fetal Down's syndrome are characterized by:

- Low MSAFP (maternal serum alpha-fetoprotein) levels, approximately 0.7 MoM (multiples of median).
- Low unconjugated estriol levels, approximately 0.8 MoM.
- High hCG levels, approximately 2.0 MoM.

The triple test is done between 15-18 weeks for the diagnosis of Down's syndrome.

Various antenatal screening tests for aneuploidy are as follows:

First-trimester screening with biochemical analytes (↓) PAPP-A and (↑) hCG and USG measurement of NT (↑) can improve the detection rate (87%) of Down's syndrome. For confirmation, a prenatal genetic study (CVS, amniocentesis or cordocentesis) has to be performed.

Second-trimester screening (quad screening) at 15–18 weeks: (↓) MSAFP, (↓) uE₃, (↑) inhibin A and (↑) hCG can detect trisomy 21 in 85% of cases with a false-positive rate of 0.9%. When it is combined with first-trimester screening, the detection rate is increased to 95%.

Dual test (1sttrimester)	Triple test (2ndtrimester)	Quadruple test (2ndtrimester)
Beta HCG	Beta HCG	Beta HCG
PAPPA	Unconjugated estriol	Unconjugated estriol
This can be combined with nuchal translucency by USG	MSAFP	MSAFP
	Seruminhibin A	

Solution to Question 15:

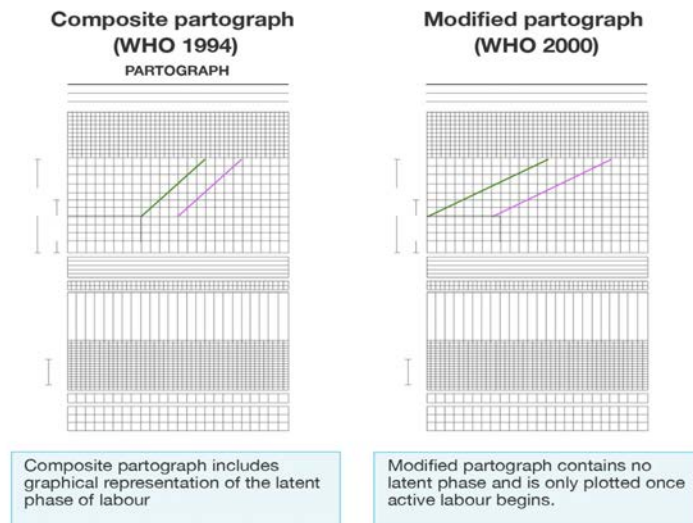
In a partograph, labor is graphed and analysis includes use of action lines and alert lines.

A partograph was designed by WHO for use in developing countries. Partograph is similar for nulliparas and multiparas.

Active phase starts at 3cm cervical dilatation and progress should be no slower than 1cm/hour.

A 4 hour wait is recommended before intervention when the active phase is slow.

In the modified partograph, there is no plotting of the latent phase. The partograph is plotted once active labor begins.



Solution to Question 16:

Naloxone is a narcotic antagonist capable of reversing respiratory depression induced by opioid narcotics.

It acts by displacing the narcotic from specific receptors in the central nervous system.

Naloxone:

It is a competitive antagonist on all types of opioid receptors. It is devoid of any kind of agonistic activity even at high doses. Injected intravenously (0.4–0.8 mg) it promptly antagonizes all actions of morphine: analgesia is gone, respiration is not only normalized but stimulated—probably due to sudden sensitization of the respiratory centre to the retained CO₂, or it may be a manifestation of acute withdrawal; pupils dilate. However, sedation is less completely reversed.

Naloxone is inactive orally because of high first-pass metabolism in the liver. Injected i.v. it acts in 2–3 min. The primary pathway of metabolism is glucuronidation. Plasma $t_{1/2}$ is 1 hour in adults and 3 hours in newborns. Duration of action is 1–3 hrs.

Adverse effects of naloxone are uncommon, and may include a rise in BP and pulmonary edema; Naloxone 0.4 mg i.v. precipitates morphine withdrawal in dependent subjects: the syndrome lasting 2–3 hours.

Uses of naloxone:

- drug of choice for morphine poisoning (0.4–0.8 mg i.v. every 2–3 min: max 10 mg)
- reversing neonatal asphyxia due to opioid use during labour (10 µg/kg in the cord)
- treat the overdose with other opioids and agonist-antagonists (except buprenorphine)
- to reverse respiratory depression due to intraoperative use of opioids: 0.1–0.2 mg i.v. (this dose usually preserves analgesia in the postoperative period)

- tried as an adjunct to intraspinal opioid analgesia: reverses respiratory depression without abolishing pain relief

Naloxone is contraindicated in newborns of narcotic-addicted mothers as it can lead to neonatal abstinence syndrome (NAS).

NAS is a drug-withdrawal syndrome that most commonly follows in-utero exposure to maternal opioids. It also can complicate exposure to alcohol or benzodiazepines. The syndrome shows hypertonia, autonomic instability, irritability, poor sucking reflex, and seizures. Care includes close observation and pharmacotherapy with morphine and methadone. Naloxone is not recommended as it can precipitate withdrawal symptoms.

Solution to Question 17:

The pregnant lady can continue the same diet since there are no additional calorie requirements in the first trimester.

Pregnancy requires an additional 80,000 kcal, mostly during the last 20 weeks. To meet this demand, a caloric increase of 100 to 300 kcal/d is recommended during pregnancy. This greater intake, however, should not be divided equally during the course of pregnancy. The Institute of Medicine (2006) recommends adding 0, 340, and 452 kcal/d to the estimated nonpregnant energy requirements in the first, second, and third trimesters respectively.

Whenever caloric intake is inadequate, protein is metabolized rather than being spared for its vital role in fetal growth and development.

Solution to Question 18:

In Turner's syndrome, most affected individuals have a 45,X karyotype, while others have mosaic karyotypes (45,X/46,XX; 45,X/46,XY). The presence of Y chromosome material is associated with a 12% risk of a gonadoblastoma. If a Y chromosome is identified, laparoscopic prophylactic gonadectomy is recommended at the time of diagnosis to eliminate the risk of malignancy.

- Gonadoblastomas can be precursors to germ cell malignancies, such as dysgerminomas (most commonly), teratomas, embryonal carcinomas, or endodermal sinus tumors.
- The karyotype should be determined in any individual with delayed puberty and increased basal FSH concentrations.
- Regardless of the karyotype, the individual with hypergonadotropic hypogonadism has some form of ovarian "failure"

Note: Prader staging system is used for the degree of virilisation of the external genitalia

Prader 0: Classical / typical female external genitalia

Prader 1: Female external genitalia with clitoromegaly

Prader 2: Clitoromegaly with partial labial fusion forming a funnel-shaped urogenital sinus

Prader 3: Increased phallic enlargement. Complete labioscrotal fusion forming a urogenital sinus with a single opening

Prader 4: Complete scrotal fusion with urogenital opening at the base or on the shaft of the phallus.

Prader 5: Classical / typical male external genitalia

Solution to Question 19:

The most commonly used fixative used for pap smear is 95% ethyl alcohol.

Historically ether was used along with alcohol as a fixative for pap smear. Nowadays a fixative ethanol spray is available and can be used conveniently in an office set-up.

For further reading on the development of Pap smear technique refer <https://www.karger.com/Article/PDF/457827>

Solution to Question 20:

Pubococcygeus is the deepest part of the levator ani muscle. Injury to it can cause enterocele, cystocele, and urethral descent.

Levator ani muscles are the most important muscles in the pelvic floor and maintain a constant state of contraction. They provide a stable floor and support the weight of the abdominopelvic contents against intra-abdominal forces. Weakening of levator ani and pelvic fascia from stretching or tearing during childbirth, may decrease support for the vagina, bladder, uterus, or rectum or alter the position of the neck of the bladder and urethra.

The levator ani muscle is a complex unit, which consists of 3 muscle components namely

- Pubococcygeus (anterior fibers)
- Pubovaginalis - elevates the urethra and hence contributes to urinary continence
- Puboperinealis - draws the perineal body towards the pubic symphysis
- Puboanalis - elevates the anus and keeps the urogenital hiatus narrowed
- Puborectalis (intermediate fibers)
- Iliococcygeus (posterior fibers)

Origin: The levator ani muscle has a linear origin from the pelvic surface of the body of the pubis, a tendinous arch of obturator fascia, and the pelvic surface of the ischial spine.

Insertion:

- The anterior fibres form a sling around the prostate (levator prostatae) or vagina (sphincter vaginae) and are inserted into the perineal body (a mass of fibrous tissue) in front of the anal canal.
- The intermediate fibres (puborectalis) form a sling around the anorectal junction to be inserted into the anococcygeal raphe (a fibrous raphe which extends from the anorectal junction to the tip

of the coccyx).

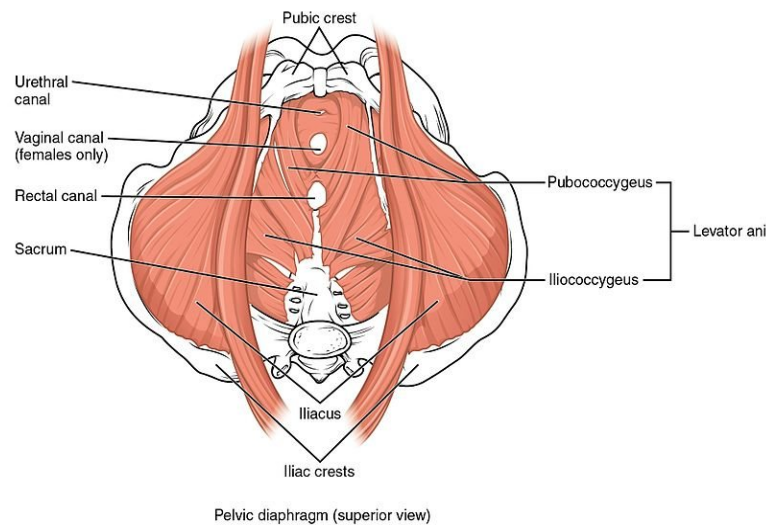
- The posterior fibres (iliococcygeus) are inserted into the anococcygeal raphe and the coccyx.

Nerve supply:

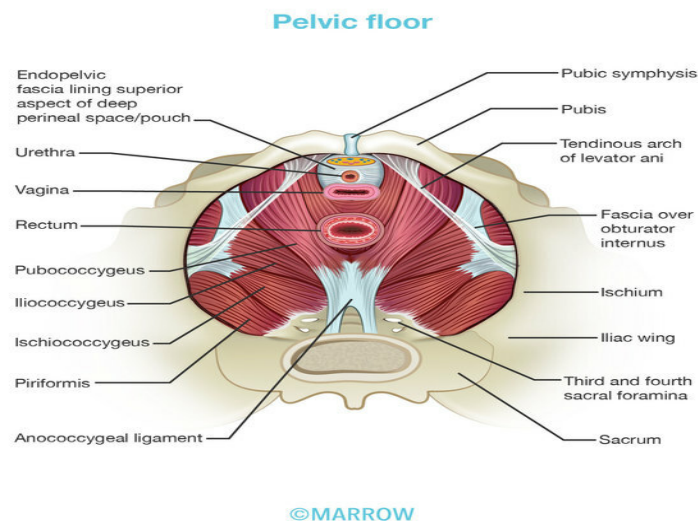
- perineal branch of 4th sacral nerve (S4) from its pelvic surface.
- perineal branch of the pudendal nerve (S2, S3) from its perineal surface.

Actions:

- The levator ani muscles of two sides together support the pelvic viscera and resist the intra-abdominal pressure during straining and expulsive efforts of the anterior abdominal wall muscles.
- They also subserve a sphincteric action on the anorectal junction to maintain the continence of faeces in both sexes and vagina in females.



The given image below shows the pelvic floor:



Solution to Question 21:

Hypertension induced by pregnancy is gestational hypertension which is characterized by BP more than 140/90 mmHg after 20 weeks of gestation in a previously normotensive woman.

The incidence of gestational hypertension and preeclampsia is remarkably increased in diabetic mothers.

40% of pregnant women with diabetes will develop preeclampsia. Chronic hypertension with overt diabetes increased the risk of preeclampsia to 60%

Hypertension is a risk factor associated with the progression of diabetic retinopathy.

The platelet count is usually found to be decreased in gestational hypertension.

Solution to Question 22:

A male with hypogonadotropic hypogonadism will have high prolactin levels.

Hypogonadotropic hypogonadism:

- Low LH & FSH
- Low testosterone
- Low testicular volume (oligospermia)
- High prolactin levels

Hypergonadotropic hypogonadism:

- High LH and FSH
- Low testosterone
- Low testicular volume (oligospermia)
- High prolactin level

Hypogonadotropic hypogonadism:

It can be congenital or acquired. Congenital hypogonadotropic hypogonadism is divided into anosmic hypogonadotropic hypogonadism (Kallmann syndrome) and congenital normosmic isolated hypogonadotropic hypogonadism (idiopathic hypogonadotropic hypogonadism).

Acquired hypogonadotropic hypogonadism can be caused by drugs, infiltrative or infectious pituitary lesions, hyperprolactinemia, encephalic trauma, pituitary/brain radiation, exhausting exercise, abusive alcohol or illicit drug intake, and systemic diseases such as hemochromatosis, sarcoidosis and histiocytosis X.

The clinical characteristics of hypogonadotropic hypogonadism are androgen deficiency and a lack/delay/stop of pubertal sexual maturation. Low blood testosterone levels and low LH and FSH levels confirm the diagnosis. A prolonged stimulated intravenous GnRH test can be useful. In

Kallmann syndrome, cerebral MRI can show an anomalous morphology or even absence of the olfactory bulb.

Therapy for hypogonadotropic hypogonadism depends on the patient's desire for future fertility. Hormone replacement with testosterone is the classic treatment for hypogonadism. Androgen replacement is indicated for men who already have children or have no desire to induce pregnancy, and testosterone therapy is used to reverse the symptoms and signs of hypogonadism. Conversely, GnRH or gonadotropin therapies are the best options for men wishing to have children.

Solution to Question 23:

HPV nonavalent vaccine (Gardasil 9 or HPV9) is active against HPV 6,11,16,18,31,33,45,52,58.

Currently, HPV nonavalent vaccine has replaced Gardasil which is a quadrivalent vaccine acting against 6,11,16,18.

Vaccines available for HPV:

Brand name	Cervarix	Gardasil	Gardasil-9	Cervavac
Characteristic and HPV subtypes	Bivalent:16,18	Quadrivalent:6,11,16,18	Nonavalent:6,11,16,18,31,33,45,52,58	Quadrivalent:6,11,16,18
Protects against:	CINCA cervix	Anogenital wartsCINCA cervix	Anogenital wartsCINCA cervixVulval intraepithelial neoplasiaVaginal intraepithelial neoplasiaAnal cancerOropharyngeal cancerHead and neck cancers	Anogenital wartsCINCA cervix
Dose	0.5 ml,IM.	0.5 ml,IM.	0.5 ml,IM.	0.5 ml,IM.
Ideal Age range:	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	The targeted age group is 9-26 years

Solution to Question 24:

The obstetric score is G4P1A2L1.

Gravida and para refer to pregnancies and not to babies.

Gravida: Total number of pregnancies including the present one.

Para: Total number of pregnancies crossing the period of viability, which is 28 weeks in India.

As such, a woman who delivers twins in the first pregnancy is still a gravida 1 and para 1. A pregnant woman with a previous history of 2 abortions and 1 term delivery can be expressed as 4th Gravida but primipara.

It is customary in clinical practice to summarise the past obstetric history by 2 digits

First digit: Number of viable births

Second digit: Number of abortions

Both digits connected with a "+" sign affixing the letter P.

Thus, P1+2 denotes patient had 1 viable birth and 2 abortions.

In some centres, it is expressed by 4 digits connected by dashes, P(A-B-C-D)

A: Number of term pregnancies (37-42 weeks)

B: Number of preterm pregnancies (28 to less than 37 weeks)

C: Number of miscarriages (less than 28 weeks)

D: Number of children alive at present.

Solution to Question 25:

The partograph is plotted in regular intervals after 4 cm dilatation.

In a partograph, the labor process is divided into

- Latent phase ends when cervix dilated upto 4 cm
- Active phase begins with cervical dilatation of ≥ 6 cm and ends at 10 cm dilatation at the rate of 1 cm/hour.

The latent phase of labour is slow and extremely variable, while the active phase of labour is fairly predictable in terms of duration. Even though the active phase of labor is from more than 6cm dilatation, partograph is marked from 4 cm dilatation.

Solution to Question 26:

Fetal alcohol syndrome is associated with microcephaly rather than macrocephaly.

- High levels of alcohol ingestion during pregnancy can be damaging to embryonic and fetal development.
- The presence of at least one of the below characteristics is defined as fetal alcohol syndrome:
- Growth restriction before and/or after birth
- Facial anomalies: small palpebral fissures, indistinct or absent philtrum, epicanthic folds, flattened nasal bridge, short length of nose, thin upper lip, lowset and unparallel ears, retarded midfacial development.
- CNS dysfunction: Microcephaly, mental retardation, abnormal neurobehavioral development (attention deficit with hyperactivity)

Other features:

- Incomplete holoprosencephaly

- Agenesis of corpus callosum and the cerebellar vermis.
- Cardiac anomalies :VSD>ASD ,tetralogy of fallot.

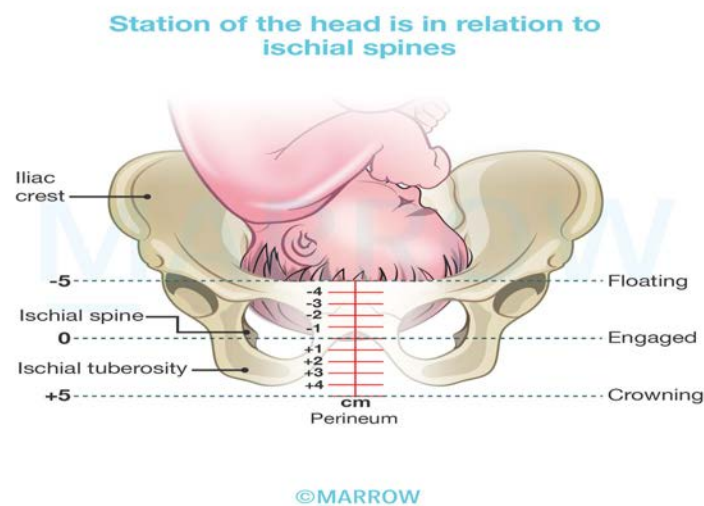
Solution to Question 27:

The station is just below the ischial spine.

Station of the head is in relation to ischial spines as spines are the most prominent bony projections felt on internal examination.

The level of ischial spines is the halfway between the pelvic inlet and outlet. This level is known as station zero (0).

The levels above and below the spines are divided into fifths to represent centimeters. The station is said to be "0" if the presenting part is at the level of the spines. The station is stated in minus figures, if it is above the spines (–1 cm, –2 cm, –3 cm, –4 cm and –5 cm) and in plus figures if it is below the spines (+1 cm, +2 cm, +3 cm, +4 cm and +5 cm).



Stations of the presenting part

Solution to Question 28:

The ureteric injury is most commonly associated with Wertheim hysterectomy than other hysterectomies due to the radical nature of the procedure.

Overall incidence is 0.5 -1 percent of all pelvic operation.

Procedure	Incidence of Ureteric injury
Abdominal hysterectomy	0.04-3

Procedure	Incidence of Ureteric injury
Vaginal hysterectomy	0.02-0.47
Subtotal hysterectomy	0.03
Wertheim's hysterectomy	1-30

Solution to Question 29:

The given clinical situation likely suggests Cortical Vein Thrombosis.

Option A - Subarachnoid hemorrhage

There are no focal deficits hence unlikely.

Option B - Meningitis

- It would have presented with neck rigidity / photophobia.
- Moreover , presence of early papilledema goes against acute meningitis.

Option C - Acute Migraine

- Migraine does not present with seizures.
- Hence most likely diagnosis is Cortical Vein Thrombosis.

Cortical Vein thrombosis

Risk factors:

- Pregnancy and puerperium
- Oral contraceptives
- Pro thrombotic conditions

Clinical presentation

- Highly variable
- Head ache is the most common symptom.
- With or without
- Vomiting
- Papilledema
- Focal deficits

Investigation

- Brain MRI with MR venography – most sensitive
- CT – normal in 30 % of cases.

Solution to Question 30:

Anti-D should be given postnatally within 72 hours & all women who deliver Rh D-positive baby should have quantification of fetomaternal hemorrhage to guide appropriate dose of anti D prophylaxis.

RCOG and NICE guidelines for Anti-D prophylaxis:

1. All Rh-negative women should be offered Anti-D

First trimester indications: (50 mcg)

- Chorionic villus sampling
- Miscarriage
- Termination of pregnancy
- Ectopic pregnancy

2nd & 3rd-trimester indications (300 mcg)

- Amniocentesis, Cordocentesis
- Obstetric hemorrhage
- Abdominal trauma
- External cephalic version

2. All Rh D-negative women should be given a single dose of 1500 units (300 mcg) of anti D between 28 - 30 weeks of pregnancy.

3. Postnatally within 72 hours

4. All women who deliver Rh D-positive baby should have quantification of fetomaternal hemorrhage to guide appropriate dose of anti D prophylaxis.

Solution to Question 31:

• Since LH and FSH are elevated above the normal value, the possibility of a pregnancy can be ruled out. Therefore no need to do a urine pregnancy test.

• Hence the next step in management would be to check serum estradiol levels to evaluate amenorrhea.

In pregnancy,

- hCG stimulates Relaxin which inhibits LH.
- Inhibin A suppresses FSH.
- Therefore, LH and FSH are low.

Evaluation of amenorrhea :

Categories of amenorrhea based on Gonadotrophin and Estrogen levels.

- Hypergonadotropic hypogonadism

- Hypogonadotropic hypogonadism
- Eugonadotropic hypogonadism

The given clinical picture fits into hypergonadotropic hypogonadism.

Hypergonadotropic hypogonadism describes any process in which:

- (1) Ovarian function is decreased or absent (hypogonadism) and
- (2) Due to absent negative sex-steroid feedback, LH and FSH, have increased serum levels.

It is also called Premature Ovarian Failure.

Hormonal levels in POF

- Increased LH, FSH
- Low serum estradiol levels
- High levels of serum prolactin

Hence, the next best step would be to test serum estradiol levels.

Differential Diagnosis of Premature Ovarian Failure
Genetic Chromosomal Normal karyotype Gonadal dysgenesis Specific gene defects Fragile X (FMR1 premutation) Galactosemia Iatrogenic Ovarian surgery Gonadal radiation Systemic chemotherapy Autoimmune disease Toxins Viruses Miscellaneous

Solution to Question 32:

The above case suggests *Trichomonas vaginalis* infection and the treatment of choice is metronidazole.

The greenish-colored vaginal discharge along with the strawberry cervix is characteristic of infection with *Trichomonas vaginalis* which can be an STD.

The treatment regimen includes:

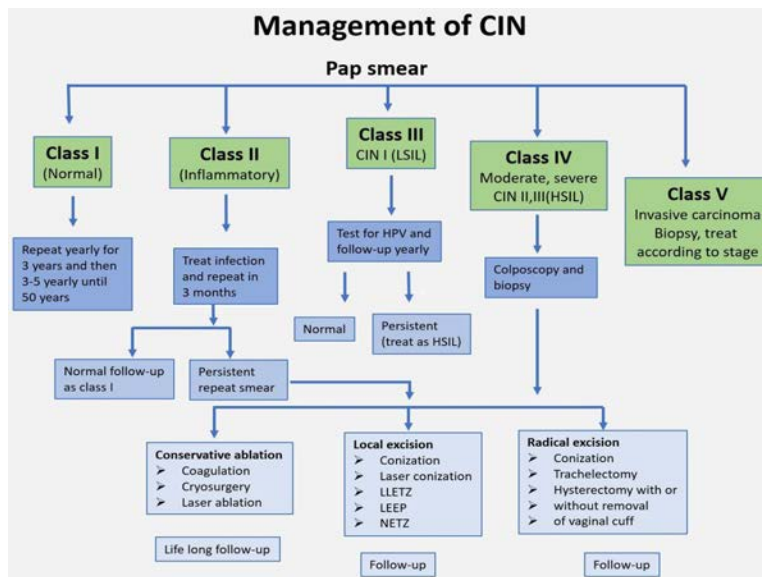
- Metronidazole 2 g orally in a single dose (or 400 mg 3 times/day for 5 days) is the standard treatment. It should be taken at night, after meals to avoid vomiting. If this too causes vomiting, or in resistant cases consider the use of alternative drugs.
- The alternative regimen includes tinidazole 2g as a single dose or secnidazole in a single dose of 2g. Single-dose therapy ensures better compliance.
- Both partners need to be given treatment at the same time to prevent the risk of recurrence.

The given image below shows strawberry cervix in trichomonas infection:



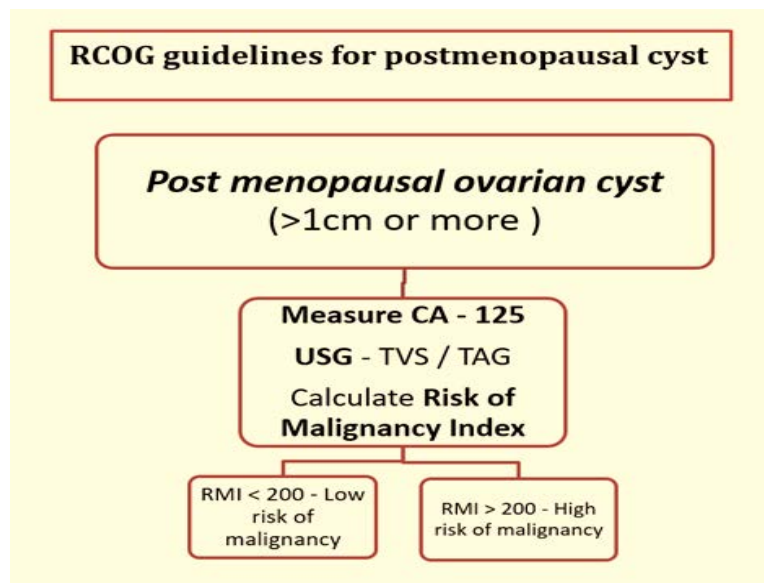
Solution to Question 33:

Carcinoma in situ is classified as Cervical Intraepithelial Neoplasia-III and the next step in management for such lesions is Colposcopy and biopsy.



Solution to Question 34:

Doing a CA-125 level measurement is the next best step according to the latest RCOG guidelines for Post-menopausal Ovarian cyst.



Initial steps in management include the following:

- Estimate CA – 125 level
- USG
- Calculating Risk of Malignancy index
- $RMI = U \times M \times Ca-125$
- U – Ultrasound score
- M – Menopausal status
- Ca 125 – Serum levels

Option B : CT scan is not recommended in initial evaluation.

Option C : Laparotomy/laparoscopy only indicated if size ≥ 5 cm or $RMI \geq 200$.

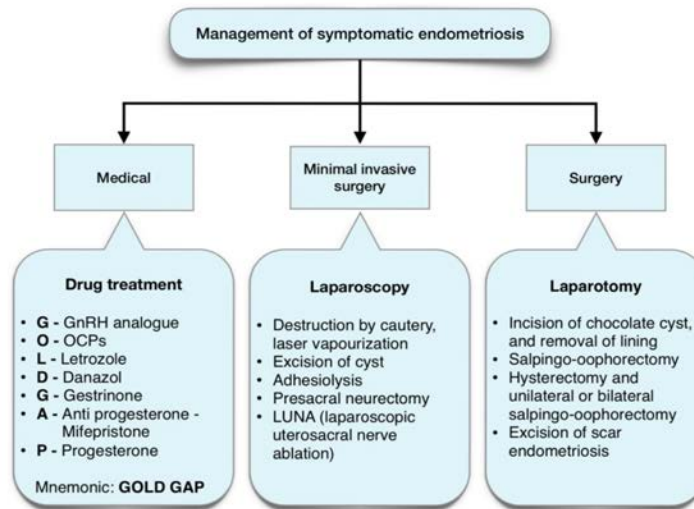
Solution to Question 35:

In endometriotic lesions, histology represents its high estrogen activity.

Endometriosis is defined by the presence of “ectopic” endometrial tissue at a site outside of the uterus. Endometriosis is a proliferative hormone-dependent disease of the childbearing period. It is extremely rare before menarche and disappears after menopause.

The endometriotic implants show certain differences when compared to normal endometrium. There is increased production of estrogen by endometriotic stromal cells due to high levels of the enzyme aromatase. This is absent in normal endometrial stroma.

Management of endometriosis:



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Question 1:

Identify the instrument.



- a) Episiotomy scissors
- b) Curved artery
- c) Curve-bladed Mayo
- d) Mosquito forceps

Question 2:

At term pregnancy, the normal amniotic fluid appearance is _____.

- a) Clear with white specks
- b) Cloudy without white specks
- c) Clear without white specks
- d) Cloudy with white specks

Question 3:

Identify this condition.



- a) Twin to Twin Transfusion Syndrome
- b) Monochorionic diamniotic pregnancy
- c) Monochorionic monoamniotic pregnancy
- d) Dichorionic diamniotic pregnancy

Question 4:

A lady with 7 weeks amenorrhea presented with slight vaginal spotting. Crown Rump Length was 5 mm with a well-formed gestational sac with a calculated gestational age of 5.6 weeks on TVS. What is the next line of management?

- a) Wait and watch for 4 weeks
- b) Repeat TVS after one week
- c) Serum HCG levels
- d) Medical intervention

Question 5:

Which of the following would be considered as an absolute contraindication for insertion of copper T to a patient?

- a) Unmotivated patient
- b) Menorrhagia
- c) Previous ectopic pregnancy

- d) Previous history of abortion**

Question 6:

A 76 year-old female presented with a non-healing ulcer on the labia majora for a period of 6 months, measuring 2 cm x 3 cm. No lymph nodes were palpable. Biopsy shows squamous cell carcinoma. How do you manage this case?

- a) Total vulvectomy with unilateral LN dissection**
- b) Partial vulvectomy with ipsilateral inguinofemoral LN dissection**
- c) Simple vulvectomy**
- d) Chemoradiation with resection**

Question 7:

A female comes to Gynecology OPD for pre-conceptual counseling, with 2 prior second trimester abortions. What is the next investigation you will advise?

- a) Pelvic sonography**
- b) Hysterosalpingography**
- c) Endometrial Biopsy**
- d) Karyotyping of the parents**

Question 8:

A pregnant female with known cardiac disease presents in the first trimester with history of warfarin embryopathy. What should be advised now?

- a) Continue warfarin throughout the pregnancy**
- b) Replace warfarin with heparin in first trimester**
- c) Give acicoumarin**
- d) Use LMW heparin**

Question 9:

A 25-year-old lady with submucosal fibroid was undergoing myomectomy. The surgeon was using 1.5% glycine as irrigating fluid for the cavity. During the surgery, the nurse informs that

there is a 500ml deficit of the irrigating fluid. What is the next step to be done?

- a) Stop the surgery
- b) Change the fluid to normal saline
- c) Continue the surgery with careful monitoring of fluid status
- d) Give furosemide to the patient and continue surgery

Question 10:

The most important marker of male fertility in semen analysis is?

- a) Morphology
- b) Concentration
- c) Motility
- d) Sperm Count

Question 11:

A 20 year old girl presents with primary amenorrhea. On examination breasts were found to be poorly developed, vagina was present. Ultrasound revealed normal uterus but gonads weren't detected. Further evaluation done showed 46XY karyotype. The most probable diagnosis?

- a) Androgen insensitivity syndrome
- b) Turner syndrome
- c) True hermaphrodite
- d) Swyer's syndrome

Question 12:

The results of semen analysis of a young male have been given below. What is the diagnosis?
5 million/ml concentration, 50% motility, 4% normal morphology

- a) Oligospermia
- b) Teratozoospermia
- c) Asthenozoospermia
- d) Oligoasthenoteratozoospermia

Question 13:

A woman presents with prolonged labor in emergency. She is being taken for Cesarean section. Which of the following is the correct position in which the nurse should keep the patient on the OT table?

- a) Supine with wedge under right hip
- b) Semi fowlers
- c) Trendelenburg with legs stirrup
- d) Prone position

Question 14:

A woman with 31 weeks of gestation presents with lower abdominal pain. The clinician is concerned if the patient is in preterm labor. Which of the following is the most appropriate test in this scenario?

- a) Alpha fetoprotein
- b) Beta HCG
- c) Fetal fibronectin
- d) PAPP-A

Question 15:

Identify the procedure done in the labor room?



- a) Amniocentesis
- b) MCA doppler
- c) Cardiotocography
- d) Amniotic fluid index determination

Question 16:

A G3P2Lo presents with 9 weeks of amenorrhea. She has a history of conization being done one year ago and is currently on follow-up, with no recurrence on PAP smear. She also has the history of preterm labor at 30 and 32 weeks. What is the next step of management?

- a) USG to see cervical length
- b) Cervical Cerclage
- c) Complete Bed rest
- d) Abdominal cerclage

Question 17:

What is the minimum endometrial thickness on transvaginal ultrasonography, above which you perform a biopsy in a case of postmenopausal bleeding?

- a) 3mm
- b) 4mm
- c) 7mm
- d) 9mm

Question 18:

A patient has been on MALA-N for 6 months. She is compliant, but she complained of missed periods. What is the next step?

- a) Transvaginal ultrasound
- b) Reassure her that it's normal to have delayed period
- c) Beta HCG estimation
- d) Serum FSH and LH

Question 19:

A pregnant woman delivered a baby with the normal expulsion of an intact placenta. After half an hour she started bleeding per vaginal. On examination, she was hypotensive and a boggy mass was palpated per abdomen. USG showed retained placental tissues. What is the likely diagnosis?

- a) Placenta membranacea
- b) Placenta succenturiata
- c) Circummarginate placenta
- d) Circumvallate placenta

Question 20:

HPV nonavalent vaccine is active against which of the following serotypes?

- a) 6, 11, 16, 18, 20, 23, 35, 42, 52
- b) 6, 11, 16, 19, 23, 33, 45, 58
- c) 6, 11, 16, 18, 32, 34, 45, 52, 58
- d) 6, 11, 16, 18, 31, 33, 45, 52, 58

Question 21:

Sentinel lymph node biopsy is most useful in which of the following gynecological malignancies?

- a) Carcinoma vagina
- b) Carcinoma vulva
- c) Carcinoma cervix
- d) Carcinoma endometrium

Question 22:

Which of the following is not included in the active management of the third stage of labor to prevent postpartum hemorrhage?

- a) Uterotonic within 1 minute of delivery
- b) Delayed clamping, cutting and ligation of cord
- c) Gentle massage of uterus
- d) Controlled cord traction

Question 23:

Which of the following changes can be seen in the quadruple screening for Down syndrome?

- a) Elevated β -hCG, PAPP-A and lower estriol, AFP
- b) Elevated β -hCG, Inhibin and lower estriol, AFP
- c) Lower β -hCG, PAPP-A and elevated estriol, AFP
- d) Lower β -hCG, Inhibin and elevated estriol, AFP

Question 24:

All of the following can be indicators of choriocarcinoma except:

- a) Increased number of theca lutein cysts
- b) Persistently enlarged uterus
- c) Plateau of high serum hCG
- d) Suburethral nodule

Question 25:

Which of the following hormones will be secreted in high amounts by the ovarian granulosa cells during the follicular phase of the menstrual cycle?

- a) FSH
- b) Estrogen
- c) LH
- d) Progesterone

Question 26:

Which of the following statements is true regarding the instrument shown below?



- a) A and B
- b) B and C
- c) A and C
- d) Only C

Question 27:

A young female has history of a spontaneous abortion. She has had amenorrhea since then. Her FSH now is 6 mIU/ml. Which of the following can be a cause for her amenorrhea?

- a) Pituitary dysfunction
- b) Ovarian dysfunction
- c) First pregnancy
- d) Uterine synechiae

Question 28:

Which among the following is an ideal treatment for a 55-year-old female with simple hyperplasia of the endometrium with atypia?

- a) Simple hysterectomy
- b) Medroxy progesterone acetate
- c) Levonorgestrel
- d) Radiotherapy

Question 29:

Which of the following is an irreversible method of contraception?

- a) Laparoscopic tubal ligation
- b) Copper-T
- c) Implanon
- d) LNG-IUD

Question 30:

What is the most useful test in vesicovaginal fistula?

- a) Cystoscopy
- b) 3 swab test
- c) MRI
- d) Cystometry

Question 31:

A 40-year-old woman has cervical intraepithelial neoplasia III. What would be the management?

- a) Hysterectomy
- b) Trachelectomy
- c) Conization
- d) Colposcopy with LEEP

Question 32:

A pregnant woman has gestational diabetes mellitus (GDM) and her blood glucose levels are well controlled on insulin. What will be the appropriate gestational age for her delivery?

- a) 37 weeks
- b) 39 weeks
- c) 38 weeks

d) 40 weeks

Question 33:

A female presents with 35 weeks and 3 days of gestation, AFI = 5 and there is no end-diastolic blood flow. What is the management?

- a) Immediate termination
- b) Give surfactant and wait till 36 weeks
- c) Do Doppler everyday to check for reversion
- d) Terminate after 37 weeks

Question 34:

Which of the following heart sounds would be normal in pregnancy?

- a) S₃
- b) S₄
- c) Fixed splitting of S₂
- d) Pericardial knock

Question 35:

A pregnant woman with cardiac disease is on warfarin therapy. When will the final switch to heparin be advised?

- a) At the onset of labour
- b) 34 weeks
- c) 36 weeks
- d) 40 weeks

Answer Key

Question No.	Correct Option
1	a

2	d
3	c
4	b
5	a
6	b
7	a
8	b
9	c
10	a
11	d
12	a
13	a
14	c
15	c
16	a
17	b
18	c
19	b
20	d
21	b
22	c
23	b
24	a
25	b
26	c
27	d
28	a
29	a
30	a
31	d
32	b
33	a
34	a
35	c

Detailed Explanations

Solution to Question 1:

The given instrument is Episiotomy scissors.

The episiotomy scissors is angulated to prevent extensions of the perineal tears into the anal margins and anal sphincters and it also prevents complete perineal tears.

Episiotomy is a planned incision made on the posterior vaginal wall and the perineum during the second stage of labor. It is done mostly to avoid spontaneous perineal injuries to the muscles and fascia at the time of delivery.

The ideal time to perform episiotomy is when the perineum is bulged thin during contraction, just prior to crowning. The most commonly performed episiotomy incision is mediolateral, which prevents any rectal involvement even if the tear extends posteriorly. Other types of episiotomy include: Median, lateral and J shaped.

Curved Mayo (option C) is used to cut thicker tissues.

Mosquito forceps (option D) is used to grasp the delicate tissues & blood vessels.

The table below explains major differences between median and mediolateral type of episiotomy.

Midline Episiotomy	Mediolateral Episiotomy
Surgical repair is easy	Surgical repair is difficult
Blood loss is lesser	Blood loss is more
Healing is superior	Healing is not so good
Dyspareunia is rare	Dyspareunia is comparatively more
Postoperative comfort is maximum	Postoperative discomfort is more
Extension, if occurs, may involve the rectum	Relative safety from rectal involvement from extension

Solution to Question 2:

At term, the amniotic fluid is cloudy and contains a large number of white flakes of vernix.

Amniotic fluid appearance is determined by the proportion of cloudiness and the number of flakes released from the fetal skin over the period of pregnancy. This is summated by the amniotic fluid macroscore.

At the end of the 32nd week of pregnancy, the amniotic fluid is clear and does not have any flakes (Option C). By the 36th week, the amniotic fluid starts to have some flakes (Option A) and gradually we will begin to see traces of cloudiness also. By 42nd week i.e. term of pregnancy, the amniotic fluid has a large number of flakes and is very cloudy (Option D).

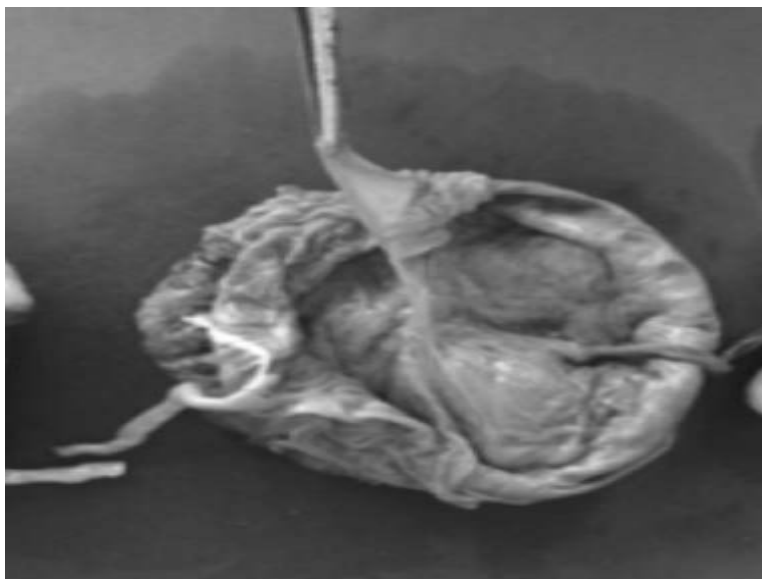
Solution to Question 3:

The given specimen depicts a monochorionic monoamniotic pregnancy as:

- 2 umbilical cords are arising from 1 placenta (monochorionic)
- No intervening amniotic membrane is attached to the placenta. This means that both fetuses must have been inside a single amniotic cavity (monoamniotic).

Option B: In Monochorionic diamniotic pregnancy, on gross examination, there will be 1 placenta and intervening 2 layers of amnion attached to the placenta. In this type of pregnancy, each fetus will have its own amniotic cavity (diamniotic) with a shared placenta (monochorionic). This is seen if the fertilized zygote divides into 2 between 4-8 days.

The below given gross image of monochorionic diamniotic pregnancy shows two umbilical cords arising separately from the placenta with the intervening amniotic membrane held by the forceps.

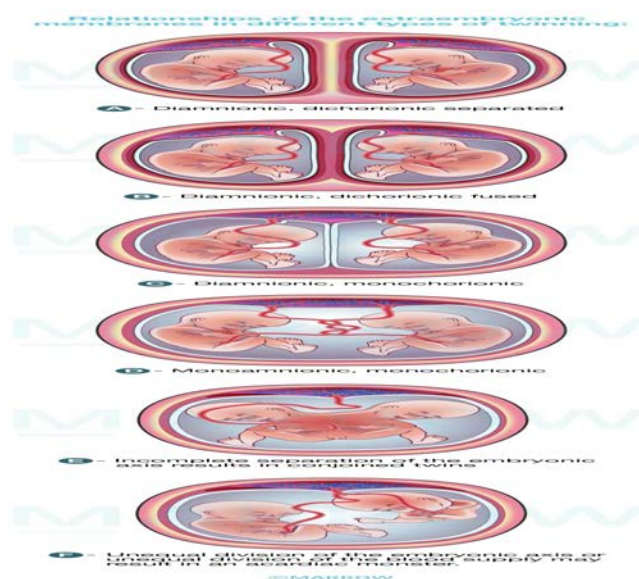


Option D: In dichorionic diamniotic pregnancy each fetus has it's own amnion & chorion. The placentas here may be fused or separate. This is seen in dizygotic twins as well as monozygotic twins that divide within 0-4 days.

Option A: Twin to twin transfusion syndrome (TTTS) most commonly complicates monochorionic (single placenta) twin pregnancy. In TTTS blood is transfused from the donor twin to the recipient twin through the abnormal vascular connections within the placenta leaving the donor anemic while the recipient becomes polycythemic.

The box below describes the type of amnion and chorion seen in different types of twins depending on the time of division.

Dizygotic twins	Dichorionic diamniotic
Monozygotic twins- division occurs within 0-4 days 4-8 days 8-12 days >13 days	Dichorionic diamniotic Monochorionic diamniotic Monochorionic monoamniotic Monochorionic monoamniotic conjoined twins



Solution to Question 4:

The given clinical scenario of a patient with amenorrhea presenting with spotting, raises a suspicion of possible abortion and in the given case the next line of management would be to repeat TVS after one week.

In a normal intrauterine pregnancy a gestational sac will be visible on TVS by 5 weeks of LMP and the embryonic poles can be visualized by 5.5 weeks. The cardiac activity of the embryo can be first visualized on TVS around 5.5-6 weeks, when the crown rump length (CRL) is around 5mm.

In the event that, the embryonic poles are not visible on TVS, even when the gestational sac diameter has reached $\geq 25\text{mm}$ or the cardiac activity is not visualized even at $\text{CRL} \geq 7\text{mm}$ a diagnosis of missed abortion can be made. The values, mean sac (gestational sac) diameter $\geq 25\text{mm}$ and $\text{CRL} \geq 7\text{mm}$ are known as critical cut off values which represent the upper limit beyond which if a structure is not visualized, it is abnormal.

Hence, Missed Abortion should be suspected if, on USG

- $\text{CRL} \geq 7\text{ mm}$ and no heart beat is recorded (CRL-crown rump length)
- $\text{MSD} \geq 25\text{ mm}$ and no embryo is seen (MSD-mean sac diameter)

In this patient the CRL is still at 5 mm and hence, a diagnosis of missed abortion cannot be made. Thus, the next line of management would be to repeat TVS after one week.

Structure	Critical cut-off to visualize the structure on USG	Structure visible on TVS at
Gestational sac in uterus(intrauterine pregnancy)	On TVS : 2000 IU of beta HCG On TAS : 6500 IU of beta HCG	4 weeks 3 days of LMP to 5 weeks
Yolk sac	-	5 weeks

Structure	Critical cut-off to visualize the structure on USG	Structure visible on TVS at
embryonic/fetal poles	Mean sac diameter ≥ 25 mm	5 weeks - 5 1/2 weeks
Cardiac activity	CRL ≥ 7 mm	5 1/2 weeks - 6 weeks

Solution to Question 5:

An unmotivated patient or someone who does not consent to the procedure should never be taken up for a procedure.

Contraindications for use of copper intrauterine devices (IUDs)

- Uterine abnormality with a distorted uterine cavity
- Active uterine and pelvic infection
- Known or suspected uterine or cervical malignancy
- Genital bleeding of unknown etiology
- Mucopurulent cervicitis
- Wilson disease
- A previously placed IUD that has not been removed
- An inability to place or retain the device

Note:

An unmotivated patient is an absolute contraindication. Out of the given options, this would be the best answer.

Solution to Question 6:

According to the recent FIGO guidelines, the patient's lesion measuring 2 cm x 3 cm with no palpable lymph nodes falls under stage Ib vulval cancer and its management is partial vulvectomy with ipsilateral inguinofemoral lymph node dissection.

For stage IB cancers, radical partial vulvectomy is recommended for the primary tumor and as the women with stage IB cancer can have diseased lymph nodes even without palpable enlargement of nodes, Sentinel Lymph Node Biopsy (SLNB) and/ inguinofemoral lymphadenectomy is performed.

Ipsilateral dissection of lymph nodes is performed if the vulvar lesion is present laterally, ≥ 2 cm beyond the midline. In cases where the vulvar lesion lies within 2 cm from the midline, bilateral inguinofemoral Lymph Node dissection will be performed.

In this patient, the best management would be to perform Partial vulvectomy with ipsilateral inguinofemoral LN dissection among the given options.

Solution to Question 7:

The most common cause of second trimester abortions are uterocervical structural anomalies and the next investigation advised in this patient would be pelvic sonography to look for these anomalies.

Although, Hysterosalpingography (Option B) can be used in the patient during the interconceptional period to diagnose internal anatomical abnormalities of the uterus and cervix, it is an invasive procedure and hence is reserved for use only after noninvasive procedures such as sonography is done. Hysterosalpingography (HSG) has limited use in diagnosing the external uterine contour and it is contraindicated in pregnancy.

Pelvic sonography is a non-invasive investigation that can be used to detect certain anatomical anomalies leading to abortions. A 3D sonography will provide better images of uterine and cervical contour than a 2D sonography. This investigation can also be done in pregnant female.

Karyotyping (Option D) is done to rule out any suspicion of chromosomal abnormalities which, more commonly cause first trimester abortions.

NOTE: If a suspicion of cervical incompetence (H/O 1 painless second trimester loss) is present, a TVS is performed in a pregnant female between 18-24 weeks and the cervical length and the cervical shape is determined.

- Normal cervical length-3-4cm ; In cervical incompetence ≤ 2.5 cm
- Normal cervix shape-Tshape; In cervical incompetence U shape

In the presence of ≥ 2 painless second trimester abortions, a history based diagnosis of cervical incompetence is made and the pregnant female can undergo prophylactic cervical cerclage with progesterone supplementation.

Some Causes of Midtrimester (second trimester) abortions

Fetal Anomalies- Structural, Chromosomal
Maternal Disorders- Autoimmune, Infections, Metabolic disorders
Placental Causes- Abruption, previa
Vasculopathy, Chorioamnionitis
Uterine Defects- Congenital anomalies, submucosal Leiomyomas, cervical incompetence

Solution to Question 8:

In the first trimester, warfarin has to be replaced with heparin due to its potential teratogenic effects such as miscarriage, stillbirths, achondroplasia and stippled epiphyses.

Low molecular weight heparin (LMWH) must replace warfarin and shouldn't be added along with the warfarin (Option D).

The detailed guideline for anticoagulation in pregnancy is as follows:

- LMWH: low molecular weight heparin
- EDD: expected date of delivery
- UFH: unfractionated heparin
- PPH: postpartum hemorrhage

Period of pregnancy	Anticoagulation of choice
First trimester	If dose of warfarin < 5mg/day – continue warfarin till 36 weeks If dose of warfarin > 5mg/day or warfarin embryopathy – replace it with heparin / LMWH
From 12 weeks to 36 weeks	warfarin
At 36 weeks	Switch to LMWH
1 week before EDD / 3 days before EDD	Switch to UFH (as LMWH have longer T _{1/2} – chances of PPH)
On the day of delivery	Stop all anticoagulants
6 hrs after vaginal delivery / 6-12 hrs after cesarean	Restart UFH and warfarin. When INR is 2.5 to 3 withdraw UFH & continue warfarin

Solution to Question 9:

This is a young patient with no comorbidities, subjected to an electrolyte poor fluid for irrigation. Since the deficit is 500 mL, it is recommended to continue the surgery with careful monitoring of fluid status.

The criteria for terminating the procedure for

- Electrolyte deficient fluid is— a deficit of 1 L
- Electrolyte rich fluid is— a deficit of 2.5 L

Glycine irrigation causes hyponatremia. The earliest symptom of hyponatremia is nausea which typically requires fluid absorption (deficit) of at least 1000 mL in women undergoing surgical procedures.

There are two types of distension medias

Electrolyte deficient fluid	Electrolyte rich media
1.5% glycine 3% sorbitol 5% mannitol	Ringer lactate Normal saline
Instruments that can be used— monopolar / bipolar instruments	Only bipolar instruments

Electrolyte deficient fluid	Electrolyte rich media
Complications: Water intoxication Hyponatremia (except mannitol)	Causes Shock & burns when used with monopolar instruments

Solution to Question 10:

Morphology of sperms is the most important marker of male fertility.

While sperm concentration and progressive motility have value for distinguishing infertile men, strict sperm morphology is the most important discriminating value compared to other parameters.

Solution to Question 11:

The given clinical scenario of a 46 XY individual with female internal & external genitalia and poorly developed breasts, presenting with primary amenorrhea points towards the diagnosis of Swyer's syndrome.

Turner syndrome (option B) is a disorder of 45 XO individuals with short stature. The gonads in these individuals are streak ovaries.

Androgen Insensitivity Syndrome (option A) is a disorder of 46 XY individuals with complete insensitivity to testosterone. These individuals have female external genitalia with an absent uterus. Secondary sexual characteristics like the development of breasts are adequate. However, pubic and axillary hairs are scanty.

True hermaphrodite (option C) can be a 46 XX or 46 XY individual with both ovaries and testis. These individuals have both male and female internal genitalia with ambiguous external genitalia.

Swyer's syndrome is a type of male pseudohermaphroditism where there is an inactivation of the SRY gene that leads to formation of dysgenetic testis.

Thus, in the absence of hormones; AMH & testosterone the Mullerian duct grows and the Wolffian duct regresses forming female internal genitalia such as uterus (infantile type).

These patients also lack the DHT hormone and hence present with poorly developed breasts (Tanner staging 1 or 2), scanty pubic, axillary hair and female external genitalia. These individuals are of normal stature with increased LH & FSH.

Management includes gonadectomy as soon as diagnosed and estrogen + progesterone for breast development.

AMH: Anti Mullerian hormone

DHT: dihydro testosterone

Solution to Question 12:

The given scenario of decreased sperm concentration with normal range of motility and morphology points to the diagnosis of oligospermia.

Oligospermia is defined as sperm concentration of <15 million/ml.

It is further subdivided into mild-moderate oligospermia when the concentration is 10-15 million sperms /ml & severe oligospermia when the concentration is <5 million sperms/ml.

Teratozoospermia or teratospermia (option B) is defined as Increased proportion of morphologically abnormal sperms.

Asthenozoospermia or asthenospermia (option C) is used to define sperms with diminished motility.

Term	definition
azoospermia	Semen with absent sperms
aspermia	Both semen and sperms are absent
leukocytospermia	Increased WBC count in the semen
necrospermia	Nonviable sperms

Solution to Question 13:

The position of the patient in the Cesarean section is supine, with a wedge under the right hip.

The wedge beneath the right hip creates a left lateral tilt to aid venous return and avoid hypotension.

Solution to Question 14:

Fibronectin detection in cervicovaginal secretions before membrane rupture is a possible marker for impending preterm labor.

Fetal fibronectin:

- Fetal fibronectin is a glycoprotein present in high concentrations in maternal blood and in amniotic fluid.
- Fetal fibronectin levels are measured using ELISA, and values exceeding 50 ng/mL are considered positive.

Screening tests in pregnancy:

1st Trimester:

Dual test + nuchal translucency

DUAL test	Down's syndrome	Trisomy 18
PAPP-A (pregnancy associated plasma protein A)	↓	↓
β-HCG	↑	↓

2nd Trimester:

Integrated screening:

Dual screening + nuchal translucency + quad test

Quadruple test	Down's syndrome	Trisomy 18
α fetoprotein	↓	↓
β-HCG	↑	↓
Estriol (unconjugated E ₃)	↓	↓
Inhibin A	↑	↓

Solution to Question 15:

The given procedure is cardiotocography (CTG; electronic fetal monitor).

CTG, is a type of intrapartum fetal monitoring where the fetal heartbeat and the uterine contractions are recorded simultaneously.

The CTG graph can be differentiated from non stress test (NST) graph by the presence of two graphical representations in CTG:

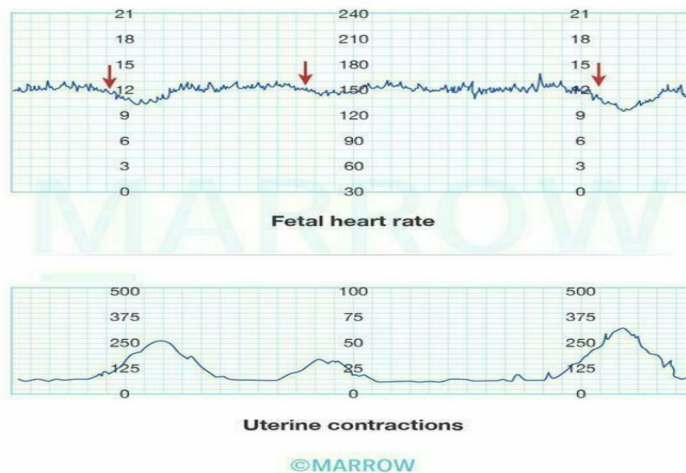
- FHR represented by the upper graph
- Uterine contractions represented by the lower graph.

NST has a single graphical representation.

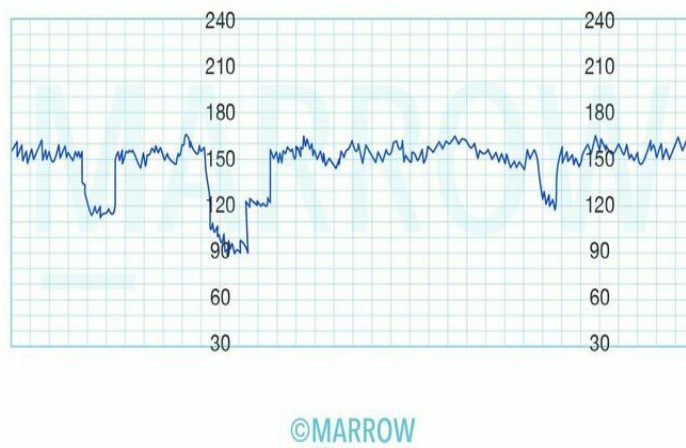
A normal CTG graph is defined by the following parameters:

- Baseline FHR-110-160
- Beat to beat variability of 5-25 bpm
- Two or more Acceleration during 20 minute period
- No late deceleration
- No variable deceleration
- Early deceleration +/-

The below given depicts a CTG graph



The below given image shows a NST graph



Solution to Question 16:

The next step of management in this patient, is to do USG to measure the cervical length.

In the given clinical scenario the patient has undergone conization which is a risk factor for cervical incompetence. Hence, it is important to measure the cervical length to diagnose cervical incompetence in this patient. If the cervical length is ≤ 2.5 cm then a cervical cerclage can be performed.

Cervical cerclage (Option B) is performed as the next step in management when the patient presents with ≥ 2 second trimester pregnancy losses (History based diagnosis of cervical incompetence). Since, the history in this patient is of preterm labour, the first step would still be

to do a TVS to measure the cervical length for diagnosing. Cerclage only comes after diagnosis. Cervical incompetence is one of the important causes second trimester (mid trimester) abortions. Risk factors for the incompetence include:

- Past h/o surgery on cervix such as conization, loop electrosurgical excision procedure (LEEP)
- Cervical trauma during labour, instrumental delivery

Diagnosis of cervical incompetence in a pregnant female:

- H/O ≥ 2 second trimester abortions (history based diagnosis) $\rightarrow$ management is by cervical cerclage+progesterone
- H/O 1 second trimester abortion $\rightarrow$ TVS done between 18-24 weeks to measure the cervical length $\rightarrow$ USG based diagnosis of cervical incompetence is made if there is h/o 1 or more second trimester abortions with TVS showing cervical length ≤ 2.5 cm.

TVS findings in cervical incompetence include:

- Cervical length ≤ 2.5 cm (Normal: 3-4cm)
- Shape of cervix becomes U shaped
- Diameter of internal os ≥ 2 cm

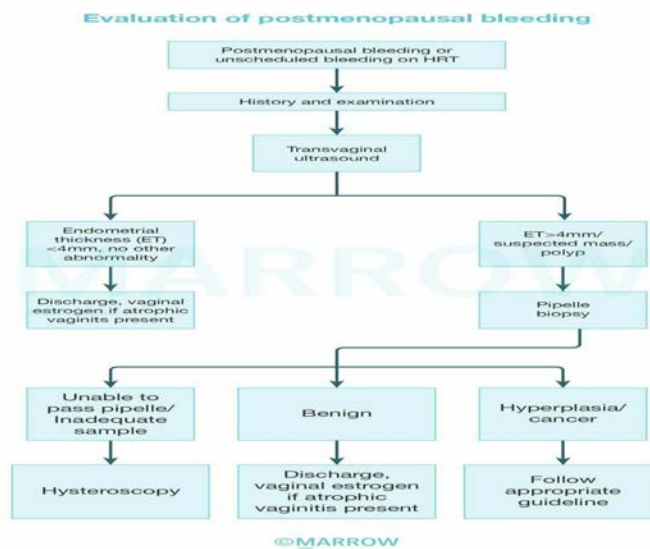
Solution to Question 17:

A finding of endometrial thickness ≥ 4 mm (according to ACOG) in a postmenopausal women requires further evaluation, such as endometrial biopsy.

If the endometrial thickness is ≤ 4 mm in a post menopausal female, malignancy is rare and hence, can be managed by tranexamic acid. If the bleeding still persists, the patient needs to undergo fractional curettage and hysteroscopy.

The algorithm for next step for abnormal uterine bleeding in different age groups is as follows:

Age group	Endometrial thickness on TVS	Action
Age <40 years	>12 mm	Endometrial sampling/Endometrial biopsy
Age >40 years or Age >45 years	-	Endometrial sampling/Endometrial biopsy is done irrespective of usg findings
Post menopausal women	≥ 4 mm	Endometrial sampling/Endometrial biopsy
Post menopausal women	<4 mm	Reassurance+tranexamic acid If, bleeding still persists then do endometrial biopsy



Solution to Question 18:

In the event of missed periods while on contraceptives, Beta-HCG estimation should be done as the next step to rule out pregnancy.

Serum beta HCG can detect a pregnancy as early as 5 days before the first missed period. This can be done by either serum beta hcg estimation or by urine pregnancy test which estimates the beta-hcg excreted in urine. If the patient is not pregnant she can continue taking the combined pill.

Mala N is a combined oral contraceptive pill that is distributed for free by the Indian government. It contains 0.15 mg levonorgestrel and 30 µg ethinyl estradiol. It contains 21 contraceptive pills + 7 iron tablets. It is started on the first day of each cycle.

The combined oral pills act by suppressing pituitary hormones and preventing ovulation. The progestogen also causes atrophic changes in the endometrium and prevents implantation. It also causes thickening of the cervical mucus, making it difficult for the sperms to penetrate.

The menstrual side effects includes breakthrough bleeding, menorrhagia and rarely post-pill amenorrhea when the patient is on drug for more than 6 months. Post pill amenorrhea is seen in <1% of patients and should be considered only after ruling out pregnancy.

Pregnancy rate with the combined oral pill is 0.1 per 100 woman-years.

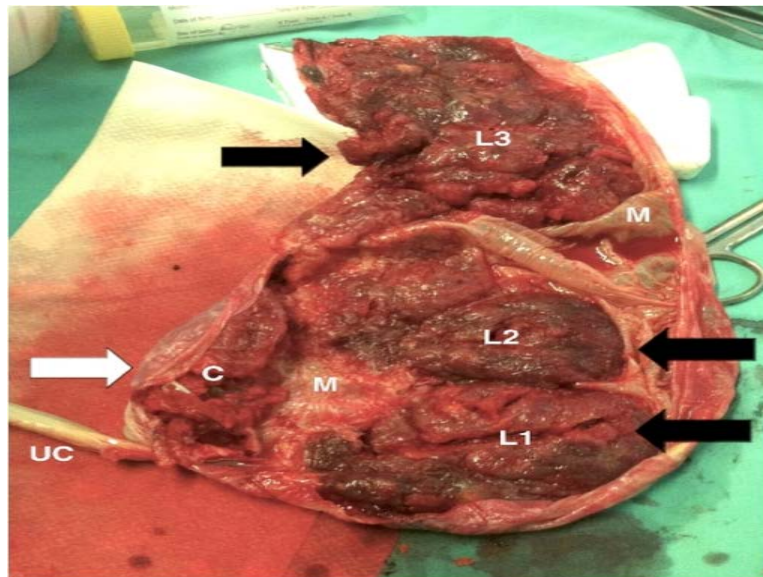
Solution to Question 19:

The reason for the postpartum hemorrhage in the given patient is likely due to the succenturiate lobe being torn away and left behind at the time of placental delivery, as seen with cases of Placenta succenturiata.

Placenta succenturiata refers to the abnormal placenta where an additional lobe/lobes of placental tissue are located a few centimeters away. The ancillary lobes are smaller in size when compared with the main placental tissue (note: Bilobed placenta has two lobes of equal sizes). These lobes have vessels that course through the membranes connecting them to the main placental tissue.

Placenta succenturiata can be associated with complications like vasa previa (fetal hemorrhage), postpartum hemorrhage due to retained placental tissues, infection such as endometritis, Subinvolution of uterus and Placental polyp formation.

An image of this variant is given below where L1,2,3 depicts smaller lobes, UC is the umbilical cord, and M is the intervening membrane.



The below given image shows a Bilobed or Bipartite or duplex placenta which is different from placenta succenturiata. The lobes in bilobed placenta are nearly equal in size. The umbilical cord is attached into a connecting chorionic bridge or into the intervening membranes in between the two placental lobes.



Other options:

Option A: Placenta membranacea is an unduly large and thin placenta that develops from both the chorionic frondosum and the chorion leave. It is associated with an increased risk of placenta previa or accreta and retained placenta and its complications.

Circummarginate placenta (option C) and Circumvallate placenta (option D) are a type of extrachorial placentation.

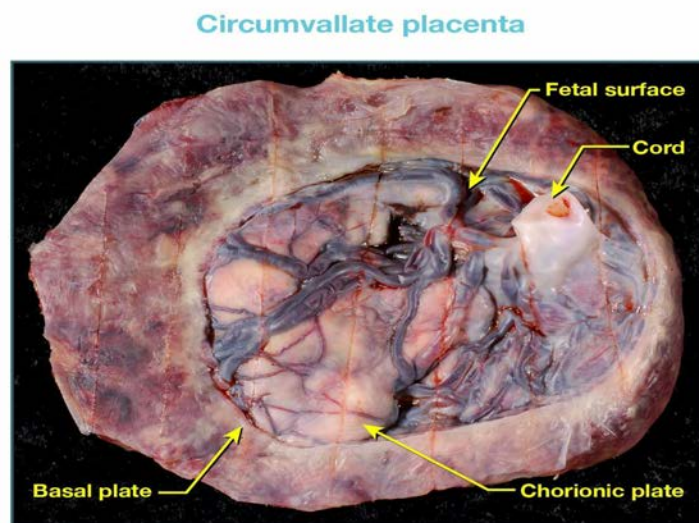
In extrachorial placentation, the chorionic plate is smaller than the basal plate. The chorionic plate does not extend upto the placental margin. The membranes (amnion and chorion) are rolled back upon themselves to form a ring leaving a rim of bare placental tissue (extrachorial portion)

Circummarginate placenta-A thin fibrous ring is present at the margin of the chorionic plate where the fetal vessels appear to terminate.

Circumvallate placenta - The fetal surface is divided into a central depressed zone surrounded by a thickened ring which is composed of a double fold of amnion and chorion. Vessels radiate from the cord insertion as far as the ring and then disappear from the view.

Complications associated with circumvallate placenta are preterm delivery, abortions, growth retardation, antepartum hemorrhage, retained placenta, and hydorrhea gravidarum.

A specimen of circumvallate placenta is shown below:



Solution to Question 20:

HPV nonavalent vaccine (Gardasil 9 or HPV9) is active against HPV 6,11,16,18,31,33,45,52,58.

Currently, HPV nonavalent vaccine has replaced Gardasil which is a quadrivalent vaccine acting against 6,11,16,18.

Vaccines available for HPV:

Brand name	Cervarix	Gardasil	Gardasil-9	Cervavac
Characteristic and HPV subtypes	Bivalent:16,18	Quadrivalent:6,11,16,18	Nonavalent:6,11,16,18,31,33,45,52,58	Quadrivalent:6,11,16,18
Protects against:	CINCA cervix	Anogenital wartsCINCA cervix	Anogenital wartsCINCA cervixVulval intraepithelial neoplasiaVaginal intraepithelial neoplasiaAnal cancerOropharyngeal cancerHead and neck cancers	Anogenital wartsCINCA cervix
Dose	0.5 ml,IM.	0.5 ml,IM.	0.5 ml,IM.	0.5 ml,IM.
Ideal Age range:	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	The targeted age group is 9-26 years

Solution to Question 21:

Among the given gynecological malignancies, sentinel lymph node biopsy (SLNB) is most useful in vulval carcinomas.

This is because women with earlier stages of vulval cancer can have diseased lymph nodes even without palpable enlargement of nodes. Hence, sentinel lymph node biopsy helps in early diagnosis lymph node spread which can be managed by lymph node dissection.

A sentinel lymph node is a physiologically, the first lymph node to receive tumor lymphatic drainage.

Malignancies where sentinel lymph node biopsy is commonly done are :

- Vulval carcinoma
- Breast carcinoma
- Penile carcinoma
- Malignant melanoma

Lymphatic mapping is similar technique to SLNB that can also be used in vulval cancers.

Solution to Question 22:

As per the latest guidelines, uterine massage is no longer a core component of Active Management of the Third Stage of Labor (AMTSL) that is performed for the prevention of postpartum hemorrhage (PPH).

Uterine massage or fundal massage is not recommended as an intervention to prevent PPH in women who have already received prophylactic uterotonic such as oxytocin (core component of AMTSL). Hence, it is now an optional component of AMTSL that is done in patients with uterine atony as a part of the management of PPH.

AMTSL is recommended by the WHO to prevent postpartum hemorrhage (PPH). Uterotonics are the most important components in AMTSL to do so.

The currently recommended components of AMTSL are as follows:

Note: In the event that uterine atony is diagnosed during the assessment of uterine tone, uterine massage can be performed to control PPH.

1.	Injecting uterotonics within 1 minute of delivery of the baby or immediately after delivery of the anterior shoulder	Most important step Oxytocin 10 IU im/iv infusion is the preferred uterotonic. (Not recommended by WHO-carboprost)
2	Delayed cord clamping	Done within 1-3 mins of delivery 80 ml of additional blood goes to the fetus Prevents fetal anemia
3	Delivery of placenta by controlled cord traction: Modified Brandt Andrews technique	The uterus is pushed upwards and backward The traction of cord is downwards and forwards
4	Intermittent assessment of uterine tone	Done every 15 mins for 2 hours

Solution to Question 23:

Quadruple screening is a second-trimester multiple marker screening test used for the detection of chromosomal anomalies like Down syndrome. It is performed between 15-21 weeks.

Pregnancies with Down syndrome are characterized by elevated β -hCG, elevated Inhibin A, reduced estriol and reduced AFP levels.

First-trimester aneuploidy screening combines two maternal serum analytes (Dual screening test) with a sonographic measurement. It is performed between 11-14 weeks.

An increased nuchal translucency indicates an increased possibility of Down's syndrome.

	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect
MSAFP	Low	Low	High
β hCG	High	Low	Normal
Unconjugated estriol	Low	Low	Normal
Inhibin A	High	Low	Normal

Dual test	Down's syndrome	Trisomy 18 & Trisomy 13
PAPP-A (pregnancy associated plasma protein A)	↓	↓
β-HCG	↑	↓

Solution to Question 24:

Increased number of theca lutein cysts is not an indicator of choriocarcinoma. Increased size or persistence of theca lutein cysts indicates choriocarcinoma.

Plateau of high serum β-hCG level for four measurements for ≥3 weeks is an indicator of choriocarcinoma.

Risk factors for malignant change are:

- Molar pregnancy - complete moles have 15-20% risk, partial moles have 1-5% risk
- Patient's age ≥ 40 years or <20 years
- Parity ≥3
- Serum β hCG >1,00,000 IU/mL
- Uterine size >20 weeks
- Previous history of molar pregnancy
- Large theca lutein cysts >6 cm in diameter

Post-evacuation uterine sonographic surveillance can be used as a predictor of subsequent neoplasia. Findings indicative of malignant transformation are:

- Suburethral nodules
- Myometrial nodules
- Hypervascularity.

Solution to Question 25:

During the follicular phase of the menstrual cycle, estrogen is secreted in high amounts by the ovarian granulosa cell.

Follicular phase (days 0–14)

- A primordial follicle develops to the Graafian stage, with atresia of neighboring follicles.
- LH and FSH receptors are up-regulated in theca and granulosa cells.
- Estradiol levels increase and cause the proliferation of the uterus.
- FSH and LH levels are suppressed by the negative feedback effect of estradiol on the anterior pituitary.

- Progesterone levels are low.

Ovulation (day 14)

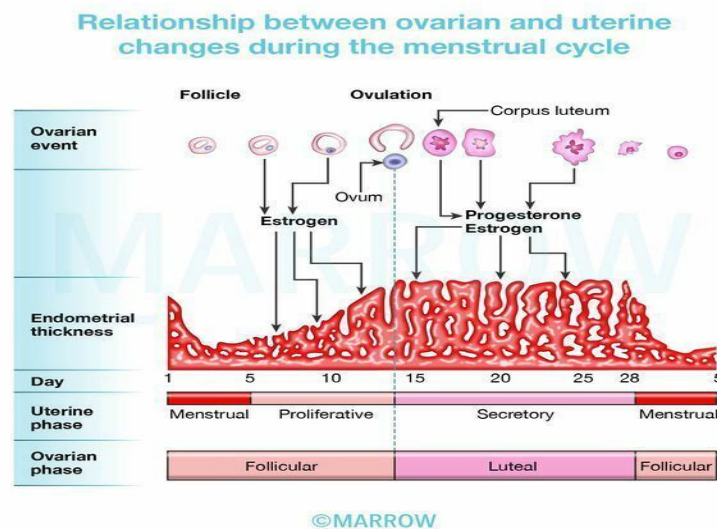
- Occurs 14 days before menses, regardless of cycle length.
- A burst of estradiol synthesis at the end of the follicular phase has a positive feedback effect on the secretion of FSH and LH (LH surge).
- Ovulation occurs as a result of the estrogen-induced LH surge.
- Estrogen levels decrease just after ovulation (but rise again during the luteal phase).
- Cervical mucus increases in quantity; it becomes less viscous and more penetrable by sperm.

Luteal phase (days 14–28)

- The corpus luteum begins to develop, and it predominantly synthesizes progesterone.
- Vascularity and secretory activity of the endometrium increase to prepare for receipt of a fertilized egg.
- Basal body temperature increases because of the effect of progesterone on the hypothalamic thermoregulatory center.
 - If fertilization does not occur, the corpus luteum regresses at the end of the luteal phase. As a result, estradiol and progesterone levels decrease abruptly.

Menses (days 0–4)

- The endometrium is sloughed because of the abrupt withdrawal of estradiol and progesterone



Solution to Question 26:

The given image shows a pneumatic compression device. It is used in women who are not on any thromboprophylaxis and planned for cesarean section (statement A) and it is discontinued once the patient starts ambulating post-surgery (statement C).

The device is used because pregnancy and surgery increase the risk of thromboembolism.

According to recent guidelines, patients with no additional risk factors and not placed on any pharmacological thromboprophylaxis are placed on pneumatic compression devices before cesarean delivery. Patients undergoing cesarean delivery with additional risk factors are put on both pneumatic compression devices and pharmacological therapy (UFH or LMWH).
UFH-unfractionated heparin, LMWH-lower molecular weight heparin.

Indications of pharmacological thromboprophylaxis:

- Inherited coagulation disorder
- Overweight or obese patient
- Heart failure
- Family history of deep vein thrombosis or pulmonary embolism

Solution to Question 27:

The development of uterine synechiae is the most likely cause of amenorrhoea in the given clinical scenario.

Certain types of spontaneous abortions such as inevitable, incomplete and missed abortions might require medical assistance for complete removal of products of conception including procedures such as dilatation and curettage.

Vigorous uterine curettage in early pregnancy damages the basal layer of endometrium leading to scarring and adhesions in the uterus that interferes with normal endometrial growth and shedding. This causes the clinical feature of secondary amenorrhea as seen in this patient.

These intrauterine adhesions are known as uterine synechiae or Asherman syndrome.

As there is no defect in the hypothalamo-pituitary-ovarian axis in the Asherman syndrome, the levels of FSH remain within the normal range of 4 - 25 IU/L as seen in this patient.

In the case of Pituitary dysfunction (option A) the FSH levels are usually decreased.

In ovarian dysfunction (option B) the negative feedback of ovarian hormones (estrogen) on pituitary and hypothalamus is lost. Hence, the levels of FSH are increased (≥ 25 IU).

Asherman syndrome is characterised by a spectrum of intra uterine scarring including thin adhesions, dense bands, or complete obliteration of the uterine cavity. It occurs due to endometrial damage following vigorous curettage, usually in miscarriage, or elective abortion. Damage may also occur from other uterine surgeries such as myomectomy or cesarean delivery or from an infection related to an intrauterine device.

Depending on the degree of scarring, patients may present with amenorrhea, hypomenorrhea or recurrent pregnancy loss due to inadequate placentation.

Asherman syndrome shows multiple filling defects on hysterosalpingography (HSG) or saline-infusion sonography. Definitive diagnosis requires hysteroscopy. Hysteroscopic lysis of adhesions is the preferred surgical treatment for the syndrome.

Solution to Question 28:

Simple hysterectomy is the best treatment in a case of atypical endometrial hyperplasia in a peri-menopausal and post-menopausal female.

Endometrial hyperplasia with atypia carries 29% risk of developing into endometrial carcinoma and hence, must be treated with total abdominal hysterectomy (simple hysterectomy).

The presence of any concomitant endometrial cancer must be ruled out by doing fractional curettage (or dilatation and curettage) before doing a hysterectomy.

If a premenopausal woman does not desire to undergo hysterectomy, progesterone can be given and the patient is followed up every 3-6 months. If the atypia persists during the follow-up hysterectomy should be performed.

The table below summarizes the management of patients with endometrial hyperplasia

Management of Endometrial hyperplasia	
Hyperplasia without atypia	
Pre-menopausal women	Progesterone therapy
Postmenopausal women: Simple hyperplasia without atypia Complex hyperplasia without atypia	Follow up without therapy Progesterone therapy
Hyperplasia with atypia	
Pre-menopausal women	If patient wishes to preserve uterus, give High-dose progesterone therapy If atypia persists, then perform Total abdominal hysterectomy
Peri-menopausal women	Total abdominal hysterectomy
Post-menopausal women	Total abdominal hysterectomy + BSO (bilateral salphingo-oophorectomy)

Solution to Question 29:

Among the given options, Laparoscopic tubal ligation is an irreversible method of contraception.

Tubal sterilization can be achieved by three methods which are:

- Electrocoagulation is used for the destruction of a segment of the tube
- Mechanical occlusion of the tube using a silicone rubber band (Falope ring) or a spring-loaded clip (Hulka Clemens clip)
- Suture ligation with or without tubal segment excision such as Pomeroy's method, Uchida's method, Irving's method, etc.

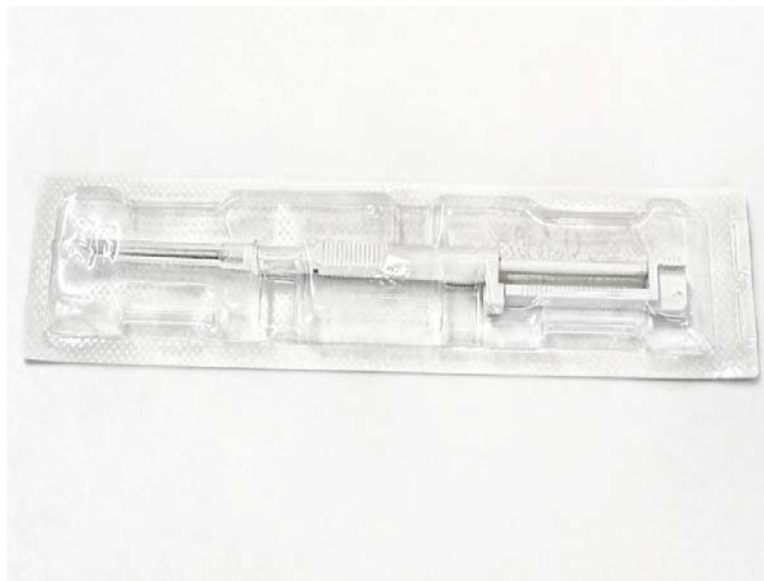
Intrauterine devices such as copper T and LNG-IUDs and implants belong to the category of long acting reversible contraception.

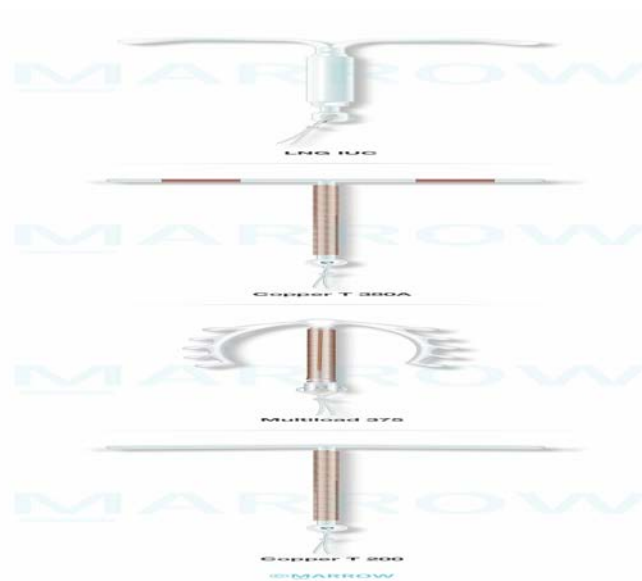
In copper-T (Option B) a copper wire is wrapped around a polypropylene vertical frame. The copper in the device causes an intense local inflammatory response in the uterus that leads to lysosomal activation and activation of other inflammatory processes that are spermicidal. The Cu-T 380A is approved for 10 years of continuous use.

LNG-IUD (Option D) is a T-shaped polyethylene structure enclosed by a cylinder containing levonorgesterl. This hormone is released continuously through the permeable cylinder. The progestin makes the endometrium atrophic and stimulates the production of thick cervical mucus that blocks sperm penetration into the uterus. It also decreases tubal motility, thereby preventing fertilization.

Nexplanon has now replaced implanon (Option C) which was a previously used etonogestrel implant. Nexplanon is a single-rod progestin etonogestrel (68 mg) containing subdermal implant that provides contraception over 3 years. Its mechanism of action is by suppression of ovulation, thickened cervical mucus, and endometrial atrophy. Irregular bleeding is an important side effect of the implant. This method of contraception is reversible in nature.

Image below shows implanon:





Solution to Question 30:

Cystoscopy is the most useful test and also the investigation of choice for vesicovaginal fistula (VVF).

Cystoscopy or cystourethroscopy helps to visualize the size of the fistulas, the exact location of multiple fistulas and their proximity to the ureteral orifice. It can also assess the bladder mucosa for any injuries.

The 3-swab test (Option B) is done to identify the types of genitourinary fistula.

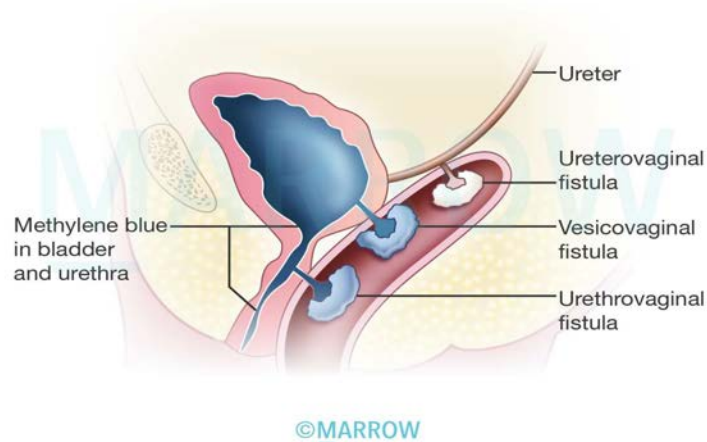
In this test, three sterile pieces of cotton swabs are serially placed in the vaginal cavity, while 50-100ml of methylene blue solution is instilled in the bladder using a catheter. The cotton swabs are placed one at the vault, one at the middle, and one just above the introitus.

Interpretation:

- Uppermost swab gets wet and not stained - the leak is from the ureter, as seen in ureterovaginal fistula
- Middle swab gets stained - the leak is from the bladder, as seen in vesicovaginal fistula
- Lowermost swab gets stained- the leak is from the urethra, as seen in urethrovaginal fistula

The below image shows the 3-swab test.

Three swab test for genitourinary fistula



Option C: MRI is usually not needed for the diagnosis of VVF. They can be used to study vesicouterine fistulas.

Option D: Cystometry is done to measure the capacity of the bladder. It is used to investigate urge incontinence.

Solution to Question 31:

Colposcopy with LEEP would be appropriate management for CIN III.

Loop electrosurgical excision procedure (LEEP) is an opd procedure done under colposcopic view and local anesthesia. Here, a low voltage current is passed through the loop and a cone of tissue is removed from the cervix. This procedure causes minimal bleeding and the tissue removed is sent for histopathological examination (HPE). This is the treatment of choice for CIN 2&3 of any age and parity.

Solution to Question 32:

The most appropriate time for the delivery in this patient would be 39 weeks as she has good glycemic control with insulin.

The timing for delivery in a pregnant woman with GDM would vary depending on the control of glucose levels & mode of treatment.

Appropriate timing for delivery in gestational diabetes	
Well-controlled with diet	39 weeks to 40 weeks +6 days

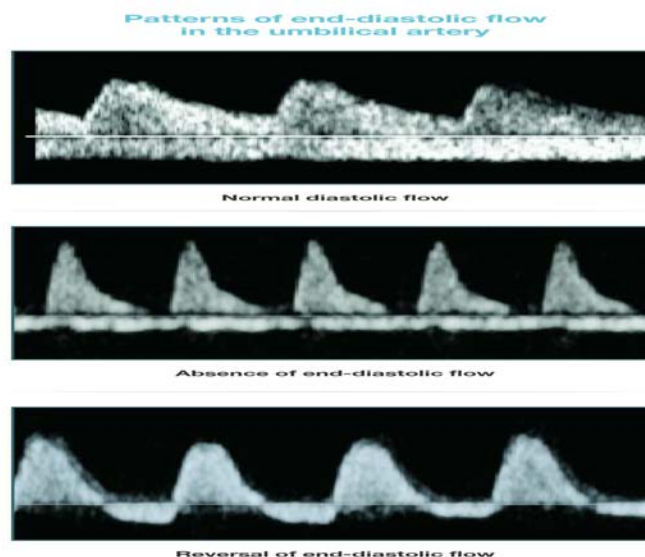
Appropriate timing for delivery in gestational diabetes	
Well-controlled with insulin/oral agent	39 weeks to 39 weeks + 6 days
Co-morbidity or suboptimal control	Induction before 39 weeks
With poor glycemic control	Reasonable to deliver at 37 to 38 weeks + 6 days of gestation In extreme cases where all efforts for glycemic control have failed induction prior to 37 weeks (no clear cut guideline)

Solution to Question 33:

The given case scenario is suggestive of fetal compromise. The management in this patient is immediate termination of pregnancy. Since this pregnancy has crossed 34 weeks, it can be terminated by cesarean section.

Abnormal flow patterns in the umbilical artery:

- Reduced end-diastolic flow indicates uteroplacental insufficiency (S/D ratio increases)
- Absent end-diastolic flow suggests fetal compromise, hence the pregnancy is terminated by C-section if ≥ 34 weeks.
- Reverse end-diastolic flow suggests impending doom of death, hence the pregnancy is terminated by C-section if ≥ 32 weeks.



Solution to Question 34:

Among the given options, the presence of third heart sound (S₃) is normal in pregnancy.

The third heart sound (S₃) occurs due to rapid diastolic filling that is seen secondary to the physiological hypervolemia that occurs during the pregnancy.

Other normal auscultatory findings include:

- First heart sound is loud with an exaggerated split
- Second heart is loud but normal
- Third heart sound is loud and easily heard
- Systolic murmur of grade <3/6
- Mammary murmur or continuous machinery murmur
- Apex beat shifted to 4th intercostal space

Solution to Question 35:

The final switch to heparin in a pregnant woman is advised at 36 weeks of pregnancy and is continued till 7 days postpartum.

This switch is made from warfarin to heparin at 36 weeks, as warfarin is a strong anticoagulant which can cause post partum hemorrhage in the mother at the time of labour and can also cross the placenta into the fetal blood and can increase the risk of intracranial hemorrhage in the fetus

The below given algorithm is followed for administration of anticoagulants in a pregnant female:

LMWH- low molecular weight heparin

UFH- Unfractionated heparin

Period of pregnancy	Anticoagulation of choice
First trimester	If dose of warfarin < 5mg/day –continue warfarin till 36 weeks If dose of warfarin > 5mg/day or warfarin embryopathy present –replace it with heparin / LMWH
From 12 weeks to 36 weeks	warfarin
At 36 weeks	Switch to LMWH
1 week before EDD / 3 days before EDD	Switch to UFH (as LMWH have longer T _{1/2} – increased chances of PPH)
On the day of delivery	Stop all anticoagulant
6 hrs after vaginal delivery / 6-12 hrs after cesarean	Restart UFH and Warfarin. When INR is 2.5 to 3 withdraw UFH & continue warfarin

Obstetrics and Gynecology AIIMS 2019 (May & Nov)

Question 1:

A 40-year-old woman comes with a history of postcoital bleeding and a visible mass seen in the cervix per speculum. The next step to be done:

- a) Conization
- b) Cryotherapy
- c) Targeted biopsy
- d) Pap smear

Question 2:

In preterm labor, which drug has a neuroprotective role?

- a) MgSO_4
- b) Nifedipine
- c) Ritodrine
- d) Isoxsuprine

Question 3:

The correct statement regarding the physiological changes in the respiratory system in pregnancy?

- a) Minute volume decreases by 50%
- b) Residual volume increases
- c) FRC decreases
- d) Total lung capacity increases

Question 4:

On finding a congenital heart block in a fetus, which of the following conditions would you evaluate for in the mother?

- a) APLA syndrome

- b) Congenital heart defects
- c) SLE
- d) Hemolytic anaemia

Question 5:

A 33 Weeks multigravida woman presents with epigastric pain, headache, visual symptoms, and proteinuria - 3+. The next step of management would be?

- a) Immediate c - section
- b) Dexamethasone for fetal lung maturity
- c) Induction - vaginal delivery
- d) MgSO₄ IV

Question 6:

All of the followings are seen in GDM except:

- a) Malformations
- b) Polyhydramnios
- c) Fetal Macrosomia
- d) Neonatal Hypoglycemia

Question 7:

Hormone replacement therapy is not indicated for:

- a) Hot flashes
- b) CAD Prophylaxis
- c) Osteoporosis
- d) Premature menopause

Question 8:

The risk of fetal varicella syndrome is maximum when the mother is infected between ____.

- a) 6 to 10 weeks

- b) 8 to 12 weeks
- c) 13 to 20 weeks
- d) 20 to 28 weeks

Question 9:

The loading dose of MgSO_4 in pre-eclampsia is?

- a) 4 ml MgSO_4 (50% w/v) in 16 ml NS
- b) 8 ml MgSO_4 (50% w/v) in 12 ml NS
- c) 12 ml MgSO_4 (50% w/v) in 8 ml NS
- d) 10 ml MgSO_4 (50% w/v)

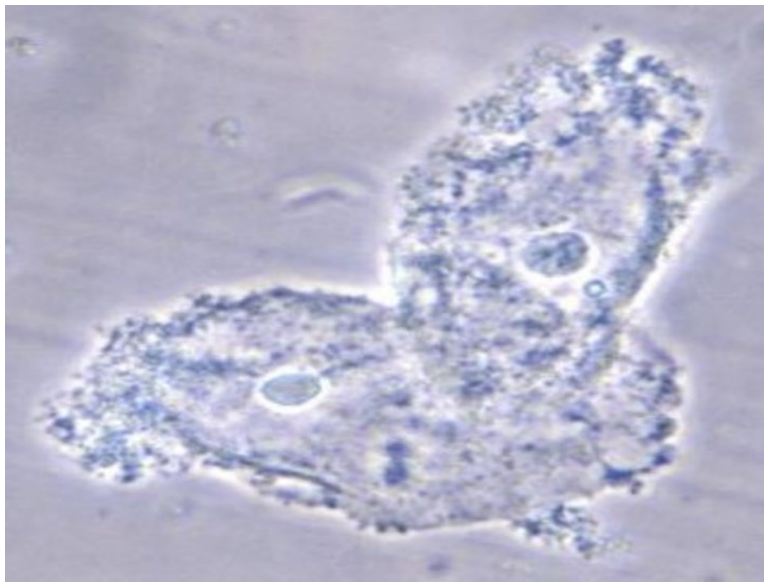
Question 10:

Drug of choice for treatment of pregnant woman with bacterial vaginosis?

- a) Metronidazole
- b) Flucanazole
- c) Clindamycin
- d) Erythromycin

Question 11:

A 40-year-old female presented with a history of profuse vaginal discharge. The discharge has a fishy odour. Which of the following best describes the microscopic finding given below?



- a) Epithelial cell with bacteria attached on cell surface
- b) Bacteria engulfed by macrophages
- c) Neutrophil engulfed bacteria
- d) Bacteria arranged in clusters

Question 12:

The following instrument is used for _____



- a) VIA
- b) VILI

- c) PAP smear
- d) Colposcopy

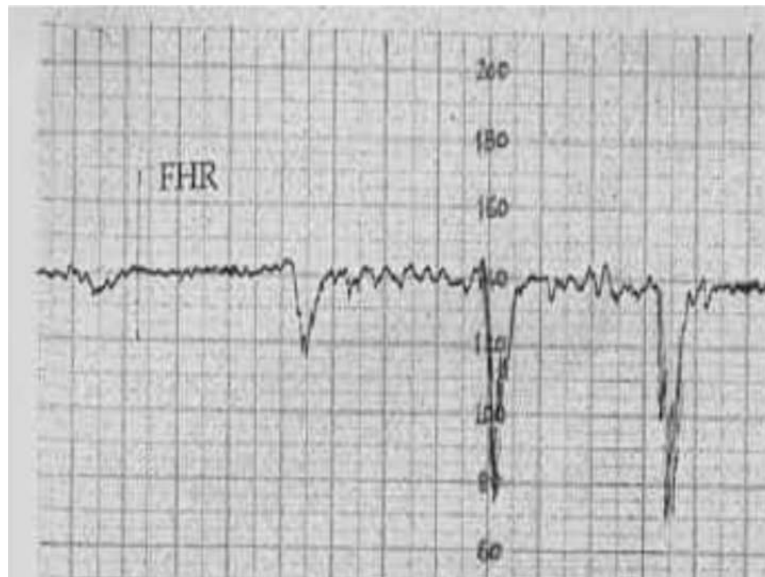
Question 13:

Anti-D vaccine is not given in which of the following conditions?

- a) Amniocentesis at 16 weeks
- b) Manual removal of placenta
- c) Intrauterine transfusion at 28 weeks
- d) MTP at 63 days

Question 14:

What does this tocograph show?



- a) Deceleration
- b) Acceleration
- c) Normal
- d) Deceleration followed by acceleration

Question 15:

A 16-year-old unmarried female has come to the gynecology OPD for cervical cancer vaccination. Which vaccine is preferred?

- a) Gardasil
- b) Biovac
- c) Rabivac
- d) Tvac

Question 16:

Which of the following agents cause hydrops fetalis?

- a) CMV
- b) Parvovirus
- c) Toxoplasma
- d) HIV

Question 17:

The given instrument is used in:



- a) Vaginal hysterectomy
- b) Cesarean section
- c) Manchester operation
- d) Fothergill's surgery

Question 18:

A case of gestational trophoblastic neoplasm is detected to have metastasis in the lungs. It is considered as stage _____.

- a) I
- b) II
- c) III
- d) IV

Answer Key

Question No.	Correct Option
1	c
2	a
3	c
4	c
5	d
6	a
7	b
8	c
9	b
10	a
11	a
12	c
13	c
14	a
15	a
16	b
17	b
18	c

Detailed Explanations

Solution to Question 1:

A targeted biopsy is the next line of management in the patient with a visible cervical mass and postcoital bleeding. A pap smear is done when there are symptoms with no identifiable lesion.

Other options:

Option A: Conization is an invasive procedure where the cervix is cut in a cone-shaped manner. It is both diagnostic and therapeutic. It is indicated for diagnosis in women with HSIL or adenocarcinoma in situ and may be considered under the following conditions:

- The limits of the lesion cannot be visualized with colposcopy.
- SCJ (squamocolumnar junction) is not seen in colposcopy.
- Colposcopy is positive for CIN 2/CIN 3.
- Lack of correlation between cytology, biopsy, and colposcopy.
- Suspected micro invasion in cytology, colposcopy, or biopsy.
- Colposcopist is unable to rule out invasive cancer.

Hence it is very clear that conization is not indicated for all symptomatic women and it should be done following either a PAP, colposcopy or a biopsy with an abnormal or inconclusive result.

Option B: Cryotherapy is ideal only for small superficial lesions. It is indicated for biopsy-proven lesions. Cryotherapy is appropriate for CIN 1 which has persisted for more than 24 months and is confined to the ectocervix.

Option D: The false negative rate of pap smear in the presence of invasive cancer is 50% and a negative pap test should never be relied on in a symptomatic patient, especially one with a visible lesion.

CIN: Cervical intraepithelial neoplasia, HSIL: High squamous intraepithelial lesion, LSIL: Low squamous intraepithelial lesion

Solution to Question 2:

In preterm labor, MgSO₄ (magnesium sulfate) is considered to have a neuroprotective role. Studies have demonstrated that magnesium exposure (compared with no exposure) significantly lowers the risks for cerebral palsy.

The combination of corticosteroids and magnesium sulfate has been associated with lower rates of neurodevelopmental impairment and death compared to corticosteroids alone.

The proposed mechanisms of action include:

- reduced presynaptic release of glutamate
- blockade of glutamatergic NMDA receptors
- potentiation of adenosine action
- improved calcium buffering by mitochondria
- blockage of calcium entry via voltage-gated channels

MgSO₄ is used as an anticonvulsant in the management of eclampsia and as prophylaxis in severe preeclampsia. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L. Measurement of serum magnesium levels is indicated to monitor magnesium toxicity (if serum creatinine > 1.0 mg/dL).

Loss of patellar reflex is the first sign of MgSO₄ toxicity and occurs when the plasma magnesium level reaches 10 mEq/L. Elevated magnesium levels above 10 mEq/L weaken breathing, while 12 mEq/L or higher levels can lead to respiratory paralysis and arrest. Other signs of magnesium toxicity include urine output <30 mL/h, chest pain, and heart block visual changes, flushing, muscle paralysis, and drowsiness.

In suspected toxicity, MgSO₄ therapy should be stopped and a slow IV infusion of 1g calcium gluconate 10 mL (10% solution) is given.

Other options:

Options B, C, and D: Nifedipine (calcium channel blocker), and ritodrine, isoxsuprine (beta-adrenergic agonists) are used as tocolytics in preterm labor.

Solution to Question 3:

Functional residual capacity (FRC) decreases by 20 to 30 percent (400 to 700 mL) during pregnancy.

FRC = Expiratory Reserve Volume (ERV) + Residual Volume (RV). ERV decreases by 15 to 20 percent (200 to 300 mL). RV decreases by 20 to 25 percent (200 to 400 mL).

FRC and RV decline due to diaphragm elevation. Significant reductions occur by the sixth month with a progressive decline across pregnancy.

The increased minute ventilation during pregnancy is influenced by various factors. These include the stimulatory action of progesterone, reduced expiratory reserve volume, and compensatory respiratory alkalosis. Total lung capacity (FRC + IC) decreases in pregnancy.

Anatomical changes such as the elevation of the diaphragm, an increase in the subcostal angle, and an expansion of the transverse diameter of the thoracic cage are seen.

Respiratory changes in pregnancy are given below:

(Mnemonic: TIMM increases in pregnancy)

Increase	Decrease	Unaffected
Tidal volume	Functional residual capacity	Respiratory rate
Inspiratory capacity (IC)	Expiratory reserve volume	Vital capacity
Minute O ₂ uptake	Residual volume	Inspiratory reserve volume
Minute ventilation	Total lung capacity	Lung compliance
Peak expiratory flow rate and airway conductance	Total pulmonary resistance	

Solution to Question 4:

On finding a congenital heart block in the fetus, the mother should be evaluated for systemic lupus erythematosus (SLE).

Fetal and neonatal heart blocks occur due to myocarditis and fibrosis in the region between the AV node and the bundle of His. Congenital heart block primarily develops in fetuses of women with antibodies to the SS-A or SS-B antigens, and up to 50 percent of women with SLE test positive for these antibodies. So in pregnancies with either of these antibodies, fetal cardiac monitoring is performed between 18 and 26 weeks of gestation.

The cardiac lesion is permanent, and in most cases, a pacemaker is required. Maternal hydroxychloroquine therapy is associated with a reduced incidence of fetal heart block, particularly in women who have previously had an affected child.

Neonatal lupus erythematosus is thought to be caused by the transplacental passage of maternal antibodies. It is characterized by skin lesions known as lupus dermatitis in newborns, as well as various hematological and systemic abnormalities. In some cases, congenital heart block may also occur. Thrombocytopenia and hepatic involvement are also seen.

The most common cutaneous finding in neonatal lupus erythematosus is an erythematous, slightly scaly eruption on the face and periorbital skin (raccoon sign/owl eye/eye mask). The eruption can be exacerbated by UV exposure. The rash improves over the first few months of life and usually resolves without scarring by 12 months of age.

Avoidance of sunlight should be advised and low-potency topical steroids may be of benefit.

The pregnancy outcomes are best in women with:

- Dormant lupus activity for at least 6 months before conception
- No lupus nephritis manifestation in the form of proteinuria or renal dysfunction
- Absent anti-phospholipid syndrome or lupus anticoagulant
- Superimposed preeclampsia does not develop

Solution to Question 5:

The given clinical scenario of a 33 weeks pregnant woman with epigastric pain, headache, visual symptoms, and proteinuria 3+ indicates severe preeclampsia. The next step of management would be iv administration of MgSO₄ for the prevention of seizures.

Preeclampsia can be classified as severe if any of the following are present:

- BP $\geq$ 160/110
- Headache
- Visual disturbance
- Upper abdominal pain

- Elevated creatinine, oliguria
- Thrombocytopenia ($<1,00,000$ cells per mm³)
- Pulmonary edema
- Marked elevation in serum transaminases
- Fetal growth restriction

Dexamethasone (option B) is also given in severe preeclampsia <34 weeks for fetal lung maturity. But the administration of MgSO₄ before corticosteroids would be the next best step for the prevention of seizures.

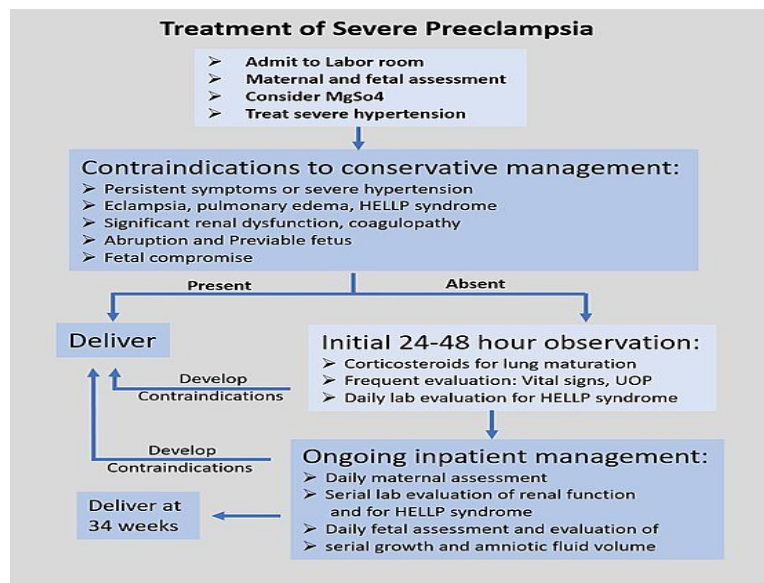
Immediate delivery (options A and C) is not indicated in this case as the patient doesn't have any contraindication for conservative management. Conservative management is contraindicated in the following cases:

- Persistent symptoms, refractory to treatment or severe hypertension
- Eclampsia, pulmonary edema, HELLP syndrome
- Significant renal dysfunction or coagulopathy
- Placental abruption
- Nonreassuring fetal status
- Fetal demise
- Disseminated intravascular coagulation
- Stroke, myocardial infarction

MgSO₄ regimens for management of eclampsia and severe preeclampsia:

Magnesium sulfate is discontinued either 24 hours after delivery or 24 hours after the onset of convulsions (whichever is later).

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4g (20% solution) IV over 3-5 minutes followed by 10g (50%), deep IM (5 gm in each buttock)	5g (50%) IM 4 hourly in the alternate buttocks
Intravenous (Zuspan or Sibal)	4g IV slow over 15-20 minute	1-2g/h IV infusion



Solution to Question 6:

Malformations are not associated with gestational diabetes mellitus (GDM). Malformations occur due to the disruption of organogenesis, usually occurring in the first trimester. The abnormal glycemic status in GDM manifests only in the second trimester.

Overt diabetes (pregestational diabetes) is associated with higher rates of malformations compared to the general population. Neural tube defects (encephalocele, meningomyelocele, anencephaly), transposition of great vessels (TGA), ventricular septal defect (VSD), atrial septal defect (ASD), hypoplastic left heart, aortic stenosis and coarctation of the aorta are commonly seen.

Caudal regression syndrome is rare, but a specific anomaly seen in infants of diabetic mothers. Small left colon syndrome is a rare anomaly that develops in the second and third trimesters.

Other options:

Option B: In GDM, increased blood glucose in the mother leads to fetal hyperglycemia. This can cause polyhydramnios due to increased fetal urine production.

Option C: Fetal hyperinsulinemia causes increased hepatic glucose uptake and glycogen synthesis, accelerated lipogenesis, and augmented protein synthesis resulting in macrosomia

Option D: Fetal hyperglycemia in GDM triggers fetal hyperinsulinemia, resulting in neonatal hypoglycemia

Solution to Question 7:

Hormone replacement therapy is not indicated in coronary artery disease prophylaxis. There is currently insufficient evidence to recommend the initiation or continuation of either estrogen therapy (ET) or estrogen-progestin therapy (EPT) for the prevention of primary or secondary

coronary heart disease (CHD).

In an 18-year reevaluation of WHI (women health initiative) participants, it was found that cardiovascular, cancer, and all-cause mortality rates were essentially the same in both the hormone therapy (HT) and placebo groups.

Currently, hormone therapy (HT) is only indicated for the treatment of bothersome vasomotor symptoms (VMS) and genitourinary symptoms, prevention of osteoporosis, and treatment of hypoestrogenism caused by hypogonadism, surgical menopause, or premature ovarian insufficiency.

Estrogen should not be used in women with the following conditions:

- undiagnosed abnormal genital bleeding
- known, suspected, or history of breast cancer
- known or suspected estrogen-dependent neoplasia
- active or prior venous thromboembolism
- active or prior stroke, myocardial infarction, or transient ischemic attack (TIA)
- known thrombophilia, liver dysfunction, or disease
- untreated hypertension
- known hypersensitivity to the ingredients of the estrogen preparation
- known or suspected pregnancy

Risks of HRT:

- Endometrial cancer
- Breast cancer
- Ovarian cancer
- Thromboembolism
- Lipid profile dysfunction
- Gall stones, liver dysfunction

Solution to Question 8:

The risk of developing fetal varicella syndrome (congenital varicella syndrome) is highest when the mother develops varicella infection between 13 to 20 weeks.

Congenital varicella syndrome occurs in 2% of infants with maternal varicella between 13 and 20 weeks of gestation and in 0.4% of infants before 13 weeks of gestation. It is rare after 20 weeks.

Congenital varicella syndrome is characterized by cicatricial skin scarring in a zoster-like pattern (zig-zag scarring), and limb hypoplasia (shown below). They can also present with neurological abnormalities (microcephaly, cortical atrophy, mental retardation), ophthalmic features (chorioretinitis, microphthalmia, cataract), and renal and autonomic system abnormalities (hydronephrosis, neurogenic bladder). Low birth weight is also common.

Infants born to mothers with varicella in the period from 5 days prior to delivery up to 2 days after delivery are at high risk for severe neonatal varicella. They acquire the infection transplacentally.

Since the mother has not developed a strong antibody response, the infant lacks the protective effect of maternal anti-varicella-zoster virus (VZV) antibodies. Therefore varicella-zoster immune globulin (VZIG) should be administered as soon as possible after birth to newborns whose mothers have varicella 5 days before and up to 2 days after the delivery.

Since perinatally acquired varicella can be potentially life-threatening, the infant should typically be treated with intravenous acyclovir (10 mg/kg every 8 hours) when lesions appear.

For pregnant women with uncomplicated primary varicella or herpes zoster that does not require hospitalization, the recommended treatment is oral acyclovir (800 mg five times daily for seven days).



Solution to Question 9:

The loading dose of MgSO_4 is 8ml MgSO_4 (50% w/v) in 12ml NS. For loading dose IV infusion of 4g of 20% solution is needed.

One 2mL vial has 1g MgSO_4 . Therefore, to make a 20% solution containing 4g of MgSO_4 , we would need to take 8 mL of 50% magnesium sulfate, and dilute it with 12 mL NS. Therefore the final 20mL solution (8+12) would have 4g of MgSO_4 , making it a 20% solution.

MgSO_4 is used as an anticonvulsant in the management of eclampsia and as prophylaxis in severe preeclampsia. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L. Measurement of serum magnesium levels is indicated to monitor magnesium toxicity if serum creatinine ≥ 1.0 mg/dL.

MgSO_4 regimens for management of eclampsia and severe preeclampsia:

Repeat injections of magnesium sulfate are continued only if the patellar reflex is present, respiratory rate is ≥ 12 , and urine output is ≥ 30 mL/h. Magnesium sulfate is discontinued

either 24 hours after delivery or 24 hours after the onset of convulsions (whichever is later).

Loss of patellar reflex is the first sign of MgSO₄ toxicity and occurs when the plasma magnesium level reaches 10 mEq/L. Elevated magnesium levels above 10 mEq/L weaken breathing, while 12 mEq/L or higher levels can lead to respiratory paralysis and arrest.

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4 g (20% solution) IV over 3-5 minutes followed by 10 g (50%), deep IM (5 gm in each buttock)	5 g (50%) IM 4 hourly in the alternate buttock
Intravenous (Zuspan or Sibai)	4g IV slow over 15-20 minute	1-2 g/h IV infusion

Solution to Question 10:

The preferred treatment for bacterial vaginosis in pregnancy is either oral metronidazole 500 mg BD for 7 days (or intravaginal metronidazole for 5 days). Treatment is typically reserved for symptomatic women.

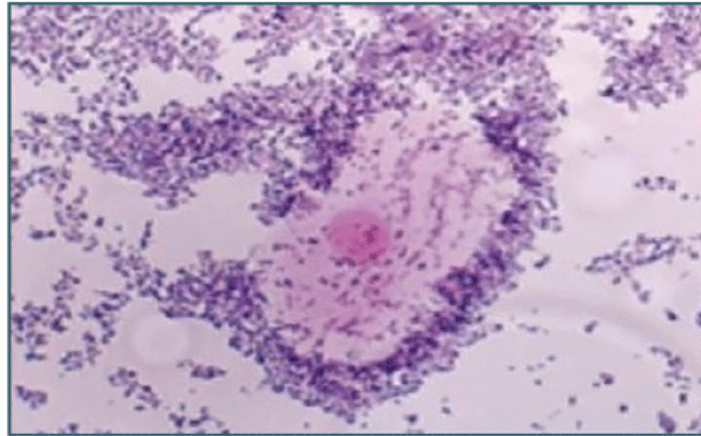
An alternative option is oral clindamycin, 300 mg BD for 7 days (or intravaginal clindamycin ovules for 3 days).

Bacterial vaginosis (BV) is characterized by a decrease in lactobacilli and an overgrowth of anaerobic bacterial species, including *Gardnerella*, *Prevotella*, *Mobiluncus*, and *Bacteroides*. Most women with BV are asymptomatic, but those with symptoms may experience a foul-smelling, thin vaginal discharge.

The clinical diagnosis of BV is based on the presence of three out of four Amsel criteria: elevated vaginal pH (>4.5), thin non-inflammatory discharge, >20% clue cells, and a fishy odor (positive whiff test) after adding 10% KOH (potassium hydroxide) to a vaginal sample.

BV is associated with adverse pregnancy outcomes such as preterm birth, preterm premature rupture of membranes, and postpartum endometritis. It also increases susceptibility to sexually transmitted infections, including HIV.

The image below shows a vaginal epithelial cell, with adherent bacteria (clue cell) :



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Solution to Question 11:

The given clinical scenario of women with fishy odor vaginal discharge is suggestive of bacterial vaginosis (BV). The microscopic finding is vaginal epithelial cells with bacteria attached to their surface called clue cells, creating a poorly defined cellular border. They are the most reliable indicators of BV.

Bacterial vaginosis (BV) is characterized by a decrease in lactobacilli and an overgrowth of anaerobic bacterial species, including *Gardnerella*, *Prevotella*, *Mobiluncus*, and *Bacteroides*. Multiple or new sexual partners, having female partners, and engaging in oral sex are associated with a higher risk of BV. They may experience a foul-smelling (fishy odor) thin vaginal discharge.

The clinical diagnosis of BV is based on the presence of three out of four Amsel criteria: elevated vaginal pH (>4.5), thin noninflammatory discharge, $>20\%$ clue cells, and a fishy odor (positive whiff test) after adding 10% potassium hydroxide to a vaginal sample. creating a poorly defined cellular border.

BV is associated with adverse pregnancy outcomes such as preterm birth, preterm premature rupture of membranes, and postpartum endometritis. It also increases susceptibility to sexually transmitted infections, including HIV.

The recommended treatment options in nonpregnant women include oral metronidazole, metronidazole gel, or clindamycin cream. Alternative options include oral secnidazole, tinidazole, or clindamycin. Partner treatment is not needed.

Solution to Question 12:

The image shown is that of Ayre's spatula which is used for preparing PAP smear. The Papanicolaou test (Pap test) is a screening test for cervical cancer.

The squamocolumnar junction is scraped using Ayre's spatula by rotating it around the cervix. The obtained scrapings are spread evenly onto a glass slide and immediately fixed in a solution of 95% ethyl alcohol and ether. After 30 minutes of fixation, the slide is air-dried and then stained with Pap or a short stain method.

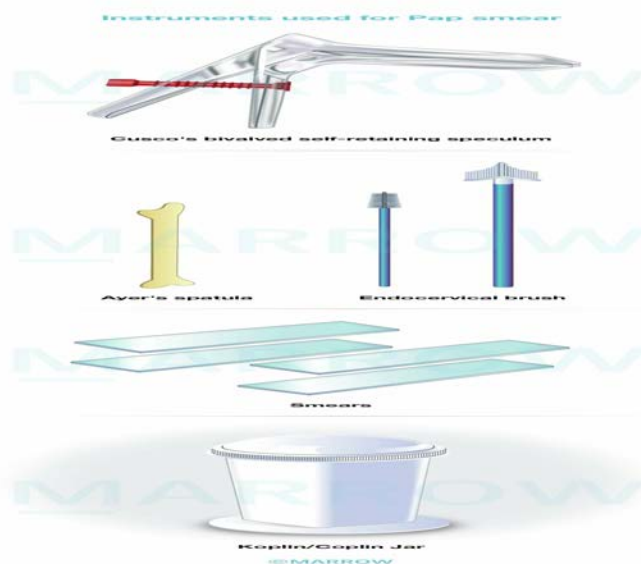
Prerequisites for Pap smear testing:

- Should be done prior to vaginal examination.
- Lubricants should not be used.
- The patient should not be menstruating during the pap test.
- The ideal time is around ovulation.
- The patient should not have had intercourse in the 24hrs prior to the test.

Screening guidelines:

- Time for initiation: 21 years, regardless of the age of first sexual intercourse.
- Women aged 21-29 years: Pap test every 3 years
- Women who are aged 30-65 years: Pap test + HPV testing (co-test) every 5 years or Pap alone every 3 years.
- Women should stop having cervical cancer screening after the age of 65 years. If suspicious PAP, screening till 75 years.
- HIV+ women: annual checkup until three consecutive negative tests followed by testing every 3 years for life.

Instruments used for Pap smear are given below:



Solution to Question 13:

The anti-D vaccine is not given in intrauterine transfusion at 28 weeks. Intrauterine transfusion is done in alloimmunized pregnancies, where the maternal immune system (Rh negative) has already been exposed to the fetal blood (Rh positive).

The purpose of the anti-D vaccine is to prevent the development of maternal antibodies against fetal red cells, which is already occurring in alloimmunized pregnancies.

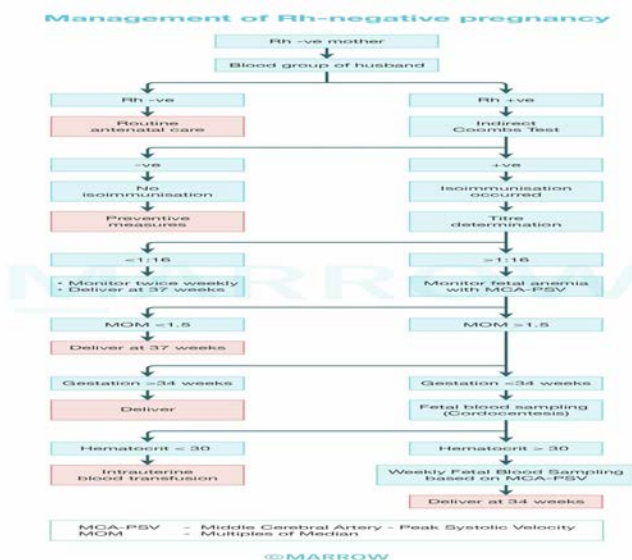
Anti-D vaccine is administered only in ICT-negative non-immunized pregnancies.

Intrauterine transfusion is done when the peak systolic velocity of the middle cerebral artery (MCA) exceeds 1.5 multiples of the median or if hydrops develop or if the fetal hematocrit is $\leq 30\%$. This indicates that the mother already has antibodies against the fetus's red blood cells, making the administration of anti-D unnecessary.

The indications for administering anti-D are:

- Pregnancy loss: spontaneous or elective abortion (option D) and ectopic pregnancy.
- Fetal death: Regardless of the trimester.
- Procedures: Such as amniocentesis (option A), chorionic villus sampling, fetal blood sampling, and molar pregnancy evacuation.
- Other situations: Antepartum bleeding (including threatened abortion), delivery (vaginal or cesarean), external cephalic version, placental abruption, manual removal of placenta (option B), and trauma to the abdomen during pregnancy.

All Rh D-negative women should be given a single dose of 300 mcg of anti-D at 28 weeks of pregnancy (ICT should be negative).



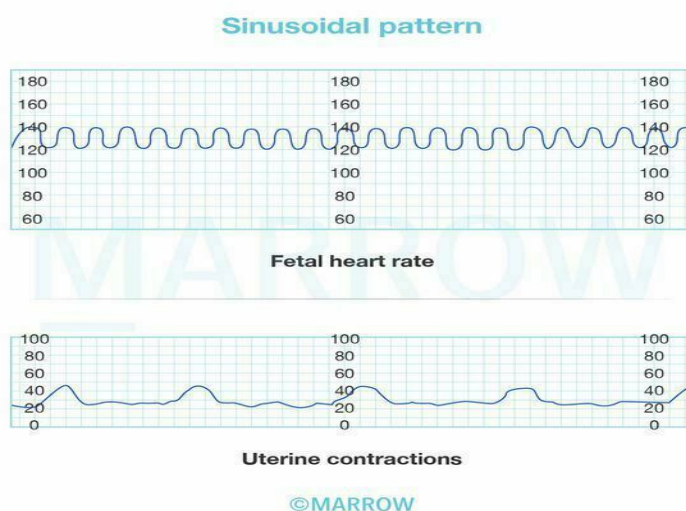
Solution to Question 14:

The given topograph shows a drop in fetal heart rate below the baseline indicating decelerations. Deceleration is the decrease in fetal heart rate below the baseline by 15 bpm for more than 15s.

Following are the various types of decelerations seen on CTG and their clinical significance:

The CTG given below shows a sinusoidal pattern. It resembles a sine wave. It is often associated with fetal anemia, fetomaternal hemorrhage, and fetal acidosis.

Feature	Early deceleration	Late deceleration	Variable deceleration
The pattern of heart rate fall	The maximum dip of the deceleration corresponds with the peak of the contraction.	Onset, maximum dip, and recovery occur after the onset, peak, and recovery of contraction respectively	Abrupt fall in the heart rate. Decelerations are variable in respect of the size, shape, depth, duration, and timing of uterine contractions.
Time taken to reach the maximum dip of deceleration	>30seconds	>30seconds	<30 seconds
Cause	Fetal head compression	Fetal hypoxia due to maternal hypotension, placental insufficiency, and excessive uterine activity	Umbilical cord compression



Solution to Question 15:

Gardasil is a vaccine administered to provide protection against cervical cancer.

Gardasil is a quadrivalent vaccine providing protection against HPV serotypes 6,11,16 and 18. It can be given to both males and females and provides protection against anogenital warts, CIN (cervical intraepithelial neoplasia), and Ca cervix.

Types of available HPV vaccines:

Cervavac is India's first indigenously prepared quadrivalent HPV (qHPV) vaccine against HPV strains 6, 11, 16, and 18.

As per the December 2022 WHO updated dose schedules on HPV vaccines, the recommendations for different age groups are as follows:

- 1 or 2 doses schedule for the primary target of girls aged 9–14 years.
- 1 or 2 doses schedule for young women aged 15–20 years.
- 2 doses with a 6-month interval for women older than 21 years.
- 3 doses (or at least 2 doses) for immunocompromised individuals/HIV infected.

Brand name	Cervarix	Gardasil	Gardasil-9	Cervavac
Characteristic and HPV subtypes	Bivalent: 16,18	Quadrivalent: 6,11,16,18	Nonavalent: 6,11,16,18,31,33,45,52,58	Quadrivalent: 6,11,16,18
Protects against:	CINCA cervix	Anogenital wartsCINCA cervix	Anogenital wartsCINCA cervixVulval intraepithelial neoplasiaVaginal intraepithelial neoplasiaAnal cancerOropharyngeal cancerHead and neck cancers	Anogenital wartsCINCA cervix
Dose	0.5 ml, IM.	0.5 ml, IM.	0.5 ml, IM.	0.5 ml, IM.

Other options:

Option B: Biovac A is a freeze-dried live attenuated vaccine indicated for the prevention of hepatitis A in persons of age 1 year or older.

Option C: Rabivac is a monovalent inactivated rabies vaccine for cats and dogs.

Option D: Tvac is a Tetanus vaccine.

Solution to Question 16:

Parvovirus B19 is the most common infectious cause of nonimmune hydrops fetalis. Parvovirus B19 infects fetal erythroid progenitor cells and causes anemia. Severe anemia leads to high-output cardiac failure and nonimmune hydrops fetalis.

Hydrops is diagnosed by identifying two or more fetal effusions, such as pleural, pericardial, or ascitic effusions, or the presence of one effusion along with anasarca. Hydrops can be seen in association with red cell auto immunization, which is termed immune hydrops. Otherwise, it is nonimmune.

Nonimmune hydrops is more common than immune hydrops. Causes of nonimmune hydrops include cardiac pathology (Ebstein anomaly), chromosomal (Turner's syndrome), hematological (hemoglobinopathies), lymphatic abnormalities (cystic hygroma), and thoracic abnormalities.

Infectious causes of nonimmune hydrops are parvovirus B19 (most common), syphilis, cytomegalovirus, toxoplasmosis, rubella, enterovirus, varicella, herpes simplex, coxsackievirus, listeriosis, leptospirosis, Chagas disease, and Lyme disease.

The following image shows a stillborn baby with hydrops fetalis:



Solution to Question 17:

The image shown is that of a Doyen's retractor which is used in caesarean sections.

It is used to retract the abdominal wall as well as the bladder for proper exposure of the lower uterine segment during cesarean sections.

Solution to Question 18:

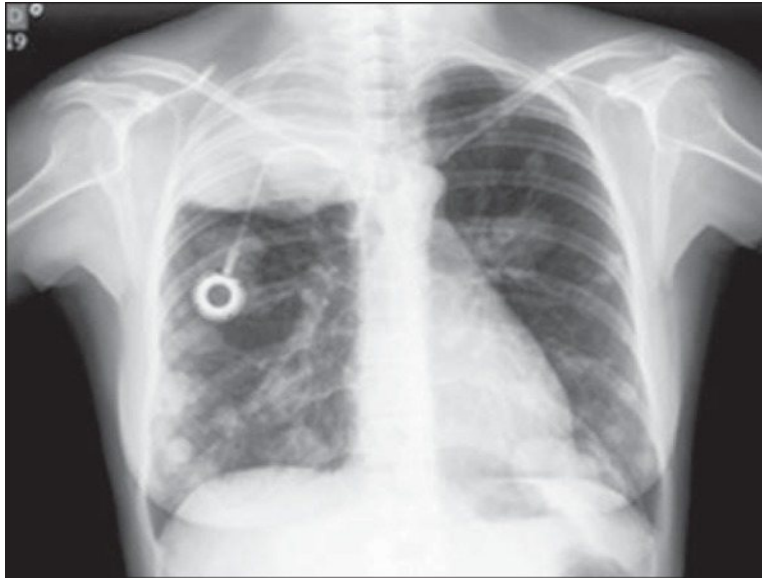
Metastases in the lungs are considered stage III GTN (gestational trophoblastic neoplasia) in FIGO classification.

International Federation of Gynecology and Obstetrics (FIGO) staging for GTN is as follows:

The chest X-ray below shows lung metastasis - stage III GTN.

Stage	Description
I	GTN confined to the uterus
II	GTN extends outside of the uterus but is limited to the genital structures (adnexa, vagina, broad ligament)
III	GTN extends to the lungs, with or without known genital tract involvement

Stage	Description
IV	Allothermetastatic sites i.e. kidney, liver, brain, bowel, etc.



Gestational trophoblastic neoplasia (GTN) refers to the aggressive invasion of the endometrium and myometrium by trophoblast cells. It includes invasive mole, gestational choriocarcinoma, placental site trophoblastic tumors, and epithelioid trophoblastic tumors. GTN is diagnosed based on elevated β -hCG levels. FIGO criteria for diagnosis of GTN:

- Persistence of β -hCG level plateau in 4 measurements for 3 weeks or longer
- Rise in β -hCG levels in 3 or more consecutive measurements done weekly, over a period of 2 weeks or longer
- Elevated β -hCG level for 6 months or more
- Histologic diagnosis of choriocarcinoma

Treatment for low-risk disease is chemotherapy with methotrexate, while those with high-risk disease are treated with etoposide, methotrexate, and dactinomycin alternating with cyclophosphamide and vincristine (EMA/CO) chemotherapy.

FIGO staging of gestational trophoblastic neoplasia



Stage I:
Confined to the uterus



Stage II:
Extends outside the uterus
but limited to genital structures



Stage III:
Extends to the lungs,
with or without
genital tract involvement



Stage IV:
All other metastatic sites

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Obstetrics and Gynecology AIIMS 2020 (May and Nov INI-CET)

Question 1:

The following groups of people must be screened for Ca cervix:

- a) More than 65 years old
- b) Less than 21 years old
- c) 21 to 65 year old females
- d) Teenagers

Question 2:

Oral contraceptives act by all of the following mechanisms except:

- a) Inhibit implantation
- b) Cervical mucus thickening
- c) Inhibit ovulation
- d) Increase GnRH release

Question 3:

A 30-year-old woman is diagnosed with a high-grade squamous intraepithelial lesion (HSIL) of the cervix and no obvious macroscopic growth. What is the next line of management?

- a) Hysterectomy
- b) Conization
- c) Colposcopy and biopsy
- d) Follow-up after 6 months

Question 4:

A patient with 6 weeks amenorrhea and a positive pregnancy test came with complaints of lower abdominal pain, moderate bleeding PV. She was hemodynamically stable. On USG, there was no gestational sac in the uterus. Ovaries were normal and no free fluid in the pouch

of Douglas. What is the next line of management?

- a) Declare as complete abortion and follow-up in 15 days
- b) Admit and observe
- c) Diagnostic laparoscopy
- d) Follow-up with beta-HCG

Question 5:

A hypertensive pregnant woman with abdominal pain, bleeding per vaginum, and loss of fetal movements. What is the probable diagnosis?

- a) Abruptio placenta
- b) Placenta previa
- c) Hydramnios
- d) Preterm labour

Question 6:

A 30-year-old woman visits an infertility clinic. She has a regular 28-day cycle. How should she be evaluated?

- a) Serum progesterone on day 14
- b) Serum progesterone on day 21
- c) Serum LH on day 14
- d) Serum LH on day 21

Question 7:

A 50-year-old woman presents with abnormal uterine bleeding for two months. What is the next line of management?

- a) LNG- IUCD
- b) POP for 3 months
- c) Hysterectomy
- d) Endometrial sampling

Question 8:

A patient with 12 weeks amenorrhea presents with vaginal bleeding. On examination, the cervical os is open and the uterus is of 14 weeks size. USG shows a snowstorm appearance. The next best step in the management is?

- a) Medical management
- b) Follow up with beta-hCG
- c) Suction and evacuation
- d) Hysterectomy

Question 9:

A 45-year-old woman is found to have a fibroid of 14 weeks size. She has no specific symptoms. What is the next step in management?

- a) Hysterectomy
- b) Myomectomy
- c) Observation
- d) Medical management

Question 10:

All of the following structures are cut in mediolateral episiotomy except?

- a) Transverse perineal muscles
- b) Obturator muscle
- c) Bulbospongiosus
- d) Transverse perineal branches of pudendal nerves

Question 11:

Raise in ESR during pregnancy is due to?

- a) Albumin
- b) Platelet
- c) Fibrinogen
- d) Anti-thrombin antibody

Question 12:

Which of the following criteria is used for diagnosing abdominal pregnancy?

- a) Spiegelberg's criteria
- b) Rubin's criteria
- c) Studdiford's criteria
- d) Godin's criteria

Question 13:

Which of the following is not a component of fetal biophysical profile?

- a) Fetal movement
- b) Fetal breathing
- c) NST
- d) Contraction stress test

Question 14:

All of the following can be used for the treatment of PCOD except?

- a) Clomiphene citrate
- b) Letrozole
- c) Ulipristal
- d) Laparoscopic ovarian drilling

Question 15:

Which of the following is not a component of the Bishop score?

- a) Cervical length
- b) Type of maternal pelvis
- c) Cervical position
- d) Dilatation of cervix

Question 16:

A nursing assistant posted in the obstetric care unit was asked to monitor the progress of labour. After how much dilatation of the cervix, should she start the documentation?

- a) 2cm
- b) 5cm
- c) 4cm
- d) 6cm

Question 17:

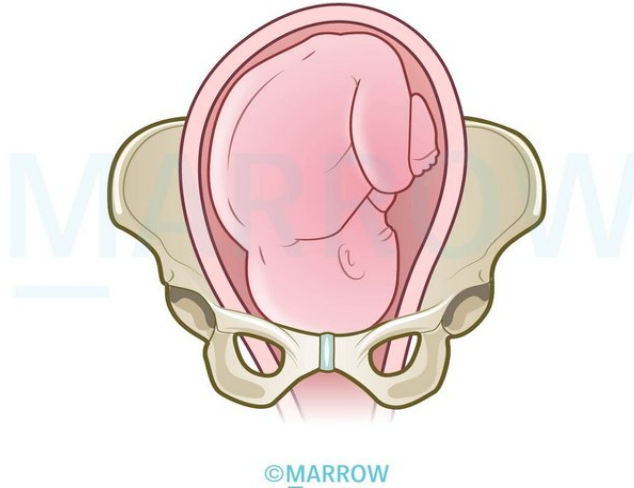
Identify the grip shown in the image below:



- a) Pawlik's grip
- b) Lateral grip
- c) Pelvic grip
- d) Fundal grip

Question 18:

Identify the fetal position in the given image.



- a) Right occipito-posterior
- b) Left occipito-anterior
- c) Left occipito-posterior
- d) Right occipito-anterior

Question 19:

A midwife is examining a pregnant woman in labour when she noticed that the cord is prolapsed. What should be the next immediate step of management?

- a) Gently replace the cord inside the vagina
- b) Inform the doctor and shift to OT
- c) Put her in Trendelenberg position
- d) Do vaginal examination and find the fetal presentation

Question 20:

A 45-year-old patient comes with repeated infections of trichomonas vaginitis, her physician advises for a Pap smear. She asks you regarding the purpose of this test. What would be your appropriate response?

- a) It is a screening test for cervical cancer
- b) It is a diagnostic test for trichomonas infection
- c) It helps to diagnose cancer in her genital tract

- d) It helps to diagnose other associated genital infections in her

Question 21:

Modified Robson's classification class 1 includes:

- a) Nulliparous, singleton, cephalic, induced labour
- b) Nulliparous, singleton, cephalic, spontaneous labour
- c) Multiparous without previous CS, spontaneous labour
- d) Multiparous without previous CS, induced/CS before labour

Question 22:

A pregnant woman has cramps in her legs. What will you suggest her to relieve the pain?

- a) Extend knee and dorsiflex foot
- b) Flex knee and dorsiflex foot
- c) Extend knee and plantar flex foot
- d) Flex knee and plantar flex foot

Question 23:

A woman in labour presented to the ER. A vaginal breech delivery was planned for her. What statement regarding the management is false?

- a) Breech identified for the first time during delivery is a contraindication for vaginal delivery
- b) Breech delivery is done in an hospital where a facility for emergency cesarean delivery is available
- c) Continuous FHR monitoring is required for breech delivery during labour
- d) Should be done by a skilled obstetrician

Answer Key

Question No.	Correct Option
1	c

2	d
3	c
4	d
5	a
6	b
7	d
8	c
9	c
10	b
11	c
12	c
13	d
14	c
15	b
16	b
17	b
18	d
19	c
20	a
21	b
22	a
23	a

Detailed Explanations

Solution to Question 1:

Screening for cervical cancer is done in females aged 21 to 65 years.

*Three consecutive negative cytologic tests or two consecutive negative co-tests within the past 10 years, with the most recent test occurring within the past 5 years

Age Group	Screening Recommendations
<21 years	Do not screen
21-29 years	Cytologic testing (Pap smear) every 3 years

Age Group	Screening Recommendations
30-65 years	Preferred: Cytologic and HPV co-testing every 5 years Alternative: Cytologic testing every 3 years
>65 years	Discontinue screening if sufficient negative screening results* present, and if there is no history of HSIL, adenocarcinoma in situ, or cancer.
Women who have undergone total hysterectomy with removal of cervix	Discontinue screening

Solution to Question 2:

Oral contraceptive pills (OCPs) do not cause an increase in GnRH, they inhibit the release of GnRH.

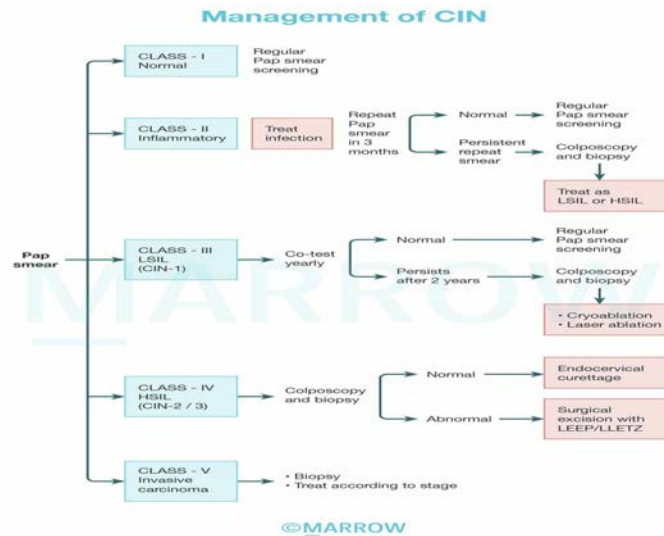
OCPs act via the following mechanisms:

- Inhibiting ovulation - Both estrogen and progesterone act on the hypothalamic-pituitary axis to inhibit the release of GnRH (and thus subsequently LH and FSH). Seen in combined oral pills.
- Preventing implantation - Under the effect of OCPs, endometrial stroma undergoes hypoplasia, with atrophy of glands. This makes it unfavorable for implantation of the embryo. Seen primarily with mifepristone, but is a secondary mechanism of action with progesterone-only pills as well.
- Cervical mucus becomes thick and scanty under the effect of progesterone. This makes it difficult for sperm to enter. Seen in progesterone-only pills.

Solution to Question 3:

In a woman diagnosed with a high-grade squamous intraepithelial lesion (HSIL), the next step in management is to perform a colposcopy and biopsy.

Patients who are found to have HSIL are further subjected to biopsy. This is done by colposcopic study for the visual inspection with acetic acid (VIA) for aceto-white areas and taking biopsies of abnormal areas.



Solution to Question 4:

The patient presents with a history of 6 weeks of amenorrhea, a positive urine pregnancy test, abdominal pain and moderate vaginal bleeding. However, the ultrasound examination does not reveal a gestational sac in the uterus, which could indicate either an ectopic pregnancy or an abortion. In such cases, the next line of management is to follow up with beta-HCG (human chorionic gonadotropin) levels.

Beta-HCG levels above 1500 mIU/mL are considered the discriminatory value, beyond which a gestational sac should be visible on transvaginal ultrasound.

If the gestational sac is not visible on ultrasound and the beta-HCG levels are above 1500 mIU/mL, it suggests an ectopic pregnancy. On the other hand, if the beta-HCG levels are below 1500 mIU/mL, they can be repeated after 48 hours and interpreted as follows:

- Decreasing HCG levels between day 1 and day 3 suggest a possible abortion.
- If the HCG levels rise during this period but does not double, it indicates a likely ectopic pregnancy.

Note: While some institutions set the discriminatory threshold for HCG levels at ≥ 1500 mIU/mL, others use ≥ 2000 mIU/mL.

Solution to Question 5:

The given clinical picture with abdominal pain, bleeding per vaginum, and loss of fetal movements in a hypertensive pregnant woman points to the diagnosis of placental abruption.

Placenta previa (option B) refers to when the placenta is implanted partially or completely over the internal os and it usually presents with painless vaginal bleeding.

Hydramnios (option C) refers to a state where the volume of amniotic fluid exceeds 2000mL and it does not present with bleeding per vaginum.

In preterm labor (option D) fetal movements are not compromised.

Abruptio placenta is the partial/total placental separation from its implantation site before delivery. The bleeding may be:

- Revealed (blood dissects downwards between membranes and uterine wall to escape through the cervix)
- Concealed (blood accumulates between the detached placenta and uterus)

Hypertension in pregnancy is the most important risk factor for abruptio placenta. Patients often present with sudden onset abdominal pain, vaginal bleeding, uterine tenderness, and frequent contractions.

The hemorrhage into decidua basalis causes separation and compression of the placenta. This leads to a compromise of circulation to the fetus, leading to fetal distress.

Clinical classification of placental abruption:

Grade 1	Grade 2	Grade 3
Vaginal bleeding is slight	Vaginal bleeding mild to moderate	Bleeding is moderate to severe or concealed.
Uterus irritable, minimal tenderness	Uterine tenderness is always present	Uterine tenderness is marked.
Maternal BP and fibrinogen unaffected	Maternal pulse increased, BP maintained	Shock is pronounced
FHS good	Fetal distress present	Fetal death is the rule
Maternal shock absent	Maternal shock absent	Associated coagulation defect or anuria may complicate

Solution to Question 6:

Serum progesterone on day 21 should be evaluated in this woman to check for anovulatory cycles.

Anovulation is the major cause of female infertility. Therefore, tests are done to rule out anovulatory defects in a woman presenting with infertility.

After ovulation, the plasma concentration of progesterone increases and reaches its peak during the mid-luteal phase. Serum progesterone levels are assessed on day 8 and day 21 of a 28-day cycle. If the progesterone value increases from less than 1 ng/mL to greater than 6 ng/mL, it indicates ovulation. Subsequently, as the corpus luteum degenerates, progesterone levels decline.

If the plasma progesterone level remains low during the mid-luteal phase, it suggests a corpus luteal phase defect and indicates the need for hormonal therapy.

Daily estimation of serum LH (options C and D) around the midcycle can detect the LH surge. Ovulation coincides about 10 to 12 hours after the LH peak.

Solution to Question 7:

The next line of management in a 50-year-old woman with abnormal uterine bleeding is endometrial sampling.

Indications for endometrial sampling:

- Any female $\geq$ 45 years with abnormal uterine bleeding
- A female of age $\leq$ 45 years having exposure to excessive estrogen such as in PCOD, severe obesity, tamoxifen treatment
- Women with failed medical management and persistent bleeding
- Transvaginal ultrasound findings $\geq$ 12 mm in premenopausal women and $\geq$ 4 mm in postmenopausal women.

Endometrial sampling may be done by either curettage or aspiration biopsy by pipelle or vibra aspirator.

Note: Transvaginal ultrasound is a better first-line investigation to assess AUB. Since TVS is not included in the options, endometrial sampling is the next best answer. Assessing the reason for AUB and ruling out malignancy in the postmenopausal age group are the initial steps before prescribing any treatment.

Solution to Question 8:

The clinical scenario of vaginal bleeding, an enlarged uterus exceeding the corresponding period of amenorrhea, and a snowstorm appearance on USG strongly indicates a diagnosis of a hydatidiform mole. Suction and evacuation is the next best step in the management of this patient.

Hydatidiform mole is an abnormal condition of the placenta, characterized by partly degenerative and partly proliferative changes in the chorionic villi. It may be either partial or complete.

The patient presents with a characteristic history of abdominal pain, vaginal bleeding or spotting, and vomiting during the first trimester. Also, there may be a history of passing grape-like vesicles vaginally. On examination, the size of the uterus is larger for gestational age and the fetal heart sounds are absent.

A USG scan shows a snowstorm appearance which is characteristic of a complete mole.



Suction and evacuation is the treatment of choice. Hysterectomy (option D) is done only if there is uncontrolled hemorrhage or perforation during surgical evacuation or if the age of the patient is above 35 years and has completed her family.

Serum hCG levels (option B) are checked within 48 hours of evacuation and are used as a baseline for future comparisons. It is then checked every week till it normalizes, which usually takes around 4 to 8 weeks. After the hCG values return to normal, a regular follow-up is done monthly for 6 months.

Solution to Question 9:

In this 45-year-old woman (>40 years) with an asymptomatic fibroid of 14-week size, the next best step in management would be observation.

Fibroids/leiomyomas are slow-growing benign tumours thus regardless of their size asymptomatic fibroids are managed conservatively with observation and annual checkups.

Indications for the treatment of fibroids are:

- Abnormal uterine bleeding
- A rapid increase in the growth of fibroid
- Infertility
- Chronic pelvic pain/ bulk symptoms

Other options:

Option A: Hysterectomy is usually reserved as the definitive treatment of choice for women who have completed their family and do not wish to preserve their fertility.

Option B: Myomectomy is a surgical procedure that involves removing fibroids while preserving the uterus. It is indicated in an infertile woman or a woman desirous of childbearing and wishing to retain the uterus.

Option D: Medical therapy aims to control menorrhagia in cases of symptomatic fibroids, improve haemoglobin levels before surgery, and shrink the fibroids before procedures such as myomectomy.

Solution to Question 10:

All the mentioned structures are cut in mediolateral episiotomy except for the obturator muscle.

Episiotomy refers to the surgically planned incision on the perineum and the posterior vaginal wall during the second stage of labor. The objective is to enlarge the vaginal introitus so as to facilitate easy and safe delivery of the fetus and to minimize overstretching and rupture of the perineal muscles and fascia, which might end up in complete tears.

In mediolateral episiotomy, the incision is made downwards and outwards from the midpoint of the fourchette either to the right or to the left. The other types of episiotomy include median, lateral, and J shaped episiotomy.

Structures cut in mediolateral episiotomy are:

- Posterior vaginal wall
- Superficial and deep transverse perineal muscles (option A)
- Bulbospongiosus and part of levator ani (option C)
- Fascia covering these muscles
- Transverse perineal branches of pudendal vessels and nerves (option D)
- Subcutaneous tissue and skin

Differences between midline and mediolateral episiotomy include:

Midline Episiotomy	Mediolateral Episiotomy
Surgical repair is easy	Surgical repair is difficult
Blood loss is lesser	Blood loss is more
Healing is superior	Healing is not so good
Dyspareunia is rare	Dyspareunia is comparatively more
Postoperative comfort is maximum	Postoperative discomfort is more
Extension, if occurs, may involve the rectum	Relative safety from rectal involvement from extension

Solution to Question 11:

Raised ESR in pregnancy is due to an increase in fibrinogen and globulin levels.

Pregnancy is a hypercoagulable state. The fibrinogen level is raised by 50% during pregnancy. As a result of rise in fibrinogen and globulin level and diminished blood viscosity, erythrocyte sedimentation rate (ESR) gives a much higher value (fourfold increase) during pregnancy.

Total plasma protein increases during pregnancy. However, due to hemodilution, plasma protein concentration falls. This results in diminished viscosity of the blood and reduced colloid osmotic tension. Because of marked fall in albumin level and only slight rise of globulin (mainly alpha-globulin), the normal albumin: globulin ratio of 1.7:1 is diminished to 1:1.

Solution to Question 12:

Studdiford's criteria is used in the diagnosis of primary abdominal pregnancy. It includes:

- Presence of normal tubes and ovaries with no evidence of recent or past injury/pregnancy.
- Absence of uteroperitoneal fistula.
- Presence of a pregnancy related exclusively to the peritoneal surface and early enough to eliminate the possibility of secondary implantation after primary tubal nidation.

Other options:

Option A: Spiegelberg's criteria is used in the diagnosis of ovarian pregnancy:

- Tube on the affected side must be intact.
- The gestation sac must be in the position of the ovary.
- The gestation sac is connected to the uterus by the ovarian ligament.
- Ovarian tissue must be found on its wall on histological examination.

Option B: Rubin's criteria is used in the diagnosis of cervical pregnancy:

- Soft, enlarged cervix equal to or larger than the fundus.
- Uterine bleeding following amenorrhea, without cramping pain.
- Products of conception are entirely confined within and firmly attached to endocervix.
- A closed internal cervical os and a partially opened external os.

Option D: Godin's sonographic criteria is used for the diagnosis of a cesarean scar pregnancy:

- Empty uterine cavity
- Empty cervical canal
- A gestational sac in the anterior part of the uterine isthmus
- Absence of healthy myometrium between the bladder and gestational sac

Solution to Question 13:

The contraction stress test is not a component of the biophysical profile.

Biophysical profile is a multi-parametric screening test to assess uteroplacental insufficiency. It consists of five components and the observations are made for 30 minutes.

Each parameter has a maximum score of 2 and a minimum score of 0. So the total maximum score will be 10. A score of 4 or less indicates fetal acidemia.

Interpretation and management of BPP scores:

The modified biophysical score has only 2 components i.e NST and amniotic fluid index (AFI). It is considered abnormal (nonreassuring) when the NST is non-reactive and/or the AFI is ≤ 5 .

Note: The contraction stress test (oxytocin challenge test) is an invasive method to assess fetal well-being during pregnancy. When there is an alteration in FHR in response to uterine contractions, it suggests fetal hypoxia.

Parameters	Minimal Normal Criteria
Non-stress Test (NST)	≥ 2 accelerations within 20-40 minutes.
Fetal breathing movements	≥ 1 episode of rhythmic breathing lasting > 30 seconds.
Gross body movements	≥ 3 discrete body/limb movements
Fetal muscle tone	≥ 1 episode of active extension (limb or trunk) with return of flexion.
Amniotic fluid volume	A pocket of amniotic fluid that measures at least 2 cm in two perpendicular planes (2×2 cm pocket)

BPP score	Interpretation	Management
10	Normal, nonasphyxiated fetuses	Repeat test weekly
8 with normal AFV	Normal, nonasphyxiated fetuses	Repeat test weekly
8 with decreased AFV	Suspicion of chronic fetal asphyxia	If ≥ 36 weeks, delivery is done
6	Possible fetal asphyxia	If AFV is abnormal-Deliver > 36 weeks. If AFV is normal- deliver ≥ 37 weeks.
4	Probable fetal asphyxia	If ≥ 32 weeks, deliver
0-2	Almost certain fetal asphyxia	Delivery

Solution to Question 14:

Ulipristal is not used for the treatment of polycystic ovarian disease (PCOD). It is a synthetic progesterone hormone receptor modulator used as an emergency contraceptive pill.

PCOD is a complex multisystemic endocrine disorder affecting women of reproductive age.

Rotterdam criteria is used to diagnose PCOD. The presence of two of the three components of this criteria establishes the diagnosis.

- Oligo/amenorrhoea, anovulation, infertility
- Hirsutism/acne
- Ultrasound findings:
- Enlarged ovaries with ovarian volume will be more than 10 mL
- 12 or more small follicles each of 2-9 mm in size placed peripherally

Clomiphene (option A) is a selective estrogen receptor modulator that is used as the first line of treatment in managing infertility. Clomiphene with dexamethasone improves the fertility rate. In resistant cases, tamoxifen or letrozole (an aromatase inhibitor) (option B) can be tried.

Surgical management of PCOD includes laparoscopic ovarian drilling (option D). Indications of surgery are:

- Failed medical therapy
- Hyperstimulation occurs
- Infertile women
- Previous pregnancy losses

Solution to Question 15:

The type of maternal pelvis is not a component of the Bishop score.

The Bishop score is a scoring system used to assess labor induction outcomes. It is based on the station of the head, dilation, effacement (or length), position, and consistency of the cervix noted by doing a vaginal examination.

Note: In the modified Bishop score (1991), the effacement is replaced by the cervical length in cm.

Solution to Question 16:

The WHO Labour Care Guide begins the documentation of labour progression at 5 cm of cervical dilatation.

The WHO Labour Care Guide is a tool designed to help health personnel involved in labour care. It aims to

- Guide the monitoring and help in the documentation of the progress of labour.
- Guide the health professionals to offer supportive care throughout labour.

- Note: Previously the answer to the question was 4 cm (according to the modified WHO patograph). However, this information is outdated, therefore the answer to this question has been revised per the latest WHO guidelines.

WHO modified partograph	WHO Labour Care Guide
Documentation of labour progression begins at 4 cm of cervical dilatation.	Documentation of labour progression begins at 5 cm of cervical dilatation.
Fixed 1 cm/hour 'alert' and 'action' lines	Time-based cut-offs for each centimetre of cervical dilatation to guide the obstetrician to take action. For example, at 5 cm of cervical dilatation, if there is no progress > 6 hours, action needs to be taken.
Monitoring of labour is restricted to the first stage only.	Monitoring of the second stage of labour is included
No section on supportive care interventions	Addition of a section on supportive care of the mother- labour companionship, pain relief, oral fluid intake and posture
Documentation of strength, duration, and frequency of uterine contractions	Documentation of only the duration and frequency of uterine contractions

[illegible]

Solution to Question 17:

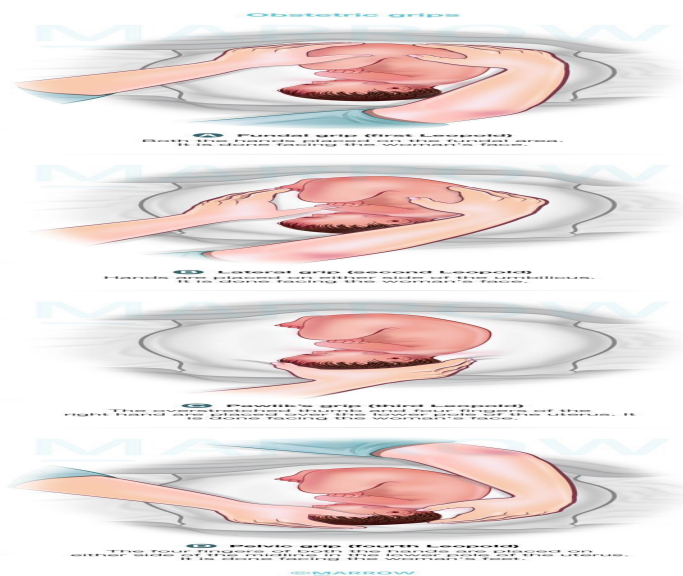
The Leopold manoeuvre shown in the image is the lateral grip. It assesses fetal orientation.

Leopold's manoeuvres help assess the fetal pole, lie, presenting part, engagement, and attitude of the fetal head. They are as follows:

- Fundal grip (first Leopold):
 - Assesses the uterine fundus.
 - Helps in identification of fetal lie and determination of fetal pole at the fundus.
 - The breech is felt as a large, irregular mass.
 - The head feels hard and round and is more mobile.
- Lateral grip (second Leopold):
 - Assesses the fetal orientation.
 - The palms are placed on either side of the maternal abdomen, and gentle but deep pressure is exerted.
 - The back is felt on one side as a hard, resistant structure.
 - The fetal extremities are felt on the other side as numerous small, irregular, mobile parts are felt.
 - Fetal orientation can be determined by noting whether the back is directed anteriorly, transversely, or posteriorly.
 - The position of the anterior shoulder is mainly assessed by this grip.
- Pawlik's grip / 1st pelvic grip (third Leopold):
 - Assesses the presenting part.
 - The thumb and fingers of one hand grasp the lower portion of the maternal abdomen just above the symphysis pubis.
 - If the presenting part is not engaged, a movable mass will be felt, usually the head.
- Deep pelvic grip / 2nd pelvic grip (fourth Leopold):
 - Assesses the engagement and attitude of the fetal head.
 - The examiner faces the mother's feet, and the fingertips of both hands are positioned on either side of the presenting part.
 - They exert inward pressure and then slide along the axis of the pelvic inlet.

Leopold	Grip	Importance
First	Fundal	To know which fetal pole occupies the uterine fundus.

Leopold	Grip	Importance
Second	Lateral/Umbilical	To know the fetal back and extremities
Third	Pawlik	To know the presenting fetal part in respect to the true pelvis
Fourth	Pelvic	To determine the degree of descent. If the head has already descended into the pelvis, the anterior shoulder or the space created by neck may be differentiated readily from the head



Solution to Question 18:

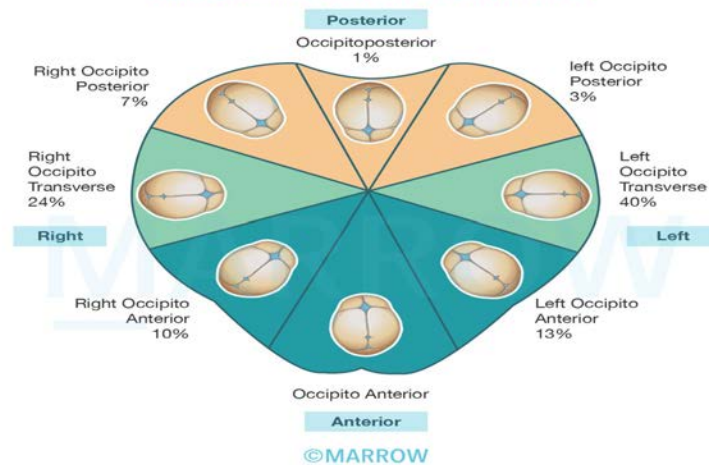
In the given vertex presentation, the fetal position is right occipitoanterior.

Fetal position refers to the relation of the denominator (arbitrary bony fixed point on the presenting part), with respect to different quadrants of the maternal pelvis. In vertex presentation, the denominator is taken as the occiput (which is close to the triangular posterior fontanelle).

To determine the fetal position the following two steps are followed:

- Determine the direction in which the occiput is facing:
- Towards the pubic symphysis - occipito-anterior
- Towards the sacral promontory - occipito-posterior
- Midway between pubic symphysis and sacral promontory - occipito-transverse
- Determine if the occiput is facing the left or the right in relation to the maternal pelvis

The position and relative frequency of the vertex at the onset of labor



Solution to Question 19:

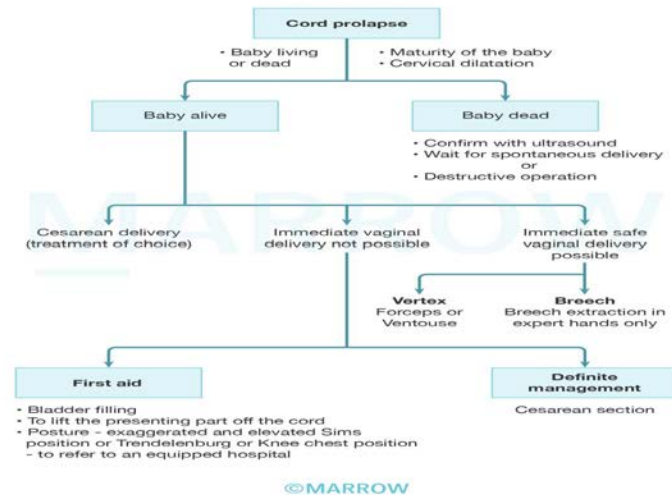
In this scenario the midwife should put the pregnant women in a Trendelenburg position or in an exaggerated and elevated Sims' position with a pillow or wedge under the hip or thigh.

Caesarean section is the recommended treatment of choice for cord prolapse. Prior to performing the procedure, several measures can be taken to alleviate cord compression. These measures involve:

- Manually elevating the fetal presenting part using two fingers or the entire hand through the vagina
- Positioning the patient in steep Trendelenburg, exaggerated Sim's position, or knee-chest position
- Filling the bladder with 500–700 cc or more of saline (known as the Vagos method)

After the diagnosis of cord prolapse is made, it is crucial not to attempt cord replacement. Such attempts are not only ineffective but also result in the inevitable rupture of the membranes, leading to cord prolapse.

Scheme of Management of Cord Prolapse



Solution to Question 20:

A pap smear is a screening test for cervical cancer. It is not a diagnostic test (option C). The presence of abnormal cells/ a positive test requires further evaluation by colposcopy, cervical biopsy, or fractional curettage.

If trichomonas is reported from Pap testing, it should be confirmed by saline microscopic preparation (option B).

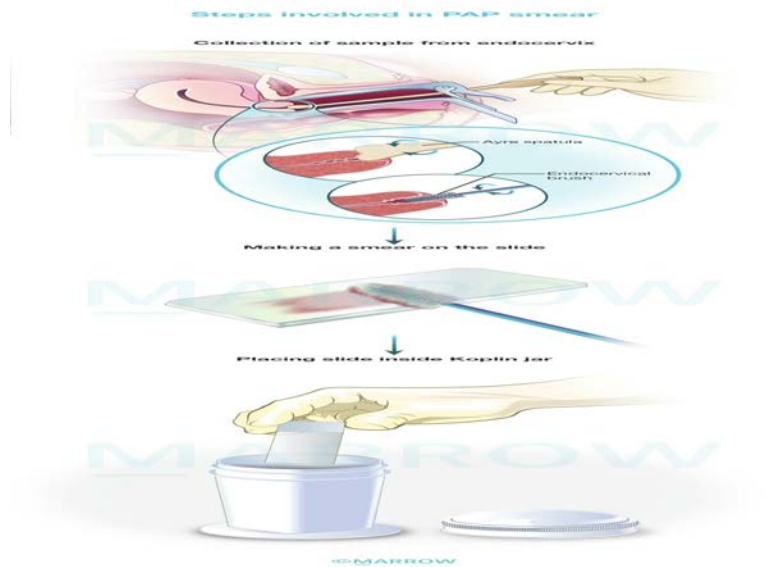
Cervical cancer screening should start at age 21. Women aged 21-29 should have a Pap smear every 3 years, while women aged 30-65 should have both HPV and Pap co-test every 5 years. Women should stop having cervical cancer screening after the age of 65 years. If suspicious PAP, screening till 75 years. HIV+ women should have annual checkups until three consecutive negative tests followed by testing every 3 years for life.

Pap testing should be done prior to vaginal examination. Lubricants should not be used and the patient should not be menstruating during the pap test. The patient should not have intercourse 24hrs prior to the test. The ideal time is around ovulation.

Procedure:

- The patient is put in a dorsal position, and the labia are parted
- Cusco's self-retaining speculum is introduced
- The squamocolumnar junction is scraped with Ayer's spatula:
- Bifid end - ectocervical sample
- Blunt end - vaginal cytology
- Endocervix is scraped with a cytobrush
- The sample is spread on a glass slide
- The slide is immediately fixed using 95% ethyl alcohol

- Stained with Papanicolau stain or short-duration Papanicolau stain (SPS).



Solution to Question 21:

Class 1 of modified Robson's classification includes nullipara, equal to or ≥ 37 weeks, single, cephalic, spontaneous labor.

It is the most widely accepted classification system available for analyzing characteristics of delivering women worldwide. It consists of 10 patient population categories. The basis for these categories include:

- Parity
- Onset of labor and mode of delivery
- Gestational age

- Fetal presentation
- No. of fetuses

The modified Robson system classification is as follows:

Class 1	Nullipara, equal to or >37 weeks, single, cephalic, spontaneous labor
Class 2	Nullipara, equal to or >37 weeks, single, cephalic2a:induced labor2b:CS before labor
Class 3	Multipara, equal to or >37 weeks, single, cephalic, spontaneous labor (excludes previous CS)
Class 4	Multipara, equal to or >37 weeks, single, cephalic4a:induced labor4b:CS before labor
Class 5	Multipara, previous CS, equal to or >37 weeks, single, cephalic
Class 6	Nullipara, single, breech
Class 7	Multipara, single, breech (including previous CS)
Class 8	Multiple pregnancies (with or without previous CS)
Class 9	Singleton pregnancy, oblique/transverse lie (with or without previous CS)
Class 10	Single, cephalic <37 weeks (including previous CS)

Solution to Question 22:

Extension at the knee joint and dorsiflexion at the ankle joint helps in stretching the calf muscles which relieves the pain.

Leg cramps are minor ailments during pregnancy mostly due to a deficiency of diffusible serum calcium or elevation of serum phosphorus. Management includes massaging the leg, stretching exercises, application of local heat, and supplementation of vitamin B1 and calcium.

Solution to Question 23:

Breech identified for the first time during delivery is not a contraindication for normal delivery.

Vaginal breech delivery is considered when:

- Fetal weight is between 1.5 kg and 3.5 kg
- There is an adequate pelvis
- The presentation is frank breech
- The fetal head is flexed
- There are no medical or obstetric complications
- Facilities for emergency cesarean section are available (anesthetists, neonatologists) (option B)
- Facilities for continuous labor monitoring are available (option C)
- A skilled obstetrician is present (option D)
- Detailed informed consent has been obtained, explaining the risks of vaginal breech delivery

Cesarean delivery for breech presentation is considered in the following cases:

- Lack of operator experience
- Patient's request for cesarean delivery
- Prior perinatal death or neonatal birth trauma
- Large fetus: > 3800 to 4000 g
- Severe IUGR, term weight < 2.5 kg to 2.8 kg
- Incomplete breech presentation
- Oligohydramnios
- Fetal anomaly incompatible with vaginal delivery
- Hyperextended head
- Contracted pelvis
- Prior cesarean delivery
- Preterm breech
- Twins, with the first baby in the breech position

Obstetrics and Gynecology NEET 2018

Question 1:

In which trimester does acute fatty liver manifest most commonly during pregnancy?

- a) First trimester
- b) Second trimester
- c) Third trimester
- d) Both (a) and (b)

Question 2:

A 55-year-old lady comes with abnormal uterine bleeding. A hysterectomy was performed and the following specimen was retrieved. What is the diagnosis?



- a) Leiomyoma
- b) Endometrial carcinoma
- c) Endometrial polyp
- d) Adenomyosis

Question 3:

The ligament that maintains anteversion of the uterus during pregnancy is:

- a) Round ligament
- b) Cardinal ligament
- c) Uterosacral ligament
- d) Pubocervical ligament

Question 4:

Quadruple testing is done during:

- a) 9-11 weeks
- b) 12-14 weeks
- c) 15-20 weeks
- d) 22-28 weeks

Question 5:

What is the ideal number of antenatal visits?

- a) 12-14
- b) 6-8
- c) 7-9
- d) 10-11

Question 6:

A woman is diagnosed to be positive for HIV in her first trimester. What is the appropriate management?

- a) Start ART immediately and continue till 6 weeks after delivery
- b) Start ART after 1st trimester and continue life-long
- c) Start ART immediately and continue life-long
- d) Start ART after 1st trimester and continue till 6 weeks after delivery

Question 7:

The fate of the Mullerian duct is determined by the:

- a) X chromosome
- b) Y chromosome
- c) 1st chromosome
- d) 2nd chromosome

Question 8:

Which of the following is the first step after fitting the pessary to check if the pessary is adequately retained?

- a) Asking the patient to void and see if the pessary is in place
- b) Seeing if the patient feels any discomfort, which means that the pessary is retained
- c) Asking the patient to perform the Valsalva manouvre and see if the pessary is in place
- d) Being able to admit two fingers between the ring and the pessary.

Question 9:

A pregnant woman presents with obstructed labor and is grossly dehydrated. Investigations reveal fetal demise. What will be the management?

- a) Craniotomy
- b) Decapitation
- c) Cesarean section
- d) Forceps extraction

Question 10:

The best time to identify the type of twin pregnancy on ultrasound is:

- a) 18-22 weeks
- b) 24-28 weeks
- c) 11-14 weeks
- d) 14-18 weeks

Question 11:

Which of the following statements is false about the umbilical cord?

- a) It has one artery and two veins
- b) It contains a gelatinous substance called Wharton's jelly
- c) Presence of a single umbilical artery is associated with congenital anomalies
- d) Normal average umbilical cord length is 50-60 cms

Question 12:

Which of the following statements is false about non-invasive prenatal testing?

- a) It uses cell-free circulating fetal DNA in the maternal serum
- b) It is used for the diagnosis of trisomy 13, 18 and 21
- c) It can be performed after 10 weeks of gestation
- d) It is done after the routine first trimester ultrasound assessment

Question 13:

Which of the following is not a cause of secondary PPH?

- a) Retained cotyledon
- b) Endometritis
- c) Placental polyp
- d) Placenta previa

Question 14:

A 16-year-old girl with primary amenorrhea has the following features. What is the most probable diagnosis?

- a) Testicular feminizing syndrome
- b) Gonadal dysgenesis
- c) Kallmann syndrome
- d) Turner syndrome

Question 15:

Fimbriectomy is also known as:

- a) Pomeroy technique
- b) Uchida's procedure
- c) Kroener's procedure
- d) Irving's procedure

Question 16:

Which of the following cannot cross the placenta?

- a) 1 and 2
- b) Only 3
- c) 2 and 3
- d) 3 and 4

Question 17:

When is dilatation and curettage contraindicated?

- a) Pelvic inflammatory disease
- b) Ectopic pregnancy
- c) Menstrual irregularities
- d) Endometriosis

Question 18:

A 45-year-old lady complains of post-coital bleeding. She has a positive pap smear. What is the next line of management?

- a) Colposcopy directed biopsy
- b) Cone biopsy
- c) Repeat pap smear
- d) Hysterectomy

Question 19:

In superfecundation, which of the following is seen?

- a) Fertilisation of 2 ova released at the same time by sperms released at single intercourse
- b) Fertilisation of 2 ova released in the same cycle by sperms released at intercourse on 2 different occasion
- c) Both of the above
- d) None of the above

Question 20:

All are true regarding peripartum cardiomyopathy except _____

- a) Development of cardiac failure within 5 months of delivery
- b) Left ventricular diastolic dysfunction
- c) Absence of identifiable cause
- d) Dilated left ventricle

Question 21:

Which of the following is not the mechanism of action of progesterone only pills?

- a) Thicken the cervical mucus
- b) Prevents ovulation
- c) Kills the sperms in the cervical canal
- d) Inhibits penetration of sperm

Question 22:

Which of the following is not considered as a factor for a high-risk pregnancy?

- a) Obesity
- b) Polyhydramnios
- c) Anaemia
- d) Previous manual removal of placenta

Question 23:

Long-standing pelvic inflammation may lead to which of the following conditions?

- a) Pyometra
- b) Uterine polypsis
- c) Pseudocyesis
- d) Endometrial hyperplasia

Question 24:

Which among the following is the most common presentation of postpartum nerve injury?

- a) Facial palsy
- b) Wrist drop
- c) Foot drop
- d) Claw hand

Question 25:

What is altered in thyroid profile throughout pregnancy?

- a) Free T₃ level
- b) Free T₄ level
- c) Total T₃ level
- d) TSH level

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	c
5	a
6	c
7	b

8	c
9	c
10	c
11	a
12	b
13	d
14	b
15	c
16	c
17	a
18	a
19	b
20	b
21	c
22	d
23	a
24	c
25	c

Detailed Explanations

Solution to Question 1:

Acute fatty liver of pregnancy (AFLP) occurs most commonly in the third trimester or early postpartum period.

Risk factors include Pre-eclampsia/HELLP syndrome, male fetus, previous history of AFLP, and multiple gestations.

AFLP (acute yellow atrophy or acute fatty metamorphosis) is the most common cause of acute liver failure in pregnancy. It usually occurs in the later stages of pregnancy, particularly in the third trimester. It occurs due to fetal deficiency of long-chain acyl-coenzyme A dehydrogenase (LCHAD) or poor placental mitochondrial function.

The patients present with persistent nausea and vomiting, progressive jaundice, and epigastric pain. Features like acute hemolysis, hypoglycemia, hypertension, proteinuria, thrombocytopenia, and edema can also be present. Symptoms may progress to fulminant hepatic failure with encephalopathy, disseminated intravascular coagulation, and death.

Liver function tests such as AST, ALT, and serum bilirubin levels are raised. Hypofibrinogenemia, hypocholesterolemia, and prolonged clotting time can be seen. An elevated d-dimer is seen. Microscopy reveals a characteristic microvesicular fatty infiltration.

The condition worsens till delivery after which the hepatic dysfunction resolves. A trial of labor by induction or delivery by Cesarean is done. Intensive supportive measures including blood transfusion should be provided.

The clinical and laboratory findings of acute liver diseases in pregnancy are as follows:

The condition worsens till delivery after which the hepatic dysfunction resolves. Appropriate management is immediate termination of pregnancy irrespective of gestational age by induction or delivery by Cesarean (for obstetric reasons) is done. Intensive supportive measures including blood transfusion should be provided.

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
Onset in pregnancy	Mid to late	Late	Late	Variable
Differentiating features	Epigastric pain, hypertension, elevated indirect bilirubin	Pruritus, elevated direct bilirubin	Signs of liver failure like hypoglycemia	Fever
AST (U/L)	≤300	≤200	200-800	2000+
Bilirubin (mg/dL)	1-4	1-5	4-10	5-20
Platelets	↓↓	Normal	↓↓	↓
Fibrinogen	Normal	Normal	↓↓↓	Normal
D-dimer	↑	Normal	↑	Normal
Prothrombin time	Normal	Normal	↑↑	↑
Hemolysis	↑↑	No	↑↑↑	No

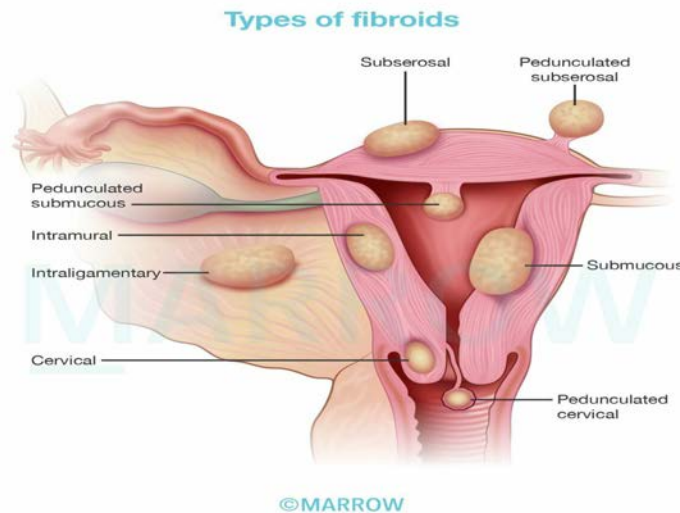
Solution to Question 2:

The given cut section image showing multiple nodules, well-circumscribed, firm, and white-greyish with a whorled appearance is Leiomyoma (Fibroid).

Leiomyoma is the most common benign and the most common pelvic tumor in females and is more common in obese females (hyper-estrogenism). They are mostly asymptomatic and the most common symptom is menorrhagia.

Leiomyomas can also present with bleeding areas, cystic degeneration, and calcifications. The uterine cavity is compressed.

The image below different types of uterine fibroids classified according to their locations:



Sub-mucous fibroids are the most symptomatic type and intra-mural fibroids are the most common type of uterine fibroids.

Other options:

Option B: Endometrial cancer presents as a polypoid mass protruding into the uterine cavity or as a diffuse thickening of the endometrium. The cut surface may appear white to yellow and may show areas of necrosis or hemorrhage.

Option C: Endometrial polyps are usually small and may be single or multiple. Grossly, they appear as soft, red to pink, often pedunculated (attached by a stalk) masses protruding into the uterine cavity. It presents with heavy cyclic or intermenstrual bleeding, or without complaints as an incidental finding on imaging. Estrogen and progesterone have been implicated in the growth of endometrial polyps, and higher receptor levels are seen compared with the normal endometrium.

Option D: In Adenomyosis (Endometriosis interna), the uterus may appear bulky and symmetrically enlarged. On the cut section, the myometrium may appear thickened and may show tiny cystic spaces filled with old blood (chocolate cysts). The preferred treatment is hysterectomy. Localised excision or levonorgestrel IUCD is tried in younger women.

The image below shows both the gross and histologic appearances of adenomyosis:



Solution to Question 3:

The round ligament helps maintain the anteversion of the uterus during pregnancy.

However, the anteversion in the non-pregnant uterus is maintained by both the uterosacral ligament and round ligament.

Anteflexion of uterus is at the level of internal os.

(Mnemonic - V for version, V for vagina)

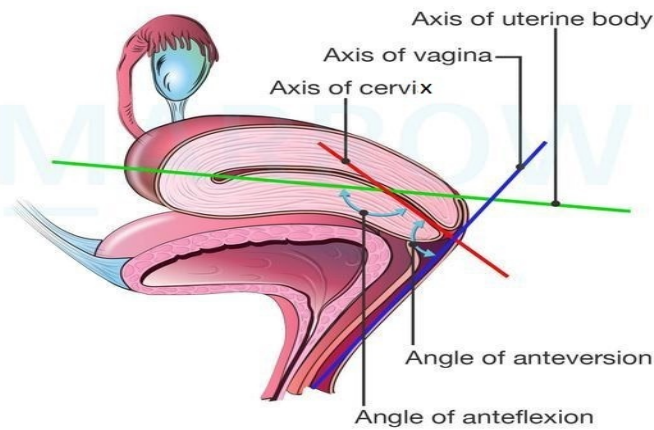
The uterus normally lies in a position of anteversion and anteflexion.

In anteversion, the external os is directed downward and backward, so that on vaginal examination, the fingers find that the lowest part of the cervix is the anterior lip.

When the uterus is retroverted, the cervix is directed downward and forward, and the lowest part of the cervix is either the external os or its posterior lip.

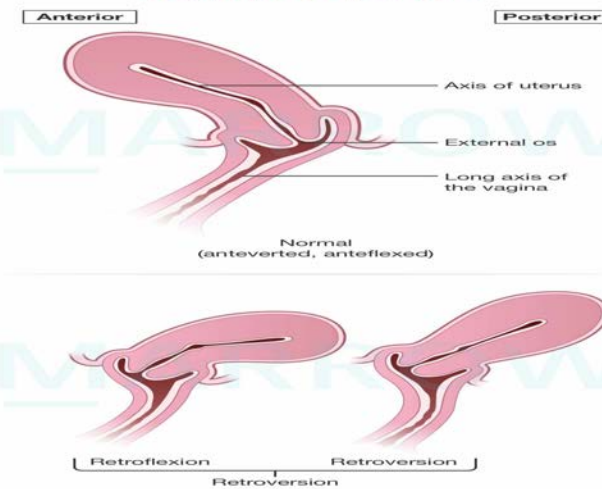
Angle of anteversion	Angle of anteflexion
Angle between the cervix and vagina	Angle between cervix and uterus
90°	120°

Anteversion and antelexion of the uterus



©Marrow

Positions of the uterus



©Marrow

Solution to Question 4:

The quadruple screening test is ideally done in the second trimester, during 15-20 weeks of gestation. The most accurate values are seen during 16-18 weeks of gestation.

It is the measurement of maternal serum alpha-fetoprotein (MSAFP), unconjugated estriol, hCG, and inhibin-A.

In Down's syndrome (trisomy 21) or Edward's syndrome (trisomy 18), quadruple-marker screenings do not provide any added benefit over first-trimester screenings. It is generally used as a stand-alone test if screening is done for the first time in the second trimester or if first-trimester screening is not available.

	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect	Trisomy 13/ Patau syndrome
MSAFP	Low	Low	High	Normal
β hCG	High	Low	Normal	Normal
Unconjugated estradiol	Low	Low	Normal	Normal
Inhibin A	High	Normal/Low	Normal	High

Solution to Question 5:

The ideal number of antenatal visits is 12-14.

The antenatal visits should ideally be: monthly once up to 28 weeks, once in two weeks between 28-36 weeks, and once weekly visits from 36 weeks onwards.

This translates to a total of 12-15 visits.

The WHO increased the minimum number of antenatal visits from 4 to 8 visits to reduce perinatal mortality.

In the Indian scenario, the earlier minimum of 4 visits were implemented as follows:

- 1st visit: within 12 weeks (preferably as soon as pregnancy is suspected)
- 2nd visit: between 14-26 weeks
- 3rd visit: between 28-34 weeks
- 4th visit: between 36 weeks and term

A booked case is one that has at least 3 antenatal visits with two visits in the third trimester.

Solution to Question 6:

ART should be initiated immediately and continued lifelong in all pregnant and breastfeeding women with HIV. This is regardless of their WHO clinical stage and CD4 cell count.

*Note: Pearl attached below has been updated according to the NACO ART guidelines 2021. Efavirenz has been replaced by Dolutegravir as per the new guidelines.

Solution to Question 7:

The fate of the Mullerian duct is determined by the presence or absence of the Y chromosome as follows:

- In presence of the Y chromosome, the SRY gene (Sex-determining Region of the Y chromosome) produces the protein called the testis determining factor which leads to the gonads developing as

the testes. The Sertoli cells of the testis will then produce a peptide hormone, the Müllerian inhibiting substance (MIS), also known as Anti-Müllerian hormone (AMH), and cause the Müllerian ducts to regress.

- In absence of the Y chromosome and the SRY gene consequently, the gonads will develop as ovaries and the Mullerian duct will differentiate into the fallopian tubes and the upper part of the vagina. The WNT4 gene on the 1st chromosome promotes female sex development.

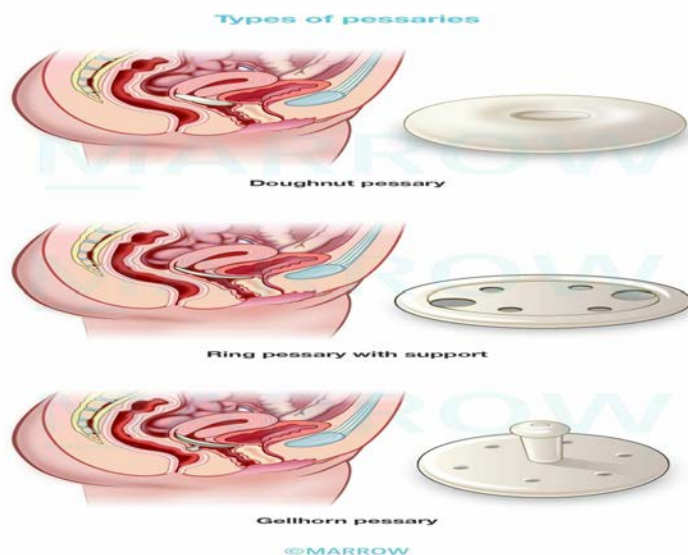
Solution to Question 8:

The first step after fitting the pessary to check if the pessary is adequately retained is to ask the patient to perform Valsalva's maneuver or cough and see if the pessary is in place.

There is some level of descent of the pessary with these maneuvers is to be expected, but the pessary should return to its normal position when these efforts are stopped. If the pessary descends to the introitus with these maneuvers, then a larger size pessary should be tried.

The pessary is well-fitted if you are able to slide the ring of the pessary up and down along the vaginal sidewall. It is well-fitted if you are able to insert one finger between the vaginal sidewall and the edge of the pessary. When the labia are separated, a well-supported pessary is not usually visible.

The patient should be comfortable, and able to void urine with the pessary in place before leaving the clinic.



Solution to Question 9:

The best management in the given clinical scenario is cesarean section.

Obstructive labor management involves administering intravenous fluids to correct dehydration, providing antibiotics to prevent infection, and promptly performing a cesarean section to relieve

the obstruction. If the fetus is alive, an immediate cesarean section is typically performed. Management of obstructed labor never involves a wait-and-watch approach or augmenting with oxytocin.

Clinical features of obstructed labor include maternal exhaustion, dehydration, tachycardia, tachypnea, genital sepsis, postpartum hemorrhage, and shock.

On P/A examination the upper uterine segment is tender and tonically contracted. The lower uterine segment is thin and stretched. A groove/depression is felt between the upper uterine segment and the lower uterine segment. This is called a Bandl's ring.

Complications of obstructed labor include rupture of the uterus and vesicovaginal fistula.

Solution to Question 10:

The best time to identify the type of twin pregnancy on ultrasound is 11-14 weeks.

Twin pregnancies should ideally be dated when the crown-rump length (CRL) measurement is between 45 and 84 mm (i.e. 11 + 0 to 13 + 6 weeks of gestation).

Chorionicity should be ideally determined between 11 + 0 and 13 + 6 weeks of gestation (the earliest time to identify chorionicity is 7 weeks). This is done using the membrane thickness at the site of insertion of the amniotic membrane into the placenta. The T sign (in monochorionic twins) or lambda sign (in dichorionic twins) can be seen here. As the determination of chorionicity is most accurate when the amnion and chorion have not yet been fused, the first trimester scan in twin pregnancy is highly important.

Markers of dichorionicity include 4 layers of membranes between the twins, membrane thickness ≥ 2 mm, different sex of twins, 2 different placentas, and positive twin peak sign.

Women with an uncomplicated dichorionic twin pregnancy should have a first trimester scan, a detailed second trimester scan, and scans every 4 weeks thereafter.

Women with uncomplicated monochorionic twins should have a first trimester scan and then scans every 2 weeks from 16 weeks onwards. This is done in order to detect TTTS and TAPS in a timely manner.

Solution to Question 11:

The umbilical cord normally has two arteries and one vein. The presence of a single umbilical artery is the most common vascular anomaly of the cord. It is associated with congenital anomalies such as genitourinary and cardiac anomalies.

The umbilical cord is a tube that connects the fetus to the placenta. It contains umbilical vessels and an obliterated allantois duct, all surrounded by Wharton's jelly and contained within an outer layer of amnion. Maternal blood enters the fetus through the umbilical vein, and blood exits the fetus via the two umbilical arteries. Wharton's jelly is a gelatinous mucopolysaccharide that protects the umbilical vessels. Local collections of Wharton's jelly produce false knots in the cord.

The average length of the umbilical cord is between 40-70 cm at term. A long cord can form a loop around the fetus' neck and can cause asphyxia. A short cord can result in cord rupture during labor.

Solution to Question 12:

Non-invasive prenatal testing (NIPT) is used for the screening of trisomy 13, 18 and 21 and Turner's syndrome. It is not a diagnostic test.

It is a novel genetic test that uses cell-free circulating fetal DNA in the maternal serum for screening of chromosomal anomalies.

DNA from the fetus is found circulating in the maternal blood in two forms - either inside the intact fetal cells or as cell-free DNA from the breakdown of placental fetal cells. In NIPT, the cell-free chromosome fragments are measured and the quantitative differences from normal fetuses are noted.

NIPT can be performed after 10 weeks of gestation and is recommended after the routine first-trimester ultrasound assessment in high-risk pregnancies or if either 1st or 2nd-trimester screening becomes positive. Its disadvantage is that it is very expensive and it can be used for screening of only a few chromosomal anomalies such as trisomy 13, 18, and 21 and it doesn't analyse the entire karyotype.

Solution to Question 13:

Placenta previa does not cause secondary postpartum hemorrhage. It causes primary postpartum hemorrhage.

Types of PPH:

Primary/early: Hemorrhage occurring within 24 hours of childbirth.

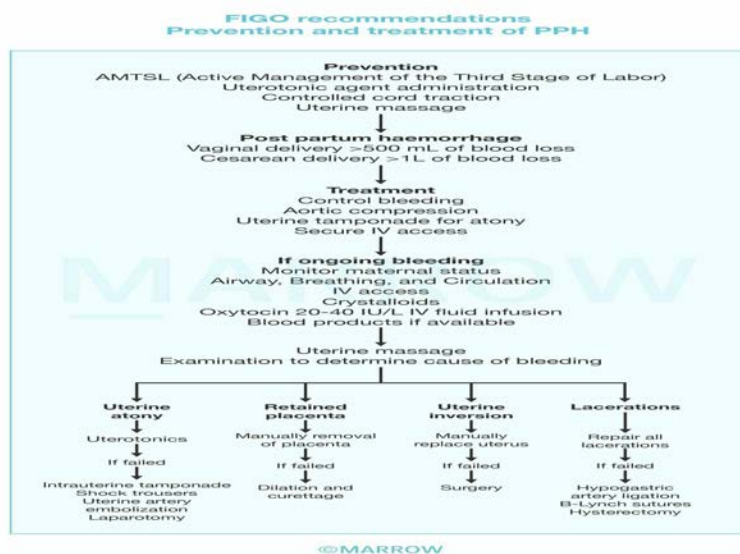
Secondary/ late: Hemorrhage occurring after 24 hours and up to 12 weeks postpartum.

Causes and risk factors for PPH:

The image below shows the prevention and management of PPH:

Primary PPH	Secondary PPH		
Atonic PPH	Traumatic PPH	Coagulopathy	
Primiparity or grand multiparity	Instrumental delivery	Sepsis	Retained bits of cotyledon or membranes
Placental abnormalities like placenta previa/accreta/increta/percreta/abruptio placenta	Precipitate labour	HELLP syndrome	Secondary haemorrhage from Caesarean section wound

Primary PPH	Secondary PPH		
Overdistended uterus as in multi-gestational pregnancy, hydramnios, macrosomia	Macrosomia	IUD	Puerperal uterine inversion
Induction of labour with prostaglandins or oxytocin	Episiotomy		Abnormal involution of placental site
Prolonged labour			Infected fibroid
Precipitate labour			Endometritis
Previous h/o PPH			Placental polyp
Obesity			Carcinoma of cervix
Age >40 years			Uterine av fistula



Solution to Question 14:

The scenario of a 16-year-old girl with absent secondary sexual characters, normal stature, with XX karyotype and high FSH and LH is suggestive of pure gonadal dysgenesis.

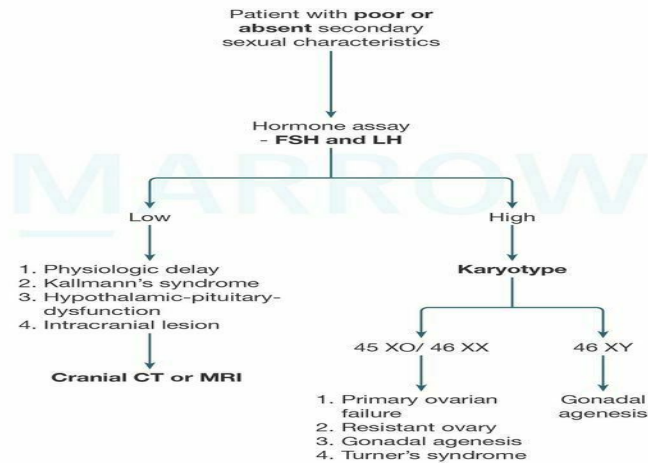
46 XX and 46 XY gonadal dysgenesis without the phenotype of Turner's syndrome (short stature, webbed neck) is described as pure gonadal dysgenesis. It is characterized by the presence of primary amenorrhea with or without normal secondary sexual characteristics.

Option A: Testicular feminizing syndrome occurs in karyotype XY. Secondary sexual characters are present and the uterus is absent.

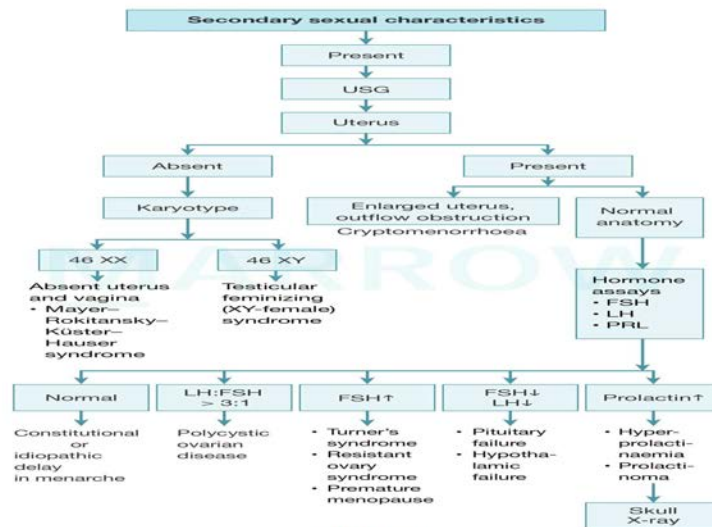
Option C: Kallmann syndrome is a form of hypogonadotropic hypogonadism. GnRH synthesis is reduced, therefore FSH and LH are low, and secondary sexual characters are absent.

Option D: Turner syndrome is associated with short stature and karyotype Xo. Secondary sexual characters and the uterus is present, but the ovaries are streaked.

Evaluation of primary amenorrhea



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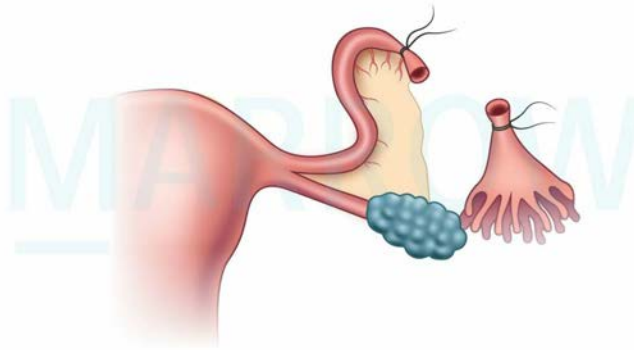


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Solution to Question 15:

Fimbriectomy is also known as Kroener's procedure. This procedure involves ligation of the distal ampulla of the tube with two permanent sutures. Then, resection of the infundibulum of the tube is done.

Fimbriectomy

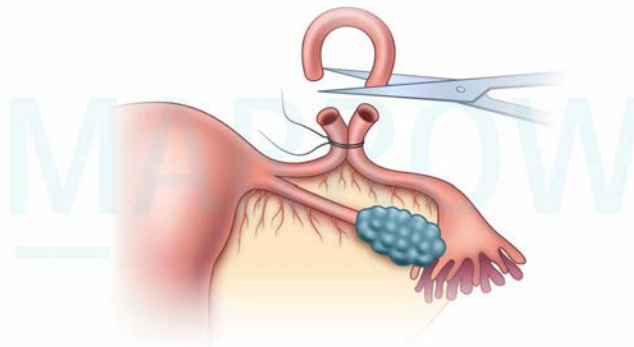


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Some more techniques of permanent sterilization include:

Pomeroy method - The fallopian tube is identified on each side, brought out through the incision. The middle portion is formed into a loop which is tied at the base with catgut and excised. This is the most commonly used technique.

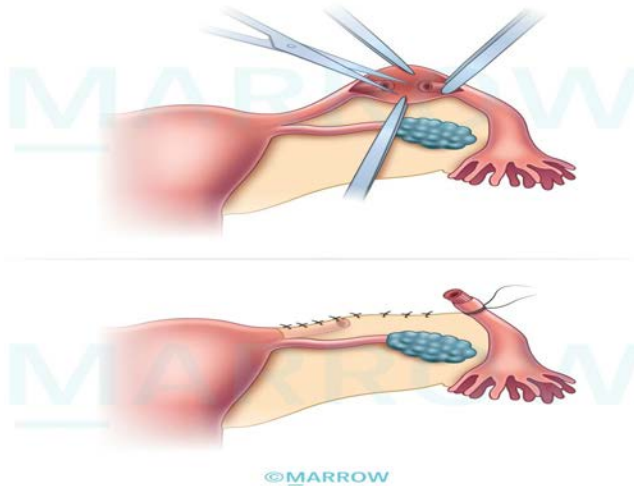
Pomeroy



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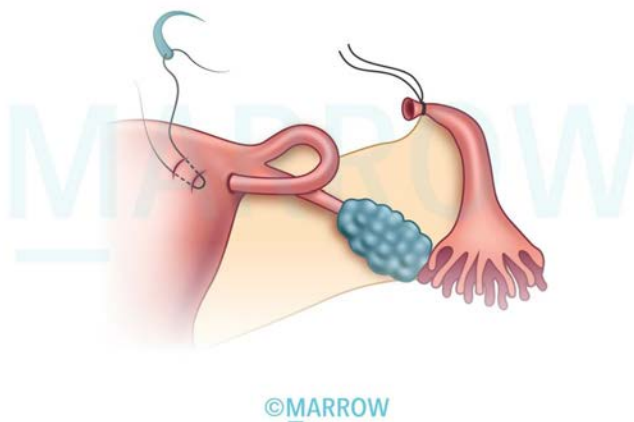
Uchida's method - The tubal serosa is stripped off the muscular layer in the midsegment of the tube. The muscle layer is excised. The proximal end is ligated and buried in the broad ligament. This minimal excision of the tube preserves the potential for tuboplasty.

Uchida technique

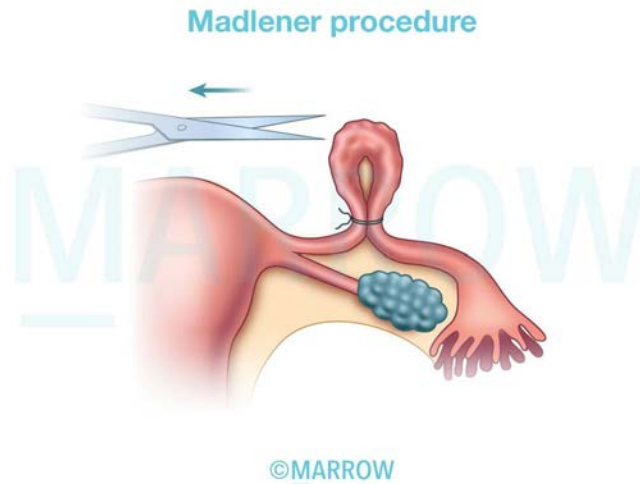


Irving's procedure - The mid-portion of the tube is ligated and the intervening portion is excised. The proximal end is buried in the myometrium and the distal end is buried in the broad ligament. It is a reliable method but irreversible and may require a laparotomy incision.

Irving procedure



Madlener's procedure - A loop of the fallopian tube is crushed and then ligated with a nonabsorbable suture.



Solution to Question 16:

Albumin and IgM cannot cross the placenta.

High molecular weight substances such as proteins (e.g., albumin, IgM, IgA) cannot cross the placenta, except IgG.

Substances which can cross the placenta are as follows:

Note: Maternal hormones like insulin, steroid hormones, and thyroid hormones cross the placenta very slowly, and only in amounts enough to support fetal development. However, calcitonin and parathormone do not cross the placenta.

Substances	Mechanism
O ₂ CO ₂ WaterElectrolytesAnesthetic agents	Simple diffusion
Immunoglobulin G(molecular weight 160,000 Da)	Endocytosis via Fc receptors in the trophoblast
Glucose	Facilitated diffusion via glucose transport proteins (GLUTs)
Amino acids	Transport-protein-mediated active transport
Lipids:GlycerolEssential fatty acidsLDL	Endocytosis via specific receptors
Minerals:IodineIron (unidirectional from mother to fetus)CalciumPhosphorus	Carrier-mediated active transport
Vitamins:Vit AVit CVit D	Carrier-mediated active transport

Solution to Question 17:

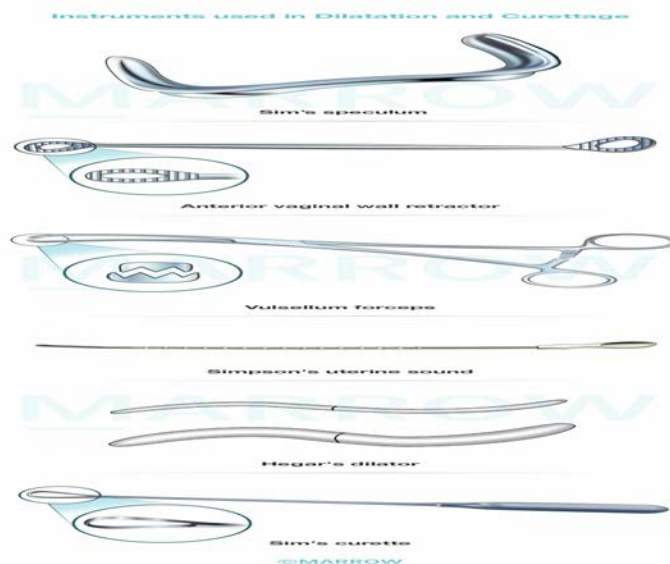
Dilatation and curettage (D&C) is contraindicated in pelvic inflammatory disease (PID) as it increases the risk of ascending genital infection.

D&C is a minor gynecological procedure of dilating the cervix and curetting the endometrial tissue from the uterine cavity. Diagnostic indications of D&C include AUB, secondary amenorrhea to detect tubercular endometritis, postmenopausal bleeding to rule out endometrial cancer or hyperplasia, and infertility due to genital TB.

Therapeutic indications of D&C include missed abortion, incomplete abortion, retained products of conception, and endometrial polypectomy.

Absolute contraindication of D&C includes Suspected/ desired pregnancy and lower genital tract infection. However, D&C is performed in some cases with genital infections like septic abortion (where retained pieces of placenta act as a nidus and need to be removed) and pyometra (as there is cervical obstruction with superinfection of menstrual blood and cellular debris, D&C is appropriate to remove the obstruction as well as the source of infection)

Complications of D&C include ascending infection, cervical tear with bleeding, incompetent os, uterine perforation, and ashermann syndrome.



Solution to Question 18:

The appropriate management of a 45-year-old with a positive pap smear and post-coital bleeding is a colposcopy-directed biopsy.

A per speculum examination must be done whenever a female comes with post-coital bleeding in order to evaluate for carcinoma of the cervix. If growth is visible, a punch biopsy should be done, and else if no growth is visible, then a pap smear is done. A pap smear is a screening test for cervical cancer. The presence of abnormal cells/ a positive test requires further evaluation by

colposcopy, cervical biopsy, or fractional curettage.

Cervical cancer screening should start at age 21. Women aged 21-29 should have a pap smear every 3 years, while women aged 30-65 should have both HPV and pap co-test every 5 years. Women should stop having cervical cancer screening after the age of 65 years. If a pap smear is suspicious, then screening is continued till 75 years. HIV+ve women should have annual checkups until three consecutive negative tests followed by testing every 3 years for life.

Pap testing should be done prior to vaginal examination. Lubricants should not be used and the patient should not be menstruating during the pap test. The patient should not have intercourse 24hrs prior to the test. The ideal time is around ovulation.

Solution to Question 19:

Superfecundation is the fertilization of two different ova released in the same cycle, by two separate acts of coitus within a short period of time.

Superfecundation is the fertilization of two different ova in the same menstrual cycle (as suggested by the same gestational age) by separate acts of coitus. It can occur even after artificial reproductive therapy. Hence, after embryo transfer, intercourse must be avoided. Heteropaternal superfecundation is used to refer to instances of two different males fathering fraternal twins. This leads to the possibility of twins also being half-siblings. This may give rise to the doubt of adultery and infidelity.

Superfetation is the fertilization of two ova released in different menstrual cycles. This is not seen in humans. The nidation and development of one fetus over another fetus is theoretically possible until the decidual space is obliterated by 14-16 weeks of pregnancy.

Solution to Question 20:

Left ventricular systolic dysfunction is a feature of peripartum cardiomyopathy (also known as postpartum cardiomyopathy) and not diastolic dysfunction.

Peripartum cardiomyopathy (PPCM) is a type of dilated cardiomyopathy. It develops during the last month of pregnancy or within the first 5 months following delivery.

The predisposing factors include Cardiac gene mutations involving several genes like TTNC1, TTN, and STAT3, Twin pregnancy/ multifetal gestation, Smoking, and Preeclampsia. Other proposed causes include viral myocarditis, abnormal immune response to pregnancy, malnutrition, inflammation, etc.

PPCM is associated with increased secretion of prolactin by the pituitary and antiangiogenic molecule sFlt-1 (soluble fms-like tyrosine kinase) by the placenta at term gestation.

Pearson's diagnostic criteria for PPCM:

- Development of cardiac failure in the last month of pregnancy or within 5 months after delivery
- Absence of an identifiable cause for the cardiac failure
- Absence of recognizable heart disease prior to the last month of pregnancy, and

- Left ventricular systolic dysfunction demonstrated by echocardiographic criteria, such as ejection fraction $\leq 45\%$ or fractional shortening $\leq 30\%$ along with a dilated left ventricle.

Management involves treating the features of heart failure. Bromocriptine has been found useful in this condition. Many women recover within 6 months of delivery, while some continue to have residual effects.

Solution to Question 21:

Minipills or progesterone-only pills (POP) do not kill sperm in the cervical canal. Progesterone-only pills act by thickening the cervical mucus (the main mechanism of all POPs except cerazette) that inhibits penetration of sperms, inhibiting tubal motility, and suppressing ovulation.

Other advantages of POPs include a reduction in the incidence of pelvic inflammatory diseases and sexually transmitted diseases (due to thickened cervical mucus).

Due to the inhibition of tubal motility, the absolute risk of all pregnancies including ectopic pregnancy decreases, while the relative risk of ectopic pregnancy increases. But a previous history of ectopic pregnancy is not a contraindication for POP.

POPs do not affect clotting, lipid metabolism, and carbohydrate metabolism. So they can safely be used by women who are obese, breastfeeding, hypertensive, have SLE or a history of thromboembolism, diabetic with vascular disease. Significant adverse effects of POPs are irregular bleeding (mostly scanty), and secondary amenorrhea after longtime usage as it causes endometrial atrophy.

Contraindications for POPs include undiagnosed vaginal bleeding, pregnancy, known or suspected breast cancer, and benign or malignant liver tumors.

Solution to Question 22:

Previous manual removal of the placenta does not indicate a high-risk pregnancy.

The high-risk pregnancies according to the Pregnancy Risk Assessment Monitoring System by the CDC are:

Current pregnancy complications	Obstetric history	Other disorders
Preterm labor (<37 Wks) and Post-term (> 42 wk)	Intrauterine growth restriction	Cardiovascular factors like chronic hypertension, severe heart failure, and renal disorders
Preterm premature rupture of membranes	Stillbirth	Metabolic disorders like Obesity, Insulin-dependent diabetes, Previous endocrine ablation, Thyroid disorders

Current pregnancy complications	Obstetric history	Other disorders
Moderate to severe preeclampsia	Late abortion (16–20 wk)	Hematological problems like sickle cell disease, severe anemia Thrombophilia
Polyhydramnios (hydramnios) or oligohydramnios	Fetal exchange transfusion (Rh incompatibility)	Infections (malaria, HIV) and Autoimmune disorders
Uterine rupture	Multifetal pregnancy	Epilepsy
Antepartum hemorrhage (Placenta previa, and Abruptio placentae)	Previous brachial plexus injury	Anatomic abnormalities like uterine malformation and cervical incompetence
Fetal tachycardia or bradycardia for >30 min	Abnormal fetal position	Exposure to teratogens like alcohol, drugs, smoking
Fetal weight < 2.5 kg or >4kg	Previous h/o myomectomy, repair of complete perineal tear, vesicovaginal fistula, and stress incontinence	Family history of diabetes, hypertension or multiple pregnancy and congenital malformation
Chorioamnionitis		Low socioeconomic status
Prolapsed cord		Abnormal cervical cytologic findings
Fetal acidosis pH ≤7		

Solution to Question 23:

Long-standing pelvic inflammation may lead to pyometra. This refers to the collection of pus in the uterine cavity.

The initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, skene glands, endometrium, and fallopian tubes. Unilateral pain and swelling over the labia may be caused by the inflammation of the Bartholin glands.

Depending on the extent of tissue involved, PID can be classified into five stages

- Stage I - Acute salpingitis without peritonitis - No adhesions
- Stage II - Acute salpingitis with peritonitis - purulent discharge
- Stage III - Acute salpingitis with superimposed tubal occlusion or tubo ovarian complex
- Stage IV - Ruptured tubo-ovarian abscess
- Stage V - Tubercular salpingitis

Long-standing pelvic inflammation may also lead to:

- Dyspareunia
- Infertility
- Ectopic pregnancy

- Formation of adhesions like hydrosalpinx, pyosalpinx, tubo-ovarian abscess
- Chronic pelvic pain

Note: Chronic pelvic inflammation leads to dense adhesions between the uterus and the appendages. Thus, it becomes difficult to define the uterus from the pelvic masses. They form a hard mass which is called the frozen pelvis.

The most common cause of the spread of PID: Ascending infection along with sperms

The only PID that does not spread with the sexual route is genital TB. It has a hematogenous spread.

The most common PID in virgin females is genital TB.

Investigation of choice in PID: Abdominal USG

The surest sign of salpingitis is pus extruding from the fimbrial end of the tube.

Solution to Question 24:

The most common postpartum nerve injury is an injury to the peroneal nerve causing foot drop.

The lateral femoral cutaneous nerve is the most commonly injured nerve after vaginal delivery followed by femoral neuropathies.

Obstetrical neuropathies are due to the pressure on branches of the lumbosacral nerve plexus during labor. They can manifest as complaints of intense neuralgia or cramp-like pains extending down one or both legs as soon as the head descends into the pelvis. Footdrop can be secondary to injury at the level of the lumbosacral plexus, sciatic nerve, or common fibular (peroneal) nerve. Injury of the iliohypogastric and ilioinguinal nerves may occur with cesarean delivery, which may manifest as an inguinal hernia and loss of sensation at the pubic region respectively.

Note: Since there are no options relating to intense neuralgia in the lower limb, foot drop becomes the best possible answer.

Solution to Question 25:

Total T3 levels are altered throughout pregnancy.

Total T3 and total T4 increase due to stimulation of thyroid tissue by hCG (α -subunit of hCG is similar to TSH) during the early period of gestation. However, there is no change in free T3 and T4 due to an increase in thyroid-binding globulin (TBG).

This leads to a transient fall in TSH (but remains in the normal range) during the early weeks of gestation. The level of TSH becomes normal during the second and third trimesters.

The hormonal changes during pregnancy are as follows:

Increased	Decreased	Unchanged
Growth Hormone	LH (Luteinizing hormone)	FreeT3, T4

Increased	Decreased	Unchanged
Insulin(due to insulin resistance)	FSH (Follicle-stimulating hormone)	TSH
Prolactin	DHEAS (Di-hydro androstosterone)	ADH (Anti- diuretic hormone)
TotalT ₃ , T ₄		
Aldosterone		
Testosterone, Androstenedione, and cortisol		

Obstetrics and Gynecology NEET 2019

Question 1:

In which trimester does acute fatty liver manifest most commonly during pregnancy?

- a) First trimester
- b) Second trimester
- c) Third trimester
- d) Both (a) and (b)

Question 2:

Which of the following is true about vaginal changes in a normal pregnancy?

- a) Decreased number of Lactobacilli
- b) Increased glycogen metabolism in the epithelium
- c) pH $\geq$ 6.8
- d) Epithelial thinning

Question 3:

You discover that there is evidence of cardiac malformation in a fetus during a routine USG scan at 16 weeks. What is the minimum level of fasting blood sugar at which you can suspect overt diabetes?

- a) 106 mg/dL
- b) 126 mg/dL
- c) 116 mg/dL
- d) 130 mg/dL

Question 4:

A 45-year-old woman presented with a lower abdominal mass and lower abdominal discomfort. On examination, the uterus appears uniformly enlarged and adnexa is free. MRI shows the presence of a myometrial cyst. What is the most probable diagnosis?

- a) Endometriosis
- b) Adenomyosis
- c) Leiomyoma
- d) Endometrial hyperplasia

Question 5:

Which of the following decreases in pregnancy?

- a) Respiratory rate
- b) Vital capacity
- c) Functional residual capacity
- d) Inspiratory capacity

Question 6:

Which of the following is not a contraindication for the induction of labor?

- a) Pelvic tumor
- b) Herpes infection
- c) Heart disease
- d) History of lower transverse cesarean section in last three pregnancies

Question 7:

The given image shows what grade of placenta previa?



- a) IV
- b) III
- c) II
- d) I

Question 8:

All are normal physiological changes in pregnancy except:

- a) Left axis deviation
- b) Mild ST changes in inferior leads
- c) Atrial and ventricular premature contractions
- d) Pansystolic murmur

Question 9:

A pregnant woman with a fibroid uterus develops severe acute pain abdomen at 28 weeks of gestation. There is no history of fever and the WBC count is 8000 cells/mL. The most likely diagnosis is:

- a) Red degeneration of fibroid
- b) Preterm labour
- c) Torsion of fibroid
- d) None of the above

Question 10:

What is the first line of treatment of mastitis in a lactating mother?

- a) Dicloxacillin
- b) Cefazolin
- c) Ceftriaxone
- d) Ampicillin

Question 11:

Major contributor for amniotic fluid post 20 weeks is?

- a) Fetal urine
- b) Fetal skin
- c) Fetal lung fluid
- d) Maternal plasma

Question 12:

What will be the level of the anti-Mullerian hormone in low ovarian reserve?

- a) <1
- b) 1-4
- c) >7
- d) >10

Question 13:

The maximum dose of PGF₂ alpha in postpartum hemorrhage is ____.

- a) 0.25 mg
- b) 2 mg
- c) 20 mg
- d) 200 mg

Question 14:

Gestational trophoblastic neoplasm does not include

- a) Partial mole
- b) Choriocarcinoma
- c) Placental site trophoblastic tumor
- d) Invasive mole

Question 15:

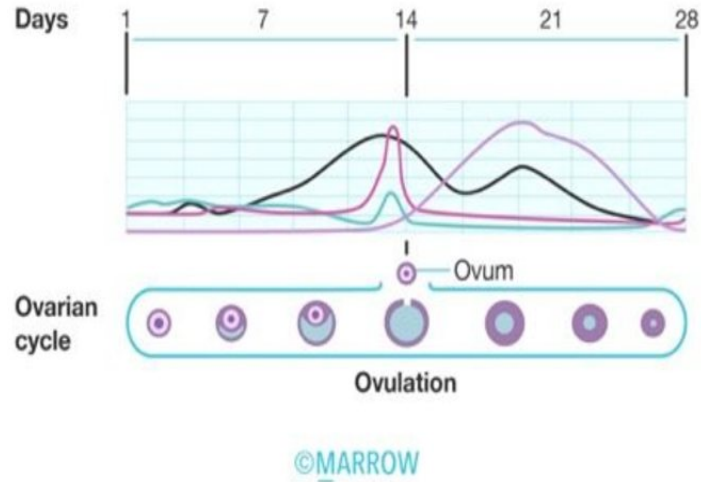
A baby was born at 34 weeks of pregnancy, the appearance of the baby is given below. What is the mother suffering from ?



- a) Gestational diabetes
- b) Rubella
- c) Hypothyroidism
- d) Gestational hypertension

Question 16:

The hormone acting on post ovulatory endometrium is :



- a) Luteinizing hormone
- b) Follicular stimulating hormone
- c) Progesterone
- d) Oestrogen

Question 17:

Which of the following is an absolute contraindication to OCP use?

- a) Chronic renal disease
- b) Thromboembolism
- c) History of amenorrhea
- d) Diabetes mellitus

Question 18:

What is the chromosome number of a partial hydatidiform mole?

- a) 46 XX
- b) 45 XY
- c) 69 XY
- d) 69 XXY

Question 19:

A 55-year-old lady presents with third degree uterine prolapse. Which of the following is the best management for her?

- a) Fothergill repair
- b) Sling operation
- c) Vaginal hysterectomy with pelvic floor repair
- d) Shirodkar procedure

Question 20:

What is the dose of Ulipristal used in emergency contraception?

- a) 30 mg
- b) 300 mg
- c) 60 mg
- d) 600 mg

Question 21:

What is the ideal time for nuchal translucency evaluation?

- a) 6-8 weeks
- b) 11-14 weeks
- c) 14-16 weeks
- d) 16-20 weeks

Question 22:

Magnesium sulphate has no role in:

- a) Prevention of seizures in severe pre-eclampsia
- b) Prevention of recurrent seizures in eclampsia
- c) Prevention of RDS in premature baby
- d) Reducing contractility of uterus

Question 23:

Which of the following is seen in the vertex presentation with complete flexion attitude of head?

- a) Suboccipito-bregmatic diameter
- b) Suboccipito-frontal diameter
- c) Occipito-frontal diameter
- d) Occipito-posterior position

Question 24:

A pregnant lady with a history of recurrent abortions is now diagnosed to have antiphospholipid syndrome. What would be the best mode of treatment for her?

- a) Aspirin only
- b) Aspirin+heparin
- c) Aspirin+heparin+steroids
- d) Aspirin+steroids

Question 25:

A 22-year-old pregnant lady presented with a fever and was started on antibiotics. The fever did not subside. On further investigation, she was diagnosed to have septic pelvic thrombophlebitis. Which among the following is the next course of treatment for her?

- a) Stop antibiotics and start heparin
- b) Antibiotics+Heparin
- c) Surgical embolectomy
- d) Hysterectomy

Question 26:

What is the treatment of choice for a woman with atrophic vaginitis with complaints of itching?

- a) Steroids
- b) Topical estrogen

- c) Antihistamines only
d) None of the above

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	b
5	c
6	c
7	b
8	d
9	c
10	a
11	a
12	a
13	b
14	a
15	a
16	c
17	b
18	d
19	c
20	a
21	b
22	c
23	a
24	b
25	b
26	b

Detailed Explanations

Solution to Question 1:

Acute fatty liver of pregnancy (AFLP) occurs most commonly in the third trimester or early postpartum period.

Risk factors include Pre-eclampsia/HELLP syndrome, male fetus, previous history of AFLP, and multiple gestations.

AFLP (acute yellow atrophy or acute fatty metamorphosis) is the most common cause of acute liver failure in pregnancy. It usually occurs in the later stages of pregnancy, particularly in the third trimester. It occurs due to fetal deficiency of long-chain acyl-coenzyme A dehydrogenase (LCHAD) or poor placental mitochondrial function.

The patients present with persistent nausea and vomiting, progressive jaundice, and epigastric pain. Features like acute hemolysis, hypoglycemia, hypertension, proteinuria, thrombocytopenia, and edema can also be present. Symptoms may progress to fulminant hepatic failure with encephalopathy, disseminated intravascular coagulation, and death.

Liver function tests such as AST, ALT, and serum bilirubin levels are raised. Hypofibrinogenemia, hypocholesterolemia, and prolonged clotting time can be seen. An elevated d-dimer is seen. Microscopy reveals a characteristic microvesicular fatty infiltration.

The condition worsens till delivery after which the hepatic dysfunction resolves. A trial of labor by induction or delivery by Cesarean is done. Intensive supportive measures including blood transfusion should be provided.

The clinical and laboratory findings of acute liver diseases in pregnancy are as follows:

The condition worsens till delivery after which the hepatic dysfunction resolves. Appropriate management is immediate termination of pregnancy irrespective of gestational age by induction or delivery by Cesarean (for obstetric reasons) is done. Intensive supportive measures including blood transfusion should be provided.

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
Onset in pregnancy	Mid to late	Late	Late	Variable
Differentiating features	Epigastric pain, hypertension, elevated indirect bilirubin	Pruritus, elevated direct bilirubin	Signs of liver failure like hypoglycemia	Fever
AST (U/L)	≤300	≤200	200-800	2000+
Bilirubin (mg/dL)	1-4	1-5	4-10	5-20
Platelets	↓↓	Normal	↓↓	↓
Fibrinogen	Normal	Normal	↓↓↓	Normal
D-dimer	↑	Normal	↑	Normal
Prothrombin time	Normal	Normal	↑↑	↑
Hemolysis	↑↑	No	↑↑↑	No

Solution to Question 2:

There is an increase in the metabolism of glycogen in the vaginal epithelium during normal pregnancy.

The changes in the vagina during pregnancy include:

- Increased vascularity and hyperemia resulting in a violet/purplish color characteristic of Chadwick's sign
- Vaginal epithelium undergoes marked proliferation (option D) with the extension of the squamocolumnar junction downwards
- Vaginal pH decreases and becomes more acidic (option C) and varies from 3.5-6.0.
- Increase in the number of *Lactobacillus acidophilus* (option A). It metabolizes the glycogen stores and causes an increased production of lactic acid. Therefore, the pH becomes acidic.
- Increased volume of cervical secretion during pregnancy results in the production of thick white vaginal discharge.

Solution to Question 3:

Overt diabetes is when fasting blood sugar is ≥ 126 mg/dl. Overt diabetics are women who are known to have diabetes before pregnancy.

A patient is said to have overt diabetes if:

- Fasting blood sugar ≥ 126 mg/dL
- A random blood sugar of ≥ 200 mg/dL
- HbA1c of $\geq 6.5\%$

The maternal complication of diabetes:

- increased Miscarriage and preterm labor.
- increased incidence of preeclampsia
- polyhydramnios
- prolongation of labor due to shoulder dystocia, operative interference, and cephalo-pelvic disproportion.

Fetal and Neonatal complications of diabetes:

- fetal macrosomia (birth weight > 4 kg)
- congenital malformations - Musculoskeletal, urogenital, and CNS anomalies.

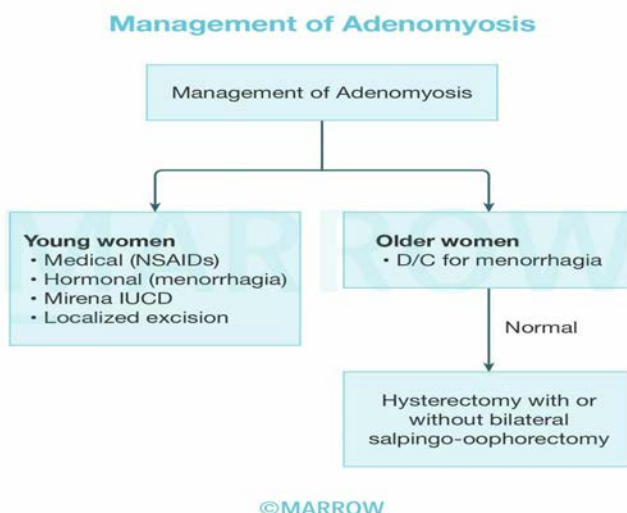
Solution to Question 4:

Multiparous women > 40 yrs, presenting with abdominal mass (free from adnexa) and myometrial cysts on MRI, point towards a diagnosis of adenomyosis.

Adenomyosis refers to the growth of the endometrium (both stromal and glandular components) directly into the myometrium. It is predominantly seen in multiparous women above the age of 40 years. Patients generally present with menorrhagia, dysmenorrhea, and sometimes dyspareunia. On examination, there is a gross symmetrical enlargement of the uterus corresponding to 12-14 weeks size. The uterus is soft and tender on bimanual examination.

Transvaginal ultrasonography (TVS) reveals a salt-and-pepper appearance of the myometrium and the presence of sub-endometrial striations producing the Venetian blind appearance. MRI can also be done for diagnosis. However, the only definitive mode of diagnosis is histopathological examination.

Since most women affected by the condition are past the childbearing age, total hysterectomy is the treatment of choice. Localized excision/Levonorgestrel containing IUCD can be tried in younger women.



Other options:

Option A: Endometriosis refers to the presence of endometrial tissue outside the endometrial lining (uterus). They usually present as dysmenorrhea and lower dull abdominal pain. Vaginal examination reveals a tender fixed retroverted uterus. They can be managed medically (oral contraceptives, aromatase inhibitors, etc) and surgically (if failed medical management). Abdominal hysterectomy with bilateral salpingo-oophorectomy is performed in advanced cases.

Option C: Leiomyoma (fibroid) is the commonest benign tumor arising from the uterus. They present as abnormal uterine bleeding, infertility, and abdominal pain. On bimanual examination, the uterus is enlarged and the mass moves with the cervix. It can be managed medically (mifepristone or newer progesterone) and surgically. A myomectomy is performed for large fibroids with significant symptoms.

Option D: Endometrial hyperplasia can be simple, glandular, or atypical. They are seen in women with estrogen excess states like obesity, PCOD, or on tamoxifen therapy. They usually present as abnormal uterine bleeding (AUB).

Solution to Question 5:

Functional residual capacity decreases in pregnancy.

The changes in the respiratory system during pregnancy are:

(Mnemonic: TIMM increases in pregnancy)

Increase	Decrease	Unaffected
Tidal volume Inspiratory capacity Minute ventilation Minute O ₂ uptake	Functional residual capacity Expiratory reserve volume Residual volume Total lung capacity	Respiratory rate Vital capacity Inspiratory reserve volume

Solution to Question 6:

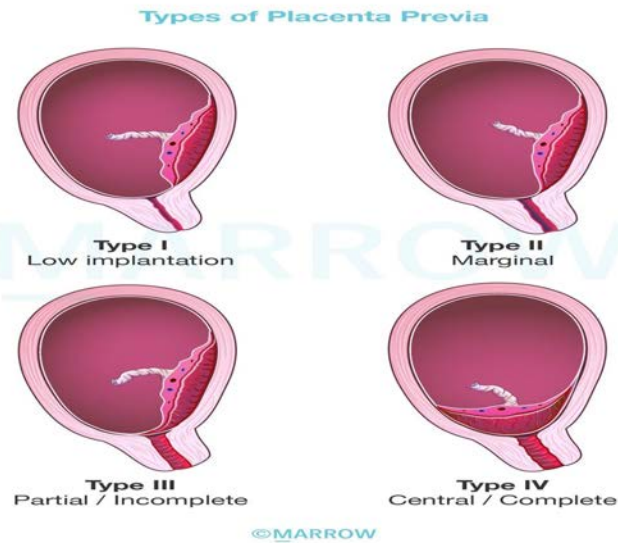
Heart disease is currently not considered a contraindication for the induction of labor. Vaginal delivery is preferred in such patients.

Solution to Question 7:

The given image shows grade III placenta previa.

Placenta previa refers to the placenta implanted (partially/completely) over the lower uterine segment (over or adjacent to the internal os). Grades of placenta previa are:

- Type—I (Low-lying): The major part of the placenta is attached to the upper segment and only the lower margin encroaches onto the lower segment but not up to the os.
- Type—II (Marginal): The placenta reaches the margin of the internal os but does not cover it.
- Type—III (Incomplete or partial central): The placenta covers the internal os partially (covers the internal os when closed but does not entirely do so when fully dilated).
- Type—IV (Central or total): The placenta completely covers the internal os even after it is fully dilated.



They typically present with sudden onset, painless vaginal bleeding. On vulval inspection, bright red blood is seen oozing from the cervical opening. Transvaginal ultrasound (TVS) is used to confirm the diagnosis and position of the placenta.

Management is dependent on the presentation of the patient. If there is a need for fetal maturity, expectant management (advocated by McAfee and Johnson) is carried out without increasing undue maternal stress. Definitive management by an elective cesarean section.

Solution to Question 8:

Physiological changes in pregnancy

Clinical examination

- Jugular venous distension and venous hum
- Mammary soufflé: Continuous murmur at 2nd to 4th intercostal space
- Occasional S3 at Mitral area
- Aortic or pulmonary flow murmurs at aortic/pulmonary areas
- S2 P increased; S2 split
- S1 M increased and widely split

Electrocardiography

- T wave inversion, biatrial enlargement
- Dysrhythmias
- As the diaphragm is elevated in advancing pregnancy, there is an average 15-degree left-axis deviation in the electrocardiogram (ECG), and mild ST changes may be seen in the inferior leads.
- Atrial and ventricular premature contractions are relatively frequent
- Pregnancy does not alter voltage

- Tricuspid regurgitation
- Increased left atrial end-diastolic dimension and
- Increased left ventricular mass

Solution to Question 9:

The given scenario of severe acute abdominal pain without fever/leucocytosis in a pregnant woman with a fibroid is suggestive of fibroid torsion.

The incidence of fibroid in pregnancy is about 1 in 1,000. Due to increased vascularity in pregnancy, the fibroid increases in size and becomes softer. The torsion is seen in the subserous pedunculated fibroid where the venous return is blocked at its site of attachment in the uterus. This leads to severe acute abdominal pain. Due to the complication being more local than systemic, fever or leucocytosis are generally not seen.

A transvaginal ultrasonogram is immediately done to detect the position and size of the fibroid, which is followed by laparotomic myomectomy as a part of emergency surgical management.

Other options:

Option A: Red degeneration of fibroid is seen commonly in the second half of pregnancy or puerperium. It is due to a hemorrhagic infarction of the tissue and typically presents as acute focal abdominal pain with fever and leucocytosis. Treatment is usually conservative with bed rest, analgesics, and sedatives.

Option B: Preterm labor is where the labor starts before 37 completed weeks. It is characterized by true labor pain. The abdominal pain is slowly increasing in intensity and duration and is also accompanied by show (expulsion of the cervical mucus plug). There are intervals between episodes of contractions where the patient doesn't complain of abdominal pain.

Solution to Question 10:

Dicloxacillin is the first line of treatment for mastitis in lactating mothers and should be taken for 10-14 days.

Mastitis refers to the infection of the breast parenchymal tissue. The most common organism causing breast infection is *Staphylococcus aureus*. It is common in new mothers at the beginning of lactation who are inexperienced with breastfeeding or during weaning when nipple trauma occurs due to the infant's growing teeth.

Clinical features of mastitis include:

- Fever and malaise
- Localized pain and tenderness
- Purulent nipple discharge
- Redness and swelling of the affected breast
- Positive fluctuation test

A culture from nipple discharge or test-needle aspiration of the cellulitic area is done to confirm the organism and its antibiotic sensitivity. An ultrasonogram of the involved breast is done to exclude abscess. They are treated with antibiotics and adequate breast support. Choice of antibiotics in lactational mastitis:

- Dicloxacillin 500 mg 4 times a day is to be started empirically.
- Erythromycin is given to women who are penicillin-sensitive.
- If the infection is caused by resistant, penicillinase-producing *Staphylococcus* species then vancomycin, clindamycin, or trimethoprim-sulfamethoxazole is given for at least 10 days.

Solution to Question 11:

Fetal urine is the major contributor to amniotic fluid post 20 weeks.

Amniotic fluid is present in the amniotic cavity which is the space between the epiblast (ectoderm) and trophoblast. The volume of the amniotic fluid increases with pregnancy and reaches a maximum at term. It is contributed by:

- Early pregnancy - ultrafiltrate of maternal plasma (option D).
- Second trimester - the extracellular fluid that diffuses through the fetal skin (option B) and thus reflects the composition of fetal plasma.
- After 20 weeks - keratinization of fetal skin prevents diffusion across the skin. Hence, fetal urine forms amniotic fluid.

Note: Fetal kidneys start producing urine at 12 weeks (8–11 weeks), and by 18 weeks, they produce 7 to 14 mL per day.

Solution to Question 12:

In low ovarian reserve, the level of anti-Müllerian hormone (AMH) will be <1 ng/mL.

AMH is a glycoprotein secreted by the granulosa cells of the primordial follicle of the ovary. It helps in oocyte maturation and mono-follicular development. AMH is also used in assessing ovarian reserve.

In women, AMH is always absent during embryogenesis and is secreted by the granulosa cells after puberty. It helps in follicular development and oocyte maturation. Estimation of serum AMH is used in the study of ovarian reserve in female infertility and secondary amenorrhoea.

In men, AMH starts to be secreted by the Sertoli cells in the testes by the 7th week of intrauterine life. It continues until puberty. It inhibits the development of the Müllerian system. The absence of AMH results in the development of hermaphroditism.

AMH levels in ng/ml	Interpretation
2–6.8	Normal

AMH levels in ng/ml	Interpretation
<1	Poor ovarian reserve
>10	PCODHyperstimulation syndrome

Solution to Question 13:

The maximum dose of prostaglandin F2 alpha in postpartum hemorrhage is 2 mg.

Prostaglandins are paracrine/autocrine hormones and act locally at the site of production. Main sources for:

- PGF₂α - Decidua and promotes myometrial contractility
- PGE₂ - fetal membranes (amnion) and induces labor and promotes cervical effacement and dilatation
- PGI₂ - Myometrium

Carboprost tromethamine (hemabate) is the 15-methyl derivative of prostaglandin F2 alpha. It is approved for the treatment of uterine atony at a dose of 0.25 mg IM. This dose can be repeated if necessary at 15- to 90-minute intervals up to a maximum of eight doses (2 mg).

It is the best drug to control PPH. Is also used as an abortifacient in the second trimester.

Contraindications include asthmatic women and those with suspected amniotic fluid embolism, as they cause pulmonary airway and vascular constriction.

Solution to Question 14:

A partial mole is not a gestational trophoblastic neoplasm (GTN).

Gestational trophoblastic disease (GTD) is the term used to describe a group of tumors due to abnormal trophoblast proliferation. Malignant forms of GTD are termed gestational trophoblastic neoplasia (GTN).

Modified WHO Classification of GTD

Molar Pregnancies	Trophoblastic tumors
Hydatidiform mole (benign) - Partial and Complete Invasive mole (malignant)	Choriocarcinoma (malignant) Placental site trophoblastic tumor (malignant) Epithelioid trophoblastic tumor (malignant)

Solution to Question 15:

The image provided depicts a baby with macrosomia (large for gestational age), the most probable cause of which is gestational diabetes (GDM) in the mother.

GDM in a mother is defined as carbohydrate intolerance with onset/first recognition during the present pregnancy. Diabetes in Pregnancy Societies of India (DIPSI) criteria is used to diagnose GDM (Venous plasma ≥ 140 mg % after 2h of ingestion of 75g of glucose).

Maternal Complications of GDM:

- During pregnancy - Recurrent spontaneous abortion, preterm labor, urinary tract infection, polyhydramnios, and increased incidence of preeclampsia
- During labor: Prolongation of labor (shoulder dystocia) and postpartum hemorrhage

Neonatal complications -

- Electrolyte imbalances - hypoglycemia, hypocalcemia, hypokalemia, hypomagnesemia
- Respiratory system - respiratory distress syndrome and delayed lung maturity
- Hematological system - polycythemia and single umbilical artery
- Renal system - hydronephrosis (renal agenesis) and polycystic kidneys
- Cardiovascular system - VSD, ASD, Coarctation of the aorta, and TGA
- Gastrointestinal system - duodenal atresia, tracheoesophageal fistula (TEF), and necrotizing enterocolitis
- Central Nervous system - anencephaly, microcephaly, neural tube defects, caudal regression syndrome, and sacral agenesis

Other options:

Option B: Congenital rubella syndrome presents with the classical triad of deafness, cataract, and congenital heart disease (the most common being - patent ductus arteriosus). Eye findings also include salt and pepper retinopathy as shown below.

Option C: Congenital hypothyroidism is the most common preventable cause of mental retardation and presents as 6 P's - pot-bellied, pale, puffy-faced child with protruding umbilicus, protuberant tongue, and poor brain development

Option D: Gestational hypertension primarily occurs due to the failure of the invasion of extravillous trophoblasts into the spiral artery. It is defined as a rise in blood pressure $\geq 140/90$ mmHg after 20 weeks of pregnancy (without proteinuria). It usually leads to intrauterine growth restriction (IUGR) and fetal death.

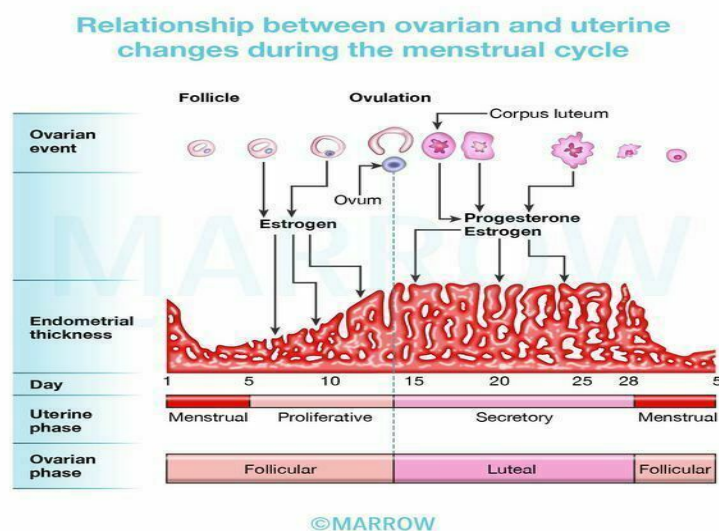
Solution to Question 16:

The hormone acting on the post-ovulatory endometrium is progesterone.

Endometrium undergoes changes (endometrial cycle) during the menstrual cycle due to the action of various hormones. The basal zone (lower 1/3rd of the endometrium in contact with the myometrium) is uninfluenced by hormones and shows no change during the cycle. The rest

(functional zone) shows the following changes:

- Regenerative phase (first 2 to 3 days)- the regeneration of glands and the stromal cells from the basal zone. Endometrial thickness is about 2mm.
- Proliferative phase (5 or 6th day to 14th day) - estrogen (option D) causes the glands to become more tubular and perpendicular to the surface and increases the thickness to 10-12mm on the 14th day (at the time of ovulation)
- Secretory phase(15th day till menstrual bleeding) - post ovulation, estrogen-primed endometrium is acted by progesterone (from corpus luteum) and causes increased blood flow (spiraling of arteries) and increases tortuosity of glands (saw-toothed/cork-screwed glands). The thickness of the endometrium is maximum (12mm) in this phase.
- Menstruation (last 4 or 5 days) - estrogen and progesterone are withdrawn due to the regression of corpus luteum (lack of fertilization). This results in the shedding of the functional zone (menstrual bleeding).



Other options:

Option A: The luteinizing hormone (LH) is secreted by the anterior pituitary and is crucial for ovulation to occur on the 14th day of the menstrual cycle.

Option B: Follicular stimulating hormone (FSH) is secreted by the anterior pituitary. The ovarian follicle with the most sensitivity to FSH becomes the dominant follicle for the current cycle and the rest undergo atresia.

Solution to Question 17:

Thromboembolism is an absolute contraindication to OCP use.

Oral contraceptives (OCP) are preparations of estrogen or progesterone or a combination of both. Estrogen in OCPs suppresses gonadotropin-releasing hormone (GnRH) and follicle-stimulating hormone (FSH) whereas progesterone suppresses luteinizing hormone (LH), thereby inhibiting

ovulation.

Though OCPs are generally used to prevent pregnancy, they are also indicated for the treatment of dysmenorrhoea, endometriosis, uterine fibroid, and dysfunctional uterine bleeding (DUB)

Contraindications of OCPs:

- Absolute contraindications (WHO category 4)-
- History of thromboembolism- DVT, stroke
- Heart diseases - Valvular or ischemic
- Severe hypertension
- Liver diseases like hepatic adenoma, and hepatic carcinoma
- Breastfeeding
- During pregnancy
- Estrogen dependant neoplasms like - breast cancer
- Relative contraindications -
-

WHO category-2 (benefits outweigh the risks) - cancer cervix, cervical intraepithelial neoplasia (CIN), and in patients with sickle cell anemia, disease of the gallbladder, or diabetes mellitus (option D).

- WHO category-3(risks outweigh benefits) - patients with benign liver tumors, unexplained vaginal bleeding, or history of smoking >20 cigarettes/day.

Other options:

Option A: Minipills (progestin-only pills) could be used safely in renal disease as they reduced clotting risks and is not an absolute contraindication.

Option C: A history of amenorrhea is investigated based on the algorithm for primary or secondary amenorrhoea. OCPs are used to regularise menstruation in polycystic ovarian syndrome (PCOD) which is a cause of secondary amenorrhoea.

Solution to Question 18:

Partial hydatidiform mole has a 69XXY or 69XYY karyotype.

Hydatiform mole belongs to the benign group of gestational trophoblastic diseases (GTD). It is an abnormality in chorionic villi making them partly proliferative and degenerative, thereby resulting in the formation of multiple small cysts. Based on the portion of the chorionic villi involved, they could be complete (diffused) or partial (focal)

Solution to Question 19:

In the clinical scenario of a 55-year-old woman with third-degree uterine prolapse, the ideal management is vaginal hysterectomy with pelvic floor repair.

Uterine prolapse is the descent of the uterus from its normal position which usually occurs due to laxity of ligaments that offer support, seen often due to childbirth, aging, or trauma. The uterine descent is clinically categorized into:

- First degree: the external os remains above the vaginal introitus
- Second degree: The external os protrudes below the vaginal introitus but the uterine body remains inside the vagina
- Third degree (Syn: Complete prolapse): The uterine, cervix, body, and fundus descend to lie outside the introitus
- Procidentia involves prolapse of the uterus with the entire vagina everted.

The patient usually presents with a sense of heaviness in the pelvic region with lower back discomfort. Based on the structures involved the patient can additionally have urinary symptoms (like incontinence and urgency), bowel disturbances (constipation and tenesmus), or sexual discomfort during intercourse.

Symptomatic uterine prolapse is managed surgically with different procedures based on the profile of the patient. Conservative management is done for old asymptomatic women with mild degree prolapse.

A vaginal hysterectomy is recommended for patients who satisfy the following criteria:

- >40 years of age
- Have completed their families
- Do not wish to preserve menstrual and child-bearing function

Other options:

Option A: Fothergill repair or Manchester operation is a procedure done to preserve the reproductive function and vaginal prolapse with supravaginal cervix elongation. It initially involves preliminary curettage with low amputation of the cervix. After Mackenrodt's ligament is plicated, anterior colporrhaphy and colpoperineorrhaphy are performed.

Option B: Sling operation (Purandare's operation) is indicated in congenital or nulliparous prolapse without cystocele. The procedure involves the extraperitoneal passage of rectus sheath strips joined to the anterior surface of the cervix by silk sutures.

Option D: Shirodkar procedure is a type of abdominal sling surgery done in young women to retain menstrual and reproductive function. It involves using a sling to fix the cervix to the lumbosacral fascia.

Solution to Question 20:

The dose of Ulipristal in emergency contraception is 30 mg to be taken within 72 hrs (upto 120hrs).

Methods of Emergency contraception

Drug	Dose
Levonorgestrel	0.75mg stat and after 12 hours
Ethinyl estradiol 50 µg + Norgestrel 0.25 mg	2 tab stat and 2 tab after 12 hours
Conjugated estrogen	15 mg BD x 5 days
Ethinyl Estradiol	2.5 mg BD X 5 days
Copper IUDs	Insertion within 5-7 days after accidental unprotected exposure. Not suitable for rape victims
Centchroman	2 tablets(60 mg) to be taken twice at an interval of 12 hours within 24 hours of intercourse.
Ulipristal	30 mg within 72 hours

Solution to Question 21:

Nuchal translucency evaluation, a component of first-trimester aneuploidy screening, is done between 11-14 weeks.

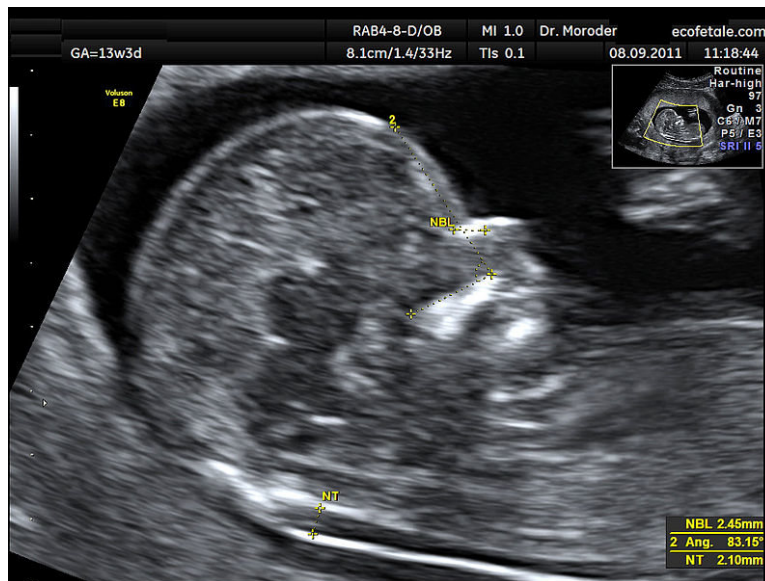
It represents the maximum thickness of the subcutaneous translucent area between the skin and soft tissue, overlying the fetal spine, at the back of the neck. When the nuchal translucency is increased, the risk of fetal aneuploidy and structural anomalies such as heart defects is significantly elevated.

First trimester aneuploidy screening combines two maternal serum analytes and a sonographic measurement:

- Human chorionic gonadotropin (hCG)
- Pregnancy-associated plasma protein A (PAPP-A)
- Nuchal translucency (NT)

It is performed between 11 and 14 weeks of gestation. In cases of fetal Down syndrome, the first-trimester serum-free β -hCG level is higher and the PAPP-A level is lower. With trisomy 18 and trisomy 13, levels of both analytes are lower.

Note: Nuchal fold is a normal fold of skin seen at the back of the fetal neck during the second trimester of pregnancy. Increased nuchal fold thickness is associated with fetal aneuploidy.



Solution to Question 22:

Magnesium sulfate has no role in the prevention of respiratory distress syndrome (RDS) in a premature baby, where there is a deficit of surfactant (which results in alveolar collapse).

Magnesium in the $MgSO_4$ competes with calcium on voltage-gated calcium channels, thereby altering its uptake and release. Functions of $MgSO_4$ include :

- Membrane protector and neuro stabilizer. It is used in the prevention of cerebral palsy in very low birth weight neonates.
- Reduction of motor endplate sensitivity to acetylcholine, and thus is used as an adjunct in the management of acute severe asthma in children and adults
- Blocking of neuronal calcium influx, and is thus used in the conservative management of pre-term labor (tocolytic, option D)
- Inducing cerebral vasodilatation, dilating the uterine arteries, increasing the production of endothelial prostacyclin, and inhibiting platelet activation. Hence it is used in the management of pre-eclampsia (option A) and to prevent progression to eclampsia (option A).

Solution to Question 23:

Suboccipito-bregmatic diameter is seen in vertex presentation with a complete flexion attitude of the head.

Diameters of the fetal skull:

Diameters	Attitude of the head
Suboccipitobregmatic	Complete flexion
Suboccipito-frontal	Incomplete flexion
Occipitofrontal	Marked deflexion
Mento-vertical	Partial extension
Submentovertical	Incomplete extension
Submentobregmatic	Complete extension

Solution to Question 24:

The best mode of treatment for a pregnant woman with antiphospholipid syndrome (APS) is aspirin and heparin.

APS is a prothrombotic disorder (autoantibody-mediated thrombophilia) affecting both arterial and venous systems.

Patients present with recurrent pregnancy loss due to thrombosis of uteroplacental vessels, poor placental perfusion, and prevention of normal decidual changes after zygote implantation. They may also present with preterm delivery, pre-eclampsia, intrauterine growth restriction (IUGR), and fetal death.

Criteria for APS include:

- at least one otherwise unexplained fetal death at or beyond 10 weeks;
- at least one preterm birth before 34 weeks gestation because of eclampsia, severe preeclampsia, placental insufficiency; or
- at least three unexplained consecutive spontaneous abortions before 10 weeks.

Aspirin (60-80mg daily) with heparin (5000-10,000 U subcutaneously every 12h) is the most effective therapy for APS. Treating APS with only aspirin (option A) is not recommended as heparin is crucial in preventing thrombus formation at the decidual-thrombotic interface, Corticosteroids (option C, D) should not be used in primary APS (without an associated connective tissue disorder) but are indicated in patients with pre-existing systemic lupus erythematosus (SLE) and APS (secondary APS).

Solution to Question 25:

Antibiotics plus heparin is the next course of treatment for septic pelvic thrombophlebitis.

Septic pelvic thrombophlebitis (SPT) refers to the pelvic infection with thrombosis of one/both ovarian veins. The pathogenesis of this disease is due to the spreading uterine infection

(secondary to trauma of delivery or surgery) combined with the hypercoagulable state of the pregnancy (thrombus formation).

The patients usually present with constant lower abdominal pain and occasionally with a palpable tender sausage-shaped mass (most diagnostic finding, but rare).

SPT is usually suspected when the fever fails to respond to antibiotic therapy. It can be confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) of the abdomen and pelvis.

Patients are treated with a combination of heparin and broad-spectrum antibiotics.

Anticoagulation is done to prevent further thrombosis and reduce the spread of septic emboli.

Note: The use of anticoagulation is controversial as not all clinicians favor it. A small randomized study did not demonstrate a benefit, but the majority accept short-term anticoagulation as a part of the treatment regimen.

Solution to Question 26:

Atrophic vaginitis is treated with topical estrogen vaginal cream.

Atrophic vaginitis (senile vaginitis) is seen in postmenopausal women due to the thinning of vaginal mucosa (lack of estrogen). They present as discomfort (itching) and dryness in the vulva with dyspareunia. Occasionally, they may also complain of yellowish or blood-stained vaginal discharge.

The possibility of a coexisting carcinoma of the body or the cervix has to be excluded before treating it with estrogen supplements. Though topical application is the initial treatment, systemic estrogens (hormone replacement therapy) can be considered based on the patient's profile and the severity of the symptoms.

Other options:

Option A: Steroids are not indicated in the management of atrophic vaginitis as the inflammation is due to a lack of estrogen. Furthermore, they may predispose the patient to vaginal infections.

Option C: Antihistamines antagonize histamine receptors and prevent the allergic response. Since atrophic vaginitis is not an allergy response, they are not indicated in the treatment.

Obstetrics and Gynecology NEET 2020

Question 1:

The remnants of Wolffian ducts in females are located in_____.

- a) Pouch of Douglas
- b) Leaves of broad ligament
- c) Uterovesical pouch
- d) Iliac fossa

Question 2:

In which part of the fallopian tube does fertilization occur?

- a) Ampulla
- b) Isthmus
- c) Interstitial
- d) Infundibular

Question 3:

A patient came to the gynaecology OPD with complaints of foul-smelling frothy vaginal discharge and intense itching. On examination, the cervix was spotted and had the appearance of a strawberry. Identify the causative organism.

- a) *Trichomonas vaginalis*
- b) *Gardnerella vaginalis*
- c) *Neisseria gonorrhoea*
- d) *Candida albicans*

Question 4:

Identify the contraceptive shown in the given image.



- a)** Male condom
- b)** Female condom
- c)** Diaphragm
- d)** Cervical cap

Question 5:

A woman presents to the OPD with complaints of amenorrhoea for the past year. She has a history of postpartum hemorrhage and lactational failure. What is the probable diagnosis?

- a)** Sheehan's syndrome
- b)** Rotor syndrome
- c)** Prolactinoma
- d)** Acromegaly

Question 6:

An 18-year-old girl presents with a history of low-grade fever, weight loss, abdominal pain and amenorrhoea for six months. On examination, a pelvic mass is felt on the left side and signs of ascites are positive. What is your diagnosis?

- a)** Tuberculosis with tubo-ovarian mass
- b)** Ovarian malignancy
- c)** Ectopic pregnancy

- d) Submucous fibroid

Question 7:

In modern obstetrics, which of the following investigations is used for monitoring the fetus of an Rh-negative sensitized mother?

- a) Biophysical profile
- b) Fetal blood sampling
- c) Amniospectrometry
- d) MCA doppler peak systolic velocity

Question 8:

What is the causative agent for granuloma inguinale?

- a) Calymmatobacterium granulomatis
- b) Chlamydia trachomatis
- c) Ureaplasma urealyticum
- d) Treponema pallidum

Question 9:

Which of the following is the causative organism of vulval warts?

- a) Human papilloma virus
- b) Herpes simplex virus
- c) Human immunodeficiency virus
- d) Cytomegalovirus

Question 10:

A 22-year-old pregnant woman at 13 weeks of gestation informs you about accidental exposure to radiation from a chest X-ray at 7 weeks of gestation. What should be the next step advised to her?

- a) Continue pregnancy
- b) Terminate pregnancy

- c) Pre-invasive diagnostic test
- d) Chromosome screening

Question 11:

After successful delivery of the baby, the obstetrician witnesses that the episiotomy performed has extended backwards beyond the perineal body. Which among the following structures is likely to be injured in this case?

- a) External anal sphincter
- b) Ischiocavernosus
- c) Bulbospongiosis
- d) Urethral sphincter

Question 12:

A 16 year old girl presents to the OPD with a history of primary amenorrhoea and cyclical pain. Perineal examination reveals a bulge protruding from the vagina. Identify the cause of this presentation:

- a) Transverse septum in the proximal vagina
- b) Complete vaginal atresia
- c) Imperforate hymen
- d) Cervical agenesis

Question 13:

A pregnant woman presents for a routine antenatal checkup at 18 weeks of gestation. On abdominal examination, the uterine size was found to correspond to 16 weeks of gestation. On USG, oligohydramnios was found. Which of the following is a likely cause?

- a) Renal agenesis
- b) Fetal anemia
- c) Barter syndrome
- d) Anencephaly

Question 14:

A 24-year-old woman visits the infertility clinic with a complaint of recurrent abortions. Upon investigations, she is found to have a septate uterus. Which of the following corrective procedures has the best obstetric outcome?

- a) Jones metroplasty
- b) Strassmann metroplasty
- c) Tompkins metroplasty
- d) Transcervical resection of septum

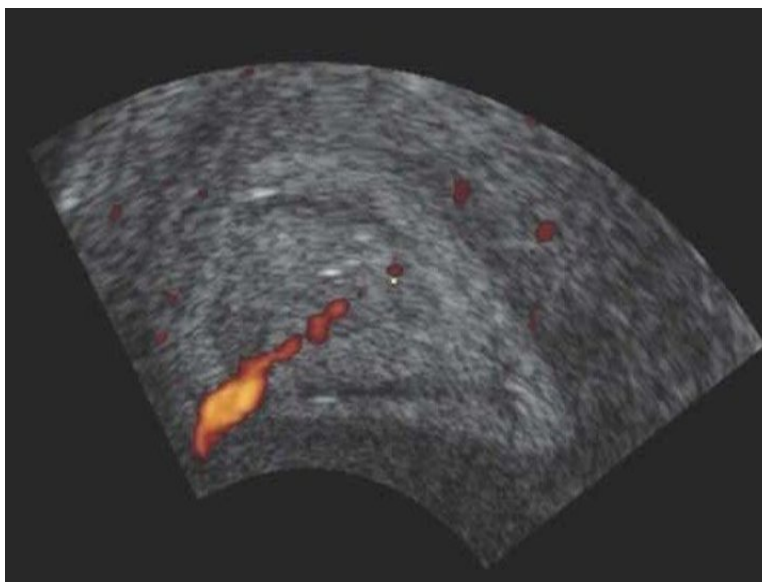
Question 15:

A 33-year-old female patient comes with complaints of menorrhagia for the past 3 months. The duration and regularity of her menstrual cycles are normal. Non-hormonal management has been tried without success. What is the most appropriate next step?

- a) Uterine artery embolisation
- b) Endometrial ablation
- c) Hormonal therapy
- d) Hysterectomy

Question 16:

A 60-year-old woman presented with post-menopausal bleeding. Ultrasonography revealed a mass with feeding vessel as shown in the image below. What is the most likely diagnosis?



- a) Fibroid uterus
- b) Endometrial polyp
- c) Endometrial carcinoma
- d) Leiomyosarcoma

Question 17:

A 35-year-old primigravida who conceived after in-vitro fertilization therapy came for an antenatal checkup at 38 weeks of gestation. Her obstetric history revealed that she was pregnant with DCDA twins. On examination, the first twin was found to be in a breech position and the maternal blood pressure was found to be greater than 140/90 mm Hg on two occasions with grade 1+ proteinuria. What should be done next?

- a) Monitor BP and terminate pregnancy if BP rises
- b) Immediate Caesarean section
- c) Terminate pregnancy at 40 weeks of gestation
- d) Induce labour with PGE₂ gel

Question 18:

Which of the following effects of puberty are not mediated by estrogen?

- a) Hair growth
- b) Cervical mucus secretion
- c) Vaginal cornification
- d) Menstruation

Question 19:

A 35-year-old woman comes with complaints of warty lesions on the vulva. On examination, multiple warty growths on the vulva and around the anal canal are seen. This condition is most likely seen in?

- a) Condyloma accuminata
- b) Molluscum contagiosum
- c) Chancroid
- d) Herpes

Question 20:

A patient is planned to undergo hysteroscopic polypectomy using bipolar electrocautery. What is the ideal agent for distention?

- a) Glycine
- b) Normal saline
- c) CO₂
- d) 70% dextrose

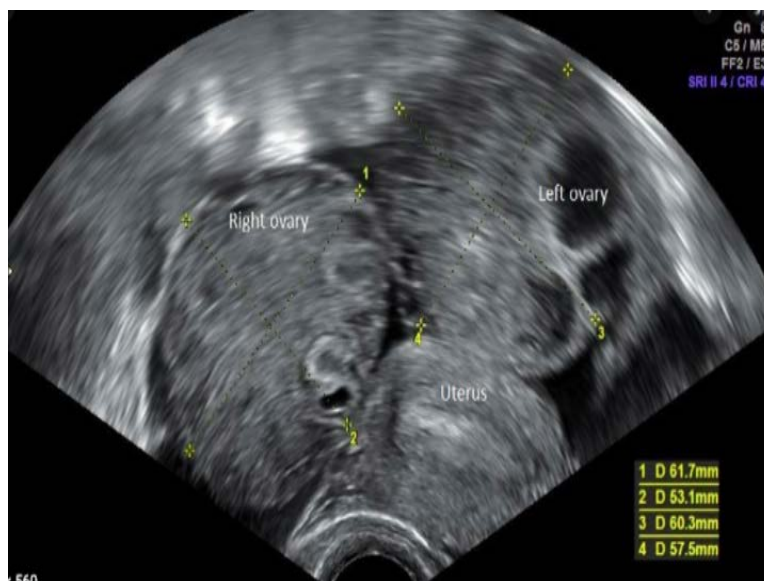
Question 21:

A 52-year-old woman diagnosed with carcinoma cervix stage 2B was advised radiotherapy. Which of the following statements regarding radiotherapy is false?

- a) Point B is 3 cm lateral to point A
- b) Point A is 2 cm above the external os and 5 cm lateral to the internal uterine canal
- c) Point A receives a dose of 7000 cGy
- d) Point B receives a dose of 6000 cGy

Question 22:

A woman with a history of infertility is on treatment with human menopausal gonadotropin. She presented with the following finding in the USG. What is the most likely diagnosis?



- a) Ovarian hyperstimulation syndrome
- b) Theca lutein cyst
- c) Polycystic ovarian syndrome
- d) None

Question 23:

A lady who is 38 weeks pregnant comes to the OPD for a routine checkup. She has a history of normal twin delivery at term 4 years ago. What is her gravida and para score?

- a) G2P2
- b) G2P1
- c) G3P2
- d) G3P1

Question 24:

Identify the finding seen in this hysterosalpingogram:



- a) Bilateral hydrosalpinx
- b) Extravasation into venous system
- c) Normal HSG
- d) Bilateral cornual block

Question 25:

Identify the maternal condition that can lead to the findings in the baby as shown below.



- a) Pre-gestational diabetes
- b) Gestational diabetes mellitus
- c) Effect of valproate
- d) Hypertensive on ACE inhibitors

Question 26:

A woman with endometrial carcinoma is undergoing radiotherapy. Which of the following is true?

- a) Intensity is inversely proportional to squared distance
- b) Small blood vessels are most radioresistant
- c) Rapidly proliferating cells are most radioresistant
- d) Small bowel is most radioresistant

Question 27:

Identify the use of the instrument given below:



- a) For tubal ligation
- b) For ligation of uterine artery during hysterectomy
- c) Ovarian cystectomy
- d) Ectopic pregnancy

Question 28:

All of the following are contraceptive methods used post-coitus except:

- a) Danazol
- b) CuT200
- c) RU486
- d) Combined high dose estrogen-progestin pills

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	a

6	a
7	d
8	a
9	a
10	a
11	a
12	c
13	a
14	d
15	c
16	b
17	b
18	a
19	a
20	b
21	b
22	a
23	b
24	a
25	a
26	a
27	a
28	a

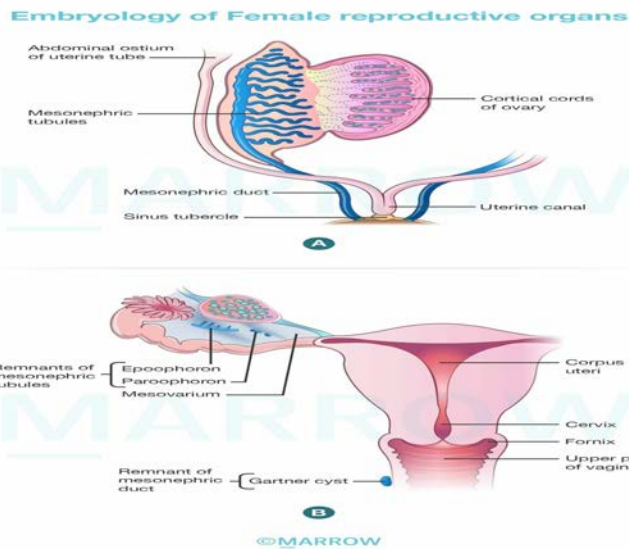
Detailed Explanations

Solution to Question 1:

The Gartner's duct is a remnant of the Wolffian ducts in women. These are located in the broad ligament and the vagina.

The Gartner's duct is an embryological remnant of the mesonephric duct/Wolffian duct formed during the development of the urinary and reproductive organs. It runs parallel to, but below, the fallopian tube in the mesosalpinx. The duct passes downwards by the side of the uterus to the level of the internal os where it passes into the tissues of the cervix. It then runs forwards to reach the anterolateral aspect of the vaginal wall and may reach as far down as the hymen.

The duct can sometimes form a cyst in the broad ligament or in the vagina. It is called Gartner's cyst. It is a benign condition and can be treated with marsupialization.



Solution to Question 2:

Fertilization usually occurs in the ampulla of the fallopian tube.

During ovulation, the ovum is released and it enters the fallopian tube. It is propelled towards the uterine cavity by the peristaltic action of the fallopian tubes, prostaglandins in the male seminal fluid and oxytocin. It can only be fertilized within 24 hours of ovulation. If coitus takes place during this period, the spermatozoa swimming up into the tubes meet the ovum in the ampulla.

Solution to Question 3:

In the given clinical scenario, *Trichomonas vaginalis* is the likely causative agent.

Vaginal itching and foul-smelling frothy vaginal discharge are seen in trichomonas vaginitis.

The given image shows colpitis macularis or strawberry cervix, which is a characteristic finding of trichomoniasis. This is due to microscopic, punctate haemorrhages of the cervix.



Solution to Question 4:

The given image shows a female condom. It consists of a pouch made of polyurethane which is 17 cm in length with a flexible ring at each end. The inner ring at the closed end is smaller than the outer ring.

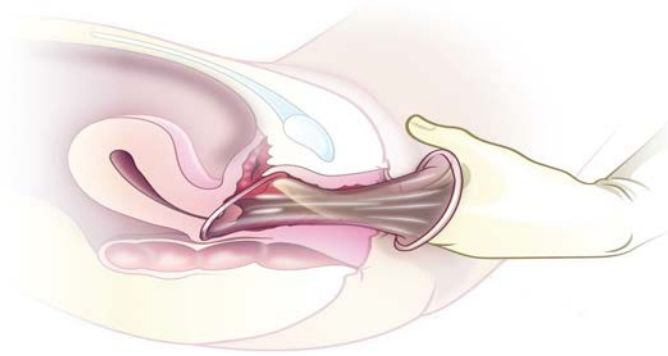
The inner ring is inserted at the apex of the vagina and the outer ring remains outside. It covers the cervix, the entire vagina, and external genitalia. Hence, it provides protection against HIV and STDs. It has a failure rate of 5–15 per 100 woman-years, which is higher than that of a male condom.

It can be fitted by the patient hours before intercourse, and easily removed soon after. It covers the cervix, vagina, and external genitalia and is protective against sexually transmitted diseases. It can also be worn in the puerperal period, unlike the cervical diaphragm.

The female condom should not be used along with a male condom because if worn together they may slip, tear, or become displaced.

The given image below shows the insertion of a female condom:

Insertion of the female condom



©MARROW

Solution to Question 5:

The clinical scenario points to a diagnosis of Sheehan's syndrome.

Sheehan's syndrome (SS) is postpartum hypopituitarism secondary to the necrosis of the pituitary gland. It usually results from severe hypotension or shock caused by massive hemorrhage during or after delivery. Patients with SS have varying degrees of anterior pituitary hormone deficiency. The hyperplastic enlargement of the pituitary, which occurs normally during pregnancy, increases the risk of infarction.

Affected women may have persistent hypotension, tachycardia, hypoglycemia, and lactation failure. The most prominent symptoms are lactation failure (due to a deficiency of prolactin) and postpartum amenorrhea (due to a deficiency of gonadotropins). Because deficiencies of some or all pituitary-responsive hormones may develop after the initial insult, Sheehan syndrome can be heterogeneous and may not be identified for years.

It can present acutely as pituitary apoplexy, an endocrine emergency, characterized by a sudden onset of headache, nausea, visual deficits, and in severe cases, severe hypoglycemia, hypotension and shock, CNS hemorrhage, and death.

Also, it can present late with the following features:

- Loss of pubic and axillary hair
- Hypothyroidism due to decreased TSH secretion
- Adrenal insufficiency due to low ACTH

Treatment is by hormone replacement - first glucocorticoids, later - thyroid hormone, sex steroids, vasopressin and growth hormones are given. If there is evidence of progressive visual loss, cranial nerve involvement or loss of consciousness, immediate surgical decompression must be carried out. Fertility can be achieved in these women with hMG and hCG. Clomiphene is ineffective.

Option B: Rotor syndrome is a type of hereditary hyperbilirubinemia with autosomal recessive inheritance. Affected individuals have mild jaundice and hyperbilirubinemia.

Option C: Prolactinoma or pituitary adenoids is the condition of hyperprolactinemia diagnosed by a prolactin level $>200 \mu\text{g/L}$. It usually presents with infertility, amenorrhea, and galactorrhea.

Option D: Acromegaly is characterized by the signs and symptoms pertaining to excessive growth hormone secretion.

Solution to Question 6:

The given clinical scenario with abdominal pain, amenorrhoea, pelvic mass and ascites is suggestive of genital tuberculosis.

Genital tuberculosis:

It is caused by *M. tuberculosis* and almost always occurs secondary to primary infection elsewhere in the extragenital states. The sites and frequency of involvement are as follows:

- Fallopian tubes (90-100%)
- Endometrium (50-60%)
- Ovaries (20-30%)
- Cervix (5-15%)
- Vulva and vagina ($< 1\%$)

Presentation is variable and it can be asymptomatic in 10-20% of cases, infertility, menstrual irregularities, abdominal pain and mass, vaginal discharge, fistula formation, ectopic pregnancy and ascites. The mass is made up of rolled-up omentum that is adherent to the uterus and adnexa.

Constitutional symptoms such as fever, weakness, anorexia or night sweats may be present.

Investigations:

- CBC, ESR and Mantoux test
- Hysterosalpingography shows a lead-pipe appearance of the fallopian tube, calcification of the tube, beading and variation in the filling, cornual block, jagged fluffiness of tubal outline and tobacco pouch appearance of hydrosalpinx and pyosalpinx.
- Laparoscopy establishes the diagnosis of genital TB. It shows tuberculous lesions and perihepatic adhesions (if present).

Differential diagnosis:

- Ovarian cyst, broad ligament cyst or encysted fluid
- Pelvic inflammatory disease
- Ectopic pregnancy
- Carcinoma cervix
- Elephantiasis of the vulva
- Pregnancy
- Puberty menorrhagia and postmenopausal bleeding

- Fungal infection and sarcoidosis

Management:

It is treated with anti-tubercular therapy for 6 months (DOT strategy):

Indications and contraindications for surgery:

Types of surgery include vulvectomy (if vulval hypertrophy is present), removal of the adnexal mass, drainage of pyometra and fistula repair.

TAH+BSO is rarely required nowadays. Tuboplasty is contraindicated.

Option B: Ovarian tumors usually don't affect menstrual cycles. Though virilizing tumors can cause amenorrhoea, other features of virilization such as hirsutism, flattening of breasts, and enlargement of the clitoris are generally present.

Option C: Though ectopic pregnancy may be suspected here, the tube is likely to rupture with exacerbation of pain, bleeding, and peritonitis before six months of gestation. Additionally, the absence of vaginal bleeding/spotting, and the presence of weight loss makes the diagnosis unlikely.

Option D: Submucous fibroid is associated with menorrhagia.

First 2 months	Next 4 months
Isoniazid 15 mg/ kg body weight Rifampicin 450-600 mg Pyrazinamide 30 mg/body weight Ethambutol 30 mg/ kg body weight 3 times a week.	Isoniazid 15 mg/kg body weight Rifampicin 450-600 mg 3 times a week

Indications	Contraindications
Unresponsiveness of the disease in spite of adequate anti-tubercular chemotherapy. Tubercular pyosalpinx, ovarian abscess or pyometra. Persistent menorrhagia and/or chronic pelvic pain causing deteriorating health status	Presence of active tuberculosis in extragenital site Favorable response to antimicrobial therapy with the diminishing size of the pelvic mass The accidental discovery of tubercular tubo-ovarian mass on laparotomy in a young patient.

Solution to Question 7:

MCA Doppler peak systolic velocity (PSV) is used for the monitoring of fetuses in Rh-negative sensitized mothers.

The middle cerebral artery is used as other fetal vessels require a larger insonating angle and confers significant measurement error.

MCA PSV is enhanced due to increased cardiac output and decreased viscosity of fetal blood, which permits the reliable, noninvasive detection of fetal anemia in cases of blood-group

alloimmunization.

If the MCA PSV exceeds 1.5 MoM (multiples of the median) and the gestational age is less than 34 or 35 weeks, fetal blood sampling should be considered. MCA Doppler PSV has replaced amniocentesis as the preferred test for the detection of anemia.

Solution to Question 8:

The causative agent for granuloma inguinale (donovanosis) is *Calymmatobacterium granulomatis*. It is now called *Klebsiella granulomatis* and is a gram-negative bacillus causing chronic ulcerative infection of the vulva.

It is highly contagious and transmitted through repeated sexual or non-sexual contact. The incubation period is 1-12 weeks.

Clinical features:

It begins as a painless nodule which later ulcerates to form multiple beefy red painless ulcers that tend to coalesce; the vulva is progressively destroyed, and minimal adenopathy may occur.

Diagnosis:

Microscopic examination of smears from the lesion/ biopsy specimens reveals pathognomonic intracytoplasmic Donovan bodies and clusters of bacteria with a bipolar (safety pin) appearance. Blue-black stained organisms are seen in the cytoplasm of mononuclear cells.

Management:

Azithromycin 1 g orally once per week or 500 mg daily for at least 3 weeks and until all lesions have completely healed.

The given image below shows the donovanosis in a female patient with a beefy red ulcer:

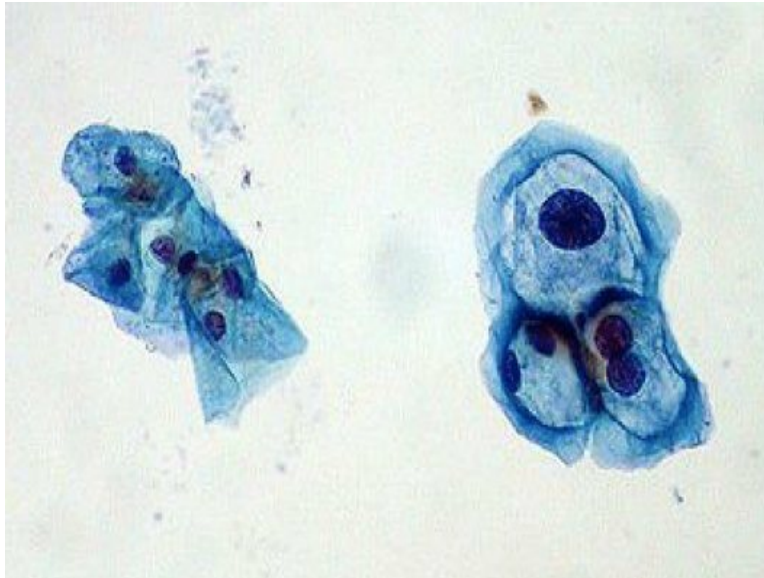


Solution to Question 9:

The causative agent of vulval warts is the human papillomavirus (HPV).

Genital warts/venereal warts/condyloma acuminata are verrucous growths that affect the skin of the labia majora, perineum, perianal region and vulva. The lesions may be discrete or coalesce to form cauliflower-like growth. The disease is sexually transmitted. The growth of the warts is favoured by vaginal discharge, the use of oral contraceptives, and pregnancy.

HPV causes structural changes in the epithelial cells of the skin called koilocytes, and their presence is characteristic of HPV infection. The koilocytes have large hyperchromatic nuclei and a perinuclear halo, as shown below.



The diagnosis is made by colposcopic studies. The lesions on the vulva are stained by toluidine blue. The abnormal areas of the skin retain the blue dye. These areas are then biopsied for histopathological examination.

The given image below shows condyloma acuminata:



Solution to Question 10:

In the given clinical case, the patient should be advised to continue the pregnancy.

The radiation dose to the fetus during the declared term of pregnancy should be less than 1 mSv. In chest X-ray PA view, the absorbed dose of radiation by the fetus is less than 0.1 mSv. Hence, pregnancy can be continued.

An X-ray has a very limited role in obstetric imaging. There are very few indications where an X-ray will be useful:

- Fetal: congenital malformation of the fetus and neonates, skeletal malformations, birth injuries like fractures or dislocations.
- Maternal: Patients having cardiopulmonary disease may require an X-ray chest during pregnancy and that should be done beyond 12 weeks.

Radiation can cause fetal morbidity (FGR, genetic mutations, neurologic abnormalities, mental retardation, childhood leukemia) and mortality. The risks are high with radiation after the first 2 weeks postconception and within the first trimester (period of organogenesis).

Radiation during pregnancy can damage the early phase of the embryo but the risk can be minimized by using the '10-day rule'. This rule states that no woman should be exposed to an X-ray for a nonurgent indication outside 10 days from her last period during the reproductive period unless the pregnancy is excluded.

The given table below shows the absorbed radiation by the fetus in different diagnostic radiation procedures:

1 cGy = 10 mSv

Procedure	Dose (c Gy)
Abdominal X-ray	0.1-0.3
Pelvic X-ray	0.5-1.1
Chest X-ray	<0.001
Abdominal CT	0.4-0.8
Chest CT	0.002-0.02
Abdominopelvic CT	2.5-3.5
Ventilation scan	0.007-0.05
PET scan	1-1.5

Solution to Question 11:

External anal sphincter is the structure injured in an episiotomy extended posteriorly beyond the perineal body.

Perineal tear have been divided into four degrees:

- First degree- laceration is restricted to the skin of the fourchette
- Second degree- muscles of the perineal body are torn through
- Third degree- the tear extends partially backward through the external sphincter of the anus
- Fourth degree- the external anal sphincter is torn and anal mucosa is also involved

A central tear of the perineum is when the head penetrates first through the posterior vaginal wall, then through the perineal body and appears through the skin of the perineum. It usually occurs in patients with a contracted outlet.

Solution to Question 12:

A history of cyclical abdominal pain, primary amenorrhea and a bulging vaginal membrane suggest cryptomenorrhoea caused by an imperforate hymen.

Imperforate hymen is a urogenital anomaly which occurs due to failure of canalization of the inferior end of the vaginal plate. It usually remains unrecognized until puberty. It can lead to mucocolpos in early infancy.

Patients present with amenorrhea, cyclic abdominal pain, urinary symptoms, and a hypogastric lump on abdominal examination. A bluish bulging membrane at the vaginal opening is noted on genital examination and the Valsalva maneuver.

Ultrasound of the pelvis reveals normal internal genitalia and hematocolpos. Laparoscopy should be performed along with hymenectomy to exclude endometriosis.

Most common causes of cryptomenorrhea:

- Intact/imperforate hymen - a bluish convex bulge is seen
- Transverse vaginal septum - a pinkish concave covering is seen

Ultrasound pelvis confirms the diagnosis. A simple cruciate incision of the hymen allows the free drainage of the collected menstrual blood and restores the reproductive capacity in most cases.

Solution to Question 13:

Among the given options, renal agenesis is the likely cause. It causes oligohydramnios in the early second trimester.

Fetal urine production starts between 8-11 weeks of gestation but becomes a major component of amniotic fluid only in the second trimester. Thus, oligohydramnios is not seen until after 18 weeks of gestation in fetuses with lethal renal abnormalities.

Oligohydramnios occurs when the amniotic fluid index (AFI) is ≤ 5 cm or the single deepest vertical pocket is ≤ 2 cm. AFI between 5-8 cm is known as borderline oligohydramnios.

Diagnosis of oligohydramnios:

- Uterine size is much smaller than the period of amenorrhea
- Less fetal movements
- The uterus is 'full of fetus' due to scanty liquor
- Malpresentation (breech) is common
- Evidence of intrauterine growth retardation of the fetus is seen
- Sonographic diagnosis is made when the largest liquor pool is less than 2 cm

Other options:

Fetal anemia, Bartter syndrome and Liddle syndrome usually lead to polyhydramnios and not oligohydramnios.

Solution to Question 14:

Transcervical hysteroscopic resection of the septum has the best obstetric outcome.

The procedure involves the resection of the septum under hysteroscopic guidance. Hence, it doesn't scar the uterus and avoids the necessity of cesarean sections for future pregnancies improving obstetric outcomes.

Metroplasty is the surgical resection of the intrauterine septum. This is done by approaching the uterus transabdominally via 3 methods:

- Jones metroplasty
- Strassmann metroplasty
- Tompkins metroplasty

All of these methods are not preferred nowadays as they cause uterine scarring.

Solution to Question 15:

Hormonal therapy is the next best step in the given scenario of menorrhagia that is unresponsive to non-hormonal therapy.

Menorrhagia/heavy menstrual bleeding is defined as heavy but regular menstrual bleeding, ≥ 80 mL of blood loss, or ≥ 8 days of bleeding per menstrual period. It can be acute (≤ 6 months) or chronic (≥ 6 months).

It is managed as follows:

- Medical - first line of treatment.
- NSAIDs

- Tranexamic acid
- OCPs/progestogens - if non-hormonal management fails
- Mirena IUCD
- GnRH agonists
- Blood transfusion, if required
- Surgical - if medical management fails, depending on the diagnosis:
 - Endometrial ablation - hysteroscopic, TCRE, RITEA, microwave, NovaSure, etc (option B)
 - Uterine artery embolization (option A)
 - Uterine tamponade
 - Hysterectomy - if all minimally invasive methods fail (option D)

Solution to Question 16:

Visualization of a mass with a single feeding vessel on a color Doppler is characteristic of an endometrial polyp. In contrast, submucous leiomyomas often receive blood flow from multiple vessels (option A).

Endometrial polyps are soft, fleshy growths within the uterus that consist of endometrial glands, fibrous stroma, and surface epithelium. Risk factors for endometrial polyps include increasing age, obesity, and tamoxifen use.

They are usually asymptomatic. When symptomatic, they present with heavy menstrual bleeding, prolonged bleeding, intermenstrual bleeding, or infertility.

The main diagnostic tools for endometrial polyps include transvaginal ultrasound (TVS) with applied color Doppler, saline infusion sonography, and hysteroscopy.

The risk factors for the malignant transformation include postmenopausal status, age older than 60, and abnormal bleeding. Other factors such as larger polyp size (≥ 1.5 cm), tamoxifen use, diabetes, and obesity may also contribute to a higher risk.

Hysteroscopic polypectomy may be considered most effective for symptomatic women or those with risk factors for malignant transformation.

Solution to Question 17:

Immediate cesarean section is done as the first twin is in breech presentation. Uncomplicated dichorionic diamniotic twin pregnancies are terminated at 38 weeks of gestation.

Timing of delivery depends on the type of twin pregnancy:

The mode of delivery depends on the 1st twin:

After delivery of 1st twin, if the 2nd twin is in breech, assisted breech delivery is done. If the 2nd twin is in a transverse lie, the internal podalic version is done under GA in the OT followed by

breech extraction.

Complications in attempting vaginal delivery in non-vertex presentation (i.e) transverse and breech:

- Interlocking of twins
- Fetal head trapping above the cervix
- Umbilical cord prolapse

According to ACOG criteria (2019), preeclampsia requires proteinuria of grade 2+ or more. In the case of pre-eclampsia, termination of pregnancy must be done at 37 weeks. According to the updated criteria, this patient does not have pre-eclampsia. Hence, a Caesarean section can be done at 38 weeks as for any uncomplicated DCDA twin pregnancy.

Type of twin pregnancy	Timing of delivery
Dichorionic diamniotic	38 weeks
Monochorionic diamniotic	34 - 37 weeks + 6 days
Monochorionic monoamniotic	32 - 34 weeks by Caesarean section

1st twin presentation	Mode of delivery
Vertex	Vaginal
Breech	Cesarean section
Transverse	Cesarean section

Solution to Question 18:

Pubarche or the development of axillary and pubic hair is not mediated by estrogen. It occurs due to testosterone in both sexes. Around 50% of testosterone comes from the ovaries and the rest from the adrenal gland.

Puberty in girls involves breast development (thelarche), an increase in adrenal androgen secretion (adrenarche), and the beginning of menstrual cycles (menarche).

Estrogen has the following actions during puberty:

- Breast development
- Development of the reproductive tract
- Starting of menstruation (option D)
- Increased fat in the hips
- Regulation of pubertal growth spurt

- Vaginal cornification (option C)
- Cervical mucus production (option B)
- Epiphyseal closure
- Maintenance of bone formation - acts against parathyroid hormone.

Solution to Question 19:

Multiple warty growths on the vulva (shown below) and around the anal canal are suggestive of condyloma acuminata (genital warts). HPV 6 and 11 are associated with the development of condyloma acuminatum. They are sexually transmitted.

The verrucous growths may appear discrete or coalesce to form large cauliflower-like growths. Infections are often transient in young women but can persist and progress to cancer in older women and immunocompromised individuals. Histopathology will show koilocytes, perinuclear halo, multinucleation, and acanthosis.

Treatment options include podophyllin, trichloroacetic acid, and surgical removal. The use of podophyllin cream is contraindicated in the first trimester of pregnancy. Larger lesions can be removed with a diathermy loop or laser ablation. Cesarean section is advised for women with genital warts during early labor to prevent the risk of transmitting laryngeal papillomatosis to the newborn.



Option B: Molluscum contagiosum presents as a crop of small domed vesicles with central umbilication (shown below). It is caused by the pox virus and it spreads by close sexual or nonsexual contact and by autoinoculation.



Option C: Chancroid appears as small papules that develop into painful ulcerating pustules. They are associated with unilateral inguinal lymphadenopathy. It is caused by *H. ducreyi*.

Option D: Herpes appears as multiple crops of vesicles and ulcers.

Solution to Question 20:

Normal saline is the ideal fluid media used for bipolar electrocautery as it can function regardless of the fluid electrolyte content.

Normal saline and lactated Ringer solutions are commonly used isotonic, electrolyte-containing fluids in operating rooms for diagnostic hysteroscopy procedures. These solutions can cause volume overload and, if rapidly absorbed in large volumes, may lead to pulmonary edema.

Isotonic, electrolyte-containing fluids are contraindicated for use with monopolar electrosurgical devices because they conduct electrical current, dissipate instrument energy and render the instrument ineffective. Low-viscosity, electrolyte-poor fluids such as 1.5-percent glycine, 3-percent sorbitol, and 5-percent mannitol are preferred as they are non-conductive and can be safely used with monopolar electrosurgical energy devices.

The primary risk of using hypotonic, hypoosmolar solutions is the intravascular absorption of these fluids, which can lead to acute hyponatremia and hypo osmolality.

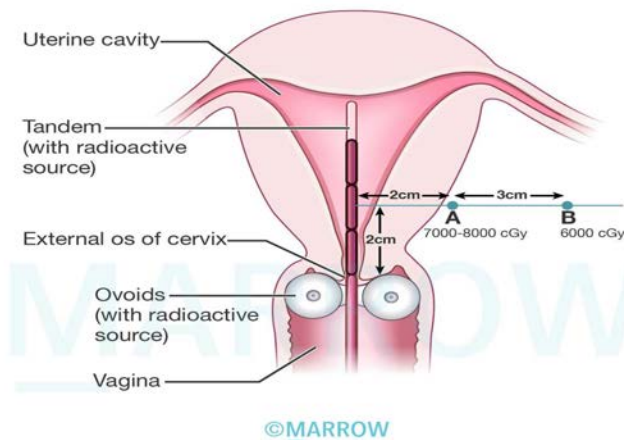
The criteria for terminating the procedure for electrolyte-deficient fluid is a deficit of 1 L and electrolyte-rich fluid is a deficit of 2.5 L.

Solution to Question 21:

Point A is 2 cm above the external os and 2 cm lateral to the internal uterine canal.

	Point A	Point B
Location	2cm above and 2cm lateral to the external os	2cm above and 5cm lateral to the external os
Structure present	Paracervical/Parametrial lymph node	Obturator lymph node
Dose of radiation	7000-8000 cGy	6000 cGy

Radiotherapy for carcinoma cervix



Solution to Question 22:

The image showing enlarged ovaries with multiple large cysts along with the history of treatment for infertility with human menopausal gonadotropin suggests the diagnosis of ovarian hyperstimulation syndrome (OHSS).

Human menopausal gonadotropin (HMG) is a medication that is composed of FSH and LH. It is used to stimulate the release of ova in women who do not ovulate spontaneously. It can lead to ovarian hyperstimulation syndrome.

Ovarian Hyperstimulation Syndrome (OHSS) is a condition associated with ovarian enlargement due to exogenous gonadotropin therapy. Symptoms include abdominal pain, distention, ascites, gastrointestinal problems, respiratory compromise, oliguria, and thromboembolism. It can occur during ovulation induction or in early pregnancies conceived through exogenous ovarian stimulation.

Predisposing factors for OHSS include multi-follicular ovaries (such as in PCOS), young age, high estradiol levels during ovulation induction, and pregnancy.

Vasoactive substances produced by the corpus luteum, such as vascular endothelial growth factor (VEGF) and angiotensin II, are believed to contribute to enhanced capillary permeability, resulting in the loss of fluid, protein, and electrolytes into the peritoneal cavity and hemoconcentration.

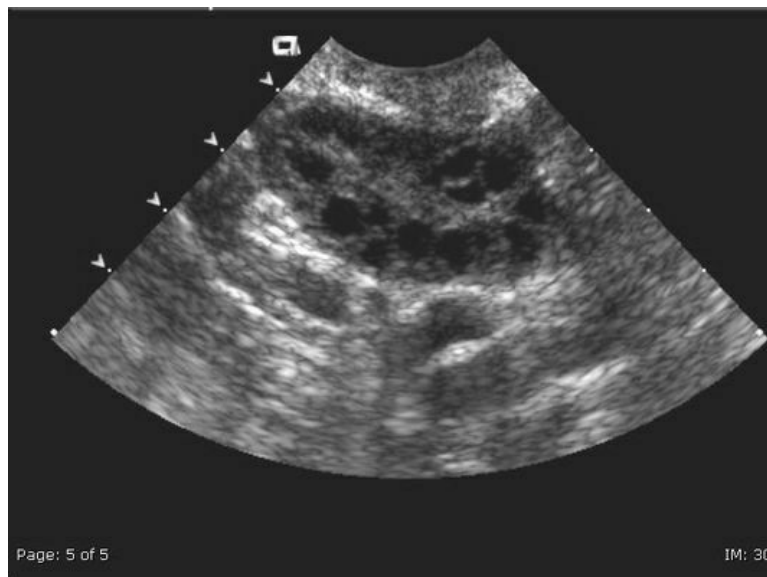
Sonographic examination reveals enlarged ovaries with numerous follicular cysts and ascites.

Maintaining fluid balance with isotonic fluids like normal saline is crucial. Strategies to prevent OHSS during exogenous ovulation include reducing follicular stimulation (low FSH dose), coasting (withholding FSH administration one or more days prior to the hCG trigger), prophylactic volume expanders, and substituting hCG for FSH towards the end of ovarian stimulation.

Other options:

Option B: Theca lutein cysts are a rare form of follicular cysts characterized by luteinization and hypertrophy of the theca interna layer. Both ovaries are enlarged and cystic. This condition is commonly associated with gestational trophoblastic disease, multifetal gestation, and placentomegaly.

Option C: Polycystic ovarian syndrome shows 12 or more small follicles each 2-9 mm in size placed peripherally (necklace pattern) (shown below). The patient presents with menstrual irregularities, hirsutism, acne, etc.



Solution to Question 23:

The woman has had two pregnancies in total - the previous twin pregnancy and the current pregnancy, hence the gravida score is 2. The first pregnancy resulted in delivery at term (crossed viable period), hence her para score is 1 (parity doesn't include current pregnancy). Hence, her obstetric score would be G2P1 (Gravida - 2, Para - 1).

Multifetal pregnancies count as one for both gravida and para. A pregnant woman who delivers twins for the first time is expressed as primipara.

The obstetric score is used to summarise the past obstetric history as follows:

- G - Gravidity, denoting the total number of pregnancies, including the present one
- P - Parity, denoting the total number of pregnancies crossing the period of viability. This includes both live births and stillbirths. The current pregnancy is not included.

- A - Abortion is the termination of pregnancy, either spontaneous or induced, before fetal viability.
- L - Living children.

The GTPAL system includes the following:

- Gravida
- Term (>37 weeks)
- Preterm (20w - 36w 6d)
- Abortions (<20 weeks)
- Number of living children

Note: The period of viability is when the fetus is capable of independent existence. The WHO defines it to be 24 weeks while the Indian definition is 28 weeks.

Solution to Question 24:

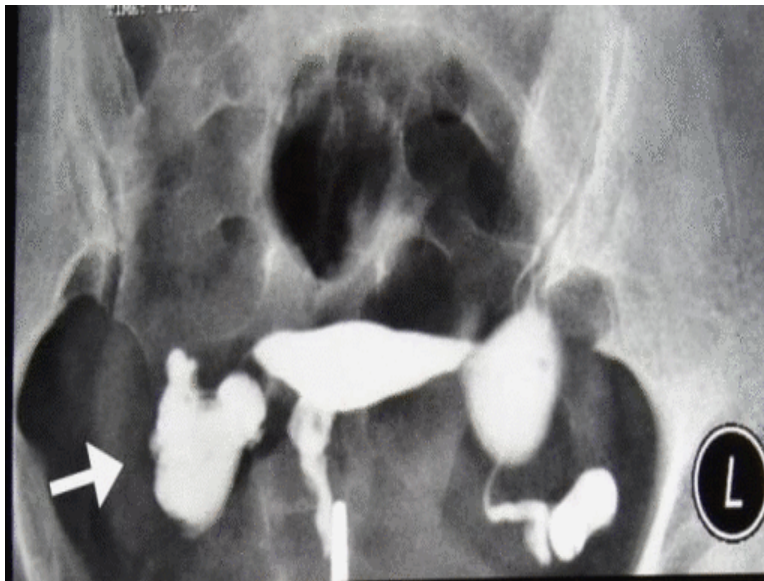
The image showing bilateral ballooned-out tubes filled with contrast is suggestive of bilateral hydrosalpinx.

Hydrosalpinx is a chronic cystic swelling of the fallopian tube caused by distal tubal obstruction, usually bilateral. It is associated with conditions like PID and endometriosis. Grossly, the fimbriae and tubal ostia are replaced by a smooth, dilated end. The ballooned elongated tubes are filled with clear fluid.

Ultrasound reveals a cystic, elongated structure with thin walls, incomplete septa, and anechoic contents. Hysterosalpingography shows dilated, clubbed fallopian tubes filled with contrast. The interstitial end of the tube is open whereas the fimbrial end is indrawn and closed. This permits the dye to enter but not exit the tube during hysterosalpingography.

Laparoscopic treatment options for those not seeking to preserve fertility include adhesion lysis and salpingectomy. For infertile women, laparoscopic neo-salpingostomy is done for mild tubal disease. In vitro fertilization (IVF) may be a better option for severe tubal disease.

The image given below shows bilateral hydrosalpinx:



Solution to Question 25:

The image showing a poorly developed lower limb suggests the diagnosis of caudal regression sequence/sacral agenesis seen in pregestational diabetes. It is the most specific anomaly seen in diabetes.

The caudal regression sequence is a rare malformation, frequently associated with maternal pre-gestational diabetes. It consists of a spectrum of structural defects of the caudal region, including incomplete development of the sacrum and the lumbar vertebrae.

Elevated maternal HbA1c levels correlate with fetal malformations. It occurs approximately 200 times more frequently in infants of diabetic mothers than in other infants.

Overt diabetes (pregestational diabetes) is associated with higher malformation rates than the general population. Neural tube defects (encephalocele, meningocele, anencephaly), transposition of great vessels, ventricular septal defect, atrial septal defect, hypoplastic left heart, aortic stenosis and coarctation of the aorta are commonly seen.

Small left colon syndrome is a rare anomaly that develops in the second and third trimesters.

Option B: Malformations are not associated with gestational diabetes mellitus (GDM). Malformations occur due to the disruption of organogenesis, usually occurring in the first trimester. The abnormal glycemic status in GDM manifests only in the second trimester.

Option C: Valproate leads to neural tube defects and not sacral agenesis.

Option D: ACE inhibitors can cause oligohydramnios.

Solution to Question 26:

The intensity of radiation is inversely proportional to the squared distance. This is known as the inverse square law in radiotherapy.

Chemoradiation is the mainstay of treatment of stage III and IV endometrial cancers. They are treated with brachytherapy followed by external radiation. Brachytherapy is a form of radiation therapy in which the radiation source is placed close to the tumor.

Other options:

Option B: Larger blood vessels are more radioresistant than microvasculature.

Option C: Permanent non-dividing cells are more radioresistant than rapidly proliferating cells.

Option D: Small intestinal epithelial cells (intestinal crypts), which divide rapidly with no differentiation are the most radiosensitive.

Solution to Question 27:

The given instrument is a fallope ring applicator. It is used in tubal ligation.

Falope ring sterilization is accomplished by applying silicone rubber bands to the fallopian tubes. The major advantage is the elimination of thermocoagulation. The occlusion is only in a short segment and can allow for tubal reconstruction.

Solution to Question 28:

Danazol is not used as a post-coitus contraceptive.

Danazol, a synthetic derivative of ethinyl testosterone, acts by inhibiting pituitary gonadotropins. It possesses mild anabolic, anti-estrogenic, and anti-progestational properties. It is used in abnormal uterine bleeding, endometriosis, and cyclic mastalgia.

Emergency or post-coital contraceptives are used to prevent pregnancy after unprotected intercourse. They are interceptives. Their main action is to make the endometrium unsuitable for implantation. They can also delay or inhibit ovulation, and prevent fertilization of the egg by sperm.

Methods of emergency contraception:

Drug	Dose
Levonorgestrel	0.75mg stat and after 12 hours or a single dose of 1.5mg stat. Taken within 72 hours can extend upto 120 hours
Ethinyl estradiol 50 µg + Norgestrel 0.25 mg (Yuzpe method)	2 tab stat and 2 tabs after 12 hours
Conjugated estrogen	15 mg BD x 5 days
Ethinyl Estradiol	2.5 mg BD X 5 days

Drug	Dose
Copper IUDs	Insertion within 5 days after accidental unprotected exposure. Not suitable for rape victims
Centchroman	2 tablets (60 mg) to be taken twice at an interval of 12 hours within 24 hours of intercourse.
Ulipristal	30 mg up to 5 days
RU486 (Mifepristone)	Single dose of 25 to 50mg

Obstetrics and Gynecology INICET 2021

Question 1:

A pregnant woman at 36 weeks gestation is found to have high blood pressure recording and a urine protein of 3+. Concomitant presence of which of the following symptoms would make you suspect imminent eclampsia?

- a) 2 and 4 only
- b) 1 and 2 only
- c) 1, 2, 3
- d) 1, 2, 3, 4

Question 2:

What is the probable diagnosis from the given clinical image?



- a) Cervical carcinoma
- b) Uterine inversion
- c) Uterocervical prolapse
- d) Fibroid polyp

Question 3:

Which of the following is not used to treat PCOD?

- a) Clomiphene citrate
- b) Metformin
- c) Tamoxifen
- d) OCP

Question 4:

A 26-year-old woman who has had 2 children through normal delivery presents with a dull aching pain over the abdomen, with irregular menses. Per vaginal examination revealed a mass felt through the posterior fornix. A USG showed a clear ovarian cyst of size 5 x 5 cm. What is the next best step?

- a) Measure CA 125 levels
- b) Follow up after 6 weeks
- c) Laparoscopy
- d) CT pelvis

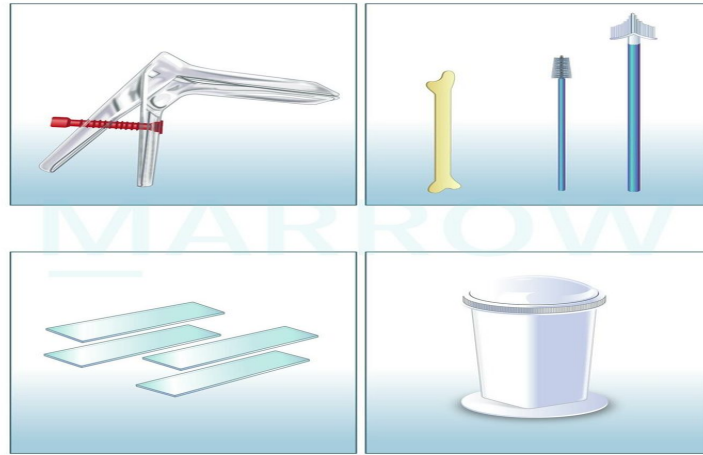
Question 5:

Which of the following is not included in the quadruple screening test?

- a) Inhibin B
- b) Unconjugated Estriol (E₃)
- c) AFP
- d) hCG

Question 6:

The following set of instruments are used for which procedure?



©MARROW

- a) Pap smear
- b) Hysteroscopy
- c) Biopsy
- d) Dilatation and curettage

Question 7:

What is the fixative used for a pap smear?

- a) Formalin
- b) Air dried
- c) Normal saline
- d) 95% ethanol

Question 8:

A doctor tells a 31-year-old woman, who is 20 weeks pregnant, that her baby has a developmental defect. She has come to you for a second opinion. The following USG finding is seen. Identify the defect.



- a) Anencephaly
- b) Omphalocele
- c) Cystic hygroma
- d) Encephalocele

Question 9:

Arrange the following in sequential order with regards to the steps of collection of samples for pap smear testing:

- a) 1, 2, 4, 3
- b) 2, 1, 3, 4
- c) 3, 1, 2, 4
- d) 1, 2, 3, 4

Question 10:

Which of the following is a part of AMTSL?

- a) Uterotonics after delivery of placenta
- b) Controlled cord traction
- c) Uterine massage
- d) Early cord clamping

Question 11:

Which of the following is not a cause of secondary amenorrhea?

- a) Turner's mosaic
- b) Kallman syndrome
- c) Sheehan's syndrome
- d) Asherman's syndrome

Question 12:

What should be the minimum percentage for normal sperm morphology in a seminal fluid analysis?

- a) 10%
- b) 4%
- c) 6%
- d) 8%

Question 13:

Which of the following best measures ovarian reserve?

- a) Luteal phase LH
- b) Post luteal phase LH
- c) FSH in the follicular phase
- d) Anti-Mullerian hormone (AMH)

Question 14:

An 18-year-old girl presented with amenorrhoea. On examination, she had normal genitalia & normal breast development. Her tests reveal a 5 alpha-reductase deficiency. What could be the probable diagnosis?

- a) Concealed sex
- b) True hermaphroditism
- c) Female pseudohermaphroditism
- d) Male pseudohermaphroditism

Question 15:

Match the following syndromes with their respective karyotypes.

- a) 1-D, 2-C, 3-B, 4-A
- b) 1-B, 2-C, 3-D, 4-A
- c) 1-A, 2-B, 3-C, 4-D
- d) 1-D, 2-A, 3-C, 4-B

Question 16:

Which of the following complication is specific to Monochorionic twins?

- a) Selective fetal growth restriction
- b) Polyhydramnios
- c) Twin to twin transfusion syndrome
- d) Neural tube defects

Question 17:

Match the following disorders causing secondary amenorrhoea with their respective lab findings.

- a) 1-C, 2-B, 3-D, 4-A
- b) 1-D, 2-B, 3-C, 4-A
- c) 1-C, 2-D, 3-B, 4-A
- d) 1-A, 2-B, 3-D, 4-C

Question 18:

Which of the following instruments are used in the caesarean section?

- a) 1, 2, 5, 6
- b) 1, 2, 3, 5
- c) 1, 2, 4, 6
- d) 1, 2, 3, 6

Question 19:

What is the most likely diagnosis?



- a) Vulval hematoma
- b) Condyloma accuminata
- c) Vulval cancer
- d) Bartholin cyst

Question 20:

What is the normal uterocervical length?

- a) 3 cm
- b) 5 cm
- c) 7 cm
- d) 10 cm

Question 21:

Which of the following investigations is used to diagnose unicornuate uterus?

- a) 2, 4
- b) 1, 2

- c) 1, 3
- d) 2, 3

Question 22:

Which of the following symptoms are seen in endometriosis?

- a) 3, 4
- b) 2, 3
- c) 1, 2
- d) 1, 2, 4

Question 23:

A female patient having frothy vaginal discharge was found to have a strawberry cervix. Which of the following is the drug of choice?

- a) Metronidazole
- b) Fluconazole
- c) Tetracycline
- d) Cefixime

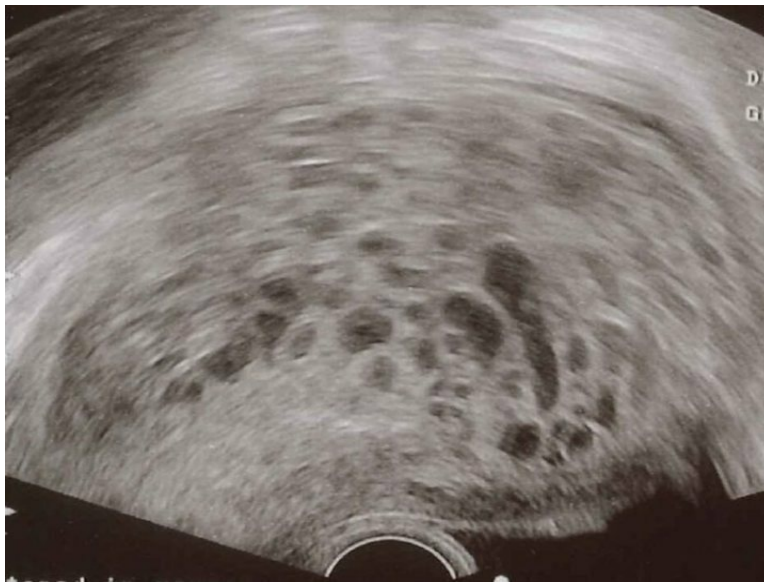
Question 24:

According to IADPSG, what is the cut-off value of fasting blood glucose for the diagnosis of gestational diabetes mellitus?

- a) ≥ 92 mg/dL
- b) ≥ 95 mg/dL
- c) ≥ 106 mg/dL
- d) ≥ 90 mg/dL

Question 25:

A primigravida in her 10th week of gestation presents with spotting. On examination, the uterus corresponds to 12 weeks. Transvaginal ultrasound was done and it is given below. What is your diagnosis?



- a) Blighted ovum
- b) Missed abortion
- c) Ectopic pregnancy
- d) Hydatidiform mole

Question 26:

A 62-year-old patient with a history of diabetes and hypertension came with post-menopausal bleeding. Which of the following you will do to evaluate the cause?

- a) Pap smear
- b) Visual inspection with iodine
- c) Colposcopy
- d) Endometrial sampling

Question 27:

Which of the following is recommended by the WHO for the management of the second stage of labor?

- a) Mandatory episiotomy
- b) Manual perineal support and warm compresses
- c) Maintain head in deflexion and deliver
- d) Deliver in lithotomy position only

Question 28:

A primigravida is in labour. Her per-vaginal examination revealed a posterior cervix with 5 cm cervical length, 1 cm dilatation, soft consistency, and head at -1 station. Calculate the modified Bishop score.

- a) 8
- b) 5
- c) 3
- d) 0

Question 29:

A couple presents to the infertility clinic with a normal hysterosalpingogram and a semen analysis report with azoospermia. Which of the following is the next best test to differentiate between an obstructive and a testicular cause of azoospermia?

- a) Serum FSH levels
- b) Serum testosterone levels
- c) Testicular biopsy
- d) Serum LH level

Question 30:

What is the dose of dexamethasone given for fetal lung maturity?

- a) 2 doses of 6mg given 12 hours apart
- b) 2 doses of 12mg given 24 hours apart
- c) 4 doses of 6 mg given 12 hours apart
- d) 4 doses of 12mg given 12 hours apart

Question 31:

What is the dose of carbetocin used in the prevention of postpartum hemorrhage?

- a) 50 mcg
- b) 150 mcg

- c) 200 mcg
- d) 100 mcg

Question 32:

Milk production in pregnancy is inhibited by :

- a) Low luteinizing hormone
- b) High estrogen
- c) Human somatomammotropin
- d) Low thyroid-stimulating hormone

Question 33:

Which of the following emergency contraceptive is used in India?

- a) Misoprostol 400 mcg
- b) Ethinyl estradiol 600 mcg
- c) Ulipristal 30 mg
- d) Levonorgestrel 1.5 mg

Question 34:

Which of the following is not a long-acting reversible contraceptive method?

- a) Copper T
- b) Implanon
- c) Laparoscopic tubal ligation
- d) Combined oral contraceptives

Question 35:

Which class of ovarian tumor does clear cell carcinoma of the ovary belong to?

- a) Epithelial cell tumor
- b) Germ cell tumor
- c) Sex cord stromal tumor

- d) Granulosa cell tumor

Question 36:

Asherman syndrome is most commonly associated with

- a) Oligomenorrhea
- b) Hypomenorrhea
- c) Dysmenorrhea
- d) Menorrhagia

Question 37:

A teenage girl presented with irregular cycles and increased facial hair. Her ovaries showed increased volume. Which of the following are used in the first line treatment?

- a) 1,2,3
- b) 2,3,4
- c) 1,3,4
- d) 1,2,4

Question 38:

Identify the instrument given below.



- a) Cusco's speculum
- b) Sims' speculum
- c) Proctoscope
- d) Doyen's retractor

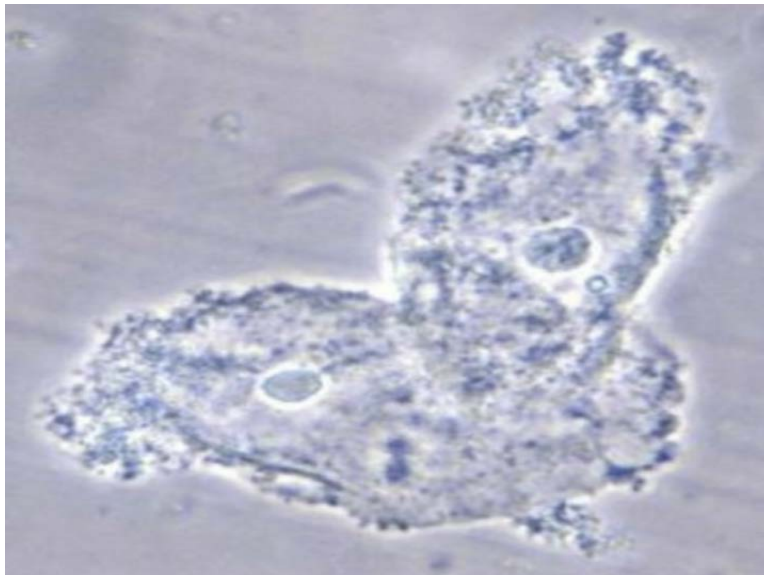
Question 39:

A 17-year-old girl was brought by her mother to the outpatient department with primary amenorrhea. She had normal breast development and pubic hair. On examination, the vagina was blind-ending and the uterus was not palpable. What is the most probable diagnosis?

- a) Mullerian agenesis
- b) Gonadal dysgenesis
- c) Turner syndrome
- d) Testicular feminization syndrome

Question 40:

A 24-year-old female presents with complaints of foul-smelling vaginal discharge. Wet smear of the discharge is shown below. Drug of choice to treat this condition is _____



- a) Clindamycin
- b) Metronidazole
- c) Cotrimoxazole

d) Azithromycin

Question 41:

Match the following :

- a) 1-A, 2-C, 3-B, 4-D
- b) 1-C, 2-A, 3-B, 4-D
- c) 1-C, 2-A, 3-D, 4-B
- d) 1-A, 2-C, 3-D, 4-B

Question 42:

Which of the following modalities are used for the diagnosis of placenta accreta?

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 4 only

Answer Key

Question No.	Correct Option
1	d
2	c
3	c
4	b
5	a
6	a
7	d
8	a
9	d
10	b
11	b
12	b

13	d
14	d
15	a
16	c
17	c
18	c
19	c
20	c
21	d
22	d
23	a
24	a
25	d
26	d
27	b
28	b
29	a
30	c
31	d
32	b
33	d
34	c
35	a
36	b
37	b
38	a
39	a
40	b
41	b
42	b

Detailed Explanations

Solution to Question 1:

The clinical case of a pregnant women at 36 weeks of gestation with hypertension and proteinuria points to a diagnosis of pre-eclampsia. Certain symptoms if present might suggest impending

eclampsia. These include 1, 2, 3, 4.

Pre-eclampsia is defined as hypertension with blood pressure $\geq 140/90$ mm Hg along with proteinuria that develops after 20 weeks of gestation in a previously normotensive and non-proteinuric woman. Preeclampsia when complicated with generalized tonicclonic convulsions and/or coma is called eclampsia.

Ominous symptoms of impending eclampsia:

- Headaches - usually over the occipital or frontal region
- Visual disturbances - These could include scotomas. These are due to spasm of retinal vessels following retinal infarction, occipital lobe damage secondary to vasogenic edema, or retinal detachment.
- Epigastric or right upper quadrant pain. This is due to hepatocellular necrosis, ischemia, and edema that stretches the Glisson capsule. This is usually seen along with elevated serum hepatic transaminase levels.
- Sudden and generalized or persistent edema may indicate imminent eclampsia. Visible edema over the ankles despite rest, especially on rising from the bed in the morning hours is pathological.
- Disturbed sleep
- Diminished urinary output - less than 400 mL in 24 hours is considered a poor sign ominous

Other symptoms include:

- Renal or cardiac involvement
- Obvious fetal-growth restriction
- Thrombocytopenia - usually due to microangiopathic hemolysis.

It is essential that the patient should be warned and educated against these symptoms. The presence of these symptoms may warn the obstetrician and possibly avert the progression into eclampsia.

Solution to Question 2:

In the provided image, it is evident that both anterior and posterior vaginal walls protrude completely outside the vaginal orifice, and the entire uterus is positioned below this level, indicating a case of procidentia/ grade IV utero-cervical prolapse.

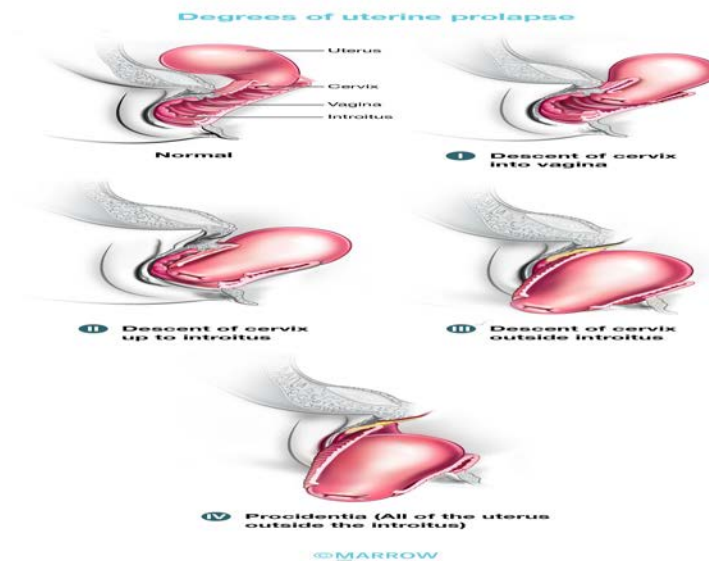
Utero-cervical prolapse is commonly seen in the elderly age group due to weak pelvic muscles.

Differential diagnoses of genital prolapse include:

- Vulval cyst and Gartner cyst.
- Cyst of the anterior vaginal wall - tense with well-defined margins and cannot be reduced with pressure.
- Urethral diverticula - small and located low in the anterior vaginal wall. Urethroscopy aids in diagnosis.

- Congenital elongation of the cervix - vaginal portion of the cervix is elongated without accompanying vaginal wall prolapse. The fornices are deep.
- Cervical fibroid polyps - cervix is located high up and a rim of the cervix is felt above the pedicle of the cervical polyp.
- Chronic inversion of the uterus- identified by the higher position of the cervix and the inability to define the uterus on palpation. Diagnosis is confirmed by the use of a uterine sound and further examination with ultrasound and laparoscopy, which reveal a depression at the fundus and an absence of the uterine fundus in the pelvis.
- Rectal prolapse.

The image below depicts the various degrees of genital organ prolapse:



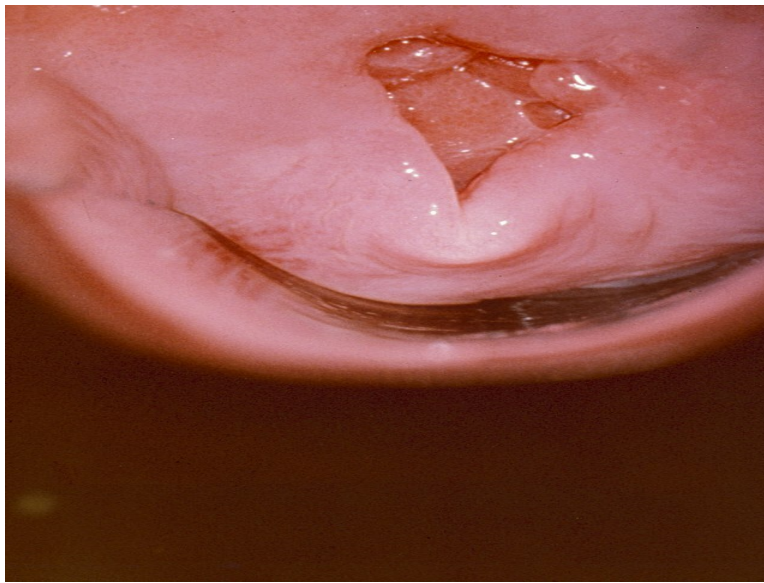
Other options:

Option A: Cervical carcinoma

It commonly presents with post-coital bleeding or postmenopausal bleeding. When they present as CIN lesions, they can be identified by colposcopy but mass lesions will present with fungating, cauliflower, or punctate hemorrhagic lesions.

The provided image shows a normal transition zone, in the given setting without VIA or colposcopy it is difficult to diagnose the lesion as CA cervix.

The image below shows characteristic hemorrhagic and cauliflower lesions seen in the CA cervix:



Option B: Uterine inversion:

Acute uterine inversion is a rare and life-threatening post-partum complication. Uterine inversion is associated with primiparity, the use of oxytocin, macrosomia, and fundal insertion of the placenta. Premature traction on the umbilical cord and fundal pressure before placental separation are the usual direct causes. During per abdomen examination fundus of the uterus is felt to be absent. Care for acute uterine inversion involves pain management, resuscitation, and replacement of the inverted uterus before edema sets in by the O'Sullivan method.

The image below image shows uterine inversion which can be identified by a completely inverted uterus.



Option D: Fibroid polyp

They are leiomyomas that present as polyps. A fibroid polyp is generally firm, permanent, paler than a mucus polyp, and is of various sizes. Sonosalpingography is also a very good technique for detecting submucous fibroid polyp.

Note:

Cervical fibroid polyps can be easily identified as the cervix is high up in its normal anatomical position unlike in the given image. The image attached below shows degenerated fibroid polyps.



Solution to Question 3:

Tamoxifen is the best answer as it is not commonly used in the treatment of polycystic ovarian disease (PCOD). Tamoxifen is a selective estrogen receptor modulator (SERM).

Treatment of PCOD:

- Weight reduction alone is beneficial in mild hirsutism.
- Lifestyle changes, such as smoking cessation, are recommended.
- Fourth-generation oral contraceptive pills (OCP) containing ethinyl estradiol and drospirenone are used for treating menstrual abnormalities, reducing acne, and hirsutism.
- OCP with cyproterone acetate is prescribed for PCOS patients with hirsutism.
- Patients with PCOD are prone to developing insulin resistance, which can be controlled with metformin. Acarbose can be used as an alternative if metformin is contraindicated.
- Drugs used in the management of infertility in women with PCOS include letrozole and clomiphene citrate. FOGSI, in 2024, updated letrozole as the first-line drug for induction of ovulation in patients with PCOS. However, the FDA approval for the same is still pending.

Surgical options include laparoscopic ovarian drilling or puncture of not more than four cysts in each ovary. This is done either by laser or by unipolar electrocautery. It restores endocrine levels and improves fertility. Surgery is indicated only if:

- Medical therapy fails
- Hyperstimulation occurs
- Infertile women

- Previous pregnancy losses

Note: In cases where clomiphene and dexamethasone fail to induce ovulation, tamoxifen can be considered as an alternative treatment for infertility associated with PCOD. However, it is not recommended as the initial choice. It's important to note that tamoxifen carries a significant risk of endometrial hyperplasia and can also result in unexpected pregnancy with potential teratogenic effects when used for induction.

Solution to Question 4:

A clear ovarian cyst with size less than 7 cm in a woman of reproductive age group is most likely to be a functional ovarian cyst. The next best step in the management of this case is to follow up after 6–12 weeks and perform a repeat USG.

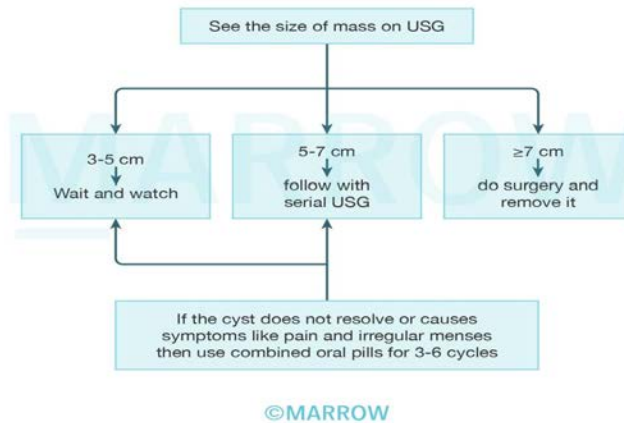
A simple unilocular cyst under 7 cm is often a functional cyst and should be observed. Most functional cysts resolve spontaneously within 4-6 months. A follow-up ultrasound will detect any persistent cyst, which may necessitate laparoscopic evaluation. To expedite resolution or to treat symptoms like pain and irregular menses, oral combined pills may be prescribed for 3-4 months in women of reproductive age.

Features of a functional cyst:

- Related to temporary hormonal variation
- Benign and rarely become complicated
- Usually 6–8 cm in diameter
- Usually asymptomatic
- Unilocular
- Contains clear fluid, unless hemorrhage occurs
- Lining epithelium corresponds to the functional epithelium of the unit from which they arise

Management of functional cysts in women of child-bearing age group is summarized in the flowchart attached below.

Management of benign ovarian mass in women of reproductive age group



Other options:

Option A: Tumour markers such as CA-125 and CEA are particularly useful in postmenopausal women suspected of having a malignant epithelial cell tumor.

Option C: If the cyst does not resolve spontaneously it can be removed laparoscopically or by laparotomy.

Option D: Imaging modalities such as CT scan and MRI help to detect ovarian neoplasms and assist in staging of ovarian cancers.

Solution to Question 5:

Inhibin B is not a component assayed under the quad-screening test. Dimeric inhibin-A is evaluated in this screening test.

Quadruple testing is done in the second trimester between 15-21 weeks of gestation. It comprises:

- Maternal serum AFP
- Human chorionic gonadotropin
- Unconjugated estriol (E3)
- Dimeric inhibin A

The following table summarizes the findings of the quad-screening test in various conditions.

Various antenatal screening tests for aneuploidy are as follows:

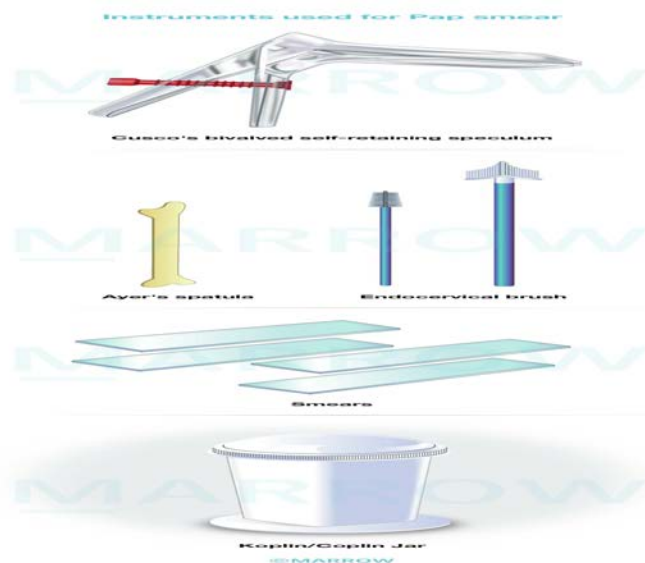
Parameter	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect
MSAFP	Low	Low	High
β hCG	High	Low	Normal

Parameter	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect
Unconjugated estriol	Low	Low	Normal
Inhibin A	High	Normal/Low	Normal

Dual test (1st trimester)	Triple test (2nd trimester)	Quadruple test (2nd trimester)
1. Beta HCG 2. PAPP-A (Pregnancy-associated plasma protein A) This can be combined with nuchal translucency by USSG	1. Beta HCG 2. Unconjugated estriol 3. MSAFP	1. Beta HCG 2. Unconjugated estriol 3. MSAFP 4. Serum inhibin

Solution to Question 6:

The given set of instruments are commonly used for the collection of pap smears. The given image includes Cusco's bivalved self-retaining speculum, Koplin's jar, smear slides, Ayer's spatula, and endocervical brush.



Pap/Papanicolaou test is a screening test only. The presence of abnormal cells/ a positive test requires further evaluation by colposcopy, cervical biopsy, or fractional curettage.

Screening guidelines:

- Time for initiation: 21 years, regardless of the age of first sexual intercourse.
- Women aged 21-29 years: Every 3 years (earlier it was done annually)
- Women who are aged 30-65 years: Pap test + HPV testing every 5 years.

- Women should stop having cervical cancer screening after the age of 65 years. If suspicious PAP, screening till 75 years.
- HIV+ women: annual checkup until three consecutive negative tests followed by testing every 3 years for life.
- HPV DNA testing in combination with PAP improves the sensitivity of screening.

Prerequisites:

- Should be done prior to vaginal examination.
- Lubricants should not be used.
- The patient should not be menstruating during the pap test.
- The ideal time is around ovulation.
- The patient should not have had intercourse in the 24hrs prior to the test.

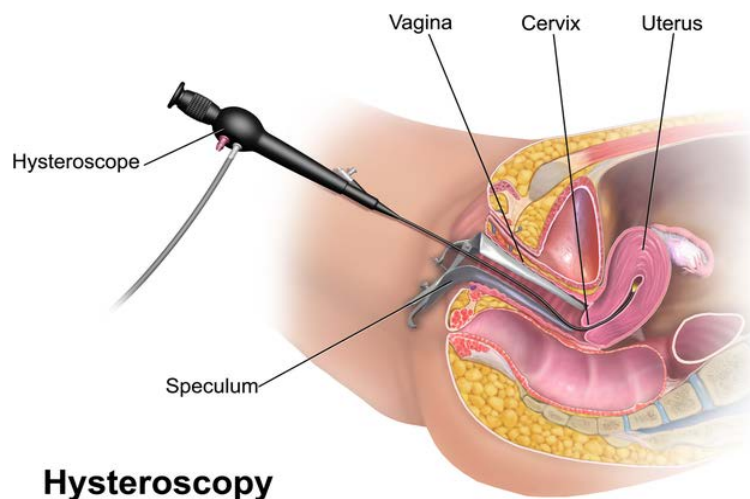
Procedure:

- The patient is put in a dorsal position & the labia are separated.
- Cusco's self-retaining speculum is introduced/posterior vaginal wall retractor is used
- The squamocolumnar junction is identified and scraped with Ayre's spatula.
- Endocervix is scraped with a cytobrush.
- Samples are spread separately on 2 different glass slides.
- The slide is immediately fixed using 95% ethyl alcohol.
- Stained with Pap stain or Short stain.

Other options:

Option B: Hysteroscopy is the process of inspection of the uterine cavity.

The image shows the procedure of hysteroscopy and the common instruments used for the same.



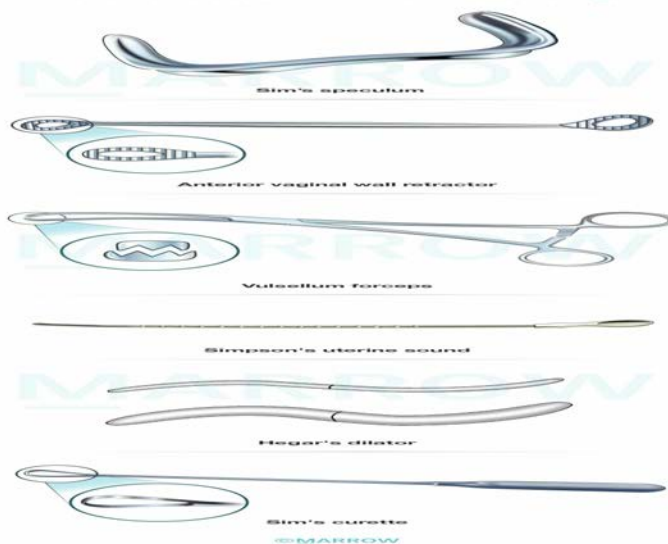
Option C: To take biopsy we commonly use Gellhorn's punch biopsy forceps as shown in the image below. It is used to take biopsies from the cervix.

Gellhorns punch biopsy forceps



Option D: Instruments used in dilatation and curettage are shown below.

Instruments used in Dilatation and Curettage



Solution to Question 7:

The most common fixative used in pap smear is 95% ethanol (ethyl alcohol).

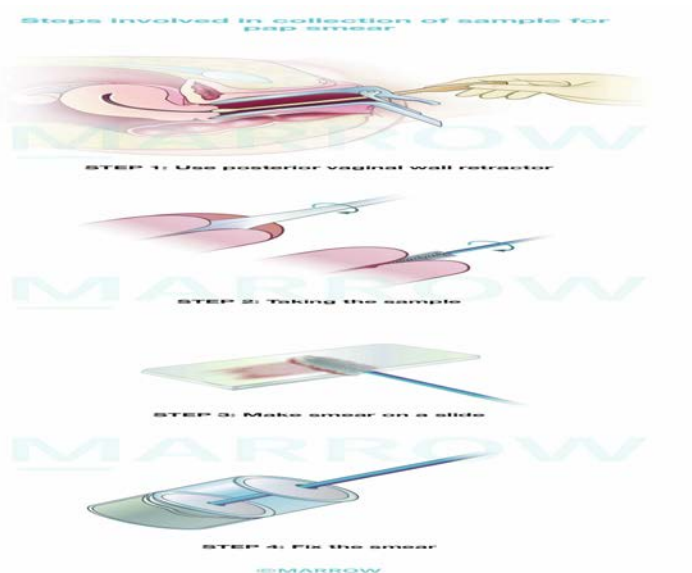
Pap smear is a screening test for cervical cancer. Cervical cancer screening should start at age 21. Women aged 21-29 should have Pap smear every 3 years, while women aged 30-65 should have both HPV and Pap co-test every 5 years.

Before the Pap test:

- The Pap test should not be deferred due to unexplained discharge or unscheduled bleeding as they may be signs of cervical or other genital tract cancers.
- Ideally scheduled to avoid menstruation. It is best performed around the time of ovulation.
- Patients should abstain from vaginal intercourse, douching, tampon use, and intravaginal medications for a minimum of 24 to 48 hours prior to the test.
- Avoid touching the cervix before performing the Pap test, as dysplastic epithelium may get removed.
- Ideally, confirmed cervicitis or vaginitis should be treated prior to the test

Procedure:

- The patient is put in a dorsal position and Cusco's self-retaining speculum is introduced.
- The squamocolumnar junction is scraped with Ayre's spatula completing at least one full rotation. It predominantly samples the ectocervix.
- Endocervix is scraped with a cytobrush, which is rotated only one-quarter to one-half turn after the ectocervix has been sampled.
- A broom simultaneously samples both endo- and ectocervix. Five rotations in the same direction are recommended, since reversing direction may cause loss of cellular material.
- Samples are spread separately on 2 different glass slides and fixed using 95% ethyl alcohol.
- Stained with Pap stain or Short stain.



Solution to Question 8:

The given ultrasound image shows an open defect in the calvaria and skin leading to the exposure of the cranial neural tube. This is consistent with the diagnosis of anencephaly.

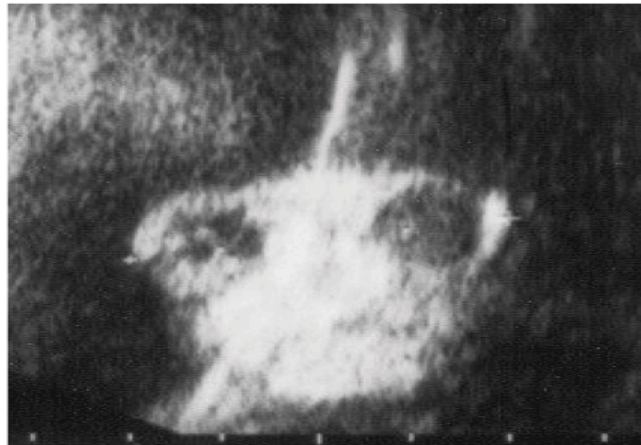
Anencephaly results from incomplete closure of the neural tube. It is one of the earliest detectable congenital defects that can be detected as early as 10 to 11 weeks of gestation. Screening can also

be done by serum estimation of maternal serum alpha-fetoprotein (MSAFP). The following are the radiological signs seen in anencephaly.

- Frog eye sign - Absence of cranial calvaria, prominent orbits thereby resembling the face of a frog.
- Shower cap sign - The absence of the cranium could lead to the protrusion of a disorganized mass of brain tissue. This resembles a shower cap giving rise to a characteristic triangular facial appearance.

Many neural-tube defects can be prevented with preconceptional supplementation of folic acid. A dose of 4 mg daily is recommended in subsequent pregnancies.

Anencephaly - Frog-eye sign

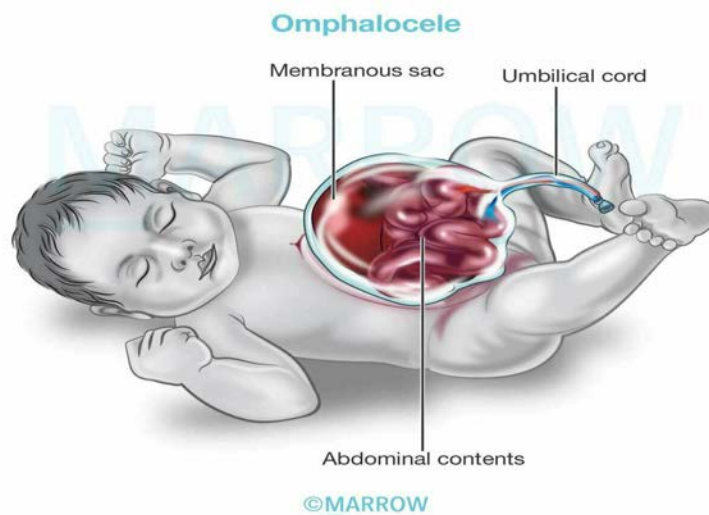
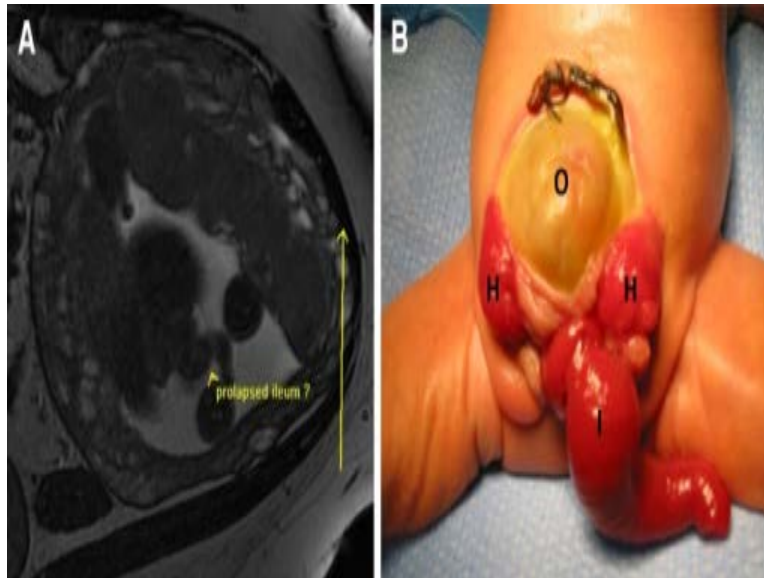


Other options:

Option B: Omphalocele is a midline anterior abdominal wall defect through which the solid abdominal viscera and/or bowel may herniate. The extruded abdominal contents are covered in a sac. These are due to the persistence of physiological herniation of the gut or as a part of

the OEIS complex (omphalocele, exstrophy, imperforate anus, and spinal abnormality) and cloacal exstrophy. The 'elephant's trunk' sign on ultrasound, due to the prolapsed terminal ileum is said to be pathognomonic for cloacal exstrophy of the OEIS complex.

The first image below shows a patient with cloacal exstrophy showing the elephant trunk sign representing the intussuscepted ileum. The second image shows omphalocele while the third image shows gastroschisis (a close differential diagnosis for omphalocele). The fourth image demonstrates an ultrasound image of the fetus suggestive of an omphalocele (asterisk)





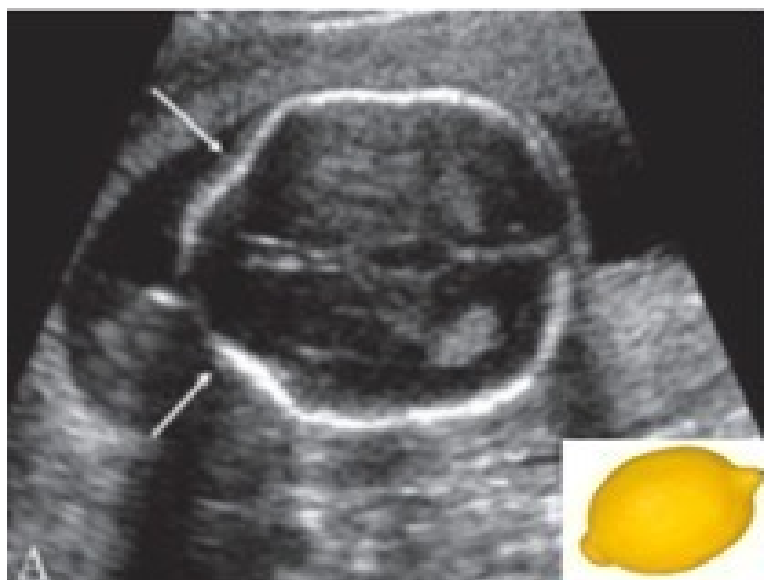
Option C: Cystic hygroma is a congenital multiloculated lymphatic lesion that is classically found in the left posterior triangle of the neck. It occurs as a result of sequestration or obstruction of developing lymph vessels. It must be carefully differentiated from nuchal translucency (NT) as both may appear similar. This defect can be picked up in a first-trimester ultrasound.

The image below shows a newborn with cystic Hygroma on the right side of the neck and cystic hygroma of the neck in the fetus.



Option D: Encephalocele is caused due to defects in the skull through which the meninges and/or neural tissue herniate. The squamous part of the occipital bone is the most frequently affected bone. In meningoencephalocele or occult spina bifida with Chiari malformations, the scalloping of the frontal bones is observed in the ultrasound as the lemon sign while anterior curvature of the cerebellum with effacement of the cisterna magna in the posterior fossa forms the banana sign.

The image shows a newborn with encephalocele. The second and third images below show the banana and the lemon sign respectively.



Solution to Question 9:

The correct sequence of the procedure involved in the collection of a sample for a pap smear is 1, 2, 3, 4.

The Pap/Papanicolaou smear is used for screening cervical neoplasia.

Procedure:

- The patient is put in a dorsal position & the labia are separated.
- Cusco's self-retaining speculum is introduced/posterior vaginal wall retractor is used
- The squamocolumnar junction is identified and scraped with Ayre's spatula.
- Endocervix is scraped with a cytobrush.
- Samples are spread separately on 2 different glass slides.
- The slide is immediately fixed using 95% ethyl alcohol.
- Stained with Pap stain or Short stain.

Solution to Question 10:

Controlled cord traction during placental delivery is a component of active management of third stage labor (AMTSL).

AMTSL is recommended by the WHO to prevent postpartum hemorrhage (PPH).

The recommended components of AMTSL are as follows:

Uterine massage or fundal massage is not recommended as an intervention to prevent PPH in women who have already received prophylactic uterotonics such as oxytocin (core component of AMTSL). Hence, it is now an optional component of AMTSL that is done in patients with uterine atony as a part of the management of PPH.

Component		Remarks
1.	Injecting uterotonics within 1 minute of delivery of the baby or immediately after delivery of the anterior shoulder before the delivery of the placenta	Most important step Oxytocin 10 IU im/iv infusion is the preferred uterotonic. (Not recommended by WHO-carboprost)
2	Delayed cord clamping	Done within 1-3 mins of delivery 80 ml of additional blood goes to the fetus Prevents fetal anemia

Component		Remarks
3	Delivery of placenta by controlled cord traction: Modified Brandt Andrews technique	Done only in settings where skilled birth attendants are available The uterus is pushed upwards and backward The traction of cord is downwards and forwards
4	Intermittent assessment of uterine tone	Done every 15 mins for 2 hours

Solution to Question 11:

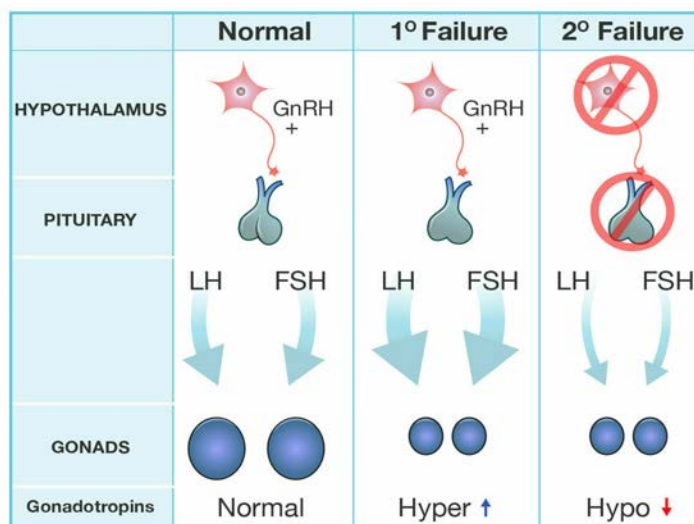
Kallman syndrome is not a cause of secondary amenorrhea. It is a cause of primary amenorrhea.

Primary amenorrhea is defined as the absence of menstruation by 14 years without the development of secondary sexual characteristics or by 16 years regardless of the presence of secondary sexual characteristics. Kallmann syndrome presents with primary amenorrhea due to hypogonadotropic hypogonadism from deficiency of GnRH. It is associated with olfactory dysfunction and anosmia.

Secondary amenorrhea is the absence of menses for more than 3 months in girls or women who previously had regular menstrual cycles or 6 months in girls or women who had irregular menses. Sheehan's syndrome (postpartum hypopituitarism secondary to the necrosis of the pituitary gland) and Asherman's syndrome (formation of intrauterine synechiae, usually following endometrial destructive procedures like curettage or ablation) are causes of secondary amenorrhea.

Note: Turner's syndrome is a cause of primary amenorrhea. Turner's mosaic may present either with primary or with secondary amenorrhea. So, of the given options, the best answer is Kallman syndrome.

The image attached below summarizes the pathology involved in various types of amenorrhea.



Solution to Question 12:

4% is the minimum percentage for normal sperm morphology in a seminal fluid analysis.

Morphology of sperms is the most important marker of male fertility. Characterization of normal sperm morphology shows improved correlation with fertilization rates during IVF cycles.

The criteria requires careful analysis of the following:

- The shape and size of the sperm head,
- The relative size of the acrosome in proportion to the head,
- And characteristics of the tail, including length, coiling, or presence of two tails.

Significantly decreased fertilization rates are observed when normal morphology of the sample falls below 4 percent.

Abnormal sperm morphology is termed teratospermia or teratozoospermia.

Solution to Question 13:

Antimüllerian Hormone (AMH), secreted by the granulosa cells, is the best marker of ovarian reserve as it can be tested on any day of the cycle.

AMH is secreted by the Sertoli cells in the testis and granulosa cells in the ovary. In males, it inhibits the development of the Müllerian system and its absence can result in hermaphroditism. In the female, it helps in follicular development and oocyte maturation. The estimation of serum AMH is used in the study of ovarian reserve in an infertile woman and a woman with secondary amenorrhoea. The levels of AMH vary minimally across the cycle and they strongly correlate with the number of ovarian primordial follicles. This makes AMH is the best predictor for measuring ovarian reserve. AMH levels can be affected by the use of oral contraceptives and tobacco smoking.

Values and interpretation:

- Normal value: 2–6.8 ng/mL
- Poor ovarian reserve : <1 ng/mL
- PCOD and hyperstimulation syndrome: >10 ng/mL

Tests for ovarian reserve: Ovarian reserve means the capacity of the ovary to produce eggs. The other indicators include:

- FSH levels are measured in the early follicular phase (Day 2 or 3 of the menstrual cycle) - High FSH means the ovarian reserve is getting depleted (FSH is highest in postmenopausal women). FSH (values >25 mIU/ml) levels are higher than normal in women with a reduced ovarian reserve and ovarian hypofunction.
- Serum oestradiol values on day 3 of the cycle. Serum oestradiol <75 pg/mL indicates ovarian hypofunction.

- LH - values <10 IU/L indicates poor reserve
- Antral follicular count - counts less than 4–5 on days 2–5 denotes poor response to hormones.
- Inhibin B - values <40 pg/mL indicates poor reserve

Solution to Question 14:

A girl presenting with amenorrhea, normal female external genitalia and breasts with 5 alpha-reductase deficiency is suggestive of male pseudohermaphroditism.

In pseudohermaphroditism, the sex glands are of a certain sex while the external genitalia is of the opposite sex.

Male pseudohermaphroditism or male intersex is an association of male gonads (XY genotype) with female external genitalia. The production of testosterone from the testes requires 5 alpha-reductase. This enzyme converts testosterone into DHT (dihydrotestosterone), which is capable of acting on peripheral target tissues to produce the male phenotype. The absence of this enzyme results in failure of the testis to descend to the scrotum and ill development of the penis, thereby resulting in the female phenotype. Many individuals develop their attitudes and psyche according to their sex of rearing (as a female). Rehabilitation to their gender roles is the best line of management.

Other options:

Option A: Concealed sex refers to the deliberate concealment of sex usually done by criminals by changing clothes or by other methods to avoid detection. This can be detected by physical examination.

Option B: In true hermaphroditism the glands of both the sexes are present in the same individual. Such cases are very rare. In most cases, the accessory sex gland is atrophic and lacks functional activity.

Option C: Female pseudohermaphrodites are genetic females (XX genotype) with male/ambiguous external genitalia due to the deficiency of 21 alpha-hydroxylase enzymes resulting in an excess of androgens. This causes clitoromegaly resembling the penis, and the fused labioscrotal folds resemble a rudimentary scrotum. Management includes administration of steroids to control excess ACTH, estrogen to confer the secondary sexual characteristics, and surgical restoration of the external genitalia to a feminine pattern.

Solution to Question 15:

The correct answer is 1-D, 2-C, 3-B, 4-A.

Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome, commonly referred to as Mullerian agenesis, refers to the failure of development of the uterus, cervix, and upper part of the vagina in a female with 46 XX karyotype. These individuals have normally functioning gonads and external genitalia.

Condition	Karyotype
1. Turner's syndrome	D. 45 + XO
2. Swyer's syndrome	C. 46+ XY
3. Androgen insensitivity syndrome	B. 46+ XY
4. MRKH (Mayer-Rokitansky-Kuster-Hauser) syndrome	A. 46 + XX

Complete androgen insensitivity syndrome/testicular feminizing syndrome:

In this syndrome, the karyotype is XY and thus, the patient is a genotypic male. Pathology is due to the defective response of the androgen receptors to androgens. Findings include:

- Primary amenorrhea → Due to the absence of uterus and cervix.
- Normally developed breasts → Due to the conversion of the excessive androgens into estrogen.
- Swelling in the inguinal region → Suggestive of an underdeveloped ectopic testis, caused by loss of action of testosterone on its receptor.
- Absence of uterus and cervix → Due to the production of Mullerian inhibiting substance from the Sertoli cells of the testis.

Note:

There is some discrepancy among authors on this topic. Androgen insensitivity syndrome (AIS) with XY karyotype may have mostly female external sexual characteristics. According to William's Textbook of Obstetrics 26th edition, breast development occurs due to peripheral aromatization of testosterone to estrogen whereas the pubic and axillary hair remains scant or absent. But as per Shaw's Textbook of Gynecology 7th edition, XY females with AIS have normal secondary sexual characteristics.

Swyer syndrome or pure gonadal dysgenesis occurs as a result of a mutation in the SRY gene. This leads to underdeveloped, dysgenetic gonads that fail to produce androgens or anti-Mullerian hormone (AMH). The following features are seen:

- Normal prepubertal female phenotype- Absent or poorly developed secondary sexual characteristics.
- Normal Müllerian system development due to the absence of AMH. (Uterus and cervix present)
- Female external genitalia
- Presence of dysgenetic testes that cannot produce AMH or sex hormones. They have a risk of malignant transformation.

As there is gonadal dysgenesis in a patient with 46 XY karyotype, they are actually determined to be males and are a type of male pseudohermaphrodite. They present with primary amenorrhoea, absence of secondary sex characters, and female external genitalia.

Turner's syndrome is a chromosomal abnormality with partial or complete monosomy of the X chromosome (45+XO). In Turner's syndrome, gonadal dysgenesis causes primary ovarian failure

that is hypogonadism. The clinical picture of amenorrhea, short stature, webbed neck, low hairline at the back of the neck, widely spaced nipples, and lack of secondary sexual characteristics is seen in Turner's syndrome.

Solution to Question 16:

Twin to twin transfusion syndrome (TTTS) is a specific complication seen in monochorionic twins. Selective fetal growth restriction, polyhydramnios and neural tube defects are seen as complications to several conditions and are not specific to monochorionic twins.

In monochorionic diamniotic gestation, twins are separated solely by two amniotic membranes, with no intervening chorion. Monochorionic monoamniotic gestation lacks a dividing membrane.

All monochorionic placentas likely have some anastomotic connections, unique to this type. Chronic fetofetal transfusion may lead to the following clinical syndromes:

- Twin-twin transfusion syndrome (TTTS)
- Twin anemia–polycythemia sequence (TAPS)
- Twin reversed-arterial-perfusion (TRAP) sequence

Twin to twin transfusion syndrome (TTTS):

TTTS is a fetal syndrome in monozygotic twinning with two amniotic sacs and a shared chorion, allowing for anatomical sharing of fetal circulations through anastomoses of placental arteries and veins. This leads to transfusion from donor twin to recipient twin, resulting in donor anemia and restricted growth, while the recipient may suffer polycythemia, circulatory overload, and hydrops.

Features:

The Quintero staging is as follows:

- Stage I—Discordant amniotic fluid volumes+ urine visible in donor's bladder.
- Stage II—Stage I + no visible urine in donor's bladder.
- Stage III— Stage II + abnormal Doppler studies of the umbilical artery, ductus venosus, or umbilical vein.
- Stage IV— frank ascites or hydrops in either twin.
- Stage V—the demise of either fetus.

Treatment:

- Laser photocoagulation of the placental vascular anastomoses is the preferred treatment for severe TTTS - Quintero stages 2 to 4.
- Septostomy is seldom done.
- Amnioreduction is a treatment modality that involves needle drainage of excessive amniotic fluid.
- Selective feticide is done only if the growth disturbances develop before 20 weeks.
- Of all, selective fetoscopic laser photocoagulation (SFLP) by Solomon technique is the preferred treatment modality as it is associated with better survival rates and a reduced likelihood of

recurrence.

Donor twin	Recipient twin
Oliguria	Polyuria
Anemia	Polycythemia
Oligohydramnios	Polyhydramnios
Activation of Renin-angioten sin system (RAAS)	Activation of Atrial natriureti c peptide (ANP)
Decreased renal blood flow	Increased renal blood flow
Non-visible bladder	Over distended bladder, circu latory overload, hyperviscosit y syndrome, hyperbilirubine mia, hypertension
Renal failure	Congestive cardiac failure (C CF), Hydrops twin (late cases)

Solution to Question 17:

The correct answer is 1-C, 2-D, 3-B, 4-A.

Secondary amenorrhoea is the absence of menses for more than three months in girls or women who previously had regular menstrual cycles or six months in girls or women who had irregular menses.

Premature Ovarian failure or premature menopause is the development of menopause before the age of 40 years. It is diagnosed when at least 2 serum FSH levels are elevated (>40 mIU/l) when measured at least 1 month apart in a woman <40 years. The other findings consistent with POF include low estradiol and an elevated FSH/LH ratio.

Asherman's syndrome is the presence of uterine synechiae, due to the destruction of large areas of the endometrium, most commonly as a result of trauma to the endometrium following pregnancy-related curettages. Oligomenorrhoea, amenorrhoea, dysmenorrhoea, abortion, and infertility are seen with uterine synechiae. The laboratory investigations and hormonal assays are generally normal in isolated cases of uterine synechiae. Diagnosis is generally established using hysteroscopy. Moth-eaten appearance on hysterosalpingography suggests a diagnosis of Asherman syndrome. Treatment is by hysteroscopic adhesiolysis.

PCOS is a multisystem, metabolic, endocrine dysfunction that commonly occurs among women in the reproductive age group. The endocrinological changes reflecting in the assays are summarized in the pearl attached below.

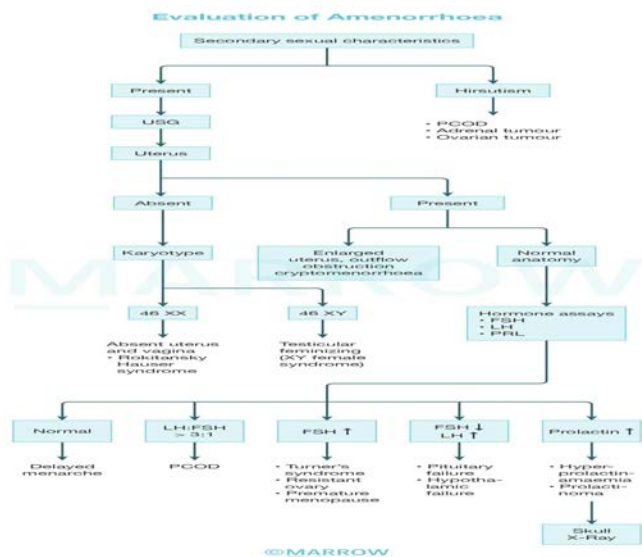
Sheehan's syndrome/ pituitary apoplexy is hemorrhagic infarction of large pituitary adenomas or pituitary infarction due to traumatic major blood loss. If this blood loss occurs during delivery, it's called Sheehan's syndrome. Affected women may present with panhypopituitarism and may have persistent hypotension, tachycardia, hypoglycemia, and lactation failure. The most prominent among symptoms are lactation failure (due to deficiency of

prolactin) and postpartum amenorrhea (due to deficiency of gonadotropins).

Condition	Lab markers
1. Sheehan's syndrome	C. Low LH, Low FSH, Low es tradiol
2. Premature Ovarian failure	D. High LH, High FSH, Low estradiol
3. PCOS	B. High LH, Low FSH, Low e stradiol
4. Asherman's syndrome	A. Normal FSH, Normal LH, Normal estradiol

Note: Estradiol level in PCOS is low in comparison to Estrone (E1).

The common differentials for the presentation for secondary amenorrhea and the lab evaluation is summarized in the flowchart shown below.



Solution to Question 18:

Of the given list/combination of instruments 1, 2, 4, 6 are commonly used in the cesarean section. Cesarean section is an operative procedure in which the fetuses are delivered through an incision on the abdominal and uterine walls.

The given image summarizes the various instruments used in the lower segment cesarean section.

Trolley with instruments prepared for cesarean section operation



- 1 - Mops (large swabs)
- 2 - Electrocautery set
- 3a - Suction tube with a cannula
- 3b - Baby suction catheter
- 4 - Towel clips
- 5 - Kidney dish
- 6 - Gauze pieces
- 7 - Lanes tissue forceps
- 8 - Needle holders
- 9 - Empty bowl
- 10 - Obstetric Forceps (Wrigley's)
- 11 - Doyen's retractor
- 12 - Kocher's clamps
- 13 - Long artery forceps
- 14 - Allis tissue forceps (long variety)
- 15 - Green Armytage forceps
- 16 - Little wood's forceps
- 17 - Allis tissue forceps (short variety)
- 18 - Artery forceps (short variety)
- 19 - Dissecting forceps toothed and non-toothed,
- 20 - Scissors (straight and curved variety)
- 21 - Knives (two)
- 22 - Bowl with povidone-iodine lotion
- 23 - Sponge holding forceps
- 24 - Suture packets

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Bard-Parker blade (scalpel with blade) has a handle with a notch to fit the blade. It is usually held by pencil grip. Commonly for abdominal incisions, we use blade no: 22. Other uses include incision of skin and sub-cutaneous tissues, cutting the clamped pedicles, and performing sharp dissections.

The given image shows various Bard-Parker blades.

Scalpel blade sizes and shapes



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Doyen's retractor is used to retract the abdominal wall as well as the bladder to facilitate better exposure of the lower uterine segment during cesarian operation. It is also used to pull back the soft tissues to widen the surgical field during laparotomy.

The image below shows Doyen's retractor.

Doyen's retractor



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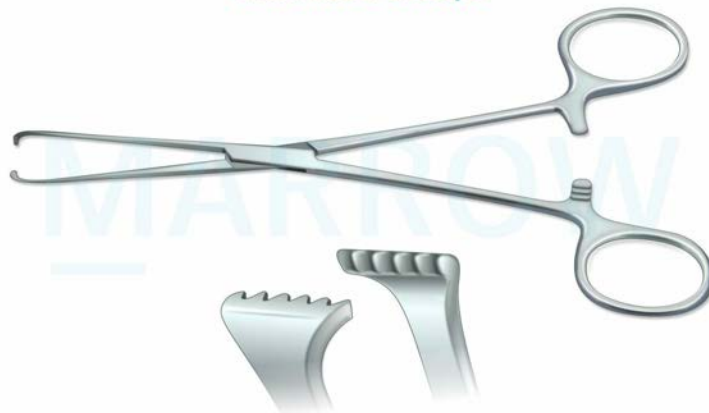
Alli's forceps: It is identified by the distal blades and a tip that has teeth that do not oppose each other. There is also a lock in the proximal part.

Uses:

- To catch hold of the margins and angles of the uterine flaps in LSCS after the delivery of the baby in settings where the Green-Armytage hemostatic clamp is not available.
- To catch hold the anterior lip of the cervix
- To hold the apex of the episiotomy wound during the repair.
- To catch hold of the margins of the peritoneum, rectus sheath, vaginal mucosa during the repair.
- To catch hold of the torn ends of the sphincter ani externus prior to suture in the repair of complete perineal tear.

The given image shows Alli's forceps.

Allis tissue forceps



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Green armytage forceps have a broad tip as shown in the image and hence are used to compress a wide area. It is used as a hemostat in caesarian sections. In LSCS (Lower segment Caesarean section), the cut uterine edges can bleed and Green Armytage forceps are applied to the edge of the incision for hemostasis.



Cusco's speculum is a self-retaining bivalve speculum used for visualization of the cervix, vaginal fornices, and for performing pap smears.

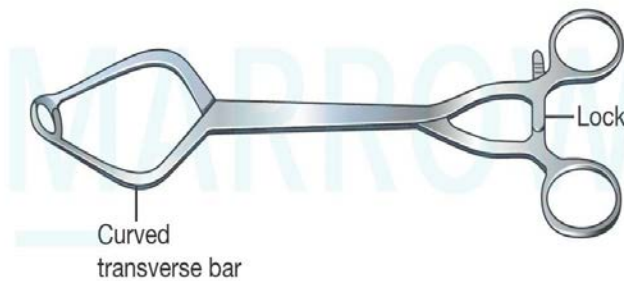
The instrument shown in the image is Cusco's speculum.



Shirodkar's uterine holding clamp has blades with curved transverse bars with gaps between them when it is locked. This gap makes it atraumatic and is used to lift, hold and steady the uterus during sling surgeries and tubal procedures.

The image below shows Shirodkar's uterine clamp.

Shirodkar's uterine holding clamp



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Solution to Question 19:

The given image shows a discrete growth on the left labia majora of the vulva. The most likely diagnosis is vulvar carcinoma.

Vulvar carcinoma:

Squamous cell carcinoma is the most common type of vulvar cancer. The most common site of vulvar cancer is labia majora & Clitoris & Labia Minora. Persistent HPV infection and chronic inflammation such as lichen sclerosis are risk factors. It is generally seen in elderly women and presents with vulvar pruritus, vulvar swelling, lump, or an ulcer. The lump is papular with a raised pigmented area. The ulcers often have everted margins with fissured, cracked, and indurated surrounding skin. They usually bleed and cause offensive vulval discharge.

The most common type of spread in vulvar cancer is lymphatic while the vascular route of spread is extremely rare in vulvar cancer. The first lymph node to be involved is the superficial inguinal lymph node and thus, a sentinel lymph node biopsy is helpful for evaluating the spread of vulval cancer. Groin node status is the best prognostic predictor.

Other options:

Option A: Vulval hematoma is a collection of blood in the vulval region. The symptoms include persistent perineal pain and rectal tenesmus. There may be retention of urine. The local examination would commonly reveal a tense swelling at the vulva which becomes dusky and purple in color. It is mostly tender to touch. A small hematoma (<5 cm) may be treated conservatively with a cold compress. Larger hematomas should be explored in the operation theatre under general anesthesia

The image shows vulval hematoma. As it is a vascular leak a blue hue can be appreciated as the blood clots.



Option B: Condyloma acuminata (venereal warts) is caused by infection with Human Papilloma Virus 6 or 11. The verrucous growths may appear discrete or coalesce to form large cauliflower-like growths. They may be asymptomatic, but can also cause discomfort discharge, or bleeding. A biopsy is necessary to rule out malignancy and the specific treatment includes electrocautery, cryosurgery, laser, or diathermy.

The image shows multiple warty growths on the vulva and around the anal canal as seen in condyloma acuminata.



Option D: Bartholin cyst is caused by obstruction of the Bartholin ducts by proteinaceous material or by inflammation from infection. Generally caused by gonorrhea and presents as a painful cystic swelling. Symptomatic or recurrent cysts may require marsupialization or gland excision. Bartholin abscess is a sequela to a cyst and it usually requires surgical drainage.

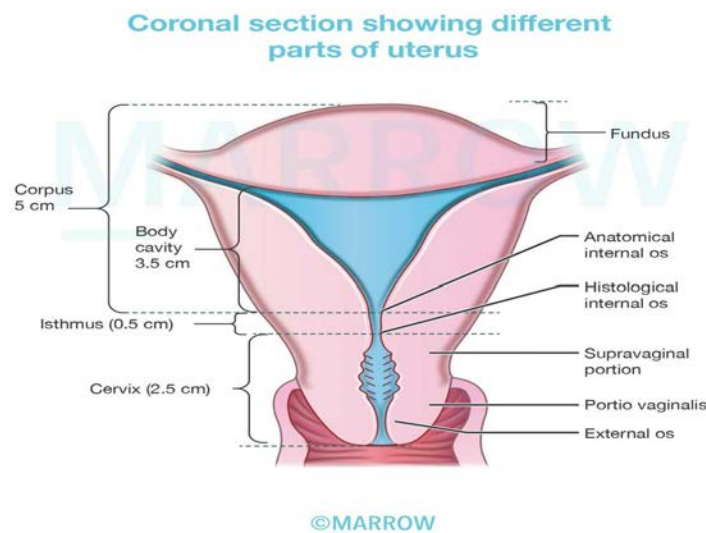
The image below shows Bartholin cyst.



Solution to Question 20:

The normal length of the uterine cavity including the cervical canal is usually 6–7 cm.

The image of the coronal section showing different parts of uterus is attached below.



Clinical application:

- Normally, the uterocervical length is measured using Sim's uterine sound.
- It is based on this length that, lengths of various intra-uterine systems/ intra-uterine devices is determined.

The image below shows Sim's uterine sound.



Solution to Question 21:

Laparoscopy and Hysterosalpingogram (HSG) can be used to diagnose unicornuate uterus.

In the unicornuate uterus, the formation of the uterus and fallopian tube occurs entirely from one Mullerian duct due to the developmental arrest of other Mullerian duct. Women with a unicornuate uterus have an increased incidence of infertility, endometriosis, and dysmenorrhea.

Evaluation:

- The initial evaluation is through sonography. It can also help pick up ipsilateral renal agenesis, which is seen in 40% of the women with unicornuate uterus.
- Saline infusion sonography (SIS) is another modality to identify Mullerian anomalies. It can be used as a screening tool for cavity evaluation as well.
- On Hysterosalpingography, the endometrial cavity usually assumes a fusiform (banana) shape, tapering at the apex and draining into a single fallopian tube. The uterus is generally shifted off the midline.
- Invasive modalities such as hysteroscopy/laparoscopy are reserved for patients with complicated Mullerian anomalies or in patients presenting with infertility to establish the diagnosis.
- These days, MR imaging is used to characterize and pre-operatively evaluate cases that are complicated.

The image below shows a hysterosalpingogram of a unicornuate uterus



Falloscopy is the inspection of the fallopian tubes through a micro-endoscope. Hysteroscopic falloscopy reveals the tubal pathology of the cornual and interstitial end of the fallopian tube.

Solution to Question 22:

The symptoms of endometriosis include 1, 2, 4. Endometriosis presents with unexplained infertility, dysmenorrhea, and vaginal bleeding.

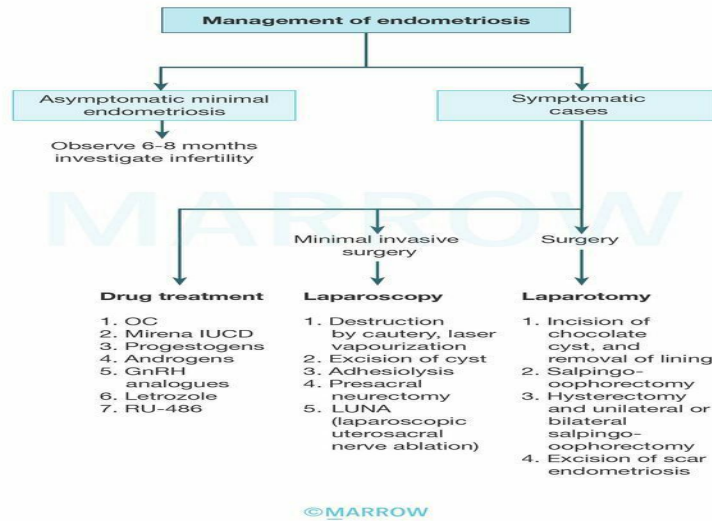
Endometriosis is the presence of endometrial tissue (glands and stroma) outside the uterus. It is an estrogen-dependent disease of the childbearing age group. The most common site of occurrence is the ovary followed by the pouch of Douglas.

Key clinical features of endometriosis include:

- Dysmenorrhoea
- Lower abdominal pain
- Infertility
- Pain during urination or defecation,
- Constipation or diarrhea
- Irregular bleeding.

The gold standard investigation for the diagnosis of endometriosis is a visual inspection by laparoscopy, preferably with histological confirmation.

The management of endometriosis is summarized in the flowchart below:



Solution to Question 23:

The given clinical scenario of frothy vaginal discharge and strawberry cervix is suggestive of trichomoniasis and metronidazole is the drug of choice for the treatment.

Trichomonas vaginalis is one of the most common causes of vaginal discharge. It is transmitted by sexual intercourse, with an incubation period of 4–21 days. The clinical features include vaginal discharge which is profuse, thin, slightly green in colour, and frothy. It may be associated with vaginal irritation, pruritus, dysuria, and increased frequency. Cervical inflammation with punctate hemorrhagic spots known as the strawberry cervix is typically seen. Motile trichomonads are seen on wet mount examination.

The below image shows the strawberry cervix seen in trichomoniasis.



Administration of metronidazole 2 g orally in a single dose or 500 mg 2 times/day for 7 days is the standard treatment. It should be taken at night and after meals to avoid vomiting. If this too causes vomiting, or in resistant cases an alternate regimen is to be considered. Both the partners need to be treated simultaneously to avoid recurrence.

The alternative regimen includes tinidazole 500 mg orally twice a day for 7 days or secnidazole in a single dose of 1000 mg daily for 2 days. The use of tinidazole is not recommended during pregnancy.

Solution to Question 24:

According to IADPSG, the cut-off value of fasting blood glucose for the diagnosis of gestational diabetes mellitus (GDM) is ≥ 92 mg/dL.

Gestational diabetes mellitus (GDM) refers to any degree of glucose intolerance having its onset or first recognition during pregnancy.

For IADPSG criteria, an oral glucose tolerance test (OGTT) is done in the fasting state using 75 g of glucose, and GDM is diagnosed if any one of the following cut-offs is met i.e.

- Fasting: ≥ 92 mg/dL
- 1 hour: ≥ 180 mg/dL
- 2 hr: ≥ 153 mg/dL

Earlier, a two-step approach was being used to diagnose GDM. In this, the first step (screening) is a 50-gram one-hour glucose challenge test (GCT). The patients having a positive result of ≥ 140 mg/dl, are candidates for the OGTT. A 100-gram, three-hour OGTT is the diagnostic test for GDM. GDM is diagnosed if two or more values are positive, as follows.

This has now been replaced by the one-step approach that omits the screening test and simplifies diagnostic testing by performing only a 75-gram, two-hour oral GTT.

Some complications of GDM:

- Maternal
- Hypertension, preeclampsia
- Hydramnios
- Increased operative interventions
- Future development of diabetes mellitus
- Fetal and neonatal
- Fetal macrosomia
- Congenital anomalies
- Metabolic abnormalities such as hypocalcemia, hypomagnesemia, hyperbilirubinemia, and neonatal hypoglycemia.
- Respiratory distress syndrome
- Subsequent childhood and adolescent obesity

Time	Carpenter and Coustan	NDDG
Fasting	95 mg/dL	105 mg/dL
1 hour	180 mg/dL	190 mg/dL
2 hour	155 mg/dL	165 mg/dL
3 hour	140 mg/dL	145 mg/dL

Solution to Question 25:

The given clinical scenario of a primigravida with spotting and an ultrasound showing a characteristic snowstorm appearance (anechoic cystic spaces) is suggestive of a hydatidiform mole (complete mole). It refers to an echogenic uterine mass that has numerous anechoic cystic spaces.

Hydatiform mole belongs to the benign group of gestational trophoblastic diseases (GTD). It is an abnormality in chorionic villi making them partly proliferative and degenerative, thereby resulting in the formation of multiple small cysts. Based on the portion of the chorionic villi involved, they could be complete (diffused) or partial (focal)

The classical clinical features of a complete mole are:

- Abnormal vaginal bleeding -Most common (90%)
- Lower abdominal pain
- Hyperemesis gravidarum due to increased β -hCG
- Features of early-onset preeclampsia
- Uterus appears large for date
- Absent fetal parts and FHS
- Expulsion of vesicular tissues (grape-like vesicles)
- Theca lutein cyst of ovaries of more than 6 cm
- Hyperthyroidism (rare)
- Serum hCG $>$ 100,000 mIU/ml
- USG - snowstorm appearance

The definitive management after diagnosis is suction and evacuation, after making the cervix favorable. Follow-up of the patients with serial hCG values is crucial.

Prophylactic chemotherapy can be started on high-risk women (WHO prognostic score $>$ 7) with methotrexate 1mg/kg/day intramuscular/intravenous on days 1, 3, 5, and 7. Patients are also advised to not get pregnant for at least a year.

Other options:

Option A: Blighted ovum (silent miscarriage) occurs when a fertilized egg implants in the uterus but doesn't develop into an embryo. It is a sonographic diagnosis where there is an absence of a fetal pole in a gestational sac with a diameter of 3 cm or more.

Option B: Missed abortion is when the fetus is dead and retained inside the uterus for a variable period. USG shows an empty sac early in the pregnancy or the absence of fetal cardiac activity and fetal movements.

Option C: Ectopic pregnancy is one in which the fertilized ovum is implanted and develops outside the normal endometrial cavity. The majority of ectopic pregnancies occur in the fallopian tube, the ampulla being the most common site.

Solution to Question 26:

Endometrial sampling is done to evaluate a patient presenting with postmenopausal bleeding.

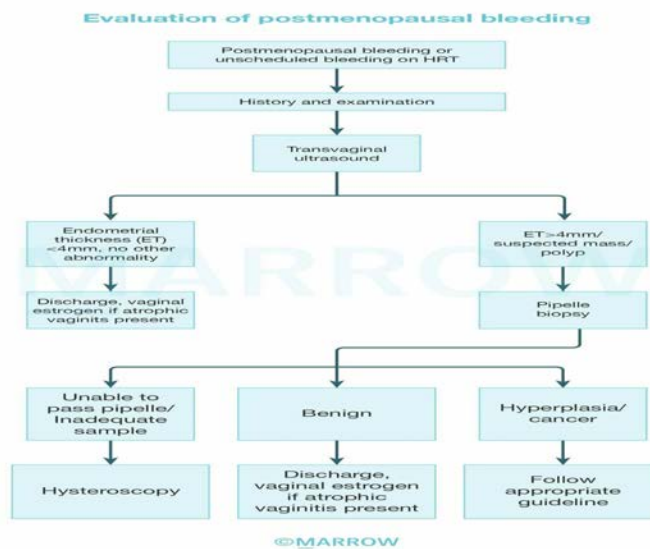
Postmenopausal bleeding is defined as any bleeding from the genital tract after 12 months of menopause in a woman not on hormone replacement. Any vaginal bleeding occurring after 6 months of amenorrhea in a menopausal woman should also be considered as postmenopausal bleeding and investigated. All postmenopausal women with unexpected uterine bleeding should be evaluated for endometrial malignancy since age is a significant risk factor for this.

Transvaginal ultrasound (TVS) is a better first-line investigation. Since TVS is not included in the options, endometrial sampling is the next best answer.

Endometrial sampling is indicated if:

- The endometrial thickness is ≥ 4 mm
- The endometrium shows diffuse or focal increased echogenicity (heterogeneity)
- The endometrium is not adequately visualized on transvaginal sonography
- The woman has persistent bleeding, as it can be a sign of endometrial cancer even when the endometrial thickness is less than 4mm.

The image below gives a summary of the evaluation of postmenopausal bleeding, as per the FOGSI guidelines.



Other options:

Option A: Pap smear should be offered to all women above the age of 21 years who are sexually active for at least 3 years. Colposcopic examination and selective biopsy are considered in patients with abnormal Pap smears.

Option B: Visual inspection with iodine is done where the facilities for Pap screening do not exist. This involves painting the cervix with Lugol's iodine. It is used to detect any abnormal cytology in the cervix and vagina, such as in cases of carcinoma cervix.

The normal vaginal and cervical epithelium contains glycogen, therefore stains mahogany brown on being painted with Lugol's iodine. The abnormal cells remain unstained. These areas need to be biopsied and sent for histopathological examination.

Option C: The colposcopic study is challenging in postmenopausal women because of narrow vaginas, senile vaginitis, indrawn and poorly visible squamocolumnar junction, and atrophic cervix.

Solution to Question 27:

Among the given options, manual perineal support and giving warm compresses are recommended by WHO in the management of the second stage of labor.

The second stage of labor refers to the time period between full cervical dilatation and the birth of the baby. For women in the second stage of labor, the following techniques to reduce perineal trauma and facilitate spontaneous birth such as perineal massage, warm compresses, and manual perineal support (hands-on guarding of the perineum) are recommended.

Option A: Mandatory episiotomy is discouraged. Episiotomy is indicated in cases of:

- Shoulder dystocia
- Breech delivery
- Fetal macrosomia

- Operative vaginal deliveries
- Persistent occiput posterior positions

Option C: Head is delivered by maintaining flexion to ensure that the small sub-occipitofrontal diameter distends the vulval outlet instead of the larger occipitofrontal diameter.

Option D: The position that is most comfortable to the mother including upright positions is recommended.

Solution to Question 28:

The Modified Bishop score of the given patient is calculated to be 5.

From the data given in the question:

- The position of the cervix is posterior, hence score = 0
- Cervical length is 5 cm, hence score = 0
- Cervical dilatation is 1 cm, hence score = 1
- Consistency of cervix is soft, hence score = 2
- The station of the head is -1, hence score = 2

Hence, the Modified Bishop score is $0+0+1+2+2 = 5$.

The Bishop score is a scoring system used to assess labor induction outcomes. It is based on the station of the head, dilation, effacement (or length), position, and consistency of the cervix noted by doing a vaginal examination.

According to the NICE guidelines, a score of ≥ 6 is considered favorable.

According to Williams Obstetrics 26th edition, a score ≥ 9 is considered favorable.

Note: In the modified Bishop score (1991) the effacement is replaced by the cervical length in cm.

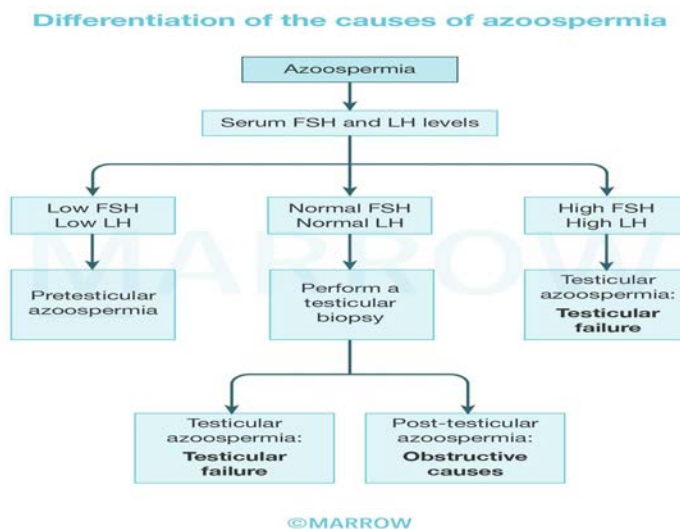
Solution to Question 29:

Serum FSH level is the next best test to differentiate between an obstructive and a testicular cause of azoospermia. FSH is the main hormone stimulating the production of sperm in the testis and has the highest specificity and sensitivity among hormonal assays while differentiating between obstructive and non-obstructive azoospermia.

The absence of sperm in semen is known as azoospermia. It can be due to:

- Pre-testicular azoospermia due to a hypothalamic or pituitary abnormality (hypogonadotropic-hypogonadism)
- Testicular failure or non-obstructive azoospermia
- Post-testicular obstruction implies normal spermatogenesis but failure to deliver the sperm into the ejaculate because of ductal obstruction.

The algorithm for differentiating the causes of normal semen volume azoospermia is as follows:



- A low FSH level indicates pituitary or hypothalamic failure
- A high FSH level denotes primary gonadal failure or testicular cause
- A normal FSH level could be due to either non-obstructive or obstructive azoospermia. A testicular biopsy is usually required in such people to provide a definitive diagnosis.

Solution to Question 30:

The dosage of dexamethasone given for fetal lung maturity is 4 doses of 6 mg, which is given 12 hours apart.

Maternal administration of glucocorticoids is advised in preterm labor when the period of gestation is less than 34 weeks. Intramuscular injection of either betamethasone or dexamethasone is given.

Dosage:

- Betamethasone (Betnesol): 2 doses of 12 mg each, given 24 hours apart.
- Dexamethasone: 4 doses of 6 mg each, given every 12 hours.

This helps in fetal lung maturation and minimize the incidence of:

- Respiratory distress syndrome
- Intraventricular hemorrhage
- Necrotizing enterocolitis

It is beneficial when the delivery is delayed beyond 48 hours after the first dose.

Solution to Question 31:

100 mcg of carbetocin is used in the prevention of postpartum hemorrhage (PPH).

The use of carbetocin (100 µg, IM/IV) is recommended for the prevention of PPH. It is a long-acting synthetic analog of oxytocin with agonist properties. It binds to oxytocin receptors in the uterine smooth muscle, and results in rhythmic contractions, increased frequency of existing contractions, and increased uterine tone.

To effectively prevent PPH, the WHO recommends the use of only one of the following uterotonic agents:

- Oxytocin - (10 IU - IM/IV)
- Carbetocin - (100 µg - IM/IV)
- Misoprostol - (400 µg or 600 µg, per-oral)
- Ergometrine/methylergometrine - (200 µg - IM/IV)
- Oxytocin and ergometrine fixed-dose combination - (5 IU, 500µg - IM)

Of these, oxytocin is used as the first-line agent in preventing postpartum hemorrhage. When oxytocin is not available, the other drugs mentioned above can be used based on the clinical condition and availability of the drug.

Solution to Question 32:

Milk production in pregnancy is inhibited by high estrogen.

The synthesis and secretion of milk from the breast alveoli is known as lactogenesis. Prolactin, an anterior pituitary hormone, is responsible for lactogenesis. In spite of high prolactin levels during pregnancy, there is inhibition of milk secretion. This is because of high levels of estrogen and progesterone circulating during pregnancy, which makes the breast tissues unresponsive to prolactin.

Following delivery, there is a fall in estrogen and progesterone levels. As a result, prolactin begins its milk secretory activity in previously fully developed mammary glands.

Other hormones involved in lactogenesis:

- Insulin
- Growth hormone
- Cortisol
- Thyroxine
- Placental lactogen aka human chorionic somatomammotropin: Has growth promoting and lactogenic effects
- Serotonin

Oxytocin is the major hormone for galactokinesis or the ejection of milk.

Solution to Question 33:

Levonorgestrel (LNG) 1.5 mg is the emergency contraceptive used in India.

Emergency contraception refers to backup methods for contraceptive emergencies which women can use within the first few days after unprotected intercourse to prevent an unwanted pregnancy.

Levonorgestrel 1.5 mg single dose can be used for emergency contraception. Alternatively, levonorgestrel 0.75 mg one tablet can be taken within 72 hours of unprotected intercourse and another tablet 12 hours later.

Note: Under the National Reproductive and Child Health Programme, the Drug Controller of India has only approved levonorgestrel tablets for use as an emergency contraceptive pill.

Other options:

Option A: Misoprostol is a prostaglandin E1 analog used in the medical termination of pregnancy.

Option B: High doses of ethinylestradiol can provide emergency contraception when used in combination with levonorgestrel. This is known as the Yuzpe regime.

Option C: Ulipristal is a synthetic progesterone hormone receptor modulator. It attaches to the progesterone receptor and prevents/ delays ovulation, and suppresses the endometrium, thereby preventing implantation. Ulipristal 30 mg should be taken within 5 days of unprotected sexual intercourse. It is approved by US-FDA and European medicine agencies but at present, it is not available in India as an emergency contraceptive pill.

Solution to Question 34:

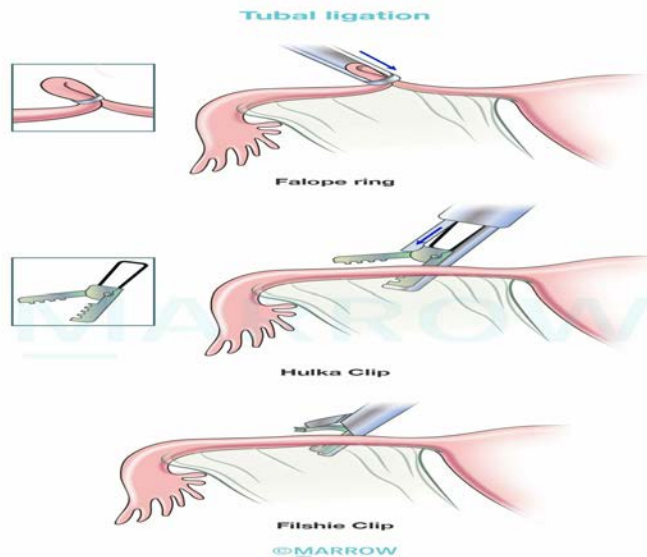
Laparoscopic tubal ligation is an irreversible method of contraception.

Sterilization is a permanent method of birth control. It includes surgical procedures like tubectomy in woman and vasectomy in man. These methods are advised for those couples, who have completed their families and do not want to bear the inconvenience or cost of the other methods of contraception, and when the other methods are contraindicated.

- Postpartum sterilization is performed in the first week of the postnatal period. It can be combined with a cesarean section
- Interval sterilization is done in non-pregnant women or at any time after 6 weeks of delivery.

Tubal sterilization can be achieved by three methods which are:

- Electrocoagulation is used for the destruction of a segment of the tube
- Mechanical occlusion of the tube using a silicone rubber band (Falope ring) or a spring-loaded clip (Hulka Clemens clip)
- Suture ligation with or without tubal segment excision such as Pomeroy's method, Uchida's method, Irving's method, etc.



Other options:

Option A: Copper T (CuT) is an effective, reversible, and long-term method of contraception. About 50 mcg of copper is eluted daily in the uterus. It can be left in place for a long time and does not interfere with sexual activity. They have an effective life of about 3–5 years. Copper T 380A, known as Paraguard, has a lifespan of 10 years.

Option B: Implanon is a subdermal implant that contains 67mg of desogestrel. It releases 30 mcg of the desogestrel daily and the effect lasts for up to 3 years. It prevents ovulation and is reversible within 1 month of removal.

Option D: Combined oral contraceptive pills (OCP) contain a mixture of ethinyloestradiol with progestogen. It acts by suppressing pituitary hormones i.e. FSH and LH peak and thereby preventing ovulation. It is a reversible method of contraception as 90% of women will have normal ovulatory cycles within 3 months of stopping OCPs.

Solution to Question 35:

Clear cell carcinoma of the ovary is a type of epithelial cell tumor.

Epithelial ovarian tumors arise from the mesoepithelial cells present on the ovarian surface. 80% of all ovarian cancers are epithelial cancers.

Clear cell carcinoma is also known as mesonephroid tumor. The histopathology of clear cell carcinoma reveals large cuboidal epithelial cells with abundant clear cytoplasm characteristically forming tubules, glands, and small cystic spaces. These cystic spaces are lined by hobnail cells. They are clear cells showing large dark nuclei protruding into the lumen. It is a highly malignant tumor.

Malignant epithelial tumors can spread widely via transcoelomic spread. It can spread through the capsule onto the peritoneal surface, omentum, intestinal viscera, and sub-diaphragmatic space. An ascitic tap will reveal blood-stained fluid with tumor cells. It may also spread to the para-aortic nodes and cause metastasis in the other ovary, liver, lungs, and gastrointestinal tract.

Solution to Question 36:

Among the given options, Asherman syndrome is most commonly associated with hypomenorrhea.

Asherman syndrome is the presence of uterine synechiae, due to the destruction of the endometrium. It can occur as a result of trauma to the endometrium following vigorous curettages such as incomplete abortion, postpartum hemorrhage, and elective abortion. Damage may also result from other uterine surgery such as metroplasty, myomectomy.

They usually present with menstrual abnormalities, infertility, recurrent pregnancy loss due to inadequate placentation. Menstrual abnormalities, including hypomenorrhea and amenorrhea, remain the most common clinical features associated with Asherman syndrome. Dysmenorrhea is occasionally present.

The diagnosis is confirmed with hysterosalpingography.

Management includes hysteroscopy, which enables adhesiolysis. Recurrence of adhesions is prevented by insertion of IUCD for 3 months and giving estrogen cyclically (21 days) for 3 months to develop endometrium.

Note:

Amenorrhea indicates the complete absence of menstruation.

Oligomenorrhoea is the term used to describe infrequent and irregular menses. In this condition, the cycle length is prolonged without affecting the duration and amount of flow.

Hypomenorrhoea refers to the condition in which the cycle length remains unaltered, however the duration of bleeding or the amount of blood loss, or both are substantially reduced.

Menorrhagia denotes excessive blood loss (increase in the duration and /or amount of bleeding) without any change in the cycle length.

Dysmenorrhoea is defined as cramping pain accompanying menstruation.

Solution to Question 37:

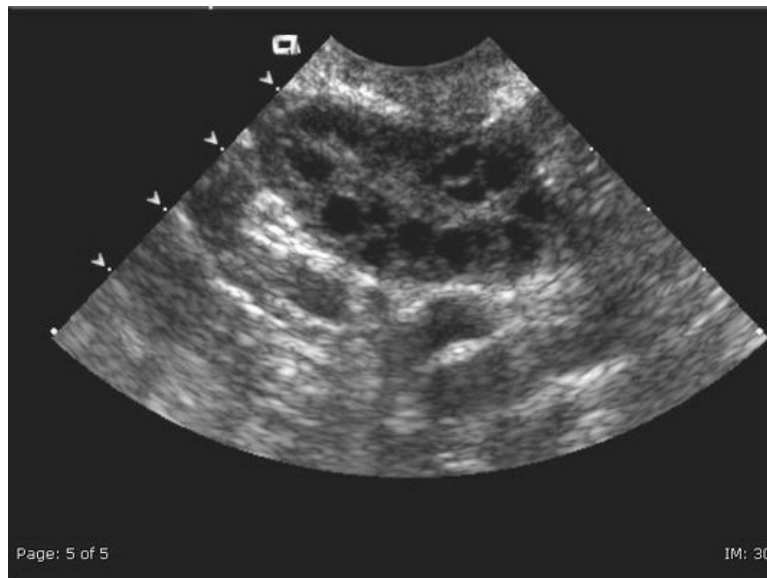
This patient's irregular menstrual cycles, increased facial hair (signifying clinical hyperandrogenism), and increased ovarian volume are suggestive of polycystic ovarian syndrome (PCOS). Anti-androgens, lifestyle modifications, and combined oral contraceptives (2, 3, and 4) constitute first-line treatment modalities for this condition. Laparoscopic drilling is a surgical procedure done when medical therapy fails.

PCOS is an endocrinopathy seen in women of reproductive age, affecting multiple systems. Rotterdam criteria is used to diagnose PCOD. The presence of two of the three components of this criteria establishes the diagnosis:

- Oligo/amenorrhoea, anovulation, infertility
- Hirsutism/acne

- Ultrasound findings:
- Enlarged ovaries with ovarian volume will be more than 10 mm³
- 12 or more small follicles each of 2-9 mm in size placed peripherally

Ultrasound image given below shows an enlarged ovary seen in PCOS with multiple follicles in the periphery.



Treatment of PCOD:

- Weight reduction alone is beneficial in mild hirsutism.
- Lifestyle changes, such as smoking cessation, are recommended.
- Fourth-generation oral contraceptive pills (OCP) containing ethinyl estradiol and drospirenone are used for treating menstrual abnormalities, reducing acne, and hirsutism.
- OCP with cyproterone acetate is prescribed for PCOS patients with hirsutism.
- Patients with PCOD are prone to developing insulin resistance, which can be controlled with metformin. Acarbosecan be used as an alternative if metformin is contraindicated.
- Infertility in PCOD patients is treated with clomiphene as the first-line drug for inducing ovulation. Dexamethasone is used in combination to improve fertility rates. Letrozole is used off-label in clomiphene-resistant cases.

Surgical options include laparoscopic ovarian drilling or puncture of not more than four cysts in each ovary. This is done either by laser or by unipolar electrocautery. It restores endocrine levels and improves fertility. Surgery is indicated only if:

- Medical therapy fails
- Hyperstimulation occurs
- Infertile women
- Previous pregnancy losses

Solution to Question 38:

The instrument shown in the image is Cusco's speculum.

Cusco's speculum is a self-retaining bivalve speculum used for visualisation of the cervix, vaginal fornices and for performing pap smear.



Other options:

Option B: Sims' speculum is used for the assessment of rectocele and cystocele. The speculum is applied against the vaginal wall and slowly retracted out until a bulge appears, indicating cystocele or rectocele. It can also be used for visualization of the vaginal walls.



Option C: A surgeon uses a lubricated proctoscope to visualize lesions in the anus, anorectal junction, and lower rectum. It is done as an OPD procedure to detect hemorrhoids, polyps, or ulcers.



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Option D: Doyen's retractor is used to retract the abdominal wall as well as the bladder to facilitate better exposure of the lower uterine segment during cesarian operation. It is also used to pull back the soft tissues to widen the surgical field during laparotomy.

Doyen's retractor



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Solution to Question 39:

The given clinical scenario of a girl with primary amenorrhea, normal breasts, and pubic hair, with a blind-ending vagina and uterus not palpable, is suggestive of Mullerian agenesis.

Absent uterus and vagina in a person with female genotype and phenotype are features of Mullerian agenesis, also known as Mayer-Rokitansky-Kuster-Hauser syndrome.

Ovaries are normal and secondary sexual characters are well developed with normal external genitalia. This condition is associated with anomalies of the spine/skeleton and the kidneys along

with the urinary tract.

Diagnosis is made with the help of pelvic sonography, MRI, and laparoscopy. Patients may have rudimentary Mullerian bulbs without an endometrial cavity. These should be surgically excised if they are symptomatic.

Other options:

Option B: Gonadal dysgenesis patients present with primary amenorrhoea, absence of secondary sex characters (poorly developed breasts), and female external genitalia.

Option C: In Turner syndrome, short stature with poor development of secondary sexual characters and short stature will be seen. It has a karyotype of 45 XO. Barr body will be absent. Female genital organs (vagina, uterus, fallopian tubes, ovaries) are present. The ovaries, though present, are atrophic.

Option D: Testicular feminization syndrome also known as androgen insensitivity syndrome has a karyotype of 46XY. Secondary sexual characters like the development of axillary hair and pubic hair are absent in these females. However, breast development is adequate due to the conversion of androgen into estrogen during puberty.

Solution to Question 40:

A female patient presenting with foul-smelling vaginal discharge and a wet smear showing clue cells points to a diagnosis of bacterial vaginosis, and the drug of choice is metronidazole.

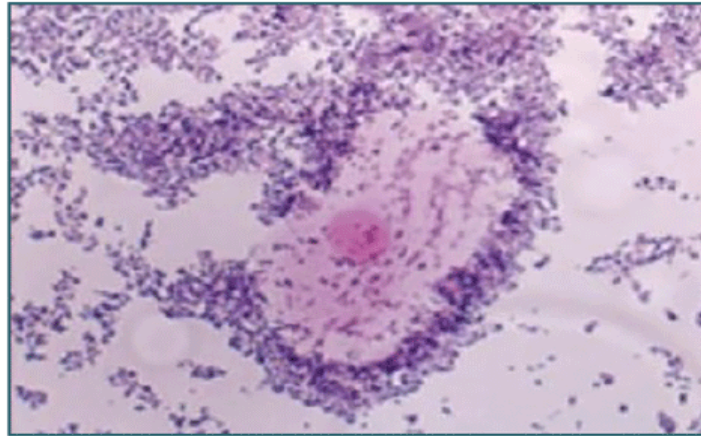
Bacterial vaginosis is a polymicrobial infection caused by *Mobiluncus curtisii*, *Gardnerella vaginalis*, *Mycoplasma hominis*, and *Haemophilus vaginalis*. It has a foul-smelling (fishy smell) vaginal discharge with minimal vulval irritation.

Amsel's criteria is helpful in diagnosing bacterial vaginosis when 3 of the following 4 are present:

- Thin, white homogenous vaginal discharge
- A pH of vaginal fluid ≥ 4.5
- Fishy odor of discharge accentuated by adding 10% KOH-Whiff test
- Presence of clue cells on microscopy: Clue cells are epithelial cells with a fuzzy border or granular cytoplasm caused by small gram-negative bacilli adhering to their surface

Treatment is oral metronidazole 500 mg BD for 7 days.

The image below shows clue cells:



©Marrow

Solution to Question 41:

Option B represents the correct answer. 1-C, 2-A, 3-B, 4-D

Endodermal sinus tumor/yolk sac tumor is a type of ovarian germ cell tumor that is seen in premenarchal girls. Schiller-Duval bodies are pathognomonic on histopathology. Alpha-fetoprotein (AFP) is commonly produced by this tumor and is used as a tumor marker.

Dysgerminoma is a type of germ cell tumor and it corresponds to the seminoma of the testis. It does not secrete either male or female sex hormones. It secretes placental alkaline phosphatase (ALP), lactate dehydrogenase (LDH), and β -hCG. Serum lactate dehydrogenase (LDH) is usually used in monitoring individuals or disease recurrence.

Choriocarcinoma is a type of mixed germ cell tumor. It is highly vascular. Histologically, the tumor shows syncytiotrophoblasts and cytotrophoblasts. It secretes large quantities of human chorionic gonadotropin (hCG) hormone, which is used as a tumor marker in the diagnosis and management of the tumor.

Malignant epithelial cell tumors of the ovary secrete CA-125. It is a glycoprotein and surface cell antigen which is secreted by malignant epithelial tumors of the ovary. A level of more than 35 U/mL suggests a malignant and residual tumor and indicates the need for chemotherapy. CA-125 is also raised in abdominal tuberculosis and endometriosis.

Ovarian tumors	Tumor marker
1. Yolk sac tumor	C. AFP
2. Dysgerminoma	A. LDH
3. Choriocarcinoma	B. Beta-HCG

Ovarian tumors	Tumor marker
4. Epithelial cell tumor	D. Ca-125

Solution to Question 42:

Among the given options, ultrasonography and MRI can be used in the diagnosis of placenta accreta.

The term placenta accreta spectrum (PAS)/morbidly adherent placenta refers to abnormal placental implantation characterized by the abnormally implanted, invasive, or adhered placenta. This is due to the absence of decidua basalis and poor development of the fibrinoid or Nitabuch layer.

Risk factors include prior cesarian delivery, prior uterine surgery, prior curettage, and endometrial ablation, associated placenta previa.

Types of the morbidly adherent placenta:

- Placenta accreta: Villi are attached to the myometrium.
- Placenta increta: Villi invade the myometrium.
- Placenta percreta: Villi penetrate through the myometrium and to or through the serosa.

Ultrasound imaging, color Doppler, and MRI have all been useful in the diagnosis of morbidly adherent placenta during pregnancy.

USG findings that suggest PAS are (1) placental lacunae, (2) thinning of the retroplacental myometrium, (3) disruption of the bladder-uterine serosal interface, (4) bridging vessels from the placenta to the bladder-serosal interface, and (5) a placental bulge that pushes outward and distorts the contour of the uterus.

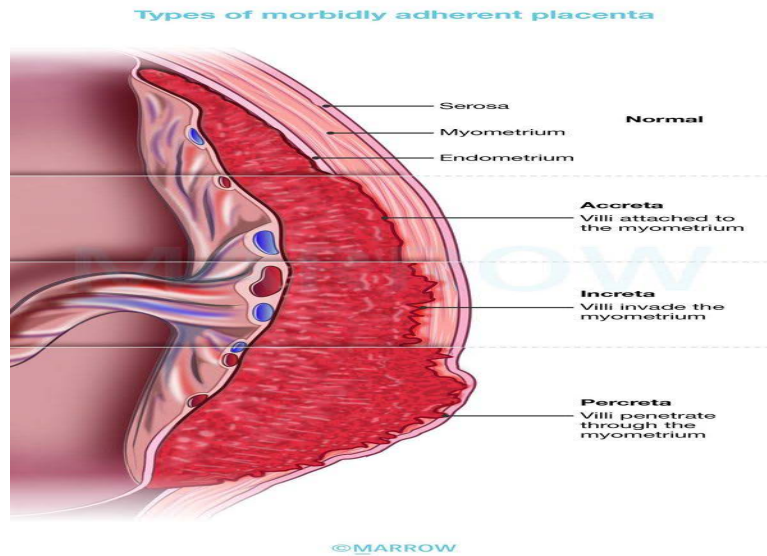
Color flow Doppler study shows hypervascularity of serosa bladder interface.

MRI reveals uterine bulging and dark intraplacental bands on T2-weighted imaging.

Management:

- Partial placenta accreta: Placental tissue is removed as much as possible. Oxytocin helps in uterine contraction and hemostasis. In cases following cesarean delivery, bleeding areas are oversewed.
- Total placenta accreta:
- Parous women- hysterectomy is done
- If the patient desires to have a child- conservative measures include incising the uterus above the placental attachment, clamping, and cutting the umbilical cord as close to its base as possible. The placenta is left behind, which may get autolyzed eventually. Uterine artery embolization or therapy with methotrexate has also been done for the conservation of the uterus.
- If placenta accreta invades the bladder, placental removal alone is avoided. It requires a hysterectomy and partial cystectomy.

The below image shows the placenta accreta, increta, and percreta.



Obstetrics & Gynecology NEET 2021

Question 1:

A pregnant woman at 36 weeks of gestation with a prosthetic valve replacement for mitral stenosis is on warfarin therapy. Her INR is 3. What is the appropriate next step in management?

- a) Stop warfarin and start LMWH
- b) Stop warfarin and start heparin
- c) Continue warfarin
- d) Stop warfarin, start LMWH and aspirin

Question 2:

A 27-year-old female patient was found to have uterus didelphys. Which of the following is not likely to be a complication of this uterine anomaly?

- a) Abortion
- b) Endometriosis
- c) Preterm labor
- d) Transverse lie

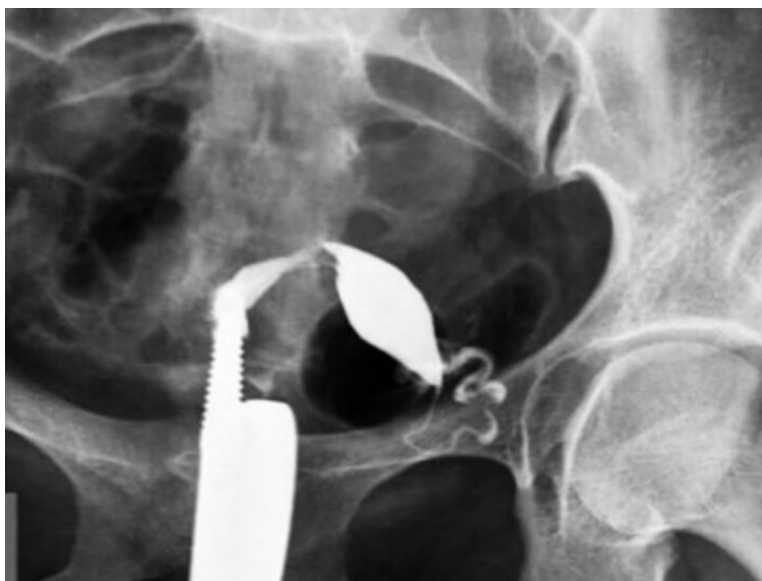
Question 3:

A woman who underwent dilatation and curettage for molar pregnancy 6 months ago presents with persistent symptoms of abdominal pain and distention. On further evaluation, the beta-hCG levels were persistently elevated and cannonball foci were found on the chest X-ray. What is the appropriate management in this case?

- a) Methotrexate with folinic acid
- b) Methotrexate
- c) EMACO regimen
- d) Hysterectomy

Question 4:

A female patient has a history of three miscarriages. Identify the investigation that was done from the image shown below.



- a) Hysteroscopy
- b) Saline infusion sonography
- c) Hysterosalpingography
- d) Transvaginal ultrasonography

Question 5:

A 56-year-old multipara woman presented with grade II/III uterine prolapse with cystocele. She complains of passing urine on coughing and sneezing. What is the type of urinary incontinence seen in this patient?

- a) Overflow incontinence
- b) Urge incontinence
- c) Stress incontinence
- d) Neurogenic bladder

Question 6:

A 32-year-old married woman who came for the evaluation of infertility was found to have a septum in her uterine cavity. What would be the appropriate management of this case?

- a) Hysterectomy

- b) Laparoscopy and metroplasty
- c) Hysteroscopy and septoplasty
- d) Laparotomy and metroplasty

Question 7:

A 30-year-old G4A3 woman had a history of miscarriage at 8 weeks, the second miscarriage at 11 weeks with no cardiac activity, and the third pregnancy loss at 24 weeks with preterm delivery due to early-onset preeclampsia. What is the most probable diagnosis?

- a) Syphilis
- b) Antiphospholipid antibody syndrome
- c) TORCH infections
- d) Gestational diabetes mellitus

Question 8:

A 26-year-old female patient came with a history of fatiguability and constipation for the past 1 year. She has a history of multiple blood transfusions after delivery 4 years ago. She has had a lactational failure and never regained her periods after the delivery. What is the most probable diagnosis?

- a) Sheehan syndrome
- b) Hypothyroidism
- c) Premature ovarian failure
- d) Delayed blood transfusion reaction

Question 9:

The ultrasound of a young woman with infertility is suggestive of a uterine anomaly. Which of the following is used to confirm the diagnosis?

- a) Hysterosalpingography
- b) Hysteroscopy + laparoscopy
- c) Laparoscopy
- d) Transvaginal sonography

Question 10:

A 26-year-old pregnant woman, on examination was found to have infra-umbilical flattening. The fetal heart sounds were heard over the lateral side. What could be the fetal presentation/position?

- a) Brow
- b) Knee
- c) Occipitoposterior
- d) Right dorso-anterior

Question 11:

A primi at term presented with labor pain and multiple painful vesicular lesions on her vulva. No rashes were found on other parts of the body. What will be the management of this case?

- a) Acyclovir and elective Cesarean section
- b) Acyclovir and induction of labor
- c) Spontaneous delivery
- d) Antiviral and normal vaginal delivery

Question 12:

A 30-year-old married woman came for an infertility evaluation. On performing further investigations, intramural fibroids of sizes 7x5cm and 5x3cm were seen at the right and left cornual ends of the uterus. Her husband's semen analysis was found to be normal. What is the appropriate management of this case?

- a) Uterine artery embolization
- b) Myomectomy
- c) Artificial reproductive techniques
- d) GnRH analogues

Question 13:

A 28-year-old married woman who is anxious to conceive presents with copious vaginal discharge, without any itching. Her last menstrual period was 13 days ago. What is the cause of her vaginal discharge?

- a) Candidiasis
- b) Bacterial vaginosis
- c) Trichomoniasis
- d) Physiological

Question 14:

A patient developed atonic postpartum hemorrhage after a normal vaginal delivery. She was shifted to the OT and the obstetrician planned ligation of vessels to arrest the bleeding. What is the order of ligation of the blood vessels supplying the uterus?

- a) Uterine artery - Ovarian artery - Internal iliac artery
- b) Uterine artery - Ovarian artery- External iliac artery
- c) Uterine artery - Ovarian artery - Vaginal artery
- d) Uterine artery - pudendal artery - Vaginal artery

Question 15:

A teenage patient presented with a history of dysmenorrhea and chronic pelvic pain. On ultrasonography, a transverse vaginal septum was seen. What is the most common complication that can be seen in this patient?

- a) Endometriosis
- b) Tubo-ovarian abscess
- c) Dermoid cyst
- d) Theca lutein cyst

Question 16:

A 28-year old pregnant woman came for a routine antenatal check-up in her second trimester. Four years back, she delivered 2 healthy full-term twins. What would her obstetric score be?

- a) G2P2
- b) G3P2
- c) G2P1
- d) G3P1

Question 17:

A 27-year-old woman who is on oral contraceptive pills for contraception has forgotten to take her pills on 4 different days in the last 2 weeks. She presents to the primary health center, asking for advice. What would you suggest to her?

- a) Take 4 pills now, continue the schedule, use barrier method of contraception
- b) Take 1 pill now, continue the schedule, and use barrier method of contraception
- c) Take 1 pill now, use barrier method, and consider emergency contraception in case of recent intercourse
- d) Discard the pack, start a new pack with alternate method of contraception

Question 18:

A 30-year-old female patient, on treatment for infertility, came with 6 weeks of amenorrhea, abdominal pain, and mild vaginal bleeding. Her beta hCG level was 2800 mIU/mL. On ultrasound, a 3cm mass was seen in the left adnexa. It contained a gestational sac with no fetal cardiac activity. What is the most appropriate management for this patient?

- a) Expectant management
- b) Single dose methotrexate
- c) Salpingectomy
- d) Milking of the tube

Question 19:

A 30-year-old woman on oral contraceptives for the past 6 months presents with 6 weeks of amenorrhea. What would be the best method of assessing the gestational age of this patient?

- a) Crown rump length by USG
- b) Adding 280 days to the LMP
- c) Fundus examination
- d) Adding 256 days to the LMP

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	c
5	c
6	c
7	b
8	a
9	b
10	c
11	a
12	b
13	d
14	a
15	a
16	c
17	d
18	b
19	a

Detailed Explanations

Solution to Question 1:

The next appropriate step in the management of a pregnant woman with a prosthetic valve on warfarin therapy at 36 weeks of gestation is to stop warfarin and start low molecular weight heparin (LMWH).

Warfarin is safe and effective in preventing thromboembolism in the mother during the 2nd trimester. However, it is teratogenic and can lead to miscarriage, stillbirths, fetal anticoagulation, and fetal embryopathy if used in the 1st trimester.

Heparin is safe for the fetus as it does not cross the placenta but is less effective than warfarin in preventing thromboembolism.

LMWH- Low molecular weight heparin, UFH- Unfractionated heparin

For patients taking LMWH, factor Xa levels should be monitored weekly.

Heparin should be discontinued just before (at least 6 hours before) delivery. If delivery occurs before discontinuation and there is extensive bleeding, then iv protamine sulfate can be given.

Warfarin or heparin anticoagulation can be restarted after 6 hours of vaginal delivery, usually without any problem.

Warfarin, LMWH, and UFH do not accumulate in breast milk and do not have an anticoagulant effect in the newborn. They are compatible with breastfeeding.

Other options:

Option B: Unfractionated heparin can also be given instead of LMWH. However, in this case, the better answer is to give LMWH; because, in clinical practice, LMWHs have a long half-life, once-daily dosing, and fewer side effects. If LMWH is unavailable or cost-prohibitive, or if anti-Xa levels cannot be monitored, continuous infusion of UFH can be used as an alternative to LMWH.

Option D: Aspirin 75-100 mg can be given daily during pregnancy. It can be continued in women with mechanical heart valves if needed for other indications.

Period of gestation	Anticoagulant
Till 12 completed weeks	LMWH or UFH
13 to 36 weeks	Warfarin
36 weeks till just before delivery	LMWH or UFH
6 hours after vaginal delivery and 24 hours after cesarean section	Warfarin, LMWH or UFH

Solution to Question 2:

Among the given options, transverse lie is not a likely complication of uterus didelphys. A transverse lie is more likely to be seen with an arcuate or sub-septate uterus.

A uterus didelphys is a Mullerian defect characterized by two separate uterine horns, each with an endometrial cavity and uterine cervix. It occurs due to the failure of fusion of the Müllerian ducts.

It is usually asymptomatic but can rarely present with dyspareunia, dysmenorrhea, hematocolpos, and hematometra. Rarely, endometriosis (option B) and genital neoplasms may be associated with uterus didelphys.

Pregnancy is possible with uterus didelphys, however, there is an increased risk of the following complications:

- Spontaneous abortion (option A)
- Preterm labor (option C)
- Breech presentation
- Cesarean delivery
- Intrauterine growth restriction
- Low birth weight (<2.5kg)
- Perinatal death

Uterus didelphys can be diagnosed with ultrasound or hysterosalpingography, but MRI provides the most accurate diagnosis.



Uterus didelphys

Solution to Question 3:

The given clinical scenario of a woman with a past history of D(&)C for molar pregnancy, now presenting with abdominal pain and distension, elevated beta-HCG levels, and cannonball metastasis to the lung is suggestive of gestational trophoblastic neoplasia (GTN).

The next step is to categorize the patient as low/high risk based on the WHO prognostic scoring system to decide the treatment modality.

WHO Prognostic Scoring System, adapted by FIGO:

Based on the available clinical information,

- Antecedent gestation: molar pregnancy, score= 0
- Interval: 6 months, score= 1
- Site of metastasis: lung, score= 0

The total score is 1, hence it is a case of low-risk GTN and the management involves treatment with methotrexate with folinic acid.

Treatment of GTN:

Low-risk GTN (FIGO Stages I-III: score ≤ 7): Single-agent chemotherapy with methotrexate (MTX) or dactinomycin (less commonly used).

- 50 mg MTX is given IM on days 1, 3, 5, and 7 with folinic acid on alternate days.

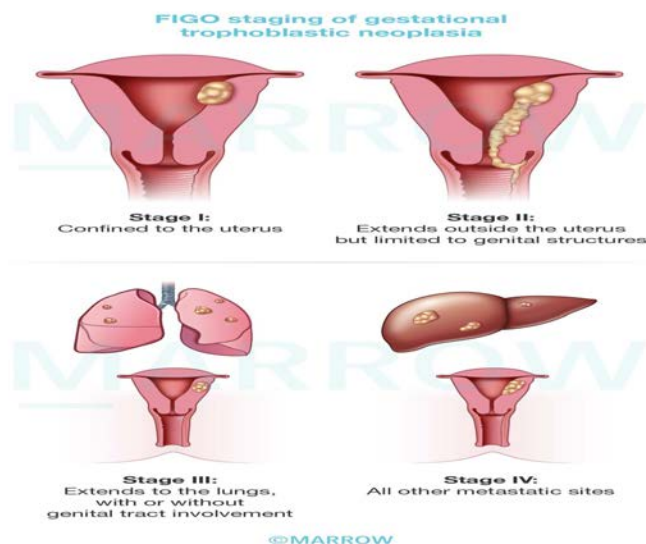
High-risk GTN (FIGO Stages II-III: score > 7 and Stage IV): Multiple agent chemotherapies, with or without adjuvant surgery or radiotherapy. The preferred multiple-agent chemotherapy regime is the EMACO regime:

- Etoposide

- Methotrexate
- Dactinomycin (actinomycin D)
- Cyclophosphamide
- Vincristine (Oncovin)

The FIGO Anatomic Staging of GTN is shown in the image below:

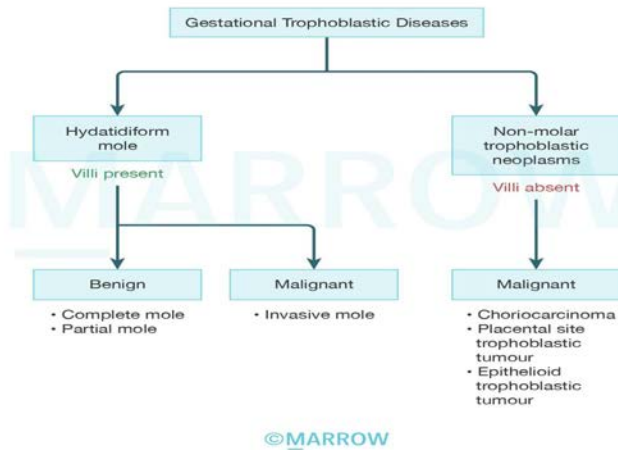
Prognostic Factor	Score			
0	1	2	4	
Age	< 40	>40	-	-
Antecedent gestation	Mole	Abortion	Term	-
Interval in months	<4	4-6	7-12	>12
Pretreatment serum hCG (IU/L)	< 103	103to < 104	104to < 105	> 105
Largest tumor size (including the uterus)	<3	3-4	≥ 5	-
Site of metastases	Lung	Spleen, kidney	GI tract	Brain, liver
Number of metastases	-	1-4	5-8	>8
Previously failed chemotherapy	-	-	single drug	≥ 2 drugs
Low-risk GTN: score ≤6High-risk GTN: score ≥7				



GTN typically develops along with or after a molar pregnancy. Rarely, it may develop after live birth, miscarriage, or medical termination of pregnancy.

The given image below shows the classification of gestational trophoblastic diseases:

Classification of Gestational Trophoblastic Diseases



Solution to Question 4:

The investigation shown in the given image is hysterosalpingography. The given hysterosalpingogram shows a unicornuate uterus.

Hysterosalpingography (HSG) is an imaging modality in which radiopaque contrast is injected through the cervical canal to assess the endometrial cavity, the fallopian tubes, and the endocervical canal.

HSG should be done between days 5 and 10 of the menstrual cycle, in the preovulatory phase, as an outpatient procedure. Leech Wilkinson cannula, Rubin's cannula, or Foley's catheter no.14 is introduced into the endometrial cavity through the cervical os, and the contrast is injected. The uterine cavity and fallopian tubes are then visualized and X-rays are obtained.

HSG can be used for the following:

- To assess tubal patency in infertility
- To detect uterine anomalies, synechiae, and polyps
- To detect incompetence of the cervical os

Contraindications to HSG:

- Acute pelvic infection
- Active uterine bleeding
- Pregnancy/ suspected pregnancy
- Contrast allergy
- Premenstrual phase

HSG is a safe procedure, however, complications such as ascending spread of infection, chemical peritonitis, allergic reaction to the dye, or pelvic endometriosis may occur.

Leech-Wilkinson cannula



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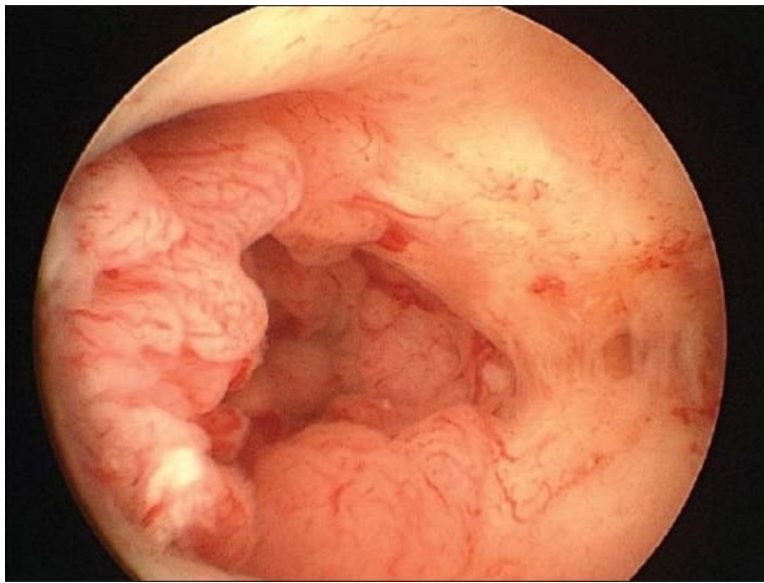
Rubin's insufflation cannula



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Other options:

Option A: Hysteroscopy is an operative procedure in which the uterine cavity can be visualized by passing a fiberoptic telescope via the cervix. It can be diagnostic or therapeutic. A hysteroscopic image of a uterus with carcinoma is shown below.



Option B: Saline infusion sonography is a procedure where normal saline is injected into the uterine cavity and the uterus is imaged with transvaginal sonography. It is shown in the image below.



Option D: Transvaginal ultrasonography is a noninvasive imaging technique using the transvaginal approach. It provides a more detailed evaluation of the pelvic organs as compared to the transabdominal approach.

Solution to Question 5:

The type of urinary incontinence present in a patient with prolapse and cystocele experiencing urination on coughing and sneezing is stress incontinence.

Stress incontinence refers to the involuntary leakage of urine when there is increased intra-abdominal pressure (eg, coughing, sneezing, laughing, Valsalva maneuver). Absence of detrusor muscle contraction is seen in urodynamic studies.

Causes:

- Age and menopause - due to estrogen deficiency
- Multiparity - due to loss of pelvic muscle tone
- Obesity
- Smoking
- Constipation, chronic cough
- Pelvic organ prolapse
- Pregnancy - due to the effect of progesterone and pressure of the gravid uterus on the bladder neck.
- Puerperium - due to descent of the bladder neck, loss of urethro-vesical angle, and reduced pelvic muscle tone
- Surgical repair of vesicovaginal fistula

Grading of genuine stress incontinence (GSI) :

- Grade I: Incontinence with only severe stress, such as coughing, sneezing, and jogging.
- Grade II: Incontinence with moderate stress, such as fast walk, going up and down the stairs.
- Grade III: Incontinence with mild stress such as standing.

Investigations are only done prior to surgery to confirm the diagnosis. They include:

- Stress test
- Cotton swab or Q-tip test
- Marshall and Bonney's test
- Urethroscopy
- Urodynamic studies.

These are not required prior to conservative management.

Conservative management includes Kegel's pelvic exercises, electric nerve stimulation, devices like pessaries, vaginal cones, and artificial sphincters. Medical treatment with estrogen creams, alpha agonists, venlafaxine, and imipramine may be useful.

Other options:

Option A: Overflow incontinence is an old term that refers to the inability to empty the bladder, with involuntary, continuous urinary dribbling. Currently, it refers to another presentation of urge incontinence.

Option B: Urge incontinence is the involuntary escape of urine when there is an imminent need to urinate. It occurs due to detrusor muscle overactivity or detrusor instability. It is seen more commonly in old women with decreased bladder capacity, decreased sensation, and CNS inhibition.

Option D: Neurogenic bladder occurs in patients with neurogenic dysfunction leading to detrusor hyperreflexia. Conditions such as multiple sclerosis, stroke, Parkinson's disease, and spinal injury give rise to this. They present with urgency or urge incontinence.

Solution to Question 6:

The appropriate management for a 32-year-old woman with a septate uterus is hysteroscopy and septoplasty.

Hysteroscopic septoplasty refers to the resection of the uterine septum under hysteroscopic guidance. The septum can be resected either with a resectoscope or a laser. It does not scar the uterus and avoids the necessity of cesarean sections for future pregnancies. There is a reduced risk of pelvic adhesions, compared with transabdominal approaches.

A septate uterus is characterized by a septum in the endometrial cavity. It is caused by the failure of the medial segments of the Müllerian ducts to regress. This creates a permanent septum within the uterine cavity.

The septum can project partially from the fundus or extend completely to the cervical os. It is the most common congenital uterine malformation in women with infertility.

Other options:

Option A: Hysterectomy is not advisable since the patient has come for infertility evaluation and desires pregnancy.

Options B & D: Laparoscopy/laparotomy and metroplasty are not preferred for the treatment of a septate uterus due to the following:

- Requirement of cesarean section for subsequent pregnancies to prevent uterine rupture.
- Postoperative pelvic adhesions leading to infertility.

Note: Jone's and Tompkin's metroplasties are surgeries for the septate uterus that involve a transabdominal approach. They cause uterine scarring and are hence not preferred.

Solution to Question 7:

The given clinical scenario of a G4A3 woman with miscarriages at 8 and 11 weeks, and preterm delivery due to preeclampsia at 24 weeks is suggestive of antiphospholipid antibody (APLA) syndrome.

APLA is an autoimmune disorder commonly associated with fetal death, preterm delivery, early-onset preeclampsia, and fetal growth restriction due to placental insufficiency and thromboses. Pregnancy loss usually occurs before 10 weeks.

Diagnosis of APLA requires at least 1 clinical criterion and 1 laboratory criterion.

- Obstetric criteria
- One or more unexplained deaths of a morphologically normal fetus at or beyond 10 weeks or
- Severe preeclampsia or placental insufficiency requiring delivery before 34 weeks, or

- Three or more unexplained consecutive spontaneous abortions before 10 weeks
- Laboratory criteria
- Presence of lupus anticoagulant, or
- Anticardiolipin antibodies, or
- Anti- β_2 glycoprotein antibody

Recurrent pregnancy loss (RPL), also referred to as a recurrent miscarriage or habitual abortion, is historically defined as 3 consecutive pregnancy losses prior to 20 weeks from the last menstrual period.

Common causes include:

- Parental chromosomal abnormalities
- Untreated hypothyroidism
- Uncontrolled diabetes mellitus
- Uterine anatomic abnormalities,
- Antiphospholipid antibody syndrome

Other options:

Option A: There is no history of STD or skin lesions suggestive of syphilis and according to Kassawitz rule, in women with untreated syphilis having a series of pregnancies, the likelihood of fetal infection in later pregnancies becomes less; usually, only the first pregnancy is an abortion, the following pregnancy losses will be stillbirths in the next trimester rather than recurrent pregnancy loss.

Option C: TORCH infections may play a role in sporadic spontaneous pregnancy loss, rather than recurrent pregnancy losses.

Option D: Gestational diabetes mellitus (GDM) refers to the onset or first recognition of diabetes during pregnancy, usually at the 24-28 weeks screening test. Women with uncontrolled overt or pregestational diabetes with elevated HbA1c or preprandial glucose in the first trimester have an increased risk for early miscarriage, but there is a history or lab findings suggestive of gestational diabetes mellitus.

Solution to Question 8:

The most likely diagnosis in a woman with lactational failure and postpartum amenorrhea, now presenting with fatigability and constipation is Sheehan syndrome. A history of multiple blood transfusions after delivery indicates major obstetric blood loss, which resulted in pituitary ischemia leading to this condition.

Sheehan syndrome is panhypopituitarism due to pituitary ischemia and necrosis following massive obstetrical hemorrhage.

The most severe form presents with shock due to pituitary apoplexy. It is characterized by a sudden onset headache, nausea, visual deficits, and hormonal dysfunction due to acute pituitary hemorrhage or infarction. Less severe forms present with anovulation and amenorrhea.

Damage to specific pituitary cell types leads to lactational failure, loss of axillary and pubic hair, and symptoms of hypothyroidism or adrenal insufficiency. Prolactin deficiency is most commonly encountered, but adrenal insufficiency is the most life-threatening complication. Hence, the adrenal function should be immediately assessed if there is any suspicion of Sheehan syndrome.

Option B: Hypothyroidism can present with vague symptoms such as fatigue, constipation, and amenorrhea. However, postpartum hypothyroidism usually presents 4-8 months after delivery after a thyrotoxic phase with palpitations and thyroid enlargement.

Option C: Premature ovarian failure presents with menopausal symptoms. It is due to primary ovarian dysfunction. It is diagnosed when serum FSH levels are >30 -40 mIU/mL on two occasions, at least one month apart.

Option D: Delayed transfusion reaction presents days to weeks after the transfusion. It may be asymptomatic or with low-grade fever. Coombs test may be positive, unconjugated bilirubin increased, and the rise in hemoglobin levels following the transfusion will be less than expected.

Solution to Question 9:

Among the given options, hysteroscopy with laparoscopy is used to confirm the diagnosis.

For infertility evaluation, hysteroscopy and laparoscopy can detect müllerian anomalies, screen for co-existent endometriosis, and exclude other tubal or uterine cavity pathology.

This approach can differentiate between a septate and a bicornuate uterus. On laparoscopy, the presence of a uterine fundus indicates that the uterus is septate, while its absence indicates it is bicornuate. The capacity of each horn in a bicornuate uterus and the depth and thickness of the uterine septum prior to surgical resection can also be assessed.

MRI is the gold standard investigation for the diagnosis of Mullerian anomalies. It clearly delineates both the internal and external uterine anatomy. Complex anomalies and common associations like renal or skeletal anomalies can also be evaluated. It is also less invasive than hysteroscopy and laparoscopy in diagnosing Mullerian anomalies.

Other options:

Option A: Hysterosalpingography (HSG) provides information about the interior contour of the uterine cavity but cannot accurately visualize the external contour of the uterus.

Option C: Laparoscopy alone as a diagnostic procedure cannot visualize the interior contour of a hollow viscus like the uterus. This can be done only with the help of hysteroscopy or imaging such as sonography, HSG, or MRI.

Option D: Transvaginal sonography is used for the initial detection of uterine anomalies causing infertility or spontaneous abortions. Complete evaluation of these anomalies is done by MRI.

Note:

- Hysteroscopy is the gold standard investigation to visualize the uterine cavity and tubal ostia. It also has therapeutic benefits in cases of endometrial polyps, submucous resection of myoma, hysteroscopic adhesiolysis, and resection of uterine septum.
- Laparoscopy is the gold standard investigation to evaluate tubal factors of infertility.

Solution to Question 10:

The presence of infra-umbilical flattening and fetal heart sounds on the lateral side indicates that the fetus is in the occipito-posterior position.

Occipito-posterior (OP) position of the fetus is when the occiput is situated posteriorly over the sacroiliac joint, or directly over the sacrum. The fetus is in vertex presentation.

Examination findings of OP position:

- On inspection, the abdomen appears flattened below the umbilicus (infra-umbilical flattening).
- Pelvic grips:
- The head is not engaged.
- The sinciput (cephalic prominence) is not prominently felt.
- Auscultation: Fetal heart sounds are heard maximally over the flanks. However, in the direct OP position, they are heard in the midline.

There are three variants of OP position:

- Left OP position (fourth position of the vertex) - When the occiput points towards the left sacroiliac joint.
- Right OP position (third position of the vertex) - When the occiput points towards the right sacroiliac joint. It is more common than the left OP position.
- Direct OP position - When the occiput points directly towards the sacrum.

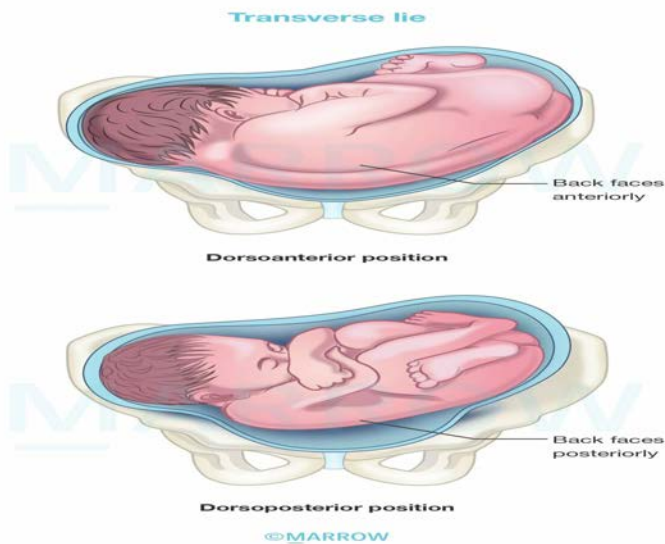
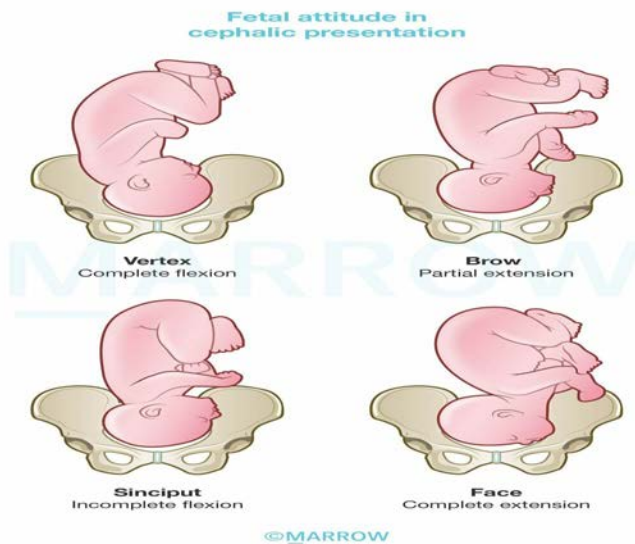
All three positions may be primary (present before the onset of labor) or secondary (develops after the onset of labor).

Other options:

Option A: Brow presentation is a rare type of cephalic presentation where the position of the head is in between full flexion and full extension. The part of the head between the orbital ridge and the anterior fontanel presents at the pelvic inlet.

Option B: Knee presentation is a type of breech presentation. Here, the thighs are extended but the knees are flexed, presenting at the pelvic inlet. The head will be palpable as a hard globular mass at the fundus.

Option D: Right dorso-anterior position is when the fetal back is directed anteriorly. This occurs in a transverse lie. On examination, the abdomen appears wide, and a hard plane of resistance is felt. The fundus extends just above the umbilicus. The head is felt in one iliac fossa and the breech is felt in the other.



Solution to Question 11:

The given clinical scenario of a primi in labor pain with multiple, painful vesicular lesions on her vulva is suggestive of an active herpes simplex virus (HSV) infection. Management involves the administration of acyclovir and elective cesarean section.

HSV infection in pregnancy can be caused by either HSV-1 or HSV-2 and transmitted by sexual contact. The infection can be primary, nonprimary, first episode, and recurrent.

HSV infection (both HSV 1 and 2) during pregnancy has a risk of vertical transmission of the infection to the fetus. The risk is higher in the primary than in the recurrent infection. The maximum risk of fetal infection is during labor, although it can occur in utero and post-natally from direct contact.

Suspicious lesions should be sent for PCR testing or culture. Antiviral treatment with acyclovir should be given if found positive. Women with no active genital lesions, asymptomatic, or those with active lesions in non-genital areas can undergo normal labor and delivery.

Management of HSV infection during pregnancy:

- Oral acyclovir or valacyclovir for primary or recurrent infections during pregnancy.
- IV acyclovir followed by oral acyclovir for severe or disseminated infection
- Oral analgesics or topical anesthetics for pain relief
- Catheterization for urinary retention

Cesarean delivery reduces the risk of vertical transmission during labor and is indicated in:

- Cases of active genital lesions - both primary and recurrent
- Prodromal symptoms like vulval itching or burning

Other options:

Option B: The presence of active lesions is a contraindication for induction of labor.

Option C: Spontaneous delivery is contraindicated in the presence of active lesions.

Option D: Even after treatment with acyclovir, normal delivery is not indicated owing to the risk of dormant viral transmission to the fetus.

Solution to Question 12:

The most appropriate management for a woman with infertility due to intramural fibroids at both the cornual ends of the uterus is myomectomy. The infertility, in this case, is due to the fibroids blocking the right and left cornua of the uterus.

Myomectomy refers to the removal of fibroids and preserving the uterus. It is indicated in women presenting with infertility, those who desire pregnancy, or those who wish to retain the uterus.

Intracavitary fibroids can be resected via the hysteroscopic approach, while subserosal or intramural fibroids require laparotomy or laparoscopic myomectomy.

Other options:

Option A: Uterine artery embolization is not done in this case, as it may decrease fertility, as a result of pelvic adhesions. It is a technique in which polyvinyl alcohol (PVA), gel foam particles, or metal coils are injected into the uterine arteries. This embolization reduces vascularity and the size of the fibroid. It is best suited to control menorrhagia in multiparous women.

Option C: Artificial reproductive techniques are not indicated in this case as infertility is seen to be due to the fibroids blocking the cornual ends of the uterus. However, it may be considered at a later stage.

Option D: GnRH analogs when used for 6 months, can reduce the tumor size by targeting the growth effects of estrogen and progesterone. It can be used in women with infertility caused by cornual fibroids. They are often used preoperatively to reduce the size of the fibroid and facilitate a hysteroscopic approach to surgery. However, it is expensive and can cause menopausal symptoms if used for >6 months. Fibroids are known to regrow 3-4 months after stopping the

therapy.

Solution to Question 13:

The given clinical scenario of a young woman with copious, non-pruritic (without itching) vaginal discharge occurring 13 days after menstrual period (period of ovulation) is most likely to be leucorrhoea (physiological).

The term leucorrhoea can be used in conditions where normal vaginal secretion is increased in amount. An increase in this normal vaginal secretion develops physiologically:

- At puberty
- At ovulation
- During pregnancy
- During the premenstrual phase, occasionally

It may be cervical or vaginal in origin. A speculum examination of the vagina can distinguish between the two. If the discharge is cervical in origin, an excessive mucoid discharge will be visible at the external os.

Normal discharge appears clear or white, without any odor or itching, and will have a pH of 3.8-4.2.

Other options:

Option A: Candidiasis is caused by infection with *Candida albicans* and presents with pruritus, burning, and dysuria with vulvar erythema, edema, and excoriations. A curdy white discharge will be present.

Option B: Bacterial vaginosis is caused by an alteration in the normal vaginal flora and presents without itching and a malodorous grey-white discharge. It is diagnosed using Amsel's criteria, where at least three of the four following criteria should be present to confirm the diagnosis:

- Homogenous vaginal discharge
- Vaginal pH $\geq$ 4.5
- Amine-like odor on adding KOH (Whiff test positive)
- Clue cells ($\geq$ 20% of cells)

Option C: Trichomoniasis is caused by *Trichomonas vaginalis* and presents with a copious, frothy, offensive yellow-green discharge with vaginal soreness or irritation and urinary frequency. On examination, the vagina looks red and inflamed. The inflamed cervix looks like a strawberry with punctate hemorrhagic spots.

Solution to Question 14:

The order of ligation of vessels to arrest bleeding in atonic postpartum hemorrhage (PPH) is the uterine artery - ovarian artery - internal iliac artery.

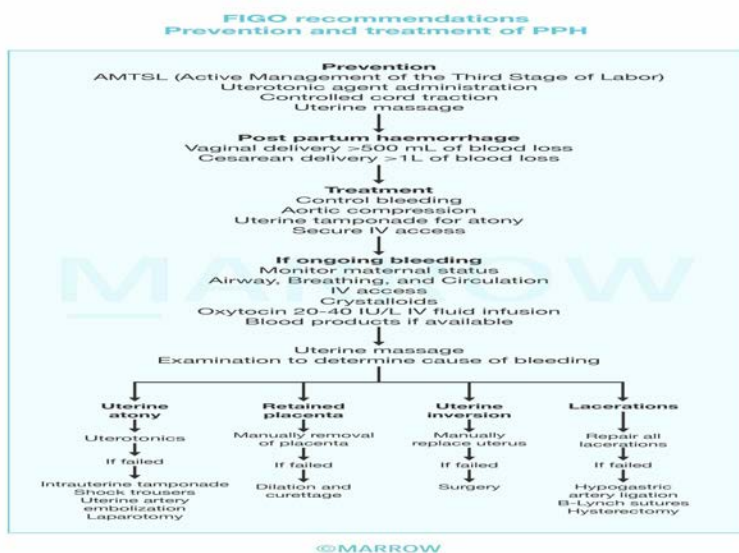
The ascending branch of the uterine artery is ligated between the upper and lower segments of the uterus, at the lateral border. If the bleeding continues, the ovarian and uterine artery anastomosis is ligated just below the ovarian ligament. If the bleeding still persists, the anterior division of the internal iliac artery is ligated 5 cm distal to the common iliac bifurcation.

Unilateral or bilateral ligation of the internal iliac artery causes a reduction in the pulse pressure in the arteries distal to the ligation. This increases the chances of hemostasis via pressure and clot formation.

Surgical methods to control PPH are outlined as stepwise uterine devascularization procedures:

- Compression sutures: B-Lynch suture
- Ligation of uterine arteries
- Ligation of ovarian and uterine anastomosis
- Ligation of internal iliac artery
- Angiographic embolization of the bleeding vessel

Pelvic packing (usually done prior to embolization) and hysterectomy are other surgical methods employed to control PPH.



Solution to Question 15:

The most common complication that can be seen in this patient is endometriosis.

Transverse vaginal septum leads to obstructed flow of menstrual blood. This may lead to retrograde menstruation, which increases the likelihood of the development of endometriosis, according to Sampson's theory of endometriosis.

Transverse vaginal septum occurs as a result of the failure of Müllerian duct fusion or failure of vaginal plate canalization. It may be obstructive or non-obstructive. The commonest site of a transverse septum is the junction between the upper and middle third of the vagina.

Patients with an obstructive or imperforate vaginal septum present with symptoms similar to the imperforate hymen. Symptoms include amenorrhoea, cyclical abdominal pain, urinary symptoms, and a palpable pelvic mass. The vagina may appear shortened and the cervix may not be identifiable. Secondary sexual characteristics appear normal. The diagnosis can be confirmed with ultrasound or MRI.

There is an increased risk of endometriosis. Other complications such as pyomucocolpos, pyometra, and pyosalpinges can also develop in infants with obstructive vaginal septums.

Treatment is by manual dilatation of the perforation or surgical excision of the septum.

Other options:

Option B: Tubo-ovarian abscess develops in pelvic inflammatory disease.

Option C: Dermoid cysts are cystic ovarian tumors that are often associated with mucinous cystadenomas.

Option D: Theca lutein cysts are found in association with hydatidiform moles, choriocarcinoma, and gonadotropin or clomiphene citrate therapy. They regress with the management of the mole and choriocarcinoma and discontinuation of gonadotropin therapy.

Solution to Question 16:

The obstetric score of a pregnant woman who previously delivered twins would be G2P1 (gravida 2 para 1).

The term gravida refers to the number of times the woman has been pregnant, irrespective of whether it crossed the period of viability. The current pregnancy is included in this count. Since she has been pregnant once before with twins and is currently pregnant now, gravida will be 2.

The term para refers to the number of previous pregnancies reaching a period of viability (gestation of 28 weeks). It does not increase to a higher number if multiples (twins/triplets) are delivered in a given pregnancy and still count as one birth. A current pregnancy, if any, is not included in this count. Therefore, para here will be 1 since she had one birth of twins and the present pregnancy is not counted.

Both gravidity and parity refer to the pregnancies and not the fetuses.

Note: The period of viability is when the fetus is capable of independent existence. The WHO defines it to be 24 weeks while the Indian definition is 28 weeks.

Solution to Question 17:

A woman who has missed 4 oral contraceptive pills in the last two weeks should take one hormonal pill as soon as possible and finish all hormonal pills in the pack as scheduled, throw away the 7 non-hormonal pills in a 28-pill pack and start a new pack the next day.

If any non-hormonal pills were missed, she can discard the remaining pills. Start the new pack of oral contraceptives and continue taking as usual.

If she vomits within 2 hours of taking a pill, another pill from the pack should be taken as soon as possible. After that, continue taking the pills as scheduled. If the vomiting or diarrhea persists for >2 days, follow the instructions for 1 or 2 missed pills as mentioned above.

The contraceptive method should be discontinued and changed in the following situations:

- Unexplained or prolonged heavy vaginal bleeding that suggests an underlying condition not related to the method
- Starting anticonvulsant medication or treatment with rifampicin
- Migraines
- Non-ambulatory conditions lasting for >1 week like major surgery, leg in a cast, etc. Combined oral contraceptives can be restarted 2 weeks after ambulation.
- Suspected heart or liver disease, hypertension, deep vein thrombosis, stroke, breast cancer, diabetes, or gall bladder disease

Missed pills	Management
Missed 1 or 2 pills/ started new pack 1 or 2 days late?	You have two options: take one hormonal pill right away, or take two pills at your usual scheduled time. The risk of pregnancy is minimal to none.
Missed 3 or more pills in the first or second week/started new pack 3 or more days late?	Take one hormonal pill immediately and then continue taking your pills as scheduled. Make sure to use a backup method of contraception for the next seven days. Additionally, if you've had sexual intercourse within the last 72 hours, consider taking emergency contraceptive pills.
Missed 3 or more pills in the third week?	Take one hormonal pill as soon as you can and complete the entire pack of hormonal pills according to the schedule. Discard the seven non-hormonal pills in a 28-pill pack. Begin a new pack the following day.
Missed any nonhormonal pills? (last 7 pills in 28-pill pack)	Dispose of any missed non-hormonal pills. Continue taking your combination oral contraceptives (COCs), one pill each day. Begin the new pack as you normally would.

Solution to Question 18:

A patient on infertility treatment, with 6 weeks of amenorrhea, abdominal pain, and vaginal bleeding with beta hCG levels of 2800 mIU/mL, and a gestational sac in the left adnexa points to the diagnosis of unruptured ectopic pregnancy and the most appropriate management is single-dose methotrexate.

A pregnancy is considered ectopic if implantation occurs at a site outside the uterine cavity. The majority of ectopic pregnancies occur in the fallopian tube, the ampulla being the most common site.

Management of ectopic pregnancy:

Expectant management: Ectopic pregnancies measuring ≤ 3 cm and with declining beta hCG levels initially ≤ 1500 mIU/mL may resolve without intervention and can be managed expectantly.

Medical management with single or multiple-dose methotrexate is preferred if:

- Hemodynamic stability
- Serum beta-hCG ≥ 1500 mIU/ml and ≤ 5000 mIU/mL
- Mean sac diameter ≤ 3.5 cm with no fetal cardiac activity detected on transvaginal ultrasound.
- Patients are willing and able to comply with post-treatment follow-up and have access to emergency medical services within a reasonable time frame in case of a deterioration in the clinical status.

Methotrexate (MTX) can be given as a single IM dose of 50 mg/m² or as a multiple-dose regimen of 1 mg/kg on days 1, 3, 5, and 7.

Surgical management with salpingectomy or salpingostomy is done when:

- Criteria for expectant and medical management are not met
- Gestational sac > 3.5 cm
- Fetal cardiac activity present
- Beta hCG > 5000 mIU/mL
- Significant pain and hemoperitoneum
- Failure of expectant or medical management
- Clinical status of patient warrants urgent intervention
- Doubtful compliance with medical management

Milking of the tube through the fimbriae carries a high risk of bleeding, damage to the tube, the persistence of trophoblastic tissue, and recurrence of ectopic pregnancy.

Solution to Question 19:

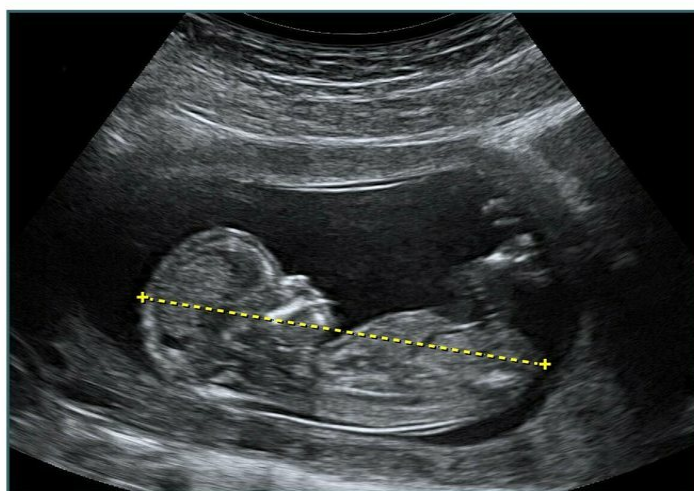
The best method of assessing the gestational age of a patient on oral contraceptives with 6 weeks of amenorrhea is by measuring the crown-rump length (CRL) using ultrasonography.

Gestational age refers to the duration of pregnancy calculated from the first day of the LMP. It is usually expressed in weeks. Following the use of oral contraceptives, ovulation may have been later than 2 weeks after the LMP. Therefore, gestational age calculated with the help of LMP will be inaccurate.

First-trimester ultrasound measurement of CRL is the best method to assess gestational age. It is measured from the vertex to the coccyx.

- $\text{CRL (in mm)} + 6.5 = \text{gestational age in weeks (or)}$
- $\text{CRL (in mm)} + 42 \text{ days} = \text{gestational age in days.}$

CRL measurement is shown in the image below:



Ultrasonographic parameters for estimation of the gestational age according to the trimester are as follows:

Other options:

Option B: Adding 280 days to the LMP will provide the expected delivery date (EDD). Adding 9 months + 7 days to the LMP or 280 days to the LMP is Naegele's formula for calculating the EDD.

Option C: Measurement of fundal height in cm (with a variation of 1-2cm) is roughly equal to the gestational age in weeks, after 24 weeks of pregnancy. This is measured from the superior border of the symphysis pubis to the top of the uterine fundus when the bladder is empty.

Trimester	Parameter
1st	Crown-rump length (CRL)
2nd and 3rd	Biparietal diameter (BPD) Abdominal Circumference (AC) Head Circumference (HC) Femur Length. (FL)

Obstetrics and Gynecology INICET 2022

Question 1:

A 25-year-old lady had an uneventful vaginal delivery. On postpartum day 1, she presented with postpartum haemorrhage and complains of visual changes and severe headache. A few hours after the complaint, she is found to be in an unresponsive state and is intubated and put on ventilation. On further examination, she has a BP of 60/40 mmHg and tachycardia, blood sugar is 48 mg/dL. Hematocrit and WBC count was normal. Which of the following drugs would help reverse this state?

- a) Inj. dobutamine
- b) Steroids
- c) Thyroxine
- d) Activated protien C

Question 2:

All are true about biophysical profile (BPP) except:

- a) Includes breathing, movements, tone, and AFI
- b) Can tell us about fetal outcome and fetal pH
- c) Score is inconsistent with fetal pH at different gestational ages
- d) Score ≤ 4 : fetal pH ≤ 7.2 ; Score 0 : 100% fetal acidemia

Question 3:

Which of the following is not true about lupus nephritis in pregnancy?

- a) Immunosuppression can be continued during pregnancy
- b) Pregnancy to be planned once the disease has been quiescent for at least 6 months and there is no evidence of renal dysfunction
- c) High dose corticosteroids for lupus flare in pregnancy is safe
- d) Ecospirin, methotrexate, cyclophosphamide, corticosteroids, azathioprine are safe in pregnancy

Question 4:

Which of the following is false regarding management of diabetes in pregnancy?

- a) Elective C-section has no role in reducing incidence of brachial plexus injury
- b) In a patient being planned for induction of labor, night dose of intermediate insulin is given as planned, and the morning dose is withheld
- c) In active labor, if RBS ≤ 70 mg%, D5 is started at 100-150 ml/hr till the RBS is ≥ 100 mg%
- d) Capillary blood glucose monitoring levels are kept at fasting- 95 mg%; 1 hr postprandial- 140 mg%; 2 hrs postprandial- 120 mg%

Question 5:

Which of the following statements regarding MgSO₄ is false?

- a) MgSO₄ causes uterine relaxation, but is dose dependant
- b) MgSO₄ is continued upto 24 hours after delivery or last attack of seizure
- c) It has inhibitory effect on cerebral cortex thereby preventing seizures and lowering the BP
- d) In renal failure, MgSO₄ loading dose can be given, but maintenance dose should be adjusted based on renal function

Question 6:

A couple already has 3 children and has been trying for another one for the last 1 year. However, the husband is diagnosed with hypogonadotropic hypogonadism. Which of the following is/are true:

- a) 1, 2, 3, 4 and 5
- b) 1, 2 and 5
- c) 2, 3, 4 and 5
- d) 1, 3 and 5

Question 7:

All of the following are theories of causation of preeclampsia except:

- a) Prostaglandins play an important role. Thromboxane A₂ decreases; Prostacyclin:Thromboxane ratio increases in preeclampsia
- b) Endothelial cell dysfunction leads to reduced nitric oxide levels and this causes the clinical presentation of preeclampsia

- c) Vasoconstriction leads to ischemia, necrosis and hemorrhage and end organ damage seen in preeclampsia
- d) Anti-angiogenic factors play an important role in preeclampsia. There is an increase in sFlt-1 and decrease in PlGF.

Question 8:

The following cost-effective investigations are routinely recommended in the screening of antenatal mothers except?

- a) Echocardiography for cardiac disease
- b) Urine analysis for bacteriuria
- c) Blood sugar levels for GDM
- d) VDRL for syphilis

Question 9:

Which of the following cerclage procedures would be done in a nonpregnant female through a transabdominal approach?

- a) Lash and Lash cerclage
- b) Benson and Durfee cerclage
- c) Wurms cerclage
- d) Laparoscopic cerclage

Question 10:

Urinary tract infection in a pregnant lady in her 2nd trimester is best defined as:

- a) Growth of 10^3 coliform in suprapubic sample
- b) Growth of 10^5 coliform in voided urine sample
- c) Frequency with dysuria with >3 WBC/HPF
- d) Frequency with dysuria with positive nitrite and leucoesterase dipstick test

Question 11:

Which of the following is incorrect regarding innervation of the uterus?

- a) Sensory level is from T10 to L1
- b) Uterine contractility is mediated by innervations from level T7 - T8
- c) In the 1st stage of labor, pain is due to the fibres at level of T10 - L1
- d) In early labor, pain is usually because of the uterine contraction

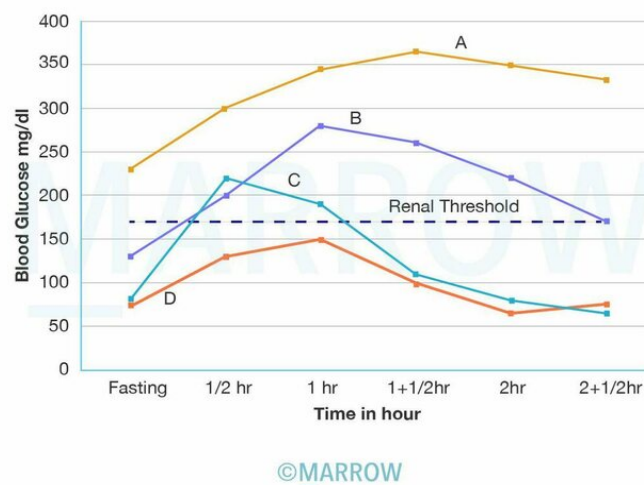
Question 12:

Seminal vesicles and vas deferens would be bilaterally absent congenitally in which of the following conditions?

- a) Cystic fibrosis
- b) Klinefelter syndrome
- c) Kallmann syndrome
- d) Kartagener syndrome

Question 13:

Which of the following graphs represents normal response to OGTT?



- a) Graph A
- b) Graph B
- c) Graph C
- d) Graph D

Question 14:

Kassowitz rule is related to:

- a) Primary syphilis
- b) Congenital syphilis
- c) Secondary syphilis
- d) Latent syphilis

Question 15:

Modified biophysical profile includes:

- a) NST and fetal tone
- b) NST and AFI
- c) NST and breathing movements
- d) NST and gross fetal movements

Question 16:

All of the following statements about anti-Müllerian hormones are true, except:

- a) It is best tested in the fasting state on day 2 of menstrual cycle
- b) Lipopeptide secreted from granulosa cell of early antral follicle
- c) Correlates with pregnancy rate of IVF cycle
- d) It decreases sensitivity of the pre-antral follicles to FSH

Question 17:

Which of the following statements is false regarding post-partum hemorrhage?

- a) Start colloids initially, 4 times the blood lost
- b) Primary PPH occurs within 24 hours of childbirth
- c) PPH is loss of 500 ml of blood in vaginal delivery
- d) Causes include retained placenta and atony

Question 18:

Recommended HPV vaccination schedule for 9 to 14 year olds according to WHO SAGE guidelines is:

- a) 1 or 2 doses
- b) 2 doses
- c) 1 dose
- d) 3 doses

Question 19:

All of the following coagulation factors increase in pregnancy except:

- a) Factor II
- b) Factor VII
- c) Factor VIII
- d) Factor XI

Question 20:

A pregnant lady at 20 weeks of gestation comes with complaints of acute onset fever, and pain in the right lumbar region. It is associated with chills, nausea, and vomiting. Urinalysis reveals the presence of E.coli and a diagnosis of acute pyelonephritis is made. Which of the following is incorrect regarding the management of acute pyelonephritis in pregnancy?

- a) Hospitalise, send urine and blood cultures and start on antibiotics.
- b) Oral antibiotics once afebrile
- c) Maintain urine output $\geq$ 50 ml/hr; Intravenous crystalloids can be given
- d) Discharge if afebrile for 72 hrs; Continue antibiotics for 5 days

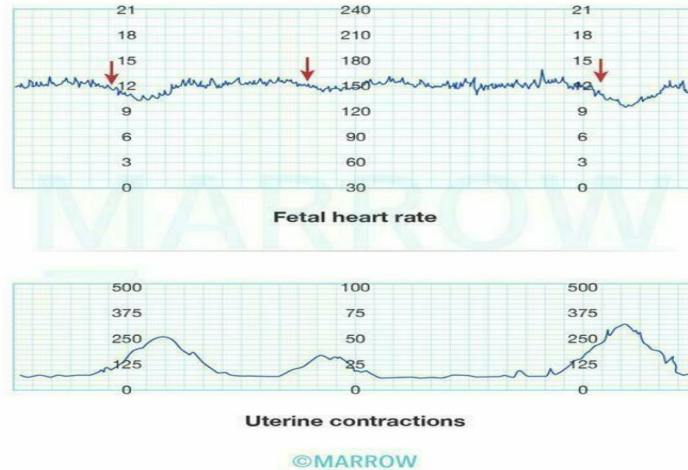
Question 21:

All of the following are the diagnostic criteria for preterm labor except?

- a) 4 contractions in 20 min and 8 contractions in 60 min with changes in cervix
- b) Cervical dilatation $\geq$ 2 cm
- c) More than 80 percent effaced cervix
- d) Cervix posterior in position

Question 22:

Which statement is false regarding late decelerations?



- a) Starts after peak of uterine contraction and ends after uterine contraction ends
- b) They are smooth and gradual
- c) At least 10 beats, 2 times in a 20 min period above baseline
- d) Not associated with accelerations

Question 23:

What is the age group distribution for vulval cancer?

- a) Not age specific
- b) 35-55 years
- c) 55-85 years
- d) Both of the above

Question 24:

The following are correct regarding normal placenta, except _____.

- a) Placenta weight at term is 470 g
- b) Round-oval with 22 cm diameter
- c) Extraplacental membranes and 2-vessel umbilical cord present

- d) Central thickness >2.5 cm

Question 25:

Arrange the following steps of dilation and curettage in sequence.

- a) 4-3-2-1
- b) 4-2-3-1
- c) 4-3-1-2
- d) 4-1-2-3

Question 26:

All are long-term reversible methods of contraception, except:

- a) Progestin injection
- b) Progestin implants
- c) Female sterilization
- d) Levonorgestrel-releasing and copper-containing intrauterine devices

Question 27:

According to new WHO guidelines, what is the minimum number of antenatal visits recommended to have a positive pregnancy outcome?

- a) 8
- b) 4
- c) 6
- d) 10

Question 28:

Identify the wrong statement regarding the period of gestation calculation.

- a) 1st trimester USG is the best assessment
- b) Calculated from the last day of the last menstrual period
- c) In an IVF pregnancy, day of embryo transfer/day of fertilization is needed to calculate POG

- d) Crown-rump length is the best measure of period of gestation in the first trimester

Question 29:

All of the following can be used in first-trimester medical termination of pregnancy (MTP), except _____.

- a) Mifepristone + misoprostol
- b) Manual vacuum aspiration
- c) Dilatation and curettage
- d) Extra-amniotic instillation of ethacridine lactate

Question 30:

A woman with eclampsia is started on magnesium sulfate. What is the first sign of magnesium sulfate toxicity?

- a) Loss of knee jerk
- b) Hypotension
- c) Respiratory depression
- d) Reduced muscle tone

Question 31:

Which of the following patients would be classified as having primary amenorrhea?

- a) 11-year-old girl with Tanner stage 1 breasts
- b) 14-year-old girl with Tanner stage 1 breasts
- c) 13-year-old girl with Tanner stage 5 breasts
- d) 12-year-old with Tanner stage 4 breasts

Question 32:

The daughter of a woman who received diethylstilbestrol during her pregnancy is at higher risk for:

- a) Squamous cell carcinoma
- b) Clear cell carcinoma

- c) Adenocarcinoma
- d) Sarcoma botryoides

Question 33:

Which of the following is not a component of biophysical profile?

- a) Amniotic fluid volume
- b) Fetal tone
- c) Daily fetal movement count
- d) Non-stress test

Question 34:

All are causes of uterine dysfunction, except _____.

- a) High fetal head station
- b) Multiparity
- c) Neuraxial block
- d) Chorioamnionitis

Question 35:

Non-immune hydrops is seen in all, except _____.

- a) Human immunodeficiency virus
- b) Herpes simplex virus
- c) Syphilis
- d) Parvovirus B19

Question 36:

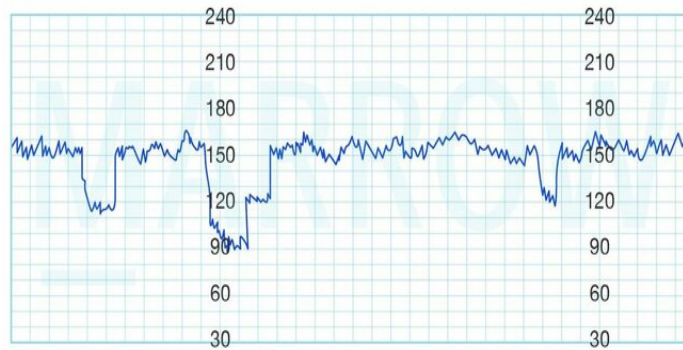
Which of the following statements is true regarding recurrent ovarian cancer?

- a) Platinum resistant if recurrence $\geq$ 12 months
- b) Partial platinum resistant if recurrence is within 6-12 months
- c) Platinum sensitive if recurrence $\geq$ 6 months

d) All of the above

Question 37:

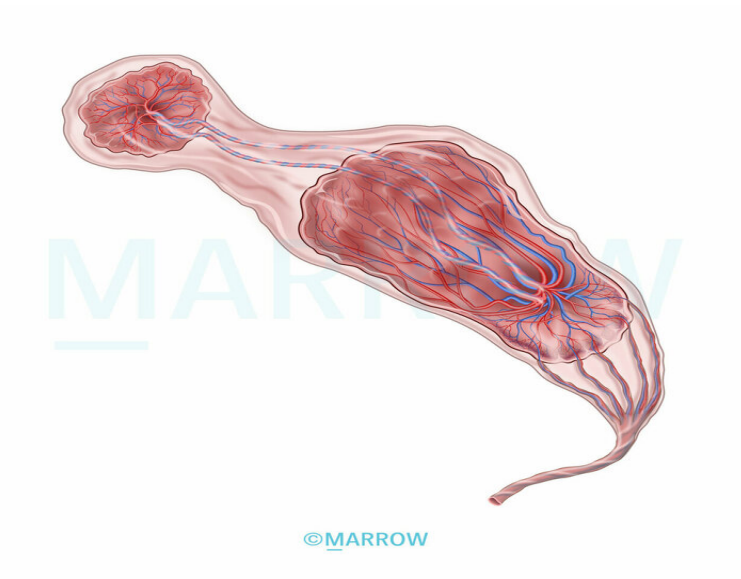
What can be said about this graph?



- a) This is a non-stress test.
- b) The baseline shows that the fetus has tachycardia.
- c) The beat to beat variability is less/inadequate.
- d) This is a uterine contraction graph.

Question 38:

The given image depicts:



- a) Succenturiate lobe
- b) Velamentous insertion of the cord
- c) Both succenturiate lobe and velamentous insertion
- d) Normal placenta

Question 39:

Which hormone is secreted by placenta?

- a) LH
- b) hCG
- c) FSH
- d) GnRH

Question 40:

Antepartum hemorrhage due to abruption shows all except:

- a) Uterine tenderness
- b) Frequent contractions
- c) Painless vaginal bleeding
- d) Mild to moderate vaginal bleeding

Question 41:

A woman presents at 8 weeks of gestation. Which of the following signs can be elicited at this gestation?

- a) 1,3
- b) 1,2
- c) 1,4
- d) 3,4

Question 42:

The non-contraceptive uses of oral contraceptives include all of the following except:

- a) Reduction in the risk of ovarian cancer
- b) Ectopic pregnancy
- c) Protection against sexually transmitted diseases including HIV
- d) Reduction in the risk of endometrial cancer

Question 43:

A 16-year-old presents with absent menses. Examination shows widely spaced nipples, webbed neck, and Tanner grade 1 breast development. USG shows streak ovaries. Investigation shows raised FSH, raised LH, and reduced estradiol. What is the diagnosis?

- a) Turner syndrome
- b) Kallmann syndrome
- c) Androgen insensitivity syndrome
- d) Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome

Question 44:

Consider the following actions under AMTSL. Arrange them in the right order:

- a) 3,1,2,4
- b) 3,2,4,1
- c) 2,4,3,1
- d) 2.4.1.3

Question 45:

A surgically resected specimen of an ovarian mass is shown below. What is the likely diagnosis?



- a) Immature teratoma
- b) Mature cystic teratoma
- c) Mucinous cystadenoma
- d) Serous cystadenoma

Question 46:

Which of the following is seen in placenta previa?

- a) Warning bleeds
- b) Dark painful bleeding
- c) Uterine tenderness
- d) Increased/large uterine size

Question 47:

A pregnant woman with 35 weeks+5 days gestation complains of epigastric pain, headache, and blurring of vision. Her blood pressure is 170/110 mmHg. She is in labor pain.

Examination of the cervix shows a 3 cm dilatation with a bag of liquor. What is the appropriate management?

- a) Conservative management
- b) Administer tocolytics and deliver at 37 wks
- c) Give steroid cover and wait until steroid cover is complete.
- d) Treat hypertension, administer MgSO₄, and consider termination of pregnancy.

Question 48:

All of the following are risk factors of vulval cancer except

- a) Vulval dystrophy
- b) Vulval hamartoma
- c) Smoking
- d) Vulval warts

Question 49:

Which of the following is true regarding perimortem cesarean section?

- a) LSCS
- b) Pfannenstiel incision
- c) Upper segment cesarean section
- d) Done in OT

Question 50:

Arrange the following in sequence.

- a) B, A, C, D
- b) B, A, D, C
- c) B, C, A, D
- d) B, D, A, C

Question 51:

Match the following

- a)** A-4, B-3, C-2, D-1
- b)** A-2, B-3, C-1, D-4
- c)** A-4, B-1, C-2, D-3
- d)** A-3, B-2, C-4, D-1

Question 52:

What is vagitus uterinus?

- a)** Cry of baby in the womb
- b)** Infection of vagina
- c)** Infection of uterus
- d)** Infection of both vagina and uterus

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	a
5	c
6	d
7	a
8	a
9	d
10	b
11	b
12	a
13	d
14	b
15	b
16	b

17	a
18	a
19	d
20	d
21	d
22	c
23	c
24	c
25	c
26	c
27	a
28	b
29	d
30	a
31	b
32	b
33	c
34	b
35	a
36	c
37	a
38	c
39	b
40	c
41	b
42	c
43	a
44	b
45	b
46	a
47	d
48	b
49	c
50	a
51	c
52	a

Detailed Explanations

Solution to Question 1:

The given history of a pregnant having postpartum hemorrhage developing headache, visual changes, and hypotension after a vaginal delivery suggests Sheehan syndrome. Steroids are the drug that can reverse this state.

Volume depletion and low cardiac output leading to critically low BP are common in acute pituitary deficiency and recovery of normal cardiovascular status is rapidly achieved under hormone and volume replacement therapy without the need for inotropes like dobutamine.

Pituitary apoplexy is a hemorrhagic infarction of large pituitary adenomas or pituitary infarction due to traumatic major blood loss. If this blood loss occurs during delivery, it's called Sheehan's syndrome. The hyperplastic enlargement of the pituitary (size of gland increases by about 125%), which occurs normally during pregnancy, increases the risk of hemorrhage and infarction.

Pituitary apoplexy presents acutely as an endocrine emergency, with sudden onset of headache, nausea, visual deficits, persistent hypotension, tachycardia, and hypoglycemia leading to shock. The most prominent symptoms are lactation failure (due to a deficiency of prolactin) and postpartum amenorrhea (due to a deficiency of gonadotropins).

Because deficiencies of some or all pituitary-responsive hormones may develop after the initial insult, Sheehan syndrome can be heterogeneous and not be identified for years. It can present late with the following features like loss of pubic and axillary hair, hypothyroidism due to decreased TSH secretion, adrenal insufficiency due to low ACTH.

Treatment is by hormone replacement - first glucocorticoids, later - thyroid hormone, sex steroids, vasopressin, and growth hormones are given. If there is evidence of progressive visual loss, cranial nerve involvement, or loss of consciousness, immediate surgical decompression must be carried out.

Solution to Question 2:

All are true about BPP (biophysical profile) except option C. BPP score is not inconsistent with fetal pH, but has a direct linear relationship with fetal pH and thus can be used to predict fetal acidemia.

BPP is a multi-parametric test to assess uteroplacental insufficiency. It consists of five components/ variables and the observations are made in a 30-minute time period.

Components of BPP:

(Mnemonic: BATMaN)

B - Breathing movements of the fetus

A - Amniotic fluid volume

T - Tone of the fetus

M - Movements of the fetus

N - Non-stress test (NST)

Each parameter has a maximum score of 2 and a minimum score of 0. So the total maximum score will be 10. A score of 4 or less indicates fetal acidemia.

Interpretation and management of BPP scores:

BPP is directly related to fetal pH and fetal outcome. A lower BPP score is associated with a poor outcome (stillbirth, perinatal mortality) and lower APGAR scores at birth.

A score of ≤ 10 is associated with a pH of ≤ 7.2 and a score of 0 is associated with 100% fetal acidemia.

Note: The modified biophysical score has only 2 components i.e. NST and amniotic fluid index only. It is considered abnormal (nonreassuring) when the NST is non-reactive and/or the AFI is ≤ 5 .

Parameters	Minimal Normal Criteria
Fetal breathing movements	≥ 1 episode lasting > 30 seconds
Amniotic fluid volume (AFV)	≥ 1 pocket measuring 2 cm in two perpendicular planes (2 $\times$ 2 cm pocket)
Fetal muscle tone	≥ 1 episode of active limb or trunk extension with the return of flexion; Eg. opening and closing of the hand, considered normal
Gross body movements	≥ 3 discrete body/limb movements
Non-stress Test (NST)	Reactive pattern

BPP score	Interpretation	Management
10	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mother or post-term pregnancy)
8 with normal AFV	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mother or post-term pregnancy)
8 with decreased AFV	Suspicion of chronic fetal asphyxia	Delivery
6	Possible fetal asphyxia	Deliver, if AFV is abnormal or > 36 weeks or if the repeat test score is ≤ 6 . If the repeat score > 6 , observation and repeat test weekly
4	Probable fetal asphyxia	Repeat testing same day If the repeat score ≤ 6 , delivery
0-2	Almost certain fetal asphyxia	Delivery

Solution to Question 3:

Option D is not true about lupus nephritis in pregnancy. Methotrexate and cyclophosphamide are teratogenic and hence they are not safe during pregnancy.

Systemic lupus erythematosus(SLE) is a chronic autoimmune disorder involving the multisystem. It affects young women of childbearing age. The renal involvement of the disease can manifest in the form of active lupus nephritis or flare-ups during pregnancy.

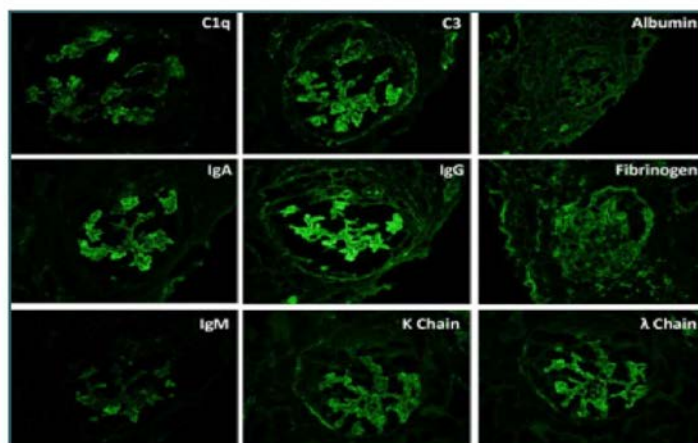
Clinical features include fever, weight loss, migratory joint pain, and myalgias. It can lead to preterm births, preeclampsia, and pregnancy loss while the infant can suffer from neonatal lupus and heart block.

Antinuclear antibodies are useful for screening the disease. Double-stranded DNA (ds DNA) is highly specific in diagnosing. Other antibodies such as lupus anticoagulant, antiphospholipid antibodies, anti-Ro, and anti-La antibodies are useful in diagnosis.

Management is as follows

- Preconceptional planning plays an important role. Conception during a period of quiescence for at least 6 months determines a successful outcome. Stable remission of renal disease for at least 6 months before conception is recommended.
- Corticosteroids are useful during episodes of lupus flare and can be used safely in pregnancy. For patients on chronic corticosteroid therapy, intravenous hydrocortisone 100 mg, three doses are given 8 hourly during labor or at the time of cesarean delivery.
- Immunosuppressants such as azathioprine can be continued during pregnancy. It may be used in case of severe SLE
- Hydroxychloroquine is found to be the best for maintenance and is safe in pregnancy. It can prevent the need for prolonged corticosteroid therapy.

Lupus nephritis - Immunofluorescence



Solution to Question 4:

False statement regarding management of diabetes in pregnancy is option A. Cesarean section is found to be protective against traumatic injuries such as brachial plexus injury, in macrosomic babies born to mothers with diabetes. An elective C-section is found to be more protective than an emergency C-section.

Almost 50 % of pregnant women with diabetes are delivered by c-section. Indications for c-section in diabetic pregnant women are-

- Fetal macrosomia of ≥ 4.5 kgs to avoid trauma to mother or neonate, and also to avoid brachial plexus injury in the baby
- Unsatisfactory progression of labor
- Complicated diabetes mellitus
- Fetal compromise during delivery
- Elderly primigravida (age > 30 years)
- Multigravida with bad obstetric history previously
- Pregnancy that is associated with complications like preeclampsia, polyhydramnios or malpresentation

Other options:

Option B: In a planned delivery, although the bedtime dose of intermediate-acting insulin is given, it is recommended that the morning dose must be withheld. This is due to reduced insulin requirements after delivery. Therefore, regular insulin is used for glycemic control during this time.

Option C: In active labor, if RBS (Random blood glucose) < 70 mg%, D5 (5% dextrose) is started at 100-150 ml/hr till the RBS is > 100 mg%. Before initiation of labor, normal saline is used for iv infusion.

Option D: Self-monitoring of capillary blood glucose is advised in a pregnant woman diagnosed with diabetes. Subcutaneous continuous glucose monitoring devices can also be used for the same. The glucose goals recommended are:

- Fasting: ≤ 95 mg/dL
- 1 hour post-prandial: ≤ 140 mg/dL
- 2 hrs post-prandial: ≤ 120 mg/dL

Solution to Question 5:

The false statement regarding MgSO_4 (Magnesium sulfate) is option C. It has an inhibitory effect on the cerebral cortex, thereby preventing seizures, but this action is not responsible for lowering blood pressure.

MgSO₄ is mainly used in preeclampsia due to its specific anticonvulsant action and neuroprotective action on the cortex. MgSO₄ acts as a membrane stabilizer. It reduces motor endplate sensitivity to acetylcholine and blocks neuronal calcium influx. It is not used to treat hypertension, although it shows some hypotensive action by decreasing vascular response to vasopressors.

MgSO₄ is used for the management of eclampsia and as prophylaxis in severe pre-eclampsia. The therapeutic level of magnesium in eclampsia is 4-7mEq/L. The two regimens commonly used are Pritchard and Sibai regimen.

Other actions:

- Cerebral vasodilatation
- Uterine artery vasodilatation
- Uterine tocolytic i.e. inhibits uterine contractility
- Increased production of endothelial prostacyclin
- Inhibits platelet activation

Other options:

Option A: MgSO₄ inhibits uterine contractility but is dose-dependent (only above serum values of 8-10 mEq/L).

Option B: MgSO₄ has to be administered up to 24 hours after the last seizure (in postpartum seizure) or delivery, whichever is later. For recurrent seizures, 2 g IV bolus is given in the above regimen.

Option D: Magnesium is cleared almost exclusively by renal excretion. But, in renal failure, the initial 4-g loading dose of magnesium sulfate can be safely administered regardless of renal function. Only the maintenance dose needs to be altered according to the renal function. This is because the loading dose is responsible for reaching therapeutic levels, but the maintenance doses are responsible for maintaining steady states.

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4g (20% solution) IV over 3-5 minutes followed by 10g (50%), deep IM (5 gm in each buttock)	5g (50%) IM 4 hourly in alternate buttocks
Intravenous (Zuspan or Sibai)	4g IV slow over 15-20 minute	1-2g/h IV infusion

Solution to Question 6:

The given patient is diagnosed with hypogonadotropic hypogonadism. True statements about the condition are 1, 3, and 5.

Hypogonadotropic hypogonadism in a male is the presence of impaired testicular function (hypogonadism), secondary to a hypothalamic-pituitary dysfunction (hypogonadotropic).

Causes include:

- Congenital
- Anosmic hypogonadotropic hypogonadism (Kallmann syndrome)
- Normosmic isolated hypogonadotropic hypogonadism (idiopathic hypogonadotropic hypogonadism)
- Acquired
- Drugs and alcohol
- Infiltrative or infectious pituitary lesions, brain radiation, trauma to the brain
- Hyperprolactinemia
- Systemic diseases such as hemochromatosis, sarcoidosis

A male with hypogonadotropic hypogonadism will have the following findings:

- Hormone levels
- Low LH (Luteinizing hormone), low FSH (follicle-stimulating hormone). This can be confirmed on GnRH stimulation test showing persistently reduced LH, and FSH levels.
- Low testosterone.
- High prolactin levels and high prolactin: testosterone ratio.
- Low testicular volume - oligospermia: the presence of less than 15 million sperm per milliliter of semen. Sperm motility is not impaired.
- MRI showing anomalous or absent olfactory bulb in Kallmann syndrome.

Treatment:

- Testosterone supplementation: Helps reverse the symptoms and signs of hypogonadism. It is advised only in males not planning to have children.
- Pulsatile GnRH or gonadotropin (FSH and hCG) therapies: In males planning to have children.
- Assisted reproductive techniques: Intrauterine insemination (IUI), Intracytoplasmic sperm injection (ICSI)

Solution to Question 7:

All of the following are theories of causation of preeclampsia except option A. There is reduced or absent prostacyclin(PGI₂) production with increased production of thromboxane (TXA₂) and reduced ratio of prostacyclin: thromboxane in preeclampsia.

The development of pre-eclampsia may be explained by:

- Failure of 2nd wave of trophoblastic invasion into uterine vessels, leading to decreased placental perfusion.
- Maternal inflammatory response directed against foreign fetal antigens, leading to immunological maladaptation.
- Maternal maladaptation to cardiovascular and inflammatory changes of normal pregnancy.

- Genetic factors and epigenetic influences.

The basic pathology is endothelial dysfunction and vascular spasm that affects vessels supplying all organs including the brain, kidney, uterus, etc. This ischemia leads to necrosis, hemorrhage, and end-organ damage of these organs.

Changes leading to vasospasm and endothelial dysfunction include:

- Imbalance of prostaglandins (Reduced vasodilatory PGI₂, increased vasoconstrictor TXA₂).
- Reduced angiotensinase activity and increased sensitivity of vessels to angiotensin II, which is a vasopressor.
- Reduced production of nitric oxide due to endothelial dysfunction. It normally acts as a vasodilator and inhibits platelet aggregation and thrombus formation.
- Increased production of endothelin 1 by endothelial cells. It is a potent vasoconstrictor.
- Increased inflammatory mediators such as tumor necrosis factor (TNF) and interleukins (IL-6) leading to endothelial injury.
- Increased lipid peroxides, reactive oxygen species (ROS) and superoxide anion radicals, leading to endothelial damage.
- Increased production of anti-angiogenic factors as compared to angiogenic factors. Two main antiangiogenic factors overproduced by the trophoblasts of women with preeclampsia are soluble fms-like tyrosine kinase I (sFlt-1) and soluble endoglin. Increased sFlt-1 inactivates and reduces the levels of free placental growth factor (PlGF) and VEGF concentrations.

All these processes lead to endothelial dysfunction and causes the clinical presentation of preeclampsia.

Solution to Question 8:

Except for echocardiography for cardiac disease, all other tests (urine analysis for bacteriuria, blood sugar levels for GDM, VDRL for syphilis) are cost-effective investigations routinely recommended in the screening of antenatal mothers.

Echocardiography is an expensive investigation and is not routinely recommended to look for cardiac abnormalities in all pregnant ladies. Indications for fetal echocardiography include:

- Abnormality in routine genetic screening such as USG to look for nuchal translucency (NT). NT is the space between the fetal skin and the underlying soft tissue near the neck of the fetus filled with fluid. NT of ≥ 3 mm is abnormal and warrants the need for detailed USG and fetal echocardiography
- Gestational diabetes mellitus for early detection of fetal anomalies. Done at 20-22 weeks.
- Heart disease in pregnancy like congenital heart disease or Marfan's syndrome needs routine echocardiography (color doppler) at mid-pregnancy to look for structural abnormalities, valve anatomy, etc.

Routine investigations during pregnancy:

- 1st trimester:

- Hemoglobin, hematocrit, blood grouping, blood glucose, and VDRL.
- Midstream urine to look for protein, sugar, pus cells, and evidence of urinary tract infections or bacteriuria by dipstick test (for nitrites).
- Serology for rubella, hepatitis B virus, and HIV.
- Genetic screening like maternal serum alpha-fetoprotein.
- Ultrasound scan to look for gross fetal abnormalities, number of fetuses, and dating of pregnancy.
- 2nd trimester:
 - Ultrasound scan (booking scan at 18-20 weeks) is done to look for cardiac activity, structural abnormalities, amniotic fluid index and placental location.
 - Triple test at 11-13 weeks for a high risk mother.
 - Oral glucose tolerance test (OGTT) at 24-26 weeks with a 1-hour 50-gram glucose load or a diagnostic 2-hour 75-gram glucose load. This is to look for gestational diabetes mellitus.
- 3rd trimester:
 - Repeat hemoglobin, hematocrit, and serology for syphilis and HIV.
 - Vaginal examination to look for Streptococcus agalactiae.

Solution to Question 9:

The cerclage procedure performed on nonpregnant females through a transabdominal approach is laparoscopic cerclage. Abdominal cerclage is conducted through either laparotomy or laparoscopy and is recommended in cases where the cervix is hypoplastic or in instances of prior failed vaginal cerclage.

Cerclage is usually done for cervical incompetence, defined as the inability to maintain pregnancy till term due to a functional or structural defect of the cervix. Cervical incompetence is diagnosed by clinical examination and ultrasonography showing the presence of short cervix (<25 mm) with or without funneling of internal os.

Prophylactic cerclage is done around 14 weeks of pregnancy or 2 weeks earlier than the previous fetal loss. Rescue or emergency cerclage is done anytime within 24 weeks of gestation. It should only be done when there is bulging of membranes through the dilated cervix.

Commonly used types of cervical cerclage during pregnancy:

- McDonald's procedure - non-absorbable (mersilene) material placed as a purse-string suture at the level of the internal os
- Modified Shirodkar's procedure - non-absorbable (mersilene) tape passed subcutaneously through a transverse anterior incision and tied posteriorly

Removal of cerclage should be done by 37 weeks of gestation. It can be removed earlier in case of preterm labor or if signs of abortion are seen.

Other options:

Option A: Lash and Lash cerclage is the first described treatment for cervical incompetence, involving the placement of stitches in the cervix through the transvaginal approach. It is now replaced by procedures such as McDonald's and Shirodkar's.

Option B: Benson und Durfee cerclage is the technique of placement of cervical cerclage through a transabdominal approach by a laparotomy. It is preferred in pregnant females when the vaginal approach is either not possible or very difficult.

Option C: Wurms cerclage is a type of cerclage used in pregnancy where monofilament material (Prolene) is used to place two higher circumferential sutures through the cervix without the need for bladder dissection.

Solution to Question 10:

Urinary tract infection (UTI) in a pregnant lady in her 2nd trimester is best defined as the growth of 10⁵ coliforms in a voided urine sample. Urine culture is the standard method to evaluate UTI pregnancy.

UTI in pregnancy is a bacterial infection caused by organisms in the normal perineal flora. It can vary from asymptomatic bacteriuria to symptomatic cystitis or pyelonephritis.

Asymptomatic bacteriuria is characterized by persistent and actively multiplying bacteria in the urinary tract without any symptoms. A positive culture is demonstrated during the antenatal visit. It can complicate into a symptomatic infection when untreated.

Symptomatic infections include

- Cystitis is characterized by dysuria, urgency and frequency along with pyuria and bacteriuria. Microscopic hematuria is also seen
- Pyelonephritis is common in the second trimester, nulliparous and young women. It is usually unilateral with fever, chills, and ache in the lumbar region. Nausea, vomiting, and costovertebral tenderness may be seen. The organisms causing pyelonephritis are *E. coli* & & *Streptococcus*, *S. aureus* & *Enterobacter*, *Klebsiella*

UTI is evaluated by blood studies including complete blood count, serum creatinine, blood urea nitrogen, urine studies and imaging. Urine studies mainly include a urine culture, urine analysis, and dipstick.

Urine culture is the standard method of evaluating UTIs in pregnancy. Positive urine culture shows a bacterial growth of >100,000 colony-forming units (CFUs) per milliliter in a voided urine sample. The presence of white blood cell (WBC) casts is suggestive of pyelonephritis. No other screening methods are as reliable as a urine culture.

Urine analysis showing positive nitrites, leukocyte esterase, WBCs (pyuria), red blood cells (RBCs), and protein are suggestive of UTI. Pyuria is seen in cystitis. Dipstick for nitrite and leukocyte esterase is used to evaluate asymptomatic bacteriuria.

Management of UTIs is as follows

- Asymptomatic bacteriuria is mainly treated with oral antibiotics such as amoxicillin, ampicillin, nitrofurantoin, and trimethoprim-sulfamethoxazole.

- Pyelonephritis is managed with intravenous hydration, and antibiotics such as ampicillin+gentamycin, cefazolin or ceftriaxone.

Solution to Question 11:

The incorrect regarding innervation of the uterus is statement B. Uterine contractility is mainly mediated by hormonal mechanisms. Estrogen, prostaglandin, and oxytocin produce contraction, while progesterone causes relaxation.

Uterine sensory supply is by ascending afferent fibers which pass through the inferior hypogastric plexus and enter the spinal cord through T10-T12 and L1 roots.

Labor pain is due to stimulation of nociceptors in the genital tract caused by ischemia.

- First stage labor: Pain is mediated by T10 to L1 spinal segments. It is caused by distension of the cervix and low uterine segments along with isometric uterine contraction.
- Second stage labor: Pain is carried by T12 to L1 and S2 to S4 spinal segments. It is caused by tissue damage in the pelvis and perineum

Analgesia in labor is achieved by intramuscular injection of pethidine 50 mg. Metoclopramide is added to prevent vomiting. Pethidine is avoided within 2 hours of delivery as it is a known respiratory depressant to the fetus. When relief of pain is required throughout labor, regional analgesia is preferred through pudendal nerve block, epidural analgesia, spinal anesthesia or combined epidural-spinal anesthesia.

Solution to Question 12:

Seminal vesicles and vas deferens would be bilaterally absent, congenitally in cystic fibrosis. This is known as the congenital bilateral absence of vas deference (CBAVD).

Cystic fibrosis is an autosomal recessive disorder that affects fluid secretion in the exocrine glands and in the epithelial lining. It is caused primarily due to a mutation in the CFTR gene, which codes for an anion channel in the apical membranes of epithelial cells. This leads to abnormally viscous secretions that obstruct organ passages, resulting in recurrent infections, pancreatic insufficiency, steatorrhea, malnutrition, hepatic cirrhosis, intestinal obstruction, and male infertility.

Infertility is due to azoospermia and CBAVD. In some patients, the bilateral absence of the vas deferens may be the only feature suggesting an underlying CFTR mutation.

On examination, a normal reproductive tract (including testis, and epididymis) is seen, with bilateral absence of vas deferens and atrophic or absent seminal vesicles. Semen analysis shows a low-volume (0.5 mL), acidic ejaculate that is devoid of fructose and seminal vesicle fluid. There may be associated renal agenesis in some patients.

Other options:

Option B: Klinefelter syndrome is a genetic disorder that results from a 47, XXY karyotype in phenotypically male patients. They present with gynecomastia, azoospermia, and infertility due

to hyalinization of the seminiferous tubules and a small testis.

Option C: Kallmann syndrome is associated with a low or absent GnRH that is characterized by anosmia, hypogonadism, infertility, cryptorchidism, delayed pubertal maturation, and neurological abnormalities.

Option D: Kartagener syndrome is an autosomal recessive disorder affecting ciliary motility; it is characterized by situs inversus, chronic sinusitis, and bronchiectasis. Male infertility may be present and is due to asthenospermia (immotile spermatozoa due to impaired sperm flagella function).

Solution to Question 13:

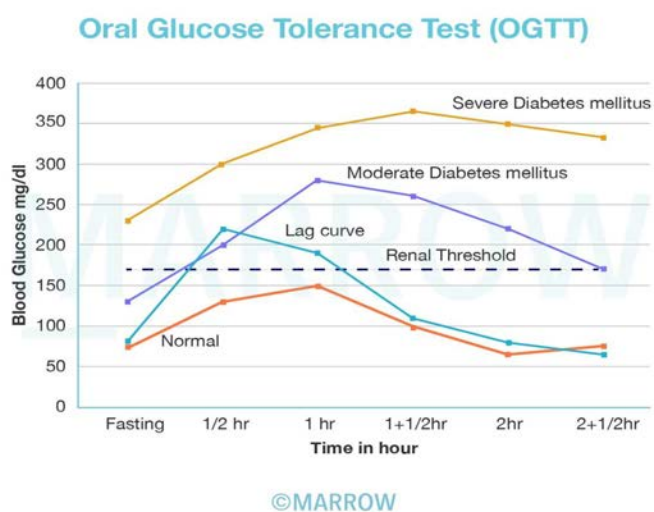
The graph representing normal response to oral glucose tolerance test (OGTT) is graph D.

Oral glucose tolerance test is routinely used to diagnose gestational diabetes mellitus. It is performed by giving a bolus of glucose (75g) to the patient, and measuring blood glucose at intervals of time to assess the tolerance and response of the islets of pancreas to glucose. 5 measurements of glucose is noted in blood and urine at half hour intervals.

Indications:

- Patients with symptoms suggestive of diabetes mellitus, but with fasting blood glucose value being borderline (between 100-126 mg/dl).
- Excessive weight gain during pregnancy, a past history of big baby (more than 4 kg) or a past history of miscarriage.
- To rule out benign renal glucosuria.

Glucose Tolerance Curve (GTC) is the graph plotted by taking plasma glucose measurements on Y-axis and plotting against time, on the X-axis.



Graph D: Normal: This is the normal response seen in a healthy individual. It is characterized by a peak glucose value reached within 1 hour, following which glucose levels reach the base value. The

levels don't cross the renal threshold at any point of time.

Graph A and Graph B: Diabetes mellitus: The graph will peak after 1-1.5 hours and does not return to fasting level within 2.5 hours. The highest value exceeds the renal threshold.

Depending on severity, the graph can show mild, moderate, and severe diabetic curve.

Graph C: Lag curve: The fasting blood glucose is normal here, but the curve rapidly rises in 1/2 to 1 hour and exceeds the renal threshold. There is a rapid and complete return to normal value in a few minutes. This is seen in hyperthyroidism, early diabetes, and sometimes during pregnancy.

Solution to Question 14:

Kassowitz rule is related to congenital syphilis.

According to Kassowitz rule, if a woman with untreated syphilis has a series of pregnancies, the likelihood of infection of the fetus in later pregnancies becomes less. Usually, only the first pregnancy is an abortion, the following pregnancy losses will be stillbirths, and later pregnancies will have alive babies with features of congenital syphilis.

Maternal syphilis can cause fetal infection (congenital syphilis) by several routes, the most common being transplacental transmission, followed by transmission during delivery or across the placental membranes. Early maternal syphilis has a higher transmission as compared to late syphilis.

Up to 70 percent of women with maternal syphilis have an adverse outcome. Most common complications include preterm labour, stillbirth, fetal-growth restriction and fetal death. Newborns can have features of infection such as jaundice, ascites, petechiae, purpura, pneumonia, myocarditis and long bone involvement.

Maternal syphilis in all trimesters should be treated with parenteral penicillin G in order to prevent the spread of infection to fetus. The treatment of choice for congenital syphilis is penicillin.

Solution to Question 15:

The modified biophysical profile includes NST (nonstress test) and AFI (amniotic fluid index).

It is a modification of the biophysical profile (BPP) test, including only 2 parameters so that it can be performed more routinely (twice a week) for surveillance and takes less time (10 minutes). Modified BPP is considered nonreassuring when the NST is non-reactive and/ or the AFI is ≤ 5 .

The nonstress test is a screening test for fetal wellness. It describes fetal heart rate acceleration (measured by a Doppler) in response to fetal movements (perceived by mother). An acoustic stimulation nonstress test is a type of NST in which a loud external sound is used to startle the fetus and provoke heart rate accelerations.

- Reactive (Reassuring) NST - Two or more accelerations of 15 seconds duration, 15 beats per minute amplitude, in a 20-minute testing time.
- Non-reactive (Non-reassuring) NST - Absence of any fetal reactivity.

BPP is a multi-parametric, screening test to assess uteroplacental insufficiency. It consists of five components/ variables and the observations are made in a 30-minute time period.

Components of BPP:

(Mnemonic: BATMaN)

B - Breathing movements of the fetus

A - Amniotic fluid volume

T - Tone of the fetus

M - Movements of the fetus

N - Non-stress test (NST)

Each parameter has a maximum score of 2 and a minimum score of 0. So the total maximum score will be 10. A score of 4 or less indicates fetal acidemia.

Interpretation and management of BPP scores:

BPP is directly related to fetal pH and fetal outcome. A lower BPP score is associated with a poor outcome (stillbirth, perinatal mortality) and lower APGAR scores at birth.

A score of ≤ 10 is associated with a pH of ≤ 7.2 and a score of 0 is associated with 100% fetal acidemia.

Note: The modified biophysical score has only 2 components i.e. NST and amniotic fluid index only. It is considered abnormal (nonreassuring) when the NST is non-reactive and/or the AFI is ≤ 5 .

Parameters	Minimal Normal Criteria
Fetal breathing movements	≥ 1 episode lasting > 30 seconds
Amniotic fluid volume (AFV)	≥ 1 pocket measuring 2 cm in two perpendicular planes (2 $\times$ 2 cm pocket)
Fetal muscle tone	≥ 1 episode of active limb or trunk extension with the return of flexion; Eg. opening and closing of the hand, considered normal
Gross body movements	≥ 3 discrete body/limb movements
Non-stress Test (NST)	Reactive pattern

BPP score	Interpretation	Management
10	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mother or post-term pregnancy)

BPP score	Interpretation	Management
8 with normal AFV	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mother or post-term pregnancy)
8 with decreased AFV	Suspicion of chronic fetal asphyxia	Delivery
6	Possible fetal asphyxia	Deliver, if AFV is abnormal or >36 weeks or if the repeat test score is ≤ 6. If the repeat score >6, observation and repeat test weekly
4	Probable fetal asphyxia	Repeat testing same day If the repeat score ≤ 6, delivery
0-2	Almost certain fetal asphyxia	Delivery

Solution to Question 16:

All of the following statements about anti-Müllerian hormones (AMH) are true, except option B. AMH is a glycoprotein secreted by the granulosa cells of the pre-antral follicles and small antral follicles of the ovary.

In males, AMH inhibits the development of the Müllerian system and its absence can result in hermaphroditism. In the female, it helps in follicular development and oocyte maturation. In the absence of AMH, growing follicles are more responsive to FSH. This is because AMH reduces the sensitivity of pre-antral follicles to FSH (option D) thereby regulating the follicular growth.

Serum AMH predicts ovarian reserve in a woman with infertility or secondary amenorrhea. Also, AMH values decrease with age and can indicate menopause, owing to the reduced ovarian reserve. This makes AMH the best predictor for measuring ovarian reserve. However, recent studies have shown intercycle variations of AMH levels, hence it is recommended to measure AMH along with antral follicle count (on USG) on days 1-3 of the cycle (option A).

It is also a useful predictor of pregnancy rate in infertile women undergoing IVF (In-vitro fertilization) treatment (option C). It helps in assessing the ovarian response to exogenous gonadotropin stimulation, blastocyst formation, and miscarriage rates.

Values and interpretation:

- Normal value: 2–6.8 ng/mL
- Poor ovarian reserve: <1 ng/mL
- PCOD and hyperstimulation syndrome: >10 ng/mL

Solution to Question 17:

The false statement regarding PPH is option A. WHO recommends intravenous isotonic crystalloids over intravenous colloids for fluid replacement in PPH. This is because colloids are

not found to be useful, and some colloids (hydroxyethyl starch) can even be harmful.

Postpartum hemorrhage is defined as a loss of more than 500 ml of blood after completion of the third stage of labor (in vaginal delivery).

PPH is blood loss of more than 1000 mL from the genital tract following cesarean section.

Depending upon the amount of blood loss, PPH can be, minor (<1L) or major (>1L). Major is further subdivided into moderate (1L-2L) and severe (>2L).

Types of PPH:

Primary/early PPH: Hemorrhage occurring within 24 hours of childbirth. Causes include

- Atony: Primi, grand multipara, over distended uterus, hydramnios, precipitate labor, prolonged labor, uterine malformations, fibroid uterus
- Trauma: Instrumental deliveries, episiotomy, precipitate labor, macrosomia
- Retained tissues: retained placenta, blood clots
- Bleeding disorders: Sepsis, Intrauterine death, HELLP syndrome.

Secondary/ late PPH: Hemorrhage occurring after 24 hours and up to 12 weeks postpartum. Causes include

- Placental polyp: Retained products undergo necrosis with fibrin deposition and eventually form a placental polyp, which on detachment from the myometrium causes brisk hemorrhage.
- Abnormal involution of the placental site.
- Retention of placental cotyledon or products of conception.
- Uterine artery pseudoaneurysm.
- Endometritis
- Coagulopathy

Resuscitative measures include

- Airway, breathing, circulation assessment
- A high concentration of oxygen (10–15 l/min) via a facemask
- Blood transfusion if clinically required
- Fluid resuscitation with warmed fluids up to 3.5 liters, starting with up to 2 liters of crystalloids such as normal saline and ringer's lactate. Colloids can be used when there is a need for packed RBCs transfusion, until blood is arranged.

Solution to Question 18:

The recommended HPV vaccination schedule for 9 to 14-year-olds according to WHO SAGE guidelines is 1 or 2 doses.

The recommendations for different age groups are as follows:

- 1 or 2 dose schedule for girls aged 9-14 years.

- 1 or 2 dose schedule for young women aged 15-20 years.
- 2 doses with a 6-month interval for women older than 21 years.
- 3 doses (or atleast 2 doses) for immunocompromised individuals.

As per WHO SAGE (Strategic Advisory Group of Experts on Immunization), a single-dose HPV vaccination schedule can provide comparable efficacy and durability of protection to a two-dose regimen. This can help in wider coverage of vaccination of all girls in low-middle income countries. Therefore, a 1 or 2-dose schedule is recommended for females aged between 9-20 years.

The available vaccinations for HPV are:

- Gardasil 9 (HPV 9) is a nonavalent vaccine active against HPV 6,11,16,18,31,33,45,52,58.
- Gardasil 4 (HPV 4) is a quadrivalent vaccine against 6,11,16,18. This is now replaced by HPV 9.
- Cervarix (HPV 2) is a bivalent vaccine active against 16, 18.

Solution to Question 19:

All the above coagulation factors (Factors II, VII, VIII) are increased in pregnancy except factor XI. Pregnancy is a hypercoagulable state. This controls bleeding and helps in hemostasis after placental separation.

Hematological changes in pregnancy:

- All clotting factors are increased, except factors 11 and 13. The levels return to normal within 2 weeks of delivery.
- Anticoagulants protein C, protein S, and antithrombin are decreased during pregnancy.
- Fibrinogen levels are increased by 50%.
- Clotting time and bleeding time remain unaffected.
- The viscosity of blood decreases due to relative hemodilution (relatively higher increase in plasma volume as compared to the increase in RBC mass).
- ESR is raised due to an increase in fibrinogen and globulin levels.
- Because of a marked fall in albumin level and only a slight rise of globulin, the normal albumin:globulin ratio of 1.7:1 is decreased to 1:1.
- Platelet counts remain the same or fall by a small amount (15%).

The most common cause of decreased platelet count in pregnancy is benign gestational thrombocytopenia. This is due to hemodilution and increased platelet consumption and the women are generally asymptomatic.

Solution to Question 20:

The incorrect statement regarding the management of acute pyelonephritis in pregnancy is option D. The patient can be discharged once afebrile for 24 hours, and antibiotics have to be continued

for 7 to 10 days.

Acute Pyelonephritis is one of the frequent and serious medical complications occurring in 2nd-trimester of pregnancy. The most common causative organism is E.coli.

Clinical features include sudden onset fever, chills, and aching pain in the lumbar region (usually unilateral). It is associated with nausea, vomiting, and anorexia. Tenderness in the costovertebral angle can be elicited on examination.

Diagnosis is confirmed by the collection of urine (preferably from catheterization) and analysis to look for leukocytes and bacteria.

Management protocol for pyelonephritis during pregnancy is

- Hospitalization is recommended for all patients. Very rare cases of uncomplicated pyelonephritis can be managed on an outpatient basis
- Urine and blood (if temperature is ≥ 39 degrees C) samples are collected for culture.
- Aggressive intravenous hydration to maintain adequate urine output is the mainstay of treatment. A urinary output ≥ 50 mL/hr is maintained with intravenous crystalloids.
- Vital signs such as blood pressure, pulse, temperature, and oxygen saturation along with urine output should be monitored for signs of worsening sepsis.
- Empirical intravenous antibiotic therapy is started with ampicillin plus gentamicin, cefazolin or ceftriaxone. It is switched to oral antibiotics once the patient is afebrile.
- In case of persistent infection or lack of clinical improvement by 48 to 72 hours, renal sonography is recommended to look for urinary tract abnormalities (like obstruction or calculi).
- The patient is discharged when afebrile for 24 hours and antimicrobial therapy is continued for 7 to 10 days.
- Urine culture is repeated 1–2 weeks after antimicrobial therapy is completed.

Solution to Question 21:

All of the following options are the diagnostic criteria for preterm labor except an posterior position of cervix (Option D). An anterior cervix favors a higher bishop's score, which means a higher chance of preterm labor as compared to the posterior cervix. The midline position of the cervix has also shown higher incidence of preterm labor as compared to posterior cervix.

Preterm labor (PTL) is defined as labor occurring between 20weeks to 37th completed week of gestation.

Diagnostic criteria of preterm labour include:

- Regular uterine contractions at least one in every 10 minutes, or 4 contractions in 20 minutes or 8 contractions in 60 min with/without pain.
- Cervix dilatation ≥ 2 cm
- Cervix effacement $\geq 80\%$
- Length of cervix (on TVS) ≤ 2.5 cm and funneling of the internal os ≥ 1 cm

- Pelvic pressure, backache, vaginal discharge, or bleeding PV

Predictors of preterm labor:

- Clinical predictors:
 - History of prior preterm birth
 - Multiple pregnancies
 - Active genital tract infection
- Biophysical predictors:
 - Uterine contractions ≥ 4 /hour
 - Bishop score ≥ 4
 - Cervical length (on TVS) ≤ 25 mm
- Biochemical predictors:
 - Fetal fibronectin (fFN) positive (> 50 ng/mL) in cervico-vaginal discharge
 - Increased levels of IL-6, IL-8, TNF-alpha

Solution to Question 22:

The false statement about late decelerations among the given options, is option C- at least 10 beats, 2 times in a 20 min period above baseline.

Typically, a late deceleration causes a fall in the fetal heart rate (FHR) by a depth of < 10 to 20 bpm below the baseline. It is usually not accompanied by accelerations (Option D).

Late decelerations are smooth, gradual, (Option B) symmetrical decline in the FHR that begins at or after the peak of uterine contraction and ends only after the contraction has ended (Option A).

It is usually indicative of uteroplacental insufficiency and fetal hypoxia. Other causes of late decelerations include hypotension from epidural anesthesia, excessive use of oxytocin, maternal hypertension, diabetes, and placental abruption.

A late deceleration plus decreased variability in FHR is predictive of fetal acidemia and neonatal morbidity.

The given CTG shows late decelerations and it's relation with the uterine contraction.



The below CTG shows early decelerations. In early decelerations, the onset, its nadir and the recovery of deceleration coincide with the beginning, peak, and ending of uterine contraction respectively.

Early decelerations are seen in fetal head compression. They typically cause a fall of ≤ 20 to 30 bpm below the baseline. These early decelerations are common during active labour and are benign in nature.



A variable deceleration is a drop in the FHR that begins with the uterine contractions onset and reaches a nadir in ≤ 30 seconds. Its depth is ≥ 15 bpm in and the deceleration lasts between 15 seconds to 2 minutes.

The slope of FHR change is erratic, giving the wave form its jagged shape. A variable deceleration is indicative of umbilical cord occlusion.

The below CTG depicts variable deceleration.



Solution to Question 23:

The age group distribution for vulval cancer is 55 to 85 years (Option C).

Vulval cancer is a malignancy of elderly women, usually in their 6th and 7th decades of life. It can also be seen in younger women who have been diagnosed with sexually transmitted infections such as HIV and HPV infections. The average age of diagnosis is 65 years.

Persistent HPV, HSV infections, smoking, chronic immunosuppression and chronic inflammation such as lichen sclerosis are risk factors. Vulvar intraepithelial neoplasia (VIN) is the preinvasive condition of the vulva.

Squamous cell carcinoma is the most common type of vulvar cancer. The most common site of vulvar cancer is labia majora > Clitoris > Labia Minora.

Vulval cancer generally presents with vulvar pruritus, vulvar swelling, a lump or an ulcer. Pain is usually a late feature of the disease.

The lump is seen as a pigmented papular raised area and the ulcers often have everted margins and usually bleed and cause an offensive vulval discharge.

The most common type of spread in vulval cancer is the lymphatic route. The first lymph node to be involved is the superficial inguinal lymph node and hence, the sentinel lymph node biopsy is helpful for evaluating the spread of vulval cancer and helps attain an adequate surgical procedure with a good prognosis.

Treatment of vulval cancer generally involves simple or radical, partial or total vulvectomy for the management of the primary tumour and inguino-femoral lymph node dissection for the involvement of the lymph nodes. This is decided based on the staging of the disease.

Solution to Question 24:

Among the given options, Extraplacental membranes and 2-veesled umbilical cord present (Option C) is the wrong statement.

At term, the normal placenta weighs 470 g (Option A), is round to oval with a 22-cm diameter (Option B) and has a central thickness of ≥ 2.5 cm (Option D). The placenta is composed of a placental disc, extraplacental membranes, and a three-vessel umbilical cord. Hence, option C is the wrong statement.

The normal human placenta is discoid, deciduate, and hemochorial.

The term deciduate means that the placenta is shed at the time of parturition. Hemochorial means that the maternal blood directly comes in contact with the fetal trophoblasts.

The placenta consists of two plates. The chorionic plate lies in the fetal surface. It is lined by the amniotic membrane and the umbilical cord is attached to this plate.

The basal plate is the maternal aspect and lies against the uterine wall. It is divided by clefts into portions termed as cotyledons.

Solution to Question 25:

The correct sequence of dilation and curettage (D&C) is 4-3-1-2.

4 - Assessment of uterine size and direction

3 - Uterine sound

1 - Serial dilation

2 - Curettage

Indications:

- Infertility
- Abnormal uterine bleeding (AUB)
- Amenorrhea
- Endometrial tuberculosis
- Endometrial carcinoma
- Postmenopausal bleeding
- Endometrial polyp
- Removal of IUD
- Incomplete abortion
- Molar pregnancy

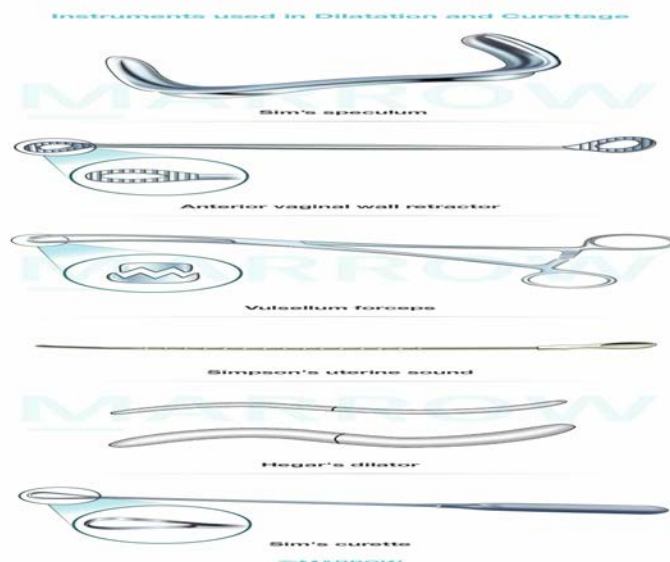
Absolute contraindication: Viable intrauterine pregnancy.

Steps:

- The patient's bladder is emptied prior to the procedure.

- The operation is performed under general anesthesia or under diazepam sedation with or without a paracervical block.
- With the patient in a lithotomy position, local antiseptic cleaning and draping are carried out.
- A bimanual examination is conducted.
- A posterior vaginal speculum is introduced.
- The anterior lip of the cervix is grasped with an Allis tissue forceps.
- A uterine sound is introduced to confirm the position and note the length of the uterocervical canal.
- The cervical canal is dilated with graduated dilators.
- When the dilator is introduced, the cervix is made steady by traction of the vulsellum (Allis's tissue forceps).
- After the desired dilatation, the uterine cavity is curetted by a uterine curette either in a clockwise or anticlockwise direction starting from the fundus down to the internal os.
- The vulsellum and the speculum are removed.
- The curetted material is preserved in 10% formol-saline, properly labeled, and sent for histological examination.

Complications: Infection, hemorrhage, cervical lacerations, uterine perforation, and post-operative uterine adhesions.



Solution to Question 26:

Female sterilization is an irreversible method of contraception.

It involves surgery and severing, destruction, or ligation of fallopian tubes through laparotomy or laparoscopy. Hence it is normally irreversible.

All other methods are reversible long-term methods of contraception.

Other options:

Option A: Progestin injections, such as NET-EN, are used both as a form of progestogen-only injectable birth control and as components of combined injectable birth control formulations. They are reversible.

Option C: Progestin implants, such as Nexplanon, are used for contraception; they are highly effective, safe, reversible, and the most convenient choice for many women.

Option D: IUCDs are a type of long-term, reversible contraception. They are T-shaped devices that are inserted into the uterine cavity. These can either be copper-containing like CuT-380A and CuT-200, or hormone-containing IUCDs like mirena.

Solution to Question 27:

The WHO recommends a minimum of 8 antenatal visits to have a positive pregnancy outcome.

According to the latest WHO guidelines the minimum number of antenatal visits has been increased from 4 to 8. This increased number of visits significantly reduces the risk of perinatal deaths and enhances maternal satisfaction.

The ideal number of antenatal visits is 12-15:

- Monthly visits up to 28 weeks
- Two weekly visits between 28-36 weeks
- Weekly visits from 36 weeks onwards

WHO ANC model	
First trimester	Contact 1: up to 12 weeks
Second trimester	Contact 2: 20 weeks Contact 3: 26 weeks
Third trimester	Contact 4: 30 weeks Contact 5: 34 weeks Contact 6: 36 weeks Contact 7: 38 weeks Contact 8: 40 weeks
Return for delivery at 41 weeks if not given birth.	

Solution to Question 28:

Gestational age or period of gestation (POG) refers to the duration of pregnancy calculated from the first day of the last menstrual period (LMP). It is usually expressed in weeks.

According to Naegele's rule, the estimated due date (EDD) is obtained by adding 7 days and subtracting 3 months from the first day of the last menstrual period (LMP). Another method

involves adding 9 months and 7 days to the LMP. Less than 5% of births follow Naegele's rule. It is based on a 28-day regular cycle.

If the cycle is shorter/longer than 28 days, corrected EDD is calculated as follows:

- For a 35-day cycle, add another 7 (35-28=7) days to the calculated EDD to get the corrected EDD.
- For a 25-day cycle, subtract 3 (28-25=3) days from the calculated EDD to get the corrected EDD.

Other options:

Option A and D: The best method of assessing the gestational age is by measuring the crown-rump length (CRL) using ultrasonography between 7 to 12 weeks of gestation. It is measured from the vertex to the coccyx. $CRL \text{ (in mm)} + 6.5 = \text{gestational age in weeks}$.

Option C: In an IVF pregnancy, the day of embryo transfer/day of fertilization is needed to calculate POG.

Solution to Question 29:

Extra-amniotic instillation of ethacridine lactate cannot be used in the first trimester. It is generally safe to use in the second trimester.

Intrauterine instillation of hyperosmotic solutions is usually employed in the second trimester:

- Extra-amniotic ethacridine lactate, prostaglandins (PGE₂, PGF₂α)
- Extra-amniotic saline infusion (isotonic) with a transcervical catheter balloon
- Intra-amniotic hypertonic urea (30%), saline (20%)

The various methods of termination of pregnancy are:

Timing	Medical methods	Surgical methods
1st trimester	Mifepristone Mifepristone and misoprostol Methotrexate and misoprostol Tamoxifen and misoprostol	Menstrual regulation Vacuum aspiration (MVA/EVA) Suction evacuation and/or curettage Dilatation and evacuation
2nd trimester	Misoprostol Mifepristone and misoprostol Dinoprostone Carboprost	Dilatation and evacuation Intrauterine instillation of hyperosmotic solutions Oxytocin infusion Hysterotomy

Solution to Question 30:

Loss of knee jerk reflex is the first sign of MgSO₄ toxicity.

MgSO₄ is used as an anticonvulsant in the management of eclampsia and for prophylaxis in severe preeclampsia. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L. Measurement of serum magnesium levels is indicated to monitor magnesium toxicity if

serum creatinine ≤ 1.0 mg/dL.

Loss of knee jerk reflex is the first sign of MgSO_4 toxicity and occurs when the plasma magnesium level reaches 10 mEq/L.

Other signs suggestive of magnesium toxicity:

- Decreased respiratory rate ≤ 16 /min
- Urine output ≤ 30 mL/h
- Chest pain and heart block
- Visual changes, flushing, muscle paralysis, and drowsiness

At ≤ 10 mEq/L, breathing weakens. At ≥ 12 mEq/L respiratory paralysis and respiratory arrest follow. At ≥ 25 mEq/L cardiac arrest occurs.

Magnesium therapy should be stopped and a slow infusion of IV calcium gluconate 10 mL (10% solution) is given. Fluid loading and forced diuresis with estimation of serum magnesium and creatinine levels are done.

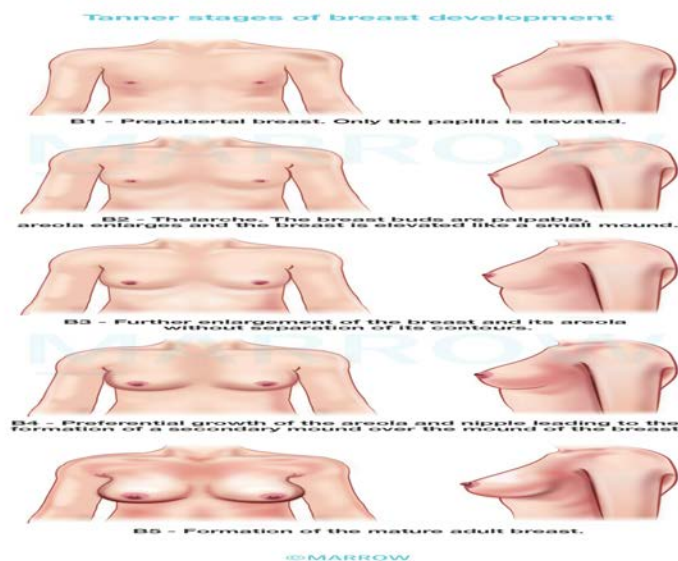
Repeat injections of magnesium sulfate are continued only if the patellar reflex is present, respiratory rate is ≥ 12 , and urine output is ≥ 30 mL/h.

Solution to Question 31:

A 14-year-old girl with Tanner stage 1 breasts would be classified as having primary amenorrhea.

Primary amenorrhea is defined as the absence of menstruation by 14 years without the development of secondary sexual characteristics or by 16 years regardless of the presence of secondary sexual characteristics.

Tanner stage 1 breasts correspond to prepubertal breasts. Breast development beyond Tanner stage 2 in girls indicates the onset of puberty.



Solution to Question 32:

The daughter of a woman who received diethylstilbestrol (DES) during her pregnancy is at a high risk of clear cell carcinoma of the vagina. Clear cell carcinoma is a type of adenocarcinoma and hence is the most specific answer.

If the female fetus is exposed to diethylstilbestrol (DES) taken by the mother during pregnancy, columnar epithelium appears in the upper two-thirds of vaginal mucosa, which can develop into vaginal adenosis and clear cell adenocarcinoma during adolescence.

Other options:

Option A: Squamous cell carcinoma arises from non-keratinized stratified epithelium and is commonly associated with human papilloma virus exposure.

Option C: Adenocarcinoma usually arise from cells of glandular tissue. It can be seen in an endometriosis foci, areas of vaginal adenosis, periurethral glands and remnants of the wolffian ducts. Vaginal adenocarcinomas occur as metastatic cancers in most patients and are rarely primary cancers. Metastases may be spread from the endometrium, breast, pancreas, kidneys and colon.

Option D: Sarcoma botryoides is the most common type of embryonal rhabdomyosarcomas in infants and children.

Solution to Question 33:

Daily fetal movement count is not a component of biophysical profile (BPP). Components of BPP are non-stress test (NST), fetal breathing movements, gross body movements, fetal muscle tone, and amniotic fluid volume.

BPP is a multi-parametric screening test to assess uteroplacental insufficiency. The observations are made in a 30-minute time period.

Each parameter has a maximum score of 2 and a minimum score of 0. So the total maximum score will be 10. A score of 4 or less indicates fetal acidemia.

Interpretation and management of BPP scores:

BPP is directly related to fetal pH and fetal outcome. A lower BPP score is associated with a poor outcome (stillbirth, perinatal mortality) and lower APGAR scores at birth.

The modified biophysical score has only 2 components: NST and amniotic fluid index.

BPP score	Interpretation	Management
10	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mothers or post-term pregnancy)

BPP score	Interpretation	Management
8 with normal AFV	Normal, nonasphyxiated fetuses	Repeat test weekly (or biweekly in diabetic mothers or post-term pregnancy)
8 with decreased AFV	Suspicion of chronic fetal asphyxia	Delivery
6	Possible fetal asphyxia	Deliver, if AFV is abnormal or > 36 weeks or if the repeat test score is ≤ 6. If the repeat score > 6, observation and repeat test weekly
4	Probable fetal asphyxia	Repeat testing same day If the repeat score ≤ 6, delivery
0-2	Almost certain fetal asphyxia	Delivery

Solution to Question 34:

Multiparity is not a cause of uterine dysfunction.

Adequate uterine contractions are required to dilate the cervix and expel the fetus. Hypotonic uterine dysfunction is a defect in the propulsive power of the uterus that results in prolonged labor. It is a common indication for cesarean section.

Causes of hypotonic uterine dysfunction:

- Neuraxial analgesia
- Chorioamnionitis
- Nulliparity
-

Mechanical disruption of myometrial function from leiomyoma or pelvic tumor

- Maternal obesity
- Uterine müllerian anomaly
- Increased maternal age
- Malposition and malpresentation of the fetus, with inadequate contact of the presenting part onto the lower uterine segment, as seen in the case of a high fetal head station

Solution to Question 35:

Human immunodeficiency virus is not a cause of nonimmune hydrops.

Hydrops is diagnosed by identifying two or more fetal effusions—pleural, pericardial, or ascites—or one effusion plus anasarca. Hydrops can be seen in association with red cell auto

immunization, which is termed immune hydrops. Otherwise, it is nonimmune hydrops.

Pathogenesis of hydrops includes decreased colloid oncotic pressure, increased hydrostatic or central venous pressure, and enhanced vascular permeability. Sonographically measured skin thickness of ≥ 5 mm constitutes edema or anasarca.

Etiology of nonimmune hydrops fetalis:

Cardiovascular

- Structural defects:
- Ebstein anomaly
- Tetralogy of Fallot with absent pulmonary valve
- Hypoplastic left or right heart
- Premature closure of ductus arteriosus
- Arteriovenous malformation (vein of Galen aneurysm)
- Cardiomyopathies
- Tachyarrhythmias
- Bradycardia

Chromosomal

- Turner syndrome (45,X), triploidy
- Trisomies 21, 18, and 13

Hematological

- Hemoglobinopathies, such as α -thalassemia
- Decreased erythrocyte production (myeloproliferative disorders)
- Fetomaternal hemorrhage

Lymphatic Abnormalities

- Cystic hygroma
- Systemic lymphangiectasis
- Pulmonary lymphangiectasis

Infections

- Parvovirus B19
- Syphilis
- Cytomegalovirus
- Toxoplasmosis
- Rubella
- Enterovirus
- Varicella

- Herpes simplex
- Coxsackievirus
- Listeriosis
- Leptospirosis
- Chagas disease
- Lyme disease

Kidney and Urinary Tract

- Kidney malformations
- Bladder outlet obstructions
- Congenital (Finnish) nephrosis
- Bartter syndrome

Placental, Twin, and Cord Abnormalities

- Placental chorioangioma
- Twin-twin transfusion syndrome
- Twin reversed arterial perfusion sequence
- Twin anemia polycythemia sequence
- Cord vessel thrombosis

Idiopathic

Solution to Question 36:

The correct statement regarding the recurrence of ovarian cancer is option C: It is classified as platinum-sensitive disease if recurrence occurs $\geq$ 6 months of primary therapy.

Gradual elevation of CA125 is considered the first sign of relapse of ovarian cancer. Tamoxifen is recommended for recurrent ovarian cancer if there is only biochemical evidence of the recurrence of ovarian cancer. Otherwise, conventional cytotoxic chemotherapy is administered.

Recurrence of ovarian cancer can be classified as platinum-refractory, platinum-sensitive, and platinum-resistant disease.

- Platinum-refractory disease includes women who progress during primary chemotherapy. It has a worse prognosis.
- Platinum-resistant disease includes women who relapse within 6 months of primary therapy. Palliative, single, non-platinum chemotherapy is recommended for platinum-refractory and platinum-resistant disease. Bevacizumab has been approved for the treatment of platinum-resistant disease. It can also be used in combination with weekly paclitaxel, doxorubicin, and topotecan.
- Platinum-sensitive disease includes women who relapse more than 6 months after primary therapy. These patients are treated with a platinum-based combination such as carboplatin and

paclitaxel or gemcitabine.

Solution to Question 37:

The given graph showing only the tracing for the fetal heart rate (FHR) is a non-stress test (NST). The NST is a screening test for fetal wellness. It describes fetal heart rate acceleration (measured by a doppler) in response to fetal movements (perceived by the mother).

- Reactive (reassuring) NST - Two or more accelerations of 15 seconds duration and amplitude of 15 beats/min, in a 20-minute testing time
- Non-reactive (non-reassuring) NST - Absence of any fetal reactivity.

Other options:

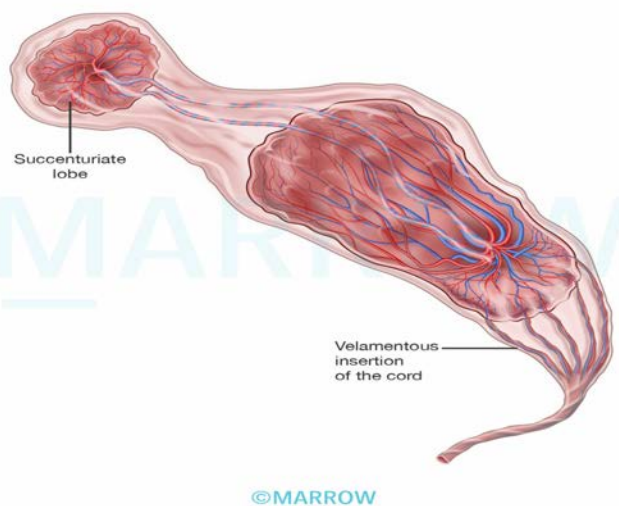
Option B: Fetal tachycardia is defined as a baseline FHR above 160 beats/min, whereas this tracing shows FHR below 160 beats/min.

Option C: Beat-to-beat variability is the change in baseline fetal heart rate (excluding the accelerations and decelerations) in response to sympathetic and parasympathetic stimulation. A beat-to-beat variability of 6-25 beats/min is considered normal. This tracing shows normal beat-to-beat variability.

Option D: Cardiotocography (CTG) includes tracings for fetal heart rate and uterine contractions.

Solution to Question 38:

The abnormal placenta shown in the image depicts both succenturiate lobe and velamentous insertion of the cord.



The umbilical cord is usually attached at or near the center of the placenta.

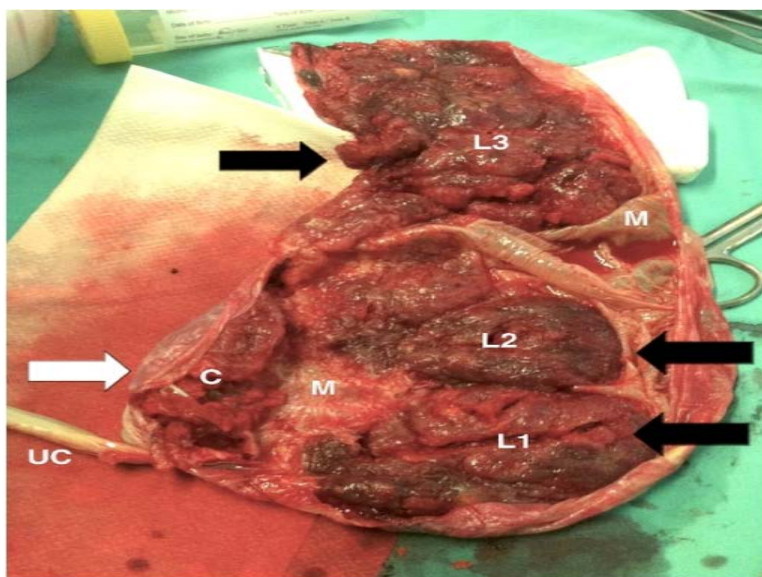
In the velamentous placenta, the cord is attached to the membranes. The cord vessels traverse between the membranes for some distance before reaching the placenta. If the leash of blood vessels traverses through the membranes overlying the internal os, in front of the presenting part, the condition is called vasa previa. The rupture of the membranes involving the overlying vessels leads to vaginal bleeding and fetal exsanguination.

The below image shows the velamentous placenta:



The succenturiate placenta is characterized by the presence of one or more small lobes of placenta placed at varying distances from the main placental margins. A leash of blood vessels connects these accessory cotyledons to the main placenta. If these vessels are absent, then it is called placenta spuria. If the succenturiate lobe is retained postpartum, there can be primary or secondary postpartum hemorrhage, subinvolution of the uterus, uterine sepsis, and polyp formation.

The below image shows a succenturiate placenta. L1, L2 and L3 depict smaller lobes, UC is the umbilical cord, and M is the intervening membrane.



Solution to Question 39:

Human chorionic gonadotropin (hCG) is the hormone secreted by the placenta.

hCG is a glycoprotein, produced by the syncytiotrophoblast of the placenta and secreted into the blood of both mother and fetus. It can be detected in the maternal serum or urine as early as 8–9 days postfertilization, with the help of radioimmunoassay. In early pregnancy, hCG concentrations in plasma double in 1.4–2 days.

Functions of hCG:

- Stimulates the secretion of progesterone by the corpus luteum: Rescue and maintenance of the corpus luteum till 6 weeks of pregnancy is the major biological function of hCG.
- Stimulate Leydig cells of the male fetus to produce testosterone.
- Stimulates both adrenal and placental steroidogenesis.
- Stimulates the maternal thyroid because of its thyrotropic activity.
- Promotes secretion of relaxin from the corpus luteum.

High levels of hCG for that particular gestational age are detected in multiple pregnancies, hydatidiform mole or choriocarcinoma, and in pregnancy with a trisomy 21 fetus (Down's syndrome). It is low in cases of ectopic pregnancy or abortion.

Other options:

Option A: LH is secreted by the gonadotrope cells of the anterior pituitary.

Option C: FSH is also secreted by the gonadotrope cells of the anterior pituitary.

Option D: Gonadotropin-releasing hormone (GnRH) regulates the synthesis and secretion of both the hormones, LH and FSH.

Solution to Question 40:

Painless vaginal bleeding is not a feature of abruption of the placenta.

Patients with placental abruption present with sudden-onset abdominal pain, vaginal bleeding, uterine tenderness, and frequent contractions. Painless vaginal bleeding can occur in placenta previa where the placenta is implanted in the lower uterine segment over or very close to the cervical os.

Abruptio placenta is the partial/total separation of a normally implanted placenta before delivery.

The bleeding may be

- Revealed (blood dissects downwards between membranes and uterine wall to escape through the cervix), or
- Concealed (blood accumulates between detached placenta and uterus), or

- Mixed

Risk factors for abruptio placenta include prior abruption history, preterm premature rupture of membranes, and pregnancy-associated hypertension. Depending upon the degree of placental abruption and its clinical effects, the cases are graded as follows:

Uteroplacental apoplexy/Couvella's uterus is a complication of severe concealed abruptio placentae. There is massive intravasation of blood into the uterine musculature up to the serous coat. It is only diagnosed on laparotomy, where a port-wine-colored uterus (diffuse or patchy) is seen. The presence of Couvella's uterus alone is not an indication of hysterectomy since it does not interfere with uterine contractions.

Other complications of abruption of placenta:

- Disseminated intravascular coagulation
- Hypovolemic shock due to torrential bleeding
- Acute kidney injury

Abruptio placenta is mostly managed by immediate delivery and management of complications such as shock with adequate blood products.

Grade 1	Grade 2	Grade 3
Slight vaginal bleeding	Mild to moderate vaginal bleeding	Bleeding is moderate to severe or concealed.
Minimal uterine tenderness	Uterine tenderness is always present	Marked uterine tenderness
Maternal BP and fibrinogen will be unaffected	Increased maternal pulse, but BP is maintained, fibrinogen levels may be decreased	BP decreases, associated coagulation defect or anuria may complicate
FHS good	Fetal distress present	Fetal death occurs
Maternal shock absent	Maternal shock absent	Pronounced shock

Solution to Question 41:

The signs that can be elicited at 8 weeks of gestation are Goodell's sign and Oslander's sign.

Goodell's sign is a sign of early pregnancy, which describes the marked softening of the cervix in contrast to the non-pregnant state. It is seen as early as the 6th week.

Oslander's sign describes the pulsations felt in the lateral fornix due to increased vascularity. It is seen around 8 weeks of gestation.

Internal ballottement can be elicited between 16–28th weeks of gestation, before which the fetus is too small and after which it is too large to displace. It may not be elicited in cases with scanty liquor amnii, or when the fetus is transversely placed.

Lightening is a sense of relief of the pressure symptoms due to the engagement of the presenting part. It is felt at about 38 weeks of gestation.

Solution to Question 42:

The non-contraceptive uses of oral contraceptives include all except protection against sexually transmitted diseases (STDs), including HIV.

Oral hormonal contraceptives can be combined oral contraceptive pills (COCPs) or progesterone-only minipills.

COCPs are a combination of ethinylestradiol at a dose of 20–30 mcg and an orally active progestogen. It is taken starting on the first day of the cycle for 21 days. They are taken at a fixed time of the day, preferably after a meal. Mala-N and Mala-D both contain levonorgestrel (0.15 mg) + ethinylestradiol (30 mcg). Mala-N is supplied free of cost in Family Planning Clinics in India. Each strip contains 21 hormonal tablets and 7 non-hormonal (iron) tablets.

The mechanism of action is the prevention of ovulation by suppression of the pituitary hormones FSH and LH. Progesterone also acts on the cervical mucus, making it thick and impenetrable by sperm. The pregnancy rate with the COCP is 0.3 per 100 woman-years.

Benefits of the COCP:

- It is effective in controlling fertility.
- It causes regular cycles and reduces menstrual blood loss and are, therefore, useful in menorrhagia and polymenorrhea.
- It lowers the incidence of benign breast neoplastic diseases, such as fibrocystic disease.
- It reduces the incidence of functional ovarian cysts and ovarian (Option A) and endometrial malignancy (Option D). The incidence of ovarian cancer is reduced by 40% and that of uterine malignancy, by 50%, if taken for 1 year; this effect lasts as long as 10 years after stoppage.
- It reduces the incidence of ectopic pregnancy (Option B) by suppressing ovulation and reducing the risk in pelvic inflammatory disease.
- It is useful in endometriosis, acne, and polycystic ovarian disease (PCOD).

Side effects:

- Higher incidence of breast cancer reported among users
- Suppression of lactation
- Occurrence of pituitary adenoma and adenoma of the liver
- Reduced carbohydrate tolerance (therefore, contraindicated in diabetic women)
- Migraine, depression, weight gain, and lethargy (due to progesterone)
- Increased risk of pulmonary embolism and cerebral thrombosis

Ninety-nine percent of women will regain normal menstrual cycles within 6 months of stopping the pills, while the return of fertility is slightly delayed.

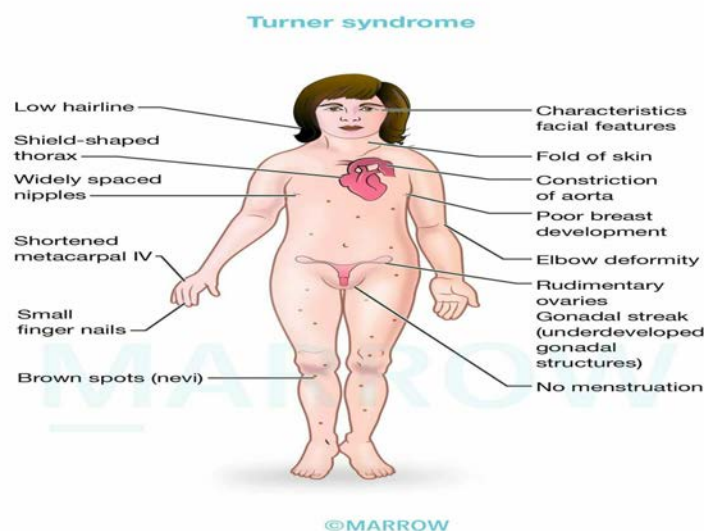
Barrier methods like condoms prevent the transmission of STDs.

Solution to Question 43:

The patient's clinical features are strongly suggestive of Turner syndrome.

Turner's syndrome is the most common sex chromosome abnormality in females. The classical karyotype is 45, XO. It is characterized by hypogonadism in phenotypic females.

The clinical features of Turner's syndrome include short stature, short webbed neck, shield chest with widely spaced nipples, primary amenorrhea, congenital heart defects, and renal anomalies. Two X chromosomes are required for the development of ovaries. The loss of one X chromosome results in streak ovaries with inadequate estrogen production. The loss of negative feedback from reduced estrogen raises the FSH and LH levels causing hypergonadotropic hypogonadism.



Other options:

Option B: Kallmann syndrome presents with delayed puberty due to hypogonadotropic hypogonadism from deficiency of GnRH. It is associated with olfactory dysfunction and anosmia.

Option C: Androgen insensitivity syndrome occurs due to a defect in the androgen receptor in a genotypic male (XY). These individuals are phenotypic females with no male or female internal genitalia. They have normal female external genitalia with an absent or shallow vagina, well-developed breasts, and scanty body hair (axillary and pubic) and are present with primary amenorrhea.

Option D: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome, commonly referred to as Mullerian agenesis, has primary amenorrhea due to failure of development of the uterus, cervix, and upper vagina. They have normally functioning gonads and external genitalia.

Solution to Question 44:

The correct sequence in AMTSL is 3,2,4,1.

AMTSL is recommended by the WHO to prevent postpartum hemorrhage (PPH). Uterotonic administration is the most important component in AMTSL. However, even before a uterotonic is administered, the birth attendant will often place their hand over the uterus to ensure that there is no undiagnosed twin.

The currently recommended components of AMTSL are as follows:

Note: Uterine massage or fundal massage is not recommended as an intervention to prevent PPH in women who have already received prophylactic uterotonics such as oxytocin (core component of AMTSL). Hence, it is now an optional component of AMTSL that is done in patients with uterine atony as a part of the management of PPH.

Component		Remarks
1.	Injecting uterotonics within 1 minute of delivery of the baby or immediately after delivery of the anterior shoulder	Most important step Oxytocin 10 IU im/iv infusion is the preferred uterotonic. (Not recommended by WHO-carboprost)
2	Delayed cord clamping	Done within 1-3 mins of delivery 80 ml of additional blood goes to the fetus Prevents fetal anemia
3	Delivery of placenta by controlled cord traction: Modified Brandt Andrews technique	Done only in settings where skilled birth attendants are available The uterus is pushed upwards and backward The traction of cord is downwards and forwards
4	Intermittent assessment of uterine tone	Done every 15 mins for 2 hours

Solution to Question 45:

The specimen showing an ovarian mass with a tuft of hair in a cystic space is a mature cystic teratoma.

Teratomas are germ cell tumors. They contain embryonic tissues leading to the differentiation of cells along these pathways. There are three classes of teratomas:

- Mature (benign) teratomas (e.g.: dermoid cyst)
- Immature (essentially malignant) teratomas (e.g.: solid teratoma)
- Monodermal (highly specialized) tumors (e.g.: struma ovarii)

Mature cystic teratomas, also known as dermoid cysts are unilocular, containing sebaceous material and hair. They have a smooth surface and are lined by squamous epithelium. Other structures like teeth, bone, cartilage, thyroid tissue, and neural tissues can also be found.

Most patients are asymptomatic. Common presentations include abdominal mass, acute pain due to torsion of the ovary, and rarely due to rupture of the cyst with spillage of contents into the

abdomen.

Treatment of mature teratomas involves excision in order to prevent complications.

Option A - Immature teratomas of the ovary are solid and trabeculated. They also contain cartilage, bone, hair, and sebaceous material. There is no structural order. They are generally malignant. Grossly, they have a smooth external surface and are solid on sectioning. There may be areas of necrosis or hemorrhage.

The following image shows both mature (left) and immature teratomas (right).



Option C- Mucinous cystadenoma is a benign tumor showing multicystic appearance, delicate septa, and the presence of glistening mucin within the cyst.

Option D- Serous cystadenoma have multicystic lesion in which papillary epithelium is contained within fibrous walled cysts or as a mass projecting from the ovarian surface.

Solution to Question 46:

Among the given options, placenta previa shows warning bleeds.

Dark painful bleeding with a tender uterus that is disproportionately enlarged is seen in placental abruption.

Placenta previa is a condition where the placenta is implanted in the lower uterine segment either over or close to the internal cervical os.

Risk factors for placenta previa

- Multigravida
- Age $\geq$ 35 years
- Smoking
- Previous cesarean section or myomectomy

- Abnormal placental size and lobes
- Prior curettage

Clinical features of placenta previa:

- Sudden-onset recurrent painless vaginal bleeding: First bleeding usually appears after the end of the second trimester or later and is usually mild and not fatal. This is called 'Sentinel bleed'. In about one third of cases, there is a history of 'warning hemorrhage' which is usually slight.
- The size of the uterus corresponds to gestational age.
- The uterus is relaxed, soft, no tenderness
- Floating head
- Fetal heart sound is present unless there is major separation and the patient is exsanguinated.
- Slowing of fetal heart rate on pressing the head down into the pelvis which recovers promptly as pressure is relieved is suggestive of a low-lying placenta especially of posterior type: Stallworthy sign.

Degree of placenta previa:

There are four types of placenta previa depending upon the degree of extension of the placenta to the lower segment

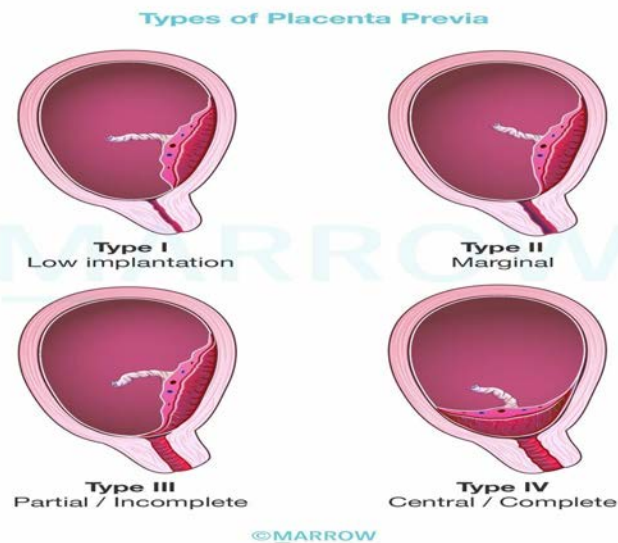
- Type—I (Low-lying): The major part of the placenta is attached to the upper segment and only the lower margin encroaches onto the lower segment but not up to the internal os.
- Type—II (Marginal): The placenta reaches the margin of the internal os but does not cover it.
- Type—III (Incomplete or partial central): The placenta covers the internal os partially (covers the

internal os when closed but does not entirely do so when fully dilated).

- Type—IV (Central or total): The placenta completely covers the internal os even after it is fully dilated.

Per-Vaginal Examination should not be done in suspected Placenta Previa, whereas it can be only done in the operation theatre with all the arrangements for C-section.

The image below shows the types of placenta previa.



Expectant management for placenta previa was advocated by Macafee and Johnson. The aim is to continue the pregnancy for fetal maturity without compromising maternal health.

Prerequisites

- Availability of blood for transfusion whenever required
- Facilities for cesarean section should be available 24 h.

Suitable cases for expectant management for placenta previa:

- Mother in good health (Hb ≥ 10 g%; hematocrit $\geq 30\%$)
- Duration of pregnancy < 37 wk
- No active vaginal bleeding
- Fetal well-being is assured (USG)

Conduct of expectant treatment

- Corticosteroids for fetal lung maturity if < 34 wk
- Betamethasone 12mg IM 24 hours apart for 2 doses or
- Dexamethasone 6mg IM 12 hours apart for 4 doses
- Bedrest
- Hb, Blood grouping and urine for protein
- Inspection of vulval pads, fetal surveillance with USG
- Supplementary hematinics
- Rh Immunoglobulin should be given to all Rh-negative (unsensitized) women
- Use of tocolytics if there are uterine contractions.

The pregnancy is carried till 37 weeks as part of expectant management. During this time, if there is fetal distress or a bleeding episode, then terminate immediately.

Solution to Question 47:

The pregnant woman exhibiting features of severe pre-eclampsia (BP of 170/110 mmHg, epigastric pain, headache, and blurring of vision) and in active labor should be managed by treating hypertension, administering MgSO₄ (magnesium sulfate), and considering termination of pregnancy.

Preeclampsia is defined as hypertension with blood pressure $\geq 140/90$ mm Hg along with proteinuria that develops after 20 weeks of gestation in a previously normotensive and non-proteinuric woman. It can be classified as severe if any of the following are present:

- BP $\geq 160/110$
- Headache
- Visual disturbance
- Upper abdominal pain
- Elevated creatinine
- Thrombocytopenia ($< 1,00,000$ cells per mm³)
- Pulmonary edema
- Marked elevation in serum transaminases

This patient has many features of severe pre-eclampsia and is in active labor. The best management for her is to admit her and bring down her BP using labetalol, nifedipine, or hydralazine and give seizure prophylaxis with magnesium sulfate in the same way as in eclampsia. Since the placenta is the main pathology causing pre-eclampsia, termination of pregnancy and delivery of the placenta is the definitive treatment.

Solution to Question 48:

Vulval hamartoma is not a risk factor for vulval cancer.

Vulval cancer is a rare genital tract cancer that may be divided into preinvasive lesions and invasive carcinoma. 90% of the invasive carcinomas are squamous cell carcinoma and are usually present on the labia majora. The risk factors for both types are:

- Chronic vulval irritation
- Immunosuppressive conditions such as pregnancy, HIV infection
- Smoking
- Condyloma (anogenital warts)
- Sexually transmitted diseases
- Dystrophies

Solution to Question 49:

An upper segment cesarean section is done in case of perimortem cesarean section.

Perimortem cesarean section is an emergency cesarean delivery of a woman who is expected to die within the next few moments or has just died. AHA 2020 Guidelines for CPR recommend that perimortem cesarean should be considered if there is no return of spontaneous circulation in 5 minutes in a pregnant woman who has experienced total cardiopulmonary arrest.

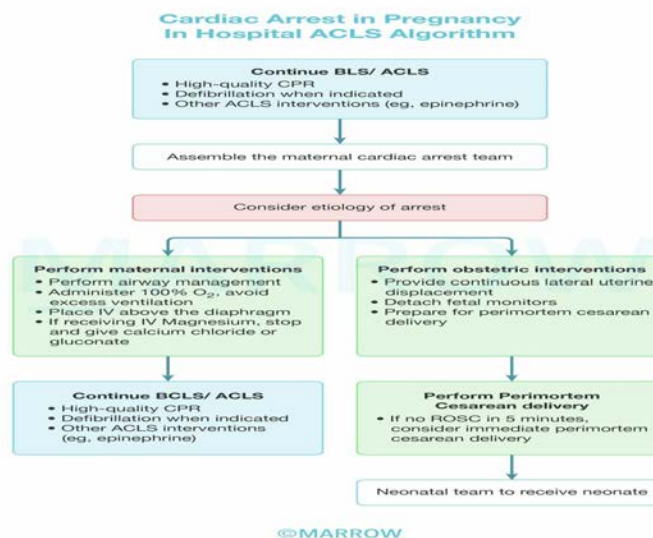
The goal of perimortem cesarean section (PMCS) is to improve maternal and fetal outcomes. PMCS improve maternal survival by decreasing compression of the IVC by the gravid uterus. This improves venous return and diaphragmatic displacement, which in turn improves respiratory dynamics. The infant may survive if PMCS is done within 10 minutes of maternal death. A classical upper-segment cesarean section with a vertical uterine incision is done.

Other options:

Option A: Upper segment cesarean section is done, not lower segment cesarean section (LSCS).

Option B: A vertical incision from the xiphoid to the symphysis pubis is done on the abdomen and not the lower segment Pfannenstiel incision.

Option D: There is no standardized guideline about where perimortem cesarean section must be performed.



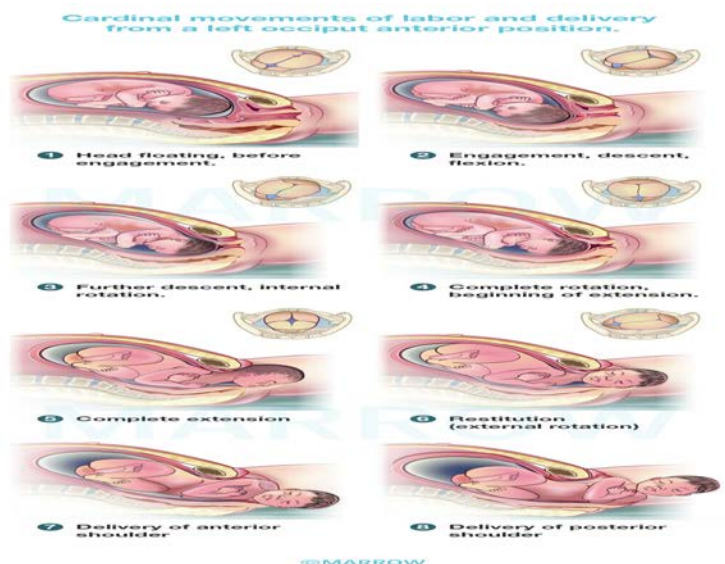
Solution to Question 50:

The cardinal movements of labor in order are engagement-crowning-restitution-external rotation.

The specific cardinal movements of labor are:

- Engagement is the mechanism by which the biparietal diameter (in occiput presentation) passes through the pelvic inlet.
- Descent is a continuous process that is completed with the expulsion of the fetus.

- Flexion of the head occurs as soon as the descending head meets resistance from the pelvic floor/pelvic walls/cervix.
- Internal rotation is the mechanism by which the occiput gradually turns away from the transverse axis. Crowning occurs after internal rotation.
- Crowning occurs when the vulval outlet is stretched by the maximum diameter of the head and there is non recession of the head after the contraction is over.
- Extension occurs when the flexed head reaches the vulva.
- Restitution is the passive movement of the head due to untwisting of the neck that occurred in internal rotation.
- External rotation is the visible movement of the delivered head occurring due to internal rotation of the shoulders.
- Expulsion is the process by which the anterior shoulder comes under the pubic symphysis to be delivered and is followed by the posterior shoulder and the rest of the fetal body.



Solution to Question 51:

The correct match is A-4, B-1, C-2, D-3

A table differentiating between the invasive prenatal diagnostic techniques is given below.

Fetoscopy is a type of micro laparoscopic technology that aids in fetal diagnosis and therapeutic interventions.

A. Amniocentesis	Amniotic fluid assessed for detection of AFP.
B. Chorionic villous sampling	Done at 8-12wks
C. Fetal blood sampling	Done at 19-20 wks

A. Amniocentesis	Amniotic fluid assessed for detection of AFP.
D. Fetoscopy	Fetal organs visualized

	Chorionic villus sampling	Amniocentesis	Cordocentesis
Time	Transcervical 10–13 weeks, Transabdominal 10 weeks to term	After 15 weeks (early 12–14 weeks)	18–20 weeks
Materials for study	Trophoblast cells	Fetal fibroblasts, fluid for biochemical study	Fetal white blood cells (others - infection and biochemical study)
Karyotype result	Direct preparation: 24–48 hours. Culture: 10–14 days	Culture: 3–4 weeks	Culture: 24–48 hours
Fetal loss	0.5–1%	0.5%	1–2%
Accuracy	Accurate; may need amniocentesis for confirmation	Highly accurate	Highly accurate

Solution to Question 52:

Vagitus uterinus is the cry of the baby in the womb.

Vagitus uterinus is the cry of an in-utero baby after the rupture of membranes, during the process of labor but before it has been delivered. In one such instance, it was described as a clear remote cry, much like a baby crying in the next room but close below their heads.

Obstetrics and Gynecology NEET 2022

Question 1:

A pregnant lady with 34 weeks of amenorrhea has the following findings: LDH - 700 IU/L, platelets - 75,000/mm³, serum bilirubin - 1.5 mg/dL, SGOT - 200 U/L, SGPT - 150 U/L, and BP - 140/96 mm Hg. Her coagulation profile and renal function tests are normal. What is the diagnosis?

- a) HELLP syndrome
- b) Acute fatty liver of pregnancy
- c) Viral hepatitis
- d) Intrahepatic cholestasis

Question 2:

A woman at 26 weeks of gestation presents for routine evaluation. On examination, fundal height corresponds to 24 weeks. Ultrasonography revealed decreased amniotic fluid. Which of the following conditions would have led to this presentation?

- a) Renal agenesis
- b) Tracheoesophageal fistula
- c) Cardiac abnormalities
- d) Ureteral stricture

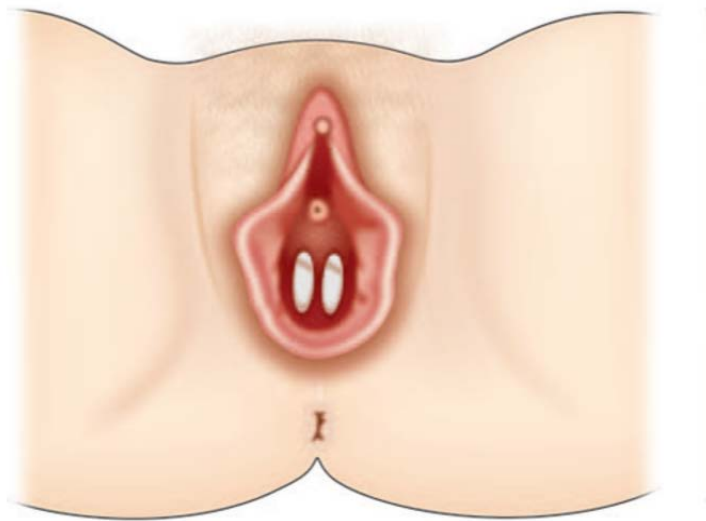
Question 3:

A type 1 diabetic mother is on magnesium sulfate infusion post-cesarean section for preeclampsia. She develops delirium and is drowsy. She has a respiratory rate of 10/min, random blood glucose level of 240 mg/dL, oliguria, and bilaterally absent knee reflex. What is the cause of her condition?

- a) Magnesium sulfate toxicity
- b) Diabetic ketoacidosis
- c) Eclampsia
- d) Diabetes insipidus

Question 4:

Identify the type of hymen shown in the image.



- a) Imperforate hymen
- b) Semilunar hymen
- c) Septate hymen
- d) Annular hymen

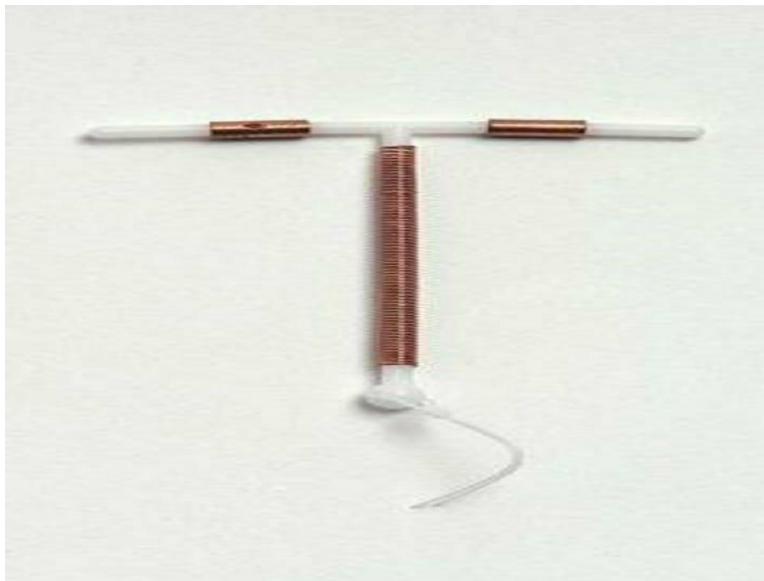
Question 5:

While discharging a patient who underwent a vesicovaginal fistula repair, which of the following would you recommend?

- a) Sexual abstinence for 3 months and avoid pregnancy for a year
- b) Sexual abstinence for 3 weeks and avoid pregnancy for 6 months
- c) Sexual abstinence for 6 weeks and avoid pregnancy for a year
- d) Sexual abstinence for 6 months and avoid pregnancy for 6 years

Question 6:

Which of the following is an absolute contraindication for the insertion of the device shown in the image below?



- a) Menstruation
- b) Trophoblastic disease
- c) Immediately after delivery
- d) Ruptured condom during intercourse

Question 7:

A 20-year-old woman is evaluated for primary infertility. Hysterosalpingography was done and reveals the finding shown below. What is the anomaly seen in the image?



- a) Septate uterus

- b) Uterine didelphys
- c) Bicornuate uterus
- d) Unicornuate uterus

Question 8:

A pregnant woman with no other comorbid conditions develops preeclampsia. She enquires about the cause of her condition. The doctor explains that it is due to the failure of the invasion of:

- a) Spiral artery by villous trophoblasts
- b) Radial artery by cytotrophoblasts
- c) Spiral artery by extravillous trophoblasts
- d) Arcuate artery by extravillous trophoblasts

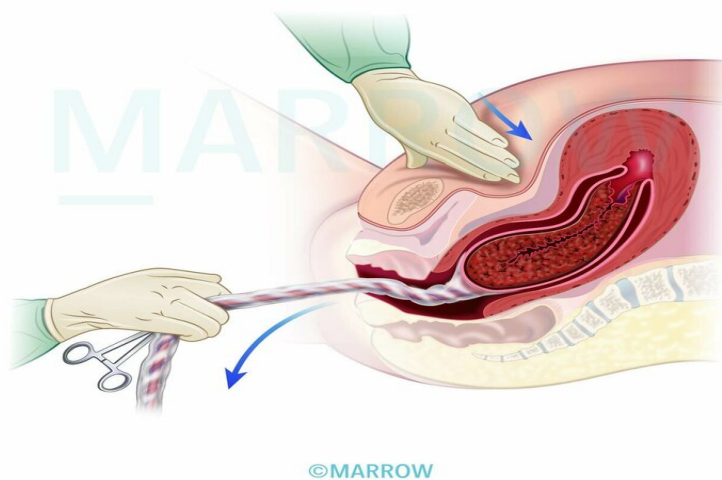
Question 9:

A woman with an obstetric score of G2P1 comes to the clinic at 14 weeks of gestation for her antenatal checkup. A uterine artery doppler was suggested by the doctor. What would it detect?

- a) Early-onset preeclampsia
- b) Late-onset preeclampsia
- c) Fetal growth restriction
- d) Placenta accreta

Question 10:

While conducting a delivery, you perform the maneuver shown below. Following this, there is incomplete separation of the placenta and massive hemorrhage. What is your next step in management?



- a) Start oxytocin infusion and wait for spontaneous delivery of placenta
- b) Uterine massage
- c) Arrange for blood and use Crede's method for placental delivery
- d) Manual removal of placenta

Question 11:

A female patient collapses soon after delivery. There is profuse bleeding and features of disseminated intravascular coagulation. Which of the following is the most likely etiology?

- a) Amniotic fluid embolism
- b) Postpartum hemorrhage
- c) Peripartum cardiomyopathy
- d) Rupture of the uterus

Question 12:

You are examining a multigravida in the second stage of labor for the past two hours. On examination, contractions are adequate, the cervix is dilated with the head at station 0 with molding 2+ and caput 2+. The sagittal suture is in the right occipitotransverse position. The fetal heart rate is 140 beats/minute. Which of the following is carried out for the management of this patient?

- a) Midpelvic forceps
- b) Vacuum-assisted delivery

- c) Wait for an hour for spontaneous labor
- d) Lower segment cesarean section

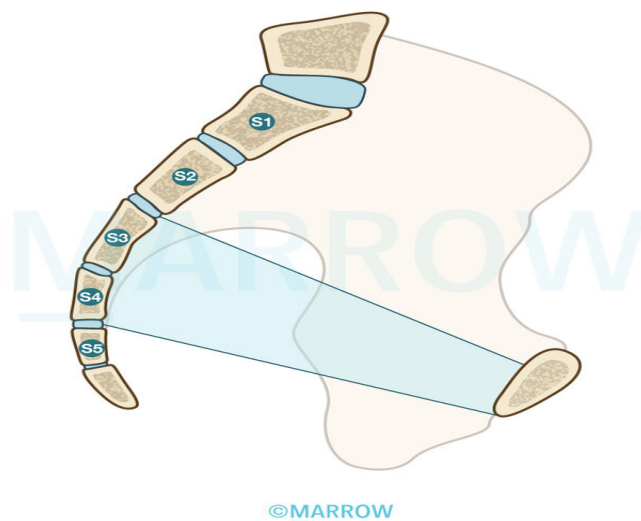
Question 13:

A primigravida presents to the labor room at 40 weeks of gestation with lower abdominal pain. She has been in labor for 3 hours. Which of the following will determine if she is in active labor?

- a) Fetal head 5/5 palpable on abdominal examination
- b) Two contractions lasting for 10 seconds in 10 minutes
- c) More than 5 cm cervical dilatation with complete effacement
- d) Rupture of membranes

Question 14:

Identify the blue-colored area shown in the image below.



- a) Obstetric outlet
- b) Anatomical outlet
- c) Pelvic inlet
- d) Midpelvis

Question 15:

What will be the level of the uterus on the postpartum day 2?

- a) Two finger breadths below umbilicus
- b) One finger breadth below umbilicus
- c) Three finger breadths below umbilicus
- d) Four finger breadths below umbilicus

Question 16:

A primigravida woman at 12 weeks of gestation comes to the antenatal clinic for nutritional advice. Which of the following will you recommend?

- a) Additional 300kcal in 2nd trimester
- b) Additional 300kcal in 1st trimester
- c) Additional 400kcal in 3rd trimester
- d) Additional 300kcal throughout the pregnancy

Question 17:

A 40-year-old G2P1 woman with 18 weeks of amenorrhea comes with a dilated cervix. The cervical length is 15 mm. In spite of explaining the risks, she insisted on cerclage. Which of the following is a contraindication for cervical cerclage?

- a) Ruptured membranes
- b) Prolapse of membranes into the vagina
- c) Fetal fibronectin positive
- d) Advanced maternal age

Question 18:

A primigravida presents to the emergency room in the early stage of labor with adequate uterine contractions. On per vaginal examination, a gynecoid pelvis is felt, the membranes are ruptured, and the vertex is felt in the right occipito-posterior position. How will you manage this patient?

- a) Vacuum-assisted delivery
- b) Cesarean section
- c) Normal vaginal delivery

- d) Forceps delivery**

Question 19:

A pregnant patient, with a history of classical cesarean section in view of fetal growth retardation in the previous pregnancy, presents to you. She is currently at 35 weeks of gestation with breech presentation. What is the next step in management?

- a) Cesarean section at 37 weeks**
- b) Advice USG and visit after 2 weeks**
- c) Internal podalic version followed by vaginal delivery**
- d) External cephalic version at 36 weeks**

Question 20:

A primigravida at 22 weeks of gestation presents to you with profuse vaginal bleeding. Her blood pressure and glucose levels are within normal limits. Placental implantation at which of the following sites can cause this?

- a) Internal os**
- b) Fallopian tube**
- c) Ovarian**
- d) Abdominal**

Question 21:

A female patient presents to you with six weeks of amenorrhea, associated with abdominal pain and vaginal bleeding with normal blood pressure. Investigations revealed beta-hCG to be 1400 mIU/mL. An ultrasound scan was done which showed a trilaminar endometrium with normal adnexa. What is the next best step in the management of this patient?

- a) Repeat beta-hCG after 48 hours**
- b) Repeat ultrasound after 3 days**
- c) Measurement of progesterone**
- d) Laparoscopy**

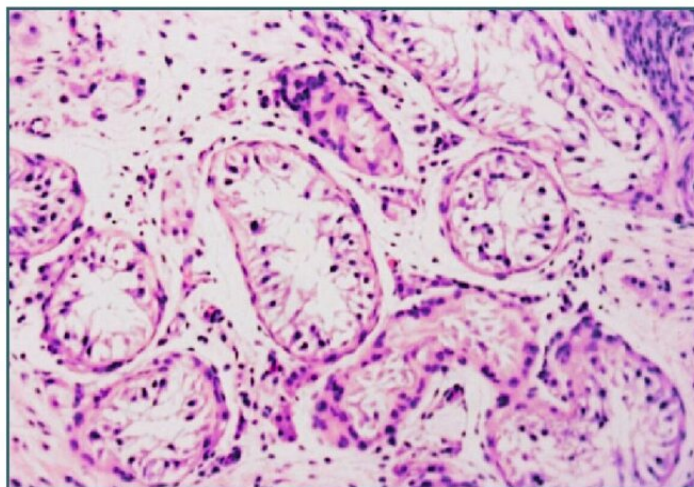
Question 22:

A 24-year-old lactating female with an 18-month-old child comes with a history of irregular, heavy bleeding seeking contraceptive advice. Which is the contraceptive of choice?

- a) Progestasert
- b) Copper-T 380A
- c) Mala N
- d) Norethisterone enanthate - depot injection

Question 23:

A 25-year-old male patient is evaluated for primary infertility. Semen analysis shows azoospermia. A testicular biopsy is done and the image is shown below. The shown finding is consistent with



- a) Sertoli cell only syndrome
- b) Testicular atrophy
- c) Benign testicular neoplasm
- d) Orchitis

Question 24:

A 20-year-old woman presented at 7 weeks of gestation, unwilling to continue the pregnancy. What are the drugs used for medical termination of pregnancy in this patient?

- a) Misoprostol and Medroxyprogesterone

- b) Misoprostol and Mifepristone
- c) Mifepristone and Methotextrate
- d) Mifepristone and Medroxyprogesterone

Question 25:

Testosterone helps in the development of various organs in the fetus. Which of the following stimulates its production?

- a) LH from maternal pituitary
- b) hCG from placenta
- c) Inhibin from corpus luteum
- d) GnRH from fetal hypothalamus

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	c
5	a
6	b
7	a
8	c
9	a
10	d
11	a
12	d
13	c
14	d
15	a
16	a
17	a
18	c

19	a
20	a
21	a
22	c
23	a
24	b
25	b

Detailed Explanations

Solution to Question 1:

The given clinical scenario with elevated liver enzymes and bilirubin, low platelet count, and hypertension indicates the diagnosis of HELLP syndrome.

HELLP is an acronym for hemolysis (H), elevated liver enzymes (EL), and low platelet count (LP). It usually occurs in patients with severe preeclampsia. The patients typically present with right upper quadrant pain, nausea, and vomiting. Hypertension and proteinuria are seen but may be absent in some cases.

In this condition, there is periportal necrosis of the liver causing an elevation in liver enzymes (AST, ALT, and LDH) and bilirubin. Tennessee criteria used to diagnose HELLP syndrome include:

- Platelet count - <1 lakh/mm³
- AST - >70 IU/L
- LDH - >600 IU/L
- Hemolysis

Prompt delivery is indicated for ≥ 34 weeks gestational age. Vaginal delivery is the preferred route of delivery if cervix is favorable. Platelet transfusion is required if counts are less than 50,000/mm³. Magnesium sulfate is given as antiseizure prophylaxis, and anti-hypertensives are given for severe hypertension.

Complications of HELLP include eclampsia, placental abruption, hepatic subcapsular hematoma, coagulopathy, stroke, and pulmonary edema.

The clinical and laboratory findings of acute liver diseases in pregnancy are as follows:

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
Onset in pregnancy	Mid to late	Late	Late	Variable
Differentiating features	-	Pruritus (elevated serum bile acids)	Signs of liver failure	Fever
AST (U/L)	≤ 300	≤ 200	200-800	2000+

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
Bilirubin (mg/dL)	1-4	1-5	4-10	5-20
Platelets	↓↓	Normal	↓↓	↓
Fibrinogen	Normal	Normal	↓↓↓	Normal
D-dimer	↑	Normal	↑	Normal
Prothrombin time	Normal	Normal	↑↑	↑
Hemolysis	↑↑	No	↑↑↑	No

Solution to Question 2:

The given clinical scenario and decreased amniotic fluid on ultrasonography are suggestive of oligohydramnios. Among the given conditions, renal agenesis is the most likely cause.

Oligohydramnios is a condition where the amount of amniotic fluid is reduced. The major source of amniotic fluid in the second trimester is fetal urine. When the fetus has renal agenesis, urine production is affected. This leads to decreased amniotic fluid/oligohydramnios.

A woman with oligohydramnios usually presents with decreased fundal height for gestational age and decreased fetal movements. On examination, fetal parts can be clearly felt because of less liquor. Fetal malpresentation and intrauterine growth retardation are seen in this condition. It is diagnosed with the help of ultrasonography when the deepest vertical pocket (DVP) of liquor is ≤ 2 cm or when the amniotic fluid index (AFI) is ≤ 5 cm at gestational age, from 28 to 40 weeks.

It can lead to abortion, deformities like wry neck, club foot, etc, fetal pulmonary hypoplasia, and cord compression. Adequate hydration and amnioinfusion can be used to manage this condition.

Polyhydramnios or hydramnios is a condition where the amount of amniotic fluid is abnormally increased. Ultrasonographic diagnosis is made when the deepest vertical pocket (DVP) is ≥ 8 cm or when the amniotic fluid index (AFI) is ≥ 25 cm.

Other options:

Option D- Ureteral stricture usually arises secondary to trauma or iatrogenic interventions like surgery.

Solution to Question 3:

The given clinical scenario and examination finding of bilaterally absent knee reflex following magnesium sulfate (MgSO_4) administration for preeclampsia suggest the cause of her condition to be magnesium sulfate toxicity.

MgSO_4 is used as an anticonvulsant in the management of eclampsia and as prophylaxis in severe preeclampsia. It can also be used as a tocolytic to stop preterm labor. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L.

Patients must be monitored for magnesium toxicity:

- Loss of deep tendon reflexes
- Decreased respiratory rate $<12/\text{min}$
- Urine output $<30 \text{ mL/h}$
- Chest pain and heart block
- Visual changes, flushing, muscle paralysis, and drowsiness

Measurement of serum magnesium levels is indicated to monitor magnesium toxicity if serum creatinine $>1.0 \text{ mg/dL}$.

Management of magnesium toxicity is as follows:

- Stop magnesium therapy
- Estimation of serum magnesium and creatinine levels
- Intravenous injection of calcium gluconate 10 mL (10% solution)

The two commonly used MgSO_4 regimens are described below:

Repeated/maintenance dose IM injections of magnesium sulfate are continued only under the following conditions:

- The patellar reflex is present
- Respiration is not depressed
- Urine output $>30 \text{ mL/h}$

Other options:

Option B: Pregnant women with diabetic ketoacidosis present with hyperventilation (Kussmaul breathing) and their blood glucose level is around 380 mg/dL .

Option C: Eclampsia patients present with generalized tonic-clonic convulsions.

Option D: Diabetes insipidus patients present with polyuria and polydipsia.

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4 g (20% solution) IV over 3–5 minutes, followed by 10 g (50%) deep IM (5 g in each buttock)	5 g (50%) IM 4 hourly in the alternate buttock
Intravenous (Zuspan or Sibai)	4 g IV slow over 15–20 minute	1–2 g/h IV infusion

Solution to Question 4:

The given image is suggestive of the septate hymen.

The hymen is the membranous vestige of the junction between the sinovaginal bulbs and the urogenital sinus, which perforates during fetal life to establish a connection between the vaginal

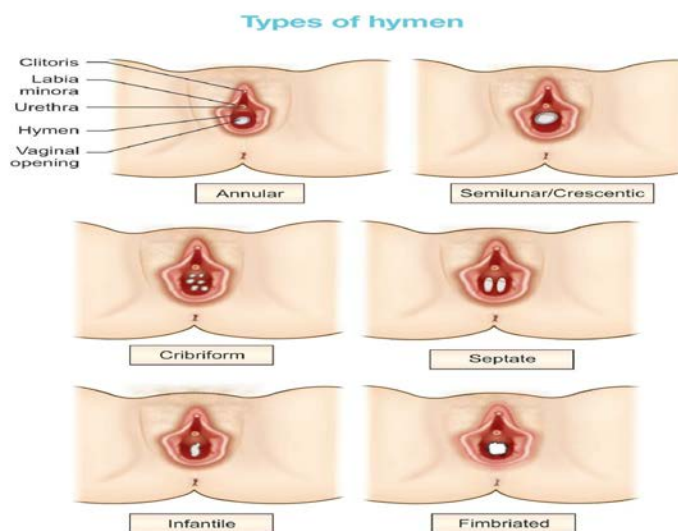
lumen and the perineum. Defective perforation of this structure may result in the following variants of hymen

- Imperforate - without any opening
- Microperforate - tiny opening
- Annular - oval central opening
- Septate - two lateral openings partitioned by a bridge of hymenal tissue
- Naviculate - boatlike opening

Patients with septate hymen will typically complain of menstrual irregularities or difficulty with tampon placement or intercourse. Under topical anesthesia, a rapidly absorbable suture or free tie can be placed at the anterior and posterior aspects of the septal tissue, and the septum is resected between the two sutures.

Some of the variants of the hymen are:

Imperforate hymen results due to failure of canalization of the inferior end of the vaginal plate. Clinical features include primary amenorrhea, cyclical abdominal pain, and bluish bulging membranes at the vaginal opening. A simple cruciate incision in the membrane will restore normal reproductive function.



Solution to Question 5:

While discharging a patient who underwent vesicovaginal fistula repair, sexual abstinence for 3 months and avoiding pregnancy for a year would be recommended.

Vesicovaginal fistula (VVF) is a type of urinary fistula which commonly arises due to obstetric trauma. Small fistulas due to difficult childbirth can heal spontaneously. Most established fistulas need to be repaired vaginally usually after 3 months to allow the inflammation to subside. Post-operatively, continuous bladder drainage for 14 days along with antibiotics is recommended. Vaginal or speculum examination and intercourse are not advised for 3 months. Avoiding

pregnancy for a year is also recommended. A cesarean section is usually done for future pregnancies.

Presenting features of VVF include constant dribbling of urine/ true incontinence. Excoriation of the vagina, vulva, and thighs can occur due to constant contact with urine. Few patients might develop bladder stones. Vaginal urine collection can be seen on retraction of the posterior vaginal wall with a speculum. A bimanual examination can be used to assess the fixity and scarring of the surrounding tissues. A cystourethroscopy can be used for localization and assessing bladder mucosa viability.

Solution to Question 6:

The given image shows a copper-containing IUCD (intrauterine contraceptive device). Trophoblastic disease is an absolute contraindication for the insertion of this device.

In cases of uterine invasion like trophoblastic disease, there is an increased likelihood of complications related to IUCD such as perforation, misplaced device, and bleeding.

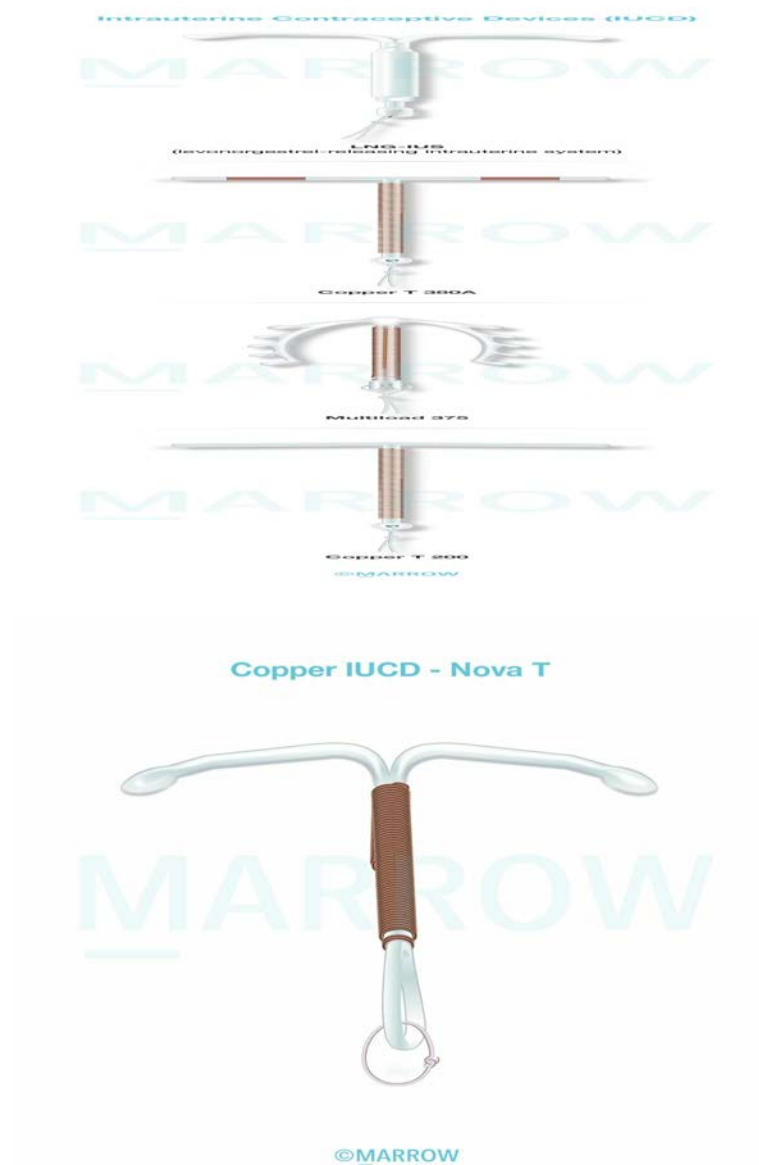
IUCDs are a type of long-term, reversible contraception. They are T-shaped devices that are inserted into the uterine cavity. These can either be copper-containing like CuT-380A and CuT-200, or hormone-containing IUCDs like Mirena.

The contraceptive action of IUCDs is because of foreign body reaction which leads to various changes in endometrial tissue that interfere with implantation. These changes include prostaglandin release that provokes uterine contractility and tubal peristalsis and leucocytic infiltration in the endometrium. It also interferes with the migration of spermatozoa.

The contraindications for insertion of IUCD include:

- Trophoblastic disease
- Distorted uterine cavity (fibroids, uterine anomalies)
- Ongoing pelvic infection
- Undiagnosed genital tract bleeding
- Suspected pregnancy
- Severe dysmenorrhea
- STIs — current or recurrent
- Wilson disease and copper allergy, for copper-containing IUCDs
- Breast cancer, for hormone-containing IUCDs

The different types of IUCD are shown below:



The timing of insertion of IUCD is as follows:

- Interval insertion (inter-conceptional period)- preferable to insert 2–3 days after menstruation, but it can be inserted at any time during the cycle. It can also be inserted during menstruation which has certain advantages like an open cervical canal and a distended uterine cavity
- Post-partum insertion is done at least 6 weeks after delivery when the uterus involutes to the normal size
- Postplacental insertion can be done immediately following delivery of the placenta
- Inserted within 5 days of coitus as an effective method of emergency contraception (Eg: incase of ruptured condom during intercourse)

Solution to Question 7:

The anomaly seen in the given hysterosalpingogram is suggestive of a septate uterus.

A septate uterus is characterized by the presence of a septum in the endometrial cavity due to the failure of the medial segments of the Müllerian ducts to regress. This creates a permanent septum within the uterine cavity. The septum can project partially from the fundus or can extend completely to the cervical os. On HSG, the septate uterus is seen as widely diverging horns with an intercornual angle < 75 degrees.

HSG provides information about the interior of the uterine cavity and not the exterior. Therefore, it cannot reliably differentiate between a septate uterus and a bicornuate uterus. A septate uterus has a smooth fundal contour whereas a bicornuate uterus has an indented fundus.

Transvaginal 3-D ultrasonography (3D TV USG) helps to differentiate between the bicornuate uterus and a septate uterus and is recommended for the assessment of patients with a suspected uterine anomaly. MRI is an accurate investigation to diagnose the septate uterus. Hysteroscopy combined with laparoscopy can confirm both the internal and the external contour of the uterus.

Hysteroscopic septoplasty is a preferred treatment option for the septate uterus. It refers to the resection of the uterine septum under hysteroscopic guidance. There is a reduced risk of pelvic adhesions, scarring and the necessity of cesarean sections for future pregnancies with the hysteroscopic approach, compared with transabdominal approaches.

Solution to Question 8:

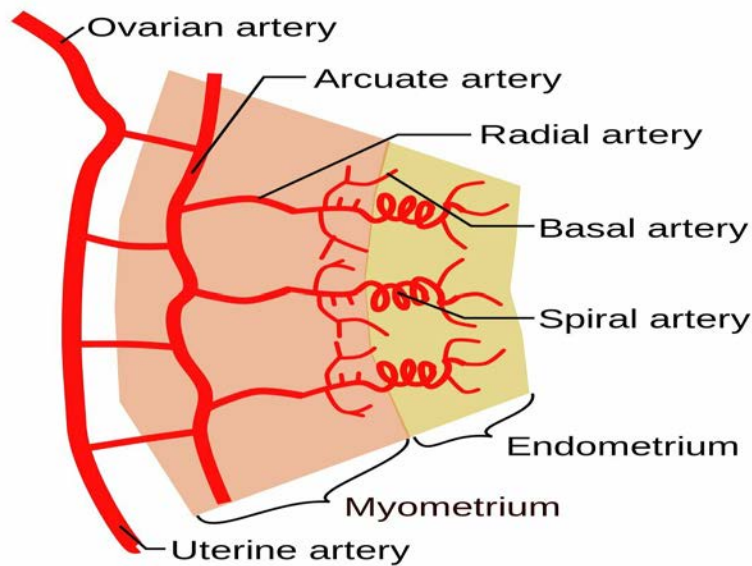
Preeclampsia occurs due to the failure of the invasion of extravillous trophoblasts into the spiral artery.

The spiral artery supplies blood to the endometrium. In normal pregnancy, there is an invasion of the extravillous trophoblasts (specifically, endovascular trophoblasts) into the walls of the spiral arteries. In the first wave of invasion (10–12 weeks), endovascular trophoblasts invade up to decidual segments. In the second wave (16–18 weeks), they invade up to the myometrial segments.

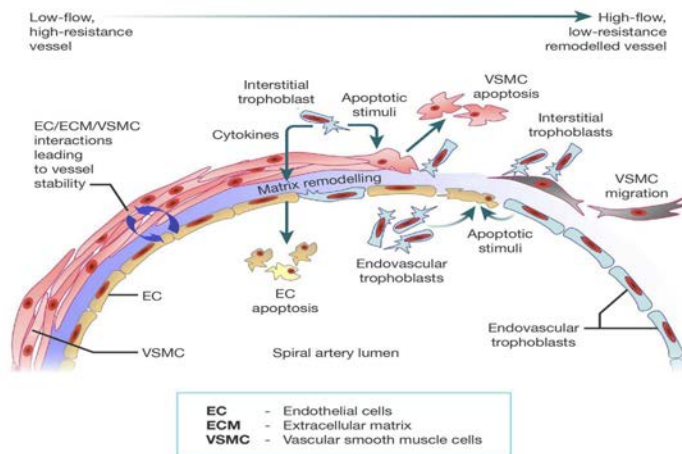
This vascular remodeling process replaces the endothelial lining and the muscular arterial wall with endovascular trophoblasts. This results in the distension of the spiral arterioles, thereby transforming them into a low resistance, low pressure, and high flow system.

In preeclampsia, the failure of the second wave of endovascular trophoblastic invasion into uterine vessels leads to decreased placental perfusion.

The images below show the sequential branching of the uterine artery and the vascular remodeling in preeclampsia respectively,



Vascular remodeling in preeclampsia



Other contributory etiological factors include genetic factors and maladaptive responses to fetal antigens and cardiovascular changes in pregnancy.

The basic pathology in preeclampsia is endothelial dysfunction leading to decreased nitric oxide production and vascular spasm. This leads to ischemic changes that result in necrosis, hemorrhage, and end-organ damage.

Changes leading to vasospasm and endothelial dysfunction include:

- Reduced vasodilatory prostaglandin PGI₂ and increased vasoconstrictor thromboxane (TXA₂)
- Reduced angiotensinase activity and increased sensitivity of vessels to angiotensin II, which is a vasopressor
- Increased production of endothelin 1, a potent vasoconstrictor
- Increased inflammatory mediators (tumor necrosis factor- α , interleukins) and reactive oxygen species (ROS) causing endothelial injury

- Increased antiangiogenic factors like soluble fms-like tyrosine kinase I (sFlt-1) and soluble endoglin (sEng)

The definitive treatment of pre-eclampsia is the termination of pregnancy at ≥ 37 weeks.

Solution to Question 9:

A uterine artery doppler would detect early-onset preeclampsia.

The development of a diastolic notch and resistance to flow in the uterine artery doppler suggests the development of gestational hypertension, pre-eclampsia, and fetal growth restriction later in the pregnancy. A high resistance index in the second trimester is associated with a sixfold increase in the incidence of preeclampsia. The trophoblastic invasion of maternal uterine vessels in the first half of pregnancy results in a decrease in vascular resistance. This can be detected by Doppler studies.

Other screening tests which can be used for the prediction of pre-eclampsia include:

- Mean arterial pressure ≥ 90 mm Hg in the second trimester
- Absence of end-diastolic frequencies in umbilical artery doppler
- Roll over test

Preeclampsia is a multi-system disorder marked by blood pressure $\geq 140/90$ mm Hg with proteinuria after the 20th week of gestation in a previously normotensive and non-proteinuric woman. Usually, there is an invasion of the extravillous trophoblasts (specifically, endovascular trophoblasts) into the walls of the spiral arteries. In the first trimester, the trophoblastic invasion up to the decidual segments occurs. The myometrial segments are invaded in the second trimester. This second wave of endovascular trophoblast migration fails to happen in preeclampsia.

The pathogenesis can be explained by an imbalance in the prostaglandins - a deficiency of PGI₂ (vasodilator) and increased thromboxane (vasoconstrictor). The deficiency of nitric oxide and elevated antiangiogenic factors are other mechanisms responsible.

Features include headache, visual disturbances, epigastric pain, and thrombocytopenia. Risk factors include young maternal age, nulliparity, metabolic syndrome, and hyperhomocysteinemia.

Other options:

Option B: Late-onset preeclampsia occurs > 34 weeks. It often has a favorable outcome.

Option C: Fetal growth restriction can be detected by abnormal uterine artery Doppler notching at 20-24 weeks of gestation.

Option D: Placenta accreta or morbidly adherent placenta arises due to partial or total absence of decidua basalis. Sonography is recommended for diagnosis. Better results can be obtained with power doppler.

Solution to Question 10:

The maneuver shown is the Brandt-Andrews maneuver used in controlled cord traction for delivery of the placenta. The clinical features point to a diagnosis of primary postpartum hemorrhage due to retained tissues and the next step in management is manual removal of the placenta.

Postpartum hemorrhage (PPH) is defined as blood loss ≥ 500 mL following the birth of the baby. It is of two types:

- Primary PPH: It occurs within 24 hours after the birth of the baby. This is again classified into third-stage hemorrhage (bleeding before placental expulsion) and true postpartum hemorrhage (bleeding after placental separation).
- Secondary PPH: It occurs after 24 hours but within the puerperal period.

The four main causes of primary PPH are atonic uterus, trauma, retained tissues, and coagulopathies.

Retained tissues such as placental bits and blood clots lead to imperfect uterine retraction. Causes for delayed placental delivery include simple adherent placenta, lower uterine segment constriction, and morbidly adherent placenta. In the presence of brisk bleeding and partial or total attachment of the placenta, manual removal is indicated. It is carried out under adequate analgesia and aseptic precautions.

If the placenta is not expelled even after 30 minutes following the birth of the baby, it is known as retained placenta. It can lead to hemorrhage, shock, and puerperal sepsis. Unseparated retained placenta is manually removed under general anesthesia.

Solution to Question 11:

The given clinical scenario with a diagnosis of disseminated intravascular coagulation (DIC) points to amniotic fluid embolism as the most likely etiology.

Amniotic fluid embolism presents with a classic triad of hemodynamic and respiratory compromise, along with DIC. Usually, it presents during the late stages of labor or immediate postpartum. Gasping for air, seizures, and cardiorespiratory arrest are other features. Clinical features are variable.

Diagnostic criteria for amniotic fluid embolism include:

- Sudden onset cardiorespiratory arrest, or both hypotension and respiratory compromise
- Overt DIC
- Onset during labor or within 30 minutes of delivery of the placenta
- No fever $\geq 38^{\circ}\text{C}$

Since there are no specific diagnostic tests, a clinical diagnosis has to be made. Other causes for a similar presentation such as myocardial infarction, pulmonary embolism, and anaphylactic shock needs to be ruled out.

Risk factors include rapid labor, meconium-stained liquor, and uterine tears. Post-term pregnancy, maternal age, and eclampsia are other predisposing conditions. The currently accepted mechanism is the entry of fetal material into the maternal circulation which leads to the activation

of proinflammatory mediator systems. This results in pulmonary vasoconstriction and hypertension. The tissue factor present in fetal material also activates factor VII resulting in DIC. The condition can be fatal and management includes cardiopulmonary resuscitation and advanced cardiac life support.

Other options:

Option B: Post-partum hemorrhage is the loss of $\geq 500\text{mL}$ of blood after the third stage of labor. Symptoms and signs of hypovolemia are commonly seen.

Option C: Peripartum cardiomyopathy presents with features of heart failure in the last month of pregnancy or within 5 months of delivery. Probable causes include viral myocarditis, malnutrition, and hormonal interactions.

Option D: Rupture of the uterus is a disruption in the continuity of all the uterine layers anytime after 28 weeks of pregnancy. Clinical features depend on the specific cause and include abdominal pain, vaginal bleeding, fainting attacks, and rapid pulse rate.

Solution to Question 12:

The given clinical scenario points to a diagnosis of deep transverse arrest. A lower segment cesarean section is carried out for the management of this patient.

In deep transverse arrest, the sagittal suture is placed in the transverse bispinous diameter. There is no descent of the head for at least 1/2-1 hour after full dilatation of the cervix. The mode of delivery is based on the fetal condition and pelvic assessment. Caput succedaneum indicates the static position of the head for a prolonged duration. Molding occurs due to the passage of the fetal head through a resistant birth passage. Grade 2 molding indicates overlapping parietal bones which can be separated easily. The presence of caput and molding of grade 2+ indicates a cephalopelvic disproportion. In such cases, a cesarean section has to be performed.

If the pelvis is adequate, application of ventouse or manual rotation and forceps application can be tried.

Deep transverse arrest, can be either due to incomplete forward rotation of the oblique occipitoposterior position or due to nonrotation in the occipitotransverse position.

Occipitoposterior position is an abnormal position of the vertex and can undergo normal labor in favorable circumstances. In some cases, it undergoes non-rotation resulting in oblique posterior arrest. If it undergoes malrotation (anterior rotation of the sinciput), it can cause a persistent occipitoposterior position. Spontaneous delivery can occur as face to pubis. If arrest occurs in this position, it is termed as occipitosacral arrest. Incomplete forward rotation, as described above, leads to a deep transverse arrest.

Features of occipitoposterior position include infra-umbilical flattening and fetal heart sounds localized to the flank. The fetal parts are more prominently felt near the midline.

Solution to Question 13:

This patient will have presented in active labor if she had more than 5 cm cervical dilatation with complete effacement

The first stage of labor is characterized by dilatation and effacement of the cervix and can be divided into latent phase and active labor. The onset of latent labor is the point at which the mother perceives regular contractions. Cervical dilatation of ≥ 5 cm in the presence of uterine contractions can be taken to reliably represent the threshold for active labor.

The first symptom to appear is intermittent painful uterine contractions followed by the expulsion of blood-stained mucus (show) per vagina. In the late first stage, the contraction comes at intervals of 3–5 minutes and lasts 45 seconds.

The second stage of labor begins with complete cervical dilatation and ends with fetal delivery.

The third stage of labor is characterized by the delivery of the placenta.

The fourth stage is a stage of observation for at least an hour after the full expulsion of the placenta.

Fetal head 5/5 on abdominal examination, signifies the degree of descent in relation to the pelvic inlet and denotes the head is floating above the brim.

Rupture of membranes at or near term signifies the labor is likely to begin soon.

More than 12 uterine contractions in 1 hour may signify labor onset. Partograph is charted every half an hour recording the number of contractions in 10 minutes and the duration of each contraction in seconds. Contractions can be classified as mild (contraction duration less than 20 seconds), moderate (between 20 and 40 seconds), and strong (more than 40 seconds).

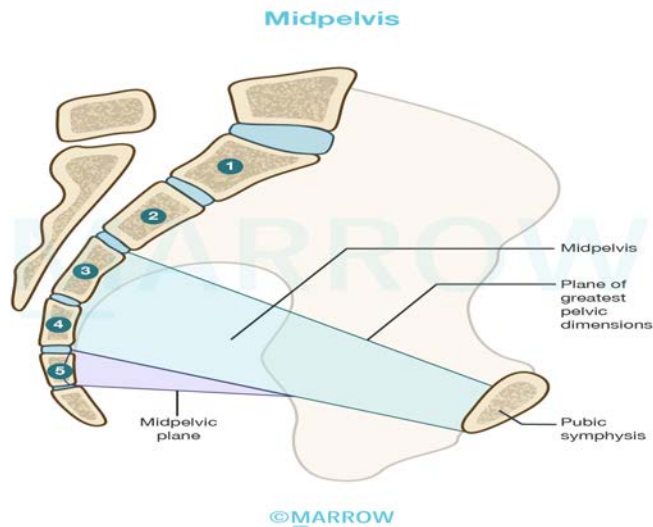
Solution to Question 14:

The blue colored area in the given image shows midpelvis.

Midpelvis is the region of the pelvis that is bounded by the plane of the greatest pelvic dimensions above and the midpelvic plane below.

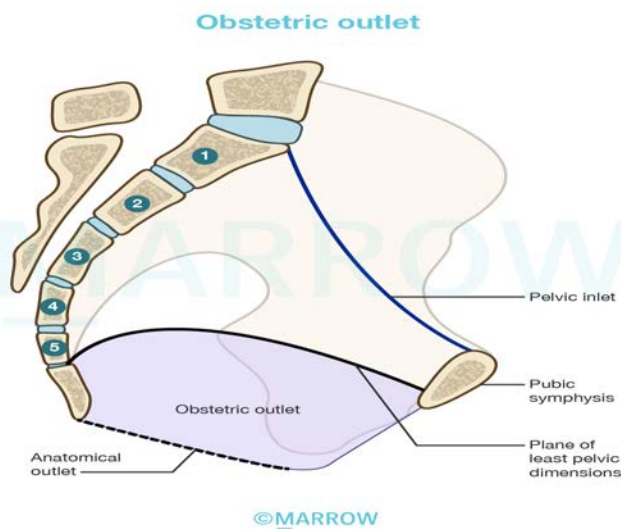
The plane of greatest pelvic dimensions (most roomy pelvic plane) extends from the middle of the posterior surface of the symphysis pubis to the junction of the second and third sacral vertebrae.

The midpelvic plane extends from the lower margin of the symphysis pubis to the tip of the sacrum at the level of ischial spines (coinciding with the plane of least pelvic dimensions), or from the lower margin of the symphysis pubis to the junction of S4 and S5, making a wedge posteriorly.

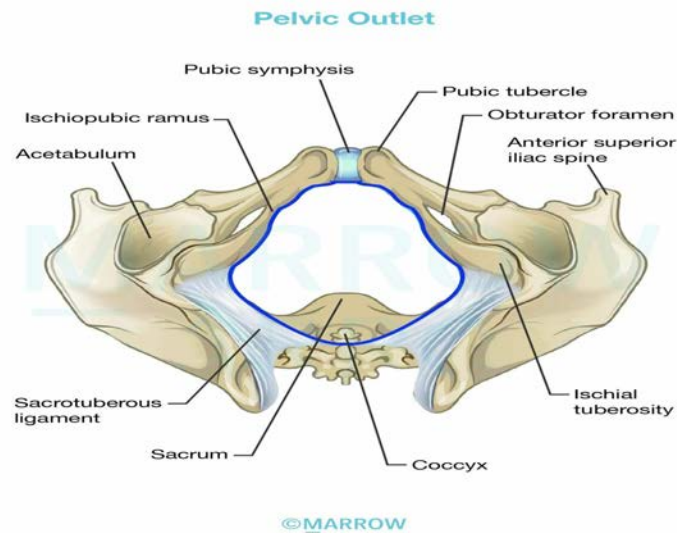


Other options:

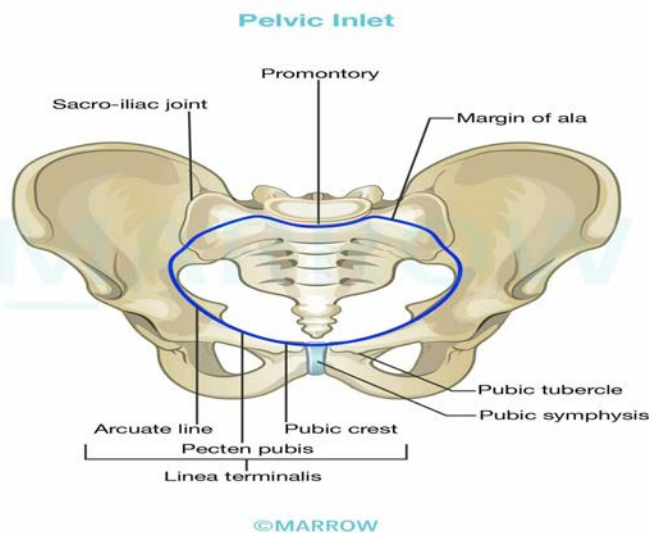
Option A: Obstetric outlet - It is the region of the pelvis that is bounded by the plane of the least pelvic dimensions above and the anatomical outlet below.



Option B: Anatomical outlet/ bony outlet - It corresponds to the bony outer margin of the pelvis. It is a diamond-shaped region formed by joining two triangular planes with a common base. The anterior triangle is bounded laterally by the ischiopubic rami and the apex is formed by the inferior border of the pubic arch. The posterior triangle is bounded laterally by the ischial tuberosity and sacrotuberous ligament and the apex is formed by the tip of the coccyx. The common base is formed by the line joining the ischial tuberosities.



Option C: Pelvic inlet - It represents the superior plane of the true pelvis. It is bounded by the horizontal pubic rami and the symphysis pubis anteriorly, the linea terminalis laterally, and the promontory and alae of the sacrum posteriorly.



Solution to Question 15:

On the second day post-delivery, the level of the uterus lies at two finger breadths below the umbilicus.

24 hours post delivery is equal to Day 0 when uterus is just below the umbilicus.

Day 1 - 1 finger breadth below umbilicus

Day 2 - 2 fingers breadth below umbilicus

The rate of involution of the uterus can be clinically assessed by noting the fundal height of the uterus in relation to the symphysis pubis.

Involution of the uterus occurs due to a decrease in the size of muscle fibers. Following delivery, the fundus lies about 5 1/2 inches above the symphysis pubis. During the first 24 hours, the level remains constant, after which there is a steady decrease in height by half an inch in 24 hours, which approximately equals two finger breadth below the umbilicus on second-day post delivery. By the end of 2nd week, the uterus becomes a pelvic organ. The rate of involution slows down after this until 6 weeks. The uterus almost becomes normal in size by 6 weeks.

Involution of the uterus occurs due to a decrease in the size of muscle fibers.

Time following delivery	Features	Weight of the uterus
Immediately following delivery	Uterus corresponds to 20 weeks gestational age. Fundus lies just below the umbilicus	1000 g
48 hours	Myometrial involution starts	-
1 week	Rapid involution of uterus	500 g
2 weeks	The uterus becomes a pelvic organ	300 g
4 weeks	Clinically, involution is complete	100 g
6-8 weeks	Uterus returns to normal (pre-pregnant) size. Sonographically, involution is complete at 8 weeks	60 g

Solution to Question 16:

For a pregnant woman, an additional 300 kcal in the 2nd trimester is recommended.

According to Williams Obstetrics, in the first trimester, energy requirements are generally the same as non-pregnant women. In the second trimester, it increases by 340 kcal and in the third, it increases by 452 kcal per day. According to ICMR (Indian Council of Medical Research), an additional intake of 300 Kcal during the second and third trimesters of pregnancy is recommended. Therefore, option A would be a better answer in this case.

Pregnancy is a period where a mother will require additional nutritional requirements to support the healthy growth and development of the fetus. According to the National Institute of Nutrition,

- Protein intake should increase by 23g per day compared to non-pregnant women
- Fat intake should be 30g per day
- Calcium intake should be 1200mg per day
- Iron intake should be 35mg per day

During lactation, ICMR recommends increasing 550 Kcal during the first six months of lactation and 400 Kcal during 7- 12 months of lactation.

Solution to Question 17:

The clinical scenario of mid-trimester cervical dilatation and a short cervix suggests cervical incompetence. Ruptured membranes is a contraindication for cervical cerclage.

Cervical incompetence or insufficiency is a functional or anatomical impairment in the retentive power of the cervix (or internal os). It is characterized by repeated, mid-trimester painless cervical dilatation. It leads to the escape of amniotic fluid followed by painless expulsion of the products of conception.

Cerclage is the procedure done to strengthen the cervix in patients with cervical incompetence. Contraindications of cervical cerclage include:

- Ruptured membranes
- Intrauterine infection
- Uterine contractions
- Vaginal bleeding
- Cervical dilatation ≥ 4 cm
- Fetal death or defect

Cervical incompetence is associated with various risk factors such as prior preterm birth, congenital uterine anomalies, iatrogenic (dilatation and curettage, induced abortion by dilatation and evacuation, vaginal operative delivery through an undilated cervix, amputation of the cervix, etc), and multiple gestations.

In the interconceptional period, diagnosis is made with the passage of number 6-8 Hegar's dilator beyond the internal os without resistance or pain. Investigations for cervical incompetence during pregnancy include:

- Per speculum examination - shows cervical dilatation with herniation of membranes
- Transvaginal ultrasound - funneling of internal os ≥ 1 cm and cervical length ≤ 25 mm

Prophylactic cervical cerclage is done between 12-14 weeks of gestation. Rescue or emergency cerclage is done when there is bulging or prolapse of membranes through a dilated cervix.

Shirodkar and McDonald's methods of cervical cerclage are the commonly performed cerclage procedures during pregnancy. In McDonald's method, a non-absorbable (mersilene) suture material is placed as a purse-string suture at the level of the internal os, whereas in the Shirodkar method, non-absorbable (mersilene) tape is used. Removal of stitch should be done at the 37th week of gestation. It can be removed earlier in case of preterm labor or if signs of abortion are seen. Laparoscopic abdominal cerclage is used for non-pregnant patients.

Other options:

Option C- Fetal fibronectin in cervicovaginal secretions is a marker for impending preterm labor before membrane rupture. Prophylactic cervical cerclage is indicated in women if ultrasonography

confirms cervical incompetence.

Option D- Advanced maternal age is not a contraindication for cervical cerclage.

Solution to Question 18:

The given features of right occipito-posterior presentation with the favorable pelvis (gynecoid) and adequate uterine contractions can be managed by normal vaginal delivery.

Right occipito-posterior position (ROP) is the most common malposition, which occurs when the occiput is situated posteriorly over the sacroiliac joint, or directly over the sacrum. It is associated with the risk of delayed engagement of the head, prolonged labor due to transverse arrest, and the need for cesarean sections.

Features of occipito-posterior position include infra-umbilical flattening and fetal heart sounds localized to the flank. The fetal parts are more prominently felt near the midline. The head engages through the right oblique diameter in ROP and the engaging AP diameter is occipitofrontal or suboccipito-frontal. In 90% of circumstances with a favorable pelvis and strong uterine contractions without any complications, it undergoes the mechanism like that of normal labor. Hence, we can wait and watch for the progress of normal vaginal delivery.

In some cases, it undergoes non-rotation resulting in oblique posterior arrest. If it undergoes malrotation (anterior rotation of the sinciput), it can cause a persistent occipito-posterior position. Spontaneous delivery can occur as face to pubis. If an arrest occurs in this position, it is termed an occipito-sacral arrest. Incomplete forward rotation leads to a deep transverse arrest.

Indications of cesarean section in occipito-posterior position include the arrest of labor, inadequate uterine contractions, fetal distress, and other obstetrical complications.

During the second stage of labor, if the anterior rotation of the occiput is completed, delivery occurs spontaneously or may require the application of ventouse or low forceps.

Solution to Question 19:

The next step in the management of this pregnancy with breech presentation and a history of previous classical cesarean delivery is a cesarean section at 37 weeks.

A classical casarean section is done by a longitudinal uterine incision. A previous classical cesarean section is considered a high-risk factor for failure of VBAC (vaginal birth after previous cesarean). Hence, it is contraindicated. There is increased risk of uterine rupture (2-9%) in such cases. In some women, classical cesarean scar can even rupture several weeks before term.

Other contraindications for VBAC include:

- ≥ 2 previous lower segment cesarean section
- Contracted pelvis or cephalo-pelvic disproportion
- Other obstetric or medical complications

- Prior uterine rupture

If the fetal buttocks or legs enters the pelvis before the head, the presentation is known as breech. Vaginal delivery of a singleton breech fetus can be considered. It can be spontaneous breech delivery, partial breech extraction, or total breech extraction. However, cesarean delivery is considered in the presence of any associated obstetric (previous cesarean delivery) or medical complications.

Other options:

Option B: An ultrasound will not change the decision regarding mode of delivery.

Option C: Internal podalic version results in a breech presentation. It is done in case of transverse lie of second baby in twin pregnancy. It is not indicated in singleton pregnancy.

Option D: External cephalic version is a manipulation through the abdominal wall that results in cephalic presentation. It is usually done at 37 weeks as it can revert back to breech position if carried out before. However, it is contraindicated when vaginal delivery is not an option and in cases of previous cesarean delivery such as in this case.

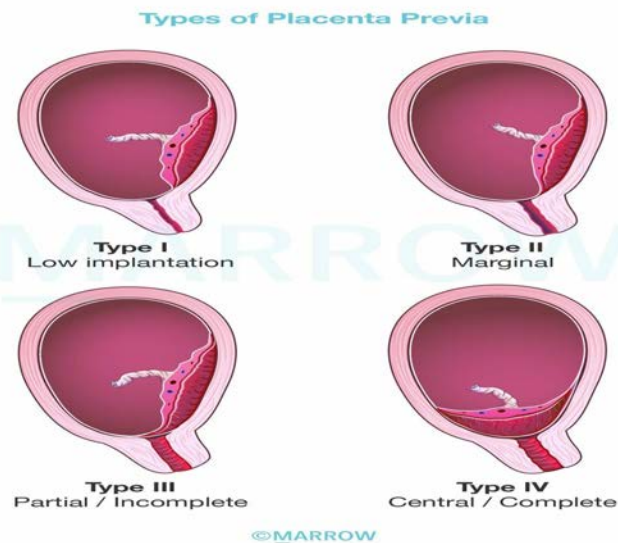
Solution to Question 20:

The given clinical scenario points to a diagnosis of placenta previa which is caused due to placental implantation at the internal os.

Placenta previa is used to describe a placenta implanted in the lower uterine segment. Painless bleeding towards the end of the second trimester or later is characteristic of the condition. It usually occurs without warning and resolves spontaneously only to recur.

There are four types of placenta previa according to the degree of placental expansion to the lower uterine segment.

- Type-I: The lower margin of the placenta reaches the lower uterine segment but not the os.
- Type-II: The placenta does not cover the internal os but reaches its margin.
- Type-III: The placenta covers the closed internal os but does not do so when fully dilated.
- Type-IV: The placenta covers the internal os even when fully dilated.



Predisposing factors include maternal age, multiparity, smoking, and leiomyomas. Previous cesarean deliveries, raised maternal alpha-fetoprotein levels, and assisted reproductive technology also increase the risk for placenta previa.

Management is individualized on a case-to-case basis. If there is no active bleeding during early gestation, close observation is recommended. Elective cesarean section is usually carried out at 38 weeks. Complications of placenta previa include primary postpartum hemorrhage, morbidly adherent placenta, and coagulation defects.

Solution to Question 21:

The given clinical vignette points to the possibility of diagnosis of either an abortion or ectopic pregnancy. The next best step in the management of this patient is to repeat beta-hCG after 48 hours for diagnosis.

The causes of bleeding in early pregnancy include abortion, ectopic pregnancy, and hydatidiform mole. The characteristic ultrasound finding of a molar pregnancy is snowstorm appearance and so it can be excluded.

A transvaginal ultrasound is done initially in such presentations to locate the gestational sac. If neither intrauterine nor extrauterine pregnancy can be confirmed, it is termed pregnancy of unknown location till definite diagnosis.

A cut-off of ≥ 1500 mIU/mL for β -hCG levels, known as the discriminatory level is used to aid in diagnosis. Serial monitoring of hCG can be used to assess fetal well-being in cases of threatened abortion. If the initial level is below this cut-off, the gestational sac location usually cannot be identified. In such situations, serial β -hCG levels are measured, and the pattern is assessed. If the value is above the discriminatory level, and an intrauterine pregnancy or abortion cannot be confirmed, then ectopic pregnancy has to be suspected.

The doubling time of hCG during early pregnancy is 1.4-2 days. Failure to double the hCG value along with an empty uterus at 48 hours is highly suggestive of ectopic pregnancy. A trilaminar

endometrial pattern can be seen in ectopic pregnancy along with an adnexal mass separate from the ovary.

Once the diagnosis of ectopic pregnancy is confirmed, administration of methotrexate in single or multiple doses can be considered. A dilatation and curettage can be used to confirm the diagnosis.

Other options:

Option C: Measurement of progesterone cannot be used as a reliable diagnostic test. In general, values ≤ 5 ng/mL point to a nonliving intrauterine or ectopic pregnancy.

Option D: Laparoscopy is the surgery of choice in hemodynamically stable patients for ectopic pregnancy.

Solution to Question 22:

Among the given options, the contraceptive of choice in this lactating mother with heavy irregular bleeding is MALA N.

MALA N is a combined oral contraceptive (COC) that contains ethinylloestradiol (EE2) in a dose of 20–30 mcg and 0.15 mg of Levonorgestrel. It acts by suppressing ovulation by suppression of pituitary hormones FSH AND LH.

COC has the lowest failure rates and helps in controlling polymenorrhea, dysmenorrhea, and menorrhagia. The risk of thromboembolism is established during the first 6 weeks postpartum (puerperal period) with COC and they are known to suppress lactation. In this clinical scenario, since the child is 18 months old (complementary feeds are usually started at 6 months and child is not entirely dependant on breastmilk for nutrition) and the woman is way past the puerperal period, it can be safely administered.

Other options:

Option A: Progestasert is a hormone-containing IUCD that acts by forming a thick mucus plug at the cervix and prevents the entry of sperms. Its usage is limited in cases with heavy unexplained bleeding.

Option B: Copper T is not used in this patient as it might worsen the symptoms of menorrhagia and may result in anemia.

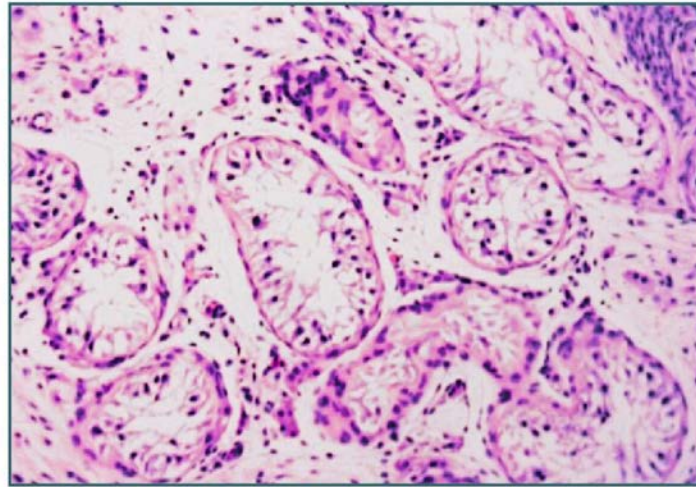
Option D: Norethisterone enanthate depot is an intramuscular injection given once in 2 months. Though it is suitable for lactating women, it has an increased risk of menstrual irregularity and osteopenia and so not preferred in this case.

Solution to Question 23:

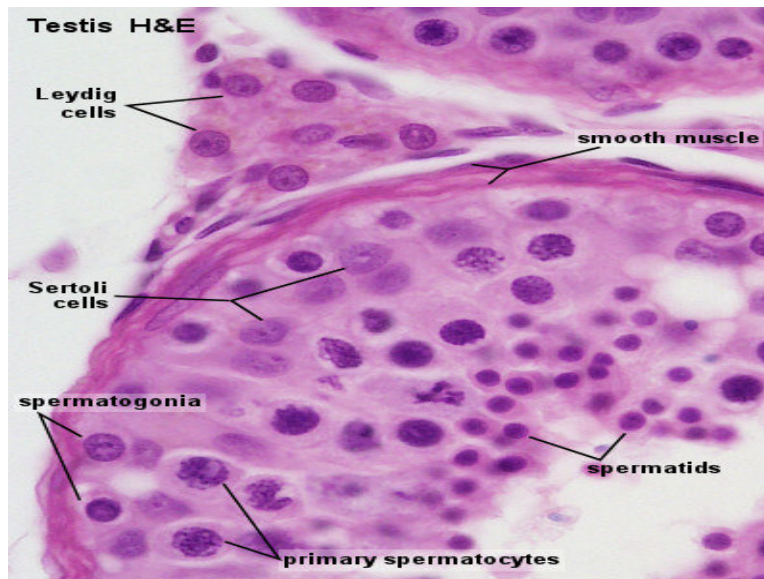
The given clinical scenario and the biopsy findings are consistent with the diagnosis of Sertoli cell-only syndrome.

Patients with Sertoli cell-only syndrome typically present between the age of 20 to 40 years with complaints of primary infertility. Diagnosis can be made on testicular biopsy findings. The histopathology of the biopsy specimen typically shows the tubules which are populated by only

Sertoli cells, with a thickening of the tubular basement membrane and absence of germ cells which is characteristic of Sertoli cell-only syndrome.



The below image shows normal histology of testis:



The Sertoli cells line the seminiferous tubules and have receptors for follicle-stimulating hormone (FSH) and testosterone. There are four sperms per Sertoli cell. It produces a testosterone-binding protein that maintains a high level of testosterone within the testis which is necessary for continuous spermatogenesis.

In Sertoli cell-only syndrome, only Sertoli cells line the seminiferous tubules of the testis, and the patients have very low or absent spermatogenesis. There is no feedback inhibition due to absent spermatozoa, which results in increased FSH levels, but testosterone levels and LH levels remain unaffected since Leydig cells are not affected.

Physical examination is normal as this condition presents with infertility without sexual abnormality. Most cases are usually multifactorial and idiopathic. Patients can still reproduce

with assisted reproductive technology.

Other options:

Option B: Testicular atrophy may be characterized by marked hyalinization and thickening of the basement membrane of the spermatid tubules with the absence of spermatids or germ cells. It may occur due to cryptorchidism, atherosclerotic changes of the supplying vessels, cachexia, and end stage of inflammatory orchitis.

Option C: Sex cord-stromal cells are benign and include Sertoli cell tumors and Leydig cell tumors. Leydig cells are characterized by the presence of crystalloids of Reinke.

Option D: Orchitis is characterized by the inflammatory infiltration of neutrophils and may progress to the formation of a frank abscess. The end-stage may lead to atrophic changes.

Solution to Question 24:

The drugs used for medical termination of pregnancy (MTP) in this patient presenting at 7 weeks of gestation are misoprostol and mifepristone.

Mifepristone 200mg is administered orally on day 1. The minimum recommended interval between mifepristone & misoprostol is 24 hours.

24- 48 hours later, misoprostol 800 mcg is administered by the vaginal, sublingual, or buccal route. Clinical confirmation of MTP is done on day 14.

The various methods of termination of pregnancy are:

Misoprostol and Methotrexate regimen can also be used but takes a longer time of action.

The Medical Termination of Pregnancy Amendment Bill, 2020 has extended the period of termination of pregnancy up to 24 weeks for special categories of women, including survivors of rape, victims of incest, and other vulnerable women (like differently-abled women, minors). The upper gestation limit is not applicable in cases of substantial fetal abnormalities.

Timing	Medical methods	Surgical methods
1st trimester(up to 12 weeks)	Mifepristone Mifepristone and misoprostol Methotrexate and misoprostol Tamoxifen and misoprostol	Menstrual regulation Vacuum aspiration (MVA/EVA) Suction evacuation and/or curettage Dilatation and evacuation - Rapid method Slow method
2nd trimester(13 - 20 weeks)	Misoprostol Mifepristone and misoprostol Dinoprostone Carboprost	Dilatation and evacuation Intrauterine instillation of hyperosmotic solutions Oxytocin infusion Hysterotomy

Solution to Question 25:

hCG (human chorionic gonadotropin) from placenta stimulates the production of testosterone in the male fetus.

hCG released from the placenta is the initial stimulus for testosterone production in the male fetus. Later by the eighth week of gestation, fetal pituitary LH (luteinizing hormone) stimulates the Leydig cells of the testes to produce testosterone.

Sex differentiation of the embryo depends on the Y chromosome, which contains the testis-determining gene called the SRY (sex-determining region on Y) gene. In a genetically male embryo (46, XY), the gonad is converted into testis under the influence of this gene. Female development occurs in the absence of this gene.

Testosterone released from the Leydig cells acts on the Wolffian duct to develop vas deferens, epididymis, and seminal vesicles. Testosterone is converted to dihydrotestosterone (DHT) with the help of 5-alpha-reductase which helps in the differentiation of male external genitalia.

The Sertoli cells of the seminiferous tubules of testes secrete antimüllerian hormone (AMH)/Müllerian-inhibiting substance (MIS) by 7 weeks of gestation. It causes regression of the Mullerian duct and prevents the development of the uterus, fallopian tube, and upper vagina.

Other options:

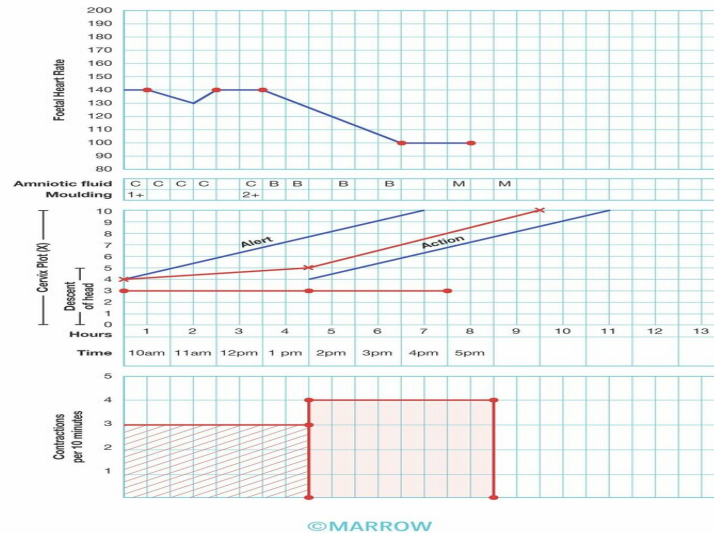
Option A: It is LH from the fetal pituitary, and not the maternal pituitary that stimulates testosterone production in the male fetus.

Option C: Inhibin A is produced by the corpus luteum and inhibin B is produced by the granulosa cells of the ovary. They inhibit FSH secretion.

Obstetrics and Gynecology NEET 2023

Question 1:

A partogram of a woman who is in labor is shown below. Oxytocin infusion is already given. How will you manage the condition?



- a) Forceps-assisted delivery
- b) Vacuum-assisted delivery
- c) Oxytocin infusion
- d) Cesarean section

Question 2:

A woman presents to you at 36 weeks of gestation with complaints of breathlessness and excessive abdominal distension. Fetal movements are normal. On examination, fetal parts are not easily felt and fetal heartbeat is heard but it is muffled. Her symphysis fundal height is 41 cm. Her abdomen is tense but not tender. What is the most likely diagnosis?

- a) Abruptio placenta
- b) Hydrocephalus of fetus
- c) Polyhydramnios
- d) Fetal-maternal ascites

Question 3:

For which of the following procedures in the given OT list, can you preferably do a hysteroscopy?

- a) Tubal ligation
- b) Asherman syndrome
- c) Endocervical polyp
- d) Subserosal fibroid

Question 4:

A primigravida presents to you with anemia early in her pregnancy. She is 7 weeks pregnant as seen on ultrasound. Her hemoglobin level is 9 g/dL. When should the iron supplements be started for her?

- a) 10 to 12 weeks
- b) 8 to 10 weeks
- c) After 14 weeks
- d) After 20 weeks

Question 5:

A 50-year-old woman presents with foul-smelling bloody discharge per vagina mixed with mucous. On examination, a necrotizing growth is seen in the cervix with lateral parametrium involvement. What is the management for this patient?

- a) Chemotherapy
- b) Brachytherapy
- c) Chemoradiation
- d) Surgery

Question 6:

A woman is set to undergo a complete laparoscopic hysterectomy. She wants to know if there are any disadvantages to this procedure. What are the disadvantages of laparoscopic over open surgery?

- a) Prolonged recovery time

- b) Not knowing the extent of thermal burns
- c) Increased bleeding
- d) Increased pain

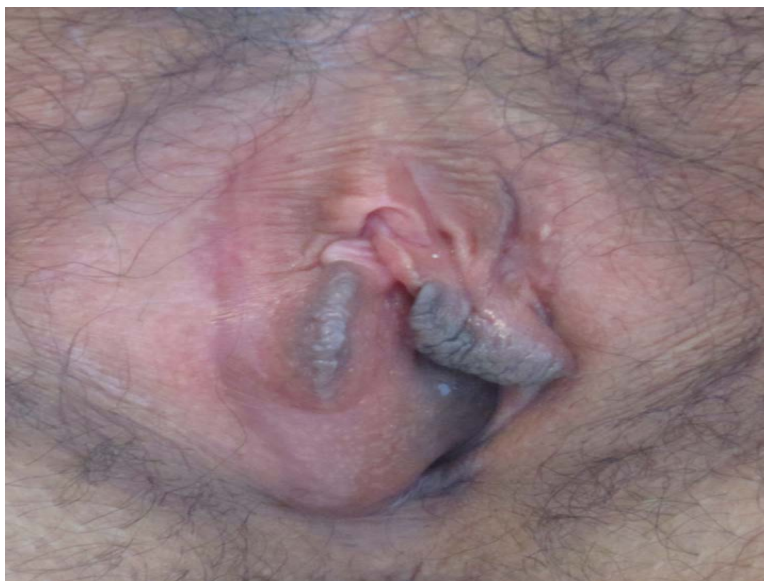
Question 7:

A female presents to you with 6 weeks of amenorrhea complaining of bleeding per vagina and slight abdominal pain. The urine pregnancy test is positive and hCG level is 2800 IU/L. A mass is seen on the left adnexa measuring 3×2.5 cm. She is hemodynamically stable. How will you manage this patient?

- a) Oral methotrexate
- b) Single-dose methotrexate injection
- c) Serial methotrexate + leucovorin rescue
- d) Salpingectomy

Question 8:

A woman comes with complaints of pain and swelling in the perineal area. She also has complaints of difficulty in walking and sitting. She gives a history of multiple sexual partners. On examination, a tender swelling is seen with redness on the labia, as shown in the image given below. What is the most likely diagnosis?



- a) Chlamydial infection
- b) Bartholin abscess

- c) Genital Tuberculosis
- d) Herpes infection

Question 9:

A 23-year-old woman accompanied by her mother-in-law comes to the infertility clinic. She has been having regular intercourse for 6 months but is not able to conceive. What is the next best step?

- a) Semen analysis for husband
- b) Reassure and review the couple after 6 months
- c) Hysterolaparoscopy
- d) Diagnostic hysteroscopy

Question 10:

A 28-year-old woman with a history of multiple sexual partners presents with lower abdominal pain for 1 month. She experiences minimal discharge and complaints of intermenstrual bleeding. What is the most likely cause?

- a) Herpes simplex
- b) Genital tuberculosis
- c) Candida
- d) Neisseria gonorrhoea

Question 11:

A 27-year-old pregnant G3P2L2 woman presents to you at 36+6 weeks. Ultrasound is done and shows the fetus is in a transverse lie. The liquor is adequate, the placenta is normal, and she has no risk factors. Both her previous deliveries were normal vaginal deliveries. How will you manage this patient?

- a) Cesarean section
- b) External cephalic version
- c) Expectant management
- d) Induction of labor

Question 12:

A 24-year-old woman who is being treated for infertility with human menopausal gonadotropin came with complaints of sudden abdominal pain, nausea, vomiting, and breathlessness. The finding of the ultrasound is shown below. What is the most likely diagnosis?



- a) Ovarian hyperstimulation syndrome
- b) Polycystic ovarian syndrome
- c) Theca lutein cysts
- d) Granulosa cell tumor

Question 13:

During a hysteroscopic removal of a submucosal fibroid, a fluid deficit of 2000 mL is estimated in a patient. What is the immediate complication of this?

- a) Acute tubular necrosis
- b) DIC
- c) Pulmonary edema
- d) Thromboembolism

Question 14:

A 54-year-old woman was diagnosed with advanced cervical cancer. She has a 14-year-old daughter. What advice would you give her daughter?

- a) Advise HPV vaccine
- b) Screen for BRCA mutation
- c) Screen for PTEN mutation
- d) Perform cervical biopsy

Question 15:

A 17-year-old girl is seen for primary amenorrhea. There is no development of breasts or hair in the pubic or axillary region. Her height is 155 cm and her weight is 48 kg. She has bilateral inguinal masses. On ultrasound examination, the uterus, fallopian tube, and ovary are absent. What is the most likely diagnosis?

- a) Complete androgen insensitivity syndrome
- b) Hypergonadotropic hypogonadism
- c) Turner syndrome
- d) Polycystic ovary syndrome

Question 16:

A 16-year-old girl presents with cyclical pelvic pain every month. She has not achieved menarche yet. On examination, a suprapubic bulge can be seen in the lower abdomen. PR examination reveals a bulging swelling in the anterior aspect. What is the most likely diagnosis?

- a) Transverse vaginal septum above the vagina
- b) Vaginal atresia
- c) Imperforate hymen
- d) Cervical agenesis

Question 17:

A pregnant woman comes to the clinic. She already has twins by normal delivery. Which of the following is the correct representation of her obstetric score?

- a) G3P2
- b) G3P1

- c) G2P1
- d) G2P2

Question 18:

A woman presents to you at 36 weeks of gestation with complaints of feeling lightheadedness and dizziness when she lies on her back. She says she feels alright if she lies on her side or when walks. What is the most likely reason behind this?

- a) Increased intracranial pressure
- b) IVC compression
- c) Heavy meals
- d) Excessive venous pooling at the feet

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	b
5	c
6	b
7	b
8	b
9	b
10	d
11	b
12	a
13	c
14	a
15	b
16	c
17	c
18	b

Detailed Explanations

Solution to Question 1:

Based on the above partogram, there is a cephalopelvic disproportion, and the management, in this case, would be to do an emergency cesarian section.

The above partogram shows that the cervix is fully dilated, molding is 2+, the station is 3+, and oxytocin infusion is already given. The patient has adequate uterine contractions, as shown. The amniotic fluid is stained with meconium, and there is a fall in fetal heart rate, which indicates that the baby is in distress. This shows that there is an arrest of the fetal head with molding, which points to cephalopelvic disproportion.

In case of uterine inertia or fetal distress, the management is an emergency cesarean section, not instrumental delivery or elective cesarean section.

Cephalopelvic disproportion is a state where the normal proportion between the size of the fetal head to the size of the pelvis is disturbed. It can either occur due to an average size baby with a small pelvis or a relatively large baby with a normal pelvis.

CPD can occur at the inlet, mid-pelvis, or at pelvic outlet. It is diagnosed at these levels by the following measurements.

Inlet CPD

- Obstetric conjugate ≤ 10 cm
- Diagonal conjugate ≤ 11 cm
- Transverse diameter ≤ 12 cm

Midpelvis CPD

- Sum of the interspinous and posterior sagittal diameters of the mid pelvis ≤ 13 cm
- Interspinous diameter ≤ 8 cm

Outlet CPD

- Interischial diameter ≤ 8 cm.

The following events may occur over the course of labor, depending on the degree of pelvic contraction.

- Early rupture of the membranes.
- Cord prolapse
- Cervical dilatation is slowed.
- Obstructed labor with features of exhaustion, dehydration, ketoacidosis, and sepsis.
- Operative interference, shock, postpartum hemorrhage, and sepsis.

The degree of disproportion must be assessed by clinical examination and supplemented by imaging pelvimetry. The moderate and the severe degrees can be managed by one of the

following:

- Induction of labor
- Elective cesarean section at term
- Trial labor

The elective cesarean section at term is commonly done in cases of a major degree of inlet contraction and in a moderate degree of inlet contraction associated with outlet contraction or complicating factors like elderly primigravida, malpresentation, post-cesarean pregnancy, etc. It is planned during the last week of pregnancy.

Solution to Question 2:

The diagnosis in this case in which symphysis fundal height (SFH) is 41 cm at 36 weeks of gestation and the woman presents with breathlessness and excessive abdominal distension is most likely polyhydramnios.

Polyhydramnios can be defined as the presence of excessive amniotic fluid, which exceeds 2,000 mL.

The sonographic diagnosis of polyhydramnios can be given as

- AFI ≥ 25 cm
- Deepest vertical pocket (DVP) measuring > 8 cm

Causes of polyhydramnios can be due to maternal and fetal conditions-

- Maternal causes include chorioangioma of the placenta, multiple pregnancies commonly in monozygotic twins, disorders of the cardiac or the renal system, and diabetes mellitus.
- Fetal causes include anencephaly, spina bifida, atretic esophagus or duodenum, facial clefts, and neck masses that lead to abnormal swallowing or hydrops fetalis.

Patients may present with dyspnea, palpitations, and pedal edema.

On examination, the following findings can be elicited:

- Markedly enlarged abdomen with flank fullness
- Tense, shiny skin with striae
- Fundal height more than the appropriate gestational age: The SFH measured in cm after 24 weeks is equivalent to the number of weeks up to 36 weeks. A variance of 2 cm is regarded as typical. The SFH in 36 weeks of gestation should have been 36 ± 2 cms, which is less than in this patient.
- Fetal parts and the presentation cannot be well-defined
- Fluid thrill will be present

On per vaginal examination, tensed bulging membranes can be felt. Investigations to confirm the diagnosis include an ultrasound of the abdomen and pelvis and an evaluation of possible causes.

Other options:

Option A - Placental abruption classically presents with sudden onset of abdominal pain, vaginal bleeding, and uterine tenderness. Frequent contractions and persistent hypertonus are the other findings. When the placenta is completely detached from the uterine wall, except at the margin, blood may collect behind the placenta. This kind of presentation is known as concealed type. In another type, blood can escape via the cervix by dissecting downward between the membranes and the uterine wall. It is called the revealed type, which is the most common type.

Option B - Hydrocephalus of the fetus: Hydrocephalus is a condition where there is an abnormal accumulation of cerebrospinal fluid within the brain, leading to increased head circumference in the fetus. Although it may lead to an increased symphysis fundal height, the clinical presentation of breathlessness and excessive abdominal distension in the mother is not typical of this condition.

Option D - Fetal-maternal ascites can present as Mirror syndrome. There is a connection between fetal hydrops and the onset of maternal edema, in which the mother exhibits the same symptoms as the fetus. Because the mother, the placenta, and the fetus all develop edema, the condition is also called triple edema. At least 20% of cases of hydrops have been observed to be complicated by mirror syndrome. Women most commonly present with edema. Other symptoms may include proteinuria, hypertension, the elevation of liver enzymes, and headache.

Solution to Question 3:

Of the following procedures given in the OT list, Asherman syndrome is the condition for which hysteroscopy would be most preferable. Hysteroscopy is used for the definitive diagnosis of Asherman syndrome.

Asherman syndrome refers to the presence of adhesion within the endometrial cavity. It is also called uterine synechiae and can be asymptomatic tiny bands or can completely or almost completely obliterate the endometrial cavity. It can occur as a result of trauma to the endometrium following vigorous uterine curettage, tuberculous endometritis, pelvic radiation, or endometrial ablation.

Depending on the degree of scarring, patients may have amenorrhea, hypomenorrhea, or recurrent pregnancy loss because of inadequate placentation. Hysterosalpingography is performed to assess the uterine cavity and tubal patency. Intrauterine adhesions are characteristically seen as irregular angulated filling defects in the uterine cavity with a classic 'moth-eaten' appearance. Definitive diagnosis (gold standard) and management are done by hysteroscopy. Hysteroscopic transection is the surgical treatment of choice.

Hysteroscopy has the advantage of being a minimally invasive procedure with lower risks and faster recovery times than traditional open surgery. It is often performed as an outpatient procedure, and patients can usually return to their normal activities within a day or two.

Other options-

Option A - Tubal ligation is done to attain permanent contraception in females, which is usually accomplished by occlusion or division of the fallopian tubes. It prevents ovum passage and fertilization. Procedures done at the time of delivery are termed puerperal sterilization. For puerperal sterilization, suture ligation with tubal segment removal is more frequently employed. A tubal sterilization procedure done at a time unrelated to recent pregnancy is called non-puerperal

tubal sterilization or interval sterilization. Currently, non-puerperal tubal sterilization is mainly accomplished via laparoscopy.

Option C - Endocervical polyps are mucinous columnar epithelium-covered benign endocervical stroma overgrowths. They often present as a single, elongated mass that is red, smooth, and projecting from the endocervical canal. Ring forceps are used to grab endocervical polyps for excision if the stalk is thin. The stalk base of the polyp is repeatedly twisted. Surgical ligation and excision of a thick pedicle may be necessary if further severe bleeding is expected. Although hysteroscopy can help visualize and assess these polyps, their removal is typically performed using forceps, either during a pelvic examination or during a colposcopy. Hysteroscopy is not the primary method for managing endocervical polyps.

Option D - Subserosal fibroids carry higher risks for incomplete resection, difficult visualization, or uterine perforation with hysteroscopic myomectomy. Hence, a number of parameters are taken into consideration while choosing a surgical strategy for subserosa or intramural myomas. Successful results and low recurrence rates are seen after laparoscopic leiomyoma excision.

Solution to Question 4:

The iron supplements for the 7-week pregnant mother presenting with anemia should be started at 8 to 10 weeks.

The Centers for Disease Control and Prevention defined anemia in pregnant women using a cut-off value. The pregnant mother is said to be anemic when the level of hemoglobin is less than 11 g/dL in the first and third trimesters and less than 10.5 g/dL in the second trimester.

Anemia in the first trimester is treated by oral iron therapy. According to the Anemia Mukht Bharat program, pregnant women with mild and moderate anemia are treated with two iron and folic acid tablets. Each tablet contains 60 mg of elemental iron and 500 mcg of folic acid. These tablets are taken daily and are supplied by the healthcare worker. Hence, in this mother iron supplementation should be started by 8-10 weeks.

Drawbacks of oral therapy are intolerance presenting as epigastric pain, nausea, vomiting, constipation, or diarrhea, and unpredictable absorption rate. The improvement in hemoglobin levels is generally evident within 3 weeks of oral iron therapy. The hemoglobin concentration is expected to rise at about 0.7 gm/100 mL/week.

Contraindications to oral therapy are intolerance, and severe anemia in advanced pregnancy. The best time to start corrective iron therapy in pregnancy depends on the individual patient's needs and the presence of anemia. For most pregnant women without anemia, it is recommended to start iron supplementation between 12-16 weeks of gestation to meet the increased iron demands during pregnancy.

If a pregnant woman is found to be anemic, the iron supplementation should be started as soon as the anemia is diagnosed, regardless of the gestational age. Early diagnosis and treatment of anemia are crucial to reduce the risk of complications for both the mother and the baby.

In the case of severe anemia or intolerance to oral iron supplements, intravenous iron therapy may be considered, depending on the clinical situation and the patient's response to oral therapy.

Solution to Question 5:

This patient with a cervical tumor involving the parametrium has stage IIB cervical carcinoma, which is treated primarily with chemoradiation.

Stage IIB cervical cancer refers to cancer that has extended beyond the cervix and upper vagina to the parametrial tissue. It has not spread to the pelvic sidewall or the lower third of the vagina. The primary management for stage IIB cervical cancer is concurrent chemoradiation therapy. This involves the combination of external beam radiation therapy, brachytherapy, and chemotherapy (usually with cisplatin). This approach has been shown to improve overall survival and reduce the risk of local and distant recurrence compared to radiation therapy alone.

Following chemoradiation, patients should be closely monitored for treatment response and potential side effects or complications. This includes regular physical examinations, Pap tests, and imaging studies as needed. The prognosis for stage IIB cervical cancer depends on various factors, such as the size of the tumor, lymph node involvement, and the patient's overall health. With appropriate treatment, the 5-year survival rate for stage IIB cervical cancer is around 58-63%.

Other options:

Option A: Chemotherapy alone: Chemotherapy alone is not the primary treatment for stage IIB cervical cancer. It may be used as an adjunct to other treatments, such as radiation therapy, but is not sufficient on its own.

Option B - Brachytherapy: Brachytherapy is a form of internal radiation therapy, in which a radioactive source is placed close to the tumor. While it is an important component of the treatment for cervical cancer, it is typically not used as a standalone treatment.

Option D - Surgery: Surgery is not the primary treatment option for stage IIB cervical cancer. In some rare cases, pelvic exenteration may be performed, but this is typically reserved for patients with recurrent or persistent disease after chemoradiation.

Solution to Question 6:

The disadvantage of laparoscopic over open surgery is not knowing the extent of the thermal burns.

Diathermy burns are a particular hazard in laparoscopic surgery due to the following reasons:

- Relative lack of visibility of the entire instrument.
- Faulty insulation of laparoscopic instruments
- Intraperitoneal contact of the diathermy with another metal instrument while activating the pedal.
- Inadvertent activation of the pedal while the diathermy tip is out of the vision of the camera.
- Retained heat in the diathermy tip touching susceptible structures, such as the bowel.

The advantages of laparoscopic over open surgery are as follows:

- Less postoperative pain (Option D)
- Less blood loss (Option C)

- Cosmetically better
- Earlier recovery and shorter postoperative stay at the hospital (Option A)
- Fewer intraoperative adhesions
- Better perception of anatomy intraoperatively

Solution to Question 7:

The clinical scenario of a patient with 6 weeks of amenorrhea, abdominal pain, and vaginal bleeding with beta-hCG levels of 2800 mIU/mL and a gestational sac in the left adnexa measuring 3 × 2.5 cm points to the diagnosis of unruptured ectopic pregnancy. The most appropriate management for this patient is a single dose of methotrexate injection.

Symptoms and signs of ectopic pregnancy are often subtle or absent in unruptured ectopic pregnancy. The classic triad includes amenorrhea, which is followed by pain and vaginal bleeding. Lower abdominal and pelvic pain is usually seen with tubal rupture. The pain is severe and frequently described as sharp, stabbing, or tearing.

Management of ectopic pregnancy:

1. Medical management is preferred in the following cases:

- Low initial serum beta-hCG level (Less than 5,000 IU/L)
- Small ectopic pregnancy size (Less than 3.5 cm)
- Absent fetal cardiac activity
- Hemodynamically stable

The most popular method of treating ectopic pregnancies is intramuscular methotrexate injection, which can be done in single-dose or multidose protocols. The dose for single-dose therapy is 50 mg/m² body surface area (BSA).

Single-dose therapy is preferred over the multidose protocol as the former offers simplicity, lower costs, and less involved post-therapy monitoring.

2. Expectant management: A low initial serum hCG concentration, a considerable reduction in levels over 48 hours, and a sonographic inhomogeneous mass instead of a tubal halo or other gestational features are predictive variables for success.

3. Surgical management with salpingectomy or salpingostomy is done when:

- Criteria for expectant and medical management are not met
- Gestational sac >3.5 cm
- Fetal cardiac activity present
- Beta-hCG >5000 mIU/mL
- Significant pain and hemoperitoneum
- Failure of expectant or medical management
- Clinical status of the patient warrants urgent intervention

- Doubtful compliance with medical management

Other options:

Option A: Oral methotrexate is not the preferred route for treating ectopic pregnancy, as it may lead to variable absorption and unreliable dosing. Intramuscular injection is the preferred route of administration for methotrexate in the management of ectopic pregnancy.

Option C: Serial methotrexate with leucovorin rescue is a multidose protocol that can be used to treat ectopic pregnancy. However, the single-dose methotrexate injection is preferred in this case due to its simplicity, lower costs, and less involved post-therapy monitoring.

Option D: Salpingectomy, the surgical removal of the affected fallopian tube, is reserved for patients who do not meet the criteria for medical management or expectant management, have significant pain and hemoperitoneum, or have failed medical management. This patient does not fulfill the criteria for surgical management.

Solution to Question 8:

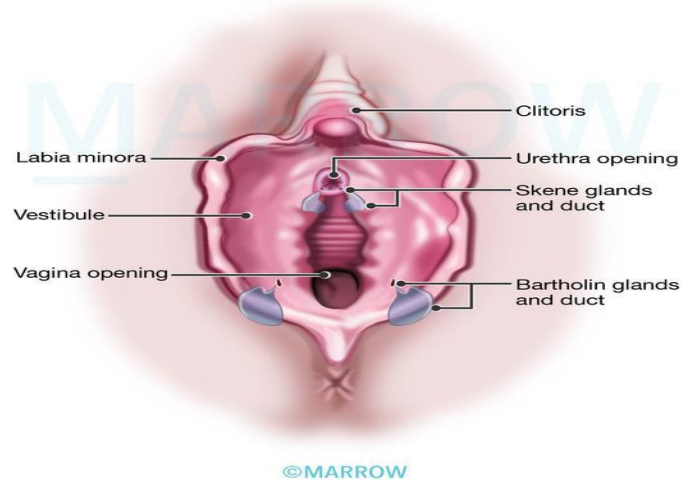
The given clinical vignette suggests the diagnosis of Bartholin's abscess.

Bartholin glands are the homologs of the male bulbourethral or Cowper glands. They open into the vestibule at approximately the 5 and 7 o'clock positions. The function of the Bartholin gland is to secrete mucus for lubrication during coitus. Bartholin cyst is caused by obstruction of the duct due to infection, trauma, mucus changes, and congenitally narrowed ducts. If the cyst gets infected, it leads to the formation of a Bartholin abscess. *Escherichia coli* is the most common causative agent, however, other gram-positive and gram-negative organisms can also cause the same.

Most Bartholin gland cysts are small and asymptomatic except for minor discomfort during sexual contact. If the cyst is larger or infected, however, patients may complain of severe vulvar pain during walking, sitting, or coitus. On examination, cysts typically are unilateral, round, or ovoid, and fluctuant or tense. Infected cysts are often erythematous and tender. Most of the cysts produce labial asymmetry.

Small, asymptomatic cysts can be managed conservatively. Large, symptomatic cysts can be managed by incision and drainage, or marsupialization, or Bartholin gland excision. Bartholin abscess can be managed by incision and drainage by word catheter placement or marsupialization. In postmenopausal women with recurrent Bartholin cysts, a suspicion of Bartholin gland cancer should arise and the cyst wall should be sent for biopsy.

Skene's glands and Bartholin's glands



Other options:

Option A - Chlamydial infections typically present with vaginal discharge, dysuria, or cervicitis rather than localized pain and swelling in the perineal area. Though Chlamydia can cause infections in the Bartholin gland, the symptoms described in the question are more suggestive of a Bartholin abscess.

Option C - Genital tuberculosis primarily affects the upper genital tract, leading to menstrual irregularities, infertility, and abdominal pain. While it can also involve the vulva and perineal area, the clinical presentation described in the question is not typical for genital tuberculosis.

Option D - Genital herpes usually presents as painful ulcers or vesicles in the genital area, rather than a tender, erythematous swelling. The clinical presentation in the question is more consistent with a Bartholin abscess.

Solution to Question 9:

The next best step would be to reassure and review the couple after 6 months.

The inability to conceive after 1 year of unprotected intercourse of reasonable frequency is described as infertility.

Infertility evaluation is offered to any couple that has failed to conceive after 1 year of regular unprotected intercourse. There is a decline in the conception rate and a rise in poor pregnancy outcomes in older women hence, the majority of experts concur that in women older than 35, evaluation may be considered after 6 months. Since the woman is 23 years old and has tried conceiving for only 6 months, it is advisable to review them in another 6 months if they fail to conceive.

Infertility workup includes the following if required:

- (1) Ovulation and evaluation for the diminished ovarian reserve-
- (2) Evaluation of Female reproductive tract abnormality-

- Hysterosalpingography
 - Sonography
 - Laparoscopy
 - Hysteroscopy- The preferred way to diagnose intrauterine anomalies is an endoscopic assessment of the uterine cavity. Both an operating room and a doctor's office can be used for hysteroscopy. It has the ability to concurrently diagnose and treat the abnormalities.
- (3) Semen analysis

Solution to Question 10:

This clinical scenario indicates infection with *Neisseria gonorrhoeae*, which has led to pelvic inflammatory disease.

Gonorrhea is a sexually transmitted disease caused by *N. gonorrhoeae*. It uses pili in adhesion to the urethra or other mucosal surfaces. The cocci penetrate the intracellular spaces to reach the subepithelial tissue by the third day of infection. The incubation period is 2-8 days.

The major symptoms include vaginal discharge, dyspareunia, and dysuria, usually without urinary frequency or urgency. The specific symptoms seen in males and females are as follows:

In females:

About half the female patients remain asymptomatic. But, patients typically present with urethritis, cervical discharge, pruritus, dysuria, and intermenstrual bleeding.

The ectocervix and vagina are not usually affected in adults as the stratified squamous epithelium is resistant to infection by the cocci. The acidic pH of vaginal secretions is also protective. The principal sites of invasion for gonococcal infection are the columnar and transitional epithelium.

The initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, skene glands, endometrium, and fallopian tubes. Unilateral pain and swelling over the labia may be caused by the inflammation of the Bartholin glands.

Pelvic inflammatory disease and salpingitis may lead to sterility. Other complications are tubo-ovarian abscesses and ectopic pregnancies. Rarely peritonitis may develop with perihepatic inflammation (Fitz-Hugh-Curtis syndrome). The patient presents with right upper quadrant pain and tenderness with abnormal liver function tests.

In males:

- Acute urethritis: This infection may extend along the urethra to involve the prostate, seminal vesicles, and epididymis.
- Genital discharge: It is mucopurulent that can be described as that resembling the 'flow of seed' (semen).
- Strictures: Chronic urethritis may lead to stricture formation.
- Water-can perineum: The infection may spread to the periurethral tissues causing abscesses and multiple discharging sinuses.

Other options:

Option A: Herpes simplex presents as painful genital ulcers.

Option B: While genital tuberculosis may present as menstrual irregularity, it is usually accompanied by systemic symptoms such as fever, malaise, and weight loss. Often with concurrent pulmonary disease. It may also present as infertility or may remain asymptomatic.

Option C: Vaginal candidiasis presents as intense pruritus and copious cottage cheese vaginal discharge.

Solution to Question 11:

A pregnant woman G3P2L2 presenting at 36+6 weeks with a transverse lie without being in labor is managed with an external cephalic version (ECV), followed by vaginal delivery.

In a transverse lie, the fetus' long axis is perpendicular to the mother's. The shoulder is typically placed above the pelvic inlet when lying transversely. The head occupies one iliac fossa, and the other by the breech. This results in a shoulder presentation.

Management of transverse lie:

1. Before labor/early in labor - If the membranes are intact (like in this case), it is recommended that ECV should be offered and attempted whenever possible.

Prerequisites for ECV - Women who have reached 37 0/7 weeks of gestation. Multiparity, unengaged presenting part, nonanterior placenta, nonobese patient, and abundant amniotic fluid improves the chances of ECV.

Absolute contraindications - When vaginal delivery is contraindicated, e.g. placenta previa

Relative contraindications - Early labor, oligohydramnios or ruptured membranes, known nuchal cord, structural uterine abnormalities, fetal growth restriction, multifetal gestation, and prior abruption of the placenta

2. Active labor - A cesarean delivery is frequently necessary for such a scenario (Option A). The lower uterine segment is not occupied by the fetal feet or head in a dorsoanterior or back down position. A vertical hysterotomy incision is usually necessary for such a position. In a dorsoposterior or backup position, one or both feet can be gripped through a small transverse incision and delivered by breech extraction.

Expectant management refers to allowing vaginal delivery to happen. It is chosen when the version fails or is contraindicated. Vaginal delivery may be considered in the case of small fetuses with congenital malformations or dead fetuses (Option C).

Transverse lie (malpresentation) is a contraindication to induction of labor (Option D).

Solution to Question 12:

The above clinical scenario and the sonogram showing enlarged ovaries having large cysts are suggestive of ovarian hyperstimulation syndrome.

Ovarian hyperstimulation syndrome (OHSS) is characterized by multiple ovarian follicular cysts accompanied by increased capillary permeability. It happens most often as a complication of ovulation induction therapy for infertility. It occurs due to the stimulation of vascular endothelial growth factor (VEGF) expression in granulosa-lutein cells by hCG.

They often present with shortness of breath, abdominal pain, and gastrointestinal disturbances like nausea, vomiting, and oliguria.

Abdominal pain occurs due to ovarian enlargement and accumulation of peritoneal fluid. Increased vascular permeability and loss of fluid, protein, and electrolytes into the peritoneal cavity cause ascites and result in hemoconcentration. Hemoconcentration causes hyperviscosity and leads to hypercoagulability.

Hyponatremia, hyperkalemia, leukocytosis, pleural effusion, and pericardial effusion can be associated with OHSS.

Some of the risk factors for OHSS:

- Polycystic ovarian syndrome
- Young age
- High estradiol during ovulation induction
- Pregnancy

This condition can be prevented by

- Using lowered FSH dose to decrease follicular stimulation
- Coasting- withholding FSH administration for 1 or more days prior to the hCG trigger injection,
- Prophylactic treatment with volume expanders,
- Substitution of hCG for FSH during the final days of ovarian stimulation.

OHSS is primarily managed with general supportive care. Paracentesis is performed to ameliorate abdominal discomfort and relieve respiratory discomfort.

Other options:

Option B- Polycystic ovarian syndrome (PCOS) is the most common endocrine disorder of the reproductive age group. Patients with PCOS present with menstrual irregularities, hirsutism, acne, infertility, sleep apnea, and depression/anxiety. These are all short-term consequences of PCOS. The long-term consequences of PCOS include diabetes mellitus, endometrial cancer, and cardiovascular disease.

According to the American Society of Reproductive Medicine (ASRM) and the European Society of Human Reproduction and Embryology (ESHRE), sonographic criteria for PCOS include

- ≥ 20 follicles and/or
- Increased ovarian volume (≥ 10 mL), or both

The ultrasound examination image shown below is of PCOS showing enlarged ovaries with multiple cysts.



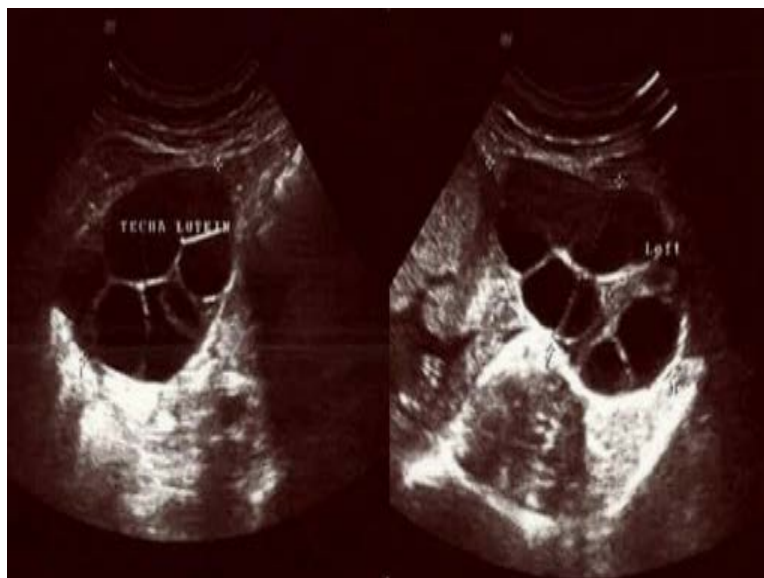
Option C - Theca lutein cysts are a type of follicular cysts. Hyperreactio luteinalis is a condition in which one or both ovaries develop multiple large theca lutein cysts. They typically occur after the first trimester due to luteinization and hypertrophy of theca interna layer in response to exceptionally high hCG or luteinizing hormone.

They are commonly seen in

- Gestational trophoblastic disease
- Multiple gestational pregnancy
- Fetal hydrops
- Placentaomegaly

Most patients are asymptomatic and the above symptoms are not seen in this condition. Surgical management is only required if it is complicated by torsion or hemorrhage.

The below image shows bilaterally enlarged ovaries with multiple theca luteal cysts as in hyperreactio luteinalis.



Option D - Most women are diagnosed with adult granulosa cell tumors after the age of 30 years, and the average age is roughly 55 years. Heavy, irregular menstrual bleeding and postmenopausal bleeding are frequent and are caused by the endometrium's extended exposure to estrogen. Breast soreness and enlargement are also seen in this condition. Instead of hormones, the mass of the ovary may be the cause of the symptoms. Abdominal pain and distension that is brought on by an expanding tumor that could potentially be hemorrhagic.

Solution to Question 13:

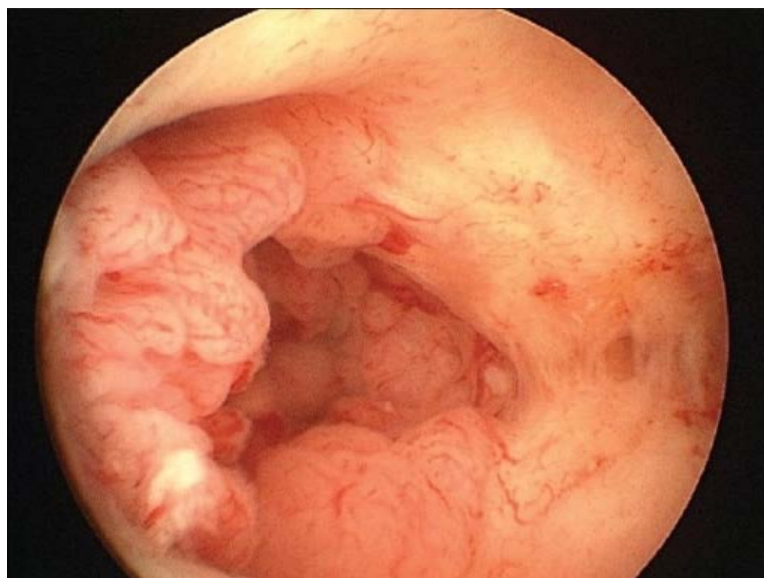
The given clinical scenario suggests the diagnosis of fluid overload, and pulmonary edema is the immediate complication anticipated in this scenario.

In the context of hysteroscopic procedures, the term fluid deficit refers to the difference between the volume of distension fluid introduced into the uterine cavity and the volume of fluid that is removed or accounted for during the procedure. A large fluid deficit indicates that a significant amount of the distension fluid has been absorbed into the patient's bloodstream.

When a large volume of fluid is absorbed during a hysteroscopic procedure, it can lead to fluid overload. This excess fluid in the bloodstream can increase the pressure in the blood vessels, particularly in the pulmonary circulation. The elevated pressure can cause fluid to leak from the small blood vessels in the lungs into the air spaces, resulting in pulmonary edema.

Pulmonary edema can lead to shortness of breath, difficulty breathing, and impaired gas exchange, as the fluid-filled air spaces in the lungs make it difficult for oxygen to pass from the air into the bloodstream. In severe cases, it can lead to respiratory failure and can be life-threatening. Prompt recognition and management of fluid overload during hysteroscopic procedures are crucial to minimizing the risk of complications like pulmonary edema.

Hysteroscopy is an operative procedure that provides the endoscopic view of endometrial cavity and tubal ostia via the cervix. It can be diagnostic or therapeutic. A hysteroscopic image of a uterus with carcinoma is shown below.



Complications of hysteroscopy include :

- Early complications:
- Anesthesia complications- air embolism can occur
- Uterine perforation
- Thermal injury with cautery and laser
- Sepsis
- Allergic reaction to dextran and glycine
- Fluid overload occurs and can lead to pulmonary edema if the fluid deficit is more than 1000 mL. It can be managed by diuretics.
- Cerebral edema
- Cardiac failure
- Late complications:
- Hematometra following cervical stenosis
- Pelvic inflammatory disease
- Uterine rupture during pregnancy
- Dysmenorrhea
- Amenorrhea

Other options:

Option A: Acute tubular necrosis (ATN) is a kidney disorder involving damage to the tubule cells of the kidneys, which can lead to acute kidney injury (AKI). While fluid overload can be a risk factor for AKI, it is not the immediate complication of a fluid deficit of 2000 mL in this clinical scenario.

Option B: Disseminated intravascular coagulation (DIC) is a serious disorder where the proteins that control blood clotting become overactive. It is not directly related to a fluid deficit during a hysteroscopic procedure and is therefore not the immediate complication in this case.

Option D: Thromboembolism refers to a blood clot that forms in a blood vessel and then breaks loose, traveling through the bloodstream and potentially causing a blockage in another part of the body. Although any surgical procedure carries a risk of thromboembolism, it is not the immediate complication related to a fluid deficit of 2000 mL in this scenario.

Therapeutic indications of hysteroscopy	Diagnostic indications of hysteroscopy
Uterine septum Asherman syndrome Retained IUCD Polyp ectomy Submucous fibroid Abnormal uterine bleeding Tubal sterilization using sclerosing agents	To study the endocervical mucosal lining Congenital malformations of the uterus Endometrial tuberculosis Asherman syndrome Misplaced IUCD Endometrial lesions Endometrial polyp Cornual tubal blockage

Solution to Question 14:

The patient's 14-year-old daughter should be advised to take the HPV vaccine.

The patient's daughter is not at a higher risk for developing cervical cancer because it does not run in families. The etiology of cervical cancer is related to human papilloma virus (HPV) infection, and immunization against the virus is protective.

There are three types of available HPV vaccines, Cervarix, Gardasil, and Gardasil-9.

Option 2: BRCA mutation causes breast cancer, it is not related to cervical cancer.

Option 3: PTEN mutation on chromosome 10 causes Cowden syndrome.

Option 4: Cervical biopsy is not indicated in this scenario.

	Cervarix	Gardasil	Gardasil-9
Characteristic and HPV subtypes	Bivalent 16, 18	Quadrivalent 6, 11, 16, 18	Nonavalent 6, 11, 16, 18, 31, 33, 45, 52, 58
Protects against	CIN, Ca cervix	Anogenital warts, CIN, Ca cervix	Anogenital warts, CIN, Ca cervix, Vulval intraepithelial neoplasia, Vaginal intraepithelial neoplasia
Dose	0.5 mL	0.5 mL	0.5 mL
Administration	0, 1, 7 months IM	0, 2, 6 months IM	0, 2, 6 months IM
Administered gender	Female and male	Female and male	Female and male
Ideal age range (Routine recommendation by CDC)	11–12 years	11–12 years	11–12 years 9–15 years - FDA approved
ACIP age recommendation in previously unimmunized women	13–26 years	13–26 years	13–26 years

Solution to Question 15:

The diagnosis based on the given clinical scenario is hypergonadotropic hypogonadism.

The clinical scenario mentions a female with primary amenorrhea with bilateral inguinal masses (undescended testis) and absent ovary pointing to the possibility of XY karyotype. Absent uterus and fallopian tubes point to the presence of a Mullerian inhibiting substance.

Hypergonadotropic hypogonadism can occur in disorders such as FSH or LH receptor mutations. Leydig cell hypoplasia is an autosomal recessive 46, XY disorder. It is caused by inactivating mutations in the LH/hCG receptor that prevents normal hormone

binding. The decrease in the number or function of Leydig cells decreases fetal testosterone production. This leads to the failure of normal male sexual differentiation.

In patients presenting with LH receptor defects, the action of the anti-mullerian hormone is normal; hence, the derivatives of the Mullerian duct are absent. Due to decreased testosterone production, wolffian duct development is impaired and the testes fail to descend. They appear female at birth and at puberty present with primary amenorrhea. They lack breast development and pubic hair development.

Other options:

Option A: Androgen insensitivity syndrome is an X-linked recessive disorder in which the androgen receptors are absent, scanty, or fail to respond to testosterone. In complete AIS, the individual is born as a phenotypically normal female but generally presents with primary amenorrhea. The external genitalia appears normal, with absent or scant pubic and axillary hair. The vagina is shortened and blind-ending with an absent uterus and ovaries. These patients may also present with a history of childhood inguinal hernias, due to the presence of undescended testis in the abdomen or in the inguinal canal. Due to the constant conversion of testosterone into estrogen, these individuals have normal breast development.

Option C: Turner's syndrome presents with a structural abnormality of the second X chromosome or has a mosaic karyotype, such as 45X/46 XX. The clinical features of Turner's syndrome include short stature, short and wide neck, broad chest, widely spaced nipples, primary amenorrhea, congenital heart defects, and renal anomalies. Two X chromosomes are required for the development of ovaries. Losing one X Chromosome results in streak ovaries and inadequate estrogen production.

Option D: Polycystic ovarian syndrome (PCOS) is the most common reproductive age group endocrine disorder. Patients with PCOS present with menstrual irregularities, hirsutism, acne, infertility, sleep apnea, and depression/anxiety. These are all short-term consequences of PCOS. The long-term consequences of PCOS include diabetes mellitus, endometrial cancer, and cardiovascular disease.

According to American Society of Reproductive Medicine (ASRM) and European Society of Human Reproduction and Embryology (ESHRE), sonographic criteria for PCOS include

- ≥ 20 follicles and/or,
- Increased ovarian volume (≥ 10 mL), or both

Solution to Question 16:

The given case scenario is suggestive of cryptomenorrhea due to an imperforate hymen.

Cryptomenorrhea is a condition where menstruation occurs but is not visible due to an obstruction of the outflow tract. The menstrual fluid gets collected in the reproductive tract.

Imperforate hymen is a urogenital anomaly. Imperforate hymen results due to failure of canalization of the inferior end of the vaginal plate. This anomaly usually remains unrecognized until puberty. It can lead to mucocolpos in early infancy. Patients present with amenorrhea, cyclic abdominal pain, urinary symptoms, and a hypogastric lump on abdominal examination. A bluish bulging membrane at the vaginal opening is noted on genital examination and the Valsalva

maneuver.

Ultrasound of the pelvis reveals normal internal genitalia and hematocolpos. Laparoscopy should be performed along with hymenectomy to exclude endometriosis.

Most common causes of cryptomenorrhea:

- Intact/imperforate hymen - a bluish convex bulge is seen
- Transverse vaginal septum - a pinkish concave covering is seen

Ultrasound pelvis confirms the diagnosis. A simple cruciate incision of the hymen allows the free drainage of the collected menstrual blood and restores the reproductive capacity in most cases.

Other options:

Option A: Transverse vaginal septum occurs due to the failure of Müllerian duct fusion or failure of vaginal plate canalization. On examination, the vagina may appear shortened and the cervix may not be identifiable. Secondary sexual characteristics appear normal. The diagnosis can be confirmed with ultrasound or MRI.

Option B: Vaginal atresia is characterized by the absence of the lower portion of the vagina with well-differentiated superior structures like the uterus. On examination, the hymenal ring is normal, proximal to which only a vaginal dimple or small pouch is found.

Option D: Cervical agenesis is the congenital absence of the cervix typically with a normal uterus. These patients also lack the upper vagina in most cases. They present with primary amenorrhea and cyclic pelvic pain. If a functional endometrium is present, a patient may have a distended uterus, and endometriosis may develop. The absence of a cervix is seen on examination.

Solution to Question 17:

The obstetric score of a pregnant woman who previously delivered twins would be G2P1.

The obstetric score is used to summarize the past obstetric history as follows:

- G - Gravidity, denoting the total number of pregnancies, including the present one. It is irrespective of the period of gestation.
- P - Parity, denoting the total number of previous pregnancies crossing the period of viability, which is 28 weeks in India.
- This includes both live births and stillbirths
- The current pregnancy is not included
- A - Abortus or the number of pregnancies that were lost for any reason, including induced abortions or miscarriages.
- L - Living children

Both Gravida and para are the numbers of pregnancies rather than babies.

Terminology

- Nulligravida: One who is not and has never been pregnant
- Primigravida: One who is pregnant for the first time

- Multigravida: One who has previously been pregnant
- Nullipara: One who has never completed a pregnancy to the stage of viability
- Primipara: One who has delivered only once (fetuses born alive or dead after becoming viable)
- Multipara: One with 2 or more pregnancies to the stage of viability or more
- Parturient: A woman in labor

Solution to Question 18:

The most likely reason a woman at 36 weeks of gestation complains of feeling lightheadedness and dizziness when she lies on her back but feels alright if she lies on her side or walks is inferior vena cava (IVC) compression.

When the patient is supine during late pregnancy, the gravid uterus has a compressive effect on the IVC. This results in opening up collateral circulation through the means of paravertebral and azygos veins. However, in some cases, the collateral circulation is unable to open up, and the heart's venous return may get severely impaired. Syncope, tachycardia, and hypotension are produced as a result. The patient is rapidly turned to a lateral posture to return the blood pressure to normal. This phenomenon is called supine hypotension syndrome (postural hypotension).

Other options-

Option A- Individuals with elevated intracranial pressure may experience papilledema along with neck stiffness and visual problems. Several herniation syndromes can result from brain matter being forced through the fixed cranial apertures as the mass grows.

Option C- Pregnant women are advised to avoid heavy meals and not to go to bed immediately after the meal. They can cause heartburn, which is common in pregnancy due to the relaxation of the esophageal sphincter.

Option D- The pressure in the femoral vein is noticeably increased, especially in the later months. The gravid uterus puts pressure on the common iliac veins, particularly on the right side due to the dextrorotation of the uterus. In a lying down position during pregnancy, the femoral venous pressure increases from 8 to 10 cm of water to around 25 cm of water, and in a standing position, it increases to approximately 80 to 100 cm of water. This excessive pooling of blood in the lower limbs explains physiological edema during pregnancy. Edema, varicose veins, piles, and deep vein thrombosis are all caused by the extensibility of the veins and blood stasis in the venous system.

Obstetrics and Gynecology INI-CET 2023

Question 1:

A lady on oral contraceptive pills presented with white discharge per vaginum. It is not associated with itching or foul smell and stains her underwear yellow in color. Examination of the discharge revealed normal lactobacilli and acidic pH. What is the most probable diagnosis?

- a) Vaginal candidiasis
- b) Increased normal vaginal discharge
- c) Bacterial vaginosis
- d) Chlamydial infection

Question 2:

What is the most common cause of puberty menorrhagia?

- a) Anovulation
- b) Endometriosis
- c) Malignancy
- d) Bleeding disorder

Question 3:

SRY region is located in:

- a) Short arm of Y chromosome
- b) Short arm of X chromosome
- c) Long arm of Y chromosome
- d) Long arm of X chromosome

Question 4:

A 32 year old female presents with grey-white vaginal discharge and on gram stain of the discharge, clue cells are seen. Clue cells are

- a) Degranulated neutrophils (pus)
- b) Epithelial cells covered by bacteria
- c) *Gardnerella vaginalis* with Bacillin rods
- d) Doderlein bacilli in the macrophage

Question 5:

Which of the following is a true association with HRT?

- a) Decreases risk of Colon cancer
- b) Decreased risk of breast cancer
- c) Decreased risk of endometrial cancer
- d) Increases risk of osteoporosis

Question 6:

All the following can result in non-immune hydrops except

- a) ABO incompatibility
- b) Parvovirus B19
- c) Chromosomal defect
- d) Alpha thalassemia major

Question 7:

In a 46XY female complete gonadal dysgenesis was noted. Complete gonadal dysgenesis is caused by what in the SRY gene?

- a) Point mutation
- b) Deletion of gene
- c) Translocation
- d) Inversion

Question 8:

Which of the following statements about primary dysmenorrhea is true?

- a) Pain begins after few hours after onset of periods and can last up to 48 hours
- b) It is dependent on psychological factors
- c) Pain begins 3 days prior to the onset of periods and is present upto the second day of the period
- d) GnRH antagonists are the first line of treatment

Question 9:

In which of the following statements will you suspect primary amenorrhea

- a) 1, 2, 3
- b) 1, 2, 3, 4
- c) 1, 2, 4
- d) 2, 3, 4

Question 10:

A 38-year-old multi gravida presents with 2 years of infertility and dyspareunia and dysmenorrhea. She has a retroverted uterus and tenderness in the pouch of Douglas. What would be the next step of management?

- a) Diagnostic laparoscopy
- b) Laparotomy
- c) HSG
- d) TVS

Question 11:

A 23-year-old female with 6 years of infertility, has a BMI of 33 and facial hair. Investigations reveal she has (PCOS) and spills from bilateral tubes. She and her husband are trying to conceive, what would be the best plan of management for this lady?

- a) Weight loss + Metformin + induction of ovulation with urinary gonadotrophins
- b) Weight loss + folic acid, Try for spontaneous induction, if failed move to induction of ovulation
- c) Metformin + finasteride + Induction of ovulation
- d) Weight loss + Metformin + Induction of ovulation with clomiphene

Question 12:

Which of the following drug is not given in endometriosis?

- a) Medroxyprogesterone acetate
- b) Cabergoline
- c) Estradiol
- d) Letrazole

Question 13:

Lady with 28 weeks of gestation presents with fever, acute pain abdomen, and mild leucocytosis. What is the most probable diagnosis?

- a) Uterine torsion
- b) Infection of fibroid
- c) Red degeneration of fibroid
- d) Preterm labor

Question 14:

Choose the incorrectly matched option.

- a) Uterine prolapse - Vaginal hysterectomy
- b) Vault prolapse - Sacrospinous ligament suspension
- c) Cystocele - Colposuspension
- d) Rectocele - Posterior colporrhaphy

Question 15:

Fructose is absent in the seminal fluid, which of the following statements is true regarding this:

- a) Congenital absence of seminal vesicles
- b) Obstruction in the ejaculatory duct
- c) Both
- d) None

Question 16:

A 15-year-old female presents with primary amenorrhea and signs of virilization. She has prepubertal development of breasts and pubic hair and is tanner stage 2. She has a blind-ended vagina and clitoromegaly. She has normal male testosterone levels. Her genotype is 46XY. What is the probable diagnosis?

- a) 5 alpha reductase deficiency
- b) 17 alpha hydroxylase deficiency
- c) Androgen insensitivity syndrome
- d) Swyer syndrome

Question 17:

Which one of the following is a virilising tumor of the ovary?

- a) Arrhenoblastoma
- b) Thecoma
- c) Brenners tumor
- d) Carcinoid tumor

Question 18:

A 28-year-old primigravida with a low-risk pregnancy presents at 39 weeks + 6 days with labor pains. Her vitals are stable. On PV, her cervix is 3 cm dilated and 60% effaced. The fetal head is at -3 station and membranes are felt as a bag of worms beneath the head. What is the appropriate course of management?

- a) Start oxytocin
- b) Rupture of membranes and confirm cord prolapse
- c) Emergency Caesarean section
- d) Wait and watch

Question 19:

A 20-year-old woman visits the OPD with complaints of irregular menses since menarche and increased facial hair. Her BMI is 29 kg/m² and her Ferriman-Gallwey score was 6. Her testosterone level was 0.36 ng/dl. Her mother is a diabetic on oral hypoglycemic agents. She should be advised the following, except?

- a) OCPs

- b) OCPs + Anti-androgens**
- c) Lifestyle modification**
- d) Counselling about risk of metabolic syndrome**

Question 20:

A 34-year-old woman at 30 weeks of gestation presents with complaints of bleeding PV and abdominal pain. She has drenched her clothes and has experienced decreased fetal movements. On examination, her cervix was 6cm dilated and 50% effaced. Her BP was 150/100mmHg, PR - 100 beats/min. Her abdomen corresponded to 34 weeks and she had a tense and tender uterus with regular contractions and no fetal heart sounds were heard. Her urine protein is 2+ and there is no active bleeding. What is the ideal management?

- a) Antihypertensives + ARM (Artificial Rupture of Membranes)**
- b) Antihypertensives + emergency caesarean section**
- c) Antihypertensives + tocolytics**
- d) Observation**

Question 21:

Which of the following vessels will serve as an alternate source of blood supply to prevent uterine ischaemia, in case the primary artery is ligated in the event of PPH?

- a) Ovarian artery**
- b) Uterine artery**
- c) Arcuate Artery**
- d) Round Ligament Artery**

Question 22:

Which of the following are responsible for increased blood volume during pregnancy?

- a) Action of estrogen and aldosterone**
- b) Increased fluid retention by the kidneys**
- c) Both A and B**
- d) Action of ADH and aldosterone**

Question 23:

Which of the following is not required to be a part of the consent for surgery in a patient who has an obstetric score of G6L1A3 with a history of two previous LSCS?

- a) Risk of bladder injury during surgery due to PAS
- b) Risk of bowel injury during surgery due to PAS
- c) Possible need for hysterectomy during surgery due to PAS
- d) Possible need for blood transfusion

Question 24:

A 30-year-old nulligravida is on DMPA and she comes 14 days past her due date for her next DMPA. What should be advised to her?

- a) Administer DMPA in the next cycle
- b) Take DMPA with back up for 7 days
- c) Ask to do UPT before taking the DMPA
- d) Take DMPA and no back up needed

Question 25:

A 67-year-old female is brought to the OPD in a poor general condition. On examination, she was found to have a right ovarian mass. Her CT scan indicated peritoneal caking. Her CA-125 levels were 1200 IU/L. What would be the management in this case?

- a) Debulking surgery followed by chemotherapy
- b) Chemotherapy followed by debulking surgery
- c) Debulking surgery followed by radiotherapy
- d) Staging laparotomy

Question 26:

Which of the following leads to an increased risk for borderline ovarian tumors?

- a) Polycystic Ovarian Syndrome (PCOS)
- b) Low parity
- c) Oral Contraceptive Pill (OCP) intake

- d) Functional ovarian cysts

Question 27:

A 28-year-old multigravida comes to the emergency at night with an inversion of the uterus following the delivery of the baby when the nurse waited till the delivery of the placenta. Her BP is 110/70 mmHg. Pulse rate 90 beats/min. What should be the management in this case?

- a) Manually remove placenta and then reposition the uterus
- b) Start oxytocin infusion
- c) Reposition the uterus without removing placenta
- d) Shift the patient to OT and reposition the uterus under anaesthesia

Question 28:

The recombinant form of vaccine to prevent cervical cancer has which of the following proteins?

- a) L1
- b) E6 and E7
- c) E1 and E2
- d) L2

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	a
6	a
7	a
8	b
9	c

10	d
11	b
12	c
13	c
14	c
15	c
16	a
17	a
18	c
19	b
20	a
21	a
22	c
23	b
24	d
25	b
26	b
27	c
28	a

Detailed Explanations

Solution to Question 1:

In the given clinical scenario history of vaginal discharge with no itching and foul smell along with normal lactobacilli count with acidic pH is suggestive of normal vaginal discharge.

Leucorrhea is defined as an excessive normal vaginal discharge.

The term leucorrhea should fulfil the following criteria:

- The excess secretion is evident from persistent vulval moistness or staining of the undergarments (brownish yellow on drying) or need to wear a vulval pad.
- It is nonpurulent and nonoffensive.
- It is nonirritant and never causes pruritus.

The physiologic basis involved in normal vaginal secretion is dependent on the endogenous estrogen level. With the rising estrogen level, there is abundant secretory activity of the endocervical glands and the superficial vaginal epithelium becomes rich in glycogen

The normal secretion is expected to increase in conditions when the estrogen levels become high. Such conditions are :

- During puberty—Increased levels of endogenous estrogen lead to marked overgrowth of the endo- cervical epithelium which may encroach onto the ectocervix producing congenital ectopy (erosion) → increased secretion.
- During menstrual cycle
- Around ovulation-Peak rise of estrogen → increase in secretory activity of the cervical glands.
- Premenstrual pelvic congestion and increased mucus secretion from the hypertrophied endometrial glands.
- Pregnancy-There is hyperestrinism with increased vascularity. This leads to increased vaginal transudate and cervical gland secretion.
- During sexual excitement, when there is an abundant secretion from the Bartholin's glands.

Other options:

Vaginal candidiasis: This condition generally presents with itching and a white, "cottage cheese"-like discharge. The absence of itching and the nature of the discharge described in the scenario rule out this option.

Bacterial vaginosis: This condition is characterized by a thin, gray, foul-smelling discharge. The fact that the discharge in this case does not have a foul smell rules out this option.

Chlamydial infection: Chlamydia infections can cause a range of symptoms, including abnormal vaginal discharge. However, many individuals with Chlamydia are asymptomatic, and the discharge associated with this condition does not typically stain underwear yellow.

Solution to Question 2:

The most common cause of puberty menorrhagia is anovulatory cycles.

Anovulatory cycles → unopposed estrogen → endometrial hyperplasia → prolonged and heavy periods.

Other causes

Endocrine dysfunction:

- Polycystic ovary syndrome (PCOS)
- Hypothyroidism or hyperthyroidism.

Hematological:

- Idiopathic thrombocytopenic purpura
- Von-Willebrand's disease leukaemia.

Pelvic tumors:

- Fibroid uterus
- Sarcoma botryoides
- Estrogen producing ovarian tumor

Pregnancy complications (abortion)

Treatment:

Reassurance and psychological support. Rest and correction of anemia is done. In refractory cases, progestogens, such as medroxyprogesterone acetate or norethisterone 5 mg thrice daily is given till bleeding stops. Usually, the bleeding is controlled within 3–7 days. Medication is continued for 21 days. The condition usually becomes normal following 2–3 courses and then normal cycles resume. In emergency, conjugated equine estrogen 20–40 mg IV is given every 6–8 hours. Once the bleeding is controlled, combined oral pills are started.

Regular menstrual cycle will be established once the hypothalamo-pituitary-ovarian axis is matured.

Solution to Question 3:

SRY region is present on the short arm of the Y chromosome (Yp11).

SRY stands for the sex-determining region on Y. It contains the SRY gene, which produces the protein called the testis determining factor. This causes the gonads to develop into the testes. If the SRY gene or Y chromosome is absent, the gonads develop as ovaries.

Solution to Question 4:

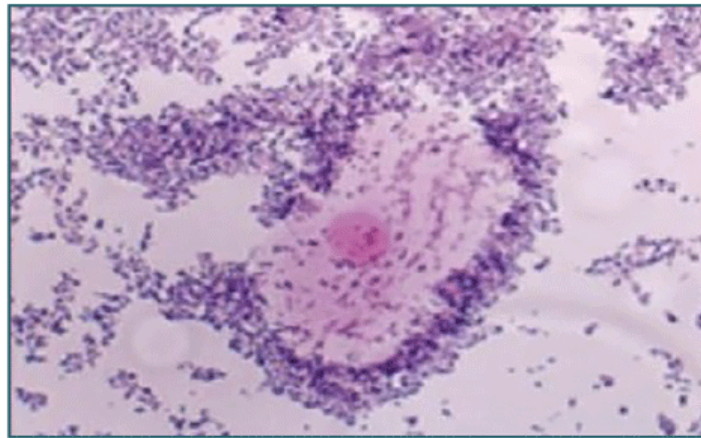
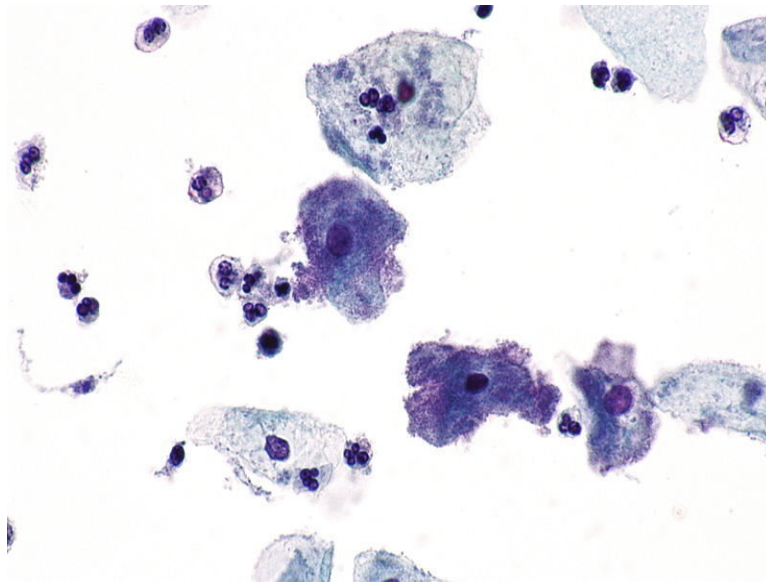
Clue cells are epithelial cells with a fuzzy border or granular cytoplasm caused by small gram-negative bacilli adhering on their surface is characteristic of bacterial vaginosis.

Bacterial vaginosis is a polymicrobial infection caused by *Mobiluncus curtisii*, *Gardnerella vaginalis*, *Mycoplasma hominis*, and *Haemophilus vaginalis*. It has a malodorous (fishy smell) vaginal discharge with minimal vulval irritation.

Amsel's Criteria is helpful in diagnosing bacterial vaginosis when 3 of the following 4 are present

- Thin, white homogenous vaginal discharge
- A pH of vaginal fluid >4.5
- Fishy odor of discharge accentuated by adding 10% KOH-Whiff test
- Presence of Clue cells on microscopy.

Treatment is oral metronidazole 500 mg BD for 7 days. Alternatively, Ampicillin 500 mg or cephalosporin 500 mg bid for 7 days is also effective.



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Solution to Question 5:

HRT decreases the risk of colon cancer.

Indications of HRT:

- Therapeutic
- Symptomatic women who suffer from estrogen deficiency
- Gonadal dysgenesis in adolescents
- Prophylactic
- High-risk cases for menopausal complications such as cardiovascular disease, osteoporosis (Option D), stroke, Alzheimer's disease, and colonic cancer

- Premature menopause, spontaneous or following surgery
- Menopause caused by radiotherapy and chemotherapy for cancer, especially alkylating agents
- Women demanding HRT as prophylaxis

Contraindications of HRT:

- Undiagnosed abnormal genital bleeding
- Known, suspected, or history of breast cancer
- Known or suspected estrogen-dependent neoplasia
- Active or prior venous thromboembolism
- Active or recent (e.g., within the past year) arterial thromboembolic disease
- Liver dysfunction or gallbladder disease
- Known hypersensitivity to the ingredients of the estrogen preparation
- Known or suspected pregnancy
- Uterine fibroids as the fibroids may enlarge in size.

Uses of HRT:

- Short term-hot flushes, vasomotor symptoms
- Dyspareunia, libido
- Urethral syndrome
- Long term-osteoporosis
- Cardiovascular disease
- Alzheimer's disease

Risks of HRT:

- Endometrial cancer(Option C)
- Breast cancer(Option B)
- Ovarian cancer
- Thromboembolism
- Lipid profile dysfunction
- Gall stones, liver dysfunction

Solution to Question 6:

ABO incompatibility is not a cause of nonimmune hydrops.

Hydrops is diagnosed by identifying two or more fetal effusions—pleural, pericardial, or ascites—or one effusion plus anasarca. Hydrops can be seen in association with red cell autoimmunization, which is termed immune hydrops. Otherwise, it is nonimmune. Pathogenesis of hydrops includes decreased colloid oncotic pressure, increased hydrostatic or central venous

pressure, and enhanced vascular permeability.

Etiology of hydrops fetalis

Cardiovascular

- Structural defects: Ebstein anomaly, tetralogy of Fallot with absent pulmonary valve, hypoplastic left or right heart, premature closure of ductus arteriosus, arteriovenous malformation (vein of Galen aneurysm)
- Cardiomyopathies
- Tachyarrhythmias
- Bradycardia, as may occur in heterotaxy syndrome with endocardial cushion defect or with anti-Ro/La antibodies

Chromosomal

- Turner syndrome (45,X), triploidy, trisomies 21, 18, and 13

Hematological

- Hemoglobinopathies, such as α_4 thalassemia
- Erythrocyte enzyme and membrane disorders
- Erythrocyte aplasia/dyserythropoiesis
- Decreased erythrocyte production (myeloproliferative disorders)
- Fetomaternal hemorrhage

Lymphatic Abnormalities

- Cystic hygroma, systemic lymphangiectasis, pulmonary lymphangiectasis

Infections

- Parvovirus B19, syphilis, cytomegalovirus, toxoplasmosis, rubella, enterovirus, varicella, herpes simplex, coxsackievirus, listeriosis, leptospirosis, Chagas disease, Lyme disease

Syndromic

- Arthrogryposis multiplex congenita, lethal multiple pterygium, congenital lymphedema, myotonic dystrophy type I, Neu-Laxova, Noonan, and Pena-Shokeir syndromes

Thoracic Abnormalities

- Cystic adenomatoid malformation
- Pulmonary sequestration
- Diaphragmatic hernia
- Hydro/chylothorax
- Congenital high airway obstruction sequence (CHAOS)
- Mediastinal tumors
- Skeletal dysplasia with very small thorax

Gastrointestinal

- Meconium peritonitis, gastrointestinal tract obstruction

Kidney and Urinary Tract

- Kidney malformations
- Bladder outlet obstructions
- Congenital (Finnish) nephrosis, Bartter syndrome, mesoblastic nephroma

Placental, Twin, and Cord Abnormalities

- Placental chorioangioma, twin-twin transfusion syndrome, twin reversed arterial perfusion sequence, twin anemia polycythemia sequence, cord vessel thrombosis

Other Rare Disorders

- Inborn errors of metabolism: Gaucher disease, galactosialidosis, GM1 gangliosidosis, sialidosis, mucopolysaccharidoses, mucopolipidoses
- Tumors: sacrococcygeal teratoma, hemangioendothelioma with Kasabach-Merritt syndrome

Idiopathic

Solution to Question 7:

Amongst the given options, a point mutation in the SRY gene is most likely to result in complete gonadal dysgenesis but this answer remains controversial as some other sources mention SRY deletion is more commonly associated. The most common cause attributable to about 85% of cases is still idiopathic.

Pure gonadal dysgenesis results from a point mutation in SRY/ SRY deletion or point mutations in another gene with testis-determining effects (DAXJ, SF-1, CBX2). This leads to underdeveloped dysgenetic gonads that fail to produce androgens or AMH. This is characterized by a normal prepubertal female phenotype and a normal mullerian system due to absent AMH.

The child usually presents with failure in the development of secondary sexual characteristics at puberty and amenorrhea. Investigations reveal the absence of estrogen production and they have elevated levels of FSH (>40 mIU/mL) and low oestradiol levels (<25 pg/mL).

Women with streak ovaries are infertile, but they can bear children with oocyte donation. They must be treated with cyclic estrogen and progestogen to promote feminization and secondary sexual characteristics and prevent osteoporosis.

Solution to Question 8:

Primary dysmenorrhea is dependent on psychological factors is the true statement.

Dysmenorrhea is defined as cramping pain that occurs during menstruation. Low social support, alexithymia, neuroticism trait, long menstrual bleeding, family history of dysmenorrhea, and high-caffeine diet are important risk factors for women with primary dysmenorrhea. It can be classified as:

A. Etiological:

- Primary dysmenorrhea is a type of dysmenorrhoea that is not associated with any pelvic pathology. It affects women in the age group of 18 to 25 years, with ovulatory cycles. It begins just before or with the onset of menstruation. (Not a few hours after as in Option A or not 3 days prior as in option C)
- Secondary (Acquired) dysmenorrhea is associated with the presence of pelvic pathologies such as fibroids, adenomyosis, pelvic inflammatory disease (PID), and endometriosis.

B. Clinical:

- Spasmodic dysmenorrhea is the most common and is characterized by cramping pain on the first and second days of menstruation.
- Congestive dysmenorrhea presents as worsening pelvic discomfort and pelvic pain, a few days before the onset of menses followed by relief of the symptoms. This is seen in PID, IUCD users, varicosities of pelvic veins, pelvic endometriosis, and fibroids.
- Membranous dysmenorrhea is rare and is characterized by crampy pain that occurs with the passage of the endometrium as a cast at the time of menstruation.

Primary dysmenorrhea can be managed with non-steroidal anti-inflammatory drugs (NSAIDs), combined oral contraceptive pills, (Option D), and progestin-only pills. GnRH agonists and androgens are also useful. Surgical management includes hysterectomy and is indicated if not responding to medical management. In secondary dysmenorrhea, treating the cause is needed.

Solution to Question 9:

Primary amenorrhea is suspected in patients who have not menstruated by age 15 or within 3 years of thelarche or has not menstruated by age 14 and shows signs of hirsutism, excessive exercise, or eating disorder. The conventional definition of primary amenorrhea is a failure of the onset of menstruation beyond 16 years of age or failure to attain secondary sexual characteristics by 13 years of age.

Secondary amenorrhea is absence of menstruation for 3 months in a female who was previously menstruating normally.

Causes of primary amenorrhea:

- Hypogonadotropic hypogonadism
- Hypothalamic and pituitary dysfunction
- Kallmann's syndrome
- CNS tumors - craniopharyngioma
- Delayed puberty
- Hypergonadotropic hypogonadism
- Primary ovarian failure
- Resistant ovarian failure
- 17-alpha-hydroxylase deficiency

- Chromosomal abnormalities
- Turner's syndrome (45 X)
- Pure gonadal dysgenesis (46 XX or 46 XY)
- Androgen insensitivity syndrome (testicular feminization syndrome) (46 XY)
- Developmental defect of the genital tract
- Imperforate hymen
- Transverse vaginal septum
- Vaginal atresia/cervical atresia
- Absent uterus in Mullerian agenesis (Mayer-Rokitansky-Küster-Hauser syndrome)
- Others:
- Uterine synechiae (congenital)
- Adrenogenital syndrome
- Cretinism
- Juvenile diabetes
- Malnutrition
- Anemia
- Weight loss
- Tuberculosis

Solution to Question 10:

This patient presenting with symptoms of dyspareunia and dysmenorrhea with a retroverted uterus and tenderness in the pouch of Douglas is most likely suffering from endometriosis. The next best step in management is to perform TVS. Laparoscopy is the gold standard for the diagnosis of endometriosis but it is not done routinely for diagnostic purposes.

Endometriosis is a condition that is characterized by the presence of endometrial tissue in ectopic sites such as:

- Ovary (chocolate cysts- left/right lower quadrant pain)
- Pouch of Douglas (pain during defecation)
- Uterosacral ligaments (chronic pelvic pain & infertility)
- Appendix (right lower quadrant pain)
- Lower segment cesarean section (LSCS) scar endometriosis
- Umbilicus
- Lung (cyclical hemoptysis)

These extra-uterine sites of endometrial tissue are influenced by cyclical changes in the hormone levels which give rise to location-specific symptoms as mentioned above.

Management:

Investigations include:

- USG: TVS will show a chocolate cyst.
- MRI
- CA-125 more than 35 U/mL
- Laparoscopy: Gold standard for the diagnosis and treatment of endometriosis. It is used to locate the sites, take a biopsy and surgically remove the deposits.

Drugs used for symptom control include the following:

- Oral contraceptive pills
- Mirena IUD
- GnRH analogs
- Letrozole

Surgical management includes:

Minimally invasive surgery may be used to destroy the tissue by cautery, laser vaporization, and lyse adhesions that have formed as a complication of endometriosis.

Major surgical measures include incision of chocolate cyst and removal of lining, salpingo-oophorectomy, hysterectomy, and unilateral or bilateral salpingo-oophorectomy.

Solution to Question 11:

In the given clinical scenario the most appropriate management would be weight loss + folic acid, try for spontaneous induction, if failed move to induction of ovulation. In obese PCOS patients, even modest weight loss (5% of body weight) can help in the restoration of ovulatory cycles in some cases. Hence spontaneous induction is tried. Folic acid is normally administered as prophylaxis before conception to prevent neural tube defects.

Polycystic ovarian syndrome

Polycystic ovarian syndrome (PCOS)/ Polycystic ovarian disease (PCOD)/ Stein-Leventhal syndrome is a multisystem disorder. It involves metabolic and endocrine dysfunctions that commonly occur among women in the reproductive age group. It has multifactorial causation, such as lifestyle changes, stress, diet, obesity, and hyperinsulinemic states.

Important clinical features of PCOS are Menstrual abnormalities (70%) - oligomenorrhea, amenorrhea, obesity, hirsutism, acne, and alopecia due to hyperandrogenism, acanthosis nigricans due to insulin resistance, and infertility.

Rotterdam criterion is used to diagnose PCOS. The presence of any two of the three components of this criteria establishes the diagnosis of PCOS.

- Oligo/ amenorrhoea, anovulation, infertility

- Hirsutism/ acne
- Ultrasound findings-
-

Enlarged ovaries with ovarian volume will be more than 10 mL

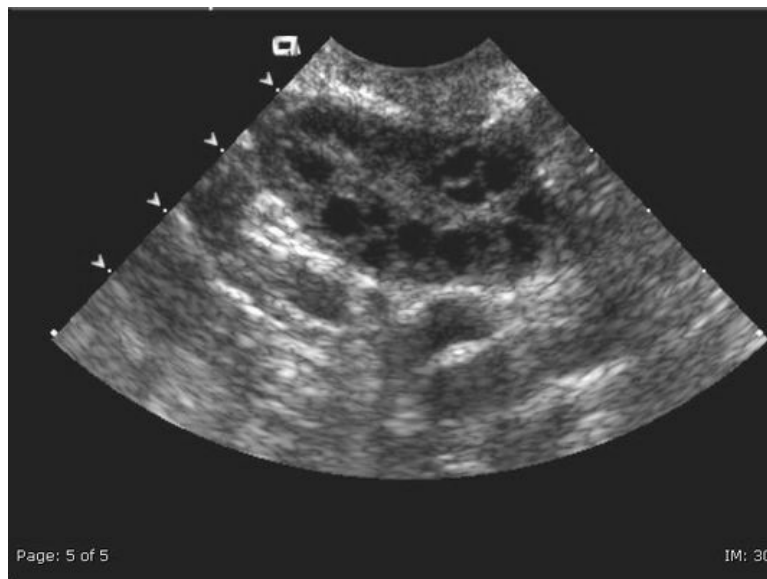
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12 or more small follicles each of 2-9 mm in size placed peripherally.

Investigations

- Hormonal studies
- USG is diagnostic of PCOS - it shows enlarged ovaries with multiple follicles in the periphery. It also rules out other pathologies like ovarian tumors, endometrial hyperplasia.
- Laparoscopy (therapeutic). It reveals bilateral enlarged ovarian cysts.

The ultrasound image below shows an enlarged ovary with multiple follicles in the periphery.



Management of PCOD is individualized. Weight reduction and lifestyle changes are the first-line therapy for PCOD. Oral contraceptive pills (OCP), preferably fourth-generation pills with 30 mcg E2 and 2-3 mg drospirenone are used to treat menstrual abnormalities. Hirsutism and acne are managed with OCP with cyproterone acetate and topical agents such as epiornithine, clindamycin lotion 1%, erythromycin gel 2%, and isotretinoin.

Patients with PCOD are more prone to developing insulin resistance which can be controlled with metformin (Option A, C and D) and acarbose. Patients presenting with infertility are treated by inducing ovulation using clomiphene as a first-line drug, along with dexamethasone and letrozole.

Surgical management includes laparoscopic ovarian drilling or puncture of not more than four cysts in each ovary.

Note:

PCOS in association with diabetes, hypertension, hyperlipidemia and cardiovascular disease presenting in reproductive age group is known as "X syndrome".

Solution to Question 12:

Estradiol is not used in the treatment of endometriosis as estrogen administration leads to further development of endometriotic lesions.

Treatment of endometriosis:

1. Medical management:

For minimal disease, the drug of choice is NSAIDs. Oral contraceptive pills are given if associated menstrual complaints are present. OCPs inhibit gonadotropin release and decrease the menstrual flow.

For moderate to severe diseases, oral progestogens are the first drugs to be started. They antagonize estrogenic effects on the endometrium, causing endometrial atrophy. The drug of choice is GnRH agonist.

Endometriosis requires angiogenesis. Cabergoline, a vascular endothelial growth factor pathway inhibitor, is an effective therapeutic option for women with chronic pain due to endometriosis. (Option B)

Mnemonic to remember drugs used: GOLD GAP

- G: GnRH analog (Leuprolide)
- O: Oral contraceptive pills
- L: Letrozole (Option D)
- D: Danazol
- G: Gestrinone
- A: Antiprogestone-Mifepristone
- P: Progesterone (Medroxyprogesterone acetate) (Option A)

2. Surgical management:

Primary surgery is preferred in infertile women because hormonal treatment would delay pregnancy.

- Conservative surgery: Endometrial deposits are destroyed by electrocautery/laser.
- Laparoscopic uterosacral nerve ablation (LUNA)
- Radical surgery: Total hysterectomy with bilateral oophorectomy.

Endometriosis is the presence of endometrial tissue (glands and stroma) outside the uterus.

Risk factors for the development of endometriosis

- Family history
- Nulliparity
- Prolonged exposure to endogenous estrogen (early menarche, late menopause)
- Estrogen therapy

- In utero exposure to diethylstilbesterol (DES)

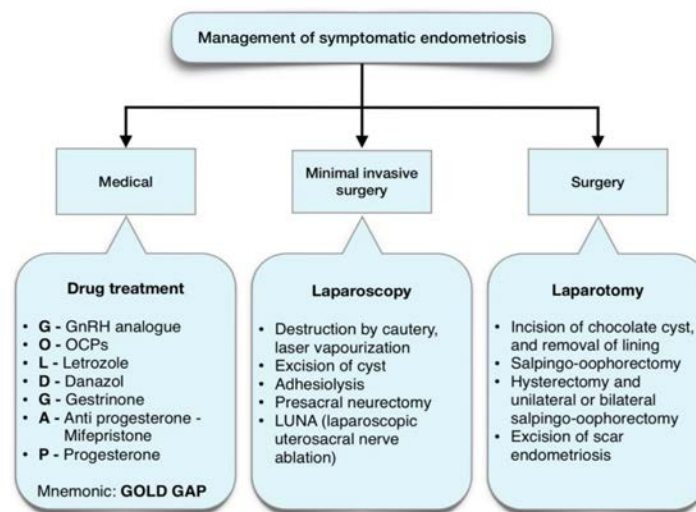
Key clinical features

- Dysmenorrhea
- Lower abdominal pain
- Infertility
- Pain during urination or defecation
- Constipation or diarrhea
- Irregular bleeding

The gold standard for the diagnosis of endometriosis has been a visual inspection by laparoscopy, preferably with histological confirmation. Laparoscopy helps detect and diagnose pelvic endometriosis, locate the site of endometriosis and staging, take a biopsy, and perform surgical ablation as a part of treatment.

Other investigations:

- CA-125 levels: raised in endometriosis.
- Monocyte chemotactic protein (MCP-1) levels: raised in the peritoneal fluid
- MRI
- Color Doppler
- Anti-endometrial antibodies



Solution to Question 13:

In the given scenario history of acute pain abdomen with fever and mild leukocytosis in a 28-week pregnant woman is suggestive of red degeneration of fibroid.

Red degeneration is the most common complication of fibroids in pregnancy. It is due to the breakdown of the fibroid resulting in ischemia and pain. Key clinical features in this condition are:

- Low-grade fever
- Tender uterus on palpation
- Elevated leukocytes
- Elevated ESR

The diagnosis is confirmed with an ultrasound. It is usually self-limiting and can be managed with non-steroidal anti-inflammatory drugs.

Other options

Option A - Uterine torsion is defined as rotation of more than 45 degrees around the long axis of the uterus. The most usual symptoms of uterine torsion are birth obstruction, abdominal pain, vaginal bleeding, shock, and urinary and intestinal symptoms. The treatment in the earlier months of pregnancy is immediate laparotomy and detorsion of the uterus. Near-term or during labor cesarean section is carried out, and elimination of the possible etiologic factors.

Option B - Infection is common in submucous and myomatous polyps if they project into the cervical canal or the vagina. An infected polyp can cause blood-stained purulent discharge. Infection is more likely in the postpartum and postabortal state.

Option D -Preterm labor is diagnosed with the help of the following criteria

- Regular uterine contractions with or without pain (at least one in every 10 minutes)
- Dilatation ($\geq$ 2 cm) and effacement (80%) of the cervix
- Length of the cervix (measured by TVS) $\leq$ 2.5 cm and funneling of the internal os
- Pelvic pressure, backache and/or vaginal discharge or bleeding.

Solution to Question 14:

The incorrectly matched option is Cystocele-colposuspension. The surgery performed for cystocele is anterior colporrhaphy.

Various surgeries were performed for the management of uterovaginal prolapse:

- Anterior colporrhaphy: This is done for cystocele or cystourethrocele.
- Posterior colporrhaphy/ colpoperineorrhaphy: This is done for posterior defects like rectocele or rectal defects with perineal laxity.
- Le Forte's repair: This is preferred only in elderly women with advanced prolapse, or those unfit for surgery. Women wanting an active sexual life are poor candidates for this procedure as it creates an adhesive area in the midline.
- Abdominal sling surgeries: Young women desirous of retaining menstrual and reproductive functions are ideal candidates. These surgeries use tapes as slings to prevent organ descent. These include:

- Abdominal wall cervicopexy (modified by Purandare and Mathre) - sling attached to cervix posteriorly near uterosacral ligaments.
- Shirodkar's sling surgery - the sling fixes the cervix to the lumbosacral fascia.
- Khanna's sling surgery - sling fixed to the anterior superior iliac spine, strengthening the Mackenrodt's ligament.
- Fothergill surgery (Manchester operation): An anterior colporrhaphy is combined with amputation of the cervix. The Mackenrodt ligaments are then sutured in front of the cervix followed by a colpoperineorrhaphy. It is suitable for women <40 years desiring reproductive function, but fertility may be reduced.
- Ward-Mayo surgery: This is a combination of vaginal hysterectomy + pelvic floor repair. It is done in women >40 years, not desiring reproductive function.
- Moschowitz surgery: This is performed for enterocele repair where the pouch of Douglas is obliterated.

Management of Vault prolapse:

- If the patient is fit for abdominal surgery: Transabdominal sacral colpopexy, i.e., suspending the vault to the sacral promontory extraperitoneally. It is best suited for younger women since coital difficulty following vaginal surgery is avoided
- If the patient is not fit for abdominal surgery: Right Transvaginal sacrospinous colpopexy. It is usually preferred in obese and elderly women, who are not fit for abdominal surgery.
- Colpocleisis is permissible if sexual activity is not desired

Solution to Question 15:

Fructose is absent in seminal fluid in case of congenital absence of seminal vesicle and obstruction of the ejaculatory duct.

The ejaculatory duct is formed by the fusion of the ductus deferens and the duct of the seminal vesicle. A major portion of the semen volume is contributed by the seminal vesicles. Seminal fluid is alkaline and protects sperm from acidity in prostatic secretions and in the vagina. Seminal fluid also provides fructose as an energy source for sperm. If seminal vesicle is absent or if there is an obstruction at the level of the ejaculatory duct then semen analysis will reveal low ejaculate volume, oligospermia or azospermia, low pH, and absence of fructose.

Solution to Question 16:

The clinical scenario points towards the diagnosis of 5 alpha-reductase deficiency.

The 5 alpha-reductase enzyme converts testosterone into a more potent form of the male sex hormone called dihydrotestosterone (DHT). Testosterone is required for the development of male internal genitalia, whereas DHT has a role in the development of male external genitalia. At puberty, testosterone is responsible for aggressive psychosexual behavior, deepening of the voice, increased muscle mass, and initiating spermatogenesis.

In 5 alpha reductase deficiency, there is a failure of conversion of testosterone to dihydrotestosterone. As testosterone levels are normal there is a normal development of Wolffian ducts and thus male internal genitalia are normal. However, some cases may have a blind-ending vagina.

As levels of DHT are low the child will have a female-looking external genitalia at birth. These children tend to be raised as females till puberty, by which they start exhibiting virilization which is caused by testosterone. Hence clitoromegaly is seen. There is a lack of breast development as testosterone is not converted to estrogen. Pubic hair is scanty as its development depends on both testosterone and dihydrotestosterone.

Investigations reveal an increase in the ratio of testosterone to DHT after hCG administration.

If the child wants to be raised as a female, then the testes have to be removed before the child attains puberty and virilization of the child. Estrogen therapy that is appropriate for inducing and maintaining feminization should be started at the time of usual pubertal maturation or immediately after gonadectomy is performed in an adult. Surgical correction in the form of vaginoplasty is performed. If the child is raised as a male corrective procedure include hypospadias correction, correction of chordee, and reconstruction of the urethra.

Other options

Option B: 17 alpha-hydroxylase deficiency results in decreased production of testosterone by adrenal glands. However, there will be normal testosterone production from the testes. Hence depending on the testosterone and dihydrotestosterone levels the child may have female-looking external genitalia or ambiguous genitalia at birth. 17 alpha-hydroxylase deficiency also results in a decrease in 17-OH progesterone levels and cortisol levels. Hence there is an increase in ACTH due to the loss of negative feedback which increases aldosterone. Due to an increase in aldosterone, the child will have hypertension and hypokalemia.

Option C: Androgen insensitivity syndrome is an X-linked recessive disorder in which the androgen receptors are absent, scanty, or fail to respond to testosterone. Hence there will be no signs of virilization. The serum testosterone levels will be elevated. There will be normal breast development as testosterone is converted to estrogen.

Option D: Swyer syndrome results from a mutation in the SRY gene on the Y chromosome. This results in dysgenetic gonads. There is no production of AMH and testosterone and hence there is development of female internal and external genitalia. There will be no signs of virilization.

Solution to Question 17:

Arrhenoblastoma (Sertoli-Leydig cell tumor) of the ovary is a virilizing tumor of the ovary.

Sertoli-Leydig cell tumors are a type of sex-cord stromal ovarian tumors. It is most commonly seen in women of reproductive age group. These produce excess androgens and are hence known as virilizing tumors. Menstrual irregularities and signs of virilization like hirsutism and deepening of the voice are common presenting complaints. Microscopy reveals Sertoli cells and Leydig cells.

Other options:

Option B: Thecomas typically produce estrogen (female sex hormone). Excess estrogen production can lead to symptoms such as abnormal uterine bleeding, endometrial hyperplasia, or

endometrial cancer.

Option C: Brenner tumor (transitional cell tumor) is a benign, epithelial, unilateral tumor, resembling a fibroma. Histologically, Walthard cell rests and coffee bean nuclei are seen. Brenner tumors represent only 2-3% of all ovarian neoplasms, and <2% of these tumors are malignant.

Option D: In carcinoid tumors, excessive levels of serotonin is produced which causes flushing, diarrhea, wheezing, damage to the tricuspid valve, etc. The levels of 5-OH indole acetic acid (serotonin metabolite) are elevated in the urine.

Solution to Question 18:

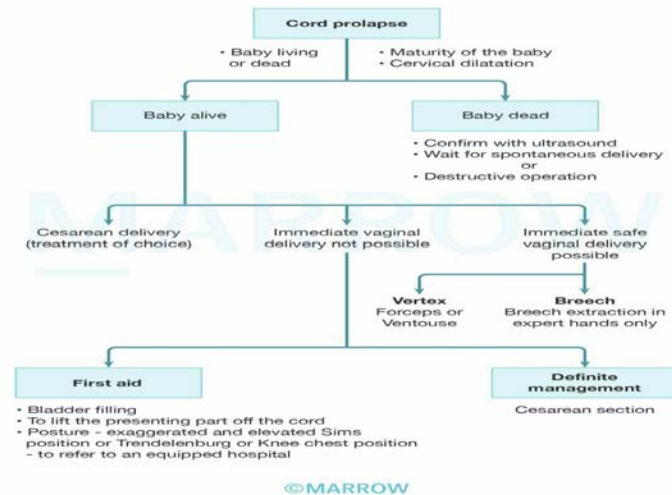
In this patient, the membranes are intact and the 'bag of worms' palpated refers to the umbilical cord within the membrane, suggestive of cord presentation. The best course of management would be to take the patient up for an emergency caesarean section.

Cord presentation refers to the presence of the umbilical cord below the presenting part over the internal os, within an intact bag of membranes, whereas in the case of a cord prolapse, the membranes are ruptured and the cord lies externally. The goal of management in the event of a cord presentation is to expedite delivery while keeping the membranes intact, and hence the best option would be to deliver the fetus via emergency LSCS. While the patient is being prepared for surgery, she may be positioned in the exaggerated Sim's position (shown below) to avoid cord compression.



In comparison, the management algorithm of cord prolapse is based on three important factors, which are: living or dead baby, maturity of the baby, degree of cervical dilatation, according to which the following steps are taken:

Scheme of Management of Cord Prolapse



Other options:

Option A: Oxytocin should not be started as the patient should not be allowed to proceed to deliver via normal vaginal delivery due to the risk of cord prolapse and fetal death.

Option B: The membranes should not be ruptured as this would lead to an overt cord prolapse and complicate the situation further.

Option D: Cord presentation is an obstetric emergency requiring immediate action.

Solution to Question 19:

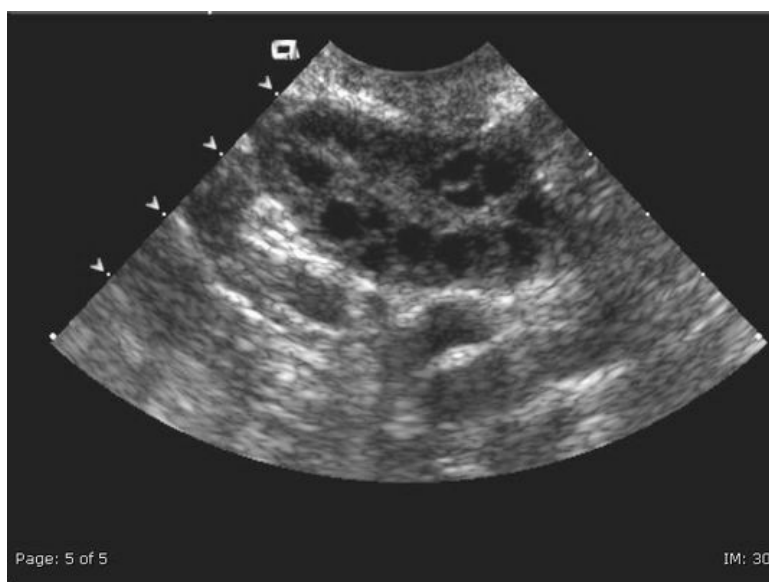
This patient's history of irregular menstrual cycles and increased facial hair and hirsutism as per the Ferriman-Gellway score are suggestive of polycystic ovarian syndrome (PCOS). Anti-androgens are not recommended in the initial therapy of PCOS.

Hormonal therapy in PCOS is initiated with combined OCPs which are useful in treating menstrual abnormalities, reducing acne, and hirsutism. Anti-androgens are recommended in PCOS only if patients have no improvement in hirsutism after 6 months of therapy with OCPs alone. Further, a serum testosterone level of 0.36 ng/dl is within the normal range (0.2-0.8 ng/dl) there is no current indication to begin anti-androgen therapy in this patient.

PCOS is an endocrinopathy seen in women of reproductive age, affecting multiple systems. Rotterdam criteria is used to diagnose PCOD. The presence of two of the three components of this criteria establishes the diagnosis:

- Oligo/amenorrhoea, anovulation, infertility
- Hirsutism/acne
- Ultrasound findings:
 - Enlarged ovaries with ovarian volume will be more than 10 mm³
 - 12 or more small follicles each of 2-9 mm in size placed peripherally

Ultrasound image given below shows an enlarged ovary seen in PCOS with multiple follicles in the periphery.



Patients with PCOS are at risk of developing metabolic syndrome (Syndrome X), characterised by abdominal obesity, high blood pressure, impaired fasting glucose, high triglyceride levels, and low HDL cholesterol levels.

Treatment of PCOD:

- Lifestyle changes (Option C) such as weight reduction, smoking cessation, are recommended and most important in preventing metabolic syndrome (Option D).
- Fourth-generation oral contraceptive pills (OCP) (Option A) containing ethinyl estradiol and drospirenone are used for treating menstrual abnormalities, reducing acne, and hirsutism.
- OCP with cyproterone acetate is prescribed for PCOS patients with hirsutism.
- Patients with PCOD are prone to developing insulin resistance, which can be controlled with metformin. Acarbose can be used as an alternative if metformin is contraindicated.
- Infertility in PCOD patients is treated with clomiphene as the first-line drug for inducing ovulation. Dexamethasone is used in combination to improve fertility rates. Letrozole is used off-label in clomiphene-resistant cases.

Surgical options include laparoscopic ovarian drilling or puncture of not more than four cysts in each ovary. This is done either by laser or by unipolar electrocautery. It restores endocrine levels and improves fertility. Surgery is indicated only if:

- Medical therapy fails
- Hyperstimulation occurs
- Infertile women
- Previous pregnancy losses

Solution to Question 20:

The clinical scenario of painful antepartum haemorrhage associated with pre-eclampsia (hypertension + proteinuria) and fetal death is suggestive of Grade 2 abruptio placentae. In this case, as there is fetal death and the patient is in labor, the best approach would be to administer antihypertensives and deliver the fetus vaginally.

Abruptio placenta is the partial/total placental separation from its implantation site before delivery. The bleeding may be:

- Revealed - blood dissects downwards between membranes and uterine wall to escape through the cervix
- Concealed - blood accumulates between the detached placenta and uterus, and the uterus is larger than expected for the period of gestation
- Mixed - features of both co-existing

Hypertension in pregnancy is the most important risk factor for abruptio placenta. Spasm of the decidual spiral artery leads to endothelial damage and extravasation of blood in the placental bed, leading to fetal distress. Patients often present with sudden onset abdominal pain, vaginal bleeding, uterine tenderness, and frequent contractions. Based on the degree of placental separation, patients are clinically graded as follows:

- Grade 0 – Asymptomatic, abruption is diagnosed only after delivery by the presence of retroplacental clots.
- Grade 1 – Vaginal bleeding, mild uterine tetani. No maternal and fetal complications.
- Grade 2 – With fetal distress or death
- Grade 3 – With maternal complications like shock, renal failure, coagulopathy, or death.

The management of abruption is based on whether the patient is in labor and the fetal status, with the goal being to expedite delivery and prevent complications such as shock and coagulopathy. Immediate management begins with the administration of IV crystalloids and arrangement of blood for the patient, along with BP control.

- If the patient is in labor and the cervix is favourable, in case the fetal heart rate is reassuring, or in case the fetal has died - amniotomy and vaginal delivery.
- If the patient is not in labor or fetal heart tracing indicate fetal distress - an emergency caesarean section is indicated. (Option B)

Other Options:

Options C and D: Abruption is considered an obstetric emergency and hence tocolytics or expectant management have no place in this situation.

Solution to Question 21:

In the event of postpartum haemorrhage (PPH), bleeding occurs from the uterine artery which is the primary vessel ligated, and the inflow from the ovarian artery compensates to prevent ischaemia.

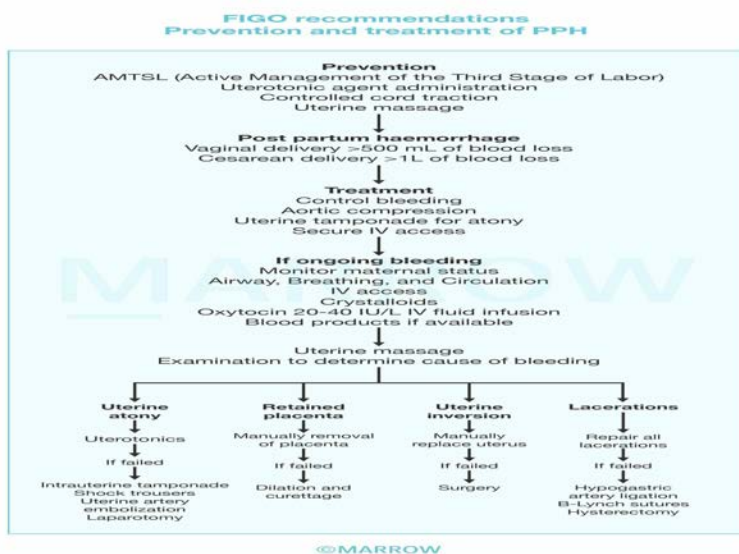
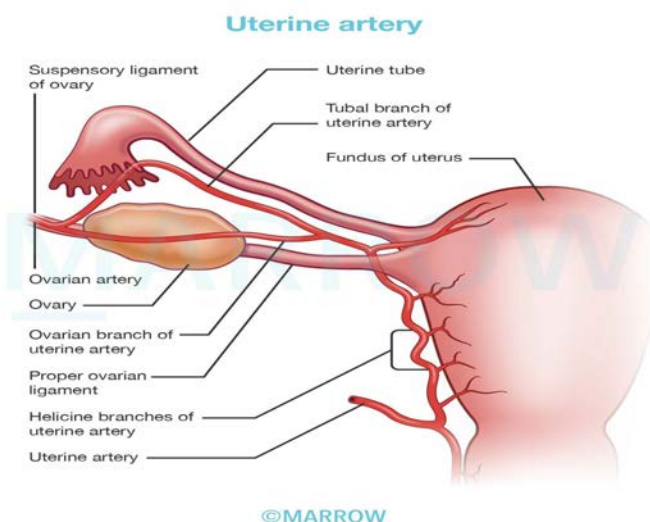
The arterial supply to the uterus is by two main arteries: the uterine artery, which is a branch of the anterior division of the internal iliac artery, and the ovarian artery which is a direct branch of the abdominal aorta. Ligating the uterine artery limits the supply from the internal iliac artery and hence the ovarian artery serves as the alternate blood supply.

The uterine artery runs from its origin towards the cervix, crosses the ureter and then ascends with a tortuous course along the lateral aspect of the uterus up to ovary, where it anastomoses with the ovarian artery. It is based on this anatomical relationship that the order of ligation of vessels to arrest bleeding in atonic postpartum haemorrhage is the uterine artery > ovarian artery > internal iliac artery.

Other Options:

Option C: The arcuate artery is a branch of the uterine artery itself.

Option D: The artery to the round ligament artery is a branch from the ovarian artery.



Solution to Question 22:

Increased blood volume during pregnancy is due to estrogen and aldosterone, both of which cause increased fluid retention by the kidneys, hence both options A and B are correct.

Increased estrogen during pregnancy directly causes fluid retention and also activates the RAAS pathway causing increased aldosterone, leading to fluid retention by the kidneys. Although there is a reset in the threshold for response to ADH, the levels of ADH remain unchanged.

Although all components of blood are increased during pregnancy, plasma volume is increased more than the RBC volume, leading to a relative hemodilution, responsible for what is termed as physiological anaemia of pregnancy. The increased plasma volume is beneficial as it reduces blood viscosity, allowing for effective gaseous exchange in placental and fetal circulation, it is protective against the possible adverse effects of change in posture causing vascular compression, and also against the possible blood loss during delivery.

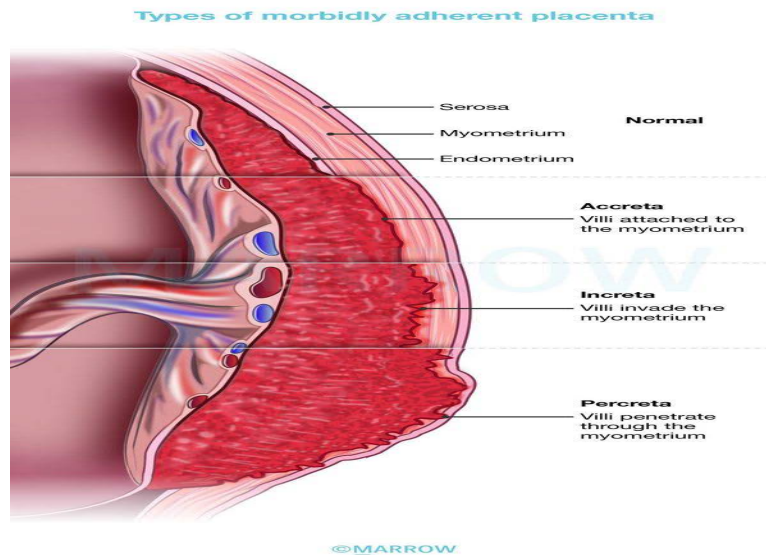
Solution to Question 23:

A patient with a history of two previous caesarean deliveries would be at a high risk of having a morbidly adherent placenta or placenta accreta spectrum (PAS). However, bowel injury is an unlikely complication of placenta accreta spectrum (PAS) and hence may not be required as part of the consent for surgery in such a patient.

The placenta accreta spectrum is characterized by abnormally implanted, invasive, or adhered placenta. This occurs due to partial or complete absence of the decidua basalis and imperfect development of the Nitabuch layer. The major risk factors include associated placenta previa and previous cesarean delivery as seen in this patient.

There are three variants of this condition based on the depth of trophoblastic growth.

- Placenta accreta, the villi are attached to the myometrium.
- Placenta increta, the villi invade into the myometrium.
- Placenta percreta, the villi penetrate the myometrium into the serosa, with possible adherence to the bladder wall



PAS can be diagnosed in the antepartum period by ultrasound which will reveal loss of normal hypoechoic retroplacental zone between the placenta and uterus, placental vascular lacunae, bridging vessels, and placental bulging leading to distortion of the contour of the uterus. MRI can be performed when the ultrasound is inconclusive. However, ultrasound remains the modality of choice for diagnosis.

Patients with PAS have a significant risk of haemorrhage and shock. Management of this condition therefore involves appropriate multidisciplinary efforts with adequate surgical, anesthetic, intensive care, and blood banking capabilities. Patients are recommended scheduled delivery between 34 0/7 to 35 6/7 weeks of gestation and the risk of hemorrhage necessitating paripartum hysterectomy and blood transfusion, or bladder injury in percreta must be properly communicated.

Solution to Question 24:

DMPA can be given as a repeat dose up to 4 weeks later from the scheduled date (within grace period) after the previous dose. But if the patient is presenting later than that, she should be given a urine pregnancy test before the next dose of DMPA.

DMPA stands for Depot Medroxy Progesterone Acetate. It is an injectable progesterone formulation available for contraception. It acts by:

- Bringing changes in the cervical mucus
- Making the endometrium hostile
- Preventing ovulation

Disadvantages include spotting and intermediate per vaginal bleeding.

DMPA guidelines:

- Injections to be provided every 3 months
- It can be given up to 2 weeks later without any other contraception.

- DMPA can be given as a repeat dose up to up to 4 weeks later from the scheduled date (within grace period) after the previous dose.



Solution to Question 25:

In the given scenario, the patient presents with poor general condition and the presence of peritoneal metastases is suggestive of Stage III ovarian cancer. The preferred management would be to initiate neoadjuvant chemotherapy with three cycles of paclitaxel + carboplatin over three weeks, followed by debulking/cytoreductive surgery, and then continue another three cycles of chemotherapy.

Epithelial ovarian cancers are treated with a multimodality approach. The usual approach in Stage I and II disease where the patient is fit for surgery, is to perform cytoreductive surgery, which also helps in determining the extent of disease, followed by adjuvant chemotherapy (Option A).

However, patients with advanced disease and poor general condition are unfit for anaesthesia and surgery at the time of presentation and have a higher operative risk, hence neoadjuvant chemotherapy prior to surgery is preferred and has been shown to reduce morbidity in such patients.

Other options:

Option C: Ovarian tumors are not radiosensitive and radiotherapy does not form part of the protocol for treatment of ovarian cancers.

Option D: Staging laparotomy is the term used to describe the procedure of surgically exploring patients with early-stage disease and would not be the approach in a patient with advanced disease in whom chemotherapy followed by debulking surgery is indicated.

Solution to Question 26:

Low parity is a risk factor for the development of borderline ovarian tumors.

Borderline ovarian tumors are ovarian epithelial tumors of low malignant potential, defined histologically by atypical epithelial proliferation without stromal invasion. They are classified as either serous or mucinous. The following are the risk factors for their development:

- low parity / unopposed estrogen
- infertility
- obesity
- smoking

Clinically patients present with symptoms similar to that of a benign ovarian tumor such as a mass per abdomen, abdominal pain and alteration in menstrual cycles. Patients require similar investigations but the diagnosis of borderline ovarian tumor is based on biopsy. Conservative surgery is usually the preferred treatment approach especially in patients who wish to retain fertility. Appendectomy is usually recommended in the mucinous variety to prevent pseudomyxoma peritonei. Patients do not require chemotherapy or radiotherapy for treatment but these tumours tend to recur and hence regular follow-up is advised.

Solution to Question 27:

This patient with a uterine inversion with stable vitals should undergo manual reposition of the uterus without removing the placenta

Uterine inversion occurs when there is excessive traction leading to the loss of the uterine convexity and protrusion of the uterus outside the vaginal introitus.

Risk factors for uterine inversion:

- Fundal attachment of the placenta
- Uterine atony
- Attempts at cord traction before placental delivery

- Adherent placenta

There are 3 degrees of uterine inversion:

- First-degree - the fundus of the uterus inverts into the uterine cavity
- Second-degree - the uterine fundus protrudes through the cervix and lies in the vagina
- Third-degree - the whole of the uterus is inverted and protrudes through the introitus

Since uterine inversion presents with profuse bleeding and a severe degree of shock, it requires prompt diagnosis and management. It is managed by promptly repositioning the uterus by exerting a firm and constant pressure on the inverted uterine fundus. The placenta should not be removed until after the replacement of the uterus if it is still attached to the uterus.

Solution to Question 28:

The recombinant HPV vaccine use virus-like particles (VLPs) that consists of the HPV L1 major capsid protein. The L1 protein self-assembles into VLPs when expressed in eukaryotic cells.

These VLPs contain the same epitopes as the HPV virion. However they do not contain genetic material and cannot transmit infection. The immunogenicity of the HPV vaccines relies on the development of conformational neutralizing antibodies to epitopes displayed on viral capsids.

The nonavalent HPV vaccine works against strains 6, 11, 16, 18, 31, 33, 45, 52 and 58. It is usually commercialized in the name of Gardasil-9. It has received expanded indication which now includes the prevention of oropharyngeal and other head and neck cancers.

As per the December 2022 WHO updated dose schedules on HPV vaccines, the recommendations for different age groups are as follows:

- 1 or 2 doses schedule for the primary target of girls aged 9–14 years.
- 1 or 2 doses schedule for young women aged 15–20 years.
- 2 doses with a 6-month interval for women older than 21 years.
-

A minimum of 2 doses and when feasible 3 doses remain necessary for those known to be immunocompromised and/or HIV-infected.

Obstetrics and Gynecology INI-CET 2024

Question 1:

Which of the following is contraindicated in PPH in a primigravida with chronic hypertension?

- a) Methyl ergometrine
- b) Misoprostol
- c) Oxytocin
- d) Prostaglandin F₂ alpha

Question 2:

A 16-year-old female came with complaints of primary amenorrhoea. She had normal secondary sexual characteristics and breast development corresponded to Tanner stage IV. Her FSH - 5.3 and LH - 12.1. Her karyotype is 46XX. What is the probable diagnosis?

- a) Kallmann syndrome
- b) Androgen insensitivity syndrome
- c) Gonadal dysgenesis
- d) Mullerian agenesis

Question 3:

Which of the following combinations is not seen in the case of twin pregnancy?

- a) Monochorionic monoamniotic (MCMA)
- b) Dichorionic diamniotic (DCDA)
- c) Monochorionic diamniotic (MCDA)
- d) Dichorionic monoamniotic (DCMA)

Question 4:

An 18-year-old girl was brought by her mother with complaints of primary amenorrhea and acute retention of urine. She had cyclical abdominal pain and on examination, a supra pubic

mass and a tense bluish bulge were seen in her vagina. What is the most appropriate cause of her symptoms?

- a) Imperforate hymen
- b) Transverse vaginal septum
- c) Vertical vaginal septum
- d) Mullerian agenesis

Question 5:

In the quadruple marker test, a 33-year-old G2P1L1 at 18 weeks of gestation was found to be at a high risk of neural tube defects and low risk of trisomy 21. What is the next investigation to be done?

- a) Amniocentesis
- b) Chorionic villus sampling
- c) Non invasive prenatal testing
- d) Anomaly Scan

Question 6:

An 18-year-old came with complaints of pain in the abdomen and bleeding PV for the last 2 days. Her last menstrual period was 6 weeks back. On examination, she has hypotension, tachycardia and cervical motion tenderness. Ultrasound of the abdomen shows free fluid in the pouch of Douglas and paracolic gutters. What is the most likely diagnosis?

- a) Placenta previa
- b) Abruptio placenta
- c) Ruptured ectopic pregnancy
- d) Missed abortion

Question 7:

A 32 year old primigravida with chronic hypertension came to the hospital at 36 weeks of gestation with headache and blurring vision. Her BP is 170/110 mm Hg. Urine analysis reveals urine albumin 3+. What is the next step to be done?

- a) Injection labetalol and magnesium sulphate and expedite the delivery
- b) Injection labetalol , antenatal steroids and immediate termination of pregnancy

- c) Injection labetalol and magnesium sulphate and deliver at 37 weeks
- d) Injection labetalol to control her blood pressure

Question 8:

A female at 40 weeks of gestation presents with labor pains and is in the second stage of labor for the last 2 hours. On examination of the abdomen, there are 4 uterine contractions per 10 minutes each lasting more than a minute. On PV examination, the fetal station is at +3 and the suture is in AP diameter and caput +. The fetal heart rate is 70 beats/ minute in the last 20 minutes. Which is the appropriate method of delivery?

- a) Outlet forceps
- b) Ventouse delivery
- c) Mid cavity forceps
- d) Low cavity forceps

Question 9:

Which of the following hormones is not responsible for the in-utero growth of the baby?

- a) Insulin
- b) Growth hormone
- c) Insulin like growth factor
- d) Thyroxine

Question 10:

A G2P1L1 delivered a baby via full-term vaginal delivery. The baby is 3 kgs and is active and plays well. Her blood group was found to be O negative and her husband's blood group was found to be B positive. She was given anti-D at 28 weeks of gestation. Which of the following is the appropriate situation to administer postpartum anti D prophylaxis?

- a) Baby is O positive and DCT is negative
- b) Baby is O negative and DCT is negative
- c) Baby is O negative and DCT positive
- d) Baby is O positive and DCT positive

Question 11:

What is the correct dose of administration of antenatal steroids for lung maturity?

- a) Betamethasone 12 mg 12 hours apart 4 doses
- b) Dexamethasone 6 mg 12 hours apart 4 doses
- c) Betamethasone 6mg 12hrly 4 doses
- d) Dexamethasone 12mg 24hrly 2 doses

Question 12:

Which of the following is not a criterion to diagnose IUGR?

- a) Estimated fetal weight $\leq$ 3rd percentile
- b) Abdominal circumference less than 3rd percentile
- c) Absent diastolic flow in umbilical artery
- d) Uterine artery PI $\leq$ 95th percentile

Question 13:

Early onset pre-eclampsia is defined before:

- a) 28 weeks
- b) 30 weeks
- c) 32 weeks
- d) 34 weeks

Question 14:

Dr Priya is conducting a talk on the screening of cervical cancer with a Pap smear. She works in a clinic in a low-resource setting and she was planning to screen a woman in her 20s. Which of the following is true regarding cervical cancer screening guidelines in her workplace?

- a) Pap smear every 3 years, starting at 21 years
- b) Co-testing (pap and Hr HPV) every 5 years starting at 25 years
- c) Hr HPV testing every 5 years starting at 30 years
- d) Pap smear every 5 years starting at 21 years

Question 15:

26-year P1L1 presents with abdominal pain. She has a past history of rheumatic heart disease for which she has undergone mitral valve replacement surgery and is on the oral tablet warfarin 3mg OD. On examination, an abdominal mass corresponding to 14 weeks was felt. On USG, an 8*6 cm unilocular cyst with a reticular pattern was observed in the left ovary. What is the next line of management?

- a) Measure CA-125
- b) Repeat USG after 6-12 weeks
- c) Surgical intervention
- d) MRI pelvis

Question 16:

When is delayed cord clamping done?

- a) Term and preterm babies irrespective of birth weights
- b) Vigorous term and preterm babies requiring resuscitation
- c) Vigorous term babies requiring resuscitation
- d) All term babies born at 37 weeks

Question 17:

A couple presents with infertility even after 2 years of unprotected intercourse. The seminal analysis reveals a volume of 2 mL, concentration of 40 million/ml, progressive motility of 55%, and morphology of 45%. The wife's hormone levels were normal with bilateral spillage of dye from both fallopian tubes on HSG. What percentage of couples can be attributed to unexplained infertility?

- a) 5%
- b) 30%
- c) 15%
- d) 25%

Question 18:

Which of the following is not a physiological change seen in pregnancy?

- a) Increase in systolic and diastolic blood pressure after 6 weeks
- b) Peripheral resistance to insulin leading to hyperinsuliemia
- c) Fasting hypoglycemia and postprandial hyperglycemia
- d) Increased plasma volume and RBC mass

Question 19:

Instrument assisted delivery is not indicated in which of the following?

- a) Fetal distress
- b) Maternal exhaustion
- c) Prolonged second stage of labour
- d) Cephalo pelvic disproportion

Question 20:

In which stage of meiosis is the primary oocyte arrested?

- a) Diplotene
- b) Leptotene
- c) Zygotene
- d) Pachytene

Question 21:

Restitution is best described as:

- a) Passive movement of head due to untwisting of neck
- b) Movement of the head by 1/8th of the circle due to torsion
- c) Stretching of the vulval outlet by the head
- d) External rotation after delivery of shoulder

Question 22:

Which of the following is NOT part of AMTSL?

- a) Uterine massage
- b) Delivery of placenta by controlled cord traction
- c) Bimanual compression
- d) Administration of oxytocin after shoulder delivery

Question 23:

A 35-year-old P2L2 woman presents with malodorous vaginal discharge and a normal cervix. The whiff test is positive. What is this condition?

- a) Candida
- b) Chlamydia
- c) Bacterial vaginosis
- d) Pelvic inflammatory disease

Question 24:

A 14-year-old came with complaints of acute urinary retention. She has not attained menarche and has cyclic abdominal pain. On genital examination, a bluish bulge is seen. What is the probable diagnosis?

- a) Transverse vaginal septum
- b) Imperforate hymen
- c) MRKH
- d) OHVIRA

Question 25:

Which of the following does not constitute a germ cell tumor?

- a) Yolk cell tumor
- b) Dysgerminoma
- c) Juvenile granulosa cell tumor
- d) Non gestational choriocarcinoma

Question 26:

Which of the following is the smallest pelvic diameter at the brim?

- a) Obstetric conjugate
- b) Diagonal conjugate
- c) True conjugate
- d) Anatomical conjugate

Question 27:

A female with abdominal pain and bleeding per vagina with 3 months of amenorrhea. On examination her BP was 90/60mmHg. Which of the following statements is/are true about ectopic pregnancy?

- a) 1, 2
- b) 3,4
- c) 1,2,3
- d) 1,2,3,4

Question 28:

A 32-year-old woman, G2P1L1, comes for a quadruple test at 16 weeks. Which of the following is not included in the quadruple test?

- a) Dimeric inhibin B
- b) Unconjugated estradiol
- c) AFP
- d) Beta hcg

Question 29:

An athlete with a BMI of 17.5 kg/m² presents with secondary amenorrhea and primary infertility. Her laboratory investigations reveal FSH 2 mIU/ml; LH 0.32 IU/L; TSH 1.2mIU/ml. Prolactin 20ng/ml. What is the underlying cause of this condition?

- a) Primary ovarian failure
- b) Hypogonadotrophic Hypogonadism
- c) Hypergonadotrophic Hypogonadism
- d) Normogonadotrophic Normogonadism

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	a
5	d
6	c
7	a
8	a
9	b
10	a
11	b
12	d
13	d
14	a
15	b
16	a
17	d
18	a
19	d
20	a
21	a
22	c
23	c
24	b
25	c
26	a
27	c
28	a
29	b

Detailed Explanations

Solution to Question 1:

Methylergometrine is contraindicated in the management of postpartum hemorrhage in a female with chronic hypertension.

Methylergometrine is an ergot derivative, which acts directly on the myometrium. It excites uterine contractions which come so frequently one after the other with increasing intensity that the uterus passes into a state of spasm without any relaxation in between. Hence, it should be used only either in the late second stage of labor(after the delivery of the anterior shoulder) or following the delivery of the baby.

Indications :

- Prophylactic - Active management of the third stage of labor as prophylaxis to excess bleeding following delivery
- Therapeutic - To stop the atonic uterine bleeding, following delivery, abortion or expulsion of hydatidiform mole. It is used at a dose of 0.2mg im/iv, repeated every 2-4 hours.

Side effects :

- Nausea and vomiting.
- Because of its vasoconstrictive action, it may precipitate a rise in blood pressure, myocardial infarction, stroke and bronchospasm.
- Repeated use may lead to gangrene of the toes due to its vasoconstrictive effect.
- Repeated use in puerperium may interfere with lactation by lowering prolactin levels.

Contraindications :

- Multiple pregnancy - If given accidentally with the delivery of the first baby, the second baby is compromised by the tetanic contractions of the uterus.
- Heart disease - Results in sudden squeezing of blood from the uterine circulation into the systemic circulation causing overloading of the right heart and failure.
- Hypertensive disorders, pre-eclampsia and eclampsia
- Rh-negative mother - More risk of fetomaternal microtransfusion.
- Vascular disease

Other options :

Option B - Misoprostol is a methyl ester of PGE₁, used in the induction of labor and in the management of postpartum hemorrhage. Side effects include fever, tachysystole (hyperstimulation), fetal heart rate changes and meconium passage. It should not be used for cases with previous cesarean birth because the risk of uterine rupture is high.

Option C - Oxytocin is used in the active management of the third stage of labor, augmentation of labor and control of postpartum hemorrhage. Contraindications for its use are hypovolemic states, obstructed labor and cardiac disease.

Option D - Prostaglandin F₂ alpha derivative (carboprost) is used in the management of postpartum hemorrhage. Contraindications for its use are asthma, active cardiac, renal, or hepatic

disease.

Solution to Question 2:

The given clinical scenario is that of a female with primary amenorrhea, preservation of secondary characteristics, and normal levels of FSH and LH, with a karyotype of 46XX. These are features of Mullerian agenesis.

Congenital absence of both the uterus and vagina is termed Mullerian agenesis or Mayer-Rokitansky-Küster-Hauser syndrome. Females have a shallow vaginal pouch (1 to 2 inches deep). The uterus, cervix, and upper part of the vagina are absent. Typically, normal ovaries persist, as they originate from the genital ridge. Individual presents with primary amenorrhea, have a normal female appearance, secondary sexual characteristics, and 46 XX karyotype. Most patients with Mullerian agenesis have only small rudimentary Mullerian bulbs without endometrial activity.

With Mullerian agenesis, traditional conception is not possible, but pregnancy may be achieved using oocyte retrieval, fertilization, and gestational surrogacy.

Individuals with Mullerian agenesis or hypoplasia have associated congenital renal or skeletal anomalies. Common renal anomalies include ectopic, horseshoe or pelvic kidney and collecting system anomalies. Skeletal malformations include spina bifida, sacralization, sacral bone lumbarization, and cervical vertebral anomalies.

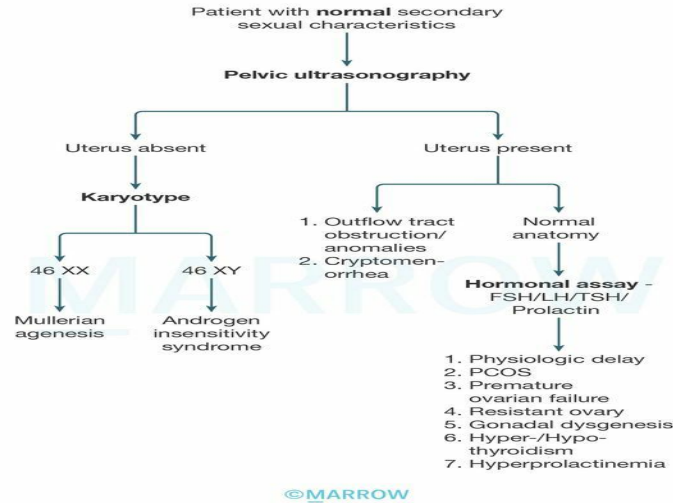
Other options:

Option A - Kallmann syndrome presents with primary amenorrhea due to hypogonadotropic hypogonadism from deficiency of GnRH. The FSH and LH levels are lower than normal, leading to poor development of secondary sexual characteristics. It is associated with olfactory dysfunction and anosmia.

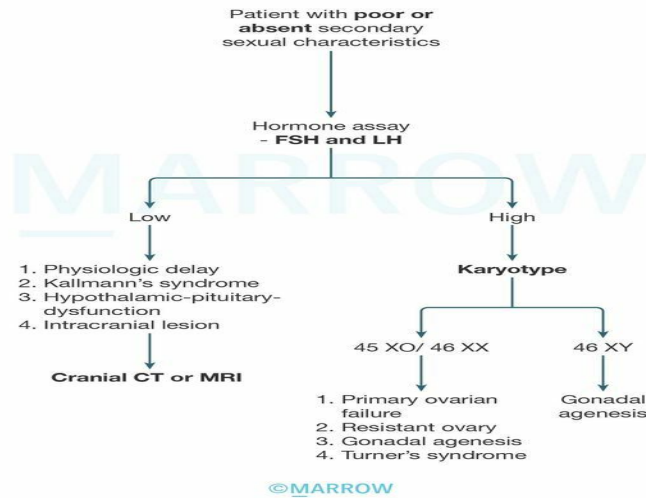
Option B - Androgen insensitivity syndrome occurs due to a defect in the androgen receptor in a genotypic male (XY). These individuals are phenotypic females with no male or female internal genitalia. They have normal female external genitalia with an absent or shallow vagina, well-developed breasts, and scanty body hair (axillary and pubic) and present with primary amenorrhea.

Option C - Gonadal dysgenesis in the form of 45XO (Turner syndrome) or 46XY (Swyer syndrome) presents with poor development of secondary sexual characteristics. Since the gonads are nonfunctional, there is a loss of negative feedback with increased levels of FSH and LH.

Evaluation of primary amenorrhea



Evaluation of primary amenorrhea



Solution to Question 3:

Dichorionic monoamniotic (DCMA) combination is not seen in twin pregnancies.

The table below describes the types of amnion and chorion seen in different types of twins, depending on the time of division.

In monochorionic diamniotic gestation, the twins are separated only by the two amniotic membranes, with no intervening chorion. So, a thin 'T sign' is formed at the point at which the amnions meet the placenta. It is best seen in the latter part of the first trimester or at the beginning of the second trimester. In monochorionic monoamniotic gestation, there is an absence of a dividing membrane.

Time of division after fertilization	Type of twin
0 - 4 days (morula)	Dichorionic diamniotic (DCDA)
4 - 8 days (blastocyst)	Monochorionic diamniotic (MCDA)
8 - 12 days (implanted blastocyst)	Monochorionic monoamniotic (MCMA)
>13 days (formed embryonic disc)	Conjoined twins



In dichorionic gestation, the twins are separated by amniotic membranes and an intervening chorion. So, a thicker twin peak sign or lambda (λ) sign is formed at the point at which the amnions meet the placenta. It is best seen in the first trimester.



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Solution to Question 4:

The features of primary amenorrhea, cyclical abdominal pain, acute retention of urine along with a tense bluish bulge make the diagnosis of imperforate hymen more likely.

Imperforate hymen is a type of Mullerian defect, of urogenital origin. Failure of canalization leads to the formation of mucocolpos; this may be recognized in early infancy. However, the anomaly often continues unrecognized until puberty. It presents with amenorrhoea in the presence of secondary sexual characters, cyclic abdominal discomfort, retention of urine and a suprapubic bulge caused by hematocolpos. Examination of the external genitalia reveals the presence of a telltale bluish bulging membrane in the region of the hymen.

A simple cruciate incision followed by an excision of the hymenal tags allows the drainage of the retained menstrual blood. The operation should be performed under aseptic conditions and under an adequate antibiotic cover to avoid any ascending infection.

Other options :

Option B - Transverse vaginal septum is a much rarer condition that occurs due to faulty fusion or canalization of the urogenital sinus and the Müllerian ducts. The majority of the cases involve the upper portion of the vagina. It can be differentiated from imperforate hymen by the presence of a short vagina and the absence of a bulging membrane.

Option C - Vertical vaginal septum occurs when the distal parts of the Müllerian ducts fail to fuse. On examination, there is a longitudinal anteroposterior septum which may be partial or complete, extending right down to the vaginal outlet. It may be associated with a double uterus and double cervix.

Option D - Mullerian agenesis is characterized by congenital absence of the uterus and vagina (small rudimentary uterine bulbs are usually present). The female presents with primary amenorrhea and preservation of female secondary sexual characteristics.

Solution to Question 5:

For a female with a quadruple test showing a low risk of trisomy 21, the next investigation to be done is an anomaly scan, as part of routine antenatal care.

Prenatal genetic counseling involves screening and diagnostic methods. The various screening methods are as follows

The values of the parameters in trisomy 21 are as follows :

PAPP-A - low, MSAFP - low, unconjugated estriol - low, NT - increased, Beta hCG - high, inhibin A - high.

If the screening tests report a high risk of aneuploidy, then the confirmatory diagnostic test is performed, depending on the trimester.

- Chorionic villus sampling - Transcervical - 10-13 weeks, transabdominal - 10 weeks to term
- Amniocentesis - After 15 weeks

- Cordocentesis - 18-20 weeks

Since the screening test report in the question mentions a low risk of trisomy 21, the female should be followed up with an anomaly scan, which is performed as a part of the antenatal care at 18-20 weeks. It gives a detailed survey of fetal anatomy, placental localization and the integrity of the cervical canal.

Other options :

Option A - Amniocentesis is a confirmatory karyotyping test performed if the initial screening test reveals a high risk of aneuploidy.

Option B - Chorionic villus sampling (CVS) is also a confirmatory test, hence done if there is a high risk of aneuploidy. Moreover, CVS is not performed at 18 weeks.

Option C - Non-invasive prenatal testing utilizes cell-free fetal DNA in the maternal blood. However, it is a screening test. Since the female here has already undergone a screening test, there is no benefit in subjecting her to a second screening test.

Dual test (1st trimester)	Triple test (2nd trimester)	Quadruple test(2nd trimester)
1. Beta HCG2. PAPP (Pregnancy-associated plasma protein A)This can be combined with nuchal translucency by USSG	1. Beta HCG2. Unconjugated estriol3. MSAFP	1. Beta HCG2. Unconjugated estriol3. MSAFP4. Serum inhibin A

Solution to Question 6:

The given clinical scenario with a triad of abdominal pain, amenorrhea and vaginal bleeding, along with the ultrasound findings of free fluid, is suggestive of a ruptured ectopic pregnancy.

An ectopic pregnancy is one in which the fertilized ovum is implanted and develops outside the normal endometrial cavity, with the Fallopian tubes being the most common site. The outcomes of such a pregnancy are varied with tubal rupture and massive intraperitoneal hemorrhage being one of them.

Clinical features :

- The incidence is maximum between the age of 20 and 30 years, the maximum fertility period. The onset is acute. The classic triad of symptoms is - abdominal pain (100%), preceded by amenorrhea of 6-8 weeks(75%), and lastly, the appearance of vaginal bleeding (70%).
- Abdominal pain is the most constant feature. Pain is located in the lower abdomen. Shoulder tip pain (25%) (referred pain due to diaphragmatic irritation from hemoperitoneum) may be present.
- Vomiting, fainting attack. Syncopal attack (10%) is due to reflex vasomotor disturbances following peritoneal irritation from the hemoperitoneum.
- Pallor and features of shock - feeble pulse with tachycardia, hypotension, cold clammy extremities.

- Pelvic examination - Vaginal mucosa is blanched white; uterus seems normal in size or slightly bulky; extreme tenderness on fornix palpation(75%).

Investigations :

Estimation of β -hCG - The findings suggestive of ectopic are a lower concentration of hCG compared to normal intrauterine pregnancy, and doubling in plasma failing to occur in 2 days.

Sonography: Transvaginal sonography (TVS) is more informative. The diagnostic features are

- Absence of intrauterine pregnancy with a positive pregnancy test.
- Fluid (echogenic) in the pouch of Douglas.
- Adnexal mass clearly separated from the ovary (Blob sign).

Color Doppler sonography can identify the placental shape (ring-of-fire pattern) and enhanced blood flow pattern outside the uterine cavity.

Management of ruptured ectopic :

The principle in the management is resuscitation and laparotomy, and not resuscitation followed by laparotomy. Salpingectomy is the definitive surgery.

Other options :

Option A - Placenta previa is a type of antepartum hemorrhage occurring after 28 weeks of pregnancy. It presents as sudden onset painless, recurrent vaginal bleeding. USG findings reveal placenta occupying the lower segment of uterus.

Option B - Abruptio placenta is another form of antepartum hemorrhage occurring due to premature separation of the normally situated placenta. The majority of cases occur in the third trimester. The clinical features vary depending on whether the hemorrhage is concealed or revealed and include abdominal pain, vaginal bleeding. Ultrasound reveals a collection of blood in the retroplacental, subchorionic or preplacental region.

Option D - Missed abortion refers to when the fetus is dead and retained inside the uterus. Clinical features include the persistence of brownish vaginal discharge and subsidence of pregnancy symptoms. Examination findings are uterus size smaller than the period of gestation and closed cervical os. USG reveals an empty sac or a fetus without cardiac activity.

Solution to Question 7:

The given scenario is suggestive of chronic hypertension with superimposed severe pre-eclampsia. The next step in the management is to give injection labetalol, and magnesium sulfate and expedite the delivery, irrespective of the duration of gestation.

The criteria for diagnosis of superimposed pre-eclampsia are :

- New onset proteinuria
- Aggravation of hypertension
- Development of HELLP syndrome
- Development of headache, scotoma, and epigastric pain.

Pre-eclampsia is further divided into :

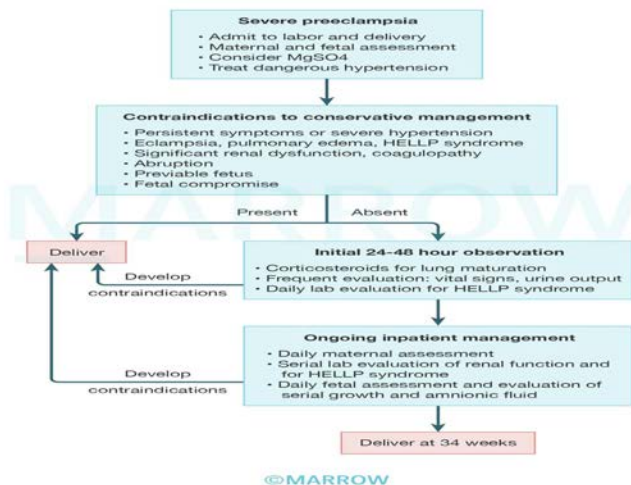
- Nonsevere pre-eclampsia: Blood pressure $\geq 140/90$ mmHg but $< 160/110$ mmHg without significant proteinuria
- Severe pre-eclampsia: Persistent increase of systolic blood pressure above 160 mmHg or diastolic above 110 mmHg, proteinuria, oliguria (< 400 mL in 24 hours), platelet count < 1 lakh/cu.mm, serum creatinine level of ≥ 1.1 mg/dL, persistent severe epigastric pain, marked elevation of liver enzyme levels, convulsions or visual disturbances, intrauterine growth restriction of the fetus, and pulmonary edema.

In a patient with severe pre-eclampsia, ie, systolic BP ≥ 160 mm Hg or diastolic ≥ 110 mm Hg, immediate delivery after single-dose steroids is indicated regardless of the period of gestation. Concurrently, injection labetalol is given to reduce the blood pressure and injection magnesium sulfate is given for seizure prophylaxis.

Patients with pre-eclampsia may be expectantly managed up to 34 weeks of gestation. After 34 weeks pregnancy, may be prolonged up to 37 weeks if there is adequate BP control, and delivery is recommended at 37 weeks of gestation. However, in the following events, pregnancy is terminated regardless of the period of gestation:

- Uncontrolled severe hypertension
- Eclampsia
- Pulmonary edema
- HELLP Syndrome
- Renal dysfunction
- Coagulopathy (disseminated intravascular coagulation)
- Abruptio
- Fetal compromise

Clinical management algorithm for severe pre-eclampsia at < 34 weeks



Solution to Question 8:

In the given scenario, outlet forceps would be the best choice of instrumental delivery as the fetus is at +3 station, with suture in AP diameter and has caput.

The criteria for the application of different types of forceps are as follows :

Outlet forceps :

- Scalp is visible at the introitus without separating the labia
- Fetal skull has reached the level of the pelvic floor
- Sagittal suture is in direct anteroposterior diameter or in the right or left occiput anterior or posterior position, but rotation is ≤ 45 degrees
- Fetal head is at or on the perineum

Low forceps :

- Leading point of the fetal skull (station) is at +2 cm or more but has not yet reached the pelvic floor, and

(a) Rotation is $\leq 45^\circ$, or (b) Rotation is $> 45^\circ$

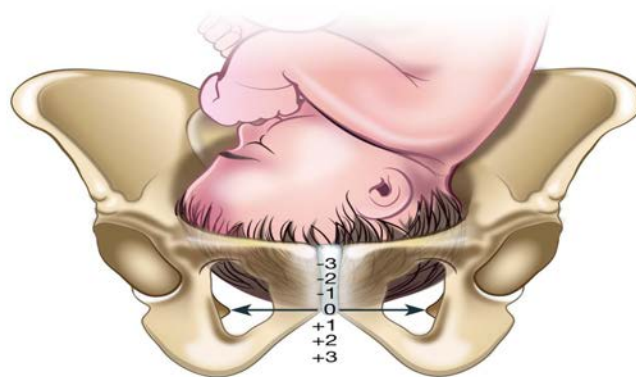
Mid forceps :

- Fetal head is engaged. Head is 1/5 palpable per abdomen but the station is above +2 cm but not above the ischial spines (station 0).

High forceps :

- Head is not engaged. This type of forceps is not used in current obstetrics.

Note : Ventouse is not the best choice for a fetus with caput, as there is a risk of scalp avulsion. Also, the vacuum cannot be adequately generated owing to the presence of soft tissue swelling.



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Solution to Question 9:

Growth hormone does not play a role in the in-utero growth of a baby. It is, however, essential for postnatal growth.

Normal fetal growth is characterized by cellular hyperplasia followed by hypertrophy and lastly by hypertrophy alone. The fetal growth increases linearly until the 37th week. It is controlled by genetic factors in the first half and by environmental factors in the second half of pregnancy. Imprinted genes (maternal or paternally acquired alleles) primarily control fetal growth.

The important physiological factors are race (European babies are heavier than Indians); sex (male baby weighs >female); parental height and weight (tall and heavier mothers have heavier babies); birth order (weight rises from first to second pregnancy) and socioeconomic factors (heavier babies in social class I and II).

Fetal growth is predominantly controlled by insulin-like growth factor -1 (IGF-1), insulin and other growth factors. Maternal and fetal thyroid hormones also play a role in the normal development of virtually all fetal tissues, especially the brain.

Solution to Question 10:

Post-partum anti-D prophylaxis is given if the baby is O positive and DCT (direct Coombs' test) is negative.

To prevent active immunization of Rh-negative yet unimmunized, Rh anti-D immunoglobulin (IgG) is administered intramuscularly to the mother following childbirth. The mode of action is antibody-mediated immune suppression.

Anti-D immunoglobulin should be administered within 72 hours or preferably earlier following delivery or abortion. It should be given provided the baby born is Rh-positive and the direct Coombs' test is negative. This reduces the alloimmunization rate by 90%. Use of anti-D immunoglobulin at 28 weeks gestation additionally reduces third-trimester alloimmunization to 0.1%.

Dose: Anti-D-gammaglobulin is administered intramuscularly to the mother (300 µg) following delivery. All Rh-negative unsensitized women should receive 50 µg of Rh-immunoglobulin IM within 72 hours of induced or spontaneous abortion, ectopic or molar pregnancy or chorionic villus sampling in the first trimester. Women with pregnancy beyond 12 weeks should have a full dose of 300 µg. Standard 300 µg of anti-D immunoglobulin IM dose will protect a mother from fetal hemorrhage of up to 30 mL of fetal whole blood or 15 mL of fetal RBCs.

Other options:

Anti-D prophylaxis is futile if -

- the baby has Rh negative blood group - there is no risk of alloimmunization when the Rh antigen is absent
- if the DCT of the baby's blood is positive - it means that alloimmunization has already occurred. Such babies are managed with other treatment options like exchange transfusion.

Solution to Question 11:

The correct dose of administration of antenatal steroids for lung maturity is dexamethasone 6 mg 12 hours apart 4 doses.

Maternal administration of corticosteroids is advocated where the pregnancy is less than 34 weeks. This helps in fetal lung maturation. This is beneficial when the delivery is delayed beyond 48 hours of the first dose. The benefit persists as long as 18 days. Either betamethasone 12 mg IM 24 hours apart for two doses or dexamethasone 6 mg IM every 12 hours for 4 doses is given. Betamethasone is the steroid of choice (RCOG–2004).

Antenatal steroids also reduce the incidence of respiratory distress syndrome, necrotizing enterocolitis, intraventricular hemorrhage and patent ductus arteriosus.

Risks of antenatal corticosteroid use:

- Premature rupture of the membranes especially with evidence of infection, as the infection may flare-up
- Insulin-dependent diabetes mellitus where patients need insulin dose readjustment
- Transient reduction of fetal breathing and body movements.

Solution to Question 12:

Uterine artery pulsatility index (PI) < 95th percentile is not a criterion used to diagnose intrauterine growth restriction (IUGR).

Fetal growth restriction (FGR)/IUGR is said to be present in those babies whose birth weight is below the 10th centile of the average for the gestational age.

The causes of fetal growth restriction are divided into :

- Maternal - constitutional, nutritional, anemia, hypertension, chronic renal disease, alcohol, smoking
- Fetal - Structural anomaly, chromosomal anomaly, infections, multiple pregnancy
- Placental -placenta previa, abruption, circumvallate placenta
- Unknown

Diagnosis :

Clinical -

- Symphysis fundal height (SFH) measurement in centimeters closely correlates with gestational age after 24 weeks. A lag of 3 cm or more suggests growth restriction.
- Maternal weight gain remains stationary or, at times, falling
- Measurement of the abdominal girth showing stationary or falling values

Ultrasound -

Various criteria and thresholds have been used to define FGR. These have included estimated fetal weights (EFW) below the 3rd, 5th, or 10th percentiles; similar abdominal circumference (AC)

percentiles; a specified decline in the EFW percentile or AC percentile over serial assessments; and various abnormal Doppler findings.

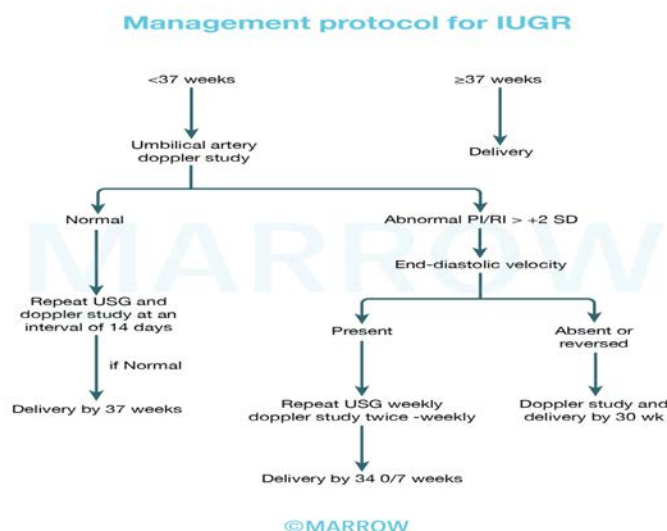
Currently, the ACOG (2021) and the Society for Maternal-Fetal Medicine (2020) recommend defining FGR as either an EFW $\leq$ 10th percentile for gestational age or an AC $\leq$ 10th percentile for gestational age.

AC is the single most sensitive parameter to detect FGR.

Doppler velocimetry - The umbilical artery Doppler study should be the primary surveillance tool in FGR (RCOG).

- In normal pregnancy, the systolic/diastolic (S/D) ratio, pulsatility index (PI) and resistance index (RI) decrease as the gestational age advances. Values greater than 2SD indicate reduced diastolic velocities and increased placental vascular resistance. These are associated with FGR and intra-uterine fetal hypoxia.
- Reduced or absent or reversed end-diastolic velocity (AREDV) in the umbilical artery indicates fetal jeopardy and poor perinatal outcome.
- Umbilical venous pulsations indicate inefficient cardiac output with a rise in central venous pressure $\rightarrow$ impending cardiac failure.
- Middle cerebral artery (MCA): Increased diastolic velocity (brain-sparing effect) is observed in FGR. This is due to cerebral vasodilatation in response to hypoxemia. Abnormal MCA PI, suggests fetal compromise.
- Doppler cerebroplacental ratio (CPR): It is calculated by dividing MCA PI by UA PI. In FGR, placental resistance (UA PI) increases and fetal brain vascular resistance (MCA PI) decreases resulting in a decrease in CPR index. CPR is more sensitive compared to UA or MCA Doppler alone.

Management is as outlined below :



Solution to Question 13:

Early onset pre-eclampsia is defined before 34 weeks.

Pre-eclampsia is characterized by hypertension and end-organ dysfunction that typically develops after 20 weeks of gestation in a previously normotensive individual.

Features of pre-eclampsia:

- Hypertension:

Blood pressure $\geq 140/90$ mmHg on 2 occasions at least 4 hours apart, or $\geq 160/110$ mmHg in a single measurement.

- Proteinuria:

≥ 300 mg of protein in a 24-hour urine collection.

Protein/creatinine ratio ≥ 0.3 .

Dipstick reading of $\geq 1+$ (though less specific).

- End-Organ Dysfunction (may occur even without proteinuria):

Renal: Elevated serum creatinine > 1.1 mg/dL or doubling of baseline.

Hepatic: Elevated liver transaminases (AST/ALT).

Hematologic: Thrombocytopenia ($< 100,000/\mu\text{L}$).

Neurologic: Severe headache, visual disturbances (e.g., scotoma, blurred vision).

Pulmonary: Pulmonary edema.

Early-onset pre-eclampsia:

- Occurs before 34 weeks of gestation.

- Associated with more severe complications, including fetal growth restriction and placental insufficiency.

- Requires careful monitoring and often early delivery if maternal or fetal conditions worsen.

Late-onset pre-eclampsia:

- Occurs at or after 34 weeks of gestation.

- Typically milder, with fewer complications.

Management

- Mild pre-eclampsia: Close monitoring, antihypertensives (e.g., labetalol, nifedipine), and maternal-fetal surveillance.

- Severe pre-eclampsia: Hospitalization, magnesium sulfate for seizure prophylaxis, and delivery planning (often the definitive treatment).

- Timing of Delivery:

Recommended at ≥ 37 weeks for mild pre-eclampsia.

Earlier delivery for severe cases.

Solution to Question 14:

In the given scenario, cervical cancer is screened by performing a pap smear every 3 years, starting at 21 years.

Pap smear is a screening test. In case of positive test further investigations should be done.

Pap Smear Schedule

- Time for initiation: 21 years, regardless of the age of first sexual intercourse.
- Women aged 21-29 years: Every 3 years(earlier it was done annually)
- Women aged 30-65 years: Pap test + HPV testing every 5 years.
- Women should stop having cervical cancer screening after the age of 65 years. If suspicious PAP, screening till 75 years.
- HIV+ women: annual checkup until three consecutive negative tests followed by testing every 3 years for life.
- HPV DNA testing in combination with PAP improves the sensitivity of screening.

Ayres spatula and endocervical brush are the instruments used for the pap test.



Ayres spatula:

- One end is blunt and the other end is bifid.
- Blunt end is used to take samples for vaginal cytology.
- Bifid end is used to take Pap smear

Endocervical brush



Solution to Question 15:

The USG findings of a unilocular cyst with reticular pattern indicate a functional hemorrhagic cyst. Hence, the next line of management is to repeat USG after 6-12 weeks.

Although the cyst size is >7 cm, invasive treatments are not suitable immediately because of her current use of warfarin which needs time to be tapered. Also, as most functional cysts resolve spontaneously over time, repeat USG might show a decrease in cyst size.

Management of ovarian cysts in reproductive age and perimenopausal women is mentioned in the table below:

Other Options:

Option A: Measure CA-125: In this case, it might not be the best first step because CA-125 is not specific and its level can be influenced by multiple factors. Additionally, the cyst described here is more likely to be benign (reticular pattern suggesting blood), such as a functional ovarian cyst or follicular cyst.

Option C: Surgical Intervention: Surgical intervention may be required if there are signs of torsion, rupture, or malignancy. However, with a simple, unilocular cyst without these features, surgery is usually not warranted immediately, especially as the patient is on warfarin and needs to be tapered. This approach would be more appropriate if the cyst showed signs of complications or did not resolve after monitoring.

Option D: MRI Pelvis: An MRI can offer more detailed imaging if there is concern about the nature of the cyst, but it is typically reserved for complicated cysts (e.g., solid components, suspicious features). It's not the first-line investigation in this case.

Size	Management
< 5 cm	Wait & watch; no additional treatment required

Size	Management
5cm - 7 cm	Follow up with serial TVS in 6-12 weeks; if cyst persists continue TVS on yearly basis
> 7cm	MRI pelvis; surgical evaluation

Solution to Question 16:

Delayed cord clamping (DCC) is done in term and preterm babies irrespective of birth weight.

DCC involves waiting at least 30–60 seconds or more after birth before clamping the umbilical cord. It is recommended for both term and preterm babies as part of standard delivery care.

Indications of DCC:

- Term infants: DCC improves iron stores and reduces the risk of anemia in infancy.
- Preterm infants: DCC increases blood volume and reduces the risks of:
- Intraventricular hemorrhage
- Necrotizing enterocolitis
- Provides stabilization benefits by enhancing transitional circulation

Babies irrespective of birth weights: The benefits of DCC apply regardless of the baby's weight.

Contraindications of DCC:

- Babies requiring immediate resuscitation: In emergencies, immediate neonatal stabilization takes priority, and DCC may not be feasible.
- Maternal or placental complications: Conditions like placental abruption or postpartum hemorrhage may necessitate immediate cord clamping for maternal safety.

Other options:

Option B and C: Vigorous term and preterm babies requiring resuscitation: Resuscitation needs immediate intervention, which may preclude delayed cord clamping.

Option D: All term babies born at 37 weeks: While DCC is recommended for term infants, the guideline includes preterm babies too, so this option is incomplete. It also fails to exclude babies requiring resuscitation.

Solution to Question 17:

25% of couples can be attributed to unexplained infertility.

Unexplained infertility refers to cases where no clear cause of infertility can be identified after a thorough evaluation of both partners. In the scenario provided, the male partner has normal seminal parameters, and the female partner has normal hormonal levels and patent fallopian

tubes, making this a case of unexplained infertility.

Some causes of unexplained infertility include:

- Ovulatory Dysfunction
- Impaired sperm function despite normal semen analysis
- Tubal or Uterine Factors
- Endometriosis
- Antisperm antibodies or autoimmune issues.
- Lifestyle and Environmental Factors: Stress, smoking, diet, or environmental toxins.

Unexplained infertility is diagnosed when standard evaluations yield normal results:

- Semen analysis showing normal volume, concentration, motility, and morphology.
- Normal hormonal tests, basal body temperature, or ovulation predictor kits.
- Hysterosalpingography (HSG) or saline sonography shows no blockages.
- Transvaginal ultrasound or hysteroscopy ruling out structural abnormalities.

If no cause is identified after these tests, the couple is categorized as having unexplained infertility.

Management includes:

1. Expectant Management

- Suitable for younger couples (<35 years) with a short duration of infertility.
- Lifestyle modifications (e.g., weight management, smoking cessation, stress reduction).

2. Medical :

- Ovulation Induction: Clomiphene citrate or Letrozole.
- Gonadotropins: Injectable hormones (FSH, LH) for controlled ovarian stimulation.

3. Intrauterine Insemination (IUI)

4. Assisted Reproductive Technology (ART)

- In Vitro Fertilization (IVF)
- Intracytoplasmic Sperm Injection (ICSI)

Solution to Question 18:

Increase in systolic and diastolic blood pressure after 6 weeks is not a physiological change seen in pregnancy.

During early pregnancy, systemic vascular resistance decreases, leading to a drop in blood pressure (both systolic and diastolic), which typically reaches its lowest point in the second trimester. Blood pressure gradually returns to pre-pregnancy levels in the third trimester.

An increase in blood pressure during early pregnancy is not physiological and could indicate an underlying pathology (e.g., chronic hypertension or gestational hypertension later in pregnancy).

Cardiovascular parameters that increase in pregnancy are:

- Cardiac output
- Stroke volume
- Heart rate

Cardiovascular parameters that decrease in pregnancy are:

- Peripheral vascular resistance (due to smooth muscle relaxation by progesterone)
- Both diastolic BP and systolic BP (decrease in DBP > SBP)
- Mean arterial BP
- Artery venous O₂ gradient

Other Options:

Option B: Peripheral resistance to insulin leading to hyperinsulinemia: During pregnancy, placental hormones like human placental lactogen (hPL), cortisol, and progesterone induce insulin resistance to ensure adequate glucose availability for the fetus. This leads to compensatory hyperinsulinemia.

Option C: Fasting hypoglycemia and postprandial hyperglycemia: Fasting hypoglycemia occurs due to increased maternal glucose utilization by the fetus. Postprandial hyperglycemia is a result of insulin resistance and slower glucose uptake.

Option D: Increased plasma volume and RBC mass: Plasma volume increases by about 40-50%, and red blood cell (RBC) mass increases by 20-30%, leading to hemodilution and the physiological anemia of pregnancy.

Solution to Question 19:

Instrument-assisted delivery is not indicated in cephalo pelvic disproportion. Ruling out CPD is a prerequisite before attempting instrument-assisted delivery.

Cephalo-pelvic disproportion (CPD) is a condition in which the baby's head is too large, or the pelvis is too small, for the baby to pass through the birth canal. In such cases, instrument-assisted delivery is not indicated because vaginal birth is not feasible. The appropriate management for CPD is usually a cesarean section to avoid complications for both mother and baby.

Conditions where instrument-assisted delivery may be indicated:

- Fetal distress: If the baby is showing signs of distress (such as decelerations in the heart rate), an instrument-assisted delivery may be needed to expedite delivery and reduce the risk of further harm. (Option A)
- Maternal exhaustion: In cases of maternal fatigue, particularly after a prolonged second stage of labor, instruments may be used to assist in delivery if the mother is unable to continue pushing effectively. (Option B)

- Prolonged second stage of labor: If the second stage of labor (when the cervix is fully dilated and the baby is descending) is prolonged, instruments can be used to assist in the delivery of the baby, especially if there are signs of fetal distress or maternal exhaustion. (Option C)

Prerequisites for instrumental vaginal delivery:

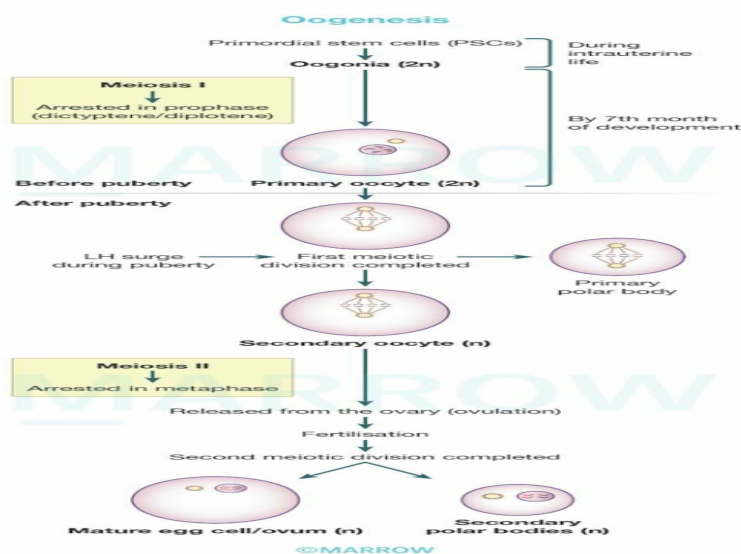
- Ruptured membranes vertex presentation
- Cephalopelvic disproportion not suspected
- No fetal coagulopathy or bone demineralization disorder
- Completely dilated cervix
- Precisely assessed fetal head position
- Engaged head
- Experienced operator

Solution to Question 20:

The primary oocyte is arrested in prophase I of meiosis during a specific substage called diplotene.

Primary oocytes initiate meiosis I, but are arrested in prophase I before birth.

The arrest persists until puberty, when hormonal signals trigger the resumption of meiosis in a selected number of oocytes during each menstrual cycle.



Meiosis I (Reduction Division): Homologous chromosomes separate.

- Prophase I: Chromosomes pair, cross over.
- Metaphase I: Homologous pairs align at the equator.
- Anaphase I: Homologous chromosomes separate.

- Telophase I: Two haploid cells form.

Meiosis II (Equational Division): Sister chromatids separate.

- Prophase II: Chromosomes condense.
- Metaphase II: Chromosomes align at the equator.
- Anaphase II: Sister chromatids separate.
- Telophase II: Four genetically unique haploid cells form.

Prophase I consists of several substages (leptotene, zygotene, pachytene, diplotene, and diakinesis). In primary oocytes:

- Leptotene: Chromosomes condense and become visible. (Option B)
- Zygotene: Homologous chromosomes pair up through synapsis. (Option C)
- Pachytene: Crossing over occurs between non-sister chromatids. (Option D)
- Diplotene stage: The homologous chromosomes are connected at chiasmata (sites of crossing over), and the oocyte enters a prolonged resting phase. At ovulation, the primary oocyte completes meiosis I and becomes a secondary oocyte, which then gets arrested in metaphase II of meiosis until fertilization.
- Diakinesis: Chromosomes fully condense; nuclear envelope breaks down.

Solution to Question 21:

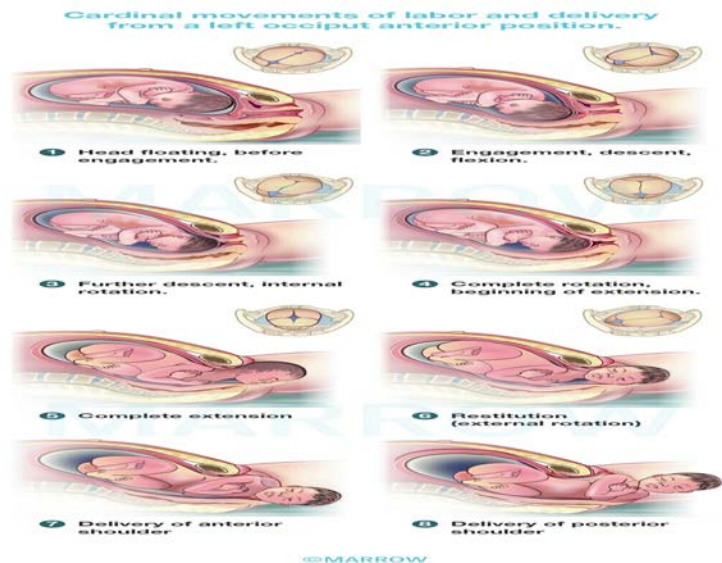
Restitution describes a passive adjustment of the head to untwist the neck after internal rotation.

The most common position of the fetus is the left occipito-lateral position where the vertex occupies the left anterior quadrant of the mother's pelvis.

During internal rotation, there will be anterior rotation of the occiput by $\frac{2}{8}$ th of a circle, placing the occiput behind the symphysis pubis. There is always an accompanying movement of descent with internal rotation. (Option B)

Torsion of the neck occurs during internal rotation of the head. The neck cannot withstand such a high degree of torsion, so the shoulders will rotate in the same direction to the extent of $\frac{1}{8}$ th of a circle, placing the shoulders in the oblique diameter with one-eighth of the torsion still left behind. Torsion refers to the twisting forces applied to the fetal head and neck during internal rotation to align the head with the birth canal.

In restitution, these twisting forces are released, allowing the head to return to its anatomical alignment with the shoulders.



Other Options:

Option B: Movement of the head by 1/8th of the circle due to torsion: While the head typically does rotate by about 1/8th of a circle (45°) during restitution, this is a passive as the tension in the neck resolves. The phrase "due to torsion" is misleading because torsion refers to twisting forces exerted during internal rotation, not restitution. Restitution is about the release of torsion, not its application.

Option C: Stretching of the vulval outlet by the head: This describes a process that occurs during the crowning stage of labor, not during restitution. At crowning, the fetal head stretches the vulval outlet as it emerges. Restitution occurs after the head is delivered and involves the passive rotation of the head to align with the shoulders.

Option D: External rotation after delivery of shoulder: Once the shoulders are delivered, the head undergoes a second adjustment to align with the shoulders as they rotate to fit through the maternal pelvis. This is an active movement that happens post-delivery of the shoulders, while restitution occurs before shoulder delivery.

Solution to Question 22:

Bimanual compression is not part of AMTSL.

AMTSL is recommended by the WHO to prevent postpartum hemorrhage (PPH). Uterotonics are the most important components in AMTSL to do so.

Note: Uterine massage or fundal massage is not recommended as an intervention to prevent PPH in women who have already received prophylactic uterotonics such as oxytocin (core component of AMTSL). Hence, it is now an optional component of AMTSL that is done in patients with uterine atony as a part of the management of PPH. (Option A)

1.	Injecting uterotonics within 1 minute of delivery of the baby or immediately after delivery of the anterior shoulder (Option D)	Most important step Oxytocin 10 IU im/iv infusion is the preferred uterotonic. (Not recommended by WHO- carboprost)
2.	Delayed cord clamping	Done within 1-3 mins of delivery 80 ml of additional blood goes to the fetus Prevents fetal anemia
3.	Delivery of placenta by controlled cord traction: Modified Brandt Andrews technique (Option B)	The uterus is pushed upwards and backward The traction of cord is downwards and forwards
4.	Intermittent assessment of uterine tone	Done every 15 mins for 2 hours

Solution to Question 23:

The patient with malodorous vaginal discharge and a positive whiff test, strongly suggests bacterial vaginosis.

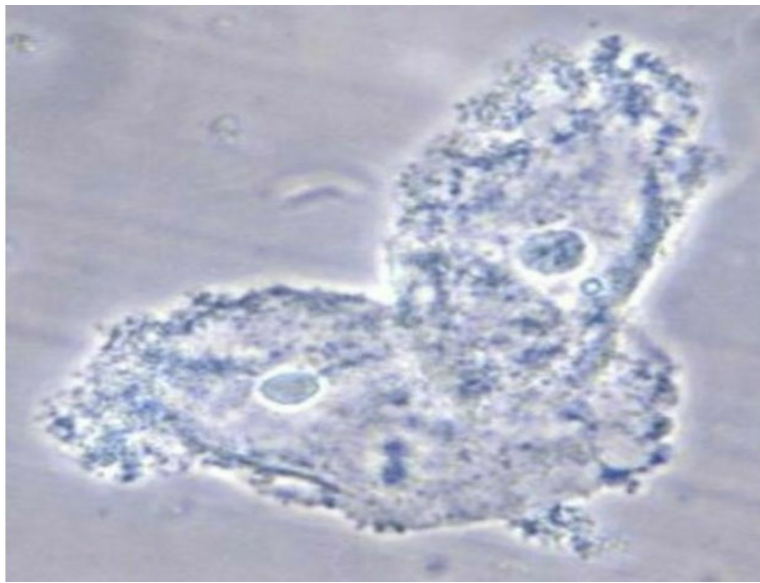
Bacterial vaginosis (BV) is caused by an imbalance in the vaginal microbiome, characterized by the overgrowth of anaerobic bacteria (*Gardnerella vaginalis*) and a decrease in the normal protective lactobacilli. Bacterial vaginosis is a polymicrobial infection characterized by malodorous (fishy smell) vaginal discharge. Causative organisms include:

- *Gardnerella*,
- *Mobiluncus*,
- *Mycoplasma hominis*

Bacterial vaginosis is diagnosed using Amsel's criteria:

Amsel's Criteria: (any 3 should be present)

- Thin, white homogenous vaginal discharge
- pH of Vaginal fluid > 4.5
- Fishy odor of discharge accentuated on adding 10% KOH (Whiff test)
- Presence of clue cells on microscopy. Many epithelial cells present with granular cytoplasm caused by small gram-negative bacilli adhering on their surface, these cells are called clue cells.



Treatment is oral metronidazole 500 mg BD for 7 days. Alternatively, Ampicillin 500 mg or cephalosporin 500 mg bid for 7 days is also effective.

Other options:

Option A: Candida: Presents with thick, white "cottage cheese-like" discharge, intense itching, and no fishy odor. Whiff test is negative.

Option B : Chlamydia: Often asymptomatic or causes mucopurulent cervical discharge and friable cervix. Does not cause a fishy odor or a positive whiff test.

Option D : Pelvic Inflammatory Disease (PID): Causes pelvic pain, fever, and cervical motion tenderness. Not typically associated with malodorous discharge or a positive whiff test.

Solution to Question 24:

This patient presenting acute urinary retention, cyclic abdominal pain, bluish bulge on genital examination and primary amenorrhea (no menarche), likely has imperforate hymen.

Imperforate hymen results due to failure of canalization of the inferior end of the vaginal plate. This anomaly usually remains unrecognized until puberty. It can lead to mucocolpos in early infancy.

Key Features of Imperforate Hymen:

- Cyclic abdominal pain: Caused by retained menstrual blood (hematocolpos) due to the obstruction.
- Primary amenorrhea: Menstruation cannot occur due to the blockage.
- Bluish bulge: Hematocolpos causes distension of the vaginal canal, which can present as a bluish bulge at the hymen.
- Urinary retention: The distended vagina (hematocolpos) can compress the bladder or urethra, leading to urinary retention.

Ultrasound pelvis confirms the diagnosis.

Management: A simple cruciate incision of the hymen allows the free drainage of the collected menstrual blood, and leads to normal reproductive function.

Other Options:

Option A: Transverse vaginal septum: Would also cause primary amenorrhea and hematocolpos. Typically presents without a bluish bulge, as the obstruction is deeper in the vagina. Extensive dissection and vaginoplasty is required.

Option C: MRKH: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome causes primary amenorrhea due to absent uterus and upper vagina. There is no cyclic pain or bluish bulge, as menstruation cannot occur.

Option D: OHVIRA: Obstructed hemivagina and ipsilateral renal anomaly: Rare condition with an obstructed hemivagina. Often associated with renal anomalies and a functional contralateral uterus, but it presents with cyclic pain later after menarche onset, not primary amenorrhea.

Solution to Question 25:

Juvenile granulosa cell tumor does not constitute a germ cell tumor. It is a pure-sex cord tumor.

Germ cell tumors:

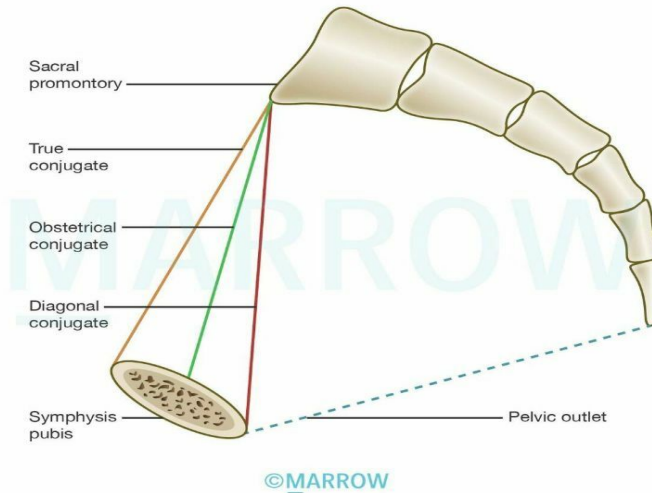
- Dysgerminoma
- Endodermal sinus tumor (Yolk sac tumor)
- Embryonal carcinoma
- Polyembryoma
- Non-gestational choriocarcinoma
- Teratoma (Mature/immature)
- Mixed forms

Solution to Question 26:

The obstetric conjugate is the smallest pelvic diameter at the brim.

It is the distance between the midpoint of the sacral promontory to the prominent bony projection in the midline on the inner surface of the symphysis pubis. It is the shortest anteroposterior diameter in the anteroposterior plane of the inlet and measures 10 cm. It represents the narrowest fixed diameter of the pelvic inlet that the fetal head must navigate. It cannot be measured directly during a pelvic exam but is estimated using the diagonal conjugate.

Diameters of the pelvic inlet



Other Options:

Option B: Diagonal Conjugate: It is the distance between the lower border of the symphysis pubis to the midpoint on the sacral promontory. It measures 12 cm. Obstetric conjugate is derived by subtracting 1.5–2 cm from the diagonal conjugate depending upon the height, thickness and inclination of the symphysis pubis.

Option C and D : True conjugate: Also called anatomical conjugate, is the distance between the midpoint of the sacral promontory to the inner margin of the upper border of the symphysis pubis and measures 11 cm.

Solution to Question 27:

An ectopic pregnancy is one in which the fertilized ovum is implanted and develops outside the normal endometrial cavity.

Clinical features :

The classic triad of symptoms is - abdominal pain (100%), preceded by amenorrhea of 6-8 weeks(75%), and lastly, the appearance of vaginal bleeding (70%).

Abdominal pain is the most constant feature. (Statement 1) Pain is located in the lower abdomen. Shoulder tip pain (25%) (referred pain due to diaphragmatic irritation from hemoperitoneum) may be present.

Pelvic examination - Vaginal mucosa is blanched white; uterus seems normal in size or slightly bulky; extreme tenderness on fornix palpation(75%).

Risk factors of ectopic pregnancy:

- Previous ectopic pregnancy
- Tubal corrective surgery, tubectomy
- Assisted reproductive technology

- Documented tubal pathology
- Pelvic inflammatory disease (Chlamydia infection and salpingitis isthmica nodosa), Genital tuberculosis, sexually transmitted diseases (Statement 2)
- Endometriosis, Pelvic adhesions
- Prior abortion, Prior Caesarean delivery
- Multiple sexual partners
- Progesterone-only pills (POPs)

Diagnosis:

- Serum β -hCG – concentrations are higher in viable IUPs than in ectopic pregnancies and IUPs that are destined to abort, and measurement of serum β -hCG may be useful in a patient with abdominal pain and bleeding who has a serum hCG level below that expected for the gestational age. However, the definition of low β -hCG is unclear as there is no specific value to diagnose ectopic pregnancy. (Statement 3)
- β -hCG - lower concentration of hCG and doubling in plasma failing to occur in 2 days.
- Transvaginal sonography (TVS) features are absence of intrauterine pregnancy with a positive pregnancy test.
- Color Doppler sonography can identify the placental shape (ring-of-fire pattern).

Management of ectopic pregnancy:

1. Medical management is preferred in the following cases:

- Low initial serum beta-hCG level (Less than 5,000 IU/L)
- Small ectopic pregnancy size (Less than 3.5 cm)
- Absent fetal cardiac activity
- Hemodynamically stable

2. Expectant management: A low initial serum hCG concentration, a considerable reduction in levels over 48 hours, and a sonographic inhomogeneous mass instead of a tubal halo or other gestational features are predictive variables for success.

3. Surgical management with salpingectomy or salpingostomy is done when:

- Gestational sac $\geq$ 3.5 cm
- Fetal cardiac activity present
- Beta-hCG $\geq$ 5000 mIU/mL
- Significant pain and hemoperitoneum

Oral methotrexate is not the preferred route for treating ectopic pregnancy, as it may lead to variable absorption and unreliable dosing. Intramuscular injection is the preferred route of administration for methotrexate in the management of ectopic pregnancy. (Statement 4)

Solution to Question 28:

Dimeric inhibin B is not included in the quadruple test.

The quadruple test, performed during the second trimester (15–22 weeks, ideally around 16–18 weeks), includes the following markers:

- Alpha-fetoprotein (AFP)
- Unconjugated estriol (uE₃)
- Human chorionic gonadotropin (hCG)
- Inhibin A

The quadruple test is primarily used for screening chromosomal abnormalities such as Down syndrome (trisomy 21), Edward syndrome (trisomy 18), and neural tube defects.

The following table summarizes the findings of the quad-screening test in various conditions.

Parameter	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect
MSAFP	Low	Low	High
β hCG	High	Low	Normal
Unconjugated estriol	Low	Low	Normal
Inhibin A	High	Normal/Low	Normal

Solution to Question 29:

The low FSH and LH levels, along with the low BMI, suggest hypogonadotropic hypogonadism.

BMI of 17.5 kg/m² indicates low body weight, which can lead to hypothalamic suppression and a disruption in the normal feedback mechanisms that regulate the menstrual cycle. FSH (2mIU/ml), LH (0.3IU/L): These values are low, indicating that the pathology lies at the level of the hypothalamus or pituitary, rather than the ovaries. In hypogonadotropic hypogonadism, there is insufficient secretion of gonadotropins (FSH and LH), which are needed for ovarian function. TSH (1.2mIU/ml) and Prolactin (20 ng/ml) are normal, which rules out thyroid dysfunction or hyperprolactinemia as the cause of the amenorrhea.

Causes of hypogonadotropic hypogonadism:

- Kallman's syndrome, Prader-Willi syndrome
- Pituitary tumors, apoplexy
- Anorexia nervosa, excess exercise
- Irradiation, Infection, Trauma

Exercise-induced hypogonadotropic hypogonadism (EIHG): intense and prolonged physical activity leads to a decrease in the function of the hypothalamic-pituitary-gonadal (HPG) axis, resulting in lower levels of sex hormones. This occurs due to reduced secretion of gonadotropin-releasing hormone (GnRH), which subsequently lowers levels of luteinizing

hormone (LH) and follicle-stimulating hormone (FSH).

Causes: Excessive physical training, especially in endurance sports. Energy imbalance (low calorie intake vs. high energy expenditure). Psychological stress and overtraining.

Symptoms:

- Men: Decreased libido, erectile dysfunction, fatigue, and muscle weakness.
- Women: Irregular or absent menstrual cycles (amenorrhea).

Management includes rest and reduced exercise intensity. Adequate caloric intake to correct energy deficits. EIHG is a reversible condition with lifestyle modifications and is most commonly seen in athletes or individuals with excessive physical demands.

Other Options:

Option A: Primary ovarian failure (Premature ovarian insufficiency) is associated with elevated FSH and LH levels due to a lack of negative feedback from the ovaries, which is not the case here (FSH and LH are low).

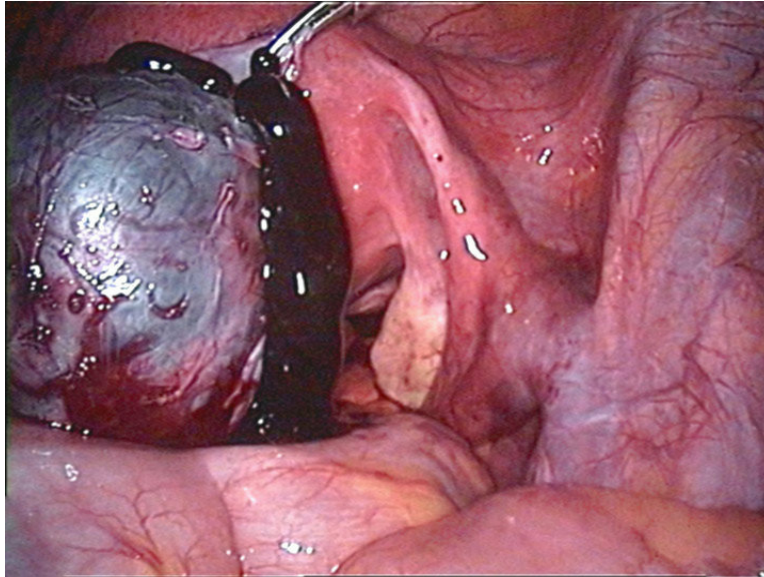
Option C: Hypergonadotropic Hypogonadism refers to primary ovarian failure, where FSH and LH levels are elevated due to the lack of ovarian response. The low levels of FSH and LH here rule out this condition.

Option D: Normogonadotropic Normogonadism would indicate normal levels of FSH, LH, and estrogen, typically seen in conditions like polycystic ovary syndrome (PCOS). However, in this case, the low FSH and LH levels suggest a hypothalamic or pituitary issue.

Obstetrics and Gynecology NEET 2024

Question 1:

Identify the image shown below :



- a) Ovarian torsion
- b) Germ cell tumor
- c) Ovarian cyst
- d) Chocolate cyst of ovary

Question 2:

A 14-year-old rape victim is 22 weeks pregnant and she would like to terminate her pregnancy. Which of the following is done under the MTP Act?

- a) Opinion of 1 doctor is enough upto 24 weeks
- b) Termination allowed between 20-24 weeks only if detrimental to mother's life
- c) Cannot be terminated
- d) Can terminate pregnancy upto 24 weeks for sexual assault

Question 3:

A ventouse vacuum was placed at the flexion point. What is the correct statement regarding flexion point?

- a) 3 cm from posterior fontanelle
- b) Midpoint of anterior and posterior fontanelle
- c) 3cm from anterior fontanelle
- d) 6 cm from posterior fontanelle

Question 4:

A primigravida who is a known case of bronchial asthma develops PPH. Which of the following needs to be avoided in this patient?

- a) Carboprost
- b) Oxytocin
- c) Mifipristone
- d) Methyl ergometrine

Question 5:

A patient with a carcinoma cervix lesion showing cervical growth extending to the pelvic side wall along with the presence of hydronephrosis. What is the stage of the tumour?

- a) II a
- b) II b
- c) III B
- d) III a

Question 6:

A 60-year-old female presented with a mass per vagina, as shown in the image. What is the diagnosis?



- a) Uterovaginal prolapse grade 1
- b) Uterovaginal prolapse grade 2
- c) Uterovaginal prolapse grade 4
- d) Uterovaginal prolapse grade 3

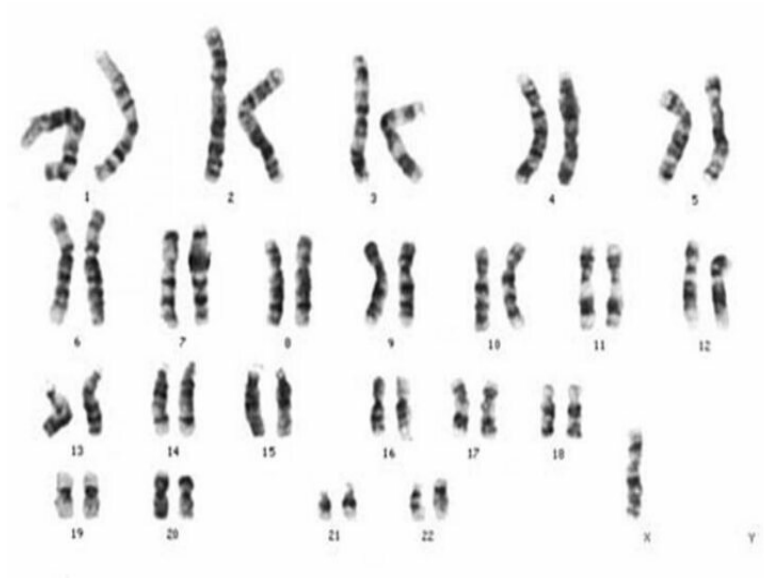
Question 7:

A woman at 37 weeks of gestation comes to the OPD for an antenatal checkup. She is on warfarin for prosthetic valves. What would be the management of this patient?

- a) Change to LMWH
- b) Continue Warfarin
- c) Immediate LSCS
- d) Induction of labour

Question 8:

An 18-year-old girl with short stature has primary amenorrhea and has not attained puberty yet. Identify the chromosomal defect as shown in the image.



- a) Turner's syndrome
- b) Klinefelter's syndrome
- c) Kallman syndrome
- d) AIS

Question 9:

The illustration given below is used for which female genital pathology?

anterior vaginal wall Aa	anterior vaginal wall Ba	cervix or cuff C
genital hiatus gh	perineal body pb	total vaginal length tvL
posterior vaginal wall Ap	posterior vaginal wall Bp	posterior fornix D

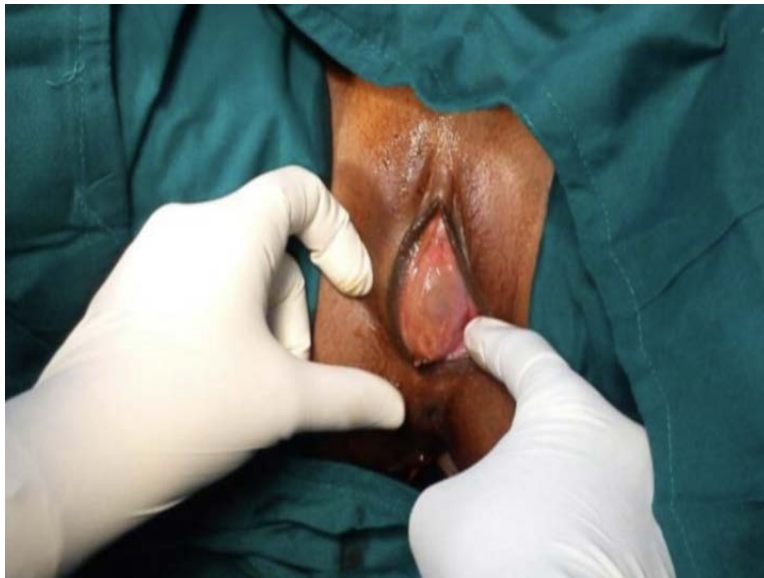
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- a) Prolapse
- b) Vulval cancer

- c) Carcinoma cervix
- d) Endometriosis

Question 10:

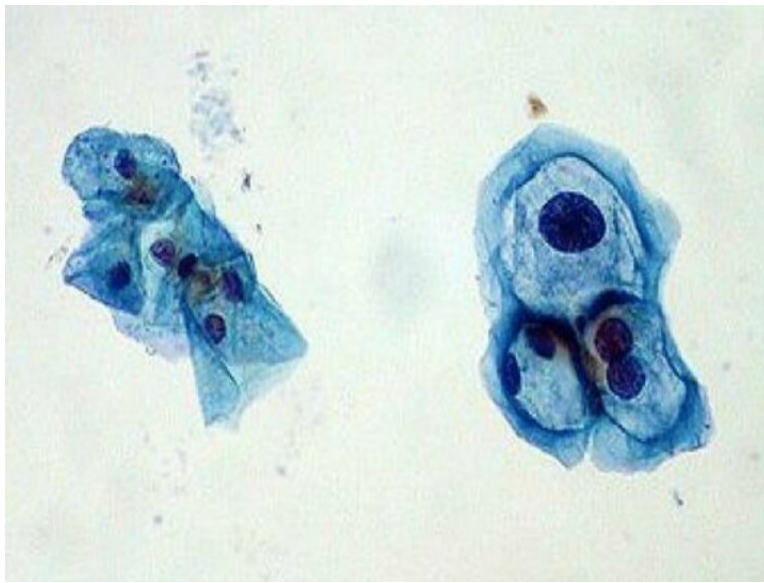
A teenage girl presents with primary amenorrhea, normal breast and pubic hair development with normal uterus & ovaries. What is the karyotype?



- a) 46 XY
- b) 45 XO
- c) 47 XXY
- d) 46 XX

Question 11:

A female complains of dysuria since 4 days associated with pelvic pain and abnormal vaginal discharge. Pap smear image is given below. What is the diagnosis?



- a) Trichomonas
- b) Human papilloma virus
- c) Cytomegalovirus
- d) Candida

Question 12:

A 28-year-old lady with 27 weeks of gestation has multiple congenital malformations in the fetus on radiological investigations. The parents want to terminate the pregnancy for which they went to the MTP board. Who is not a part of the MTP board?

- a) Gynaecologist
- b) Radiologist
- c) Paediatrician
- d) Lawyer

Question 13:

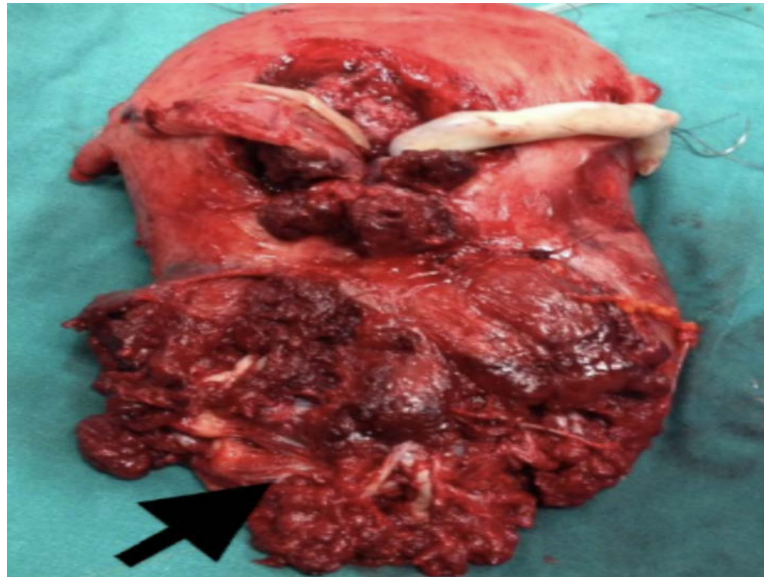
A woman at 30 weeks of gestation presents with pain and redness in the calf region. A diagnosis of DVT is made. Which of the following would you use in the management of the patient?

- a) LMWH
- b) Warfarin
- c) Fondaparinux

d) Rivaroxbaran

Question 14:

A G2P1L1 woman with a previous history of breech delivery underwent an LSCS which was uneventful. Now, she was brought to the ER with an antepartum haemorrhage and was shifted for LSCS during which she lost a lot of blood. An obstetric hysterectomy had to be carried out. An image of the uterus is given below. What is the diagnosis?



- a) Placenta previa
- b) Abruptio placenta
- c) Placenta accreta syndrome
- d) Uterine rupture

Question 15:

A patient with 32 weeks of gestation had tachycardia and a blood pressure of 160/110 mm of Hg. On examination, she had pedal edema and lab investigations showed proteinuria. Pritchard regimen with magnesium sulphate was started. What is the indication to stop MgSO_4 ?

- a) Urine output is 60 ml in 4 hours
- b) Respiratory rate $\geq 18/\text{min}$
- c) Knee jerk present
- d) Visual abnormalities

Question 16:

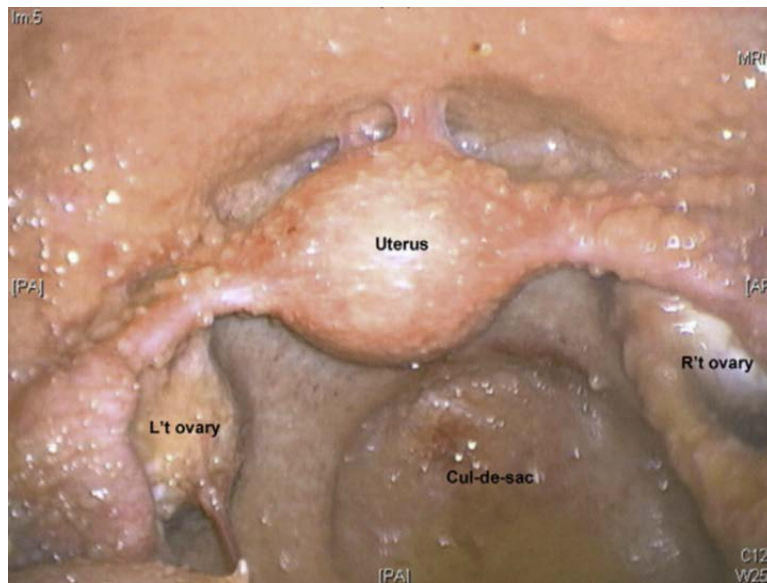
A G5P4 female comes for her routine scan. She is already a mother of 4 girl children and wants a boy this time. She is now interested to know the sex of the current child. According to PCPNDT act, what should you do as a radiologist?



- a) Check for congenital anomalies of the fetus but not look for the sex
- b) Check for congenital anomalies, the sex of the child and tell her if it is a female child
- c) Check for congenital anomalies, the sex of the child and do not tell her
- d) Only check routine ANC and do not do USG

Question 17:

A P1L1 woman, with a history of uneventful previous delivery, presents with secondary infertility and amenorrhea for 2 years. The laparoscopy performed is shown below. What is the likely diagnosis?



- a) Tuberculosis
- b) Endometriosis
- c) Multiple fibroids
- d) Sarcoidosis

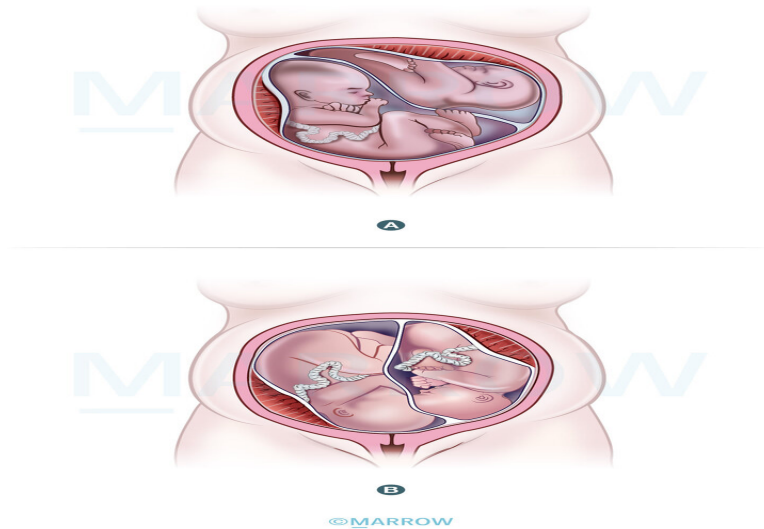
Question 18:

A 45-year-old female came with complaints of spotting and foul-smelling discharge. On per speculum examination, cervical erosions were present. Which of the following is the most appropriate investigation required?

- a) Liquid based cytology, Cervical biopsy, HPV antigen testing
- b) Pap smear, Cervical biopsy, HCV testing
- c) Pap smear, Cervical biopsy, HBV DNA
- d) Pap smear, Electrophoresis, HSV DNA

Question 19:

In the absence of other obstetric complications in twin pregnancies A and B. What is the preferred mode of delivery for A and B respectively?



- a) A. Vaginal delivery B. LSCS
- b) A. LSCS B. Vaginal delivery
- c) Both LSCS
- d) Both vaginal delivery

Question 20:

A P1L1 woman had delivered a baby 8 hrs before by vaginal delivery. What is the earliest time to insert an IUCD?

- a) After 3 months
- b) Within 48 hours
- c) After 6 weeks
- d) Between 48 hours - 6 weeks

Question 21:

A pregnant woman presents with a history suggestive of hyperthyroidism at 8 weeks of POG. What is the drug of choice?

- a) Methimazole
- b) Carbimazole
- c) Propylthiouracil
- d) Propranolol

Question 22:

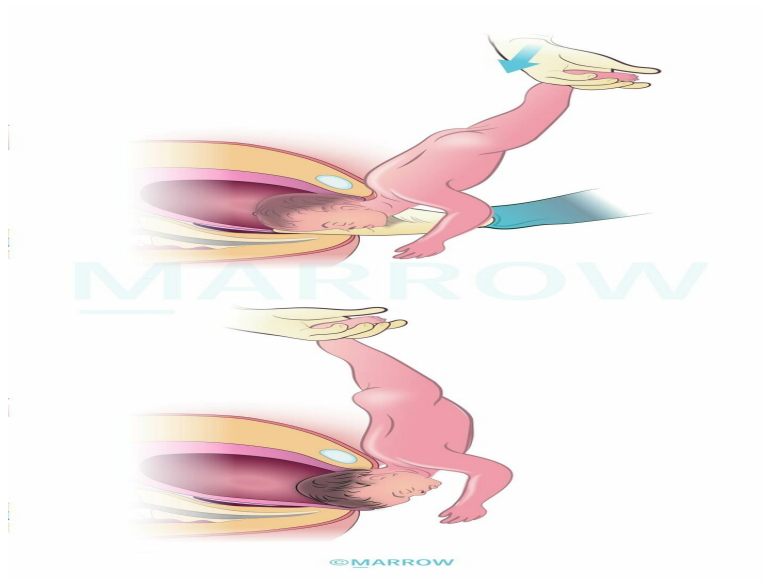
A 42-year-old woman, P2L2, who had her last pregnancy 20 years ago, presents with heavy bleeding. A smooth, round, globular structure is observed on examination. On bimanual examination, the uterus is described as multiparous and firm, and the cervix admits one finger. What is the likely diagnosis?



- a) Hypertrophied Cervix
- b) Myomatous Polyp
- c) Polypoid adenocarcinoma
- d) Placental Polyp

Question 23:

The method shown in the image is:



- a) Pinard's Maneuver
- b) Mauriceau Smellie Veit Method
- c) Burns Marshall Technique
- d) Ritgen Maneuver

Question 24:

A woman presents with lower abdominal pain. Transvaginal ultrasound (TVS) performed over two days reveals a left adnexal cystic mass with a cogwheel appearance, incomplete septa, and no vascular flow on Doppler imaging. What is the most likely diagnosis?



- a) Hydrosalpinx

- b) Malignant ovarian cyst
- c) Follicular cyst
- d) Hemorrhagic cyst

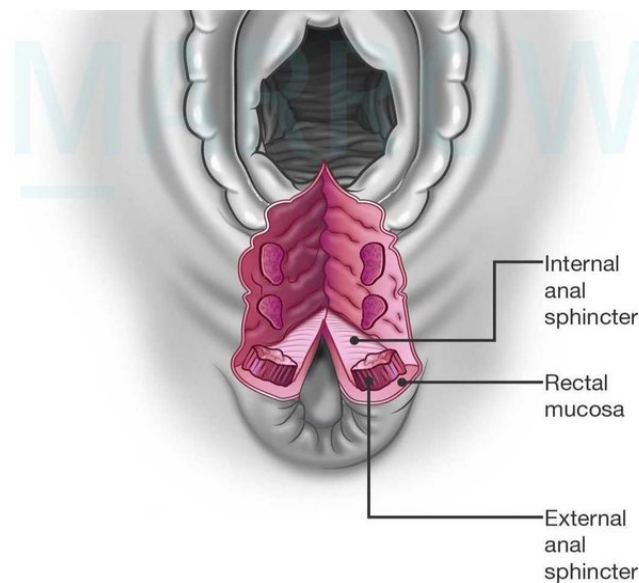
Question 25:

A woman undergoing infertility treatment is found to have bilateral tubal blockage on hysterosalpingography. What is the most appropriate next step in her management?

- a) Intrauterine insemination (IUI)
- b) In vitro fertilization (IVF)
- c) Laparoscopic chromopertubation
- d) Ovulation induction

Question 26:

A primigravida experienced precipitous labor while being transported in an ambulance. Upon arrival at the hospital, the given injury is observed. What type of injury is most likely?



- a) 2nd degree perineal tear
- b) 4th degree perineal tear
- c) 3rd degree perineal tear
- d) 1st degree perineal tear

Question 27:

What is the most common site for tubal ligation and the reason for ligation at this site?

- a) Ampulla; as it has more circular muscles than longitudinal muscles
- b) Ampulla; as it is the widest part
- c) Isthmus; as it has uniform muscle thickness
- d) Isthmus; as it is the narrowest part

Question 28:

A 30-year-old female, P2L2, presents with a mass per vaginum. On examination, infravaginal elongation of the cervix is noted. How will you treat this condition?

- a) McCall culdoplasty
- b) Le Forte's surgery
- c) Fothergill surgery
- d) Latzko procedure

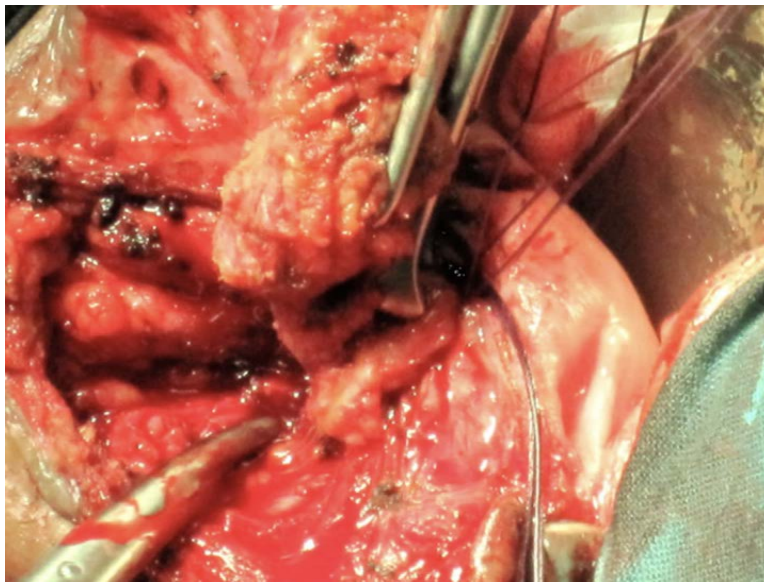
Question 29:

A 28-year-old P2L2 woman presents with complaints of a mass descending through the vagina and wishes to retain her fertility. She underwent an abdominal sling procedure with mesh placement. Which is the correct site of attachment of the mesh?

- a) Symphysis Pubis
- b) Ischial Tuberosity
- c) Ischial Spines
- d) Sacral Promontory

Question 30:

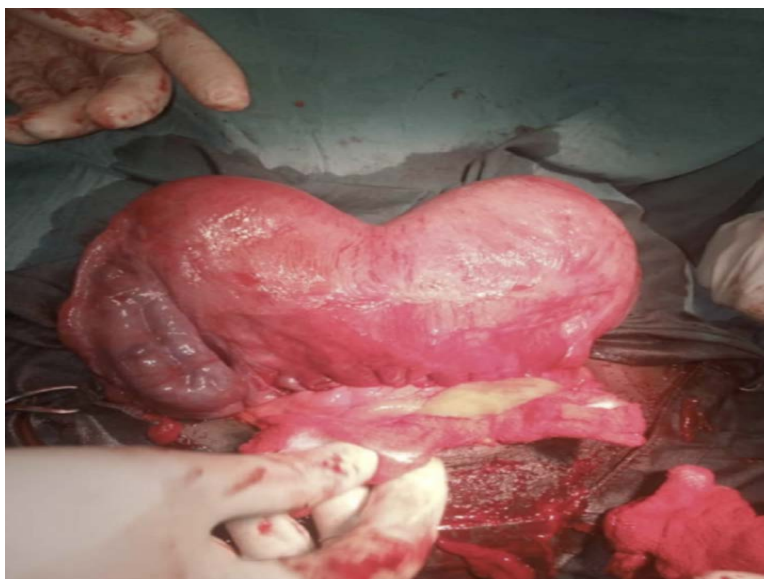
A woman who had a mediolateral episiotomy during normal labor presents with pain at the episiotomy site, which worsens during her menstrual periods. The findings noted on episiotomy site exploration are shown in the image below. What is the most likely diagnosis?



- a) Retained Suture
- b) Melanoma
- c) Inclusion cyst
- d) Endometriosis

Question 31:

During a lower segment cesarean section (LSCS) for a patient with a breech presentation, following findings were noted as given in the image. On per vaginal examination, two cervixes are felt. What is the most likely diagnosis?



- a) Unicornuate Uterus with Rudimentary Horn

- b) Arcuate Uterus
- c) Bicornuate Uterus
- d) Uterine Didelphys

Question 32:

A 30-year-old lady, G3P2 with 10 weeks intrauterine gestation comes to OPD with an Intra Uterine Contraceptive Device(IUCD) and the tail of the IUCD is noted to be protruding from the cervical os. The most appropriate course of action is?

- a) Keep the IUCD in place and continue prophylactic antibiotics
- b) Keep the IUCD in place without any treatment
- c) Remove the IUCD immediately and continue pregnancy
- d) Terminate the pregnancy because of high risk of infection

Question 33:

A 24-year-old woman at 34 weeks of gestation is admitted for preeclampsia with severe features, and you have administered MgSO₄ as prophylaxis. Which of the following is not a part of monitoring?

- a) Routine measurement of serum magnesium levels
- b) Urine output should be $\geq$ 30 ml/hr
- c) Respiratory rate $\geq$ 12/min
- d) Deep tendon reflexes should be present

Question 34:

An unmarried girl presents with amenorrhea and abdominal pain. On genital examination, she is found to have a softened cervix. What is this sign called?

- a) Hegar's Sign
- b) Braxton Hicks Sign
- c) Oslander's Sign
- d) Goodell's Sign

Question 35:

Which of the following is not a contraindication to the procedure shown in the image?



- a) Twin pregnancy
- b) Primigravida
- c) Placenta previa
- d) Premature rupture of membranes

Question 36:

A woman presents with chronic lower abdominal pain that worsens during menstruation. Ultrasound reveals a right adnexal mass with homogeneous echoes. MRI shows a 4x5 cm T1 hyperintense lesion without fat suppression, while the T2 sequence demonstrates low signal intensity with characteristic dark shading. What is the most likely diagnosis?

- a) Cystic tumor of ovary
- b) Dermoid cyst
- c) Endometrioma
- d) Teratoma

Question 37:

A woman's vaginal sample shows an abnormal Nugent's score. What is the most likely diagnosis?

- a) Bacterial Vaginosis

- b) Herpes Infection
- c) Trichomoniasis
- d) Candida Albicans

Question 38:

After ovulation, in which phase is the released egg arrested?

- a) Primary oocyte with meiosis I completed
- b) Secondary oocyte arrested in meiosis II
- c) Primary oocyte arrested in meiosis I
- d) Secondary oocyte with meiosis II completed

Question 39:

A 45-year-old woman presented with menorrhagia. A hysterectomy was performed, and the specimen is shown below. What is the most likely diagnosis?



- a) Intramural Fibroid
- b) Myomatous Polyp
- c) Endometrioma
- d) Adenomyoma

Question 40:

A 55-year-old obese woman with a history of diabetes and hypertension presents with postmenopausal bleeding. Endometrial sampling was performed and it revealed a positive result for PTEN, CTNNB1, KRAS, and MSI markers. What is the most likely diagnosis?

- a) Mixed Müllerian Tumor
- b) Leiomyosarcoma
- c) Endometrioid Cancer
- d) Serous Cystadenoma of Ovary

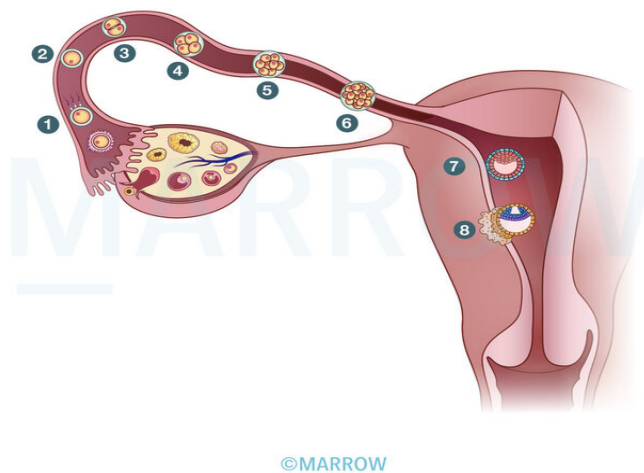
Question 41:

A 21-year-old woman at 10 weeks of gestation, comes to you for medical termination of pregnancy. Which method would you prefer for this patient?

- a) Dinoprostone
- b) Mifepristone and Misoprostol
- c) Oxytocin infusion
- d) Intrauterine instillation of hyper-osmotic saline

Question 42:

What is the time taken for the embryo to travel from point 1 to 8?



- a) 2 days
- b) 8 days
- c) 5 days
- d) 14 days

Question 43:

A 28-year-old woman, G4P3L3, with 1 neonatal death (NND) and 2 previous lower segment cesarean sections (LSCS), presents at term with an active genital herpes infection. What is the preferred mode of delivery?

- a) Elective LSCS
- b) Classical Cesarean section
- c) Trial of labor (VBAC)
- d) Induction of labor

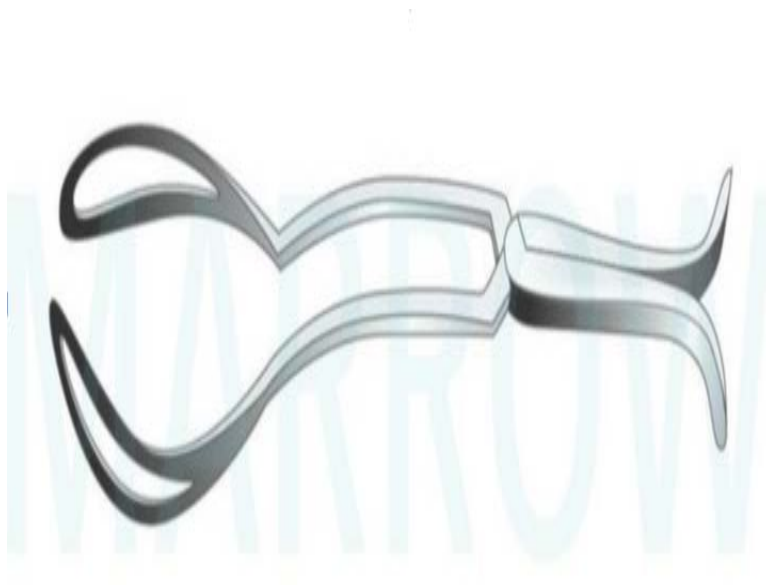
Question 44:

A woman presents with lower abdominal pain. Ultrasound reveals a right adnexal mass with septations. Which tumor markers would be most appropriate to order in this case?

- a) CA125, AFP, HCG, CEA
- b) CA125, AFP, HCG, LDH
- c) CA125, AFP, CA15-3
- d) CA125, AFP, CA19-9

Question 45:

Identify the forceps typically used for delivering the aftercoming head in a breech delivery:



- a) Wrigley's forceps
- b) Kielland's forceps
- c) Das's forceps
- d) Piper's forceps

Answer Key

Question No.	Correct Option
1	d
2	d
3	a
4	a
5	c
6	c
7	a
8	a
9	a
10	d
11	b
12	d
13	a
14	c

15	a
16	a
17	a
18	a
19	b
20	b
21	c
22	b
23	c
24	a
25	c
26	b
27	c
28	c
29	d
30	d
31	d
32	c
33	a
34	d
35	b
36	c
37	a
38	b
39	a
40	c
41	b
42	b
43	a
44	b
45	d

Detailed Explanations

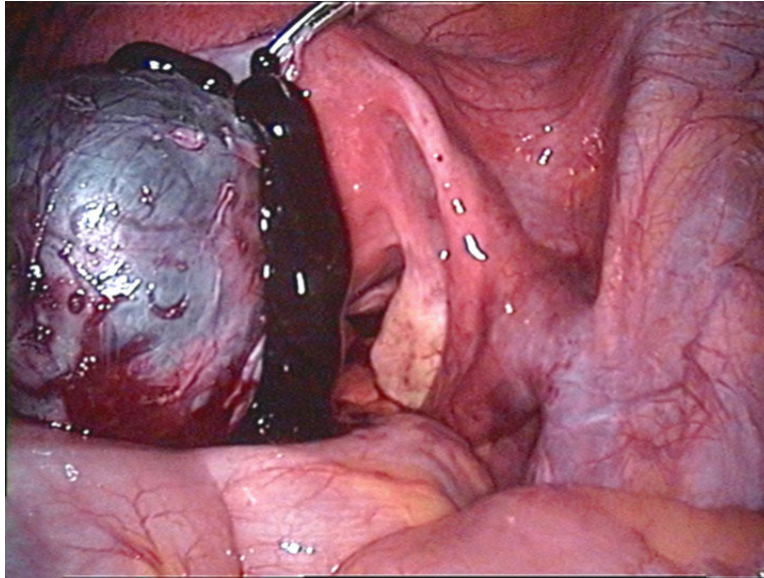
Solution to Question 1:

The given image shows a chocolate cyst of the ovary.

Chocolate cysts of the ovaries represent the most important manifestation of endometriosis. Ovaries are frequently involved bilaterally. It is lined by columnar epithelium.

Beneath the epithelium, large cells with brown cytoplasm called pseudoxanthoma cells are present. These are large macrophages or scavenger cells and their brown colouration is due to ingested blood pigments such as haemosiderin. They are polyhedral in shape and resemble lutein cells.

Chocolate cyst of ovary:

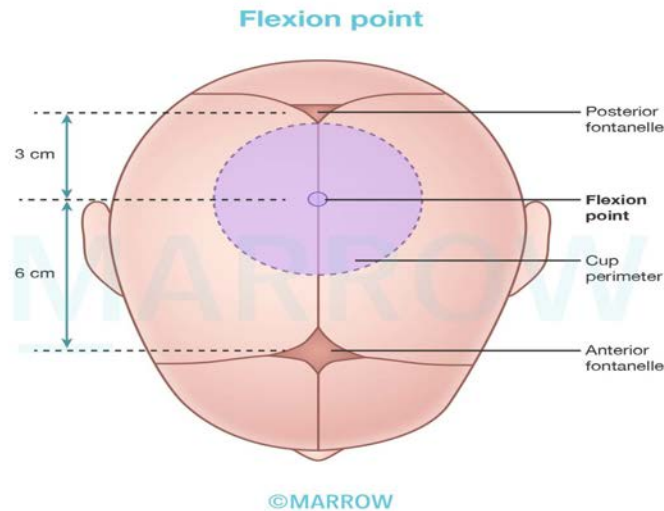


Solution to Question 2:

According to the MTP Act with 2021 amendments, pregnancy can be terminated upto 24 weeks of gestation in victims of rape and incest, fetal anomalies, or congenital malformations.

Solution to Question 3:

An important step in vacuum extraction is proper cup placement over the flexion point which is a point on the sagittal suture that lies 3 cm in front of the posterior fontanelle.



Advantages of correct positioning of the cup in vacuum extraction:

- Maximum traction
- Minimum cup detachment
- Twisting of the fetal head is avoided
- The smallest head diameter is delivered through the pelvic outlet
- The success rate is improved
- Fetal scalp injury rates are lowered
- Perineal trauma is reduced because the smallest fetal head diameter distends the vulva

Solution to Question 4:

The drug carboprost is contraindicated in patients with bronchial asthma.

Carboprost is a 15-methyl PGF₂ alpha derivative used as a 2nd line drug in the treatment of PPH that is unresponsive to oxytocin. The normal dosage is 0.25mg. The maximum dose is 2mg.

Contraindications of carboprost:

Absolute	Relative
Asthma Pulmonary hypertension Suspected amniotic fluid embolism	Pre-eclampsia Renal disease Liver disease Heart disease

Other options :

Option B: Oxytocin is the first-line drug to prevent and treat PPH according to WHO guidelines. Carbetocin, which is a long-acting analogue of oxytocin, can also be used.

Option C: Mifepristone (RU486) is a progesterone antagonist, which blocks the hormonal action on the endometrium, causing the sloughing and shedding of decidua, preventing implantation.

Option D: Methyl ergometrine is contraindicated in hypertensive patients. It is given as an intramuscular injection or slow intravenous to avoid sudden hypertension.

Solution to Question 5:

Cervical cancer with hydronephrosis is classified as Stage III B according to FIGO staging.

Cervical cancer is the second most common gynaecological cancer. The FIGO staging uses clinical, radiological, or pathological findings to stage cervical cancer from Stage I to IV.

Stage III cervical carcinoma involves the lower 1/3rd of the vagina, with or without pelvic wall extension, hydronephrosis, and lymph node involvement. It is further staged as follows:

- Stage III A: Lower 1/3rd of the vagina involvement, no pelvic wall extension
- Stage III B: Pelvic wall involvement with or without hydronephrosis or a non-functioning kidney
- Stage III C: Involvement of the pelvic and/or paraaortic lymph nodes
- Stage III C1: Metastasis to pelvic lymph node only
- Stage III C2: Metastasis to para-aortic lymph nodes

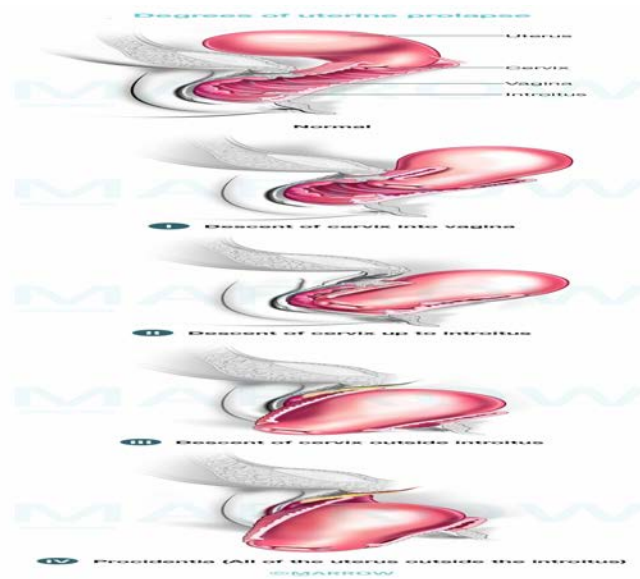
Note: If the carcinoma had extended beyond the true pelvis, involving adjacent pelvic organs like the bladder or rectum, it would've been classified as Stage IV A.

Solution to Question 6:

In the given image, the uterus is visible outside the introitus suggesting the diagnosis of grade IV utero-vaginal prolapse.

The descent of one or more genital organs below their normal anatomical position is called pelvic organ prolapse (POP). It occurs when the muscles and tissues that support the pelvic organs become weakened or damaged, leading to one or more organs bulging into the vaginal canal.

The image below summarises the various degrees of genital organ prolapse.



Types of vaginal prolapse-

- Anterior vaginal wall: prolapse in the upper two-thirds wall is cystocele and in the lower one-third is urethrocele.
- Posterior vaginal wall: prolapse in the upper two-thirds wall is enterocele and in the lower one-third is rectocele.

Image given below shows cystocele.



Treatment options may include pessary, pelvic floor exercises, or surgery depending on the severity of prolapse.

Solution to Question 7:

The next appropriate step in the management of a pregnant woman with a prosthetic valve on warfarin therapy at 37 weeks of gestation is to stop warfarin and start low molecular weight heparin (LMWH).

Warfarin is safe and effective in preventing thromboembolism in the mother during the 2nd trimester. However, it is teratogenic and can lead to miscarriage, stillbirths, fetal anticoagulation, and fetal embryopathy if used in the 1st trimester.

Heparin is safe for the fetus as it does not cross the placenta but is less effective than warfarin in preventing thromboembolism.

LMWH- Low molecular weight heparin, UFH- Unfractionated heparin

For patients taking LMWH, factor Xa levels should be monitored weekly.

Heparin should be discontinued just before (at least 6 hours before) delivery. If delivery occurs before discontinuation and there is extensive bleeding, then iv protamine sulfate can be given. Warfarin or heparin anticoagulation can be restarted after 6 hours of vaginal delivery, usually without any problem.

Warfarin, LMWH, and UFH do not accumulate in breast milk and do not have an anticoagulant effect in the newborn. They are compatible with breastfeeding.

Period of gestation	Anticoagulant
Till 12 completed weeks	LMWH or UFH
13 to 36 weeks	Warfarin
36 weeks till just before delivery	LMWH or UFH
6 hours after vaginal delivery and 24 hours after cesarean section	Warfarin, LMWH or UFH

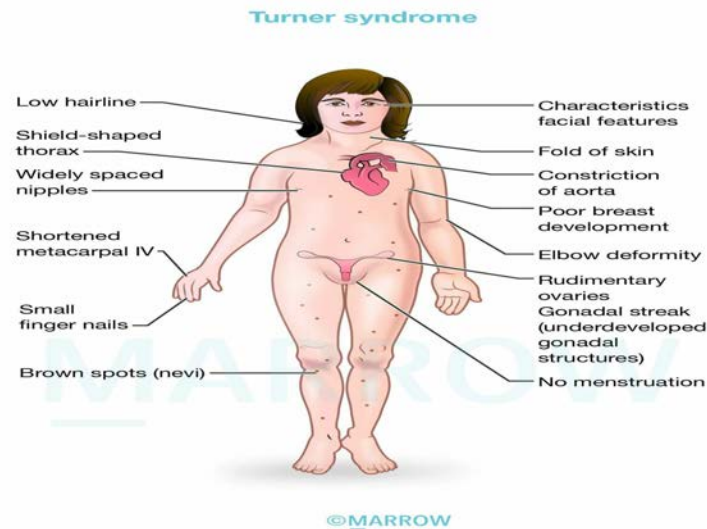
Solution to Question 8:

The given karyotyping image shows a “45 XO” abnormality, which is seen in Turner's syndrome. Clinical features of Turner's syndrome include short stature, low posterior hairline, broad chest, widely spaced nipples, coarctation of the aorta, bicuspid aortic valve, cubitus valgus, and pigmented nevi.

Turner's syndrome is the most common sex chromosome abnormality in females. The classical karyotype is 45, XO. Other variants have structural abnormalities of X chromosomes or mosaic patterns. Some patients may present during infancy. They mainly present with oedema of the dorsum of the hand and foot due to lymphatic stasis. The most important cause of increased mortality in children with Turner's syndrome is cardiac abnormalities.

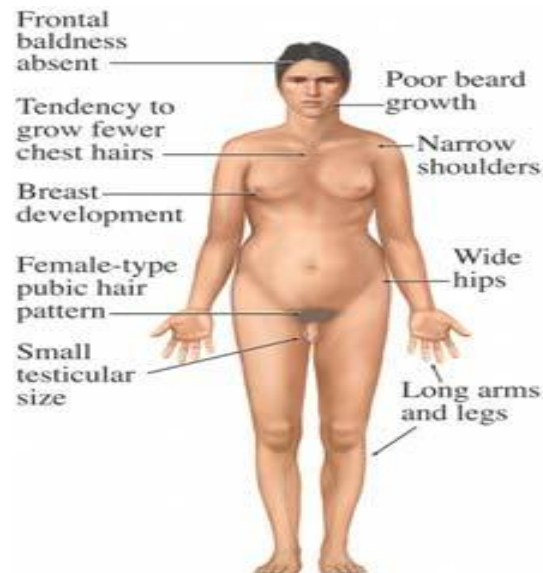
Turner's syndrome is a common cause of primary amenorrhea. During embryogenesis, fetal ovaries develop normally and the absence of a second X chromosome leads to accelerated loss of normally formed oocytes. This gets completed by the age of 2. The ovaries are reduced to atrophic fibrous bands without ova and follicles. These are called streak ovaries.

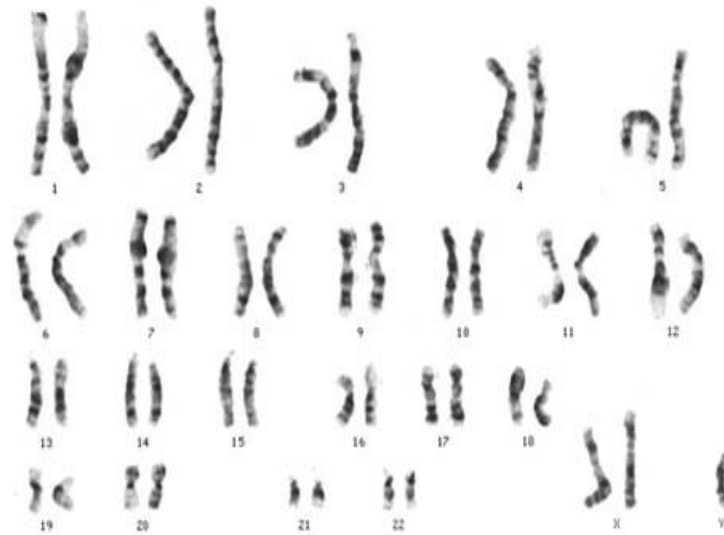
At puberty, they fail to develop secondary sexual characteristics.



Other options:

Option B: Eunuchoid body habitus with abnormally long legs is usually seen in Klinefelter Syndrome (47, XXY). The image below shows the clinical features and karyotyping of Klinefelter Syndrome respectively. Note the two X and one Y chromosomes.





Option C : Kallman's syndrome is a hypothalamic disorder that occurs due to defective gonadotropin-releasing hormone (GnRH).

It is mainly associated with anosmia or hyposmia or hypoplasia. Other associated features such as colour blindness, optic atrophy, deafness, cleft palate, renal abnormalities, and cryptorchidism may be present.

Option D : Androgen insensitivity syndrome is an X-linked recessive disorder in which the androgen receptors are absent, scanty, or fail to respond to testosterone. In complete AIS, the individual is born as a phenotypically normal female but generally presents with primary amenorrhea.

The external genitalia appears normal, with absent or scant pubic and axillary hair. The vagina is shortened and blind-ending with an absent uterus and ovaries. These patients may also present with a history of childhood inguinal hernias, due to the presence of undescended testis in the abdomen or in the inguinal canal. Due to the constant conversion of testosterone into estrogen, these individuals have normal breast development.

Solution to Question 9:

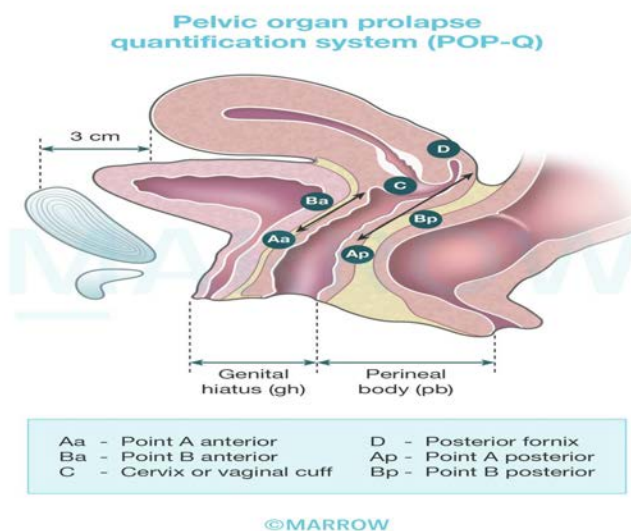
The given image shows POP-Q (pelvic organ prolapse quantification) system which was described by the International Continence Society for the prolapse of pelvic organs and has the following features:

- The hymen is taken as a fixed point
- Six reference points are measured, using a scaled spatula:
 - Two on the anterior vaginal wall
 - Two on the apical vagina
 - Two on the posterior vaginal wall
- Three measurements: The genital hiatus, perineal body, and total vaginal length.

- The points above the hymen are described as minus and points below as plus.

Staging of POP Q:

Stage	Description
Stage 0	No descent of pelvic organs.
Stage 1	The leading edge of the prolapse is >1 cm above the level of the hymen.
Stage 2	The leading edge of the prolapse is <1 cm proximal or distal to the hymen.
Stage 3	The leading point of prolapse is >1 cm below the hymen but protrudes no further than 2 cm less than the total length of the vagina.
Stage 4	Complete eversion of vagina



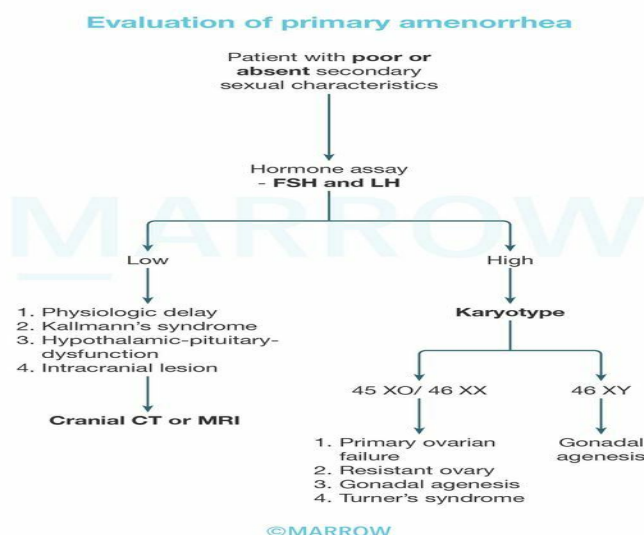
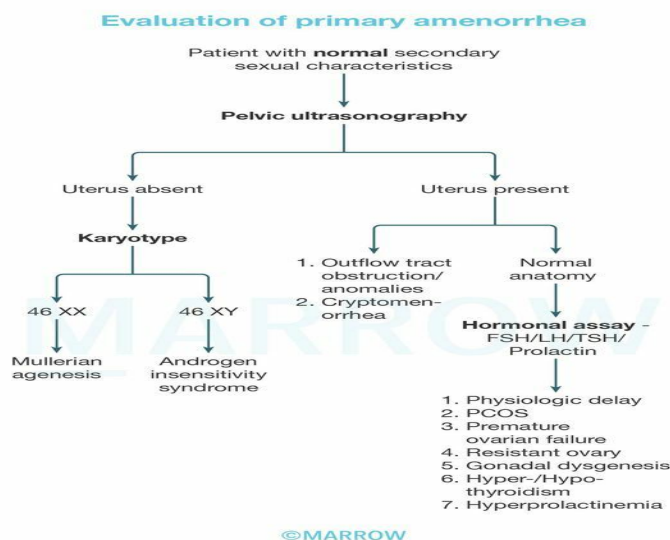
Solution to Question 10:

The given clinical scenario is that of a female with primary amenorrhea, preservation of secondary characteristics with normal uterus & ovaries and an image showing imperforate hymen. The karyotype is normal, i.e, 46 XX karyotype.

Other options:

Option A & B: Gonadal dysgenesis in the form of 45XO (Turner syndrome) or 46XY (Swyer syndrome) presents with poor development of secondary sexual characteristics. Since the gonads are nonfunctional, there is a loss of negative feedback with increased levels of FSH and LH.

Option C: Klinefelter syndrome, characterized by an XXY genotype, is commonly associated with males having atrophic testes with small penis. Other features include tall stature, gynecomastia, lack of secondary sexual characteristics, and infertility.



Solution to Question 11:

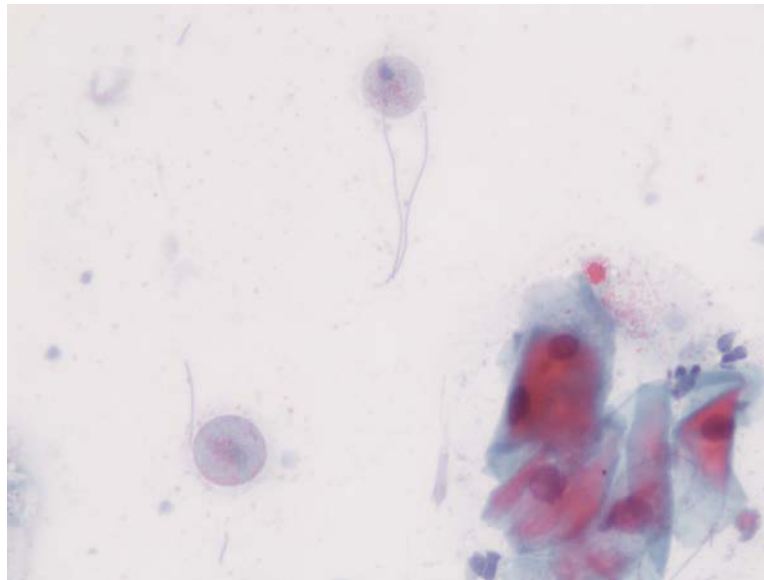
This patient with symptoms of dysuria, pelvic pain, and abnormal vaginal discharge and the above pap smear shows large hyperchromatic nuclei and a perinuclear halo which is suggestive of koilocytes, which is a characteristic feature of infection caused by human papillomavirus (HPV).

Other features that can also be seen in histology are:

- Perinuclear halo
- Multinucleation
- Orangeophilic(Keratinisation) cytoplasm
- Acanthosis
- Chronic inflammatory infiltrate

Other options:

Option A: *Trichomonas vaginalis* is flagellated protozoa that causes trichomonas vaginitis. The pap smear shows a greyish-blue pear-shaped structure without the flagella as shown in the image below. The hanging drop test will reveal the presence of an actively motile pear-shaped flagellate organism.



Option C: Cytomegalovirus infection may be seen in pap smears rarely. Pap smear shows large, basophilic intranuclear inclusions or intracytoplasmic eosinophilic inclusions in endocervical glandular epithelial cells.

Option D: *Candida albicans* causes candidal vaginitis. On pap smear, It shows pseudohyphae and yeast cells and reactive squamous epithelial cells appear in the form of "shish kebab" as shown in the image below.



Solution to Question 12:

A lawyer is not a part of the medical board. According to the MTP Act with 2021 amendments, a medical board consisting of an obstetrician, a paediatrician, a radiologist and a person appointed by the state.

Solution to Question 13:

The pregnant woman with DVT at 30 weeks of gestation will be managed by low molecular weight heparin.

LMWH is the drug of choice in the management of DVT in any trimester of pregnancy, as it does not cross the placenta to cause fetal malformations. It has better bioavailability, longer half-life, predictable dose response and reduced adverse effects like osteoporosis and thrombocytopenia.

LMWH is given subcutaneously twice a day for up to 3- 6 months, followed by a prophylactic dose for the remaining pregnancy and post-partum period. After the symptoms subside, graduated compression stockings can be used for two years to reduce post-thrombotic syndromes like chronic leg paresthesia, intractable oedema and leg ulcers.

Other options:

Option B- Warfarin is used in the second trimester in patients with mechanical valves even though it is teratogenic, as LMWH does not provide adequate therapeutic effects and is associated with increased maternal thromboembolic events.

Option C and D- Fondaparinux and rivaroxaban are not typically used in pregnant women due to insufficient data about their safety during pregnancy. Fondaparinux is reserved for cases with heparin-induced thrombocytopenia and for those who do not tolerate heparin.

Solution to Question 14:

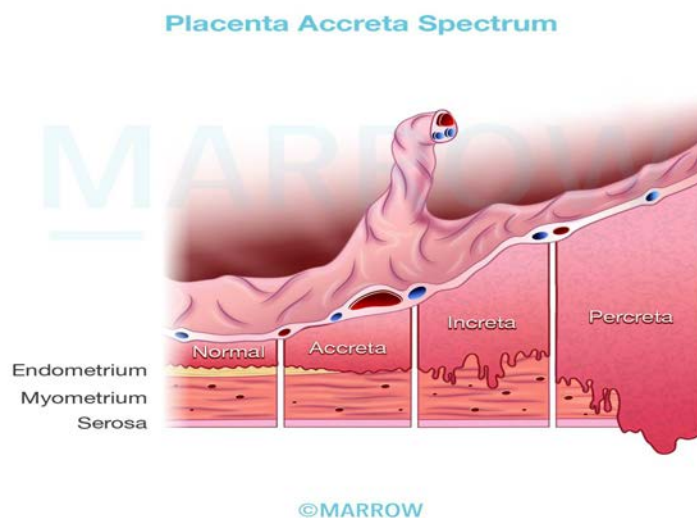
The given clinical scenario of this woman with previous LSCS, bleeding per vaginum during pregnancy with hysterectomy image showing placenta adherent to the uterine wall suggests that it is a case of placenta accreta syndrome.

Placenta accreta is described as an abnormally implanted, invasive, or adhered placenta. It is due to the partial or total absence of the decidua basalis and imperfect development of the Nitabuch layer.

Nitabuch layer is a layer of fibrin between the compact endometrium and the cytotrophoblastic shell in the placenta. This layer is absent in the placenta accreta.

There are 3 variants of placenta accreta syndrome. They are :

- Placenta accreta - villi are attached to the myometrium.
- Placenta increta - villi invade the myometrium.
- Placenta percreta - villi penetrate through the myometrium and to or through the serosa.



Solution to Question 15:

A urine output of 60 ml in 4 hours (≤ 30 ml/hr) is an indication to stop MgSO_4 to prevent magnesium toxicity.

MgSO_4 is used as an anticonvulsant in the management of eclampsia and as prophylaxis in severe preeclampsia. It can also be used as a tocolytic to stop preterm labour. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L.

Patients must be monitored for magnesium toxicity:

- Loss of deep tendon reflexes
- Decreased respiratory rate ≤ 12 /min

- Urine output ≥ 30 mL/h
- Chest pain and heart block
- Visual changes, flushing, muscle paralysis, and drowsiness

Measurement of serum magnesium levels is indicated to monitor magnesium toxicity if serum creatinine ≥ 1.0 mg/dL.

Management of magnesium toxicity is as follows:

- Stop magnesium therapy
- Estimation of serum magnesium and creatinine levels
- Intravenous injection of calcium gluconate 10 mL (10% solution)

The two commonly used MgSO₄ regimens are described below:

Repeated/maintenance dose IM injections of magnesium sulfate are continued only under the following conditions:

- The patellar reflex is present
- Respiration is not depressed
- Urine output ≥ 30 mL/h

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4 g (20% solution) IV over 3–5 minutes, followed by 10 g (50%) deep IM (5 g in each buttock)	5 g (50%) IM 4 hourly in the alternate buttock
Intravenous (Zuspan or Sibai)	4 g IV slow over 15–20 minutes	1–2 g/h IV infusion

Solution to Question 16:

According to PCPNDT act, the radiologist should check for congenital anomalies of the fetus but not look for the sex.

Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994 is an act of the Parliament of India enacted to stop female foeticides and arrest the declining sex ratio in India.

The few basic requirements of the Act are:

Registration done under Section (18) of the PC-PNDT Act.

Written consent of the pregnant woman and there should be prohibition of communicating the sex of fetus under Section 5 of the Act.

Maintenance of records should be done as provided under Section 29 of the Act.

Creating public awareness at large by placing the board of prohibition on sex determination.

Initiatives Against Sex-Selective Abortions :

Beti Bachao Beti Padhao: Launched by the Government of India in 2015, it aims to address the issue of declining child, sex ratio and empower girls through awareness generation, education, and protection.

The National Plan of Action for Children, 2016 : It identifies the elimination of gender-biased sex selection as one of the key priority areas for children's rights and well-being.



Beti Bachao, Beti Padhao - "Save the daughter, educate the daughter" is a campaign of the Government of India that aims to generate awareness and improve the efficiency of welfare services intended for girls.

The objectives of BBBP are :

- To prevent gender-biased sex-selective elimination
- To ensure survival and protection of the girl child
- To ensure education and participation of the girl child

Solution to Question 17:

The clinical scenario of secondary infertility with previous uneventful delivery and the laparoscopy image showing multiple granulomatous foci which are tuberculous lesions point towards the diagnosis of genital tuberculosis.

Genital tuberculosis is almost always secondary to a primary focus elsewhere. The most frequent location is the lungs (50%), although it can also come from lymph nodes (40%), kidneys, joints, the gastrointestinal system, or as part of a broad miliary infection. It mostly spreads via the haematogenous route or lymphatics sometimes.

In Genital tuberculosis, the leading presenting complaints include infertility, abdominal pain, vaginal discharge, and suspicion of neoplasm. 10-15% of women are found to be asymptomatic as well.

The fallopian tubes are the most common site to be affected. The disease subsequently spreads to other genital organs and the peritoneum.

Investigations:

1. Complete blood count, erythrocyte sedimentation rate (ESR), enzyme-linked immunoabsorbent assay (ELISA) tests - IgG and IgM, and Mantoux tests are done to evaluate the case of genital tuberculosis. The nucleic acid amplification technique (NAAT) can detect tuberculosis within a few hours compared to culture.

2. Hysterosalpingography (HSG) - If genital tuberculosis is suspected, HSG should not be performed because of the risk of spread of infection.

If performed in an asymptomatic woman, the following patterns may be seen:

- The tobacco-pouch appearance of hydrosalpinx and pyosalpinx with blocked fimbrial end.
- Rigid non-peristaltic pipe-like tube (lead-pipe appearance).
- Beading and variation in filling defect.
- The jagged fluffiness of the tubal outline.
- Golf club appearance with slight or moderate dilatation of the ampullary portion of the fallopian tube.

3. The investigation of choice in genital tuberculosis is an endometrial biopsy. Endometrium tissue is obtained by D&C/hysteroscopy-directed biopsy. The endometrial scrapings are used for histopathology, PCR testing, and AFB smear and culture.

Diagnostic laparoscopy is widely used to establish the diagnosis of genital tuberculosis/ abdominal pelvic tuberculosis. Tuberculous lesions can be seen on the parietal peritoneum, intestinal serosa, omentum, surface of the uterus, and fallopian tubes.

Solution to Question 18:

Liquid-based cytology, Cervical biopsy, and HPV antigen testing (Option A) are the most appropriate investigations required in this patient with spotting, foul discharge, and cervical erosion which are the features suggestive of cervical cancer.

In liquid-based cytology, a smeared plastic spatula is placed in the liquid fixative (methanol) which removes blood, mucus, and inflammatory cells from epithelial cells. The epithelial cells are then sucked into a filter and pressed onto a glass slide to form a monolayer. This is done to screen patients for cervical cancer.

Differentiating between cervical erosion/ectropion and cervical intraepithelial neoplasia (CIN) and cervical cancer is difficult on macroscopic visualization, therefore cervical biopsy is required to rule out CIN and cervical cancer.

HPV antigen testing is required as HPV-16 is associated with anogenital warts and cervical carcinoma.

Pap/ Papanicolaou test is a screening test only. A positive test requires further evaluation - colposcopy, cervical biopsy, and fractional curettage should be done.

Solution to Question 19:

Image A shows 1st twin in the breech presentation and Image B shows 1st twin in the vertex presentation, Therefore the mode of delivery for images A and B would be LSCS and Vaginal delivery respectively. (Option B)

Timing of delivery depends on the type of twin pregnancy:

The mode of delivery depends on the 1st twin:

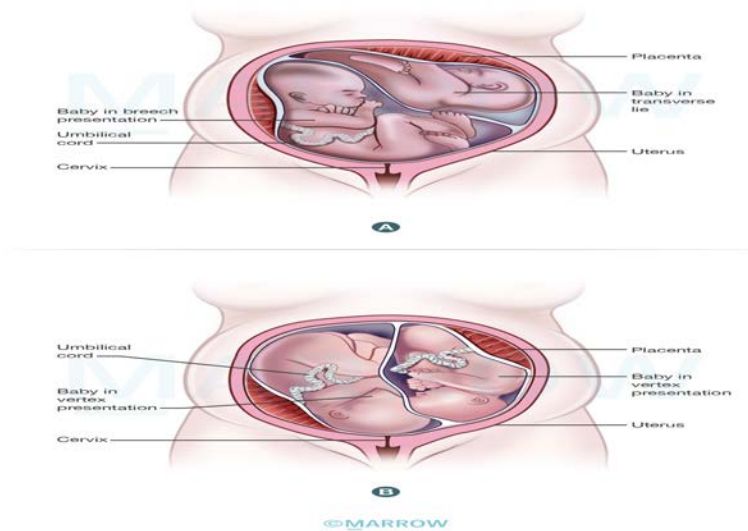
After delivery of 1st twin, if the 2nd twin is in breech, assisted breech delivery is done. If the 2nd twin is in a transverse lie, the internal podalic version is done under GA in the OT followed by breech extraction.

Complications in attempting vaginal delivery in non-vertex presentation (i.e) transverse and breech:

- Interlocking of twins
- Fetal head trapping above the cervix
- Umbilical cord prolapse

Type of twin pregnancy	Timing of delivery
Dichorionic diamniotic	38 weeks
Monochorionic diamniotic	34 - 37 weeks + 6 days
Monochorionic monoamniotic	32 - 34 weeks by Caesarean section

1st twin presentation	Mode of delivery
Vertex	Vaginal
Breech	Cesarean section
Transverse	Cesarean section



Solution to Question 20:

The earliest time to insert an IUCD is within 48 hours.

PPIUCD (postpartum IUCD) can be inserted:

- (a) post-placental (within 10 minutes of placental delivery)
- (b) within 48 hours postpartum
- (c) during cesarean section
- (d) 4-6 weeks after postpartum.

With PPIUCD, the chances of perforation are less, and the expulsion rate is slightly high.

The timing of the insertion of IUCD is as follows:

Interval insertion (inter-conceptional period)- preferable to insert 2–3 days after menstruation, but it can be inserted at any time during the cycle. It can also be inserted during menstruation which has certain advantages like an open cervical canal and a distended uterine cavity

It can also be inserted within 5 days of coitus as an effective method of emergency contraception (Eg: incase of ruptured condom during intercourse)

Solution to Question 21:

The drug of choice for the treatment of thyrotoxicosis during the first trimester of pregnancy is propylthiouracil (PTU). It is because it has lower teratogenic effects than carbimazole or methimazole.

Carbimazole and methimazole are associated with teratogenic effects such as:

- Aplasia cutis congenita

- Choanal atresia
- Treacheosophageal fistula
- Fetal goiter

However, propylthiouracil use is limited to the first trimester due to the risk of hepatotoxicity. It is replaced with carbimazole or methimazole in the 2nd and 3rd trimesters of the pregnancy.

The lowest effective antithyroid drug dose should be used throughout gestation. This is to maintain the maternal serum-free T₄ level in the upper limit of the non-pregnant normal reference range.

Propranolol is a non-selective beta blocker. It is used in thyroid storm preferentially as it blocks the peripheral conversion of inactive T₄ to active form T₃.

Solution to Question 22:

In the clinical scenario, the smooth, globular structure shown during examination as well as the presentation of heavy menstrual bleeding suggests the diagnosis of myomatous polyp.

Myomatous Polyp, also known as a fibroid polyp, is a type of benign tumor originating from the smooth muscle cells of the uterus, called leiomyomas. These polyps can protrude from the endometrial cavity or cervix and may appear as a smooth, pinkish, round mass, which can sometimes be seen on physical examination, especially when they extend into the vagina.

Fibroids can cause symptoms like heavy menstrual bleeding (menorrhagia), pelvic pain, and pressure. These polyps may remain attached to the uterine wall via a stalk, which can give them a globular appearance.

Since myomatous polyps are associated with fibroids, they tend to occur in women of reproductive age and may persist into the perimenopausal period, as seen in this 42-year-old patient.

Other options:

Option A: Hypertrophied Cervix: A hypertrophied or enlarged cervix is not typically described as a smooth, globular mass. It usually does not present with a history of heavy bleeding.

Option C: Polypoid adenocarcinoma mainly occurs due to malignant changes of uterine or endometrial polyp in post-menopausal women. Endometrial polyps, another name for uterine polyps, are soft, microscopic growths that develop inside a woman's uterus, or womb. They originate from the endometrium, the tissue that lines the uterus.

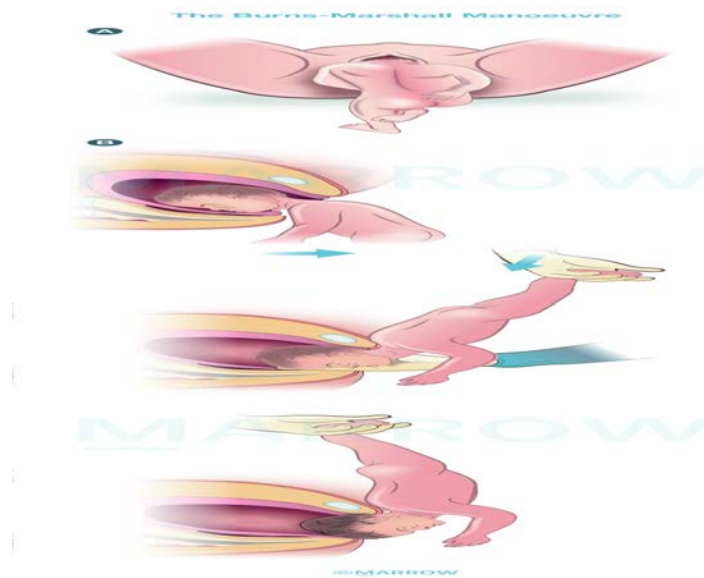
Option D: Placental Polyp: Placental polyps are remnants of placental tissue and can cause abnormal bleeding but typically do not present with the smooth, globular structure.

Solution to Question 23:

The image shows Marshall Burns's technique used in the delivery of aftercoming head in breech delivery.

Technique:

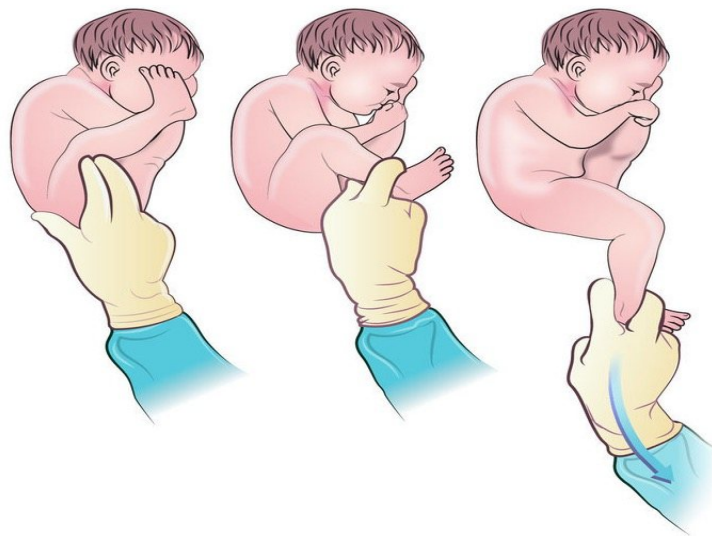
- Baby is allowed to hang by its weight till the nape of the neck is visible
- When the nape of the neck is visible under the pubic arch, the baby is grasped by the ankles
- Baby is swung in an arc in an upward and forward direction towards the mother's abdomen



Other Options:

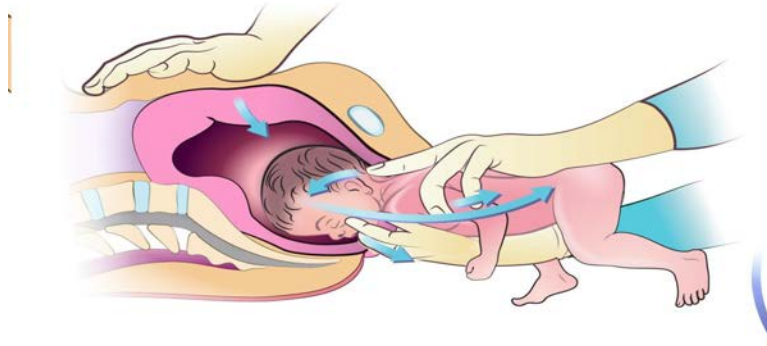
Option A: Pinard's manoeuvre used for delivery of extended legs in case of breech delivery.

Pinard's maneuver



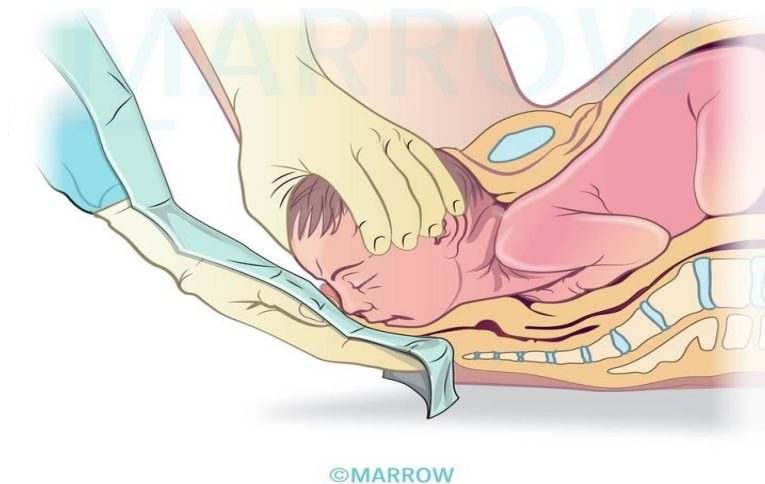
Option B: The Mauriceau-Smellie-Veit technique is another method that can be used in the delivery of the aftercoming head in case of breech presentation.

Modified Mauriceau-Smellie-Veit technique



Option D: The Ritgen manoeuvre is used for controlled delivery of the head in normal labour.

Ritgen's maneuver



Solution to Question 24:

Transvaginal ultrasound findings of a left adnexal cystic mass, with incomplete septa and no vascular flow along with the complaints of lower abdominal pain suggest the diagnosis of hydrosalpinx.

A hydrosalpinx is a chronic cystic swelling of the fallopian tube that forms after a distal tubal obstruction. The cogwheel appearance and incomplete septa described in the ultrasound can correspond to the appearance of a hydrosalpinx. The lack of vascular flow on Doppler imaging supports this diagnosis. The tubes are enormously distended due to filling with clear serous fluid and become retort-shaped. On histology, the tubal plicae appear flattened. The interstitial end of the tube is open whereas the fimbrial end is indrawn and closed. This permits the dye to enter but

not exit the tube during HSG.

It is caused due to:

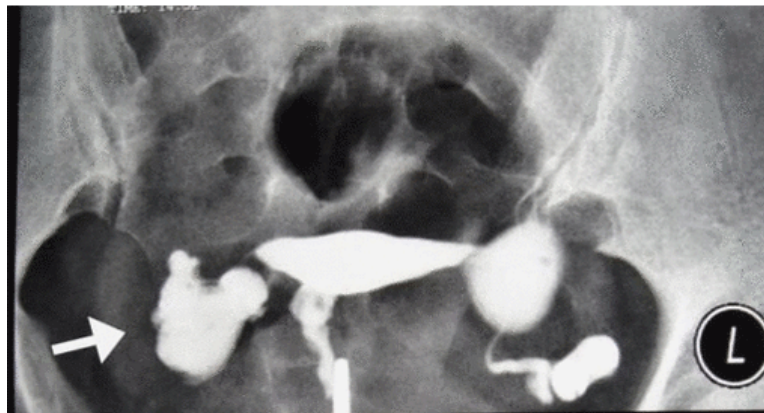
- Pelvic inflammatory disease
- Endometriosis
- Tuberculosis - tubercular salpingitis may also cause tobacco-pouch appearance, lead pipe appearance and pyosalpinx
- Fallopian tube neoplasms (rare)

It can be an incidental finding during a pelvic exam or ultrasound. The patient may be asymptomatic, or present with infertility or chronic pelvic pain.

On HSG, it appears as terminal sacculation and occlusion of both fallopian tubes with a normal uterine cavity. Usually, a thin-walled, hypoechoic, cystic structure is seen. However, it may appear as multiple hyperechoic mural nodules around the inner circumference of the tube creating a 'beads on a string' appearance.

Treatment:

- Treat the underlying cause
- Laparoscopic salpingostomy - in mild disease
- IVF recommended for women with infertility and severe hydrosalpinx



©MARROW

Other options:

Option B: Malignant ovarian cyst: Malignant ovarian cysts can appear as complex masses with irregular shapes and internal vascularity.

Option C: Follicular cyst: A follicular cyst is a common type of ovarian cyst that occurs when a follicle fails to release an egg. It typically appears as a simple cyst on ultrasound.

Option D: Hemorrhagic cyst: A hemorrhagic cyst results from bleeding into a pre-existing cyst and often shows complex internal echoes and potentially some vascular flow on Doppler imaging.

Solution to Question 25:

The most appropriate next step in management for a patient with bilateral tubal blockage on hysterosalpingography is laparoscopic chromopertubation.

Tubal contractions, particularly cornual spasms, can give an incorrect impression of proximal fallopian tube obstruction (a false-positive result) on hysterosalpingography. Laparoscopy with chromotubation is therefore considered to confirm the diagnosis of cornual spasm. Laparoscopy is both diagnostic and therapeutic in this condition.

Laparoscopic chromopertubation involves the laparoscopic visualization of pelvic structures, including the uterus, fallopian tubes, and ovaries, followed by the injection of methylene blue dye through the cervix to assess tubal patency. In addition to evaluating tubal patency, laparoscopy can help diagnose peritubal adhesions, endometriosis, and other pelvic abnormalities.

Chromotubation may be performed, in which a dilute dye is injected through a Leech-Wilkinson cannula placed against the cervix or through a balloon catheter positioned within the uterine cavity. Tubal spill is evaluated through the laparoscope. Indigo carmine dye, if available, is preferable to methylene blue, as methylene blue rarely may induce acute methemoglobinemia, particularly in patients with glucose-6-phosphate dehydrogenase deficiency. It is commonly done in the proliferative phase.

Other options:

Option A: Intrauterine insemination (IUI): IUI is used for treating infertility but relies on open fallopian tubes for natural fertilization.

Option B: In vitro fertilization (IVF): While IVF is an effective treatment for infertility due to tubal blockage, the immediate next step to confirm and evaluate the extent of tubal blockage is typically laparoscopic chromopertubation.

Option D: Ovulation induction: Ovulation induction aims to stimulate egg release but does not address tubal blockage.

Solution to Question 26:

The given image shows a perineal tear extending up to the rectal mucosa, therefore, it is a grade 4 perineal tear.

The involvement of rectal mucosa as shown in the image differentiates it from grade 3 tear. Grade 3 tear involves the vaginal mucosa, perineal skin, perineal muscles, and the anal sphincter complex (external and/or internal), but not rectal mucosa, while grade 4 tear involves all these layers including the rectal mucosa.

Minor perineal injuries are quite common, especially during the first birth. Gross injuries like third or fourth degrees of perineal tears are a result of mismanagement of the second stage of labour. This can be prevented by proper conduct of the second stage labour.

Management is as follows:

All recent tears should be repaired immediately after the delivery of the placenta. If there is a delay of 24 hours, the repair has to be withheld and the patient has to be started on antibiotics to prevent infection. Complete perineal tears that were not repaired in the first 24 hours should be operated on between 3-6 months after delivery.

In 3 months of time, any local infection that might have been present would subside, the general condition of the patient will also improve. However, if the perineal tear is untreated for more than 6 months, dense scar tissue forms and making the surgery difficult.

Solution to Question 27:

The most common site for tubal ligation is the isthmus of the fallopian tube due to its uniform muscle thickness, which facilitates effective occlusion.

The isthmus, being a narrow segment closest to the uterine wall, provides an ideal location for surgical intervention. This site ensures a more reliable and efficient ligation, minimizing the risk of complications and maximizing the success rate of the procedure. The uniformity of the muscle layer at this location allows for a consistent occlusion technique. Common procedures for tubal ligation at this site include the Pomeroy method, where a loop of the tube is tied and excised, and other techniques like the Falope ring or the electrocoagulation method.

While the ampulla, which has a larger-diameter segment lateral to the isthmus, is also a part of the fallopian tube, it is less frequently chosen for tubal ligation. Additionally, ligation at the isthmus allows for the possibility of future reversal, should the patient desire to conceive again, making it a favourable choice for women considering sterilization.

The fallopian tube consists of three layers: the mesosalpinx, an outer mesothelial layer serving as visceral peritoneum; the myosalpinx, which has an inner circular smooth muscle layer for pressure during contractions and an outer longitudinal layer for tube movement; and the endosalpinx, an inner columnar epithelium with ciliated and secretory cells, featuring longitudinal folds. Together, these layers transport the ovum efficiently toward the uterine cavity.

Note: The interstitial portion of the fallopian tube is the narrowest segment, located within the uterine wall, and it forms the tubal ostia that opens into the endometrial cavity.

Solution to Question 28:

The appropriate treatment for a 30-year-old female with infravaginal elongation of the cervix is Fothergill surgery.

Fothergill surgery, or the Manchester operation, involves performing an anterior colporrhaphy combined with the amputation of the cervix. During the procedure, the Mackenrodt ligaments are sutured in front of the cervix, and the operation concludes with a colpoperineorrhaphy. This surgical approach is particularly suitable for women under 40 years old who wish to maintain reproductive function, although it's important to note that fertility may be reduced following the procedure.

Various other surgeries were performed for the management of uterovaginal prolapse:

- Anterior colporrhaphy: This is done for cystocele or cystourethrocele.
- Posterior colporrhaphy/ colpoperineorrhaphy: This is done for posterior defects like rectocele or rectal defects with perineal laxity.
- Le Forte's repair(option B): This is preferred only in elderly women with advanced prolapse, or those unfit for surgery. Women wanting an active sexual life are poor candidates for this procedure as it creates an adhesive area in the midline.
- Abdominal sling surgeries: Young women desirous of retaining menstrual and reproductive functions are ideal candidates. These surgeries use tapes as slings to prevent organ descent. These include:
 - Abdominal wall cervicopexy (modified by Purandare and Mathre) - sling attached to cervix posteriorly near uterosacral ligaments.
 - Shirodkar's sling surgery - the sling fixes the cervix to the lumbosacral fascia.
 - Khanna's sling surgery - sling fixed to the anterior superior iliac spine, strengthening the Mackenrodt's ligament.
 - Fothergill surgery (Manchester operation)
 - Ward-Mayo surgery: This is a combination of vaginal hysterectomy + pelvic floor repair. It is done in women >40 years, not desiring reproductive function.
 - Moschowitz surgery: This is performed for enterocele repair where the pouch of Douglas is obliterated.

Management of Vault prolapse:

- If the patient is fit for abdominal surgery: Transabdominal sacral colpopexy, i.e., suspending the vault to the sacral promontory extraperitoneally. It is best suited for younger women since coital difficulty following vaginal surgery is avoided
- If the patient is not fit for abdominal surgery: Right Transvaginal sacrospinous colpopexy. It is usually preferred in obese and elderly women, who are not fit for abdominal surgery.
- Colpocleisis is permissible if sexual activity is not desired.

Other options:

Option A: McCall culdoplasty is primarily used to prevent enterocele formation and provide support to the posterior vaginal apex, particularly following vaginal hysterectomy.

Option D: The Latzko procedure is a surgical technique used to treat vesicovaginal fistula. It involves removing the vaginal epithelium around the fistula and approximating the edges to promote healing.

Solution to Question 29:

In abdominal sling procedures such as sacral colpopexy, which is used to treat pelvic organ prolapse while preserving fertility, the mesh is typically attached to the sacral promontory. This site provides strong support for the prolapsed vaginal apex and allows for a durable suspension without significant compromise to the patient's fertility.

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 - Fothergill surgery (Manchester operation): An anterior colporrhaphy is combined with amputation of the cervix. The Mackenrodt ligaments are then sutured in front of the cervix followed by a colpoperineorrhaphy. It is suitable for women <40 years desiring reproductive function, but fertility may be reduced.
 - Ward-Mayo surgery: This is a combination of vaginal hysterectomy + pelvic floor repair. It is done in women >40 years, not desiring reproductive function.
 - Moschowitz surgery: This is performed for enterocele repair where the pouch of Douglas is obliterated.

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- Colpocleisis is permissible if sexual activity is not desired

Solution to Question 30:

This patient presenting with symptoms of cyclical pain at the episiotomy site with image showing perineal scar endometriosis has ectopic endometrial tissue. Endometriosis can occur at the site of a surgical scar and typically presents with cyclical pain that corresponds with the menstrual cycle.

The extra-uterine sites of endometrial tissue are influenced by cyclical changes in the hormone levels which give rise to location-specific symptoms as mentioned above.

Management:

Investigations include:

- USG: TVS will show a chocolate cyst.
- MRI
- CA-125 more than 35 U/mL
- Laparoscopy: Gold standard for the diagnosis and treatment of endometriosis. It is used to locate the sites, take a biopsy and surgically remove the deposits.

Drugs used for symptom control include the following:

- Oral contraceptive pills
- Mirena IUD
- GnRH analogs
- Letrozole

Surgical management includes:

Minimally invasive surgery may be used to destroy the tissue by cautery, laser vaporization, and lyse adhesions that have formed as a complication of endometriosis.

Major surgical measures include incision of chocolate cyst and removal of lining, salpingo-oophorectomy, hysterectomy, and unilateral or bilateral salpingo-oophorectomy.

Other options:

Option A: Retained Sutures: Retained sutures can cause localized pain and inflammation, but this is usually accompanied by visible signs of infection or irritation rather than cyclical pain linked to menstruation.

Option B: Melanoma: Melanoma is a type of skin cancer that would usually present as a suspicious, irregularly shaped lesion rather than cyclical pain at the site of a previous surgical scar.

Option C: Inclusion Cyst: Inclusion cysts are benign and can cause localized pain or discomfort. However, they typically do not present with pain that worsens specifically during menstruation.

Solution to Question 31:

The most likely diagnosis is this patient with two cervixes on examination and two uterine domes as seen in the image is uterine didelphys. It is a congenital anomaly where there are two separate uterine cavities and two cervixes, each with its own set of endometrial lining.

Uterine didelphys is commonly associated with renal anomalies such as unilateral renal agenesis, severe renal hypoplasia, etc, and is often associated with

- Recurrent pregnancy loss
- Preterm labor
- Malpresentations
- Prematurity

Per speculum examination will reveal two cervixes in such women.

USG is a non-invasive, easily available, convenient, and inexpensive imaging modality. 3-D USG is sufficient to diagnose uterine didelphys without further imaging. The USG findings seen are:

- Two widely separated uterine horns
- A deep fundal indentation

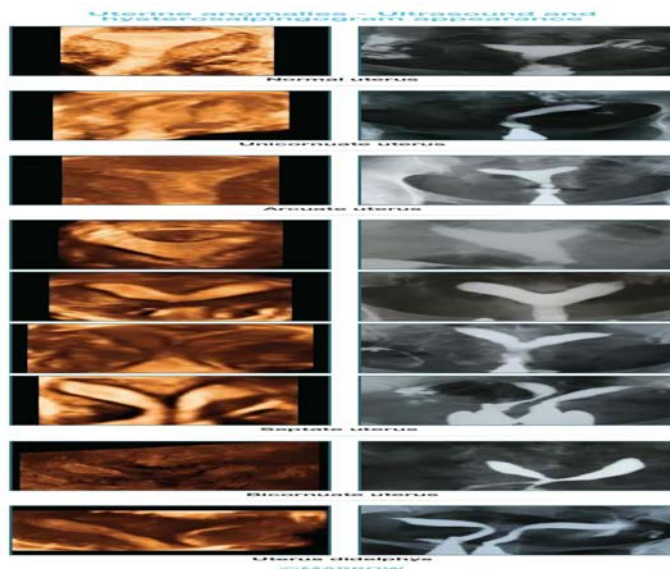
MRI is used in cases where a definitive diagnosis cannot be made based on ultrasonography. Invasive modalities such as hysteroscopy and laparoscopy are reserved for patients with complicated Mullerian anomalies.

Other options:

Option A: Unicornuate Uterus with Rudimentary Horn: This condition involves a single functioning uterine horn and a non-functional rudimentary horn. It presents with a single cervix and an abnormal horn that may cause complications.

Option B: Arcuate Uterus: An arcuate uterus has a normal single uterine cavity with a small indentation at the top of the uterus, which is a minor congenital anomaly.

Option C: Bicornuate Uterus: A bicornuate uterus has two cavities within a single uterus but typically only one cervix. It may have a partial separation or indentation at the top of the uterus but would not present with two cervixes.



Solution to Question 32:

The most appropriate course of action in this patient at 10 weeks intrauterine gestation is to Remove the IUCD immediately and continue pregnancy.

Intrauterine Devices in Pregnancy

- If IUD strings are visible and the pregnancy is less than 13 weeks, the IUD should be removed as soon as possible to prevent later septic abortion, premature rupture of the membranes, and premature birth .

- When the strings of the IUD are not visible, an ultrasound examination should be performed to localize the IUD and determine whether expulsion has occurred. If the IUD is present, there are three options for management:
- Therapeutic abortion .
- Ultrasound-guided intrauterine removal of the IUD.
- Continuation of the pregnancy with the device left in place

Note :

If the patient wishes to continue the pregnancy, ultrasound evaluation of the location of the IUD should be considered .

- If the IUD is not in a fundal location, ultrasound-guided removal using small alligator forceps may be successful.
- If the location is fundal, the IUD should be left in place. When pregnancy continues with an IUD in place, the patient must be warned of the symptoms of intrauterine infection and should be cautioned to seek care promptly for fever or flulike symptoms, abdominal cramping, or bleeding. At the earliest sign of infection, high-dose intravenous antibiotic therapy should be given and the pregnancy evacuated promptly.

Solution to Question 33:

Routine measurement of serum magnesium levels is not typically required unless clinical signs of toxicity (e.g., loss of deep tendon reflexes, respiratory depression, decreased urine output) are present.

MgSO₄ is used as an anticonvulsant in the management of eclampsia and as prophylaxis in severe preeclampsia. It can also be used as a tocolytic to stop preterm labor. The therapeutic level of magnesium in eclampsia is 4–7 mEq/L.

Patients must be monitored for magnesium toxicity:

- Loss of deep tendon reflexes (Option D)
- Decreased respiratory rate <12/min (Option C)
- Urine output <30 mL/h (Option B)
- Chest pain and heart block
- Visual changes, flushing, muscle paralysis, and drowsiness

Measurement of serum magnesium levels is indicated to monitor magnesium toxicity if serum creatinine >1.0 mg/dL.

Management of magnesium toxicity is as follows:

- Stop magnesium therapy
- Estimation of serum magnesium and creatinine levels
- Intravenous injection of calcium gluconate 10 mL (10% solution)

The two commonly used MgSO₄ regimens are described below:

Repeated/maintenance dose IM injections of magnesium sulfate are continued only under the following conditions:

- The patellar reflex is present
- Respiration is not depressed
- Urine output ≥ 30 mL/h

Regimen	Loading Dose	Maintenance Dose
Intramuscular (Pritchard)	4 g (20% solution) IV over 3–5 minutes, followed by 10 g (50%) deep IM (5 g in each buttock)	5 g (50%) IM 4 hourly in the alternate buttock
Intravenous (Zuspan or Sibai)	4 g IV slow over 15–20 minute	1–2 g/h IV infusion

Solution to Question 34:

The sign present in this girl is called Goodell's sign. Goodell's sign is a sign of early pregnancy, which describes the marked softening of the cervix in contrast to the non-pregnant state. It is seen as early as the 6th week.

Other options:

Option A: Hegar's Sign: Refers to the softening of the lower uterine segment during early pregnancy, felt during bimanual examination.

Option B: Braxton Hicks Sign: These are irregular, painless uterine contractions that can occur during pregnancy but are not associated with cervical changes.

Option C: Oslander's Sign: Oslander's sign describes the pulsations felt in the lateral fornix due to increased vascularity. It is seen around 8 weeks of gestation.

Solution to Question 35:

The procedure shown in the image is the external cephalic version (ECV). Being a primigravida (first pregnancy) is not a contraindication to ECV. The procedure can be performed in both primigravida and multigravida patients as long as other contraindications are not present.

ECV is recommended to manage transverse lie when the membranes are intact.

Prerequisites for ECV - Women who have reached 37 0/7 weeks of gestation. Multiparity, unengaged presenting part, nonanterior placenta, nonobese patient, and abundant amniotic fluid improve the chances of ECV.

Absolute contraindications - When vaginal delivery is contraindicated, e.g. placenta previa

Relative contraindications - Early labor, oligohydramnios or ruptured membranes, known nuchal cord, structural uterine abnormalities, fetal growth restriction, multifetal gestation, and prior

abruption of the placenta

Solution to Question 36:

The most likely diagnosis is endometrioma. The patient's chronic lower abdominal pain, which worsens during menstruation, and the ultrasound finding of a right adnexal mass with homogeneous echoes are highly suggestive of an endometrioma.

Endometriosis is defined by the presence of ectopic endometrial tissue at a site outside of the uterus. It is a benign condition but it is locally invasive. Some common sites of endometriosis are ovaries, pelvic peritoneum, pouch of Douglas, uterosacral ligaments, rectovaginal septum, and sigmoid colon. Endometriosis in the ovaries is called endometrioma.

Chocolate cysts of the ovaries represent the most important manifestation of endometriosis.

Patients with endometriosis mainly present with progressively increasing secondary dysmenorrhea, menorrhagia, infertility, dyspareunia, and chronic pelvic pain.

MRI is the diagnostic tool, it shows the characteristic hyperintensity on T1 weighted images and hypointensity on T2 weighted images. Laparoscopy confirms the diagnosis.

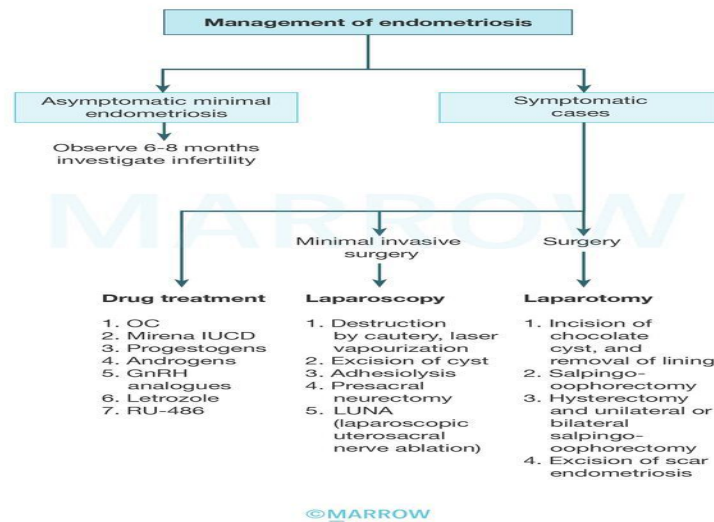
Medical management of endometriosis:

- Gestrinone is an anti-estrogenic, anti-progestational, and mildly anabolic drug that reduces sex hormone-binding globulin to release free testosterone. It also decreases the LH surge. It causes regression of lesions and amenorrhea. It is a 19-nortestosterone derivative similar to danazol but with fewer side effects.
- Combined oral contraceptives are used for symptom control and alleviating disease. These also cause menstrual cycle regulation and delay pregnancy.
- Oral progestogens exert an anti-estrogenic effect and cause decidualization and endometrial atrophy. This causes disease regression. It doesn't prevent ovulation, thus suitable for women trying to conceive.
- Mirena IUCDs reduce dysmenorrhea and menorrhagia. It is a single-sitting treatment with minimal side effects.
- Aromatase inhibitors like letrozole are anti-estrogens given for disease regression.
- Anti-inflammatory drugs like mefenamic acid and naproxen relieve dysmenorrhea.

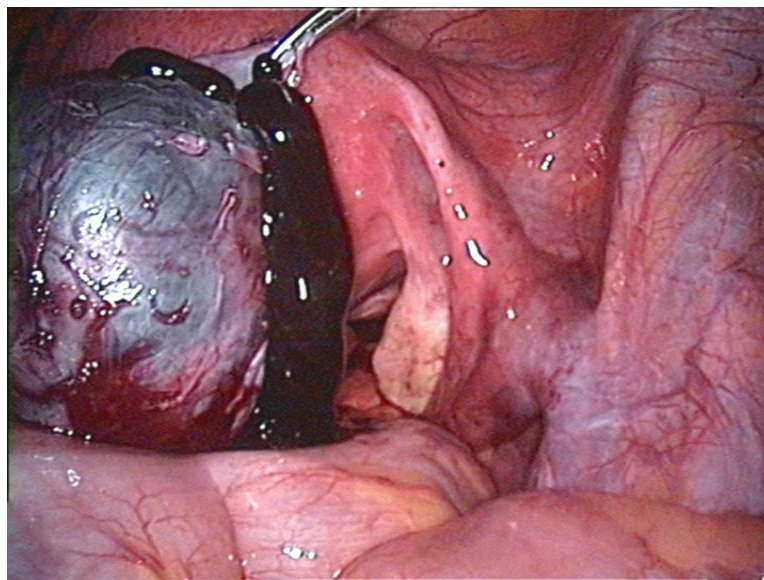
Therapeutic procedures:

- Aspiration of peritoneal fluid from the cul-de-sac removes PGE₂, relieves dysmenorrhea, and pelvic pain, and improves fertility.
- Destruction of endometriotic lesions <3cm with diathermy, Nd-YAG laser, or CO₂ vaporization can be attempted.
- Larger lesions and chocolate cysts can be excised.
- Laparoscopic breaking of adhesions relieves dysmenorrhea and chronic pelvic pain. It also restores tubal patency and improves ovulation. Pre-sacral neurectomy can be done simultaneously.

- Laser uterosacral nerve ablation (LUNA) is done for pelvic pain.



Laparoscopic image of chocolate cyst of ovary:



Solution to Question 37:

The most likely diagnosis is bacterial vaginosis.

Nugent's criteria are used to evaluate three types of bacteria in a case of suspected bacterial vaginosis: Mobiluncus, Lactobacillus, and Gardnerella/Bacteroides. Bacteria are graded through the gram stain of vaginal samples.

They are each graded on a scale of 1+ to 4+ where:

- 1+ is ≤ 1 cell per field

- 2+ is 1-5 cells per field
- 3+ is 6-30 cells per field
- 4+ is ≥ 30 cells per field

Lactobacillus and Bacteroides/Gardnerella are given scores between 0-4, but Mobiluncus is graded only from 0-2. The total scores are then calculated and interpreted as follows:

- 0-3 - Normal
- 4-6 - Intermediate bacterial count
- 7-10 - Bacterial vaginosis

Solution to Question 38:

After ovulation, the released egg is arrested in meiosis II.

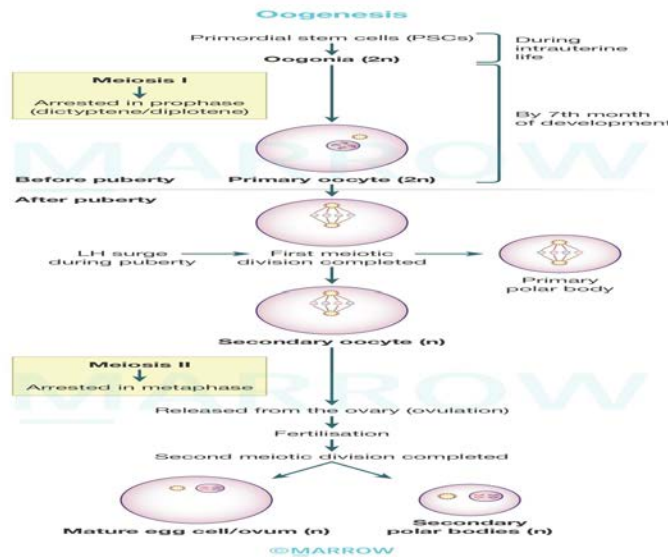
The released egg is in the secondary oocyte stage, having completed meiosis I. It then enters meiosis II, but this process is arrested at metaphase II until fertilization occurs. If fertilization takes place, meiosis II will be completed; otherwise, the secondary oocyte remains arrested in meiosis II.

Oogenesis is the process wherein oogonia differentiate into mature oocytes. It begins before birth. At birth, there are approximately 2 million primordial follicles. Primordial germ cells divide by mitosis to form oogonia.

The majority of the oogonia divide by mitosis, however, some enter meiosis I and get arrested in the diplotene stage of prophase and are now called primary oocytes. These primary oocytes remain arrested in this manner until puberty. Primary oocytes haven't completed meiosis, hence they're diploid. The primary oocyte along with surrounding epithelial cells forms the primordial follicle.

Meiosis I is completed after LH surge during puberty to form the secondary oocyte. The cell then enters meiosis II but arrests in metaphase approximately 3 hours before ovulation.

If fertilization does not occur, the secondary oocyte fails to complete the second meiotic division and degenerates about 24 hours after ovulation. If fertilization occurs, penetration by the sperm signals the secondary oocyte to complete meiosis II. This results in one mature oocyte (ovum) and another smaller cell called the second polar body that is shed off.



Solution to Question 39:

The given clinical scenario as well as the specimen suggests that this is a case of leiomyoma uteri or fibroid uterus. The specimen is of a leiomyoma.

Gross specimen appearance:

- The uterine body/corpus is enlarged and deformed.
- The cut surface shows multiple nodules, well-circumscribed, firm, and white-greyish with a whorled appearance formed by the intersecting bundles of brown muscle fibres and white fibrous tissue.
- Leiomyomas can also present bleeding areas, cystic degeneration, and calcifications. The uterine cavity is compressed.
- A typical myoma is a well-circumscribed tumor with a pseudo capsule and firm inconsistency. The cut surface has a whorled appearance.
- The vessels that supply blood to the tumor lie in the capsule and send radial branches into the tumor.
- Therefore, the central portion of the tumour receives the least blood supply and degeneration is noticeable early in this part of the tumour.
- On the other hand, calcification begins at the periphery and spreads inwards along the vessels.

Risk factors:

Increased risk of leiomyomas is seen in nulliparous women, those with obesity, and individuals in a hyper-estrogenic state, such as in conditions like endometriosis, endometrial hyperplasia, or endometrial carcinoma. Black women are also at higher risk.

Conversely, the risk is decreased in multiparous women. Recent studies have shown no significant correlation between smoking and the development of leiomyomas.

Symptoms of fibroid uterus:

Most women with leiomyomas are asymptomatic, but those affected may experience bleeding, pain, pressure, or infertility. Symptom risk increases with the size and number of myomas. Heavy menstrual bleeding (HMB) is common, linked to dilated endometrial venules and dysregulated vasoactive growth factors. When these engorged venules are disrupted during menstruation, bleeding can overwhelm normal hemostatic mechanisms, causing HMB.

Treatment:

Medical therapy for fibroids includes mifepristone, GnRh analogs, Aromatase inhibitors

Drugs to reduce the size of the fibroid

- Antiprogestosterone – Mifepristone/RU-486
- Danazol (Usually not used due to androgenic side effects)
- GnRH analogs
- Agonists – goserelin, leuprorelin, nafarelin, buserelin
- Antagonists – cetrorelix, ganirelix
- Levonorgestrel releasing intrauterine system
- Selective progesterone receptor modulator (SPRM's) – Asoprisnil

Other management options

- Uterine artery embolization
- Myomectomy
- Hysterectomy

Other Options:

Option B: Myomatous Polyp: They are typically found on the endometrial lining and would not present as a fibroid embedded in the wall of the uterus.

Option C: Endometrioma: An endometrioma is an ovarian cyst filled with endometrial tissue. It would not be found within the uterine wall but rather on the ovaries.

Option D: Adenomyoma: Adenomyoma is a localized area of endometrial tissue within the myometrium (muscle layer) of the uterus. While it can cause similar symptoms, it does not typically present as a discrete mass protruding from the uterine wall like an intramural fibroid.

Solution to Question 40:

The most likely diagnosis is endometrioid carcinoma. It is the most common type of endometrial cancer, particularly associated with genetic mutations such as PTEN, KRAS, CTNNB1 (beta-catenin), and MSI (microsatellite instability). These genetic alterations are characteristic of Type I endometrial cancers, which have a better prognosis compared to other types.

The histopathological examination (HPE) shows glandular structures that resemble normal endometrial glands but are more crowded and irregular. The cells show nuclear atypia with mitotic activity.

This cancer is generally hormone-dependent (Type I endometrial cancer) and often carries a better prognosis compared to other forms of endometrial cancer. The presence of PTEN and KRAS mutations are indicators of this type of malignancy, which often grows slowly and is diagnosed early due to the symptoms of postmenopausal bleeding.

Endometrial cancer occurs in a state of estrogen excess and is therefore mostly seen in:

- nulliparous women
- early menarche
- late menopause
- PCOS
- HRT
- estrogen-producing tumors
- Hypertension, diabetes, and obesity (adipose tissue produces estrogen E1) together form the corpus cancer syndrome.

Endometrial biopsy is the investigation of choice and if positive should be followed by hysteroscopy with fractional curettage (gold standard) for assessing grade and histology.

Other options:

Option A: Mixed Müllerian Tumor: Also known as carcinosarcoma, this tumor shows both epithelial and mesenchymal components. It is less commonly associated with the mutations listed (PTEN, KRAS, etc.) and typically presents with more aggressive features than endometrioid carcinoma.

Option B: Leiomyosarcoma: This is a malignant tumor of smooth muscle, typically arising from the myometrium, and is not associated with genetic mutations (PTEN, KRAS, CTNNB1).

Option D: Serous Cystadenoma of Ovary: A benign ovarian tumor, this condition would not typically involve the genetic markers found in endometrial carcinoma and would present differently from endometrial cancer. Serous tumors of the ovary are more often associated with TP53 mutations, not PTEN or KRAS.

Solution to Question 41:

Misoprostol and mifepristone are administered during first trimester to induce medical termination of pregnancy.

200mg of mifepristone is administered orally on day 1. The minimum recommended interval between mifepristone & misoprostol is 24 hours. 24- 48 hours later, misoprostol 800 mcg is administered by the vaginal, sublingual, or buccal route. Clinical confirmation of MTP is done on day 14.

The various methods of termination of pregnancy are:

Misoprostol and Methotrexate regimen can also be used, but take a longer time of action.

The Medical Termination of Pregnancy Amendment Bill, 2020 has extended the period of termination of pregnancy up to 24 weeks for special categories of women, including survivors

of rape, victims of incest, and other vulnerable women (like differently-abled women, minors). The upper gestation limit is not applicable in cases of substantial fetal abnormalities.

Timing	Medical methods	Surgical methods
1st trimester(up to 12 weeks)	MifepristoneMifepristoneand misoprostolMethotrexate and misoprostolTamoxifen and misoprostol	Menstrual regulationVacuum aspiration (MVA/EVA)Suction evacuation and/or curettageDilatation and evacuation - Rapid methodSlow method
2nd trimester(13 - 20 weeks)	MisoprostolMifepristone and misoprostolDinoprostone(option A)Carboprost	Dilatation and evacuationIntrauterine instillationofhypertonic solutions(option D)Oxytocin infusion(option C)Hysterotomy

Solution to Question 42:

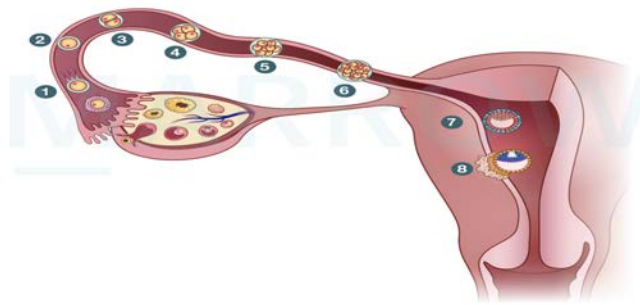
The embryo takes approximately 8 days to travel from point 1 (fertilization) to point 8 (implantation) (typically 6-7 days).

Timeline of implantation:

- 3-4 days after fertilization: The morula (12-16 cells) enters the uterus.
- 4th day after fertilization: The advanced morula reaches the uterine lumen.
- 4-5 days after fertilization: The early blastocyst stage is seen.
- 6-7 days after fertilization: The blastocyst implants in the uterus. Implantation corresponds to 20-22 days of the menstrual cycle.

The blastocyst normally implants in the functional layer of the uterine endometrium. In the secretory phase, the endometrium comprises three layers: stratum compactum, stratum spongiosum, and stratum basale. The first two layers, collectively known as the stratum functionale or functional layer, are crucial for implantation. If implantation does not occur, this functional layer is shed off during menstruation. However, if implantation occurs, the functional layer is retained as the decidua, marking the beginning of a successful pregnancy.

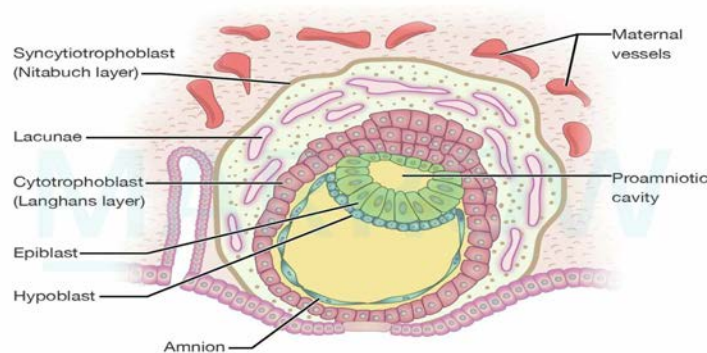
Fertilization and implantation



1	2	3-6	7	8
Fertilization	Zygote formation	Cleavage	Blastocyst formation	Implantation

©MARROW

Implantation



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Solution to Question 43:

For women with active genital herpes at the time of labor, elective cesarean section is the preferred mode of delivery to prevent neonatal herpes transmission. A standard lower segment cesarean section (LSCS) is typically performed unless there are other complications necessitating a different approach.

HSV infection in pregnancy can be caused by either HSV-1 or HSV-2 and transmitted by sexual contact. The infection can be primary, nonprimary, first episode, and recurrent.

HSV infection (both HSV 1 and 2) during pregnancy has a risk of vertical transmission of the infection to the fetus. The risk is higher in the primary than in the recurrent infection. The maximum risk of fetal infection is during labor, although it can occur in utero and post-natally from direct contact.

Suspicious lesions should be sent for PCR testing or culture. Antiviral treatment with acyclovir should be given if found positive. Women with no active genital lesions, asymptomatic, or those with active lesions in non-genital areas can undergo normal labor and delivery.

Management of HSV infection during pregnancy:

- Oral acyclovir or valacyclovir for primary or recurrent infections during pregnancy.
- IV acyclovir followed by oral acyclovir for severe or disseminated infection
- Oral analgesics or topical anesthetics for pain relief
- Catheterization for urinary retention

Cesarean delivery reduces the risk of vertical transmission during labor and is indicated in:

- Cases of active genital lesions - both primary and recurrent
- Prodromal symptoms like vulval itching or burning

Other options:

Option B: Classical Cesarean Section: Classical cesarean section involves a vertical incision on the upper part of the uterus and is generally reserved for specific cases such as abnormal uterine anatomy or placenta previa

Option C and D: Trial of Labor (VBAC) and Induction of Labor: Vaginal birth after cesarean (VBAC) and induction of labor are contraindicated in women with active genital herpes due to the risk of transmitting the virus to the newborn during vaginal delivery.

Solution to Question 44:

In cases of adnexal masses, particularly when considering the possibility of ovarian cancer or germ cell tumors, the following tumor markers are useful:

Other options:

CEA (Carcinoembryonic antigen): Used for gastrointestinal cancer.

CA15-3: Marker associated with breast cancer.

CA19-9: Marker associated with pancreatic and gastrointestinal cancers

Tumor markers	Ovarian tumors
AFP (alpha-fetoprotein)	Yolk sac tumor
LDH (lactate dehydrogenase)	Dysgerminoma
HCG (human chorionic gonadotropin)	Choriocarcinoma
CA125	Epithelial cell tumor

Solution to Question 45:

The image shows Piper's forceps, identified by its distinct perineal curve. It is used to assist in the delivery of the after-coming head of breech.

It has a cephalic curve, reverse pelvic curve, and long parallel shanks that permit the baby's body to rest against it during head delivery. It may be applied electively or when the Mauriceau maneuver cannot be accomplished easily.

Other options:

Option A: Wrigley forceps: These are short forceps generally used for low or outlet deliveries but are not suitable for delivering the after-coming head in breech deliveries.

Option B: Kielland forceps: These are rotational forceps used to assist in malpositioned heads during cephalic deliveries.

Option C: Das forceps: These forceps are used for cephalic deliveries.